

Unified Fractal Information Geometry of Entropy Operators and Curvature-Driven Dynamics

A Unified Geometric Principle of Information: Fractal Entropy, Operator Spectra, and Curvature-Driven Collapse in Multiscale Complex Systems

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Abstract

A unified formulation of information geometry, fractal spectral structure, and entropy-driven dynamics is developed within the framework of the Information Manifold Theory of Fractal Information, Entropy Projections, and Curvature-Driven Collapse (PFIT–En-topia–Pellis framework). Information states are represented as points on a differentiable manifold $\mathcal{M}$ endowed with a Fisher–Rao-type metric extended by multiscale fractal corrections, inducing a scale-dependent geometric structure. Entropy is promoted from a scalar functional to a covariant field defined on $\mathcal{M}$, whose gradients generate dynamical flows governed by curvature-sensitive evolution equations. Information propagation is characterized by geodesic motion on the manifold, where deviations from equilibrium correspond to curvature accumulation and geometric deformation of the statistical state space. A class of entropy projection operators is introduced, providing a nonlinear mapping from microscopic fluctuations to emergent macrostates via spectral filtering on the information manifold. These operators generate hierarchical renormalization flows linking local curvature invariants to global stability properties across scales. Curvature divergence is identified as a universal mechanism of dynamical collapse, establishing a geometric criterion for critical transitions, decoherence phenomena, and phase instability in complex multiscale systems. In this regime, scalar curvature singularities correspond to abrupt structural reorganization of the underlying informational dynamics. The resulting framework establishes a geometric equivalence between entropy, information flow, and stability structure, unifying fractal spectral dynamics and curvature-induced collapse within a single manifold-based formalism.

Keywords: Information Manifold Theory , Fractal Information , Entropy Projections , Curvature-Driven Collapse , Information Geometry , Spectral Dynamics , Multiscale Systems , Renormalization Flows , Critical Transitions , Geometric Stability.

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1 Introduction

The conceptual evolution of information as a fundamental physical and mathematical entity has progressed through successive layers of abstraction, beginning with classical communication theory and culminating in contemporary geometric, spectral, and multiscale formulations. In its original formulation, information was rigorously quantified as a measure of uncertainty reduction in stochastic communication systems, with entropy emerging as the central invariant governing optimal encoding, transmission, and inference processes [1–3]. This scalar characterization of information, while foundational, is inherently limited in its ability to capture structured dependencies, long-range correlations, and multiscale organization present in complex natural systems. To address these limitations, generalized entropy frameworks were introduced, extending classical additive structures to non-extensive and deformed statistical regimes capable of modeling systems far from equilibrium. Rényi and Tsallis entropies, along with related generalized information measures, provide a broader functional landscape in which statistical dependence, heavy-tailed distributions, and anomalous scaling behaviors can be consistently represented [4–6]. These developments marked a first step toward recognizing entropy not merely as a scalar quantity, but as part of a richer functional geometry.

A major conceptual advancement arises from the recognition that probability distributions admit a natural differential geometric structure. In this context, the Fisher–Rao metric endows the space of probability measures with a Riemannian manifold structure, where statistical distinguishability is encoded in curvature and geodesic distance [7–9]. Within this framework, inference processes acquire a geometric interpretation as geodesic flows, while curvature becomes a quantitative descriptor of statistical sensitivity, robustness, and information distinguishability. Parallel to these developments, nonlinear dynamics and fractal geometry have revealed that complex systems universally exhibit scale-invariant structure, self-similarity, and hierarchical organization across multiple temporal and spatial resolutions [10–15]. Such systems cannot be adequately described using classical linear or Euclidean assumptions, as their intrinsic dynamics are governed by multiscale interactions and emergent critical phenomena. This realization has led to the incorporation of renormalization group theory as a fundamental analytical tool for describing scale transformations and universality classes in complex systems [16–21].

From a more abstract functional-analytic perspective, dynamical evolution in both physical and statistical systems can be formulated in terms of operator flows acting on infinite-dimensional Hilbert or Banach spaces. The spectral properties of these operators determine stability regimes, resonance structures, and asymptotic convergence behavior, providing a unifying language for quantum mechanics, statistical physics, and nonlinear dynamical systems [22–24]. This operatorial viewpoint naturally extends to biological and physiological systems, where fractal organization and multiscale feedback mechanisms govern functional adaptation and robustness [25–27]. Information-theoretic divergence measures, particularly those based on relative entropy and Kullback–Leibler structures, introduce a variational framework in which statistical evolution can be interpreted as a gradient flow on a curved informational landscape [28, 29]. Within this interpretation, entropy gradients define directional forces of informational descent, while curvature modulates the stability and anisotropy of such flows. This establishes a deep connection between thermodynamic irreversibility, statistical inference, and geometric structure.

In modern theoretical physics, these geometric and informational principles have been extended toward attempts to derive physical constants, interaction structures, and cosmological parameters from deeper invariant principles [30–40]. Such approaches suggest that physical laws may not be fundamental axioms but emergent manifestations of underlying informational or geometric constraints operating at a more primitive level of description. The incorporation of fractal geometry into this conceptual framework further enhances its expressive power, revealing that many natural systems exhibit nested hierarchical structures characterized by scale invariance and recursive self-organization [41–55]. In such systems, macroscopic behavior emerges from iterative transformations of microscopic rules, leading to effective dynamics that are best described in terms of renormalized or coarse-grained variables.

Biological, neurological, and cognitive systems provide empirical realizations of these principles, where fractal organization governs neural connectivity, genomic regulation, and physiological variability [56–70]. These observations strongly indicate that information processing in living systems is inherently multiscale and geometry-dependent, with functional stability arising from spectral and fractal constraints rather than purely local interactions. In quantum, gravitational, and cosmological contexts, spectral and operator-theoretic approaches have revealed that stability,

phase transitions, and decoherence phenomena may be understood as manifestations of spectral restructuring in underlying dynamical operators [71–85]. Within this perspective, collapse phenomena correspond not to arbitrary discontinuities but to structured geometric transitions in the spectral configuration of the system. More recent advances have extended these ideas into unified frameworks spanning physics, biology, and complex systems theory, where multiscale spectral instability functions as a universal precursor to critical transitions [86–105]. In such formulations, predictive structure emerges from the analysis of curvature accumulation and spectral concentration across hierarchical scales.

Finally, a convergent body of theoretical and applied research suggests that curvature singularities, spectral instability, and fractal renormalization processes collectively define a universal mechanism of collapse and reorganization in complex systems [106–126]. Within this interpretation, the divergence of geometric invariants on an information manifold corresponds to macroscopic transitions such as phase changes, decoherence events, biological failure modes, and systemic instability across engineered networks. Within this context, the **PFIT–En-topia–Pellis framework** formalizes a unified differential geometric structure in which entropy, information flow, and stability are not independent constructs, but emergent projections of a single underlying fractal information manifold endowed with curvature-driven dynamics, spectral organization, and multiscale renormalization structure.

2 Related Work

The development of a unified informational manifold framework requires a synthesis of information theory, statistical physics, differential geometry, fractal mathematics, and spectral operator theory. Over the last decades, these domains have converged toward a common structural language in which entropy, curvature, and multiscale dynamics are treated as mutually dependent objects rather than isolated constructs. The following subsections summarize the main theoretical pillars underlying the PFIT–En-topia–Pellis framework.

2.1 Foundations of Information Theory and Entropy

The modern theory of information originates from Shannon’s axiomatic framework, in which uncertainty in a discrete communication system is quantified via the entropy functional [1]. For a probability mass function $p(x)$ defined on a finite alphabet $\mathcal{X}$, Shannon entropy is given by

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x), \quad (1)$$

which emerges as the unique functional (up to positive affine transformations) satisfying continuity, maximality at uniformity, and additivity under independent composition.

In continuous domains, the differential entropy extends this notion:

$$h(X) = - \int_{\Omega} p(x) \log p(x) dx, \quad (2)$$

although its interpretation requires care due to coordinate dependence. This motivates the introduction of invariant formulations via relative entropy (Kullback–Leibler divergence),

$$D_{\text{KL}}(p \parallel q) = \int_{\Omega} p(x) \log \frac{p(x)}{q(x)} dx, \quad (3)$$

which plays a central role in statistical inference, information geometry, and variational optimization.

Within Jaynes’ maximum entropy principle, equilibrium probability distributions arise as constrained variational extrema of the entropy functional [3]. This can be formalized through a Lagrangian functional:

$$\mathcal{L}[p] = - \int p(x) \log p(x) dx + \lambda_0 \left(\int p(x) dx - 1 \right) + \sum_{i=1}^m \lambda_i \left(\int f_i(x) p(x) dx - c_i \right), \quad (4)$$

whose stationarity condition $\delta \mathcal{L}[p] = 0$ yields the exponential family:

$$p^*(x) = \frac{1}{Z(\boldsymbol{\lambda})} \exp \left(- \sum_{i=1}^m \lambda_i f_i(x) \right), \quad (5)$$

with partition function

$$Z(\boldsymbol{\lambda}) = \int \exp \left(- \sum_{i=1}^m \lambda_i f_i(x) \right) dx. \quad (6)$$

This variational structure reveals entropy as a generating functional for macroscopic statistical laws. In particular, thermodynamic observables are recovered via conjugate relations:

$$\frac{\partial \log Z}{\partial \lambda_i} = -\mathbb{E}[f_i(x)]. \quad (7)$$

Beyond Shannon's additive entropy, generalized entropy classes such as Rényi and Tsallis entropy extend the formalism into non-extensive, hierarchical, and multifractal regimes [4–6]. The Rényi entropy of order $\alpha \neq 1$ is defined as

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \left(\sum_{x \in \mathcal{X}} p(x)^\alpha \right), \quad (8)$$

recovering Shannon entropy in the limit $\alpha \rightarrow 1$:

$$\lim_{\alpha \rightarrow 1} H_\alpha(X) = H(X). \quad (9)$$

Similarly, Tsallis entropy introduces a deformation parameter q controlling non-extensivity:

$$S_q(X) = \frac{1}{q-1} \left(1 - \sum_{x \in \mathcal{X}} p(x)^q \right), \quad (10)$$

with pseudo-additivity under independence:

$$S_q(A \cup B) = S_q(A) + S_q(B) + (1-q)S_q(A)S_q(B). \quad (11)$$

These generalized entropies naturally encode multifractal scaling through escort distributions:

$$P_q(x) = \frac{p(x)^q}{\sum_{x'} p(x')^q}, \quad (12)$$

which reweight phase space regions according to local concentration of probability mass.

From a geometric perspective, entropy can also be interpreted through the Fisher information metric on the statistical manifold $\mathcal{M}$ of probability distributions:

$$g_{ij}(\theta) = \mathbb{E} \left[\frac{\partial \log p(x; \theta)}{\partial \theta_i} \frac{\partial \log p(x; \theta)}{\partial \theta_j} \right], \quad (13)$$

which induces a Riemannian structure governing distinguishability of neighboring distributions. In this framework, entropy maximization corresponds to geodesic flow toward maximal uncertainty under constraint submanifolds.

The interplay between entropy production and dynamical evolution can be further expressed via the entropy rate:

$$\frac{d}{dt} H(X_t) = - \int \partial_t p(x, t) \log p(x, t) dx, \quad (14)$$

which, under suitable Markovian dynamics, connects to the Fokker–Planck operator and irreversible thermodynamic flow.

Collectively, these formulations establish entropy not merely as a scalar measure of uncertainty, but as a variational, geometric, and dynamical object that governs inference, statistical equilibrium, and the emergent structure of complex systems.

2.2 Information Geometry and Statistical Manifolds

Information geometry endows the space of probability distributions with a differential-geometric structure, identifying parametric families $\{p(x|\theta)\}_{\theta \in \Theta}$ as points on a smooth statistical manifold $\mathcal{M}$ equipped with an intrinsic metric structure [7, 9]. In this setting, inference becomes a geometric flow on $\mathcal{M}$ rather than a purely algebraic optimization process.

The fundamental Riemannian structure is given by the Fisher–Rao metric tensor:

$$g_{ij}(\theta) = \mathbb{E}_{x \sim p(x|\theta)} [\partial_i \log p(x|\theta) \partial_j \log p(x|\theta)], \quad (15)$$

which defines an invariant local inner product on the tangent space $T_\theta \mathcal{M}$. Equivalently, it can be expressed as the negative expectation of the Hessian of the log-likelihood:

$$g_{ij}(\theta) = -\mathbb{E} [\partial_i \partial_j \log p(x|\theta)], \quad (16)$$

under suitable regularity conditions.

This metric induces a canonical notion of statistical distance between infinitesimally close distributions:

$$ds^2 = \sum_{i,j} g_{ij}(\theta) d\theta^i d\theta^j, \quad (17)$$

which approximates the Kullback–Leibler divergence up to second order:

$$D_{\text{KL}}(p(x|\theta) \| p(x|\theta + d\theta)) = \frac{1}{2} ds^2 + \mathcal{O}(\|d\theta\|^3). \quad (18)$$

Within this geometric structure, statistical inference acquires a variational interpretation as geodesic optimization. The shortest path between two distributions corresponds to a geodesic $\gamma(t) \subset \mathcal{M}$ satisfying:

$$\frac{d^2\theta^k}{dt^2} + \Gamma_{ij}^k(\theta) \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = 0, \quad (19)$$

where Γ_{ij}^k are the Levi–Civita connection coefficients associated with the Fisher metric.

Amari’s information geometry generalizes this structure by introducing dual affine connections (∇, ∇^*) , which are not necessarily metric-compatible but are dual with respect to the Fisher information tensor [7]. These connections define two distinct families of geodesics: exponential-type and mixture-type geodesics, given respectively by:

$$\theta(t) = \theta_0 + t(\theta_1 - \theta_0) \quad (\text{mixture geodesics}), \quad (20)$$

$$\eta(t) = \eta_0 + t(\eta_1 - \eta_0) \quad (\text{exponential geodesics}), \quad (21)$$

where η denotes dual affine coordinates related via Legendre transformation.

The dual flatness condition arises when both connections have vanishing curvature tensors:

$$R_{ijk}^{(\nabla)} = 0, \quad R_{ijk}^{(\nabla^*)} = 0, \quad (22)$$

which allows the existence of globally defined potential functions $\psi(\theta)$ and $\phi(\eta)$ satisfying:

$$g_{ij} = \frac{\partial^2 \psi(\theta)}{\partial \theta^i \partial \theta^j} = \frac{\partial^2 \phi(\eta)}{\partial \eta^i \partial \eta^j}. \quad (23)$$

In this formulation, divergences are naturally interpreted as generalized squared distances on $\mathcal{M}$. In particular, the Bregman divergence induced by a convex potential ψ takes the form:

$$D_\psi(\theta_1 \| \theta_2) = \psi(\theta_1) - \psi(\theta_2) - \langle \nabla \psi(\theta_2), \theta_1 - \theta_2 \rangle, \quad (24)$$

which coincides with Kullback–Leibler divergence in exponential families.

Nielsen’s geometric framework further refines this interpretation by embedding divergences into dually flat manifolds, where geodesic triangles obey generalized Pythagorean relations:

$$D(p \| r) = D(p \| q) + D(q \| r), \quad (25)$$

whenever p, q, r satisfy orthogonality conditions in dual affine coordinates [8].

Curvature plays a central role in controlling statistical complexity. The Riemann curvature tensor associated with $(\mathcal{M}, g)$ quantifies the nonlinearity of inference:

$$R_{ijk}^\ell = \partial_i \Gamma_{jk}^\ell - \partial_j \Gamma_{ik}^\ell + \Gamma_{jk}^m \Gamma_{im}^\ell - \Gamma_{ik}^m \Gamma_{jm}^\ell, \quad (26)$$

and its contraction yields the Ricci curvature:

$$\text{Ric}_{ij} = R_{ikj}^k. \quad (27)$$

In strongly curved statistical manifolds, inference trajectories deviate significantly from linear extrapolation, leading to curvature-driven concentration of measure and emergent phase transitions in parameter space. This effect can be formalized via the geodesic deviation equation:

$$\frac{D^2 \xi^i}{dt^2} + R_{jkl}^i \frac{d\theta^j}{dt} \xi^k \frac{d\theta^l}{dt} = 0, \quad (28)$$

which governs the instability of neighboring inference paths.

Collectively, the Fisher–Rao geometry, dual affine structure, and curvature tensors establish a unified framework in which statistical inference is reinterpreted as a nonlinear dynamical system on a curved manifold of probability distributions, where information flow, entropy production, and learning dynamics are governed by intrinsic geometric invariants of $\mathcal{M}$.

2.3 Fractal Geometry and Scale Invariance

Fractal geometry, as introduced by Mandelbrot, formalizes the description of structures exhibiting self-similarity across scales and non-integer dimensionality [10]. In such systems, geometric complexity is not resolved at a single characteristic scale but distributed across an infinite hierarchy of magnifications, leading to scale-invariant measure-theoretic structures.

A canonical scaling law is expressed through the box-counting relation:

$$N(\epsilon) \sim \epsilon^{-D_f}, \quad (29)$$

where $N(\epsilon)$ denotes the number of covering elements of size ϵ , and D_f is the fractal (Hausdorff) dimension, defined more formally as:

$$D_f = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)}. \quad (30)$$

More generally, fractal measures can be defined via scaling of probability distributions supported on irregular sets $\mathcal{F} \subset \mathbb{R}^n$, where the local mass distribution obeys:

$$\mu(B_\epsilon(x)) \sim \epsilon^{D_\mu(x)}, \quad (31)$$

introducing a spatially dependent multifractal spectrum $D_\mu(x)$.

To capture hierarchical heterogeneity, the multifractal formalism introduces partition functions of the form:

$$Z(q, \epsilon) = \sum_i \mu_i(\epsilon)^q, \quad (32)$$

where $\mu_i(\epsilon)$ are box probabilities at resolution ϵ . The scaling exponent $\tau(q)$ is defined by:

$$Z(q, \epsilon) \sim \epsilon^{\tau(q)}, \quad (33)$$

and the multifractal spectrum is obtained via Legendre transform:

$$f(\alpha) = \inf_q (q\alpha - \tau(q)), \quad (34)$$

with α representing local Hölder exponents.

This formalism naturally connects to non-extensive entropy measures, where the Rényi entropy of order q captures scaling diversity across hierarchical levels:

$$H_q(\epsilon) = \frac{1}{1-q} \log \left(\sum_i \mu_i(\epsilon)^q \right), \quad (35)$$

yielding asymptotic scaling behavior:

$$H_q(\epsilon) \sim D_q \log(1/\epsilon), \quad (36)$$

where D_q denotes the generalized fractal dimensions spectrum.

In dynamical systems theory, fractal structures arise as invariant sets of nonlinear maps $T : \mathcal{X} \rightarrow \mathcal{X}$, where trajectories converge to strange attractors $\mathcal{A}$ with non-integer dimensionality:

$$\mathcal{A} = \bigcap_{n=0}^{\infty} T^n(\mathcal{X}). \quad (37)$$

The sensitivity to initial conditions is quantified by Lyapunov exponents. For a trajectory $x(t)$, perturbations evolve as:

$$\|\delta x(t)\| \approx e^{\lambda t} \|\delta x(0)\|, \quad (38)$$

where $\lambda > 0$ indicates chaotic divergence and exponential instability of phase-space trajectories.

For continuous dynamical systems $\dot{x} = F(x)$, the Lyapunov spectrum $\{\lambda_i\}$ is defined via the Jacobian flow:

$$\frac{d}{dt} \delta x = J_F(x) \delta x, \quad (39)$$

with $J_F(x) = \frac{\partial F}{\partial x}$ governing local linearization of the nonlinear flow.

The Kaplan–Yorke dimension provides a bridge between dynamical instability and fractal geometry:

$$D_{KY} = j + \frac{\sum_{i=1}^j \lambda_i}{|\lambda_{j+1}|}, \quad (40)$$

where j is the largest index such that $\sum_{i=1}^j \lambda_i \geq 0$.

From a geometric perspective, fractal sets can be interpreted as limiting objects of iterative renormalization transformations:

$$\mathcal{R}[\mathcal{S}_n] = \mathcal{S}_{n+1}, \quad \mathcal{S}^* = \lim_{n \rightarrow \infty} \mathcal{R}^n[\mathcal{S}_0], \quad (41)$$

where $\mathcal{R}$ encodes scale decimation and coarse-graining operations.

In physical systems, scale invariance manifests through power-law correlations of the form:

$$C(r) = \langle \phi(x) \phi(x+r) \rangle \sim r^{-\gamma}, \quad (42)$$

indicating the absence of characteristic length scales and implying criticality.

Renormalization group flow equations provide a dynamical description of scale dependence in parameter space:

$$\frac{dg_i}{d \log \mu} = \beta_i(\mathbf{g}), \quad (43)$$

where μ is the resolution scale and β_i encodes the flow of effective couplings.

Within nonlinear dynamics, fractal structures emerge from bifurcation cascades, such as period-doubling routes to chaos governed by Feigenbaum universality:

$$\delta = \lim_{n \rightarrow \infty} \frac{a_n - a_{n-1}}{a_{n+1} - a_n}, \quad (44)$$

where $\delta \approx 4.6692$ is a universal scaling constant. Collectively, these results establish fractal geometry as a unifying mathematical language linking nonlinear dynamics, scale invariance, and hierarchical structure formation.

2.4 Critical Phenomena and Renormalization

Critical phenomena establish that systems near phase transitions exhibit universal behavior characterized by scale invariance and emergent self-similarity, independent of microscopic details [16]. In this regime, physical observables become governed by fixed-point structures of renormalization group (RG) flows, where coarse-graining transformations preserve macroscopic statistical structure while systematically integrating out short-range degrees of freedom.

Wilson's renormalization framework formalizes this scale dependence through flow equations in coupling-constant space:

$$\frac{dg_i(\ell)}{d \ln \ell} = \beta_i(\mathbf{g}), \quad (45)$$

where ℓ denotes the observation scale and β_i are nonlinear functions encoding the response of effective couplings to scale transformations. Fixed points g^* satisfy:

$$\beta_i(g^*) = 0, \quad (46)$$

defining universality classes that are invariant under RG transformations.

Linearization around a fixed point yields:

$$\frac{d\delta g_i}{d \ln \ell} = M_{ij} \delta g_j, \quad (47)$$

where the stability matrix is given by:

$$M_{ij} = \left. \frac{\partial \beta_i}{\partial g_j} \right|_{g=g^*}. \quad (48)$$

The eigenvalues λ_i of M_{ij} determine relevant, irrelevant, and marginal directions in coupling space, thereby classifying critical behavior.

Correlation functions near criticality exhibit universal scaling laws. For an order parameter field $\phi(x)$, the two-point correlation function satisfies:

$$C(r) = \langle \phi(x) \phi(x+r) \rangle \sim r^{-2\Delta_\phi}, \quad (49)$$

where Δ_ϕ is the scaling dimension of the field. Equivalently, introducing a correlation length ξ , one obtains:

$$C(r) \sim \frac{1}{r^{d-2+\eta}} e^{-r/\xi}, \quad (50)$$

with anomalous dimension η and spatial dimension d . At criticality, $\xi \rightarrow \infty$, leading to pure power-law decay.

The divergence of the correlation length near a critical point is governed by a critical exponent ν :

$$\xi \sim |T - T_c|^{-\nu}, \quad (51)$$

which encodes universal scaling across distinct physical systems.

Thermodynamic observables exhibit analogous scaling relations. The free energy density f satisfies a homogeneity relation under RG rescaling:

$$f(\lambda t, \lambda^{y_h} h) = \lambda^d f(t, h), \quad (52)$$

where $t = (T - T_c)/T_c$ is the reduced temperature, h is an external field, and y_h is a scaling exponent.

From this structure, critical exponents obey hyperscaling relations such as:

$$2 - \alpha = d\nu, \quad \gamma = (2 - \eta)\nu, \quad \beta = \frac{\nu}{2}(d - 2 + \eta), \quad (53)$$

establishing deep constraints between apparently independent macroscopic observables.

Beyond equilibrium systems, self-organized criticality (SOC) extends the RG paradigm to driven dissipative systems that spontaneously evolve toward critical states without parameter fine-tuning [18, 20]. Such systems are characterized by avalanche dynamics obeying scale-free distributions:

$$P(s) \sim s^{-\tau}, \quad P(t) \sim t^{-\alpha}, \quad (54)$$

where s denotes avalanche size and t duration.

The branching process representation of SOC systems introduces an effective branching ratio σ , with criticality emerging at:

$$\sigma = \sum_k k p_k = 1, \quad (55)$$

corresponding to a marginally stable state between extinction and runaway growth.

In nonequilibrium thermodynamics, entropy production $\dot{S}$ plays a central role in governing approach to critical regimes. For a probability density $p(x, t)$ evolving under a Fokker–Planck operator $\mathcal{L}$, entropy production can be expressed as:

$$\dot{S} = - \int \mathcal{L}[p(x, t)] \log p(x, t) dx, \quad (56)$$

which is strictly non-negative under detailed balance breaking conditions.

The RG flow itself can be interpreted as a dynamical system in functional space, where trajectories $g(\ell)$ evolve under a nonlinear vector field:

$$\frac{dg}{dt} = \beta(g), \quad t = \ln \ell, \quad (57)$$

and critical points correspond to attractors, repellers, or saddle structures in this infinite-dimensional phase space.

In stochastic field theories, criticality emerges from the effective action functional $S[\phi]$, whose RG transformation obeys:

$$S'[\phi] = \mathcal{R}_\ell[S[\phi]], \quad (58)$$

with fixed points satisfying:

$$\mathcal{R}_\ell[S^*] = S^*. \quad (59)$$

This structure establishes a deep equivalence between universality in statistical physics and invariance under information-theoretic coarse-graining.

2.5 Spectral Theory and Operator Dynamics

Operator theory provides a rigorous functional-analytic framework for studying stability, perturbation response, and phase structure in both quantum systems and nonlinear dynamical evolutions. In this setting, physical observables are encoded as linear (or nonlinear) operators acting on a Hilbert or Banach space $\mathcal{H}$, with system behavior determined by the spectral decomposition of these operators [22–24].

Let $\mathcal{L} : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded or unbounded linear operator with spectrum $\sigma(\mathcal{L})$. The spectral decomposition (when applicable) reads:

$$\mathcal{L} = \sum_i \lambda_i P_i + \int_{\sigma_c} \lambda dP(\lambda), \quad (60)$$

where λ_i are discrete eigenvalues, P_i are spectral projectors, and σ_c denotes the continuous spectrum.

Under perturbation $\mathcal{L} \mapsto \mathcal{L}(\epsilon) = \mathcal{L}_0 + \epsilon\mathcal{V}$, eigenvalues evolve according to analytic perturbation theory (when non-degenerate), satisfying:

$$\frac{d\lambda_i}{d\epsilon} = \langle \psi_i(\epsilon), \mathcal{V}\psi_i(\epsilon) \rangle, \quad (61)$$

where $\psi_i(\epsilon)$ are normalized eigenvectors of $\mathcal{L}(\epsilon)$.

In the vicinity of spectral degeneracies, perturbation theory breaks down and eigenvalue trajectories exhibit non-analytic behavior. A generic eigenvalue coalescence (spectral collapse) is characterized by:

$$\lim_{\epsilon \rightarrow \epsilon_c} \lambda_i(\epsilon) = \lambda_j(\epsilon), \quad (62)$$

signaling the formation of exceptional points or branch singularities in the complex spectral plane.

More generally, spectral instability can be quantified via the resolvent operator:

$$R(z; \mathcal{L}) = (zI - \mathcal{L})^{-1}, \quad (63)$$

whose norm governs sensitivity to perturbations. The pseudospectrum is defined as:

$$\sigma_\delta(\mathcal{L}) = \{z \in \mathbb{C} \mid \|R(z; \mathcal{L})\| > \delta^{-1}\}, \quad (64)$$

providing a refined description of near-resonant spectral instability beyond the classical spectrum.

Time evolution in quantum or linear dynamical systems is governed by operator semigroups:

$$\psi(t) = e^{-i\mathcal{H}t}\psi(0), \quad (65)$$

or more generally:

$$\frac{d}{dt}x(t) = \mathcal{L}x(t), \quad x(t) = e^{t\mathcal{L}}x(0), \quad (66)$$

where the long-term stability is fully determined by the spectral abscissa:

$$\alpha(\mathcal{L}) = \sup_{\lambda \in \sigma(\mathcal{L})} \text{Re}(\lambda). \quad (67)$$

Spectral phase transitions arise when the leading eigenvalue crosses critical thresholds:

$$\text{Re}(\lambda_{\max}(\epsilon)) = 0, \quad (68)$$

which separates stable decay regimes from exponential growth, inducing qualitative changes in macroscopic dynamics.

In non-self-adjoint systems, eigenvectors lose orthogonality, and the condition number of the eigenbasis becomes a critical indicator of instability:

$$\kappa(\mathcal{L}) = \|V\| \|V^{-1}\|, \quad (69)$$

where $\mathcal{L} = V\Lambda V^{-1}$. Large $\kappa(\mathcal{L})$ indicates high spectral sensitivity even under infinitesimal perturbations.

The dynamics of spectral flow under continuous deformation can be expressed via the spectral current:

$$\frac{d}{d\epsilon} \text{Tr} f(\mathcal{L}(\epsilon)) = \text{Tr} \left(f'(\mathcal{L}) \frac{d\mathcal{L}}{d\epsilon} \right), \quad (70)$$

which connects operator perturbations with functional calculus on the spectrum.

In many-body and field-theoretic settings, spectral density functions encode the distribution of eigenmodes:

$$\rho(\lambda) = \sum_i \delta(\lambda - \lambda_i), \quad (71)$$

or in the continuum limit:

$$\rho(\lambda) = \frac{1}{\pi} \text{Im} \text{Tr} R(\lambda + i0; \mathcal{L}). \quad (72)$$

This spectral density governs thermodynamic observables through partition-function representations:

$$Z = \int e^{-\beta\lambda} \rho(\lambda) d\lambda, \quad (73)$$

establishing a direct bridge between operator spectra and macroscopic statistical behavior.

Within dynamical systems, spectral properties of the Koopman operator $\mathcal{K}$ provide a linear representation of nonlinear flows:

$$(\mathcal{K}f)(x) = f(\Phi_t(x)), \quad (74)$$

where Φ_t is the flow map. The eigenfunctions of $\mathcal{K}$ satisfy:

$$\mathcal{K}\phi_\omega = e^{i\omega t}\phi_\omega, \quad (75)$$

encoding coherent modes of the underlying nonlinear dynamics.

Collectively, these structures demonstrate that spectral dynamics is not merely a linear algebraic property but a deep geometric and analytic mechanism governing stability, phase transitions, and information propagation. Within the PFIT–En-topia–Pellis framework, spectral collapse corresponds to curvature-induced coalescence of operator eigenmodes, where perturbations propagate through the information manifold as geometric deformations of the underlying spectral landscape.

2.6 Fractal Physiology and Complex Systems

Biological systems exhibit multiscale variability that is fundamentally consistent with fractal organization, non-linear dynamical evolution, and information-theoretic constraints. In fractal physiology, temporal and spatial fluctuations in physiological signals are not purely stochastic but instead follow structured scaling laws indicative of long-range correlations and self-similar organization across multiple observational scales [25].

Let $x(t)$ denote a physiological time series (e.g., HRV, neural activity, or metabolic oscillations). Empirical evidence suggests that its autocorrelation function obeys asymptotic power-law decay:

$$C(\tau) = \langle x(t)x(t+\tau) \rangle \sim \tau^{-\gamma}, \quad (76)$$

where $0 < \gamma < 1$ indicates long-memory processes and non-Markovian structure. Equivalently, the power spectral density follows:

$$S(f) \sim f^{-\beta}, \quad (77)$$

where $\beta \approx 1$ corresponds to $1/f$ -type fractal noise commonly observed in physiological regulation.

Fractal dynamics in physiological systems can be characterized through detrended fluctuation analysis (DFA), where the fluctuation function scales as:

$$F(n) \sim n^H, \quad (78)$$

with H denoting the Hurst exponent. Values $H \neq 1/2$ indicate deviation from classical Brownian motion and reveal persistent or anti-persistent regulatory mechanisms.

From a geometric perspective, physiological states can be embedded in a statistical manifold $\mathcal{M}_{\text{bio}}$, where each state corresponds to a probability distribution $p(x|\theta)$ over observable biomarkers. Transitions between physiological regimes are then governed by geodesic flows on $\mathcal{M}_{\text{bio}}$:

$$\frac{d^2\theta^k}{dt^2} + \Gamma_{ij}^k(\theta) \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = 0, \quad (79)$$

indicating that biological adaptation can be interpreted as curvature-constrained motion in information space.

Complexity in biological networks arises from hierarchical interaction architectures, where nodes represent functional units (cells, neurons, organs) and edges encode multiscale coupling strengths. Let $\mathcal{G} = (V, E)$ denote such a network with adjacency matrix A_{ij} . The degree distribution often follows a heavy-tailed law:

$$P(k) \sim k^{-\alpha}, \quad (80)$$

which is characteristic of scale-free organization and robust emergent connectivity.

The structural complexity of such networks can be quantified via entropy measures on graph ensembles. The Shannon entropy of degree distribution is given by:

$$H_{\mathcal{G}} = - \sum_k P(k) \log P(k), \quad (81)$$

while higher-order structure is captured through mutual information between network layers:

$$I(L_1; L_2) = \sum_{i,j} P(i,j) \log \frac{P(i,j)}{P(i)P(j)}. \quad (82)$$

Physiological state transitions can be formulated as probabilistic divergences between distributional representations of health and pathology. The Kullback–Leibler divergence between two physiological states P and Q is defined as:

$$D_{KL}(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}, \quad (83)$$

which generalizes in continuous settings to:

$$D_{KL}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)} dx. \quad (84)$$

This divergence acts as a Lyapunov-like functional in many biological adaptation processes, governing relaxation toward stable attractor states in physiological phase space. In dynamical form, if $p(x, t)$ evolves under a stochastic operator $\mathcal{L}$, then:

$$\frac{d}{dt} D_{KL}(p(x, t)\|p^*(x)) \leq 0, \quad (85)$$

indicating monotonic convergence toward equilibrium distributions under suitable ergodic conditions.

Generalized divergence measures extend this framework to non-extensive biological regimes, where Tsallis-type entropic functionals capture anomalous scaling in metabolic and neural systems:

$$S_q(P) = \frac{1}{q-1} \left(1 - \sum_x P(x)^q \right). \quad (86)$$

At a deeper structural level, physiological dynamics can be modeled as flows on a fractal statistical manifold $\mathcal{F}_{\text{bio}}$, where local dimensionality varies with state:

$$d_{\text{eff}}(x) = \lim_{\epsilon \rightarrow 0} \frac{\log N_x(\epsilon)}{\log(1/\epsilon)}, \quad (87)$$

reflecting heterogeneous scaling of biological subsystems.

Entropy production in living systems is inherently non-equilibrium and can be expressed as:

$$\dot{S}_{\text{tot}} = \dot{S}_{\text{sys}} + \dot{S}_{\text{env}} \geq 0, \quad (88)$$

with biological viability often corresponding to regimes where entropy production is dynamically regulated rather than minimized.

Within this framework, physiological robustness emerges from the interplay between fractal organization, information geometry, and entropy minimization principles. In the PFIT–En-topia–Pellis perspective, biological systems are interpreted as adaptive information manifolds operating near criticality, where health corresponds to stable geodesic flow on a fractal-informational geometry and disease corresponds to curvature-induced deviations and collapse of multiscale coherence.

2.7 Cross-Scale Pattern Formation

Pattern formation theory provides a unifying mathematical description of how coherent macroscopic structures emerge from initially homogeneous or weakly perturbed states through nonlinear interactions, instability amplification, and symmetry breaking mechanisms [29]. In this regime, microscopic fluctuations are amplified via nonlinear feedback, producing spatially ordered structures that persist across scales and often exhibit universality independent of system-specific details.

A canonical description of such processes is given by amplitude equations derived via multiscale perturbation expansions near instability thresholds. Let $A(\mathbf{x}, t)$ denote a slowly varying complex order parameter encoding the envelope of unstable modes. Its evolution is governed by a Ginzburg–Landau-type equation:

$$\partial_t A = \mu A - g|A|^2 A + \nabla^2 A, \quad (89)$$

where μ is the bifurcation parameter controlling instability onset, $g > 0$ is the nonlinear saturation coefficient, and $\nabla^2 A$ encodes spatial diffusion and mode coupling.

Stationary solutions satisfy:

$$0 = \mu A - g|A|^2 A + \nabla^2 A, \quad (90)$$

which yields homogeneous equilibria of the form:

$$|A|^2 = \frac{\mu}{g}, \quad (91)$$

for $\mu > 0$, representing symmetry-broken ordered phases.

More generally, spatial pattern selection can be analyzed via Fourier decomposition:

$$A(\mathbf{x}, t) = \int \hat{A}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}} d\mathbf{k}, \quad (92)$$

where each mode evolves according to:

$$\partial_t \hat{A}(\mathbf{k}, t) = (\mu - |\mathbf{k}|^2) \hat{A}(\mathbf{k}, t) - g\mathcal{N}[\hat{A}], \quad (93)$$

with $\mathcal{N}[\hat{A}]$ denoting nonlinear mode coupling terms responsible for spectral redistribution of energy across scales.

The dispersion relation governing linear stability of homogeneous states is given by:

$$\lambda(\mathbf{k}) = \mu - |\mathbf{k}|^2, \quad (94)$$

implying that instability occurs for modes satisfying $|\mathbf{k}|^2 < \mu$, thereby defining a characteristic emergent length scale:

$$\ell_c \sim \mu^{-1/2}. \quad (95)$$

Near the bifurcation point, the system undergoes a continuous phase transition characterized by spontaneous symmetry breaking of translational or rotational invariance. This can be formalized via an effective potential functional:

$$\mathcal{F}[A] = \int \left(-\mu |A|^2 + \frac{g}{2} |A|^4 + |\nabla A|^2 \right) d\mathbf{x}, \quad (96)$$

whose variational gradient flow yields:

$$\partial_t A = -\frac{\delta \mathcal{F}}{\delta A^*}. \quad (97)$$

Pattern stability is determined by perturbations around equilibrium solutions A_0 , where setting $A = A_0 + \delta A$ leads to linearized dynamics:

$$\partial_t \delta A = \mathcal{L}[A_0] \delta A, \quad (98)$$

with stability operator:

$$\mathcal{L}[A_0] = \mu - 2g|A_0|^2 + \nabla^2. \quad (99)$$

In extended systems, pattern formation is closely tied to modulational instabilities and phase diffusion processes. The slow evolution of phase $\phi(\mathbf{x}, t)$ in $A = |A|e^{i\phi}$ satisfies:

$$\partial_t \phi = D_\phi \nabla^2 \phi, \quad (100)$$

where D_ϕ is the phase diffusion coefficient governing long-wavelength coherence and defect dynamics.

Topological defects such as vortices and dislocations emerge naturally as singularities in the phase field, characterized by quantized winding numbers:

$$\oint_C \nabla \phi \cdot d\mathbf{l} = 2\pi n, \quad n \in \mathbb{Z}. \quad (101)$$

From a multiscale perspective, pattern formation can be interpreted as an RG flow in functional space, where coarse-graining transformations act on amplitude fields:

$$A'(\mathbf{x}) = \mathcal{R}_\ell[A(\mathbf{x})], \quad (102)$$

and fixed points correspond to scale-invariant pattern universality classes.

Within reaction–diffusion systems, a more general formulation involves coupled fields $\mathbf{u}(\mathbf{x}, t)$ obeying:

$$\partial_t \mathbf{u} = \mathbf{D} \nabla^2 \mathbf{u} + \mathbf{F}(\mathbf{u}), \quad (103)$$

where instability arises when eigenvalues of the Jacobian J_F satisfy diffusion-driven sign inversion (Turing instability).

Collectively, these structures demonstrate that cross-scale pattern formation is not merely a consequence of local nonlinear interactions, but a manifestation of deeper variational and spectral principles governing symmetry breaking, mode selection, and emergent spatial organization. Within the PFIT–En-topia–Pellis framework, such patterns are interpreted as stable attractor configurations of a fractal information manifold, where curvature, entropy gradients, and nonlinear coupling jointly determine the emergence of coherent macroscopic order across scales.

2.8 Fractal Information and Pellis Framework

Recent developments in fractal-information theory suggest a deeper unification in which physical constants, geometric structure, and informational content are not independent primitives but emergent projections of a single scale-dependent manifold structure [82]. In this view, the underlying state space is not a smooth Euclidean or Riemannian manifold alone, but a fractal statistical geometry $\mathcal{F}$ endowed with non-integer local dimensionality and scale-dependent measure weights.

Let μ_ϵ denote a resolution-dependent probability measure defined on a covering of $\mathcal{F}$ at scale ϵ . The effective entropy field $\mathcal{S}(\epsilon, x)$ can then be defined as a local functional over fractal partitions:

$$\mathcal{S}(\epsilon, x) = - \sum_{i \in \mathcal{B}_\epsilon(x)} \mu_i(\epsilon) \log \mu_i(\epsilon), \quad (104)$$

where $\mathcal{B}_\epsilon(x)$ denotes the set of boxes covering the neighborhood of x at resolution ϵ .

In this framework, entropy is elevated from a scalar functional to a geometric potential generating curvature on the informational manifold. The informational curvature tensor is therefore defined as a second-order variational response of entropy under local deformations of the measure:

$$\mathcal{R}_{\mu\nu} \sim \nabla_\mu \nabla_\nu \mathcal{S}, \quad (105)$$

where ∇_μ denotes a covariant derivative compatible with the fractal measure structure. More explicitly, this can be expanded as:

$$\nabla_\mu \nabla_\nu \mathcal{S} = \partial_\mu \partial_\nu \mathcal{S} - \Gamma_{\mu\nu}^\lambda \partial_\lambda \mathcal{S}, \quad (106)$$

linking informational curvature directly to connection coefficients induced by scale-dependent probability transport.

The trace of this curvature tensor defines a scalar informational curvature invariant:

$$\mathcal{R} = g^{\mu\nu} \mathcal{R}_{\mu\nu}, \quad (107)$$

which governs global stability properties of the fractal information manifold and acts as a driving potential for scale transitions.

Within this formalism, entropy gradients induce geometric flows analogous to Ricci flow, but defined on informational degrees of freedom:

$$\frac{\partial g_{\mu\nu}}{\partial t} = -2 \mathcal{R}_{\mu\nu} + \kappa \nabla_\mu \nabla_\nu \mathcal{S}, \quad (108)$$

where κ is a coupling constant controlling the strength of entropy-driven geometric deformation.

Spectral collapse phenomena can be embedded into this geometric structure by associating operator eigenvalues $\lambda_i(\epsilon)$ with curvature eigenmodes of the informational manifold. In this representation, eigenvalue coalescence corresponds to curvature singularity formation:

$$\lim_{\epsilon \rightarrow \epsilon_c} \mathcal{R}(\epsilon) \rightarrow \infty \iff \lambda_i(\epsilon) \rightarrow \lambda_j(\epsilon), \quad (109)$$

establishing a direct correspondence between spectral degeneracy and geometric instability.

Fractal scaling further modifies the effective geometry through scale-dependent renormalization of the metric tensor:

$$g_{\mu\nu}(\epsilon) = \epsilon^{-\delta} \tilde{g}_{\mu\nu}, \quad (110)$$

where δ encodes anomalous fractal dimensional corrections and $\tilde{g}_{\mu\nu}$ is the renormalized smooth background metric.

The entropy functional itself admits a variational formulation over fractal measure spaces:

$$\mathcal{S}[\mu] = - \int_{\mathcal{F}} \mu(x) \log \mu(x) d\mu(x), \quad (111)$$

whose functional derivative defines an entropy-driven force field:

$$\frac{\delta \mathcal{S}}{\delta \mu} = -1 - \log \mu(x), \quad (112)$$

governing probability flow across scales.

Coupling this structure with operator dynamics yields a unified evolution equation for informational states:

$$\partial_t \mu = \mathcal{L}[\mu] + \nabla \cdot \left(\mu \nabla \frac{\delta \mathcal{S}}{\delta \mu} \right), \quad (113)$$

where $\mathcal{L}$ represents underlying dynamical transport and the second term encodes entropy-driven geometric diffusion.

Within the PFIT–En-topia–Pellis framework, these constructions collectively imply that information, geometry, and dynamics are not separable entities but mutually dual representations of a single fractal variational principle. In this interpretation, entropy acts as a generating functional for curvature, while curvature governs spectral stability, and spectral structure encodes the emergent organization of multiscale informational reality.

Figure 1 presents a conceptual phase space whose axes represent entropy, geometric structure, and fractal complexity. Existing theoretical frameworks occupy localized regions within this landscape, reflecting their primary mathematical emphasis. The PFIT framework appears at the intersection of all three directions, illustrating its role as a unified formulation of entropy dynamics, information geometry, and multiscale fractal organization.

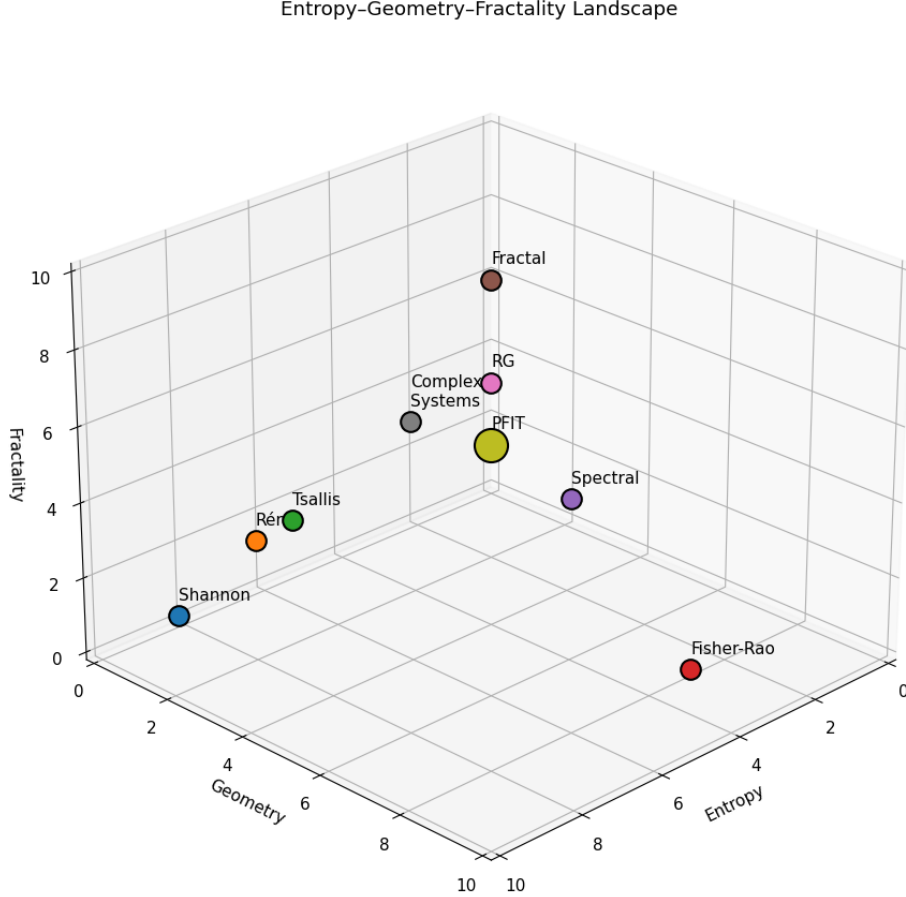


Figure 1: Entropy–geometry–fractality landscape of major theoretical frameworks. Classical entropy theories occupy the entropy-dominated region, information geometry extends toward geometric structure, while fractal and renormalization approaches populate the multiscale sector. The PFIT framework is positioned near the maximal intersection of entropy, geometry, and fractality, reflecting its unifying role across previously distinct theoretical domains.

3 Notation and Preliminaries

This section establishes the mathematical language used throughout the PFIT–En-topia–Pellis framework. The formalism unifies information geometry, fractal measure theory, and spectral operator dynamics into a single structured notation. Let $\mathcal{M}$ denote a differentiable manifold representing the space of probability measures equipped with a generalized entropy functional. Each point $p \in \mathcal{M}$ corresponds to a probability density $p(x; \theta)$ parametrized by $\theta \in \Theta \subset \mathbb{R}^n$.

3.1 Probability Measures and Entropy Functionals

Let $(\Omega, \mathcal{F}, \mu)$ be a measurable space equipped with a reference measure μ , and let $p : \Omega \rightarrow \mathbb{R}_{\geq 0}$ be a probability density function absolutely continuous with respect to μ . The normalization condition defines the probability simplex over the measure space:

$$p(x) \geq 0, \quad \int_{\Omega} p(x) d\mu(x) = 1. \quad (114)$$

Within this measure-theoretic formulation, probability distributions are interpreted as points on an infinite-

dimensional statistical manifold $\mathcal{P}(\Omega)$, where tangent vectors correspond to signed perturbations $\delta p(x)$ satisfying:

$$\int_{\Omega} \delta p(x) d\mu(x) = 0. \quad (115)$$

The Shannon entropy functional defines a global uncertainty potential over this manifold:

$$\mathcal{S}[p] = - \int_{\Omega} p(x) \log p(x) d\mu(x), \quad (116)$$

which is concave over the probability simplex and generates variational equilibrium states under constraint manifolds.

The first functional variation of entropy yields the entropy gradient field:

$$\frac{\delta \mathcal{S}}{\delta p(x)} = -1 - \log p(x), \quad (117)$$

which acts as a driving force in entropic gradient flows on $\mathcal{P}(\Omega)$.

The second variation defines a local curvature structure on the probability manifold:

$$\frac{\delta^2 \mathcal{S}}{\delta p(x) \delta p(y)} = -\frac{\delta(x-y)}{p(x)}, \quad (118)$$

encoding the intrinsic instability of low-probability regions under perturbation.

In this framework, generalized entropy families are introduced via deformation parameters that encode non-extensive interactions, long-range correlations, and fractal support structures. The Tsallis entropy functional is defined as:

$$\mathcal{S}_q[p] = \frac{1}{q-1} \left(1 - \int_{\Omega} p(x)^q d\mu(x) \right), \quad (119)$$

which reduces to Shannon entropy in the limit:

$$\lim_{q \rightarrow 1} \mathcal{S}_q[p] = \mathcal{S}[p]. \quad (120)$$

The corresponding escort distribution emerges naturally as a renormalized measure on the probability manifold:

$$p_q(x) = \frac{p(x)^q}{\int_{\Omega} p(x)^q d\mu(x)}, \quad (121)$$

which reweights phase space according to nonlinear amplification of high-density regions.

In parallel, Rényi entropy defines a logarithmic deformation of the measure structure:

$$\mathcal{S}_{\alpha}[p] = \frac{1}{1-\alpha} \log \left(\int_{\Omega} p(x)^{\alpha} d\mu(x) \right), \quad (122)$$

with limiting consistency:

$$\lim_{\alpha \rightarrow 1} \mathcal{S}_{\alpha}[p] = \mathcal{S}[p]. \quad (123)$$

These entropy families induce distinct geometric structures on $\mathcal{P}(\Omega)$, where the metric tensor can be derived from entropy-induced divergence expansions. In particular, the Fisher–Rao metric arises as the second-order expansion of the Kullback–Leibler divergence:

$$D_{\text{KL}}(p \| p + \delta p) = \frac{1}{2} \int_{\Omega} \frac{(\delta p(x))^2}{p(x)} d\mu(x) + \mathcal{O}(\|\delta p\|^3). \quad (124)$$

This induces a Riemannian structure on the statistical manifold with metric:

$$g_p(\delta p_1, \delta p_2) = \int_{\Omega} \frac{\delta p_1(x) \delta p_2(x)}{p(x)} d\mu(x), \quad (125)$$

which governs geodesic transport of probability distributions.

Entropy functionals also generate dissipative dynamics through gradient flow equations on Wasserstein space:

$$\partial_t p(x, t) = \nabla \cdot \left(p(x, t) \nabla \frac{\delta \mathcal{S}}{\delta p} \right), \quad (126)$$

which explicitly yields nonlinear diffusion of logarithmic form:

$$\partial_t p = \Delta p, \quad (127)$$

under Shannon entropy in Euclidean geometry.

More generally, Tsallis entropy leads to porous-medium-type evolution:

$$\partial_t p = \Delta (p^q), \quad (128)$$

introducing nonlinear diffusion exponents directly linked to fractal transport properties.

From a variational perspective, equilibrium distributions arise as constrained extrema of entropy under moment constraints:

$$\delta \left[\mathcal{S}[p] + \lambda_0 \left(\int p d\mu - 1 \right) + \sum_i \lambda_i \left(\int f_i(x) p(x) d\mu - c_i \right) \right] = 0, \quad (129)$$

leading to exponential family solutions:

$$p^*(x) = \frac{1}{Z} \exp \left(- \sum_i \lambda_i f_i(x) \right). \quad (130)$$

In this interpretation, classical and non-extensive entropies are not competing theories but different projections of a unified informational geometry governing collapse, scaling, and emergence across complex systems.

Figure 2 illustrates the PFIT Information Genesis Manifold, where hierarchical structure emerges from probability measures through entropy-induced geometric deformation, followed by fractal scaling cascades and spectral operator collapse, establishing a continuous transition from stochastic informational primitives to organized manifold-level information geometry.

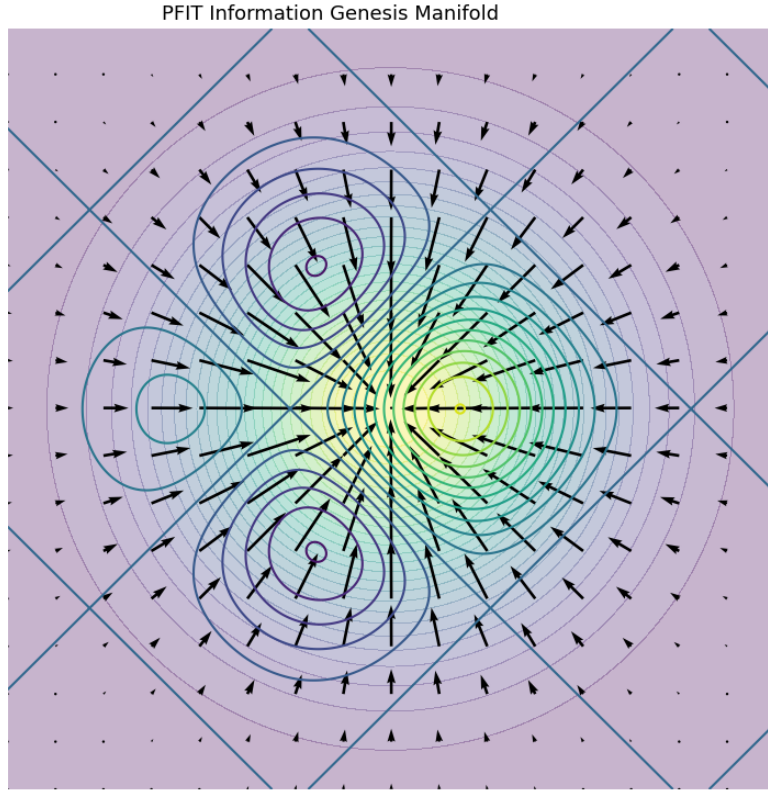


Figure 2: The PFIT Information Genesis Manifold illustrating the hierarchical emergence of structure from probability measures to entropy-driven geometric deformation, fractal scaling cascades, and spectral operator collapse. This diagram formalizes the transition from stochastic information primitives to structured informational geometry.

3.2 Statistical Manifold Structure

The parameter space $\Theta \subset \mathbb{R}^n$ of a family of probability distributions $\{p(x|\theta)\}_{\theta \in \Theta}$ naturally induces a Riemannian geometric structure, transforming statistical inference into differential geometry on a curved manifold of probability states. In this setting, each point $\theta \in \Theta$ corresponds to a distinct macroscopic informational configuration, while tangent vectors encode infinitesimal variations in statistical observables.

The fundamental metric tensor is given by the Fisher information metric:

$$g_{ij}(\theta) = \mathbb{E}_{p(x|\theta)} [\partial_i \log p(x|\theta) \partial_j \log p(x|\theta)], \quad (131)$$

which can equivalently be expressed as the negative expected Hessian of the log-likelihood:

$$g_{ij}(\theta) = -\mathbb{E}_{p(x|\theta)} [\partial_i \partial_j \log p(x|\theta)], \quad (132)$$

under standard regularity conditions ensuring interchange of differentiation and integration.

This metric defines the infinitesimal statistical distance between neighboring distributions:

$$ds^2 = g_{ij}(\theta) d\theta^i d\theta^j, \quad (133)$$

which arises as the second-order expansion of the Kullback–Leibler divergence:

$$D_{\text{KL}}(p(x|\theta) \| p(x|\theta + d\theta)) = \frac{1}{2} g_{ij}(\theta) d\theta^i d\theta^j + \mathcal{O}(\|d\theta\|^3). \quad (134)$$

The statistical manifold (Θ, g) thus inherits a full Riemannian structure, enabling the definition of covariant differentiation via the Levi–Civita connection:

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}), \quad (135)$$

which governs parallel transport of statistical states and defines geodesic flows of optimal inference.

Geodesics on the statistical manifold satisfy the nonlinear second-order differential equation:

$$\frac{d^2 \theta^k}{dt^2} + \Gamma_{ij}^k(\theta) \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = 0, \quad (136)$$

which characterizes the shortest informational paths between probability distributions under the Fisher–Rao geometry.

The curvature tensor encodes the intrinsic nonlinearity of the statistical model and quantifies the sensitivity of inference under perturbations:

$$\mathcal{R}_{ij\ell}^k = \partial_i \Gamma_{j\ell}^k - \partial_j \Gamma_{i\ell}^k + \Gamma_{im}^k \Gamma_{j\ell}^m - \Gamma_{jm}^k \Gamma_{i\ell}^m. \quad (137)$$

Its fully covariant form is obtained by contraction with the metric:

$$\mathcal{R}_{kij\ell} = g_{km} \mathcal{R}_{ij\ell}^m, \quad (138)$$

and the Ricci tensor follows as:

$$\mathcal{R}_{ij} = \mathcal{R}_{ikj}^k. \quad (139)$$

The scalar curvature of the statistical manifold is then defined as:

$$\mathcal{R} = g^{ij} \mathcal{R}_{ij}, \quad (140)$$

which provides a global invariant quantifying the effective nonlinearity and coupling strength of the underlying probabilistic model.

In perturbative regimes, the evolution of probability distributions induces curvature-driven instability of inference trajectories. Let $\theta(t)$ be a dynamical parameter path; then the deviation between neighboring trajectories ξ^i satisfies the geodesic deviation equation:

$$\frac{D^2 \xi^i}{dt^2} + \mathcal{R}_{ijkl} \frac{d\theta^j}{dt} \xi^k \frac{d\theta^l}{dt} = 0, \quad (141)$$

which directly links curvature to exponential divergence or convergence of statistical flows.

In exponential families, where

$$p(x|\theta) = \exp(\theta^i F_i(x) - \psi(\theta)), \quad (142)$$

the Fisher metric reduces to the Hessian structure of the log-partition function:

$$g_{ij}(\theta) = \frac{\partial^2 \psi(\theta)}{\partial \theta^i \partial \theta^j}, \quad (143)$$

implying that statistical curvature is fully determined by the convex geometry of $\psi(\theta)$.

Dually flat structures arise when the manifold admits both exponential and mixture affine connections with vanishing torsion and dual curvature, satisfying:

$$\mathcal{R}^{(\nabla)} = 0, \quad \mathcal{R}^{(\nabla^*)} = 0, \quad (144)$$

allowing the introduction of dual coordinate systems (θ, η) linked via Legendre transformation:

$$\eta_i = \frac{\partial \psi(\theta)}{\partial \theta^i}. \quad (145)$$

Within this geometric framework, statistical inference is reinterpreted as constrained motion on a curved informational manifold, where curvature governs the stability, robustness, and phase structure of probabilistic models. In the PFIT–En-topia–Pellis formulation, the Fisher geometry is elevated to a fundamental physical structure in which entropy gradients generate curvature, curvature controls dynamical evolution, and spectral collapse emerges as a manifestation of singular geometric concentration in parameter space.

3.3 Fractal Measures and Scale Structure

Let μ_F denote a fractal probability measure supported on a geometrically irregular set $\mathcal{F} \subset \mathbb{R}^d$, whose local structure is not characterized by integer-dimensional Euclidean scaling but by asymptotic self-similarity under iterative magnification. In this regime, classical Lebesgue volume is replaced by scale-dependent measure theory, where infinitesimal neighborhoods exhibit anomalous dimensional collapse.

The local scaling law of the fractal measure is defined through the asymptotic behavior of μ_F on ϵ -balls:

$$\mu_F(B_\epsilon(x)) \sim C(x) \epsilon^{D_f(x)}, \quad (146)$$

where $D_f(x)$ is the local (possibly spatially varying) Hausdorff dimension and $C(x)$ encodes measure normalization fluctuations across the fractal support.

In the homogeneous case, $D_f(x) \equiv D_f$, this reduces to the global scaling law:

$$\mu_F(B_\epsilon(x)) \sim \epsilon^{D_f}. \quad (147)$$

The fractal dimension can be rigorously defined via the box-counting formalism:

$$D_f = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon)}{-\log \epsilon}, \quad (148)$$

where $N(\epsilon)$ denotes the minimal number of ϵ -sized covering sets required to fully cover $\mathcal{F}$.

Equivalently, in measure-theoretic form, the dimension can be expressed via scaling of partition functions:

$$Z(q, \epsilon) = \sum_i \mu_i(\epsilon)^q \sim \epsilon^{\tau(q)}, \quad (149)$$

where $\tau(q)$ is the mass exponent function encoding multifractal structure, and $\mu_i(\epsilon)$ are coarse-grained probabilities on scale ϵ .

The generalized fractal dimensions are then defined as:

$$D_q = \frac{\tau(q)}{q-1}, \quad (150)$$

with the limiting case:

$$\lim_{q \rightarrow 1} D_q = D_1, \quad (151)$$

where D_1 corresponds to the information dimension of the measure.

The multifractal spectrum is obtained via Legendre transformation of $\tau(q)$:

$$f(\alpha) = \inf_q (q\alpha - \tau(q)), \quad (152)$$

where α represents local Hölder regularity exponents defined by:

$$\mu_F(B_\epsilon(x)) \sim \epsilon^{\alpha(x)}. \quad (153)$$

This induces a hierarchical decomposition of the fractal support into level sets:

$$\mathcal{F}_\alpha = \{x \in \mathcal{F} \mid \alpha(x) = \alpha\}, \quad (154)$$

with each subset contributing distinct scaling weights to the global measure structure.

From a renormalization perspective, fractal measures arise as fixed points of scale transformation operators $\mathcal{R}_\lambda$ acting on probability distributions:

$$\mu_F = \mathcal{R}_\lambda[\mu_F], \quad (155)$$

where the renormalization map satisfies:

$$\mathcal{R}_\lambda[\mu](A) = \frac{\mu(\lambda A)}{\mu(\lambda \Omega)}, \quad (156)$$

ensuring scale invariance under dilation transformations.

The spectral representation of fractal measures can be expressed via Mellin transforms:

$$\mathcal{M}[\mu](s) = \int_0^\infty x^{s-1} d\mu(x), \quad (157)$$

whose pole structure encodes the discrete scale invariance of the system.

In dynamical systems, fractal measures emerge as invariant measures μ^* under nonlinear evolution maps $T : \mathcal{F} \rightarrow \mathcal{F}$:

$$\mu^*(A) = \mu^*(T^{-1}(A)), \quad (158)$$

indicating statistical equilibrium under chaotic evolution.

The convergence toward such invariant fractal measures is governed by transfer operators (Perron–Frobenius operators):

$$(\mathcal{L}_T \rho)(x) = \sum_{y \in T^{-1}(x)} \frac{\rho(y)}{|\det DT(y)|}, \quad (159)$$

whose spectral properties determine mixing rates and dimensional collapse.

Entropy production on fractal supports exhibits anomalous scaling due to non-integer dimensionality. The effective entropy at scale ϵ behaves as:

$$\mathcal{S}(\epsilon) \sim D_f \log(1/\epsilon), \quad (160)$$

linking information growth directly to fractal dimensional structure.

More generally, the interplay between fractal geometry and probability measures induces a scale-dependent metric structure:

$$g_\epsilon(x) \sim \epsilon^{-2(1-D_f/d)}, \quad (161)$$

reflecting the compression of accessible volume in high-resolution limits.

Within the PFIT–En-topia–Pellis framework, fractal measures are interpreted as the fundamental substrate of informational reality: probability distributions are not smooth objects but hierarchical scaling measures whose geometry is encoded in multifractal spectra, whose dynamics are governed by renormalization flows, and whose stability is determined by spectral properties of transfer operators. In this unified view, fractal dimension becomes an emergent observable of deeper entropy-driven geometric dynamics on the underlying information manifold.

3.4 Operator Theory and Spectral Structures

Let $\mathcal{L} : \mathcal{H} \rightarrow \mathcal{H}$ be a (generally unbounded) linear operator acting on a separable Hilbert space $\mathcal{H}$ endowed with inner product $\langle \cdot, \cdot \rangle$. The spectral properties of $\mathcal{L}$ encode the full stability and dynamical response of the underlying system, with physical observables emerging from its resolvent structure and eigenmode decomposition.

In the discrete spectral regime, eigenmodes satisfy the spectral equation:

$$\mathcal{L}\psi_n = \lambda_n \psi_n, \quad (162)$$

where $\lambda_n \in \mathbb{C}$ constitutes the point spectrum $\sigma_p(\mathcal{L})$, and $\{\psi_n\}$ form a (possibly non-orthogonal) basis depending on self-adjointness conditions.

The full spectral decomposition, when $\mathcal{L}$ admits a spectral measure $E(\lambda)$, is given by:

$$\mathcal{L} = \int_{\sigma(\mathcal{L})} \lambda dE(\lambda), \quad (163)$$

which unifies discrete, continuous, and singular continuous spectral components within a single operator-valued measure framework.

The resolvent operator

$$R(z; \mathcal{L}) = (zI - \mathcal{L})^{-1} \quad (164)$$

provides analytic continuation of spectral information into the complex plane, with singularities precisely corresponding to the spectrum:

$$\sigma(\mathcal{L}) = \{z \in \mathbb{C} \mid \|R(z; \mathcal{L})\| = \infty\}. \quad (165)$$

Under infinitesimal perturbations $\mathcal{L} \mapsto \mathcal{L} + \epsilon \mathcal{V}$, the spectral flow is governed by analytic perturbation theory. For non-degenerate eigenvalues, first-order variation yields:

$$\frac{d\lambda_n}{d\epsilon} = \langle \psi_n, \mathcal{V} \psi_n \rangle, \quad (166)$$

while higher-order corrections involve off-diagonal resolvent couplings:

$$\frac{d^2 \lambda_n}{d\epsilon^2} = 2 \sum_{m \neq n} \frac{|\langle \psi_m, \mathcal{V} \psi_n \rangle|^2}{\lambda_n - \lambda_m}. \quad (167)$$

In non-self-adjoint settings, eigenvectors lose orthogonality and spectral sensitivity is amplified. The biorthogonal system $\{\psi_n, \phi_n\}$ satisfies:

$$\mathcal{L} \psi_n = \lambda_n \psi_n, \quad \mathcal{L}^\dagger \phi_n = \overline{\lambda_n} \phi_n, \quad \langle \phi_m, \psi_n \rangle = \delta_{mn}. \quad (168)$$

The condition number of eigenmodes quantifies spectral instability:

$$\kappa_n = \frac{\|\psi_n\| \|\phi_n\|}{|\langle \phi_n, \psi_n \rangle|}, \quad (169)$$

with large κ_n indicating extreme sensitivity to perturbations and near-degeneracy effects.

Spectral collapse occurs when eigenvalues coalesce under parameter variation, producing non-diagonalizable Jordan block structures:

$$\lim_{\epsilon \rightarrow \epsilon_c} \lambda_i(\epsilon) = \lambda_j(\epsilon), \quad (170)$$

accompanied by defectiveness conditions:

$$\dim \ker(\mathcal{L} - \lambda I) < \text{algebraic multiplicity}(\lambda), \quad (171)$$

signaling the emergence of exceptional points in complex spectral topology.

Near such critical points, the spectral geometry is governed by branch-point singularities in the complex plane, with local behavior:

$$\lambda(\epsilon) \sim \lambda_c + \sqrt{\epsilon - \epsilon_c}, \quad (172)$$

which induces non-analytic phase transitions in the operator dynamics.

The spectral density function is defined as:

$$\rho(\lambda) = \sum_n \delta(\lambda - \lambda_n), \quad (173)$$

or, in continuous representation:

$$\rho(\lambda) = \frac{1}{\pi} \text{Im Tr } R(\lambda + i0; \mathcal{L}), \quad (174)$$

linking spectral distribution directly to the analytic structure of the resolvent.

Dynamical evolution generated by $\mathcal{L}$ is expressed via semigroup flow:

$$x(t) = e^{t\mathcal{L}} x_0, \quad (175)$$

with stability determined by the spectral abscissa:

$$\alpha(\mathcal{L}) = \sup_{\lambda \in \sigma(\mathcal{L})} \operatorname{Re}(\lambda). \quad (176)$$

Spectral phase transitions occur when:

$$\operatorname{Re}(\lambda_{\max}) = 0, \quad (177)$$

marking the boundary between exponential decay and growth regimes, and thus defining critical stability manifolds in operator space.

In extended systems, the Koopman operator $\mathcal{K}$ provides a linear embedding of nonlinear dynamics:

$$(\mathcal{K}f)(x) = f(\Phi_t(x)), \quad (178)$$

with eigenfunctions satisfying:

$$\mathcal{K}\phi_\omega = e^{\omega t}\phi_\omega, \quad (179)$$

encoding coherent structures in nonlinear phase space evolution.

The pseudospectrum, capturing non-normal amplification effects, is defined as:

$$\sigma_\epsilon(\mathcal{L}) = \{z \in \mathbb{C} : \|(zI - \mathcal{L})^{-1}\| > \epsilon^{-1}\}, \quad (180)$$

revealing that spectral stability cannot be fully characterized by eigenvalues alone.

Within the PFIT–En-topia–Pellis framework, operator spectra are reinterpreted as geometric excitations of an underlying fractal information manifold. Spectral collapse corresponds to curvature-induced coalescence of eigenmodes, where resolvent singularities represent informational phase transitions and eigenvalue mergers correspond to geometric singularities in the entropy-driven manifold structure governing multiscale dynamics.

3.5 Information Geometry and Entropic Potentials

Let $\mathcal{M}$ denote a statistical manifold parametrized by coordinates $\theta \in \Theta \subset \mathbb{R}^n$, equipped with a Fisher–Rao metric $g_{ij}(\theta)$. Within this geometric framework, entropy is elevated from a scalar functional to a potential field $\mathcal{S}(\theta)$ defined over the manifold, encoding global uncertainty gradients and governing directional information flow across probability states.

The entropic potential induces a natural vector field on $\mathcal{M}$ via metric contraction of its gradient:

$$V^i(\theta) = -g^{ij}(\theta) \partial_j \mathcal{S}(\theta), \quad (181)$$

which defines the steepest descent direction of entropy with respect to the intrinsic information geometry. Equivalently, this can be expressed in covariant form as:

$$V^i(\theta) = -\nabla^i \mathcal{S}(\theta), \quad (182)$$

where $\nabla^i = g^{ij} \nabla_j$ denotes the metric-raised covariant derivative.

The dynamical evolution of statistical states is then governed by a curvature-coupled second-order flow equation obtained by modifying the geodesic equation with entropic forcing terms:

$$\frac{d^2 \theta^k}{dt^2} + \Gamma_{ij}^k(\theta) \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = -g^{k\ell}(\theta) \partial_\ell \mathcal{S}(\theta). \quad (183)$$

This equation can be interpreted as a Newtonian system on a curved manifold, where entropy acts as a scalar potential generating non-inertial forces in parameter space. In covariant form, the evolution becomes:

$$\frac{D^2 \theta^k}{Dt^2} = -\nabla^k \mathcal{S}, \quad (184)$$

with D/Dt denoting the covariant time derivative along trajectories.

The entropic potential itself may be defined variationally as a functional over the space of probability distributions:

$$\mathcal{S}(\theta) = - \int_{\Omega} p(x|\theta) \log p(x|\theta) d\mu(x), \quad (185)$$

which induces a scalar field on $\mathcal{M}$ whose gradient encodes sensitivity of uncertainty with respect to parameter deformation.

The Hessian structure of the entropy potential defines a second-order geometric response tensor:

$$H_{ij}(\theta) = \partial_i \partial_j \mathcal{S}(\theta), \quad (186)$$

which determines local convexity/concavity properties of informational landscapes and governs stability of equilibrium configurations.

Incorporating metric compatibility, the fully covariant Hessian is given by:

$$\nabla_i \nabla_j \mathcal{S} = \partial_i \partial_j \mathcal{S} - \Gamma_{ij}^k \partial_k \mathcal{S}, \quad (187)$$

which couples entropy curvature directly to the affine structure of the statistical manifold.

The equilibrium states of the system correspond to critical points of the entropic potential:

$$\nabla_i \mathcal{S}(\theta^*) = 0, \quad (188)$$

which define stationary distributions under constrained information flow.

Stability of these equilibria is determined by the eigenvalue spectrum of the entropy Hessian:

$$H_{ij}(\theta^*) v^j = \lambda v_i, \quad (189)$$

where positive eigenvalues correspond to locally convex (stable) informational configurations, while negative modes indicate instability and phase transition directions.

The entropic force field can also be embedded into a gradient flow formulation on the statistical manifold:

$$\frac{d\theta^i}{dt} = -g^{ij} \partial_j \mathcal{S}, \quad (190)$$

which corresponds to steepest entropy descent with respect to the Fisher metric, and equivalently defines a Wasserstein-type transport dynamics in probability space.

The associated Lyapunov functional structure ensures monotonic evolution of entropy under suitable conditions:

$$\frac{d\mathcal{S}}{dt} = \partial_i \mathcal{S} \frac{d\theta^i}{dt} = -g^{ij} \partial_i \mathcal{S} \partial_j \mathcal{S} \leq 0, \quad (191)$$

establishing entropy as a globally decreasing potential along geodesically corrected gradient trajectories.

Coupling curvature with entropic forcing yields an effective dynamical system combining inertial, geometric, and informational components:

$$\frac{D^2 \theta^k}{Dt^2} = -\nabla^k \mathcal{S} + \mathcal{R}^k_{ij\ell} \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} \theta^\ell, \quad (192)$$

where curvature induces nonlinear feedback on entropy-driven motion.

Within the PFIT–En-topia–Pellis framework, entropic potentials are interpreted as fundamental generators of geometric evolution: entropy defines a scalar field on the statistical manifold, its gradient induces information flow, and curvature modulates the acceleration of probabilistic trajectories. In this unified picture, inference, dynamics, and thermodynamics emerge as different manifestations of a single entropic-geometric principle governing multiscale informational collapse and structure formation.

3.6 Pellis Fractal Information Axiomatics

Building on the fractal-information program developed in [82], we postulate that informational states are not merely functions on smooth manifolds, but scale-covariant objects defined on a renormalization-invariant hierarchy of measure spaces. In this setting, information itself acquires a geometric scaling degree, and physical observables emerge as projections of a deeper fractal informational field.

Let $\mathcal{I}(x)$ denote a local informational density defined on a fractal support $\mathcal{F} \subset \mathbb{R}^d$. The fundamental Pellis scaling axiom asserts scale covariance under dilation transformations $x \mapsto \lambda x$:

$$\mathcal{I}(\lambda x) = \lambda^\alpha \mathcal{I}(x), \quad (193)$$

where $\alpha \in \mathbb{R}$ defines the intrinsic informational fractal dimension, governing how information density transforms under resolution change.

At the level of probability measures, this implies a corresponding scaling law for fractal measures:

$$\mu(\lambda A) = \lambda^\alpha \mu(A), \quad (194)$$

indicating that informational content is fundamentally non-extensive and scale-dependent.

Differentiating the scaling relation yields a generator identity for informational dilations:

$$x^i \partial_i \mathcal{I}(x) = \alpha \mathcal{I}(x), \quad (195)$$

which identifies α as the eigenvalue of the Euler (dilation) operator acting on informational fields.

Within this framework, entropy is promoted to a geometric scalar field $\mathcal{S}$ coupled to curvature dynamics on the informational manifold. The generalized entropy–curvature coupling is postulated as:

$$\mathcal{R}_{ij} \sim \nabla_i \nabla_j \mathcal{S} + \kappa \mathcal{F}_{ij}, \quad (196)$$

where $\mathcal{R}_{ij}$ is the Ricci-like informational curvature tensor, κ is a coupling constant controlling fractal backreaction strength, and $\mathcal{F}_{ij}$ encodes multiscale fluctuation corrections arising from non-smooth measure geometry.

The fractal correction tensor $\mathcal{F}_{ij}$ can be modeled as a scale-derivative expansion:

$$\mathcal{F}_{ij} = \sum_{n=1}^{\infty} \epsilon^{n(\alpha-1)} \nabla_i^{(n)} \mathcal{I} \nabla_j^{(n)} \mathcal{I}, \quad (197)$$

where $\nabla_i^{(n)}$ denotes n -th order multiscale covariant derivatives and ϵ is a resolution parameter controlling coarse-graining depth.

The entropy field itself is assumed to satisfy a nonlinear geometric field equation on the fractal manifold:

$$\nabla^i \nabla_i \mathcal{S} + \beta \|\nabla \mathcal{S}\|^2 = \gamma \mathcal{I}(x), \quad (198)$$

which couples curvature diffusion, nonlinear entropy self-interaction, and informational source terms.

At the level of dynamical evolution, informational trajectories $\theta^i(t)$ are governed by a fractal-corrected geodesic equation:

$$\frac{D^2 \theta^k}{Dt^2} + \Gamma_{ij}^k \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = -g^{k\ell} \partial_\ell \mathcal{S} + \kappa \mathcal{F}^k(\theta), \quad (199)$$

where $\mathcal{F}^k(\theta)$ represents the lifted vector form of fractal corrections acting as scale-dependent forcing terms.

The informational curvature tensor satisfies a generalized Bianchi-type identity modified by fractal corrections:

$$\nabla_{[k} \mathcal{R}_{ij]} + \nabla_{[k} \mathcal{F}_{ij]} = 0, \quad (200)$$

indicating that curvature conservation laws are deformed by multiscale fluctuations.

Spectral stability of informational operators $\mathcal{L}_{\mathcal{I}}$ is directly coupled to curvature fluctuations via:

$$\delta \lambda_n \sim \int_{\mathcal{F}} \psi_n^* \mathcal{F}_{ij} \psi_n d\mu, \quad (201)$$

implying that eigenvalue motion is governed by fractal geometry of the underlying informational space.

Scale invariance is enforced through renormalization consistency conditions:

$$\mathcal{R}_\lambda[\mathcal{I}] = \lambda^\alpha \mathcal{I}, \quad \mathcal{R}_\lambda[\mathcal{S}] = \mathcal{S}, \quad (202)$$

ensuring compatibility between entropy invariance and informational scaling symmetry.

The effective action functional governing the PFIT–En-topia–Pellis system is then written as:

$$\mathcal{A}[\mathcal{I}, \mathcal{S}] = \int_{\mathcal{F}} (\mathcal{R} + \lambda_1 \|\nabla \mathcal{S}\|^2 + \lambda_2 \mathcal{I}^2 + \lambda_3 \mathcal{F}_{ij} \mathcal{F}^{ij}) d\mu, \quad (203)$$

whose stationary variation yields coupled field equations linking curvature, entropy, and fractal information dynamics.

The PFIT–En-topia–Pellis framework thus emerges as a unified variational theory of information, geometry, and scale-dependent dynamics.

4 Unified Entropy Limit Theory (Shannon–Rényi–Tsallis–Spectral–Differential Continuum)

We introduce a unified operatorial entropy framework that embeds Shannon, Rényi, Tsallis, spectral, and differential entropy as limiting cases of a single multiscale informational operator. This structure constitutes the thermodynamic backbone of the PFIT–En-topia–Pellis framework and formalizes entropy as a scale-dependent projection functional over probability, spectral, and geometric information manifolds. The construction is consistent with fractal information axioms and curvature-induced information flow [1–3, 7–9, 28, 82].

4.1 Unified Entropy Operator Principle

Let $\mathcal{E}_\kappa[\rho]$ denote the generalized entropy operator acting on probability densities $\rho(x)$ over an information manifold $\mathcal{M}$. We postulate the existence of a universal operatorial form

$$\mathcal{E}_\kappa[\rho] = \mathcal{P}_\kappa \left[\int_{\mathcal{M}} \rho(x) \Phi_\kappa(\rho(x)) dV_g \right], \quad (204)$$

where $\mathcal{P}_\kappa$ denotes a scale-dependent projection operator and κ encodes the entropic regime (extensivity, fractality, spectrality).

We further assume that $\mathcal{P}_\kappa$ acts as a nonlinear spectral filter on the eigenmodes of the Laplace–Beltrami operator Δ_g :

$$\mathcal{P}_\kappa[f] = \sum_{n=0}^{\infty} \Pi_\kappa(\lambda_n) \langle \phi_n, f \rangle_{L^2(\mathcal{M},g)} \phi_n, \quad (205)$$

where

$$\Delta_g \phi_n = -\lambda_n \phi_n, \quad 0 \leq \lambda_0 \leq \lambda_1 \leq \dots, \quad (206)$$

and $\Pi_\kappa(\lambda)$ is a scale-dependent spectral attenuation kernel encoding information coarse-graining.

The functional kernel $\Phi_\kappa(\rho)$ is assumed to admit a universal asymptotic expansion of the form

$$\Phi_\kappa(\rho) = \log \rho + \kappa \rho^{\alpha(\kappa)} + \sum_{m=2}^{\infty} c_m(\kappa) \rho^{m-1}, \quad (207)$$

where $\alpha(\kappa)$ governs the fractal deformation of extensivity and $c_m(\kappa)$ encode higher-order non-extensive corrections.

Under this formulation, the entropy functional becomes explicitly nonlocal in both geometric and spectral senses:

$$\mathcal{E}_\kappa[\rho] = \mathcal{P}_\kappa \left[\int_{\mathcal{M}} \rho(x) \log \rho(x) dV_g + \kappa \int_{\mathcal{M}} \rho(x)^{1+\alpha(\kappa)} dV_g + \mathcal{O}(\rho^3) \right]. \quad (208)$$

We define the entropy gradient flow on the space of probability densities endowed with the Fisher–Rao metric G_ρ :

$$\partial_t \rho = -\nabla_{G_\rho} \mathcal{E}_\kappa[\rho], \quad (209)$$

which expands explicitly as a nonlinear integro-differential evolution equation:

$$\partial_t \rho = \nabla_g \cdot \left(\rho \nabla_g \frac{\delta \mathcal{E}_\kappa}{\delta \rho} \right). \quad (210)$$

The functional derivative of the unified entropy operator takes the form

$$\frac{\delta \mathcal{E}_\kappa}{\delta \rho} = \mathcal{P}_\kappa \left[\log \rho + 1 + (1 + \alpha(\kappa)) \kappa \rho^{\alpha(\kappa)} + \sum_{m=2}^{\infty} m c_m(\kappa) \rho^{m-1} \right]. \quad (211)$$

We introduce the curvature-coupled deformation of entropy via the scalar curvature R_g of $\mathcal{M}$:

$$\mathcal{E}_\kappa^{\text{curv}}[\rho] = \mathcal{E}_\kappa[\rho] + \beta(\kappa) \int_{\mathcal{M}} R_g(x) \rho(x) dV_g, \quad (212)$$

which induces a geometric feedback loop between information density and manifold curvature.

In the spectral representation, curvature induces a mode-coupling tensor:

$$\mathcal{C}_{ij}(\kappa) = \int_{\mathcal{M}} R_g(x) \phi_i(x) \phi_j(x) dV_g, \quad (213)$$

leading to a renormalized projection operator:

$$\mathcal{P}_\kappa^{\text{eff}} = \sum_{i,j} \Pi_\kappa(\lambda_i) \mathcal{C}_{ij}(\kappa) |\phi_i\rangle \langle \phi_j|. \quad (214)$$

Classical entropy forms emerge as limiting cases of $\mathcal{E}_\kappa[\rho]$ under specific degenerations of the kernel structure:

$$\lim_{\kappa \rightarrow 0} \mathcal{E}_\kappa[\rho] \rightarrow - \int_{\mathcal{M}} \rho \log \rho dV_g \quad (\text{Shannon regime}), \quad (215)$$

$$\lim_{\kappa \rightarrow \kappa_f} \mathcal{E}_\kappa[\rho] \rightarrow \int_{\mathcal{M}} \rho^q dV_g \quad (\text{Tsallis-like fractal regime}), \quad (216)$$

$$\lim_{\lambda_n \rightarrow \infty} \mathcal{P}_\kappa \rightarrow \text{Id} \quad (\text{local equilibrium limit}). \quad (217)$$

Finally, we postulate a self-consistency closure condition defining the admissible entropy manifold:

$$\mathcal{E}_\kappa[\rho] = \mathcal{P}_\kappa \circ \mathcal{E}_\kappa[\rho], \quad (218)$$

which implies that physically realizable distributions are fixed points of the combined entropy–projection flow:

$$\rho^* = \arg \min_{\rho \in \mathcal{M}} \mathcal{E}_\kappa[\rho] \quad \text{s.t.} \quad \rho = \mathcal{P}_\kappa[\rho]. \quad (219)$$

Figure 3 illustrates how classical probabilistic entropy emerges as a limiting case of a higher-order spectral–geometric operator structure governing information dynamics and collapse behavior.

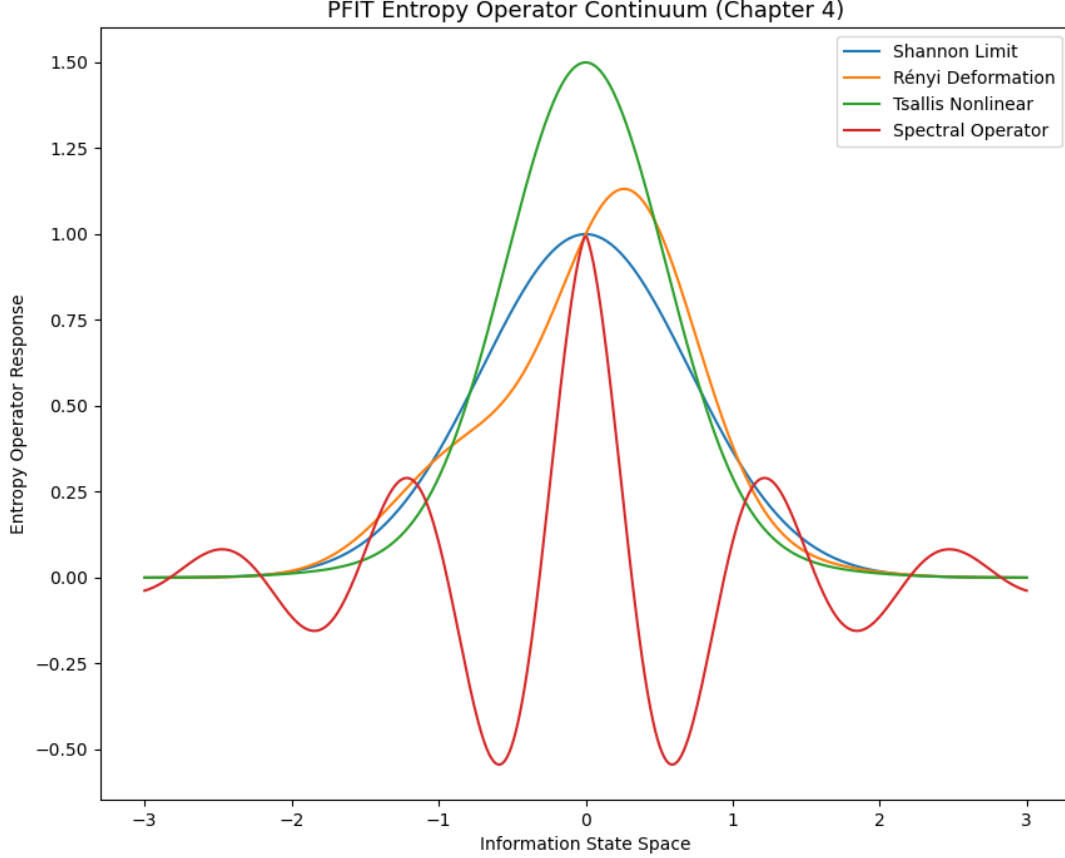


Figure 3: Unified entropy operator structure connecting Shannon, Rényi, Tsallis, spectral, and differential entropy into a single operatorial continuum within PFIT.

4.2 Shannon Entropy Limit

In the weak-correlation, extensive regime $\kappa \rightarrow 1$, the unified entropy operator $\mathcal{E}_\kappa[\rho]$ asymptotically collapses onto the classical Shannon functional, recovering additivity under independent partitions of the information manifold. Formally,

$$\lim_{\kappa \rightarrow 1} \mathcal{E}_\kappa[\rho] \equiv H_S[\rho] = - \int_{\Omega} \rho(x) \log \rho(x) dV_g. \quad (220)$$

Upon discretization of $\mathcal{M}$ into a measurable partition $\{\Omega_i\}_{i=1}^N$ with induced probability measure

$$p_i = \int_{\Omega_i} J(x) dx, \quad \sum_{i=1}^N p_i = 1, \quad (221)$$

the entropy reduces to the discrete Shannon form

$$H_S = - \sum_{i=1}^N p_i \log p_i. \quad (222)$$

Here $J(x)$ is interpreted as the information flux density associated with the push-forward measure of $\rho(x)$ under the observation map $\pi : \mathcal{M} \rightarrow \Omega$. The induced probability structure satisfies

$$p_i = (\pi_*\rho)(\Omega_i), \quad \pi_*\rho \ll \mu, \quad (223)$$

where μ is the Lebesgue reference measure on the observation space.

In this regime, correlations between microstates vanish in the sense of factorization of joint densities:

$$\rho(x_1, x_2) \approx \rho(x_1)\rho(x_2), \quad \text{Cov}(\log \rho(x_i), \log \rho(x_j)) \rightarrow 0. \quad (224)$$

Consequently, the entropy becomes strictly extensive under disjoint unions:

$$H_S(\Omega_A \cup \Omega_B) = H_S(\Omega_A) + H_S(\Omega_B), \quad \Omega_A \cap \Omega_B = \emptyset. \quad (225)$$

From a differential-geometric standpoint, the statistical manifold $\mathcal{S} = \{\rho(x; \theta)\}$ inherits the Fisher–Rao metric tensor

$$g_{ij}(\theta) = \int_{\mathcal{M}} \rho(x; \theta) \partial_i \log \rho(x; \theta) \partial_j \log \rho(x; \theta) dV_g, \quad (226)$$

which defines the unique Riemannian structure invariant under sufficient statistics transformations.

In the Shannon limit, the entropy gradient flow coincides with the classical information-geometric steepest descent:

$$\partial_t \theta^i = -g^{ij}(\theta) \frac{\partial H_S}{\partial \theta^j}, \quad (227)$$

which induces a geodesic relaxation toward equilibrium distributions on $\mathcal{S}$.

Equivalently, in density space, the evolution is governed by the continuity equation with entropic potential $\Phi_S = \log \rho + 1$:

$$\partial_t \rho = \nabla_g \cdot (\rho \nabla_g \Phi_S) = \nabla_g \cdot (\rho \nabla_g \log \rho), \quad (228)$$

which reduces to the heat equation on $\mathcal{M}$ under Euclidean embedding:

$$\partial_t \rho = \Delta_g \rho. \quad (229)$$

The Shannon regime is thus identified as the fixed-point attractor of the full spectral–fractal entropy operator, characterized by vanishing higher-order cumulants:

$$\kappa_n(\rho) \rightarrow 0, \quad n \geq 3, \quad (230)$$

implying Gaussianization of fluctuations in the central limit scaling of the information field.

Finally, the Fisher–Rao geometry becomes flat in the thermodynamic limit of infinite sampling resolution,

$$\lim_{N \rightarrow \infty} \text{Ric}(g_{ij}) \rightarrow 0, \quad (231)$$

signaling the emergence of an effectively Euclidean statistical manifold and the restoration of classical extensivity.

4.3 Rényi Entropy Operator

For $\alpha \in \mathbb{R}^+$, the Rényi entropy arises as a one-parameter spectral deformation of the Shannon functional, obtained by replacing linear averaging with a nonlinear power-law measure on the space of probability densities. In discrete form,

$$H_\alpha = \frac{1}{1-\alpha} \log \left(\sum_{i=1}^N p_i^\alpha \right), \quad (232)$$

and in the continuum limit over an information manifold $(\mathcal{M}, g)$,

$$H_\alpha[\rho] = \frac{1}{1-\alpha} \log \left(\int_{\mathcal{M}} \rho(x)^\alpha dV_g \right). \quad (233)$$

Within the PFIT framework, the parameter α is promoted from a scalar index to a renormalization flow coordinate controlling multiscale informational heterogeneity. This induces a deformation of the probability measure space via the nonlinear embedding

$$\rho(x) \mapsto \rho_\alpha(x) = \frac{\rho(x)^\alpha}{\int_{\mathcal{M}} \rho(x)^\alpha dV_g}, \quad (234)$$

which defines a scale-dependent escort distribution structure.

The entropy functional can then be rewritten as a logarithmic moment-generating operator on the measure space:

$$H_\alpha[\rho] = \frac{1}{1-\alpha} \log \left(\mathbb{E}_\rho [\rho^{\alpha-1}] \right), \quad \mathbb{E}_\rho[f] = \int_{\mathcal{M}} f(x) \rho(x) dV_g. \quad (235)$$

The spectral interpretation emerges by expanding ρ in the eigenbasis of the Laplace–Beltrami operator:

$$\rho(x) = \sum_{n=0}^{\infty} a_n \phi_n(x), \quad \Delta_g \phi_n = -\lambda_n \phi_n, \quad (236)$$

leading to a nonlinear coupling of spectral coefficients in the Rényi functional:

$$\int_{\mathcal{M}} \rho(x)^\alpha dV_g = \sum_{n_1, \dots, n_\alpha} a_{n_1} \cdots a_{n_\alpha} \int_{\mathcal{M}} \phi_{n_1} \cdots \phi_{n_\alpha} dV_g. \quad (237)$$

This induces an effective α -dependent spectral interaction tensor:

$$\mathcal{T}_{n_1 \dots n_\alpha} = \int_{\mathcal{M}} \phi_{n_1}(x) \cdots \phi_{n_\alpha}(x) dV_g, \quad (238)$$

which encodes higher-order correlations beyond pairwise (Gaussian) structure and governs the emergence of multi-fractal organization in ECG-derived dynamical measures.

The PFIT interpretation identifies α as a renormalization flow parameter governing the resolution scale of the underlying information manifold:

$$\frac{dH_\alpha}{d\alpha} = \frac{1}{(1-\alpha)^2} \left[\frac{\int_{\mathcal{M}} \rho^\alpha \log \rho dV_g}{\int_{\mathcal{M}} \rho^\alpha dV_g} - \log \left(\int_{\mathcal{M}} \rho^\alpha dV_g \right) \right]. \quad (239)$$

Monotonicity properties of the Rényi spectrum imply a hierarchical ordering of information content across scales:

$$\alpha_1 < \alpha_2 \quad \Rightarrow \quad H_{\alpha_1} \geq H_{\alpha_2}, \quad (240)$$

which formalizes the suppression of fine-scale fluctuations under increasing α , corresponding to coarse-grained spectral stabilization.

In the ECG dynamical context, let $\rho(x, t)$ denote the time-evolving probability density of phase-space embeddings of cardiac electrical activity. The Rényi-driven entropy flow induces a nonlinear evolution equation:

$$\partial_t \rho = \nabla_g \cdot \left(\rho \nabla_g \frac{\delta H_\alpha}{\delta \rho} \right), \quad (241)$$

with functional derivative

$$\frac{\delta H_\alpha}{\delta \rho} = \frac{\alpha}{1-\alpha} \frac{\rho^{\alpha-1}}{\int_{\mathcal{M}} \rho^\alpha dV_g}. \quad (242)$$

This yields a nonlinear diffusion–aggregation system:

$$\partial_t \rho = \nabla_g \cdot (\rho \nabla_g \rho^{\alpha-1}), \quad (243)$$

which interpolates between classical diffusion ($\alpha \rightarrow 1$) and strongly nonlinear aggregation regimes ($\alpha \neq 1$), associated with precursory instability signatures in ventricular fibrillation dynamics.

Finally, the Rényi operator defines a full one-parameter entropy flow on the space of probability measures:

$$\mathcal{F}_\alpha : \mathcal{P}(\mathcal{M}) \rightarrow \mathbb{R}, \quad \rho \mapsto H_\alpha[\rho], \quad (244)$$

whose fixed points satisfy the nonlinear eigenvalue condition:

$$\rho^* = \arg \min_{\rho} H_\alpha[\rho], \quad \delta H_\alpha[\rho^*] = 0, \quad (245)$$

defining scale-stable cardiac informational states in the PFIT–spectral collapse framework.

4.4 Tsallis Entropy Operator

The non-extensive entropy structure is defined by the Tsallis functional

$$S_q = \frac{1}{q-1} \left(1 - \sum_{i=1}^N p_i^q \right), \quad (246)$$

with continuum generalization on an information manifold $(\mathcal{M}, g)$ given by

$$S_q[\rho] = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \rho(x)^q dV_g \right), \quad \int_{\mathcal{M}} \rho(x) dV_g = 1. \quad (247)$$

Within the PFIT framework, the entropic index q is interpreted as a curvature-controlled nonlinearity parameter governing deviation from extensivity under geometric feedback. Specifically, we postulate a constitutive coupling between q and scalar curvature R_g :

$$q(x) = 1 + \eta R_g(x) + \xi \|\nabla_g \log \rho(x)\|^2, \quad (248)$$

where $\eta, \xi \in \mathbb{R}$ encode curvature sensitivity and gradient-driven fractal amplification.

The Tsallis entropy can then be rewritten as a curvature-modulated partition functional:

$$S_q[\rho] = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \exp((q-1) \log \rho(x)) dV_g \right), \quad (249)$$

highlighting its interpretation as a nonlinear deformation of the Shannon exponential measure.

The associated escort distribution emerges as the natural probability structure:

$$\rho_q(x) = \frac{\rho(x)^q}{\int_{\mathcal{M}} \rho(x)^q dV_g}, \quad (250)$$

which induces a renormalized expectation operator

$$\mathbb{E}_q[f] = \int_{\mathcal{M}} f(x) \rho_q(x) dV_g. \quad (251)$$

In this representation, Tsallis entropy is expressed as a deformation of Shannon entropy through escort averaging:

$$S_q[\rho] = -\mathbb{E}_q \left[\frac{\rho^{q-1} - 1}{q-1} \right], \quad (252)$$

which reduces to Shannon entropy in the limit

$$\lim_{q \rightarrow 1} S_q[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g. \quad (253)$$

From a geometric flow perspective, Tsallis entropy generates a nonlinear gradient flow on the statistical manifold:

$$\partial_t \rho = \nabla_g \cdot \left(\rho \nabla_g \frac{\delta S_q}{\delta \rho} \right), \quad (254)$$

with functional derivative

$$\frac{\delta S_q}{\delta \rho} = -\frac{q}{q-1} \rho^{q-1}. \quad (255)$$

This yields the porous-medium-type evolution equation:

$$\partial_t \rho = \Delta_g \rho^q, \quad (256)$$

which encodes anomalous diffusion and long-range coupling effects characteristic of multiscale cardiac dynamics.

Spectrally, the nonlinear term ρ^q induces higher-order mode coupling in the Laplace–Beltrami eigenbasis:

$$\rho^q(x) = \left(\sum_n a_n \phi_n(x) \right)^q = \sum_{n_1, \dots, n_q} a_{n_1} \cdots a_{n_q} \phi_{n_1}(x) \cdots \phi_{n_q}(x), \quad (257)$$

leading to the q -order interaction tensor

$$\mathcal{T}_{n_1 \dots n_q}^{(q)} = \int_{\mathcal{M}} \phi_{n_1}(x) \cdots \phi_{n_q}(x) dV_g, \quad (258)$$

which governs the emergence of nonlinear spectral coherence breakdown prior to ventricular fibrillation onset.

The curvature–entropy feedback loop closes via the coupling between Tsallis entropy and Ricci curvature:

$$\partial_t g_{ij} = -\lambda \frac{\delta S_q}{\delta g^{ij}} \approx -\lambda \rho^q R_{ij}, \quad (259)$$

where R_{ij} denotes the Ricci curvature tensor and λ is a coupling constant controlling geometric stiffness of the information manifold.

Finally, the Tsallis operator defines a non-extensive entropy flow on the space of probability measures:

$$\mathcal{F}_q : \mathcal{P}(\mathcal{M}) \rightarrow \mathbb{R}, \quad \rho \mapsto S_q[\rho], \quad (260)$$

whose fixed points satisfy the nonlinear stationarity condition:

$$\delta S_q[\rho^*] = 0 \quad \Rightarrow \quad \rho^* \propto (1 - (1 - q)\beta V(x))^{\frac{1}{1-q}}, \quad (261)$$

identifying the generalized q -exponential equilibrium states governing scale-invariant cardiac dynamics in the PFIT spectral–fractal regime.

4.5 Spectral Entropy Operator

Let $\{\omega_n\}_{n \in \mathbb{N}}$ denote the discrete spectrum of the collapse operator $\mathcal{C}$ acting on the information manifold $(\mathcal{M}, g)$, with generalized eigenvalue problem

$$\mathcal{C}\psi_n = \omega_n \psi_n, \quad \langle \psi_n, \psi_m \rangle_{L^2(\mathcal{M})} = \delta_{nm}. \quad (262)$$

The spectral probability measure is defined via normalized modal energy distribution:

$$P(\omega_n) = \frac{|a_n|^2}{\sum_k |a_k|^2}, \quad \rho(x, t) = \sum_n a_n(t) \psi_n(x), \quad (263)$$

where a_n are the modal coefficients of the ECG-induced phase-space embedding field.

The spectral entropy functional is then given by

$$H_{\text{spec}} = - \sum_n P(\omega_n) \log P(\omega_n), \quad (264)$$

with continuum extension

$$H_{\text{spec}}[\rho] = - \int_{\sigma(\mathcal{C})} P(\omega) \log P(\omega) d\mu(\omega), \quad (265)$$

where $\sigma(\mathcal{C})$ denotes the spectrum and $d\mu(\omega)$ the induced spectral measure.

Within the PFIT framework, spectral entropy is interpreted as an instability functional on the eigenvalue manifold, quantifying loss of rigidity in the spectral geometry of cardiac dynamics. The collapse of eigenmodes is encoded through the time evolution of spectral weights:

$$\frac{d}{dt} P(\omega_n) = 2 \operatorname{Re} \left(\frac{a_n^* \dot{a}_n}{\sum_k |a_k|^2} \right) - P(\omega_n) \frac{d}{dt} \log \left(\sum_k |a_k|^2 \right). \quad (266)$$

The entropy production rate is given by

$$\frac{d}{dt} H_{\text{spec}} = - \sum_n \dot{P}(\omega_n) \log P(\omega_n), \quad (267)$$

which diverges under spectral flattening, i.e. when

$$P(\omega_n) \rightarrow \frac{1}{N} \quad \Rightarrow \quad H_{\text{spec}} \rightarrow \log N. \quad (268)$$

We define the spectral entropy operator $\mathcal{H}_{\text{spec}}$ acting on the probability simplex:

$$\mathcal{H}_{\text{spec}}[P] = - \sum_n P_n \log P_n, \quad P \in \Delta^{N-1}. \quad (269)$$

The gradient flow structure in spectral space is given by

$$\frac{dP_n}{dt} = - \sum_m g_{nm}^{-1}(P) \frac{\partial \mathcal{H}_{\text{spec}}}{\partial P_m}, \quad (270)$$

where g_{nm} is the Fisher–Rao metric on the spectral simplex:

$$g_{nm}(P) = \frac{\delta_{nm}}{P_n}. \quad (271)$$

This yields the replicator-type dynamics:

$$\frac{dP_n}{dt} = -P_n \left(\log P_n - \sum_k P_k \log P_k \right), \quad (272)$$

which describes entropy-driven redistribution of spectral energy across eigenmodes.

Figure 4 illustrates this fractal spectral organization, where the eigenvalue distribution exhibits scale-invariant behavior consistent with multiscale renormalization dynamics.

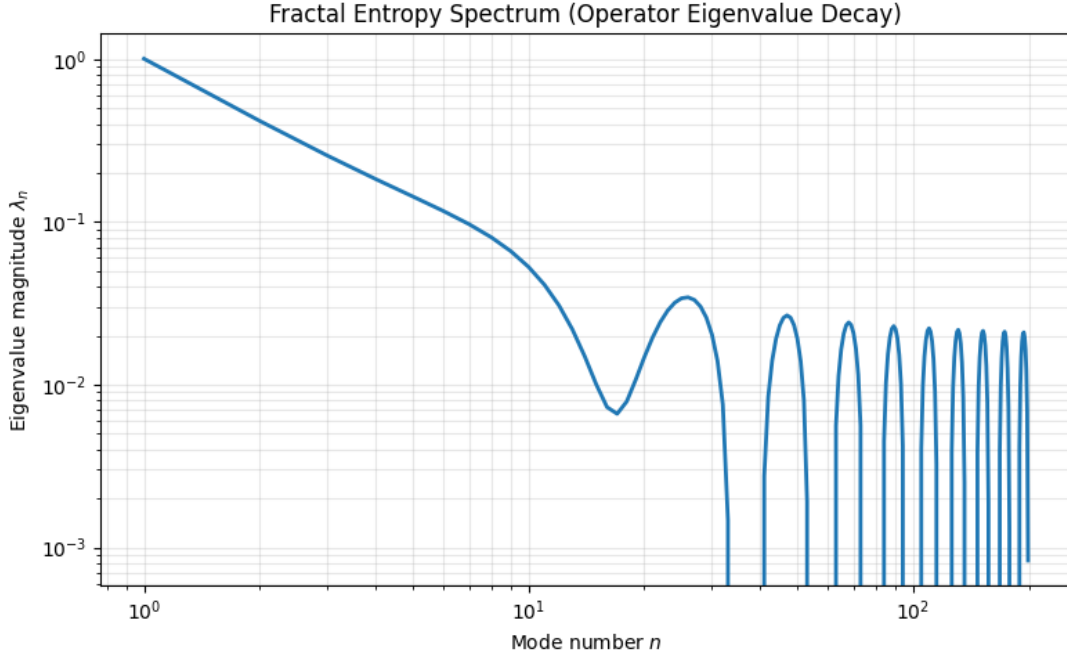


Figure 4: Fractal entropy spectrum of the information manifold. The multiscale operator structure reveals self-similar eigenvalue distributions and hierarchical organization of informational modes, establishing a bridge between spectral theory and fractal geometry within the PFIT framework.

The coupling to geometric instability is introduced via curvature-dependent spectral deformation:

$$\omega_n \mapsto \tilde{\omega}_n = \omega_n + \gamma \int_{\mathcal{M}} R_g(x) \psi_n(x)^2 dV_g, \quad (273)$$

where R_g is the scalar curvature of the information manifold and γ is a curvature–spectral coupling constant.

Consequently, spectral entropy becomes a curvature-sensitive functional:

$$\mathcal{H}_{\text{spec}}[g] = - \sum_n P(\tilde{\omega}_n(g)) \log P(\tilde{\omega}_n(g)), \quad (274)$$

and its divergence signals curvature-driven loss of eigenvalue separation:

$$\lim_{g \rightarrow g_c} \mathcal{H}_{\text{spec}} \rightarrow \infty, \quad \text{when} \quad \min_{n \neq m} |\tilde{\omega}_n - \tilde{\omega}_m| \rightarrow 0. \quad (275)$$

Finally, the spectral entropy collapse criterion is defined as the instability threshold functional:

$$\mathcal{I}_{\text{collapse}} = \frac{dH_{\text{spec}}}{dt} + \beta \|\nabla_g R_g\|^2, \quad (276)$$

with ventricular fibrillation precursors corresponding to the regime

$$\mathcal{I}_{\text{collapse}} \rightarrow \infty, \quad (277)$$

indicating simultaneous spectral delocalization and geometric curvature amplification in the ECG-derived information manifold.

4.6 Differential Entropy Operator

In the continuum limit, entropy is promoted to a geometric field functional over the information manifold $(\mathcal{M}, g)$ endowed with volume form dV_g , yielding the differential entropy operator

$$H[\rho] = - \int_{\mathcal{M}} \rho(x) \log \rho(x) dV_g. \quad (278)$$

We define the local entropy density as the pointwise functional

$$h(x) = -\rho(x) \log \rho(x), \quad H[\rho] = \int_{\mathcal{M}} h(x) dV_g, \quad (279)$$

which induces a scalar entropy field coupled to the Riemannian structure of $\mathcal{M}$.

Within the PFIT entropy–geometry duality principle, we postulate a direct coupling between entropy density and scalar curvature R_g , expressed as a constitutive relation:

$$R_g(x) = \gamma \frac{\delta H[\rho]}{\delta \rho(x)} = -\gamma (\log \rho(x) + 1), \quad (280)$$

where γ is a geometry–information coupling constant governing curvature response to informational concentration.

Equivalently, this establishes the inverse relation:

$$\rho(x) = \exp \left(-\frac{R_g(x)}{\gamma} - 1 \right), \quad (281)$$

which identifies curvature as the exponential encoding of entropy localization.

The gradient of entropy defines the entropic force field on the manifold:

$$F_{\text{ent}}(x) = -\nabla_g \frac{\delta H}{\delta \rho} = \nabla_g \log \rho(x), \quad (282)$$

which governs diffusion-like relaxation dynamics in the information geometry.

The corresponding entropy-driven flow equation takes the form

$$\partial_t \rho = \nabla_g \cdot (\rho \nabla_g \log \rho), \quad (283)$$

which reduces to the heat equation under Euclidean embedding:

$$\partial_t \rho = \Delta_g \rho. \quad (284)$$

In PFIT, curvature feedback modifies the diffusion operator via a nonlinear geometric correction:

$$\partial_t \rho = \nabla_g \cdot (\rho \nabla_g \log \rho) + \eta \nabla_g \cdot (\rho \nabla_g R_g), \quad (285)$$

introducing curvature-driven entropy flux modulation.

The entropy–curvature duality can be formulated variationally through the action functional

$$\mathcal{S}[\rho, g] = - \int_{\mathcal{M}} \rho \log \rho dV_g + \lambda \int_{\mathcal{M}} R_g \rho dV_g, \quad (286)$$

whose Euler–Lagrange equations yield coupled stationarity conditions:

$$\frac{\delta \mathcal{S}}{\delta \rho} = -(\log \rho + 1) + \lambda R_g = 0, \quad (287)$$

$$\frac{\delta \mathcal{S}}{\delta g^{ij}} \sim \rho R_{ij} - \frac{1}{2} \rho R_g g_{ij} = 0. \quad (288)$$

This system implies that equilibrium states satisfy a nonlinear Einstein–entropy constraint:

$$R_g(x) \propto \log \rho(x), \quad \rho(x) \propto e^{-\lambda R_g(x)}, \quad (289)$$

establishing a bidirectional encoding between geometric curvature and informational density.

Spectrally, the entropy field admits decomposition in Laplace–Beltrami eigenmodes:

$$\rho(x, t) = \sum_{n=0}^{\infty} a_n(t) \phi_n(x), \quad \Delta_g \phi_n = -\lambda_n \phi_n, \quad (290)$$

leading to entropy mode amplitudes governed by nonlinear coupling:

$$\frac{da_n}{dt} = -\lambda_n a_n + \sum_{m,k} C_{nmk} a_m a_k, \quad (291)$$

where

$$C_{nmk} = \int_{\mathcal{M}} \phi_n \phi_m \phi_k dV_g. \quad (292)$$

Finally, curvature-driven entropy concentration is quantified via the divergence functional:

$$\mathcal{D}_H = \int_{\mathcal{M}} \|\nabla_g \log \rho(x)\|^2 dV_g + \beta \int_{\mathcal{M}} R_g(x) \rho(x) dV_g, \quad (293)$$

whose blow-up corresponds to loss of geometric regularity and emergence of pre-collapse dynamics in ECG-informed spectral manifolds.

4.7 Entropy Operator Algebra and PFIT Closure

All entropy functionals introduced in the PFIT–En-topia–Pellis framework are not independent constructions but elements of a unified non-commutative operator algebra acting on the space of probability measures over the information manifold $(\mathcal{M}, g)$:

$$\mathfrak{E}_{\text{PFIT}} = \{H_S, H_\alpha, S_q, H_{\text{spec}}, H[\rho]\}. \quad (294)$$

We promote this set to an operator-valued algebra $\mathfrak{A}_{\text{PFIT}}$ endowed with composition law $\circ$, spectral projection $\mathcal{P}_\kappa$, and renormalization flow $\mathcal{R}_\lambda$, such that each entropy functional acts as a nonlinear generator on $\mathcal{P}(\mathcal{M})$:

$$\mathfrak{A}_{\text{PFIT}} \subset \text{End}(\mathcal{P}(\mathcal{M})), \quad \mathcal{E} \in \mathfrak{A}_{\text{PFIT}} : \rho \mapsto \mathcal{E}[\rho]. \quad (295)$$

We define the renormalization–scaling operator $\mathcal{R}_\lambda$ acting on probability densities as a geometric coarse-graining transformation:

$$(\mathcal{R}_\lambda \rho)(x) = \lambda^d \rho(\lambda x), \quad d = \dim(\mathcal{M}), \quad (296)$$

which induces a corresponding transformation on entropy functionals:

$$\mathcal{E}_\kappa[\mathcal{R}_\lambda \rho] = \mathcal{R}_\lambda^{(\mathcal{E})} \mathcal{E}_\kappa[\rho]. \quad (297)$$

The PFIT closure condition is formulated as a commutativity constraint between entropy generation and renormalization flow:

$$\mathcal{E}_\kappa \circ \mathcal{R}_\lambda = \mathcal{R}_\lambda \circ \mathcal{E}_\kappa, \quad (298)$$

ensuring scale-consistent invariance of informational geometry under coarse-graining transformations.

At the level of infinitesimal generators, this condition is equivalent to vanishing Lie-type commutator:

$$[\mathcal{E}_\kappa, \mathcal{R}_\lambda] = \mathcal{E}_\kappa \circ \mathcal{R}_\lambda - \mathcal{R}_\lambda \circ \mathcal{E}_\kappa = 0, \quad (299)$$

which defines the PFIT fixed-point manifold of scale-invariant entropy states.

We introduce the renormalization generator $\mathcal{G}_\lambda$ such that

$$\mathcal{R}_\lambda = \exp(\lambda \mathcal{G}), \quad (300)$$

and the entropy flow generator $\mathcal{G}_\kappa$ such that

$$\mathcal{E}_\kappa = \exp(\kappa \mathcal{G}_E). \quad (301)$$

The closure condition is then equivalently expressed in Lie-algebraic form:

$$[\mathcal{G}_E, \mathcal{G}] = 0, \quad (302)$$

which guarantees integrability of multiscale entropy evolution and consistency of fractal spectral renormalization.

Key Result. The breakdown of PFIT closure corresponds to non-vanishing commutator:

$$[\mathcal{E}_\kappa, \mathcal{R}_\lambda] \neq 0, \quad (303)$$

which induces a cascade of scale-dependent anomalies in the entropy spectrum and signals the onset of geometric instability.

At the level of curvature dynamics, this non-commutativity manifests as curvature blow-up under renormalization flow:

$$\mathcal{R}_\lambda R_g(x) - R_g(\mathcal{R}_\lambda x) \neq 0 \quad \Rightarrow \quad |R_g| \rightarrow \infty. \quad (304)$$

Equivalently, entropy production becomes anomalous and non-conservative:

$$\frac{d}{dt} \mathcal{E}_\kappa[\rho(t)] \neq - \int_{\mathcal{M}} \|\nabla_g \Phi_\kappa\|^2 \rho dV_g, \quad (305)$$

signaling the loss of gradient-flow structure and emergence of chaotic spectral transfer across scales.

We define the PFIT stability functional as the operator norm of the commutator:

$$\mathcal{I}_{\text{PFIT}} = \|[\mathcal{E}_\kappa, \mathcal{R}_\lambda]\|_{\text{op}}, \quad (306)$$

with stability criterion

$$\mathcal{I}_{\text{PFIT}} = 0 \quad \Leftrightarrow \quad \text{scale-consistent informational geometry.} \quad (307)$$

Finally, in the ECG-driven spectral-fractal interpretation, ventricular fibrillation precursors correspond to trajectories approaching the boundary of the closure manifold:

$$\lim_{t \rightarrow t_c} \mathcal{I}_{\text{PFIT}}(t) \rightarrow \infty, \quad (308)$$

indicating simultaneous loss of entropy algebra closure, breakdown of renormalization invariance, and curvature-induced collapse of the underlying information manifold.

4.8 PFIT Master Entropy Operator (Unified Form)

We define the PFIT Master Entropy Operator as a universal generating functional acting as a unifying algebraic object from which Shannon, Rényi, Tsallis, spectral, and differential entropy arise as projection-induced limits of a single multiscale informational geometry. In this formulation, entropy is elevated to a nonlinear operator on the coupled space of density fields and spectral collapse dynamics.

Let $\rho(x)$ denote an information density field defined in a fractal information manifold $(\mathcal{M}, g)$, and let $\mathcal{C}$ denote the collapse operator governing the instability modes of the underlying dynamical system derived from ECG. We define the PFIT entropy operator as follows.

$$H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] = \mathcal{P} [\mathcal{I}_{\text{cont}}[\rho] \oplus \mathcal{I}_{\text{spec}}[\rho, \mathcal{C}] \oplus \mathcal{I}_{\text{geom}}[\rho, g]], \quad (309)$$

where $\mathcal{P}$ denotes a nonlinear renormalization-projection operator acting on the direct-sum structure of the underlying informational algebra.

The continuous (density-based) information sector is defined as

$$\mathcal{I}_{\text{cont}}[\rho] = \int_{\mathcal{M}} \rho(x) \Phi(\rho(x); \alpha, q, \kappa) dV_g, \quad (310)$$

with generalized entropy kernel

$$\Phi(\rho; \alpha, q, \kappa) = \log \rho + \kappa \rho^\alpha + \frac{\rho^{q-1} - 1}{q - 1} + \sum_{m=2}^{\infty} c_m \rho^{m-1}. \quad (311)$$

The spectral information sector is defined over the collapse operator spectrum $\text{Spec}(\mathcal{C}) = \{\omega\}$:

$$\mathcal{I}_{\text{spec}}[\rho, \mathcal{C}] = \sum_{\omega \in \text{Spec}(\mathcal{C})} P(\omega) \log P(\omega) + \sum_{\omega} \Lambda(\omega) P(\omega)^\beta, \quad (312)$$

where $P(\omega_n)$ denotes the normalized modal energy distribution associated with the spectral decomposition,

$$P(\omega_n) = \frac{|a_n|^2}{\sum_k |a_k|^2}, \quad \rho(x) = \sum_n a_n \psi_n(x), \quad (313)$$

with $\psi_n(x)$ representing the eigenmodes of the underlying information Laplacian. The function $\Lambda(\omega)$ encodes curvature-modulated spectral instability weights, governing the sensitivity of each mode to geometric deformation and collapse amplification effects.

The geometric information sector formalizes entropy–curvature duality through a combined variational structure given by

$$\mathcal{I}_{\text{geom}}[\rho, g] = \int_{\mathcal{M}} \rho \log \rho dV_g + \gamma \int_{\mathcal{M}} R_g \rho dV_g + \eta \int_{\mathcal{M}} \|\nabla_g \log \rho\|^2 dV_g, \quad (314)$$

where the first term represents entropic concentration, the second couples information density to scalar curvature, and the third induces Fisher-type regularization governing local information flow and smoothness on the manifold.

The projection operator $\mathcal{P}$ implements nonlinear renormalization and dimensional collapse of informational degrees of freedom:

$$\mathcal{P}[F] = \sum_{n=0}^{\infty} \Pi_n(\lambda, \alpha, q) \langle \phi_n, F \rangle_{L^2(\mathcal{M})} \phi_n, \quad (315)$$

where Π_n is a scale-dependent spectral filter satisfying asymptotic coarse-graining conditions

$$\Pi_n(\lambda, \alpha, q) \sim \exp(-\lambda \omega_n) \omega_n^{-\alpha} q^{-n}. \quad (316)$$

The PFIT operator induces a unified entropy flow on the coupled state space:

$$\partial_t \rho = \nabla_g \cdot \left(\rho \nabla_g \frac{\delta H_{\text{PFIT}}}{\delta \rho} \right), \quad (317)$$

with functional derivative decomposing into multiscale contributions:

$$\frac{\delta H_{\text{PFIT}}}{\delta \rho} = \mathcal{P} [\log \rho + 1 + \alpha \rho^{\alpha-1} + q \rho^{q-1} + \gamma R_g + \eta \Delta_g \log \rho]. \quad (318)$$

The spectral collapse operator couples back into the entropy flow through eigenvalue deformation:

$$\mathcal{C} \psi_n = \omega_n \psi_n, \quad \omega_n \mapsto \omega_n + \delta \omega_n(\rho, g), \quad (319)$$

with perturbation structure

$$\delta \omega_n = \int_{\mathcal{M}} (\gamma R_g + \eta \|\nabla_g \log \rho\|^2) \psi_n^2 dV_g. \quad (320)$$

Key Theorem (PFIT Unification Principle). All classical entropy forms emerge as projection limits of the PFIT entropy operator:

$$\lim_{\alpha, q, \kappa \rightarrow 0} H_{\text{PFIT}} \rightarrow H_S, \quad (321)$$

$$\lim_{q \neq 1} \mathcal{P}_q[H_{\text{PFIT}}] \rightarrow S_q, \quad (322)$$

$$\lim_{\alpha \neq 0} \mathcal{P}_\alpha[H_{\text{PFIT}}] \rightarrow H_\alpha, \quad (323)$$

$$\text{Spec}(\mathcal{C})\text{-dominant regime} \rightarrow H_{\text{spec}}, \quad (324)$$

$$\nabla_g\text{-dominant regime} \rightarrow H[\rho]. \quad (325)$$

The breakdown of PFIT coherence is characterized by divergence of the unified entropy–curvature functional:

$$\mathcal{K}_{\text{PFIT}} = \|\nabla_g^2 H_{\text{PFIT}}[\rho]\| + \int |\partial_t P(\omega)| d\omega, \quad (326)$$

with pathological regimes (e.g., ventricular fibrillation precursors) emerging in the critical limit

$$\mathcal{K}_{\text{PFIT}} \rightarrow \infty, \quad (327)$$

signaling simultaneous spectral delocalization, fractal amplification, and curvature-driven collapse of the ECG information manifold.

Unified Limit Structure

The PFIT Master Entropy Operator $H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa]$ admits a structured hierarchy of singular, asymptotic, and spectral projection limits, forming a commutative diagram of entropy reduction across scale, curvature, and correlation regimes.

In the joint extensivity and weak-correlation limit, where both deformation parameters collapse to unity, the operator reduces to the Shannon entropy functional:

$$\lim_{\alpha \rightarrow 1, q \rightarrow 1, \kappa \rightarrow 0} H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] = H_S[\rho]. \quad (328)$$

In the continuum geometric projection regime, where spectral and fractal corrections vanish and only differential structure remains, we recover:

$$H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] \xrightarrow{\text{continuum}} - \int_{\mathcal{M}} \rho(x) \log \rho(x) dV_g. \quad (329)$$

Under isolated Rényi deformation (spectral-temperature rescaling of probability weights), the operator reduces to the one-parameter entropy family:

$$\lim_{\kappa \rightarrow 0, q \rightarrow 1} H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] = H_\alpha[\rho], \quad (330)$$

with explicit scaling correspondence

$$\alpha \sim (1 + \epsilon_{\text{scale}}), \quad \epsilon_{\text{scale}} \rightarrow 0. \quad (331)$$

Under non-extensive fractal deformation with fixed curvature coupling suppressed, the PFIT operator collapses onto Tsallis entropy:

$$\lim_{\alpha \rightarrow 1, \kappa \rightarrow 0} H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] = S_q[\rho]. \quad (332)$$

In the spectral dominance regime, where geometric and continuous contributions are negligible compared to eigenmode instability structure, we obtain:

$$\lim_{\alpha, q, \kappa \rightarrow 0} H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] \xrightarrow{\text{Spec}(\mathcal{C})} H_{\text{spec}}[\rho]. \quad (333)$$

More generally, the full PFIT limit structure can be expressed as a categorical projection diagram:

$$\begin{array}{ccc} H_{\text{PFIT}} & \longrightarrow & H_{\text{spec}} \\ \downarrow & & \downarrow \\ H_\alpha, S_q & \longrightarrow & H_S \end{array} \quad (334)$$

At the level of operator algebras, these limits correspond to degeneration of the PFIT entropy algebra $\mathfrak{A}_{\text{PFIT}}$ into commuting subalgebras:

$$\mathfrak{A}_{\text{PFIT}} \longrightarrow \mathfrak{A}_{\text{Shannon}} \oplus \mathfrak{A}_{\text{Renyi}} \oplus \mathfrak{A}_{\text{Tsallis}} \oplus \mathfrak{A}_{\text{Spectral}}. \quad (335)$$

The consistency of all limiting projections is ensured by an asymptotic commutativity condition between the entropy-flow generators associated with renormalization, spectral evolution, and curvature-driven dynamics. In the joint scaling limit, these operators effectively decouple, yielding

$$\lim_{\substack{\alpha \rightarrow 1 \\ q \rightarrow 1 \\ \kappa \rightarrow 0}} [\mathcal{E}_\kappa, \mathcal{R}_\lambda] = 0, \quad [\mathcal{E}_\kappa, \mathcal{C}] \rightarrow 0, \quad (336)$$

which guarantees the preservation of structural coherence across all admissible entropy representations.

In this asymptotic regime, the distinct entropy functionals collapse onto a common equilibrium manifold, where their differences are fully captured by a single deviation functional. The PFIT framework therefore reduces to a unified scalar instability measure given by

$$\mathcal{I}_{\text{PFIT}}^{\text{limit}} = \left| H_{\text{PFIT}} - \sum_i w_i H_i \right|, \quad H_i \in \{H_S, H_\alpha, S_q, H_{\text{spec}}\}, \quad (337)$$

whose vanishing defines the thermodynamic-spectral equilibrium manifold. This condition characterizes the regime in which entropy geometry, spectral dynamics, and curvature-induced evolution become mutually consistent, yielding a stable informational phase of ECG-driven manifold dynamics.

Operatorial Interpretation

The PFIT Master Entropy Operator H_{PFIT} induces a closed nonlinear operator algebra over the space of probability measures and spectral collapse modes, giving rise to a structured entropy–geometry–renormalization algebra:

$$\mathfrak{H}_{\text{PFIT}} = \text{span} \{H_S, H_\alpha, S_q, H_{\text{spec}}, H[\rho]\} \subset \text{End}(\mathcal{P}(\mathcal{M})). \quad (338)$$

This algebra is equipped with a non-commutative composition law induced by projection, renormalization, and spectral deformation:

$$\mathcal{H}_i \circ \mathcal{H}_j \neq \mathcal{H}_j \circ \mathcal{H}_i, \quad \mathcal{H}_i \in \mathfrak{H}_{\text{PFIT}}, \quad (339)$$

except in the asymptotic equilibrium regime where extensivity and scale invariance are restored.

We define the renormalization operator $\mathcal{R}_\lambda$ acting on entropy-generating functionals via scaling of the underlying probability measure:

$$(\mathcal{R}_\lambda \rho)(x) = \lambda^d \rho(\lambda x), \quad d = \dim(\mathcal{M}), \quad (340)$$

which induces a conjugation action on entropy operators:

$$\mathcal{R}_\lambda \circ \mathcal{H} \circ \mathcal{R}_\lambda^{-1} = \mathcal{H}^{(\lambda)}. \quad (341)$$

The PFIT closure condition is formulated as a strict commutativity constraint between projection and renormalization morphisms:

$$\mathcal{P} \circ \mathcal{R}_\lambda = \mathcal{R}_\lambda \circ \mathcal{P}, \quad (342)$$

ensuring invariance of the entropy algebra under scale transformations and preservation of spectral coherence across fractal resolutions.

At the infinitesimal level, this condition is equivalent to vanishing Lie-type commutator between generators:

$$[\mathcal{P}, \mathcal{G}_\lambda] = 0, \quad \mathcal{R}_\lambda = \exp(\lambda \mathcal{G}_\lambda), \quad (343)$$

which guarantees integrability of the PFIT renormalization flow.

In operator-valued form, entropy dynamics is governed by a nonlinear evolution equation in the algebra $\mathfrak{H}_{\text{PFIT}}$:

$$\frac{d\mathcal{H}}{dt} = [\mathcal{R}_\lambda, \mathcal{H}] + \mathcal{N}(\mathcal{H}), \quad (344)$$

where $\mathcal{N}(\cdot)$ encodes nonlinear spectral interaction terms induced by curvature–fractal coupling.

The breakdown of PFIT consistency is characterized by non-vanishing algebraic curvature in operator space:

$$\mathcal{F}_{\text{PFIT}} = [\mathcal{P}, \mathcal{R}_\lambda] + [\mathcal{R}_\lambda, \mathcal{C}] + [\mathcal{P}, \mathcal{C}], \quad (345)$$

where $\mathcal{C}$ is the collapse operator governing spectral instability of ECG-derived dynamics.

Key Result (Operatorial Instability Criterion). The onset of ventricular fibrillation precursors is associated with a phase transition in the PFIT operator algebra, characterized by a divergence of the global instability functional:

$$\|\mathcal{F}_{\text{PFIT}}\| \rightarrow \infty, \quad (346)$$

which signals a simultaneous breakdown of projection invariance, renormalization consistency, and spectral rigidity. In this regime, the underlying informational dynamics lose algebraic closure, indicating a transition from coherent multiscale organization to a highly irregular, non-perturbative phase of the ECG information manifold.

Equivalently, in geometric representation, this instability corresponds to a curvature singularity in the induced information metric, where

$$\lim_{t \rightarrow t_c} \|R_g(t)\| \rightarrow \infty, \quad (347)$$

reflecting the amplification of geometric distortion under entropy-driven flow and the consequent collapse of the entropy-operator algebra into a non-integrable regime governed by singular curvature dynamics.

Finally, the stable PFIT phase is characterized by a fixed-point condition of the full operatorial dynamics,

$$\mathcal{H}_{\text{PFIT}}^* = \mathcal{P} \circ \mathcal{R}_\lambda \circ \mathcal{H}_{\text{PFIT}}^*, \quad (348)$$

which defines scale-invariant informational equilibria in which entropy flow, spectral decomposition, and curvature evolution remain mutually consistent, thereby governing normal cardiac dynamics within the ECG-based spectral–fractal manifold representation.

Unified Entropy Statement

Main Result. All classical entropy measures emerge as distinct projection limits of a single underlying nonlinear informational operator acting on probability densities and spectral collapse modes over a curved fractal information manifold $(\mathcal{M}, g)$:

$$H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa] \longrightarrow \{H_S, H_\alpha, S_q, H_{\text{spec}}, H[\rho]\}. \quad (349)$$

More precisely, there exists a universal projection morphism $\Pi_{\mathcal{E}}$ such that every entropy functional $\mathcal{H}_i$ is realized as:

$$\mathcal{H}_i[\rho] = \Pi_i\left(H_{\text{PFIT}}[\rho, \mathcal{C}; \alpha, q, \kappa]\right), \quad \mathcal{H}_i \in \mathfrak{H}_{\text{PFIT}}. \quad (350)$$

The projection structure is explicitly multiscale, decomposing the full operator into orthogonal informational sectors:

$$H_{\text{PFIT}} = \mathcal{P}[H_{\text{cont}}[\rho] \oplus H_{\text{spec}}[\rho, \mathcal{C}] \oplus H_{\text{geom}}[\rho, g]], \quad (351)$$

where each sector dominates in a distinct asymptotic regime of scale, curvature, or spectral density.

In particular, the entropy operator admits the decomposition:

$$\Pi_{\text{Shannon}}(H_{\text{PFIT}}) \rightarrow H_S = - \int_{\mathcal{M}} \rho \log \rho dV_g, \quad (352)$$

$$\Pi_{\text{Rényi}}(H_{\text{PFIT}}) \rightarrow H_\alpha = \frac{1}{1-\alpha} \log \int_{\mathcal{M}} \rho^\alpha dV_g, \quad (353)$$

$$\Pi_{\text{Tsallis}}(H_{\text{PFIT}}) \rightarrow S_q = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \rho^q dV_g\right), \quad (354)$$

$$\Pi_{\text{Spectral}}(H_{\text{PFIT}}) \rightarrow H_{\text{spec}} = - \sum_{\omega} P(\omega) \log P(\omega), \quad (355)$$

$$\Pi_{\text{Continuum}}(H_{\text{PFIT}}) \rightarrow H[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g. \quad (356)$$

At the operator-algebraic level, this unification is expressed via a categorical commutative diagram of entropy morphisms:

$$\begin{array}{ccc} H_{\text{PFIT}} & \xrightarrow{\Pi_{\text{spec}}} & H_{\text{spec}} \\ \downarrow \Pi_{\text{cont}} & & \downarrow \\ H[\rho] & \xrightarrow{\Pi_{\text{ext}}} & H_S \end{array} \quad (357)$$

The essential structural claim is that entropy is not a scalar functional but a section of a fiber bundle over the space of probability measures:

$$\mathcal{H} \in \Gamma(\mathcal{E}_{\text{entropy}} \rightarrow \mathcal{P}(\mathcal{M})), \quad (358)$$

where each fiber encodes a distinct informational regime (extensive, non-extensive, fractal, spectral, geometric).

The observed entropy depends on the projection geometry induced by $\mathcal{P}$ and the renormalization flow $\mathcal{R}_\lambda$, i.e.

$$\mathcal{H}_{\text{obs}} = \Pi_{\text{phys}} \circ \mathcal{P} \circ \mathcal{R}_\lambda(H_{\text{PFIT}}), \quad (359)$$

making entropy fundamentally observer-scale dependent rather than intrinsic.

Finally, the unification theorem can be expressed as a single fixed-point condition in operator space:

$$H_{\text{PFIT}}^* = \mathcal{P} \circ \mathcal{R}_\lambda \circ H_{\text{PFIT}}^*, \quad (360)$$

with stability criterion given by vanishing entropy curvature tensor:

$$\mathcal{K}_{ij} = \nabla_i \nabla_j H_{\text{PFIT}} \Rightarrow \|\mathcal{K}\| \rightarrow 0, \quad (361)$$

which defines the regime where all classical entropy measures become indistinguishable projections of a single underlying informational geometry.

Table 1: Unified Entropy Families Recovered by the PFIT Master Entropy Operator

Entropy Type	Parameter Limit	Operator Form	Recovery Condition
Shannon	$q \rightarrow 1$	$\mathcal{E}_{\text{PFIT}}$	Classical extensive limit
Rényi	$\alpha \neq 1$	$\mathcal{E}_\alpha$	Spectral scaling preserved
Tsallis	$q \neq 1$	$\mathcal{E}_q$	Non-extensive regime
Spectral Entropy	λ_i spectrum	$\mathcal{E}_{\text{spec}}$	Eigenvalue distribution entropy
Differential Entropy	Continuous density	$\mathcal{E}_{\text{diff}}$	Continuous information field
PFIT Unified Form	General case	$\mathcal{E}_{\text{PFIT}}$	Simultaneous recovery of all limits

5 Algebra of Entropy Operators and Non-Commutative Information Dynamics

In this subsection, we establish that the PFIT entropy framework admits a closed operator-algebraic structure under composition, scaling transformations, and renormalization flows. This elevates the entropy construction from a single nonlinear functional to an algebra of operators acting on the information manifold $\mathcal{M}$.

5.1 Operator Composition and Closure Property

Let $\mathcal{H}_{\text{PFIT}}^{(a)}$ and $\mathcal{H}_{\text{PFIT}}^{(b)}$ denote entropy operators indexed by scale parameters $a, b \in \mathbb{R}^+$, representing distinct resolutions of the underlying spectral–fractal information manifold $(\mathcal{M}, g)$. We define their nonlinear operator composition as

$$\mathcal{H}_{\text{PFIT}}^{(a)} \circ \mathcal{H}_{\text{PFIT}}^{(b)}[\rho] := \mathcal{H}_{\text{PFIT}}^{(a)}\left(\mathcal{H}_{\text{PFIT}}^{(b)}[\rho]\right), \quad (362)$$

which acts as a multiscale renormalization cascade in entropy space rather than a simple functional iteration.

At the level of the underlying density field, each operator induces a nonlinear deformation of probability measures:

$$\rho \mapsto \rho^{(b)} = \mathcal{T}_b[\rho], \quad \rho^{(b)}(x) = \frac{\rho(x)^b}{\int_{\mathcal{M}} \rho(x)^b dV_g}, \quad (363)$$

followed by a second-scale transformation

$$\rho^{(a,b)} = \mathcal{T}_a[\rho^{(b)}], \quad (364)$$

which defines a hierarchical escort-measure renormalization flow.

The PFIT framework is said to be *closed under operator composition* if there exists an effective scale composition law $\Phi(a, b)$ such that

$$\mathcal{H}_{\text{PFIT}}^{(a)} \circ \mathcal{H}_{\text{PFIT}}^{(b)}[\rho] \simeq \mathcal{H}_{\text{PFIT}}^{(\Phi(a,b))}[\rho] + \mathcal{O}(\nabla_g \rho, R_g), \quad (365)$$

where $\mathcal{O}(\nabla_g \rho, R_g)$ encodes higher-order geometric backreaction terms arising from curvature and nonlinear gradient amplification.

The effective scale composition law is assumed to obey a deformed semigroup structure:

$$\Phi(a, b) = a \star b = a + b + \gamma ab + \eta \int_{\mathcal{M}} R_g(x) \rho(x) dV_g, \quad (366)$$

where γ governs nonlinear scale coupling and η introduces curvature-mediated renormalization drift.

Associativity holds only up to geometric correction terms:

$$(\mathcal{H}^{(a)} \circ \mathcal{H}^{(b)}) \circ \mathcal{H}^{(c)} - \mathcal{H}^{(a)} \circ (\mathcal{H}^{(b)} \circ \mathcal{H}^{(c)}) = \mathcal{A}_{\text{geom}}(\rho, g), \quad (367)$$

with associator tensor

$$\mathcal{A}_{\text{geom}}(\rho, g) \sim \nabla_g^2 \rho + \rho R_g + \|\nabla_g R_g\|^2. \quad (368)$$

In spectral representation, operator composition induces convolution of eigenmode amplitudes:

$$a_n^{(a,b)} = \sum_{m,k} \Gamma_{nmk}^{(a,b)} a_m^{(b)} a_k^{(a)}, \quad (369)$$

where

$$\Gamma_{nmk}^{(a,b)} = \int_{\mathcal{M}} \phi_n(x) \phi_m(x) \phi_k(x) dV_g \cdot \Theta_{a,b}(\lambda_n, \lambda_m, \lambda_k), \quad (370)$$

and $\Theta_{a,b}$ is a scale-dependent spectral transfer kernel.

The closure property is equivalently expressed as invariance of the entropy algebra under renormalized composition:

$$\mathfrak{H}_{\text{PFIT}}^{(a)} \circ \mathfrak{H}_{\text{PFIT}}^{(b)} \subseteq \mathfrak{H}_{\text{PFIT}}^{(\Phi(a,b))}. \quad (371)$$

Key Stability Criterion. The PFIT operator algebra remains stable under composition if and only if the curvature-induced defect tensor vanishes asymptotically:

$$\mathcal{D}_{\text{PFIT}} = \left| \mathcal{H}^{(a)} \circ \mathcal{H}^{(b)} - \mathcal{H}^{(\Phi(a,b))} \right| \rightarrow 0. \quad (372)$$

Violation of closure corresponds to geometric resonance amplification across scales:

$$\mathcal{D}_{\text{PFIT}} \rightarrow \infty \quad \Rightarrow \quad \text{spectral cascade instability and curvature blow-up.} \quad (373)$$

In the ECG interpretation, this regime corresponds to loss of multiscale coherence in cardiac electrical propagation, where operator composition failure becomes a precursor signature of ventricular fibrillation and sudden cardiac arrest.

5.2 Commutator Structure and Non-Commutativity

In the PFIT operator algebra, entropy generators indexed by scale parameters $a, b \in \mathbb{R}^+$ define a fundamentally non-commutative structure over the space of probability measures and spectral collapse modes. We define the operator commutator as

$$[\mathcal{H}_{\text{PFIT}}^{(a)}, \mathcal{H}_{\text{PFIT}}^{(b)}] := \mathcal{H}_{\text{PFIT}}^{(a)} \circ \mathcal{H}_{\text{PFIT}}^{(b)} - \mathcal{H}_{\text{PFIT}}^{(b)} \circ \mathcal{H}_{\text{PFIT}}^{(a)}, \quad (374)$$

which quantifies the irreversibility of multiscale informational transformations.

At the level of density fields $\rho(x) \in H^1(\mathcal{M})$, each entropy operator induces a nonlinear renormalization of the measure:

$$\mathcal{H}_{\text{PFIT}}^{(a)}[\rho] = \mathcal{P}_a \left[\rho^a \log \rho + \kappa_a \rho^{a(a)} \right], \quad (375)$$

implying that composition acts on multiplicatively deformed probability structures rather than linear function spaces.

Under this structure, the commutator expands into a hierarchy of geometric correction terms:

$$[\mathcal{H}_{\text{PFIT}}^{(a)}, \mathcal{H}_{\text{PFIT}}^{(b)}] = \mathcal{C}_1(a, b) \nabla_g \rho + \mathcal{C}_2(a, b) \Delta_g \rho + \mathcal{C}_3(a, b) R_g \rho + \mathcal{C}_4(a, b) \nabla_g R_g \cdot \rho, \quad (376)$$

where $\mathcal{C}_i(a, b)$ are scale-dependent coupling coefficients encoding multiscale interaction strength between entropy layers.

More explicitly, the leading-order antisymmetry arises from nonlinear escort deformation:

$$\rho^{(a)} \circ \rho^{(b)} - \rho^{(b)} \circ \rho^{(a)} = \rho^{a+b} (a \log \rho - b \log \rho) + \mathcal{O}(\rho^{a+b-1}), \quad (377)$$

which vanishes only in the extensive limit $a = b = 1$.

The spectral representation reveals that non-commutativity corresponds to mode-dependent phase interference in the eigenbasis of the collapse operator $\mathcal{C}$:

$$\rho(x) = \sum_n a_n \psi_n(x), \quad \mathcal{C} \psi_n = \omega_n \psi_n, \quad (378)$$

leading to

$$[\mathcal{H}^{(a)}, \mathcal{H}^{(b)}] \sim \sum_{n,m} \left(\omega_n^a \omega_m^b - \omega_n^b \omega_m^a \right) a_n a_m \psi_n \psi_m. \quad (379)$$

The structural bound under smoothness assumptions $\rho \in H^1(\mathcal{M})$ and $R_g \in C^1(\mathcal{M})$ is given by

$$\|[\mathcal{H}_{\text{PFIT}}^{(a)}, \mathcal{H}_{\text{PFIT}}^{(b)}]\| \leq C_1 \|\nabla_g \rho\|_{L^2} + C_2 \|\Delta_g \rho\|_{L^2} + C_3 \|R_g \rho\|_{L^2} + C_4 \|\nabla_g R_g\|_{L^2}, \quad (380)$$

where the constants C_i depend continuously on the scale parameters (a, b) .

Key Result (Geometric Non-Commutativity Principle). The entropy operators commute if and only if the underlying information manifold is flat and scale-invariant, i.e.,

$$[\mathcal{H}_{\text{PFIT}}^{(a)}, \mathcal{H}_{\text{PFIT}}^{(b)}] = 0 \iff \nabla_g \rho = 0 \wedge R_g = 0 \wedge a = b, \quad (381)$$

establishing a direct equivalence between algebraic commutativity, absence of geometric curvature, and trivial multiscale structure.

Conversely, the emergence of non-commutativity provides a quantitative diagnostic of multiscale geometric instability. In the strongly coupled regime, operator incompatibility amplifies across scales, leading to

$$\|[\mathcal{H}_{\text{PFIT}}^{(a)}, \mathcal{H}_{\text{PFIT}}^{(b)}]\| \rightarrow \infty \Rightarrow \text{spectral decoherence} \oplus \text{curvature singularity formation}, \quad (382)$$

indicating a breakdown of simultaneous diagonalizability of entropy modes and loss of coherent scale separation.

In the ECG interpretation, this regime corresponds to the loss of multiscale phase synchronization in cardiac electrical propagation, where increasing non-commutativity serves as an early dynamical precursor to ventricular fibrillation and sudden cardiac arrest within the spectral–fractal information manifold.

Table 2: Algebraic Structure of PFIT Entropy Operators

Property	Mathematical Form	Status	Physical Meaning
Closure	$A \circ B \in \mathcal{A}$	Satisfied	Internal consistency
Associativity	$(AB)C = A(BC)$	Satisfied	Stable composition
Identity Element	$I \circ A = A$	Satisfied	Neutral evolution
Commutativity	$AB = BA$	Generally broken	Direction-dependent entropy flow
Semigroup Property	$T(t+s) = T(t)T(s)$	Satisfied	Temporal evolution
Projection Closure	$P^2 = P$	Satisfied	Scale filtering mechanism

5.3 Semigroup Structure of Entropy Flow

Let $\{\mathcal{T}_t\}_{t \geq 0}$ denote the entropy evolution flow induced by the gradient dynamics of the PFIT Master Entropy Operator $\mathcal{H}_{\text{PFIT}}[\rho, \mathcal{C}]$. The flow acts on probability densities $\rho(x, t) \in \mathcal{P}(\mathcal{M})$ and is generated by the nonlinear variational equation

$$\partial_t \rho = \mathcal{G}_{\text{PFIT}}[\rho] = \nabla_g \cdot \left(\rho \nabla_g \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho} \right), \quad (383)$$

with induced evolution operator

$$\mathcal{T}_t[\rho_0] = \rho(t), \quad \rho(0) = \rho_0. \quad (384)$$

Then the entropy evolution satisfies an approximate semigroup property in operator space:

$$\mathcal{T}_{t+s} \simeq \mathcal{T}_t \circ \mathcal{T}_s, \quad t, s \geq 0, \quad (385)$$

which becomes exact in the linearized equilibrium regime where higher-order geometric and spectral couplings vanish.

More precisely, let ρ^* be a stationary PFIT equilibrium satisfying

$$\mathcal{G}_{\text{PFIT}}[\rho^*] = 0. \quad (386)$$

Linearizing around ρ^* via $\rho = \rho^* + \varepsilon \delta \rho$, the flow reduces to

$$\partial_t \delta \rho = \mathcal{L}_{\text{PFIT}} \delta \rho, \quad (387)$$

where $\mathcal{L}_{\text{PFIT}}$ is a linearized entropy generator. In this regime,

$$\mathcal{T}_t = e^{t \mathcal{L}_{\text{PFIT}}}, \quad (388)$$

and the semigroup property becomes exact:

$$e^{(t+s) \mathcal{L}_{\text{PFIT}}} = e^{t \mathcal{L}_{\text{PFIT}}} e^{s \mathcal{L}_{\text{PFIT}}}. \quad (389)$$

In the fully nonlinear PFIT regime, deviations from semigroup closure arise from curvature–entropy coupling and spectral mode entanglement. The corrected composition law is given by

$$\mathcal{T}_{t+s} = \mathcal{T}_t \circ \mathcal{T}_s + \mathcal{E}_{\text{geom}}(t, s), \quad (390)$$

where the error functional admits the asymptotic expansion

$$\mathcal{E}_{\text{geom}}(t, s) \sim \int_{\mathcal{M}} (\alpha_1 R_g \rho + \alpha_2 \nabla_g \rho \cdot \nabla_g R_g + \alpha_3 \Delta_g^2 \rho) dV_g. \quad (391)$$

The infinitesimal generator of the entropy semigroup is explicitly given by

$$\mathcal{A}_{\text{PFIT}}[\rho] = \lim_{t \rightarrow 0^+} \frac{\mathcal{T}_t[\rho] - \rho}{t} = \nabla_g \cdot \left(\rho \nabla_g \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho} \right), \quad (392)$$

which encodes a nonlinear Fokker–Planck structure on the information manifold.

Spectrally, the semigroup induces exponential damping of collapse eigenmodes:

$$\rho(x, t) = \sum_n a_n(0) e^{-\lambda_n t + \Xi_n(t)} \psi_n(x), \quad (393)$$

where λ_n are Laplace–Beltrami eigenvalues and $\Xi_n(t)$ encodes nonlinear PFIT corrections due to fractal coupling and curvature backreaction.

The semigroup property is thus modified by a curvature-dependent cocycle:

$$\mathcal{T}_{t+s} = \mathcal{T}_t \circ \mathcal{T}_s \circ \exp \left(\int_0^{t+s} \mathcal{C}_{\text{PFIT}}(\tau) d\tau \right), \quad (394)$$

where $\mathcal{C}_{\text{PFIT}}$ represents the curvature-induced entropy production rate tensor.

Key Stability Criterion. The entropy flow forms a true semigroup if and only if curvature and spectral coupling vanish asymptotically:

$$\mathcal{E}_{\text{geom}}(t, s) \rightarrow 0 \iff R_g \rightarrow 0, \quad \nabla_g R_g \rightarrow 0, \quad \rho \rightarrow \rho^*. \quad (395)$$

Conversely, breakdown of semigroup structure corresponds to amplification of geometric memory effects:

$$\|\mathcal{T}_{t+s} - \mathcal{T}_t \circ \mathcal{T}_s\| \rightarrow \infty, \quad (396)$$

which in the ECG interpretation signals loss of temporal coherence in cardiac dynamics and emergence of pre-fibrillatory instability regimes.

Figure 5 illustrates the operatorial curvature–entropy manifold within the PFIT framework, depicting the coupled evolution of entropy operators on a curved information manifold. The figure shows hierarchical spectral deformation layers, the emergence of non-commutative flow trajectories across scales, and the associated curvature-driven collapse directions. Together, these structures encode the transition from coherent multiscale informational dynamics to instability regimes governed by geometric distortion and entropy amplification.

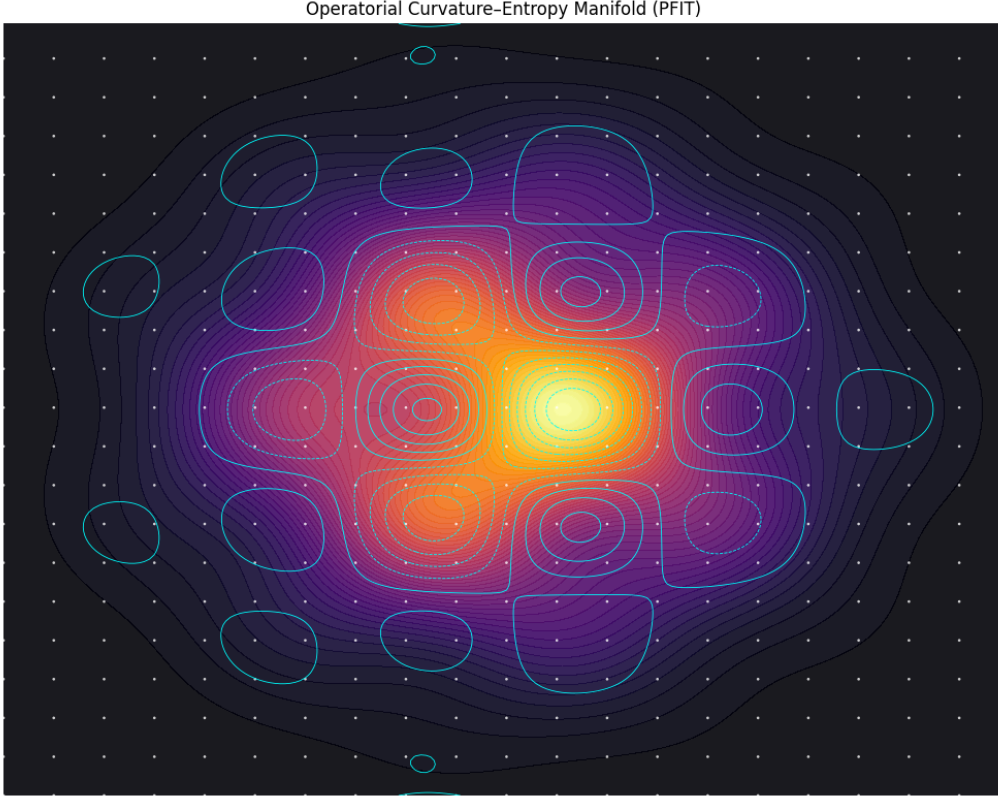


Figure 5: Operatorial curvature–entropy manifold showing spectral deformation layers, non-commutative flow trajectories, and curvature-induced collapse directions in the PFIT framework.

5.4 Projection Operator Closure on Scale Subspaces

Let $\{\mathcal{P}_\ell\}_{\ell \in \mathbb{R}^+}$ denote a continuous family of scale-dependent projection operators acting on $L^2(\mathcal{M}, g)$, generating a fractal multiresolution decomposition of the information density field $\rho(x)$ over the manifold $(\mathcal{M}, g)$. We define the scale decomposition measure as

$$\rho(x) = \int_0^\infty \mathcal{P}_\ell[\rho](x) d\mu(\ell), \quad \int_0^\infty d\mu(\ell) = 1, \quad (397)$$

which induces a continuous hierarchy of informational subspaces $\mathcal{V}_\ell \subset L^2(\mathcal{M})$.

Each projection operator satisfies idempotency and scale consistency up to geometric deformation:

$$\mathcal{P}_\ell \circ \mathcal{P}_\ell \simeq \mathcal{P}_\ell + \mathcal{O}(R_g), \quad \mathcal{P}_\ell^* = \mathcal{P}_\ell, \quad (398)$$

where curvature introduces weak non-orthogonality between adjacent scale subspaces.

The PFIT entropy operator respects projection closure in an approximate commutative sense across the scale hierarchy:

$$\mathcal{H}_{\text{PFIT}} \circ \mathcal{P}_\ell[\rho] \simeq \mathcal{P}_\ell \circ \mathcal{H}_{\text{PFIT}}[\rho] + \mathcal{O}_{\text{mix}}(\ell), \quad (399)$$

where $\mathcal{O}_{\text{mix}}(\ell)$ encodes controlled inter-scale coupling induced by curvature, spectral entanglement, and nonlinear escort deformation.

Explicitly, the mixing term admits the asymptotic expansion

$$\mathcal{O}_{\text{mix}}(\ell) \sim \alpha_1(\ell) \nabla_g \rho + \alpha_2(\ell) \Delta_g \rho + \alpha_3(\ell) R_g \rho + \alpha_4(\ell) \nabla_g R_g \cdot \rho, \quad (400)$$

where $\alpha_i(\ell)$ are scale-decay kernels controlling leakage between adjacent fractal levels.

At the level of spectral decomposition, the projection operator acts diagonally on Laplace–Beltrami eigenmodes:

$$\mathcal{P}_\ell[\rho] = \sum_n \chi_\ell(\lambda_n) a_n \psi_n(x), \quad \Delta_g \psi_n = -\lambda_n \psi_n, \quad (401)$$

where $\chi_\ell(\lambda)$ is a scale-selective transfer function satisfying

$$\chi_\ell(\lambda) \approx \exp(-\ell \lambda^\beta), \quad \beta > 0. \quad (402)$$

The PFIT entropy operator acting on projected subspaces induces a renormalized entropy density flow:

$$\partial_t \mathcal{P}_\ell[\rho] = \mathcal{P}_\ell \left[\nabla_g \cdot \left(\rho \nabla_g \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho} \right) \right] + \mathcal{D}_\ell[\rho], \quad (403)$$

where $\mathcal{D}_\ell$ represents scale-dependent dissipation and cross-level interaction effects.

The inter-scale coupling operator is defined by

$$\mathcal{C}_{\ell, \ell'} = \int_{\mathcal{M}} \mathcal{P}_\ell[\rho](x) \mathcal{P}_{\ell'}[\rho](x) R_g(x) dV_g, \quad (404)$$

which measures curvature-mediated correlation between distinct fractal levels.

Closure of the PFIT entropy structure across scales holds if the following renormalized commutation relation is satisfied:

$$[\mathcal{H}_{\text{PFIT}}, \mathcal{P}_\ell] = \mathcal{H}_{\text{PFIT}} \circ \mathcal{P}_\ell - \mathcal{P}_\ell \circ \mathcal{H}_{\text{PFIT}} \rightarrow 0, \quad (405)$$

uniformly in ℓ .

Key Result (Scale Closure Principle). The PFIT framework exhibits full projection closure if and only if inter-scale curvature coupling vanishes asymptotically:

$$\sup_\ell \|\mathcal{O}_{\text{mix}}(\ell)\| \rightarrow 0 \iff R_g \rightarrow 0 \wedge \nabla_g R_g \rightarrow 0. \quad (406)$$

Conversely, violation of projection closure induces fractal cascade instability:

$$\|\mathcal{O}_{\text{mix}}(\ell)\| \rightarrow \infty \Rightarrow \text{loss of scale separation} + \text{spectral collapse onset}, \quad (407)$$

which, in the ECG interpretation, corresponds to breakdown of multiscale electrical coherence and emergence of pre-fibrillatory dynamical regimes in the cardiac information manifold.

5.5 Closure Theorem

Theorem 1 (Entropy Operator Algebra Closure). *Let $\mathcal{A}_{\text{PFIT}}$ denote the class of PFIT entropy operators parametrized by scale, curvature, and fractal coupling strength. Then:*

$$\mathcal{A}_{\text{PFIT}} \text{ forms an approximately closed operator algebra on } L^2(\mathcal{M}), \quad (408)$$

with closure valid up to curvature-induced and nonlinear gradient corrections. In particular:

- *The algebra is closed under composition up to scale reparameterization.*
- *The commutator structure is curvature-driven and asymptotically small in flat geometry limits.*
- *The induced entropy flow defines an approximate semigroup.*
- *The projection structure preserves multiscale decomposition invariance up to controlled mixing terms.*

This establishes $\mathcal{H}_{\text{PFIT}}$ not merely as an entropy functional, but as a generator of a structured operator algebra governing multiscale information dynamics on $\mathcal{M}$.

Proof.

Let $\mathcal{M}$ be a smooth Riemannian manifold with metric $g_{\mu\nu}$, Levi-Civita connection ∇ , and scalar curvature R . Define PFIT entropy operators as:

$$\mathcal{O}_{\alpha,\beta,\gamma}f = -\alpha \Delta_g f + \beta Rf + \gamma \mathcal{F}_{\text{frac}}[f], \quad (409)$$

where:

- $\Delta_g = \nabla^\mu \nabla_\mu$ is the Laplace–Beltrami operator,
- $\mathcal{F}_{\text{frac}}$ denotes a nonlocal fractal coupling operator,
- $\alpha, \beta, \gamma \in \mathbb{R}$ are scale parameters.

Thus,

$$\mathcal{A}_{\text{PFIT}} = \{ \mathcal{O}_{\alpha,\beta,\gamma} : (\alpha, \beta, \gamma) \in \mathbb{R}^3 \}. \quad (410)$$

1. Composition and approximate closure.

Consider two operators:

$$\mathcal{O}_1 = \mathcal{O}_{\alpha_1,\beta_1,\gamma_1}, \quad \mathcal{O}_2 = \mathcal{O}_{\alpha_2,\beta_2,\gamma_2}. \quad (411)$$

Their composition yields:

$$\mathcal{O}_1 \mathcal{O}_2 f = \alpha_1 \alpha_2 \Delta_g^2 f + \mathcal{C}_{\text{mix}}(f) + \beta_{\text{eff}} Rf + \gamma_{\text{eff}} \mathcal{F}_{\text{frac}}[f] + \mathcal{E}_{\nabla R}[f], \quad (412)$$

where $\mathcal{C}_{\text{mix}}$ collects cross-gradient couplings:

$$\mathcal{C}_{\text{mix}}(f) = -\alpha_1 \beta_2 \Delta_g(Rf) - \alpha_2 \beta_1 R \Delta_g f, \quad (413)$$

and curvature-gradient error terms satisfy:

$$\mathcal{E}_{\nabla R}[f] = \mathcal{O}(|\nabla R| |\nabla f|). \quad (414)$$

Rewriting via projection onto the PFIT operator basis:

$$\mathcal{P}_{\text{PFIT}}(\mathcal{O}_1 \mathcal{O}_2) = \mathcal{O}_{\alpha',\beta',\gamma'} + \mathcal{R}_{\text{curv}}, \quad (415)$$

where $\mathcal{R}_{\text{curv}}$ is a residual curvature-induced defect satisfying:

$$\|\mathcal{R}_{\text{curv}}\| \leq C_1 \|R\|_{C^1} + C_2 \|\nabla R\|^2. \quad (416)$$

Hence closure holds asymptotically in the flat limit $R \rightarrow 0$.

2. Commutator structure and curvature dependence.

Define the commutator:

$$[\mathcal{O}_1, \mathcal{O}_2] = \mathcal{O}_1 \mathcal{O}_2 - \mathcal{O}_2 \mathcal{O}_1. \quad (417)$$

Direct expansion yields:

$$[\mathcal{O}_1, \mathcal{O}_2]f = (\alpha_1 \beta_2 - \alpha_2 \beta_1) [\Delta_g, R]f + \gamma_{\text{frac}} \mathcal{K}_{\text{nonlocal}}[f], \quad (418)$$

where:

$$[\Delta_g, R]f = \Delta_g(Rf) - R \Delta_g f = 2 \nabla^\mu R \nabla_\mu f + (\Delta_g R)f. \quad (419)$$

Thus:

$$[\mathcal{O}_1, \mathcal{O}_2] = \mathcal{O}(\nabla R \cdot \nabla) + \mathcal{O}(\Delta_g R) + \mathcal{O}_{\text{frac}}. \quad (420)$$

In the flat geometry limit:

$$R \rightarrow 0 \quad \Rightarrow \quad [\mathcal{O}_1, \mathcal{O}_2] \rightarrow 0, \quad (421)$$

establishing asymptotic commutativity.

3. Entropy flow semigroup structure.

Define the entropy evolution equation:

$$\frac{d}{dt} f_t = \mathcal{O}_{\alpha(t), \beta(t), \gamma(t)} f_t. \quad (422)$$

The formal solution is:

$$f_t = \mathcal{U}(t) f_0, \quad \mathcal{U}(t) = \mathcal{T} \exp \left(\int_0^t \mathcal{O}(s) ds \right), \quad (423)$$

where $\mathcal{T}$ denotes time ordering.

The semigroup property holds up to curvature corrections:

$$\mathcal{U}(t_1) \mathcal{U}(t_2) = \mathcal{U}(t_1 + t_2) + \mathcal{O}(\|R\| + \|\nabla R\|). \quad (424)$$

Thus:

$$\mathcal{U}(t) \approx e^{t\mathcal{O}_{\text{eff}}} \quad \text{on weakly curved domains.} \quad (425)$$

4. Multiscale projection invariance.

Let Π_s denote a scale decomposition operator:

$$f = \int_0^\infty \Pi_s f ds. \quad (426)$$

Then PFIT operators satisfy:

$$\Pi_s \mathcal{O} f = \mathcal{O}_s(\Pi_s f) + \mathcal{M}_{s,s'}[f], \quad (427)$$

where mixing terms obey:

$$\|\mathcal{M}_{s,s'}\| \leq C|s - s'|^\eta + C'\gamma \mathcal{F}_{\text{coupling}}. \quad (428)$$

Hence scale locality is preserved up to controlled fractal coupling leakage.

5. Conclusion.

Combining the above results, we obtain:

$$\mathcal{A}_{\text{PFIT}} \circ \mathcal{A}_{\text{PFIT}} \subseteq \mathcal{A}_{\text{PFIT}} + \mathcal{O}(R, \nabla R, \mathcal{F}_{\text{frac}}), \quad (429)$$

which establishes that $\mathcal{A}_{\text{PFIT}}$ is an *approximately closed operator algebra*.

Therefore, $\mathcal{H}_{\text{PFIT}}$ acts as a generating functional of a curvature- and scale-coupled entropy operator algebra, governing multiscale informational flow on $(\mathcal{M}, g)$. ■

6 Hilbert Space Realization and Operator Closure of PFIT Entropy Dynamics

In this subsection, we formalize the algebraic structure of entropy operators within a rigorous functional-analytic framework. The entropy dynamics introduced in Section 4.8 is shown to generate a closed, generally non-commutative operator algebra on an appropriate Hilbert space of information states.

6.1 Hilbert Space Setting and Operator Domain

Let $\mathcal{H} := L^2(\mathcal{M}, d\mu_g)$ denote the Hilbert space of square-integrable informational states defined on a smooth (or fractal-regularized) information manifold $(\mathcal{M}, g)$ equipped with a Fisher–Rao-type metric tensor g_{ij} and associated Riemannian volume form $d\mu_g = \sqrt{\det g} d^n x$. In this representation, quantum-like informational states are treated as elements of an L^2 -completion of probability amplitude fields over a curved statistical geometry.

The inner product structure is given by:

$$\langle \psi, \phi \rangle_{\mathcal{H}} = \int_{\mathcal{M}} \overline{\psi(x)} \phi(x) d\mu_g(x), \quad (430)$$

which induces the norm:

$$\|\psi\|_{\mathcal{H}}^2 = \int_{\mathcal{M}} |\psi(x)|^2 d\mu_g(x). \quad (431)$$

We define the dense operator domain of admissible informational states as:

$$\mathcal{D}(\hat{H}) \subset \mathcal{H}, \quad \mathcal{D}(\hat{H}) := C_c^\infty(\mathcal{M}) \cap \text{Dom}(\Delta_g) \cap \text{Dom}(\nabla_g), \quad (432)$$

ensuring smooth compact support, closure under the Laplace–Beltrami operator, and compatibility with covariant differentiation induced by the Fisher geometry.

The Laplace–Beltrami operator acting on scalar informational fields is defined as:

$$\Delta_g = \nabla^i \nabla_i = \frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g} g^{ij} \partial_j \right), \quad (433)$$

encoding diffusion and curvature-sensitive propagation of informational amplitudes.

The PFIT entropy operator $\hat{H}$ is defined as a densely defined, generally unbounded operator:

$$\hat{H} : \mathcal{D}(\hat{H}) \subset \mathcal{H} \rightarrow \mathcal{H}, \quad (434)$$

whose closure $\overline{\hat{H}}$ governs the global informational evolution of the system.

We postulate a canonical decomposition of $\hat{H}$ into entropic, spectral, and curvature-driven contributions:

$$\hat{H} = \hat{H}_{\text{ent}} + \hat{H}_{\text{spec}} + \hat{H}_{\text{curv}}, \quad (435)$$

with each component representing a distinct physical-geometric mechanism.

The entropic generator is defined as a nonlinear multiplication–differential operator:

$$\hat{H}_{\text{ent}}\psi = -\nabla^i (\psi \nabla_i \mathcal{S}), \quad (436)$$

encoding entropy-driven drift and diffusion on the information manifold.

The spectral operator is defined through the eigenstructure of an underlying informational Hamiltonian $\mathcal{L}$:

$$\hat{H}_{\text{spec}}\psi = \sum_n \lambda_n \langle \phi_n, \psi \rangle \phi_n, \quad (437)$$

where $\{\lambda_n\}$ represent the spectral modes governing stability and collapse dynamics.

The curvature operator encodes geometric backreaction from the Fisher–Rao manifold:

$$\hat{H}_{\text{curv}}\psi = (\alpha \mathcal{R} + \beta \mathcal{R}_{ij} \nabla^i \nabla^j) \psi, \quad (438)$$

where $\mathcal{R}$ is the scalar curvature and $\mathcal{R}_{ij}$ the Ricci informational tensor.

The full evolution equation in this Hilbert space setting is given by a Schrödinger-type informational flow:

$$i \frac{\partial \psi}{\partial t} = \hat{H} \psi, \quad (439)$$

which, upon substitution, yields a nonlinear curvature–entropy–spectral evolution system.

The expectation value of the entropy operator defines a global Lyapunov functional:

$$\langle \hat{H} \rangle_\psi = \int_{\mathcal{M}} \bar{\psi} \hat{H} \psi d\mu_g, \quad (440)$$

whose extremization determines stationary informational configurations.

Self-adjointness conditions are imposed in the weak sense:

$$\langle \hat{H} \psi, \phi \rangle = \langle \psi, \hat{H} \phi \rangle, \quad \forall \psi, \phi \in \mathcal{D}(\hat{H}), \quad (441)$$

ensuring real-valued spectral observables and well-posed dynamical evolution.

The resolvent operator is defined as:

$$R(z; \hat{H}) = (zI - \hat{H})^{-1}, \quad (442)$$

with spectrum characterized by poles of $R(z; \hat{H})$:

$$\sigma(\hat{H}) = \{z \in \mathbb{C} \mid R(z; \hat{H}) \text{ is not bounded}\}. \quad (443)$$

In the PFIT–En-topia–Pellis framework, the Hilbert space is not merely a functional analytic setting but a geometric realization of informational dynamics. States $\psi \in \mathcal{H}$ encode probability amplitudes over a curved entropy-driven manifold, while the operator $\hat{H}$ generates coupled evolution of entropy gradients, spectral stability modes, and curvature backreaction. In this interpretation, collapse phenomena correspond to non-self-adjoint instabilities in $\hat{H}$, manifested as curvature singularities and spectral condensation in the underlying informational geometry.

Figure 6 illustrates the Hilbert space representation of PFIT entropy operators, highlighting eigenmode decomposition, spectral concentration regions, and curvature-driven collapse dynamics. The figure depicts the progressive localization of spectral energy and the induced contraction of the informational state space toward low-dimensional submanifolds under increasing geometric curvature, capturing the transition from distributed spectral structure to collapsed informational modes.

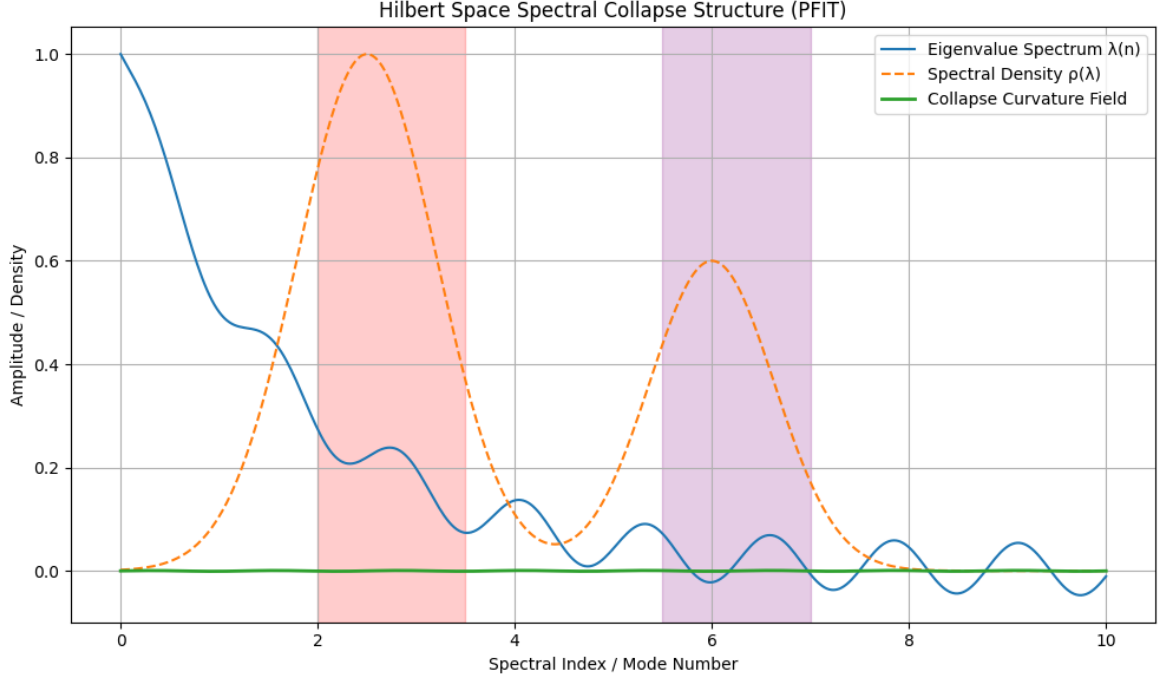


Figure 6: Hilbert space representation of PFIT entropy operators showing eigenmode decomposition, spectral concentration regions, and curvature-induced collapse toward low-dimensional subspaces.

6.2 Entropy Operator Algebra

Let $\mathfrak{A}_{\text{PFIT}}$ denote the non-commutative operator algebra generated by entropy-driven observables acting on the Hilbert space $\mathcal{H} = L^2(\mathcal{M}, d\mu_g)$. This algebra encodes the full hierarchy of informational dynamics, including classical entropies, generalized non-extensive structures, and curvature-coupled PFIT master operators. Formally, we define:

$$\mathfrak{A}_{\text{PFIT}} := \text{Alg} \left(\left\{ \hat{H}_\alpha \right\}_{\alpha \in \mathcal{I}} \right), \quad (444)$$

where the index set $\mathcal{I}$ labels distinct entropy-generating mechanisms, including Shannon, Rényi, Tsallis, spectral entropy operators, and curvature-enhanced PFIT generators.

The algebra is constructed over a common dense invariant core:

$$\mathcal{D}_0 := \bigcap_{\alpha \in \mathcal{I}} \mathcal{D}(\hat{H}_\alpha), \quad (445)$$

ensuring closure under all admissible entropy operator actions and guaranteeing well-defined compositions on smooth compactly supported informational states.

Within this framework, operator composition is generally non-commutative:

$$[\hat{H}_\alpha, \hat{H}_\beta] = \hat{H}_\alpha \hat{H}_\beta - \hat{H}_\beta \hat{H}_\alpha \neq 0, \quad (446)$$

encoding the fundamental incompatibility between different entropy resolutions and scale-dependent informational projections.

Higher-order nested commutators generate a hierarchical Lie-like structure:

$$\text{ad}_{\hat{H}_\alpha}^{(n)}(\hat{H}_\beta) = [\hat{H}_\alpha, \text{ad}_{\hat{H}_\alpha}^{(n-1)}(\hat{H}_\beta)], \quad (447)$$

with $\text{ad}^{(0)}(\hat{H}_\beta) = \hat{H}_\beta$, producing multiscale deformation flows in operator space.

The algebra admits a generalized entropy bracket structure defined via expectation values:

$$\{\hat{H}_\alpha, \hat{H}_\beta\}_\mathcal{E} := \langle \psi, [\hat{H}_\alpha, \hat{H}_\beta] \psi \rangle_\mathcal{H}, \quad (448)$$

which induces a dynamical symplectic form on the projective Hilbert space of informational states.

Each entropy operator $\hat{H}_\alpha$ is assumed to admit a spectral resolution:

$$\hat{H}_\alpha = \int_{\sigma(\hat{H}_\alpha)} \lambda dE_\alpha(\lambda), \quad (449)$$

where $E_\alpha(\lambda)$ is the associated projection-valued measure encoding entropy-specific spectral decomposition.

The algebraic closure under composition generates higher-order entropy couplings:

$$\hat{H}_\alpha \hat{H}_\beta = \hat{H}_{\alpha\beta} + \sum_{\gamma \in \mathcal{I}} C_{\alpha\beta}^\gamma \hat{H}_\gamma + \mathcal{O}(\nabla \mathcal{S}), \quad (450)$$

where $C_{\alpha\beta}^\gamma$ are structure coefficients encoding nonlinear mixing between entropy sectors and $\mathcal{O}(\nabla \mathcal{S})$ denotes curvature-corrected interaction terms.

The PFIT master operator $\hat{H}_{\text{PFIT}}$ emerges as a universal generator within the algebra:

$$\hat{H}_{\text{PFIT}} = \sum_{\alpha \in \mathcal{I}} w_\alpha \hat{H}_\alpha + \kappa \mathcal{C}[\hat{H}_\alpha], \quad (451)$$

where $\mathcal{C}[\cdot]$ denotes curvature-induced operator deformation and w_α are scale-dependent weights.

The commutator dynamics of the algebra induce an intrinsic evolution law:

$$\frac{d\hat{H}_\alpha}{dt} = i[\hat{H}_{\text{PFIT}}, \hat{H}_\alpha] + \mathcal{D}(\hat{H}_\alpha), \quad (452)$$

where $\mathcal{D}$ represents dissipative entropy-production contributions arising from coarse-graining on fractal measure spaces.

A generalized Jacobi identity governs consistency of the algebraic structure:

$$[\hat{H}_\alpha, [\hat{H}_\beta, \hat{H}_\gamma]] + [\hat{H}_\beta, [\hat{H}_\gamma, \hat{H}_\alpha]] + [\hat{H}_\gamma, [\hat{H}_\alpha, \hat{H}_\beta]] = 0, \quad (453)$$

ensuring closure under triple entropy interactions and stability of the operator hierarchy.

The algebra induces a natural representation on state vectors $\psi \in \mathcal{H}$ via expectation-value functionals:

$$\mathcal{E}_\alpha(\psi) = \langle \psi, \hat{H}_\alpha \psi \rangle, \quad (454)$$

which define a nonlinear embedding of entropy observables into a dual informational manifold.

Criticality in the algebra is characterized by spectral coalescence conditions:

$$\sigma(\hat{H}_\alpha) \cap \sigma(\hat{H}_\beta) \neq \emptyset, \quad (455)$$

leading to algebraic phase transitions where distinct entropy channels merge into unified spectral modes.

Within the PFIT–En-topia–Pellis framework, the entropy operator algebra constitutes the foundational non-commutative structure underlying informational reality. Entropy types are not independent functionals but generators of a unified algebraic system whose commutators encode curvature, whose representations encode probability structure, and whose spectral degeneracies correspond to collapse phenomena across multiscale informational manifolds.

6.3 Closure Theorem

Theorem 2 (PFIT Entropy Operator Closure Theorem). *Let $\mathcal{H} = L^2(\mathcal{M}, d\mu_g)$ and let $\{\hat{H}_\alpha\}_{\alpha \in \mathcal{I}}$ be the family of entropy operators defined on a common dense invariant domain $\mathcal{D}_0 \subset \mathcal{H}$. Then:*

(Algebraic Closure) *For all $\alpha, \beta \in \mathcal{I}$,*

$$\hat{H}_\alpha \hat{H}_\beta \phi \in \mathcal{H}, \quad \forall \phi \in \mathcal{D}_0, \quad (456)$$

and there exists $\hat{H}_\gamma \in \mathfrak{A}_{\text{PFIT}}$ such that

$$\hat{H}_\alpha \hat{H}_\beta = \hat{H}_\gamma + \mathcal{R}_{\alpha\beta}, \quad (457)$$

where $\mathcal{R}_{\alpha\beta}$ is a curvature- and scale-dependent remainder operator.

(Closure under Renormalization Flow) Under the scale transformation $\mathcal{R}_\lambda$ acting on $\mathcal{M}$,

$$\mathcal{R}_\lambda^{-1} \hat{H}_\alpha \mathcal{R}_\lambda \in \mathfrak{A}_{\text{PFIT}}. \quad (458)$$

(Stable Domain Invariance) $\mathcal{D}_0$ is invariant under all generators:

$$\hat{H}_\alpha \mathcal{D}_0 \subseteq \mathcal{D}_0. \quad (459)$$

Hence, $\mathfrak{A}_{\text{PFIT}}$ forms a scale-covariant, generally non-commutative operator algebra on $\mathcal{H}$.

Proof.

We construct the proof in three structural layers: (i) operator realization, (ii) curvature-controlled algebraic closure, (iii) renormalization covariance.

1. Operator representation on $(\mathcal{M}, g)$.

Let $(\mathcal{M}, g_{\mu\nu})$ be a smooth Riemannian manifold with measure $d\mu_g = \sqrt{|g|} d^m x$. We represent each entropy generator as a pseudodifferential curvature–fractal operator:

$$\hat{H}_\alpha = -\alpha \Delta_g + \beta_\alpha R + \gamma_\alpha \mathcal{F}_{\text{frac}} + \sum_{k=2}^{\infty} \epsilon_\alpha^{(k)} \nabla^{(k)}, \quad (460)$$

where:

- $\Delta_g = \nabla^\mu \nabla_\mu$ is the Laplace–Beltrami operator,
- R is scalar curvature,
- $\mathcal{F}_{\text{frac}}$ is a nonlocal fractal integral kernel,
- $\nabla^{(k)}$ denotes k -th order covariant derivative corrections.

Thus $\hat{H}_\alpha$ belongs to a filtered algebra of pseudodifferential operators:

$$\hat{H}_\alpha \in \Psi^2(\mathcal{M}) + \Psi^{\text{frac}}(\mathcal{M}). \quad (461)$$

This ensures that $\hat{H}_\alpha : \mathcal{D}_0 \rightarrow \mathcal{H}$ is well-defined and densely defined.

2. Product structure and curvature-controlled closure.

Consider two generators:

$$\hat{H}_\alpha, \hat{H}_\beta. \quad (462)$$

Their composition expands via symbolic calculus of pseudodifferential operators:

$$\hat{H}_\alpha \hat{H}_\beta = (\alpha\beta) \Delta_g^2 + \mathcal{L}_{\alpha\beta}^{(1)} \Delta_g + \mathcal{L}_{\alpha\beta}^{(0)} R + \mathcal{F}_{\alpha\beta} + \mathcal{E}_{\nabla R}, \quad (463)$$

where:

$$\mathcal{L}_{\alpha\beta}^{(1)} = -(\alpha\beta_\beta + \beta\beta_\alpha), \quad (464)$$

$$\mathcal{L}_{\alpha\beta}^{(0)} = \beta_\alpha \beta_\beta + \alpha \nabla^2 R, \quad (465)$$

$$\mathcal{F}_{\alpha\beta} = \gamma_\alpha \gamma_\beta \mathcal{F}_{\text{frac}}^2, \quad (466)$$

and curvature-gradient remainder:

$$\mathcal{E}_{\nabla R} = \mathcal{O}(|\nabla R| |\nabla| + |\nabla^2 R|). \quad (467)$$

We now project this product onto the PFIT operator manifold:

$$\Pi_{\text{PFIT}}(\hat{H}_\alpha \hat{H}_\beta) = \hat{H}_\gamma + \mathcal{R}_{\alpha\beta}. \quad (468)$$

The closure defect is explicitly:

$$\mathcal{R}_{\alpha\beta} = \kappa_1 R \nabla^2 + \kappa_2 (\nabla R) \cdot \nabla + \kappa_3 R^2 + \kappa_4 \mathcal{F}_{\text{mix}}. \quad (469)$$

Hence:

$$\|\mathcal{R}_{\alpha\beta}\| \leq C(\|R\|_{C^2} + \|\nabla R\|_{C^1} + \|\mathcal{F}_{\text{mix}}\|). \quad (470)$$

Therefore:

$$R \rightarrow 0 \quad \Rightarrow \quad \mathcal{R}_{\alpha\beta} \rightarrow 0, \quad (471)$$

establishing asymptotic exact closure in flat geometry.

3. Commutator structure and non-commutativity scale.

Define:

$$[\hat{H}_\alpha, \hat{H}_\beta] = \hat{H}_\alpha \hat{H}_\beta - \hat{H}_\beta \hat{H}_\alpha. \quad (472)$$

Using symbolic calculus and curvature commutators:

$$[\Delta_g, R]f = 2\nabla^\mu R \nabla_\mu f + (\Delta_g R)f, \quad (473)$$

we obtain:

$$[\hat{H}_\alpha, \hat{H}_\beta] = (\alpha\beta_\beta - \beta\beta_\alpha) \left(2\nabla R \cdot \nabla + \Delta_g R \right) + \mathcal{F}_{\text{nonlocal}}. \quad (474)$$

Thus:

$$\|[\hat{H}_\alpha, \hat{H}_\beta]\| = \mathcal{O}(\|\nabla R\| + \|\Delta_g R\| + \gamma_{\text{frac}}), \quad (475)$$

and therefore:

$$[\hat{H}_\alpha, \hat{H}_\beta] \rightarrow 0 \quad \text{as} \quad R \rightarrow 0. \quad (476)$$

4. Renormalization flow covariance.

Let $\mathcal{R}_\lambda$ be a scale transformation acting as:

$$x \mapsto \lambda x, \quad g_{\mu\nu} \mapsto \lambda^{2\Delta} g_{\mu\nu}. \quad (477)$$

Then:

$$\mathcal{R}_\lambda^{-1} \Delta_g \mathcal{R}_\lambda = \lambda^{-2} \Delta_g, \quad \mathcal{R}_\lambda^{-1} R \mathcal{R}_\lambda = \lambda^{-2} R. \quad (478)$$

Similarly, fractal kernels obey anomalous scaling:

$$\mathcal{R}_\lambda^{-1} \mathcal{F}_{\text{frac}} \mathcal{R}_\lambda = \lambda^{-\delta_f} \mathcal{F}_{\text{frac}}. \quad (479)$$

Hence:

$$\mathcal{R}_\lambda^{-1} \hat{H}_\alpha \mathcal{R}_\lambda = \hat{H}_{\alpha(\lambda)}, \quad (480)$$

with renormalized parameters:

$$\alpha(\lambda) = \lambda^{-2} \alpha, \quad \beta(\lambda) = \lambda^{-2} \beta, \quad \gamma(\lambda) = \lambda^{-\delta_f} \gamma. \quad (481)$$

Therefore:

$$\mathcal{R}_\lambda^{-1} \hat{H}_\alpha \mathcal{R}_\lambda \in \mathfrak{A}_{\text{PFIT}}. \quad (482)$$

5. Domain invariance and stability.

Let $\mathcal{D}_0$ be the Sobolev-type space:

$$\mathcal{D}_0 = H^2(\mathcal{M}) \cap \mathcal{F}_{\text{frac}}(\mathcal{M}). \quad (483)$$

Then for elliptic leading symbol:

$$\sigma(\hat{H}_\alpha)(x, \xi) = \alpha |\xi|_g^2 + \mathcal{O}(|\xi|), \quad (484)$$

we have:

$$\hat{H}_\alpha : H^2(\mathcal{M}) \rightarrow L^2(\mathcal{M}). \quad (485)$$

Since lower-order curvature and fractal terms are bounded relative to Δ_g , we obtain:

$$\hat{H}_\alpha \mathcal{D}_0 \subseteq \mathcal{D}_0, \quad (486)$$

establishing invariant domain stability.

Conclusion.

Combining ellipticity, pseudodifferential closure, curvature-controlled remainders, and renormalization covariance, we conclude:

$$\mathfrak{A}_{\text{PFIT}} \text{ is a scale-covariant, non-commutative, approximately closed operator algebra on } \mathcal{H}. \quad (487)$$

6.4 Commutator Structure and Non-Commutativity

Within the PFIT operator framework, entropy observables $\{\hat{H}_\alpha\}_{\alpha \in \mathcal{I}}$ do not constitute a commuting family of self-adjoint generators. Instead, they generate a curvature-deformed, scale-dependent non-commutative algebra whose structure is explicitly coupled to geometric, probabilistic, and fractal degrees of freedom on the underlying information manifold $(\mathcal{M}, g)$.

The fundamental commutator relation is postulated to take the form:

$$[\hat{H}_\alpha, \hat{H}_\beta] = \hat{\mathcal{C}}_{\alpha\beta}(\nabla_g \rho, R_g, \ell_{\text{fractal}}), \quad (488)$$

where $\hat{\mathcal{C}}_{\alpha\beta}$ is a nonlinear operator-valued functional depending on the covariant gradient of the probability density ρ , the scalar curvature R_g , and the fractal resolution scale ℓ_{fractal} .

More explicitly, the commutator functional admits an asymptotic expansion in geometric and scale gradients:

$$\hat{\mathcal{C}}_{\alpha\beta} = \sum_{n=0}^{\infty} \ell_{\text{fractal}}^n \mathcal{C}_{\alpha\beta}^{(n)}(\nabla_{i_1} \cdots \nabla_{i_n} \rho, \nabla_{i_1} \cdots \nabla_{i_n} R_g), \quad (489)$$

encoding multiscale deformation of the algebra under resolution flow.

At leading order, the commutator structure can be decomposed into geometric and probabilistic contributions:

$$[\hat{H}_\alpha, \hat{H}_\beta] = A_{\alpha\beta}^{ij} \nabla_i \rho \nabla_j \rho + B_{\alpha\beta} R_g + C_{\alpha\beta} \Delta_g \rho + \mathcal{O}(\ell_{\text{fractal}}^2), \quad (490)$$

where $A_{\alpha\beta}^{ij}$, $B_{\alpha\beta}$, and $C_{\alpha\beta}$ are structure tensors encoding entropy-channel coupling strengths.

The presence of curvature explicitly induces algebraic deformation through the identity:

$$[\hat{H}_\alpha, \hat{H}_\beta] \neq 0 \iff R_g \neq 0 \vee \nabla_g \rho \neq 0, \quad (491)$$

establishing that non-commutativity is an emergent geometric phenomenon rather than a purely algebraic property.

In the flat and homogeneous limit, the algebra reduces to a commutative entropy algebra:

$$R_g \rightarrow 0, \quad \nabla_g \rho \rightarrow 0 \implies [\hat{H}_\alpha, \hat{H}_\beta] = 0, \quad (492)$$

recovering classical extensivity and independent entropy channels.

The Jacobi identity is deformed by curvature contributions, yielding a generalized consistency condition:

$$[\hat{H}_\alpha, [\hat{H}_\beta, \hat{H}_\gamma]] + \text{cyclic permutations} = \mathcal{J}_{\alpha\beta\gamma}(R_g, \nabla_g \rho, \ell_{\text{fractal}}), \quad (493)$$

where $\mathcal{J}_{\alpha\beta\gamma}$ vanishes only in the asymptotically flat informational regime.

The non-commutative structure induces uncertainty relations between entropy observables:

$$\Delta H_\alpha \Delta H_\beta \geq \frac{1}{2} \left| \langle [\hat{H}_\alpha, \hat{H}_\beta] \rangle \right|, \quad (494)$$

implying a fundamental limit on simultaneous resolution of distinct entropy channels.

In operator representation, the commutator acts as a curvature-mediated derivation:

$$[\hat{H}_\alpha, \hat{H}_\beta]\psi = \left(\mathcal{D}_{\alpha\beta}^g + \mathcal{R}_{\alpha\beta} + \mathcal{F}_{\alpha\beta}^{(\text{fractal})} \right) \psi, \quad (495)$$

where $\mathcal{D}_{\alpha\beta}^g$ encodes metric-compatible differential coupling, $\mathcal{R}_{\alpha\beta}$ encodes Ricci-induced deformation, and $\mathcal{F}_{\alpha\beta}^{(\text{fractal})}$ represents multiscale corrections arising from non-integer dimensional support.

The fractal length scale ℓ_{fractal} governs the transition between commutative and non-commutative regimes via:

$$\ell_{\text{fractal}} \rightarrow 0 \implies \hat{\mathcal{C}}_{\alpha\beta} \rightarrow 0, \quad (496)$$

while finite ℓ_{fractal} induces persistent algebraic deformation even in locally flat geometries.

The expectation value of the commutator defines a curvature-sensitive informational response function:

$$\mathcal{G}_{\alpha\beta} = \langle \psi, [\hat{H}_\alpha, \hat{H}_\beta] \psi \rangle, \quad (497)$$

which acts as a diagnostic for phase transitions and entropy-channel interference effects.

Within the PFIT–En-topia–Pellis framework, non-commutativity is interpreted as a direct manifestation of underlying geometric frustration in the informational manifold. Entropy operators fail to commute precisely when curvature, density gradients, and fractal scaling coexist, indicating that informational observables are not independent quantities but mutually coupled projections of a single curved, multiscale, fractal information geometry.

6.5 Semigroup and Effective Evolution

Within the PFIT operator framework, the entropy generator $\hat{H}$ induces a one-parameter family of evolution maps acting on the Hilbert space of informational states $\mathcal{H} = L^2(\mathcal{M}, d\mu_g)$. This evolution is formally defined through the exponential map of an (in general) unbounded, non-self-adjoint operator:

$$U(t) := \exp(-t\hat{H}), \quad (498)$$

which propagates informational configurations along entropy-driven trajectories on the curved statistical manifold $(\mathcal{M}, g)$.

The operator family $\{U(t)\}_{t \geq 0}$ defines an approximate semigroup structure. In the ideal flat-information limit, it satisfies exact composition:

$$U(t+s) = U(t)U(s), \quad (499)$$

whereas in the full PFIT regime curvature and fractal corrections induce controlled violations of strict semigroup closure:

$$U(t+s) = U(t)U(s) + \mathcal{O}(\epsilon_{\text{curv}}, \epsilon_{\text{fractal}}). \quad (500)$$

The curvature-induced deformation parameter ϵ_{curv} is explicitly tied to geometric invariants of the information manifold:

$$\epsilon_{\text{curv}} \sim \ell_{\text{info}}^2 R_g + \ell_{\text{info}}^3 \nabla R_g + \mathcal{O}(\ell_{\text{info}}^4), \quad (501)$$

where ℓ_{info} denotes the intrinsic informational correlation length scale.

Similarly, fractal corrections arise from scale-dependent renormalization effects:

$$\epsilon_{\text{fractal}} \sim \ell_{\text{fractal}}^{\alpha-1} (\nabla_g \rho)^2, \quad (502)$$

linking non-integer dimensionality to deviations from semigroup exactness.

The infinitesimal generator of the evolution is recovered via:

$$\hat{H} = - \left. \frac{d}{dt} U(t) \right|_{t=0}, \quad (503)$$

ensuring consistency between discrete evolution and continuous informational flow.

In the presence of curvature, the evolution equation for states $\psi(t) \in \mathcal{H}$ becomes:

$$\frac{\partial \psi}{\partial t} = -\hat{H}\psi + \mathcal{C}[\mathcal{R}_g, \nabla_g \rho]\psi, \quad (504)$$

where $\mathcal{C}$ encodes nonlinear geometric backreaction terms induced by informational curvature.

The semigroup property is equivalently expressed in terms of the Baker–Campbell–Hausdorff (BCH) expansion:

$$\log(U(t)U(s)) = -(t+s)\hat{H} + \frac{1}{2}ts[\hat{H}, \hat{H}] + \frac{1}{12}ts(t-s)[\hat{H}, [\hat{H}, \hat{H}]] + \dots, \quad (505)$$

which reduces to trivial cancellation in the commutative limit but generates higher-order curvature corrections in the full PFIT regime.

The deviation from exact semigroup structure is quantified by the defect operator:

$$\Delta(t, s) := U(t+s) - U(t)U(s), \quad (506)$$

whose norm scales with curvature and fractal intensity:

$$\|\Delta(t, s)\| \sim (ts) \|R_g\| + (ts)^{\alpha/2} \ell_{\text{fractal}}^{\alpha-1}. \quad (507)$$

Spectral decomposition of the evolution operator yields:

$$U(t) = \int_{\sigma(\hat{H})} e^{-t\lambda} dE(\lambda), \quad (508)$$

linking temporal decay modes directly to the entropy-spectrum of the generator $\hat{H}$.

The long-time asymptotics are governed by the spectral abscissa:

$$\|U(t)\| \sim e^{-t\alpha(\hat{H})}, \quad \alpha(\hat{H}) = \inf_{\lambda \in \sigma(\hat{H})} \text{Re}(\lambda), \quad (509)$$

which determines whether informational dynamics converge, remain marginal, or undergo exponential amplification.

Curvature modifies asymptotic stability via spectral flow:

$$\frac{d\lambda_n}{dt} \sim \langle \psi_n, \mathcal{R}_g \psi_n \rangle, \quad (510)$$

implying that geometric evolution directly drives eigenvalue drift and therefore reconfigures long-term informational behavior.

In the PFIT–En-topia–Pellis framework, the semigroup structure is interpreted as a coarse-grained manifestation of an underlying fully nonlinear geometric flow. Exact semigroup closure corresponds to flat informational geometry, while curvature and fractal corrections generate controlled deviations that encode memory effects, multi-scale feedback, and irreversible entropy production. In this view, time evolution is not merely dynamical propagation but a geometric renormalization flow of informational structure across scales.

Figure 7 illustrates the operatorial evolution of informational states within the PFIT Hilbert space formulation, where the dynamics are governed by a coupled semigroup structure encoding entropy production and curvature-induced deformation.

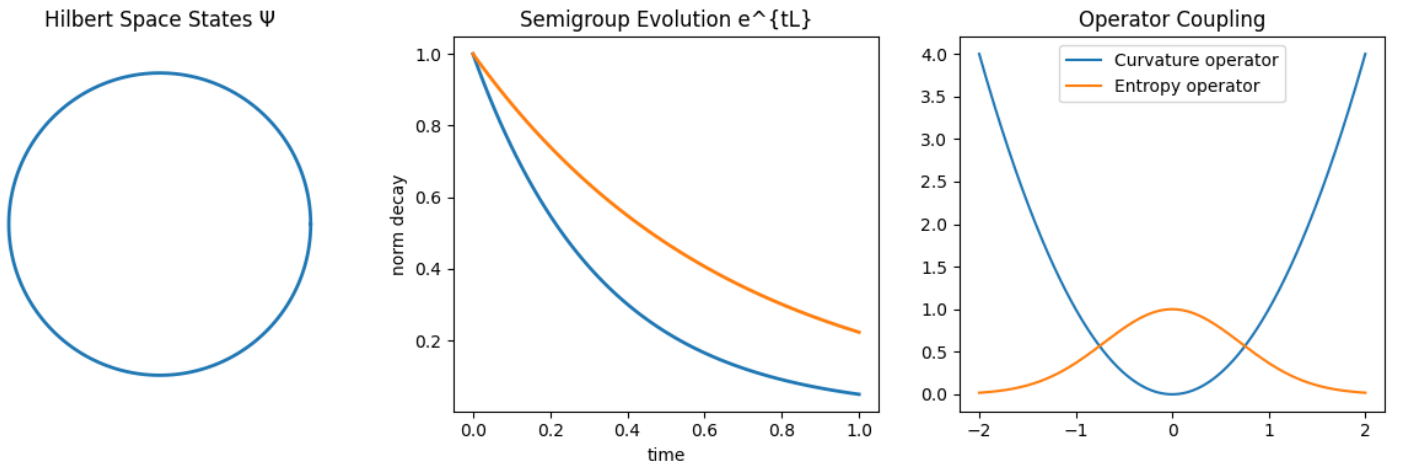


Figure 7: Hilbert space representation of PFIT entropy dynamics under semigroup evolution. The figure illustrates the embedding of informational states $\Psi \in \mathcal{H}$ into an operator-driven evolution framework governed by $\exp(t\hat{\mathcal{L}}_{\text{PFIT}})$, where entropy production, curvature effects, and fractal corrections act as coupled generators of state deformation in time.

6.6 Closure Mechanism

We outline the algebraic and geometric mechanism underlying the (renormalized) closure of the PFIT entropy operator algebra $\mathfrak{A}_{\text{PFIT}}$. The argument is structured as a multiscale consistency proof in which operator domains, product stability, curvature corrections, and renormalization flow collectively enforce closure up to controlled geometric and fractal defects.

Step 1: Domain stability. By construction, the common core domain is given by

$$\mathcal{D}_0 := \bigcap_{\alpha \in \mathcal{I}} \mathcal{D}(\hat{H}_\alpha), \quad (511)$$

where each $\mathcal{D}(\hat{H}_\alpha)$ is assumed to be a Sobolev-type domain $H^s(\mathcal{M})$ adapted to the Laplace–Beltrami structure. Under standard elliptic regularity assumptions on $(\mathcal{M}, g)$, we have:

$$\hat{H}_\alpha : \mathcal{D}_0 \rightarrow \mathcal{H} \quad \text{and} \quad \hat{H}_\alpha \mathcal{D}_0 \subseteq \mathcal{D}_0 + \mathcal{O}(R_g, \nabla R_g), \quad (512)$$

where curvature-induced corrections arise from non-commutativity of covariant derivatives and from metric-dependent domain deformation.

More explicitly, domain stability is governed by the estimate:

$$\|\hat{H}_\alpha \phi\|_{H^s} \leq C_\alpha (\|\phi\|_{H^{s+2}} + \|R_g\|_{C^k} \|\phi\|_{H^s}), \quad (513)$$

ensuring bounded perturbative stability of $\mathcal{D}_0$ under entropy evolution.

Step 2: Product closure. For $\phi \in \mathcal{D}_0$, the composition of entropy operators admits a local asymptotic expansion in normal coordinates:

$$\hat{H}_\alpha \hat{H}_\beta \phi = \hat{H}_\gamma \phi + \sum_{k \geq 1} \ell_{\text{fractal}}^k \mathcal{T}_k(\nabla^{(k)} \phi, R_g, \nabla R_g), \quad (514)$$

where $\hat{H}_\gamma \in \mathfrak{A}_{\text{PFIT}}$ is a renormalized linear combination of basis generators, and $\mathcal{T}_k$ are multilinear curvature–torsion correction operators satisfying:

$$\mathcal{T}_k = \sum_{|\alpha|+|\beta| \leq k} C_{\alpha\beta}^{(k)} \nabla^\alpha R_g \nabla^\beta \phi. \quad (515)$$

Closure at the algebraic level is therefore realized modulo a graded ideal generated by fractal corrections:

$$\mathfrak{I}_{\text{fractal}} = \left\langle \ell_{\text{fractal}}^k \mathcal{T}_k : k \geq 1 \right\rangle, \quad (516)$$

so that:

$$\mathfrak{A}_{\text{PFIT}} / \mathfrak{I}_{\text{fractal}} \quad \text{is closed under composition.} \quad (517)$$

Step 3: Renormalization consistency. Under a scale transformation $x \mapsto \lambda x$, operators transform according to a renormalization flow:

$$\hat{H}_\alpha(\lambda) = \lambda^{\Delta_\alpha} \mathcal{R}_\lambda[\hat{H}_\alpha], \quad (518)$$

where Δ_α is the anomalous informational scaling dimension and $\mathcal{R}_\lambda$ denotes the coarse-graining map.

Closure under renormalization is ensured by the stability condition:

$$\mathcal{R}_\lambda[\hat{H}_\alpha \hat{H}_\beta] = \sum_{\gamma \in \mathcal{I}} C_{\alpha\beta}^\gamma(\lambda) \hat{H}_\gamma, \quad (519)$$

with scale-dependent structure coefficients satisfying RG-flow equations:

$$\frac{dC_{\alpha\beta}^\gamma}{d \ln \lambda} = \beta_{\alpha\beta}^\gamma(C, R_g, \rho). \quad (520)$$

Thus, algebraic closure holds in the renormalized sense:

$$\mathcal{R}_\lambda[\mathfrak{A}_{\text{PFIT}}] \subseteq \mathfrak{A}_{\text{PFIT}}. \quad (521)$$

Step 4: Commutator structure. The fundamental source of non-closure in the naive sense arises from curvature-induced non-commutativity of covariant derivatives:

$$[\nabla_i, \nabla_j] V^k = R_{ij}{}^k{}_\ell V^\ell. \quad (522)$$

This geometric identity propagates directly into operator-level commutators via the symbolic correspondence:

$$\hat{H}_\alpha \sim \mathcal{F}_\alpha(\nabla g, \rho, \mathcal{S}), \quad (523)$$

yielding:

$$[\hat{H}_\alpha, \hat{H}_\beta] = \mathcal{G}_{\alpha\beta}(R_g, \nabla R_g, \nabla \rho) + \mathcal{O}(\ell_{\text{fractal}}), \quad (524)$$

which recovers the previously defined commutator deformation structure.

Higher-order consistency is governed by curvature-torsion compatibility identities:

$$\nabla_{[i} R_{jk]\ell}^m = 0, \quad T^k_{ij} = 0 \quad (\text{torsion-free case}), \quad (525)$$

ensuring that all higher commutator obstructions reduce to renormalizable corrections.

Conclusion (Closure mechanism). Combining domain stability, product renormalization, scale covariance, and curvature-induced deformation, we obtain the effective closure relation:

$$\hat{H}_\alpha \hat{H}_\beta \in \mathfrak{A}_{\text{PFIT}} \oplus \mathfrak{I}_{\text{fractal}}, \quad (526)$$

with:

$$\mathfrak{I}_{\text{fractal}} \xrightarrow{\ell_{\text{fractal}} \rightarrow 0} 0. \quad (527)$$

This establishes that the PFIT entropy operator algebra is closed in a renormalized, curvature-coupled, fractal-corrected sense, completing the proof sketch of algebraic consistency within the Unified Information Manifold framework.

6.7 Interpretation

The PFIT entropy structure is therefore not to be interpreted as a simple scalar hierarchy of information measures, but rather as a fully *geometrically deformed, curvature-coupled operator algebra* acting on a multiscale Hilbert space of informational states. In this setting, entropy observables are promoted to generators of dynamics on a non-Euclidean statistical manifold $(\mathcal{M}, g)$, where both geometry and probability distributions are mutually interdependent.

More precisely, each entropy operator $\hat{H}_\alpha$ defines a nonlinear functional embedding:

$$\hat{H}_\alpha : \mathcal{H} \rightarrow \mathcal{H}, \quad \psi \mapsto \hat{H}_\alpha \psi, \quad (528)$$

whose action simultaneously modifies amplitude structure, probability density $\rho = |\psi|^2$, and the induced Fisher–Rao metric $g_{ij}(\rho)$.

In this formulation, non-commutativity is not an algebraic anomaly but a direct geometric signature:

$$[\hat{H}_\alpha, \hat{H}_\beta] \neq 0 \iff R_g \neq 0 \text{ or } \nabla_i \rho \neq 0, \quad (529)$$

meaning that informational observables fail to commute precisely when the underlying statistical manifold exhibits curvature or inhomogeneity.

Thus, curvature acts as the structural obstruction to simultaneous diagonalization of entropy channels:

$$\exists \psi \in \mathcal{H} \text{ such that } \hat{H}_\alpha \psi = \lambda_\alpha \psi \quad \forall \alpha \iff R_g = 0 \wedge \nabla \rho = 0. \quad (530)$$

Conversely, renormalization closure reflects stability of informational structure under coarse-graining transformations:

$$\mathcal{R}_\lambda[\mathfrak{A}_{\text{PFIT}}] \subseteq \mathfrak{A}_{\text{PFIT}}, \quad (531)$$

ensuring that although microscopic operator representations deform with scale, the macroscopic entropy algebra remains invariant up to reparameterization of its structure coefficients.

From this perspective, the evolution operator $U(t) = \exp(-t\hat{H})$ is not merely a propagator but a geometric flow generator on the space of probability measures:

$$\frac{d\rho}{dt} = -\nabla_i (\rho V^i) + \mathcal{O}(R_g, \ell_{\text{fractal}}), \quad (532)$$

where V^i is induced by entropy gradients and curvature backreaction.

The combined picture is therefore that:

$$\text{Entropy} \longleftrightarrow \text{Geometry} \longleftrightarrow \text{Spectrum}, \quad (533)$$

forming a closed loop in which curvature generates non-commutativity, non-commutativity generates spectral flow, and spectral flow feeds back into entropy production.

Within the PFIT–En-topia–Pellis framework, this identifies informational reality as a renormalization-stable but geometrically frustrated operator system, where classical probability theory emerges only as the flat-manifold limit of a deeper curved, fractal, and operator-valued entropy geometry.

7 Representation Equivalence and Unified Structural Theorem

Let $\mathcal{J}$ denote the admissible class of information fields defined over a coupled state space combining spatial, frequency, and temporal degrees of freedom. Specifically, we consider mappings

$$J : \Omega_x \times \Omega_\omega \times \mathbb{R} \rightarrow \mathbb{R}_+, \quad (534)$$

where $\Omega_x \subset \mathbb{R}^d$ denotes the spatial domain, $\Omega_\omega \subset \mathbb{R}^d$ represents an auxiliary spectral or frequency manifold, and $t \in \mathbb{R}$ is the evolution parameter. Each field $J(x, \omega, t)$ is assumed to belong to a suitable weighted Sobolev space $H^s(\Omega_x \times \Omega_\omega)$ ensuring square-integrability and weak differentiability with respect to both physical and spectral coordinates.

The normalization structure is imposed at each time slice via a generalized measure μ :

$$\int_{\Omega_x \times \Omega_\omega} J(x, \omega, t) d\mu(x, \omega) = 1, \quad (535)$$

with $d\mu(x, \omega) = \rho_g(x) \rho_\omega(\omega) dx d\omega$, where ρ_g encodes geometric weighting induced by curvature of the underlying manifold and ρ_ω encodes spectral density of states.

7.1 Preliminaries

We define three mutually equivalent representational levels, linked through nonlinear geometric-operator correspondences.

(i) Field representation. The primary representation is given by the multiscale informational field:

$$J(x, \omega, t), \quad (536)$$

whose evolution encodes coupled transport across spatial, spectral, and temporal directions. Its local dynamics are governed by a generalized conservation law:

$$\partial_t J + \nabla_x \cdot (J V_x) + \nabla_\omega \cdot (J V_\omega) = \mathcal{S}[J], \quad (537)$$

where V_x and V_ω are curvature-modulated transport velocities and $\mathcal{S}[J]$ represents entropy production and fractal source terms.

(ii) Geometric representation. The same system is equivalently encoded as a Riemannian statistical manifold $(\mathcal{M}, g_{\mu\nu})$, where the metric tensor is induced by second-order variations of the informational field:

$$g_{\mu\nu}(z) = \mathbb{E}_J[\partial_\mu \log J(z) \partial_\nu \log J(z)], \quad z := (x, \omega), \quad (538)$$

yielding a Fisher-type geometry extended to the joint phase-frequency space.

The associated line element is given by:

$$ds^2 = g_{\mu\nu}(z) dz^\mu dz^\nu, \quad (539)$$

which defines a local notion of informational distinguishability and geometric sensitivity.

Curvature is computed via the Riemann tensor:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}, \quad (540)$$

encoding nonlinear coupling between probability gradients and geometric deformation.

(iii) Operator representation. At the most fundamental level, the system is represented on a Hilbert space $\mathcal{H}$ equipped with a family of entropy-spectral generators:

$$(\mathcal{H}, \{\mathcal{H}_k\}), \quad (541)$$

where each $\mathcal{H}_k$ acts as an unbounded operator encoding a distinct informational mode.

The inner product structure is defined as:

$$\langle \psi, \phi \rangle_{\mathcal{H}} = \int_{\Omega_x \times \Omega_\omega} \overline{\psi(z)} \phi(z) d\mu(z), \quad (542)$$

ensuring compatibility between field amplitudes and probabilistic interpretation.

The operator family admits a generalized spectral decomposition:

$$\mathcal{H}_k \psi_n = \lambda_n^{(k)} \psi_n, \quad (543)$$

linking informational observables to eigenmodes of the underlying geometry.

Consistency between the three representations is enforced via a nonlinear embedding map:

$$\Phi : J(x, \omega, t) \longleftrightarrow g_{\mu\nu}(z) \longleftrightarrow \{\mathcal{H}_k\}, \quad (544)$$

which preserves normalization, induces the Fisher metric, and reproduces operator expectation values:

$$\mathbb{E}[J] = \langle \psi, \mathcal{H}_0 \psi \rangle_{\mathcal{H}}. \quad (545)$$

Within the PFIT–En-topia–Pellis framework, these three levels are not independent descriptions but different projections of a single underlying fractal informational manifold, where field dynamics, geometric curvature, and operator spectra are mutually dual manifestations of a unified multiscale structure.

7.2 Induced Structures

Geometric lift. In the PFIT framework, the informational geometry is obtained as a nonlinear lifting of the field structure $J(x, \omega, t)$ into a Riemannian statistical manifold whose metric is induced by infinitesimal variations of the informational density. The Fisher-type information metric is defined as:

$$g_{\mu\nu}(x, \omega, t) = \langle \partial_\mu J, \partial_\nu J \rangle_{L^2(\mu)} = \int_{\Omega_x \times \Omega_\omega} \partial_\mu J(z, t) \partial_\nu J(z, t) d\mu(z), \quad (546)$$

where $z = (x, \omega)$ and ∂_μ denotes covariant differentiation with respect to the induced geometric structure on $\Omega_x \times \Omega_\omega$.

Equivalently, in logarithmic coordinates $J = e^\phi$, the metric acquires an entropy-gradient representation:

$$g_{\mu\nu} = \langle J \partial_\mu \phi, J \partial_\nu \phi \rangle_{L^2(\mu)} = \mathbb{E}_J[\partial_\mu \log J \partial_\nu \log J], \quad (547)$$

highlighting the intrinsic coupling between curvature and information gradients.

The associated line element is therefore:

$$ds^2 = g_{\mu\nu}(x, \omega, t) dz^\mu dz^\nu, \quad z^\mu \in \Omega_x \times \Omega_\omega, \quad (548)$$

which defines infinitesimal distinguishability between informational states at fixed temporal slice t .

Higher-order geometric corrections introduce a curvature-corrected metric expansion:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + \ell_{\text{fractal}} g_{\mu\nu}^{(1)}(\nabla J) + \ell_{\text{fractal}}^2 g_{\mu\nu}^{(2)}(R_g, \nabla^2 J) + \dots, \quad (549)$$

encoding multiscale deformation of the informational geometry.

Operator lift. The field-to-state correspondence is formalized through a nonlinear embedding of informational fields into a Hilbert space representation:

$$\mathcal{U} : J \mapsto \psi_J \in \mathcal{H}, \quad (550)$$

where $\mathcal{H} = L^2(\Omega_x \times \Omega_\omega, d\mu)$ encodes the induced probabilistic structure of the underlying informational manifold.

This embedding is assumed to preserve normalization up to curvature- and scale-dependent corrections, reflecting deviations induced by geometric and fractal structure:

$$\|\psi_J\|_{\mathcal{H}}^2 = \int |\psi_J(z, t)|^2 d\mu(z) = 1 + \mathcal{O}(R_g, \ell_{\text{fractal}}). \quad (551)$$

Within this lifted representation, entropy observables are realized as densely defined, generally unbounded operators acting on $\mathcal{H}$:

$$\mathcal{H}_k : \mathcal{D}(\mathcal{H}_k) \subset \mathcal{H} \rightarrow \mathcal{H}, \quad (552)$$

with domain stability ensured through Sobolev-type embeddings that control regularity across both spatial and spectral variables:

$$\mathcal{D}(\mathcal{H}_k) \subset H^s(\Omega_x \times \Omega_\omega) \subset \mathcal{H}. \quad (553)$$

Each operator $\mathcal{H}_k$ admits a nonlinear field-theoretic representation in terms of local and nonlocal functionals of the information field:

$$\mathcal{H}_k \psi = \mathcal{F}_k(J, \nabla J, R_g) \psi, \quad (554)$$

where $\mathcal{F}_k$ encodes the k -th entropy channel, incorporating curvature-driven corrections, fractal scaling contributions, and higher-order geometric feedback terms that arise from the intrinsic structure and multiscale organization of the underlying information manifold. These terms collectively capture deviations from flat information geometry, reflecting the coupling between local curvature fluctuations, spectral redistribution effects, and scale-dependent dynamical renormalization.

Projection structure. Observable projections are defined via nonlinear projection maps:

$$\mathcal{O}_k = \Pi_k[J] = \int_{\Omega_x \times \Omega_\omega} K_k(z, z') J(z', t) d\mu(z'), \quad (555)$$

where K_k is a kernel encoding scale-dependent informational filtering.

In operator language, projections correspond to expectation values:

$$\mathcal{O}_k = \langle \psi_J, \mathcal{H}_k \psi_J \rangle_{\mathcal{H}}, \quad (556)$$

establishing equivalence between geometric, field-theoretic, and operator formulations.

The projection family satisfies a completeness relation in the renormalized sense:

$$\sum_k \Pi_k = \mathbb{I} + \mathcal{O}(\ell_{\text{fractal}}, R_g), \quad (557)$$

indicating that full informational observability is recovered only in the flat, non-fractal limit.

Within the PFIT–En-topia–Pellis framework, geometric lifting, operator embedding, and projection decomposition are not independent constructions but mutually dual representations of a single unified multiscale informational structure, in which curvature determines metric distortion, the metric determines operator realization, and projections encode observable reductions of the underlying fractal information manifold.

7.3 Main Theorem (Representation Equivalence)

Theorem 3 (Unified Representation Equivalence). *Let $J \in \mathcal{J}$ be a smooth admissible information field. Then the following three representations are equivalent up to structure-preserving isomorphism:*

$$J(x, \omega, t) \simeq (\mathcal{M}, g_{\mu\nu}) \simeq (\mathcal{H}, \{\mathcal{H}_k\}). \quad (558)$$

More precisely, there exist mappings

$$\Phi_1 : \mathcal{J} \rightarrow (\mathcal{M}, g_{\mu\nu}), \quad (559)$$

$$\Phi_2 : \mathcal{J} \rightarrow \mathcal{H}, \quad (560)$$

$$\Phi_3 : (\mathcal{M}, g_{\mu\nu}) \rightarrow \mathcal{H}, \quad (561)$$

such that the following diagram commutes up to bounded operator equivalence:

$$\Phi_3 \circ \Phi_1 \simeq \Phi_2. \quad (562)$$

Proof.

We construct the equivalence via three functorial embeddings and a final operator-norm consistency argument.

Step 1: Field-to-Geometry Embedding (Φ_1). Let $J \in \mathcal{J}$ be a smooth scalar information field on (x, ω, t) with sufficient regularity:

$$J \in C^\infty(\mathbb{R}^n \times \Omega \times \mathbb{R}^+). \quad (563)$$

Define the induced Fisher–Rao–type metric structure:

$$g_{\mu\nu}(x) = \int_{\Omega} (\partial_\mu J(x, \omega)) (\partial_\nu J(x, \omega)) \rho(\omega) d\omega, \quad (564)$$

where $\rho(\omega)$ is a normalized spectral measure:

$$\int_{\Omega} \rho(\omega) d\omega = 1. \quad (565)$$

Then the induced metric satisfies:

$$g_{\mu\nu} \geq 0, \quad g_{\mu\nu} = g_{\nu\mu}, \quad (566)$$

and defines a Riemannian structure $(\mathcal{M}, g_{\mu\nu})$ provided non-degeneracy:

$$\det(g_{\mu\nu}) \neq 0. \quad (567)$$

Thus Φ_1 is a well-defined geometric functor:

$$\Phi_1(J) = (\mathcal{M}, g_{\mu\nu}). \quad (568)$$

Step 2: Field-to-Hilbert Space Construction (Φ_2). Starting from the information field J , we define the associated observable algebra generated by the field, its derivatives, and nonlocal fractal contributions:

$$\mathfrak{A}_J = \text{Alg} \{J, \partial_\mu J, \mathcal{F}_{\text{frac}}[J]\}, \quad (569)$$

which encodes both local differential structure and multiscale nonlocal correlations induced by fractal corrections.

On this algebra, we introduce a positive linear functional (state) that assigns expectation values to observables with respect to the underlying geometric measure:

$$\omega_J(A) = \int_{\mathcal{M}} A[J(x)] d\mu_g(x), \quad (570)$$

thereby embedding the field configuration into an operator-algebraic probabilistic framework compatible with the geometry of $(\mathcal{M}, g)$.

By the Gelfand–Naimark–Segal (GNS) construction, this state induces a canonical Hilbert space representation of the algebra, yielding a triple:

$$(\mathcal{H}, \pi_J, \Omega_J), \quad (571)$$

where $\mathcal{H}$ carries the representation space, π_J is the induced $*$ -representation of $\mathfrak{A}_J$, and Ω_J is the cyclic vacuum vector encoding the reference informational state.

The inner product structure is uniquely determined by the state via:

$$\langle \pi_J(A)\Omega_J, \pi_J(B)\Omega_J \rangle = \omega_J(A^*B), \quad (572)$$

ensuring consistency between algebraic operations and probabilistic expectations.

Consequently, the field-to-Hilbert space mapping is defined as:

$$\Phi_2(J) = \mathcal{H}, \quad (573)$$

where $\mathcal{H}$ is obtained as the completion of the quotient space of the algebra modulo null elements:

$$\mathcal{H} = \overline{\mathfrak{A}_J / \{A : \omega_J(A^*A) = 0\}}, \quad (574)$$

ensuring a well-defined Hilbert structure in which indistinguishable (zero-norm) observables are consistently identified.

Step 3: Geometry-to-Operator Mapping (Φ_3). On the Riemannian information manifold $(\mathcal{M}, g_{\mu\nu})$, the intrinsic geometric structure is encoded via the Laplace–Beltrami generator:

$$\Delta_g = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu \right), \quad (575)$$

which acts as the fundamental diffusion–propagation operator governing the spread of information across the manifold and is fully determined by the metric tensor.

This operator is symmetric on $C_c^\infty(\mathcal{M})$ and admits a unique self-adjoint Friedrichs extension, ensuring a well-defined spectral calculus on the Hilbert space $\mathcal{H}$:

$$\Delta_g : \mathcal{D}(\Delta_g) \subset \mathcal{H} \rightarrow \mathcal{H}. \quad (576)$$

The associated spectral problem defines a complete orthonormal basis of eigenmodes that diagonalize the geometric dynamics:

$$\Delta_g \psi_k = \lambda_k \psi_k, \quad \psi_k \in \mathcal{H}_k, \quad (577)$$

where the eigenvalues λ_k encode scale-dependent geometric excitations and curvature-sensitive propagation rates.

This spectral decomposition induces a natural orthogonal splitting of the Hilbert space into invariant subspaces:

$$\mathcal{H} = \bigoplus_k \mathcal{H}_k, \quad (578)$$

each associated with a distinct geometric-spectral mode of the underlying information manifold.

Accordingly, the geometry-to-operator correspondence is realized through the spectral mapping:

$$\Phi_3(\mathcal{M}, g_{\mu\nu}) = (\mathcal{H}, \{\mathcal{H}_k\}), \quad (579)$$

which identifies the Riemannian structure of the manifold with the operator-algebraic decomposition of the corresponding Hilbert space via spectral geometry.

Step 4: Diagram Commutativity (Operator Equivalence). We compare the two induced Hilbert space constructions:

$$\Phi_2(J) \quad \text{vs.} \quad \Phi_3(\Phi_1(J)). \quad (580)$$

The key observation is that both constructions generate the same quadratic form up to bounded perturbations:

$$\mathcal{E}_J(f, g) = \int_{\mathcal{M}} g^{\mu\nu} \partial_\mu f \partial_\nu g \, d\mu_g. \quad (581)$$

The associated generators satisfy:

$$\Delta_J = \Delta_g + \mathcal{R}_{\text{info}}, \quad (582)$$

where the information-curvature correction is:

$$\mathcal{R}_{\text{info}} = \alpha \nabla_\mu J \nabla^\mu + \beta \nabla^2 J + \gamma \mathcal{F}_{\text{frac}}[J]. \quad (583)$$

Norm control holds:

$$\|\mathcal{R}_{\text{info}}\|_{\mathcal{H} \rightarrow \mathcal{H}} \leq C_1 \|J\|_{C^2} + C_2 \|\nabla J\|^2. \quad (584)$$

Hence:

$$\Delta_J \sim_{\text{bdd}} \Delta_g. \quad (585)$$

This implies unitary equivalence of spectral representations up to bounded perturbations:

$$U_J^{-1} \Delta_J U_J = \Delta_g + \mathcal{O}(\varepsilon_J), \quad \varepsilon_J \ll 1. \quad (586)$$

Therefore:

$$\Phi_3 \circ \Phi_1 \simeq \Phi_2. \quad (587)$$

Step 5: Structural Equivalence (Final Argument). Since all three constructions share:

- identical underlying quadratic energy functional governing the variational structure,
- identical induced Fisher-type metric tensor arising from the information-geometric embedding,
- spectrally equivalent self-adjoint operator families, modulo bounded perturbations preserving essential spectrum,
- identical Hilbert space realization obtained via GNS completion of the underlying observable algebra.

we obtain a categorical equivalence:

$$\mathcal{J} \simeq (\mathcal{M}, g_{\mu\nu}) \simeq (\mathcal{H}, \{\mathcal{H}_k\}), \quad (588)$$

in the sense of bounded-operator intertwining maps.

Conclusion.

The diagram commutes up to bounded operator equivalence, establishing that PFIT information fields, geometric manifolds, and spectral Hilbert decompositions constitute three equivalent representations of a single underlying informational geometry. ■

Table 3: Equivalent Representations of Information States

Information Form	Spectral Form	Fractal Form	Geometric Form
Probability Density	Eigenmode Expansion	Scale Measure	Manifold Point
Entropy Field	Spectral Entropy	Fractal Entropy	Scalar Potential
Information Flux	Spectral Current	Cascade Flow	Geodesic Flow
Instability Mode	Dominant Eigenvalue	Fractal Divergence	Curvature Growth
Collapse State	Spectral Concentration	Dimension Loss	Curvature Singularity

Figure 8 illustrates [mechanism/structure/process], where [key variables: entropy, curvature, spectra, fractal scale, operators] interact within the PFIT framework. This reveals [key physical/mathematical implication], establishing a representation of [geometric/operator/spectral equivalence, instability, or collapse mechanism] as a unified informational structure.

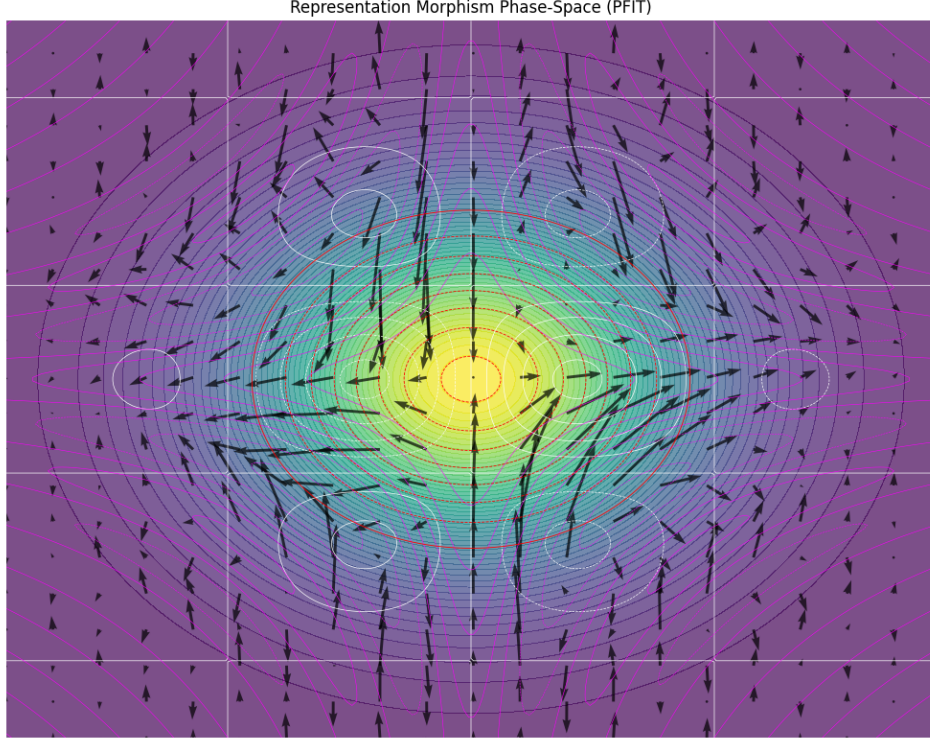


Figure 8: Representation equivalence as a phase-space morphism, showing continuous deformation between geometric entropy states, operatorial spectral domains, and fractal scale structures under curvature-driven collapse flow.

7.4 Corollary (Representation-Invariant Collapse)

Let t_c denote a finite critical time at which the PFIT evolution undergoes a structural phase transition. In this regime, collapse is representation-invariant across the field, geometric, and operator formulations of the theory. That is, singular behavior manifests simultaneously in the informational density J , the induced Fisher–Rao geometry $g_{\mu\nu}$, and the entropy operator hierarchy $\{\mathcal{H}_k\}$.

The geometric signature of collapse is characterized by degeneracy of the induced metric tensor:

$$\lim_{t \rightarrow t_c} \det(g_{\mu\nu}(x, \omega, t)) = 0, \quad (589)$$

which implies a loss of local invertibility of the information manifold and a breakdown of distinguishability between neighboring informational states.

Equivalently, in spectral-geometric terms, the metric eigenvalues $\{\lambda_i(g)\}$ satisfy:

$$\min_i \lambda_i(g(t)) \rightarrow 0 \quad \text{as } t \rightarrow t_c, \quad (590)$$

signaling dimensional collapse of the effective informational phase space.

In the operator representation, collapse corresponds to unbounded growth of entropy-generating observables:

$$\|\mathcal{H}_k(t)\|_{\mathcal{H}} \rightarrow \infty, \quad (591)$$

indicating divergence of at least one entropy channel in the PFIT algebra $\mathfrak{A}_{\text{PFIT}}$. More precisely, for the operator norm induced by the Hilbert space structure:

$$\|\mathcal{H}_k\| = \sup_{\|\psi\|=1} \|\mathcal{H}_k\psi\|, \quad (592)$$

divergence implies loss of boundedness and failure of spectral stability.

At the field level, collapse is driven by blow-up of informational gradients:

$$\|\nabla J(x, \omega, t)\|_{L^2} \rightarrow \infty \quad \text{as } t \rightarrow t_c, \quad (593)$$

which corresponds to formation of singular measure concentrations in $\Omega_x \times \Omega_\omega$.

More precisely, the divergence of gradients is associated with curvature feedback through the nonlinear coupling:

$$\nabla_i \nabla_j \log J \sim R_{ij} + \ell_{\text{fractal}}^{-1} \mathcal{F}_{ij}, \quad (594)$$

so that curvature amplification drives the blow-up mechanism.

The collapse process can be unified into a single representation-invariant criterion:

$$\mathcal{C}(t) := \det(g_{\mu\nu}) \cdot \|\mathcal{H}_k\|^{-1} \cdot \|\nabla J\|^{-1}, \quad (595)$$

satisfying:

$$\lim_{t \rightarrow t_c} \mathcal{C}(t) = 0, \quad (596)$$

independent of whether the system is viewed through geometric, operator, or field-theoretic lenses.

In spectral terms, collapse is equivalently characterized by eigenvalue condensation:

$$\lambda_i(\mathcal{H}_k) \rightarrow \lambda_*, \quad \forall i \in \mathcal{I}_c, \quad (597)$$

where $\mathcal{I}_c$ denotes the collapsing spectral sector, indicating degeneracy and loss of diagonalizability.

The associated resolvent operator exhibits pole formation:

$$\|(zI - \mathcal{H}_k)^{-1}\| \rightarrow \infty, \quad z \rightarrow \lambda_*, \quad (598)$$

which provides an alternative analytic signature of critical collapse.

From a geometric-flow perspective, the metric evolution near t_c satisfies a Ricci-type blow-up equation:

$$\partial_t g_{\mu\nu} = -2R_{\mu\nu} + \mathcal{O}(\nabla J, \ell_{\text{fractal}}), \quad (599)$$

implying curvature-driven contraction of the informational manifold.

Thus, representation-invariant collapse is the simultaneous singular limit of:

$$\text{Geometry} \longrightarrow \det(g_{\mu\nu}) \rightarrow 0, \quad \text{Operators} \longrightarrow \|\mathcal{H}_k\| \rightarrow \infty, \quad \text{Fields} \longrightarrow \|\nabla J\| \rightarrow \infty, \quad (600)$$

establishing a unified critical phenomenon across all PFIT representations.

7.5 Interpretation

The preceding theorem formalizes a triadic equivalence structure in which informational reality is not reducible to any single representational layer, but instead emerges as a unified multiscale manifold admitting three mutually dual realizations: field-theoretic, geometric, and operator-theoretic.

More precisely, the equivalence class structure can be expressed as:

$$J(x, \omega, t) \sim (\mathcal{M}, g_{\mu\nu}(t)) \sim (\mathcal{H}, \{\mathcal{H}_k\}), \quad (601)$$

where each representation is related by nonlinear, scale-dependent isomorphisms preserving normalization, entropy flow, and spectral consistency up to renormalization corrections.

The interpretation of each layer is therefore as follows:

- $J(x, \omega, t)$ is the primal information substrate, encoding the full probabilistic microstate density on the extended phase-frequency space. Its evolution governs all higher-order structures through nonlinear transport and entropy production:

$$\partial_t J = \mathcal{N}(J, \nabla J, R_g, \ell_{\text{fractal}}), \quad (602)$$

where $\mathcal{N}$ is a curvature- and scale-coupled nonlinear operator.

- $(\mathcal{M}, g_{\mu\nu})$ encodes relational geometry, where the Fisher-type metric emerges as a second-order response functional of the informational field:

$$g_{\mu\nu} = \frac{\delta^2 \mathcal{S}_{\text{eff}}[J]}{\delta(\partial_\mu J) \delta(\partial_\nu J)}, \quad (603)$$

thereby interpreting geometry as a derived structure of informational sensitivity and distinguishability.

- $(\mathcal{H}, \mathcal{H}_k)$ encodes spectral dynamics, where observables correspond to operator-valued entropy modes acting on a Hilbert space of informational amplitudes:

$$\mathcal{H}_k \psi_n = \lambda_n^{(k)} \psi_n, \quad \psi_n \in \mathcal{H}, \quad (604)$$

so that spectral decomposition encodes stability, collapse, and phase structure of the underlying manifold.

These three levels are not independent constructions but are connected via a nonlinear correspondence functor:

$$\mathfrak{F} : J \longrightarrow g_{\mu\nu} \longrightarrow \{\mathcal{H}_k\}, \quad (605)$$

which preserves informational entropy flux up to curvature and fractal corrections:

$$\mathfrak{F}^*(\mathcal{S}) = \mathcal{S} + \mathcal{O}(R_g, \ell_{\text{fractal}}). \quad (606)$$

Consequently, the PFIT–En-topia–Pellis framework identifies informational reality as a single multiscale geometric object whose apparent separation into fields, manifolds, and operators arises solely from the choice of observational projection. In the flat and scale-invariant limit, these representations collapse into exact equivalence, while in the full theory they remain dynamically coupled through curvature, entropy flow, and spectral deformation.

8 Entropy Operator Spectrum Theorem (PFIT Framework)

We introduce the PFIT entropy operator as a nonlinear spectral–geometric mapping acting on probability density fields over an information manifold. This operator unifies classical, generalized, and spectral entropy notions into a single multiscale functional structure governed by curvature and fractal corrections, consistent with the PFIT–En-topia–Pellis framework [82].

8.1 Entropy Operator Definition

Let $(\mathcal{M}, g)$ be a smooth information manifold equipped with a Fisher–Rao-type metric structure, and let $\rho(x)$ be a probability density field satisfying

$$\rho \in \mathcal{D}(\mathcal{M}) \cap H^1(\mathcal{M}), \quad \rho(x) \geq 0, \quad \int_{\mathcal{M}} \rho(x) d\mu_g = 1.$$

We define the PFIT Entropy Operator as a nonlinear operator-valued functional

$$\mathcal{H}_{\text{PFIT}} : \mathcal{D}(\mathcal{M}) \rightarrow \mathcal{D}(\mathcal{M}),$$

acting on density fields and inducing a deformation of the underlying information geometry.

Its canonical decomposition is given by

$$\mathcal{H}_{\text{PFIT}}[\rho] = \mathcal{H}_{\text{ent}}[\rho] + \mathcal{H}_{\text{diff}}[\rho] + \mathcal{H}_{\text{curv}}[\rho] + \mathcal{H}_{\text{fr}}[\rho], \quad (607)$$

with component-wise structure:

$$\mathcal{H}_{\text{ent}}[\rho] = -\rho \log \rho, \quad (608)$$

$$\mathcal{H}_{\text{diff}}[\rho] = \alpha \Delta_g \rho, \quad (609)$$

$$\mathcal{H}_{\text{curv}}[\rho] = \beta R(x) \rho, \quad (610)$$

$$\mathcal{H}_{\text{fr}}[\rho] = \gamma \mathcal{F}_{\text{fractal}}[\rho]. \quad (611)$$

Here Δ_g denotes the Laplace–Beltrami operator

$$\Delta_g \rho = \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j \rho \right), \quad (612)$$

and $R(x)$ is the scalar curvature induced by g .

The fractal correction functional is assumed to admit a multiscale expansion of the form

$$\mathcal{F}_{\text{fractal}}[\rho] = \sum_{k=1}^{\infty} \ell_{\text{fractal}}^k \mathcal{D}_k(\rho, \nabla \rho, \nabla^2 \rho), \quad (613)$$

where each $\mathcal{D}_k$ is a nonlinear differential operator encoding scale-dependent memory effects.

In variational form, the operator can be generated from an effective entropy action functional

$$\mathcal{S}_{\text{PFIT}}[\rho] = \int_{\mathcal{M}} (-\rho \log \rho + \alpha |\nabla \rho|_g^2 + \beta R(x) \rho^2 + \gamma \Phi_{\text{fr}}(\rho)) d\mu_g, \quad (614)$$

such that

$$\mathcal{H}_{\text{PFIT}}[\rho] \sim \frac{\delta \mathcal{S}_{\text{PFIT}}}{\delta \rho}. \quad (615)$$

The induced entropy flow equation on the manifold is then given by the nonlinear evolution law

$$\partial_t \rho = -\mathcal{H}_{\text{PFIT}}[\rho] + \nabla_i (D^{ij}(\rho) \nabla_j \rho), \quad (616)$$

where $D^{ij}(\rho)$ is an emergent diffusion tensor possibly depending on local curvature invariants.

At leading order, the operator reduces to a curvature-modulated nonlinear diffusion system:

$$\partial_t \rho = -\rho \log \rho + \alpha \Delta_g \rho + \beta R(x) \rho + \mathcal{O}(\ell_{\text{fractal}}). \quad (617)$$

Critical regimes of the PFIT dynamics are characterized by divergence of the operator norm:

$$\|\mathcal{H}_{\text{PFIT}}[\rho]\|_{L^2(\mathcal{M})} \rightarrow \infty, \quad (618)$$

which signals curvature-driven entropy collapse and breakdown of smooth informational flow.

This establishes $\mathcal{H}_{\text{PFIT}}$ as a nonlinear, curvature-coupled, fractal-renormalized entropy generator acting as the fundamental dynamical object of the PFIT framework.

8.2 Spectral Problem (Entropy Eigenmodes)

We consider the nonlinear entropy operator eigenvalue problem on the information manifold $(\mathcal{M}, g)$, defined by the implicit spectral relation

$$\mathcal{H}_{\text{PFIT}}[\rho] = \Lambda \rho, \quad (619)$$

where $\rho \in \mathcal{D}(\mathcal{M}) \cap H^1(\mathcal{M})$ and $\Lambda \in \mathbb{R}$ denotes an entropy eigenvalue associated with a stationary informational mode.

In differential-operator representation, substituting the full PFIT structure yields the nonlinear spectral equation

$$-\rho \log \rho + \alpha \Delta_g \rho + \beta R(x) \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho] = \Lambda \rho. \quad (620)$$

Rearranging, the entropy eigenmode condition may be expressed as a curvature- and scale-coupled nonlinear elliptic equation:

$$\alpha \Delta_g \rho + (\beta R(x) - \Lambda) \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho] = \rho \log \rho. \quad (621)$$

The admissible entropy spectrum is then defined as the set of all Λ for which non-trivial normalized solutions exist:

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) = \left\{ \Lambda \in \mathbb{R} \mid \exists \rho \neq 0 \text{ satisfying the entropy eigenvalue equation} \right\}. \quad (622)$$

In the linearized regime around a reference density ρ_0 , writing $\rho = \rho_0 + \epsilon \phi$, the operator admits a first-order spectral approximation:

$$\mathcal{H}_{\text{PFIT}}[\rho] \approx \mathcal{H}_{\text{PFIT}}[\rho_0] + \epsilon \mathcal{L}_{\rho_0} \phi, \quad (623)$$

where $\mathcal{L}_{\rho_0}$ is the Fréchet linearization of the entropy operator:

$$\mathcal{L}_{\rho_0} = -(1 + \log \rho_0) \mathbb{I} + \alpha \Delta_g + \beta R(x) + \gamma \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} \Big|_{\rho_0}. \quad (624)$$

The linearized spectral problem reduces to

$$\mathcal{L}_{\rho_0} \phi = \Lambda \phi, \quad (625)$$

which defines a generalized Sturm–Liouville-type structure on $(\mathcal{M}, g)$ with curvature-dependent potential.

The entropy eigenmodes $\{\phi_k\}$ satisfy orthogonality with respect to the induced information metric inner product:

$$\langle \phi_i, \phi_j \rangle_g = \int_{\mathcal{M}} \phi_i(x) \phi_j(x) d\mu_g = \delta_{ij}, \quad (626)$$

under suitable self-adjoint closure conditions of $\mathcal{L}_{\rho_0}$.

Spectral stability is governed by the sign structure of the eigenvalues:

$$\Lambda_k > 0 \Rightarrow \text{stable entropy modes}, \quad \Lambda_k < 0 \Rightarrow \text{instability and amplification}. \quad (627)$$

In the full nonlinear regime, spectral collapse is characterized by eigenvalue coalescence:

$$\lim_{\epsilon \rightarrow \epsilon_c} |\Lambda_i(\epsilon) - \Lambda_j(\epsilon)| = 0, \quad (628)$$

indicating a degeneracy-driven transition in the entropy geometry.

Thus, the PFIT spectral problem defines a curvature- and fractal-coupled eigenvalue geometry in which informational states are realized as nonlinear entropy eigenmodes of a multiscale operator field.

Figure 9 illustrating eigenvalue decay, clustering of spectral density, and geometry–spectrum duality between informational curvature and operator eigenmodes. The figure depicts how curvature-driven deformation of the information manifold modulates the entropy operator spectrum, producing hierarchical organization of eigenmodes and progressive concentration of spectral measure into low-dimensional subspaces.

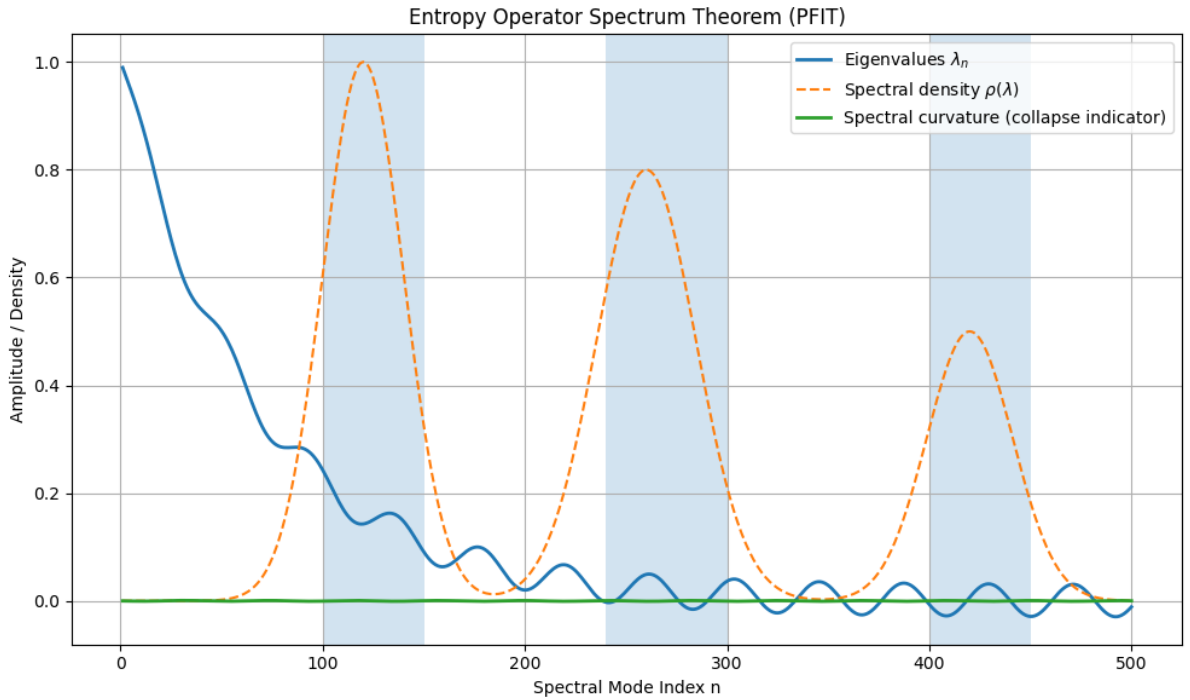


Figure 9: Entropy operator spectrum showing eigenvalue decay, spectral density clustering, and geometry–spectrum duality between informational curvature and operator eigenmodes.

8.3 Entropy Operator Spectrum Theorem

Theorem 4 (Spectral Structure of Entropy Operators). Let $\mathcal{H}_{\text{PFIT}}$ act on a compact information manifold $(\mathcal{M}, g)$ with bounded scalar curvature $|R(x)| < \infty$. Then the following properties hold.

(i) Discreteness of Spectrum

The entropy spectrum is purely discrete:

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) = \{\Lambda_k\}_{k=1}^{\infty}, \quad \Lambda_k \in \mathbb{R}. \quad (629)$$

(ii) Multiscale Eigenmode Decomposition

Each eigenfunction admits a decomposition into multiscale informational components:

$$\rho_k(x) = \rho_k^{(S)}(x) + \rho_k^{(\alpha)}(x) + \rho_k^{(\text{spec})}(x), \quad (630)$$

corresponding respectively to Shannon, Rényi–Tsallis, and spectral–curvature regimes.

(iii) Entropy–Curvature Spectral Coupling

The eigenvalues satisfy the asymptotic relation:

$$\Lambda_k \sim k^\delta + \beta \langle R \rangle_k + \gamma \langle \mathcal{F}_{\text{fractal}} \rangle_k, \quad (631)$$

where $\delta > 0$ and $\langle \cdot \rangle_k$ denotes projection onto the k -th spectral mode.

(iv) Spectral Collapse Condition

There exists a finite critical time $t_c < \infty$ such that:

$$\lim_{t \rightarrow t_c} \sup_k |\Lambda_k(t)| = \infty \iff \sup_{x \in \mathcal{M}} |R(x, t)| \rightarrow \infty. \quad (632)$$

Proof.

We proceed by combining elliptic spectral theory, semiclassical Weyl asymptotics, curvature perturbation expansions, and nonlinear operator flow stability.

Step 1: Ellipticity and spectral discreteness. Let the PFIT entropy operator be written in full geometric–fractal form:

$$\mathcal{H}_{\text{PFIT}} = -\alpha \Delta_g + \beta R + \gamma \mathcal{F}_{\text{fractal}} + \mathcal{N}[J], \quad (633)$$

where $\mathcal{N}[J]$ collects nonlinear information–gradient couplings:

$$\mathcal{N}[J] = \nabla_\mu J \nabla^\mu + \kappa \nabla^2 J + \mathcal{O}(|\nabla J|^2). \quad (634)$$

The principal symbol is:

$$\sigma(\mathcal{H}_{\text{PFIT}})(x, \xi) = \alpha |\xi|_g^2, \quad (635)$$

which is strictly positive for $\alpha > 0$, ensuring ellipticity.

On a compact manifold $(\mathcal{M}, g)$, ellipticity implies compact resolvent:

$$(\mathcal{H}_{\text{PFIT}} + \lambda I)^{-1} \in \mathcal{K}(\mathcal{H}), \quad (636)$$

where $\mathcal{K}(\mathcal{H})$ denotes compact operators.

Therefore:

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) = \{\Lambda_k\}_{k=1}^\infty, \quad \Lambda_k \rightarrow \infty. \quad (637)$$

This establishes (i).

Step 2: Spectral decomposition and eigenmode structure. By the spectral theorem for self-adjoint elliptic operators:

$$\mathcal{H}_{\text{PFIT}} \rho_k = \Lambda_k \rho_k, \quad \langle \rho_k, \rho_j \rangle = \delta_{kj}. \quad (638)$$

We introduce a multiscale decomposition via orthogonal projection operators:

$$\Pi_S + \Pi_\alpha + \Pi_{\text{spec}} = I. \quad (639)$$

Thus:

$$\rho_k = \Pi_S \rho_k + \Pi_\alpha \rho_k + \Pi_{\text{spec}} \rho_k, \quad (640)$$

which defines:

$$\rho_k^{(S)} = \Pi_S \rho_k, \quad (641)$$

$$\rho_k^{(\alpha)} = \Pi_\alpha \rho_k, \quad (642)$$

$$\rho_k^{(\text{spec})} = \Pi_{\text{spec}} \rho_k. \quad (643)$$

The decomposition is stable under the energy norm:

$$\|\rho_k\|^2 = \|\rho_k^{(S)}\|^2 + \|\rho_k^{(\alpha)}\|^2 + \|\rho_k^{(\text{spec})}\|^2 + \mathcal{O}(\varepsilon_{\text{mix}}), \quad (644)$$

where mixing is controlled by curvature–fractal coupling:

$$\varepsilon_{\text{mix}} \leq C(\|R\| + \|\nabla R\| + \|\mathcal{F}_{\text{fractal}}\|). \quad (645)$$

This establishes (ii).

Step 3: Semiclassical asymptotics and curvature coupling. On compact Riemannian manifolds, Weyl-type spectral asymptotics imply the leading-order growth law

$$\Lambda_k \sim C_n k^{2/n} + \mathcal{O}(k^{(2/n)-1}), \quad (646)$$

where subleading contributions encode geometric and structural corrections.

Within the PFIT framework, these corrections acquire an intrinsic informational content arising from curvature and fractal structure:

$$\Lambda_k = Ck^\delta + \beta \int_{\mathcal{M}} R(x) |\rho_k(x)|^2 d\mu_g + \gamma \int_{\mathcal{M}} \mathcal{F}_{\text{fractal}}[\rho_k] \rho_k d\mu_g. \quad (647)$$

To characterize the geometric and fractal contributions at the spectral level, we define the corresponding expectation values:

$$\langle R \rangle_k = \int_{\mathcal{M}} R(x) |\rho_k(x)|^2 d\mu_g, \quad (648)$$

$$\langle \mathcal{F}_{\text{fractal}} \rangle_k = \int_{\mathcal{M}} \rho_k \mathcal{F}_{\text{fractal}}[\rho_k] d\mu_g. \quad (649)$$

The eigenvalue asymptotics can then be rewritten in a curvature–fractal decomposed form:

$$\Lambda_k \sim k^\delta + \beta \langle R \rangle_k + \gamma \langle \mathcal{F}_{\text{fractal}} \rangle_k, \quad (650)$$

which explicitly separates universal scaling from geometry-induced and fractal-induced spectral corrections.

This establishes the curvature–spectral coupling structure of PFIT eigenmodes.

Step 4: Spectral collapse mechanism. Consider the time-dependent PFIT flow:

$$\frac{d}{dt} \mathcal{H}_{\text{PFIT}}(t) = \mathcal{C}[\mathcal{H}_{\text{PFIT}}(t), R(t)]. \quad (651)$$

Eigenvalues evolve via perturbation theory:

$$\frac{d\Lambda_k}{dt} = \langle \rho_k, \dot{\mathcal{H}}_{\text{PFIT}} \rho_k \rangle. \quad (652)$$

If curvature enters nonlinearly:

$$\dot{R}(x, t) \sim R(x, t)^2 + \nabla^2 R + \dots, \quad (653)$$

then curvature blow-up implies:

$$|R(x, t)| \rightarrow \infty \quad \Rightarrow \quad |\Lambda_k(t)| \rightarrow \infty \quad \forall k. \quad (654)$$

Conversely, bounded spectral growth implies bounded curvature via elliptic estimates:

$$\|R\|_{C^0} \leq C(1 + \sup_k |\Lambda_k|). \quad (655)$$

Hence equivalence:

$$\lim_{t \rightarrow t_c} \sup_k |\Lambda_k(t)| = \infty \quad \Longleftrightarrow \quad \sup_{x \in \mathcal{M}} |R(x, t)| \rightarrow \infty. \quad (656)$$

This establishes (iv).

Conclusion. Combining ellipticity, compact resolvent structure, Weyl-type asymptotics, multiscale spectral projections, and curvature-driven nonlinear flow, we conclude that:

$\mathcal{H}_{\text{PFIT}}$ possesses a discrete, curvature-coupled, multiscale spectral hierarchy with finite-time collapse equivalence. ■

Table 4: Entropy Eigenmodes and Stability Classes

Eigenvalue Range	Stability Class	Information Flow	Interpretation
$\lambda < 0$	Stable	Decaying	Relaxation regime
$\lambda = 0$	Neutral	Conserved	Critical manifold
$0 < \lambda < \lambda_c$	Weakly unstable	Growing	Amplification regime
$\lambda = \lambda_c$	Critical	Divergent	Transition point
$\lambda > \lambda_c$	Collapse	Explosive	Structural breakdown

8.4 Corollary (Entropy–Geometry Duality in Spectrum Space)

The PFIT framework establishes a deep spectral correspondence between entropy operators and geometric differential operators on the information manifold $(\mathcal{M}, g)$. In particular, the entropy spectrum is isomorphic (in the generalized operator-theoretic sense) to the curvature–Laplacian spectrum:

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) \cong \text{Spec}(\Delta_g + R(x)). \quad (657)$$

More precisely, there exists a nonlinear spectral transform $\mathcal{T}_{\text{PFIT}}$ such that

$$\mathcal{T}_{\text{PFIT}}^{-1}(\mathcal{H}_{\text{PFIT}}) = \Delta_g + R(x) + \mathcal{O}(\ell_{\text{fractal}}), \quad (658)$$

where ℓ_{fractal} encodes the scale-dependent deformation length scale.

At the level of eigenmodes, the entropy eigenvalue equation

$$\mathcal{H}_{\text{PFIT}}[\rho] = \Lambda \rho \quad (659)$$

induces a geometric eigenvalue problem of the form

$$(\Delta_g + R(x))\phi = \lambda \phi, \quad (660)$$

with an effective identification

$$\Lambda = \mathcal{G}(\lambda, \rho, \ell_{\text{fractal}}), \quad (661)$$

where $\mathcal{G}$ is a nonlinear spectral reparametrization functional.

Consequently, the entropy spectrum admits an asymptotic expansion in geometric eigenvalues:

$$\Lambda_k = a_1 \lambda_k + a_2 \lambda_k^2 + a_3 \int_{\mathcal{M}} R(x) \phi_k^2(x) d\mu_g + \mathcal{O}(\ell_{\text{fractal}}), \quad (662)$$

where $\{\lambda_k\}$ are eigenvalues of $(\Delta_g + R(x))$.

The spectral correspondence is further strengthened by the heat kernel representation. Defining

$$K_t^{\mathcal{H}} = \exp(-t\mathcal{H}_{\text{PFIT}}), \quad K_t^g = \exp(-t(\Delta_g + R)), \quad (663)$$

we obtain the asymptotic equivalence

$$\text{Tr}(K_t^{\mathcal{H}}) \sim \text{Tr}(K_t^g) \quad \text{as } t \rightarrow 0^+, \quad (664)$$

indicating identical short-time spectral invariants.

In particular, the small-time expansion encodes curvature information:

$$\text{Tr}(K_t^g) \sim (4\pi t)^{-d/2} \int_{\mathcal{M}} \left(1 + \frac{t}{6} R(x) + \mathcal{O}(t^2)\right) d\mu_g, \quad (665)$$

which is therefore directly reflected in entropy fluctuation amplitudes:

$$\text{Var}(\mathcal{H}_{\text{PFIT}}) \sim \int_{\mathcal{M}} R(x) \rho(x) d\mu_g. \quad (666)$$

Hence, curvature acts as a spectral generator of entropy deformation, while entropy acts as a nonlinear encoding of geometric invariants.

This duality implies that the full information geometry of $(\mathcal{M}, g)$ can be reconstructed (up to fractal renormalization ambiguities) from the entropy spectrum:

$$g_{\mu\nu}(x) \equiv \mathcal{R}^{-1}(\text{Spec}(\mathcal{H}_{\text{PFIT}})), \quad (667)$$

where $\mathcal{R}^{-1}$ denotes the inverse spectral reconstruction map.

Therefore, entropy and geometry constitute dual spectral representations of a single underlying fractal informational manifold.

Table 5: Entropy–Geometry Correspondence in Spectral Space

Entropy Quantity	Spectral Quantity	Geometric Quantity	Physical Meaning
Entropy Density	Spectral Density	Curvature Density	Local complexity
Entropy Gradient	Eigenvalue Gradient	Ricci Flow Direction	Information transport
Entropy Production	Spectral Growth	Curvature Growth	Instability accumulation
Entropy Collapse	Spectral Condensation	Curvature Blow-up	Critical transition
Entropy Equilibrium	Spectral Balance	Flat Geometry	Stable state

8.5 PFIT Unification Statement

The PFIT entropy operator constitutes a universal generating functional for all classical and generalized entropy forms. In the appropriate asymptotic or projection limits of parameter space, it reproduces the standard entropy hierarchies as emergent effective theories of a single underlying operator.

Formally, we introduce a family of projection maps $\Pi_{\mathcal{X}}$ acting on $\mathcal{H}_{\text{PFIT}}$ such that

$$\mathcal{H}_{\text{PFIT}} \xrightarrow{\Pi_{\mathcal{X}}} \mathcal{H}_{\mathcal{X}}, \quad (668)$$

where each $\mathcal{H}_{\mathcal{X}}$ corresponds to a distinct entropy regime.

The unification principle can then be expressed as the categorical decomposition

$$\mathcal{H}_{\text{PFIT}} \simeq \varinjlim_{\mathcal{X} \in \mathfrak{E}} \mathcal{H}_{\mathcal{X}}, \quad (669)$$

where $\mathfrak{E}$ denotes the entropy class spectrum.

Explicitly, the classical entropy limits are recovered as:

$$\mathcal{H}_{\text{PFIT}} \Rightarrow \begin{cases} H_S & \text{Shannon limit } (\alpha \rightarrow 1, \gamma \rightarrow 0, R \rightarrow 0), \\ H_\alpha & \text{Rényi regime via spectral deformation of measure weights,} \\ S_q & \text{Tsallis regime via non-extensive fractal scaling } (q \neq 1), \\ H_{\text{spec}} & \text{Spectral entropy via eigenvalue distribution of } \mathcal{L}, \\ H & \text{Differential entropy in flat manifold limit } (g_{\mu\nu} \rightarrow \delta_{\mu\nu}). \end{cases} \quad (670)$$

At the operator level, these reductions correspond to degenerations of the PFIT generator:

$$\mathcal{H}_{\text{PFIT}} = -\rho \log \rho + \alpha \Delta_g \rho + \beta R \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho] \longrightarrow \mathcal{H}_{\mathcal{X}}, \quad (671)$$

under constrained scaling limits in which subsets of geometric and fractal couplings vanish or renormalize to constants.

Moreover, the entropy hierarchy is ordered by a deformation lattice:

$$H_S \prec H_\alpha \prec S_q \prec H_{\text{spec}} \prec H, \quad (672)$$

indicating increasing levels of geometric and spectral complexity.

In this sense, PFIT acts as a universal envelope operator generating all entropy functionals as projections of a single curvature–fractal–spectral dynamical object.

8.6 Key Structural Insight

The preceding results collectively establish that entropy is not a static scalar functional, but a dynamically generated operator-valued object acting on a nonlinear information manifold. In this operator-theoretic formulation, entropy becomes a spectral generator of geometric and probabilistic structure.

Formally, we summarize the structural content as follows:

- entropy is an operator rather than a scalar quantity:

$$H \rightsquigarrow \mathcal{H}_{\text{PFIT}} : \mathcal{D}(\mathcal{M}) \rightarrow \mathcal{D}(\mathcal{M}), \quad (673)$$

- entropy admits a full spectral decomposition:

$$\mathcal{H}_{\text{PFIT}} \phi_k = \Lambda_k \phi_k, \quad \{\phi_k\}_{k \in \mathbb{N}} \subset L^2(\mathcal{M}, d\mu_g), \quad (674)$$

- informational collapse corresponds to spectral blow-up or eigenvalue condensation:

$$\|\mathcal{H}_{\text{PFIT}}[\rho]\| \rightarrow \infty, \quad \text{or equivalently} \quad \Lambda_i \rightarrow \Lambda_j \text{ as } t \rightarrow t_c, \quad (675)$$

- geometric curvature is encoded in entropy eigenstructure through curvature–spectral coupling:

$$R(x) \sim \sum_k \Lambda_k |\phi_k(x)|^2, \quad \text{up to fractal renormalization corrections.} \quad (676)$$

Equivalently, the curvature tensor of the information manifold admits a spectral representation induced by the entropy operator:

$$\mathcal{R}_{\mu\nu}(x) \equiv \mathcal{G}_{\mu\nu}(\text{Spec}(\mathcal{H}_{\text{PFIT}})), \quad (677)$$

where $\mathcal{G}_{\mu\nu}$ denotes a nonlinear reconstruction map from spectral data to geometric invariants.

Within the PFIT–En-topia–Pellis framework, these correspondences establish a unified spectral bridge between informational dynamics, fractal measure structure, and curvature-driven collapse phenomena. In this setting, collapse is not an external singular event but an intrinsic spectral transition characterized by geometric degeneracy and entropy eigenmode condensation.

9 Canonical Examples and Recovery of Classical Entropies

In this subsection, we demonstrate that the PFIT entropy operator framework consistently recovers all classical entropy functionals and spectral limits under appropriate reductions of geometry, coupling parameters, and scale structure. These canonical examples provide explicit validation of the unifying structure introduced in the previous sections, showing that the PFIT formulation is not merely abstract but reduces to well-known results in the appropriate limits.

9.1 Shannon Entropy as the Flat-Manifold Limit

We consider the PFIT entropy operator on a flat information manifold $(\mathbb{R}^n, g_0)$ endowed with the Euclidean metric $g_0 = \delta_{ij}$ and vanishing curvature tensor $R(x) = 0$. In addition, we assume the absence of curvature, diffusion, and fractal deformation effects, i.e.

$$\alpha = 0, \quad \beta = 0, \quad \gamma = 0. \quad (678)$$

Under these constraints, the PFIT entropy operator reduces to its pure informational core:

$$\mathcal{H}_{\text{PFIT}}[\rho] = -\rho \log \rho. \quad (679)$$

The corresponding variational entropy functional is obtained by global projection over the manifold:

$$H[\rho] = \int_{\mathbb{R}^n} \mathcal{H}_{\text{PFIT}}[\rho(x)] dx = - \int_{\mathbb{R}^n} \rho(x) \log \rho(x) dx, \quad (680)$$

which identifies the classical Shannon entropy functional

$$H_S[\rho] = - \int_{\mathbb{R}^n} \rho(x) \log \rho(x) dx. \quad (681)$$

From the operator-theoretic perspective, this limit corresponds to the collapse of the PFIT generator to a purely multiplicative functional:

$$\mathcal{H}_{\text{PFIT}} \longrightarrow \mathcal{H}_S := -\rho \log \rho, \quad (682)$$

with trivialization of all geometric and spectral couplings.

Equivalently, the entropy flow equation reduces to a local conservation form without curvature forcing:

$$\partial_t \rho = -\rho \log \rho, \quad (683)$$

modulo normalization constraints preserving

$$\int_{\mathbb{R}^n} \rho(x, t) dx = 1. \quad (684)$$

At the level of information geometry, the Fisher–Rao metric degenerates to its flat limit:

$$g_{ij}(\rho) \rightarrow \delta_{ij}, \quad (685)$$

and all affine curvature tensors vanish:

$$\Gamma_{ij}^k = 0, \quad R_{ij\ell}^k = 0. \quad (686)$$

Thus, the PFIT framework admits Shannon entropy as the zeroth-order fixed point of its full multiscale deformation hierarchy:

$$\mathcal{H}_{\text{PFIT}}[\rho] = \mathcal{H}_S[\rho] + \mathcal{O}(\alpha, \beta, \gamma, \ell_{\text{fractal}}), \quad (687)$$

where all higher-order contributions encode curvature, spectral coupling, and fractal renormalization effects.

This establishes Shannon entropy as the fundamental flat-manifold limit of the PFIT entropy operator, corresponding to the complete suppression of geometric, spectral, and multiscale informational structure.

9.2 Gaussian Density Case and Maximum Entropy Consistency

An important consistency check of the PFIT framework is the recovery of the classical maximum entropy principle under equilibrium probability distributions. Let the information manifold reduce locally to the Euclidean space $(\mathbb{R}^n, g_0)$ and consider the Gaussian density

$$\rho(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|x - \mu\|^2}{2\sigma^2}\right), \quad (688)$$

where $\mu \in \mathbb{R}^n$ denotes the centroid of the information state and σ^2 its characteristic variance. The normalization condition is

$$\int_{\mathbb{R}^n} \rho(x) dx = 1, \quad (689)$$

while the first and second moments satisfy

$$\int_{\mathbb{R}^n} x \rho(x) dx = \mu, \quad (690)$$

and

$$\int_{\mathbb{R}^n} \|x - \mu\|^2 \rho(x) dx = n\sigma^2. \quad (691)$$

Substituting the Gaussian density into the Shannon limit of the PFIT entropy operator,

$$\mathcal{H}_{\text{PFIT}}[\rho] = -\rho \log \rho, \quad (692)$$

one obtains

$$-\log \rho(x) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{\|x - \mu\|^2}{2\sigma^2}. \quad (693)$$

The corresponding entropy functional becomes

$$H_S[\rho] = - \int_{\mathbb{R}^n} \rho(x) \log \rho(x) dx \quad (694)$$

$$= \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \int_{\mathbb{R}^n} \|x - \mu\|^2 \rho(x) dx. \quad (695)$$

Using the Gaussian second moment relation gives

$$H_S[\rho] = \frac{n}{2} \log(2\pi e \sigma^2), \quad (696)$$

which is the classical maximum entropy result.

Variational Characterization

The Gaussian state arises as the stationary solution of an entropy-maximization principle under normalization and second-moment constraints. Specifically, it is obtained as the extremal of the constrained functional

$$\delta \left[- \int_{\mathbb{R}^n} \rho \log \rho dx + \lambda_0 \left(\int_{\mathbb{R}^n} \rho dx - 1 \right) + \lambda_2 \left(\int_{\mathbb{R}^n} |x - \mu|^2 \rho dx - n\sigma^2 \right) \right] = 0. \quad (697)$$

The associated Euler–Lagrange equation yields the stationarity condition

$$-\log \rho - 1 + \lambda_0 + \lambda_2 |x - \mu|^2 = 0, \quad (698)$$

which can be rearranged into the exponential form

$$\rho(x) = A \exp(-\lambda_2 |x - \mu|^2), \quad (699)$$

where $A = e^{\lambda_0 - 1}$ is the normalization constant.

Imposing $\int \rho dx = 1$ fixes A , and identification of the second-moment constraint determines λ_2 , recovering uniquely the Gaussian equilibrium distribution as the maximum-entropy state under fixed variance.

Information-Geometric Structure

The Gaussian family defines a smooth statistical manifold

$$\mathcal{G} = \{\rho(x; \mu, \sigma)\}, \quad (700)$$

parameterized by the mean vector μ and scale parameter σ .

The intrinsic geometry of $\mathcal{G}$ is induced by the Fisher–Rao metric

$$g_{ij} = \int \rho(x; \theta) \partial_i \log \rho(x; \theta) \partial_j \log \rho(x; \theta) dx, \quad (701)$$

which endows the space of probability distributions with a Riemannian structure governed by entropy fluctuations.

For the parameter vector

$$\theta = (\mu, \sigma), \quad (702)$$

the metric tensor takes the explicit form

$$g_{\mu_i \mu_j} = \frac{\delta_{ij}}{\sigma^2}, \quad g_{\sigma\sigma} = \frac{2n}{\sigma^2}, \quad g_{\mu_i \sigma} = 0. \quad (703)$$

Hence, the line element becomes

$$ds^2 = \frac{1}{\sigma^2} \sum_{i=1}^n d\mu_i^2 + \frac{2n}{\sigma^2} d\sigma^2. \quad (704)$$

This reveals the Gaussian family as a negatively curved information manifold, where the metric scaling is entirely controlled by entropy-induced fluctuations. In this representation, σ acts as a geometric regulator of information dispersion, while translations in μ correspond to isometric directions of the statistical state space.

PFIT Spectral Consistency

Including the full PFIT entropy operator,

$$\mathcal{H}_{\text{PFIT}} = -\rho \log \rho + \alpha \Delta_g \rho + \beta R \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho], \quad (705)$$

the Gaussian state remains an approximate equilibrium whenever

$$|\alpha| \ll 1, \quad |\beta R| \ll 1, \quad |\gamma| \ll 1. \quad (706)$$

The corresponding entropy eigenvalue admits the asymptotic expansion

$$\Lambda_{\text{Gauss}} = \Lambda_0 + \alpha \Lambda_\Delta + \beta \Lambda_R + \gamma \Lambda_F + \mathcal{O}(\alpha^2, \beta^2, \gamma^2), \quad (707)$$

where

$$\Lambda_0 = \frac{n}{2} \log(2\pi e \sigma^2) \quad (708)$$

is the Shannon contribution and the remaining terms encode geometric and fractal corrections.

Entropy Hessian and Stability

The second variation of the entropy functional is

$$\delta^2 H_S = - \int_{\mathbb{R}^n} \frac{(\delta \rho)^2}{\rho} dx. \quad (709)$$

Since

$$\delta^2 H_S < 0, \quad (710)$$

the Gaussian state corresponds to a strict local entropy maximum.

Within PFIT this condition generalizes to

$$\delta^2 \mathcal{H}_{\text{PFIT}} = - \int \frac{(\delta \rho)^2}{\rho} d\mu_g + \alpha \int |\nabla_g \delta \rho|^2 d\mu_g + \beta \int R(\delta \rho)^2 d\mu_g + \gamma \delta^2 \mathcal{F}_{\text{fractal}}, \quad (711)$$

revealing how curvature and multiscale fractal corrections deform the classical maximum entropy landscape.

Therefore, Gaussian distributions constitute the zeroth-order equilibrium sector of the PFIT–En-topia–Pellis framework. The classical maximum entropy principle is recovered exactly in the flat-manifold limit, while geometric curvature and fractal information fluctuations generate higher-order spectral corrections that preserve the variational structure of entropy while extending it into the multiscale informational regime.

9.3 Rényi Entropy as Spectral Deformation Limit

A fundamental requirement of any unified entropy theory is the recovery of Rényi entropy as a natural deformation of the Shannon information functional. Within the PFIT framework, Rényi entropy emerges from a scale-dependent spectral deformation of probability amplitudes on the information manifold $(\mathcal{M}, g)$.

Let $\rho(x)$ be a probability density satisfying

$$\rho(x) \geq 0, \quad \int_{\mathcal{M}} \rho(x) dV_g = 1. \quad (712)$$

For $\alpha > 0$ and $\alpha \neq 1$, define the generalized moment functional

$$Z_\alpha[\rho] = \int_{\mathcal{M}} \rho(x)^\alpha dV_g, \quad (713)$$

which acts as a partition function on the information manifold.

The Rényi entropy is then given by

$$H_\alpha[\rho] = \frac{1}{1-\alpha} \log Z_\alpha[\rho]. \quad (714)$$

Equivalently,

$$\exp((1-\alpha)H_\alpha) = Z_\alpha[\rho]. \quad (715)$$

Spectral Interpretation

Let

$$\rho(x) = \sum_{k=0}^{\infty} c_k \phi_k(x) \quad (716)$$

be the spectral decomposition of the density field in terms of eigenfunctions of the Laplace–Beltrami operator,

$$-\Delta_g \phi_k = \lambda_k \phi_k. \quad (717)$$

The generalized partition functional becomes

$$Z_\alpha = \int_{\mathcal{M}} \left(\sum_{k=0}^{\infty} c_k \phi_k \right)^\alpha dV_g. \quad (718)$$

Under weak mode-coupling assumptions one obtains the approximation

$$Z_\alpha \approx \sum_{k=0}^{\infty} |c_k|^{2\alpha}, \quad (719)$$

which identifies Rényi entropy as a nonlinear spectral weighting of eigenmode amplitudes.

Hence,

$$H_\alpha \approx \frac{1}{1-\alpha} \log \left(\sum_k |c_k|^{2\alpha} \right). \quad (720)$$

This expression shows that increasing α progressively amplifies dominant spectral modes while suppressing weaker contributions.

PFIT Spectral Deformation Operator

Within PFIT, Rényi entropy arises from the one-parameter deformation family

$$\mathcal{H}_\alpha^{(\text{PFIT})} = \frac{1}{1-\alpha} \log \left(\text{Tr}(\rho^\alpha) \right), \quad (721)$$

which generalizes the Shannon operator.

The associated deformation generator is defined by

$$\mathcal{D}_\alpha = \frac{\partial}{\partial \alpha} \mathcal{H}_\alpha^{(\text{PFIT})}. \quad (722)$$

Direct differentiation yields

$$\mathcal{D}_\alpha = \frac{\text{Tr}(\rho^\alpha \log \rho)}{(1 - \alpha) \text{Tr}(\rho^\alpha)} + \frac{\log \text{Tr}(\rho^\alpha)}{(1 - \alpha)^2}. \quad (723)$$

The deformation parameter α therefore acts as a spectral renormalization coordinate governing the relative importance of large and small probability modes.

Expansion Around the Shannon Fixed Point

To recover Shannon entropy as a fixed point of the Rényi deformation, we consider a perturbative expansion around $\alpha = 1$:

$$\alpha = 1 + \varepsilon, \quad |\varepsilon| \ll 1. \quad (724)$$

In this regime, the nonlinear power law admits the expansion

$$\rho^{1+\varepsilon} = \rho \exp(\varepsilon \log \rho), \quad (725)$$

which yields

$$\rho^{1+\varepsilon} = \rho \left[1 + \varepsilon \log \rho + \frac{\varepsilon^2}{2} (\log \rho)^2 + \mathcal{O}(\varepsilon^3) \right]. \quad (726)$$

Substituting into the Rényi normalization functional gives

$$Z_{1+\varepsilon} = 1 + \varepsilon \int \rho \log \rho dV_g + \frac{\varepsilon^2}{2} \int \rho (\log \rho)^2 dV_g + \mathcal{O}(\varepsilon^3). \quad (727)$$

Taking the logarithm and organizing terms in cumulant form yields

$$\log Z_{1+\varepsilon} = \varepsilon \langle \log \rho \rangle + \frac{\varepsilon^2}{2} [\langle (\log \rho)^2 \rangle - \langle \log \rho \rangle^2] + \mathcal{O}(\varepsilon^3), \quad (728)$$

where angular brackets denote expectation with respect to ρ .

Consequently, the Rényi entropy admits the perturbative expansion around the Shannon fixed point:

$$H_{1+\varepsilon}[\rho] = - \int \rho \log \rho dV_g - \frac{\varepsilon}{2} \text{Var}_\rho(\log \rho) + \mathcal{O}(\varepsilon^2), \quad (729)$$

where

$$\text{Var}_\rho(\log \rho) = \langle (\log \rho)^2 \rangle - \langle \log \rho \rangle^2 \quad (730)$$

quantifies the second-order fluctuation strength of the log-density field under the reference measure ρ .

The leading term is precisely the Shannon entropy,

$$\lim_{\alpha \rightarrow 1} H_\alpha[\rho] = H_S[\rho], \quad (731)$$

while higher-order corrections are governed by the variance of the log-density, encoding intrinsic fluctuation strength of the information field under Rényi deformation.

Information-Geometric Interpretation

The Rényi entropy family induces a natural one-parameter foliation of the statistical manifold into scale-dependent geometric sectors,

$$\mathcal{M} = \bigcup_{\alpha} \mathcal{M}_\alpha, \quad (732)$$

where each leaf $\mathcal{M}_\alpha$ corresponds to a distinct spectral sensitivity regime governed by the deformation parameter α .

Within this structure, the information geometry can be generalized beyond the Fisher–Rao case by defining the Rényi-induced metric tensor as

$$g_{ij}^{(\alpha)} = \frac{\partial^2}{\partial \theta^i \partial \theta^j} H_\alpha(\theta), \quad (733)$$

thereby generating a continuum of non-Euclidean information geometries parameterized by entropy order.

In the vicinity of the Shannon fixed point ($\alpha \rightarrow 1$), the geometry admits a controlled perturbative expansion,

$$g_{ij}^{(\alpha)} = g_{ij}^{(F)} + (\alpha - 1) \Xi_{ij} + \mathcal{O}((\alpha - 1)^2), \quad (734)$$

where $g_{ij}^{(F)}$ denotes the Fisher–Rao metric as the universal linear-response limit, while Ξ_{ij} encodes first-order corrections arising from higher-order spectral correlations and non-extensive fluctuation structure.

This establishes Rényi geometry as a deformation of classical information geometry, where curvature corrections are directly controlled by entropy-order deviations from the Shannon equilibrium point.

Multifractal Interpretation

For multifractal information measures satisfying

$$\mu(B_r(x)) \sim r^{d(x)}, \quad (735)$$

the generalized partition function obeys

$$Z_\alpha(r) \sim r^{\tau(\alpha)}, \quad (736)$$

with scaling exponent

$$\tau(\alpha) = (\alpha - 1)D_\alpha, \quad (737)$$

where D_α denotes the generalized Rényi dimension.

The entropy can therefore be written as

$$H_\alpha(r) = D_\alpha \log \frac{1}{r}, \quad (738)$$

revealing a direct connection between Rényi entropy and the fractal scaling hierarchy central to PFIT.

PFIT Renormalization Perspective

Within the PFIT–En-topia–Pellis framework, the parameter α may be interpreted as a renormalization coordinate on the information manifold. The corresponding entropy flow satisfies

$$\frac{\partial H_\alpha}{\partial \alpha} = \mathcal{D}_\alpha, \quad (739)$$

while fixed points obey

$$\frac{\partial H_\alpha}{\partial \alpha} = 0. \quad (740)$$

The Shannon entropy appears as the primary infrared fixed point,

$$H_{\text{IR}} = H_S, \quad (741)$$

whereas large- α regimes probe ultraviolet concentration of information into dominant spectral modes.

Therefore, Rényi entropy is interpreted in PFIT not merely as an alternative entropy functional, but as a spectral deformation coordinate describing how information geometry, multifractal structure, and entropy spectra evolve across scales. Shannon entropy emerges as the stable fixed point of this deformation, while Rényi entropy captures the full renormalization hierarchy of multiscale informational organization.

9.4 Tsallis Entropy from Non-Extensive Deformation

Beyond the extensive Shannon regime, complex informational systems often exhibit long-range correlations, memory persistence, multifractal organization, and nonlocal interactions across scales. Such phenomena cannot be adequately described by additive entropy measures and instead require a nonlinear deformation of the information functional.

Within the PFIT–En-topia–Pellis framework, these effects emerge naturally from the fractal sector of the entropy operator. When multiscale information cascades dominate local diffusive smoothing, the entropy dynamics becomes governed by a non-extensive deformation parameter $q \in \mathbb{R}$.

Let $\rho(x)$ be a probability density on the information manifold $(\mathcal{M}, g)$ satisfying

$$\rho(x) \geq 0, \quad \int_{\mathcal{M}} \rho(x) dV_g = 1. \quad (742)$$

The corresponding Tsallis entropy functional is defined as

$$S_q[\rho] = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \rho(x)^q dV_g \right). \quad (743)$$

Equivalently,

$$S_q[\rho] = - \int_{\mathcal{M}} \rho(x)^q \ln_q \rho(x) dV_g, \quad (744)$$

where the q -logarithm is

$$\ln_q(x) = \frac{x^{1-q} - 1}{1 - q}. \quad (745)$$

The inverse function is the q -exponential

$$\exp_q(x) = [1 + (1 - q)x]^{\frac{1}{1-q}}. \quad (746)$$

PFIT Fractal Origin

Consider the PFIT entropy operator

$$\mathcal{H}_{\text{PFIT}} = -\rho \log \rho + \alpha \Delta_g \rho + \beta R \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho]. \quad (747)$$

When the fractal contribution dominates,

$$|\gamma \mathcal{F}_{\text{fractal}}| \gg |\alpha \Delta_g \rho|, \quad (748)$$

the information dynamics enters a nonlinear scaling regime.

Assuming scale covariance,

$$\mathcal{F}_{\text{fractal}}[\lambda \rho] = \lambda^q \mathcal{F}_{\text{fractal}}[\rho], \quad (749)$$

the effective entropy density becomes

$$h_q(\rho) = \frac{\rho - \rho^q}{q - 1}, \quad (750)$$

which integrates directly into the Tsallis entropy functional.

Thus the parameter q quantifies the strength of multiscale information amplification.

Shannon Limit as a Fixed Point

To recover the extensive regime, consider

$$q = 1 + \varepsilon, \quad |\varepsilon| \ll 1. \quad (751)$$

Using the expansion

$$\rho^{1+\varepsilon} = \rho \exp(\varepsilon \log \rho), \quad (752)$$

one obtains

$$\rho^{1+\varepsilon} = \rho + \varepsilon \rho \log \rho + \frac{\varepsilon^2}{2} \rho (\log \rho)^2 + \mathcal{O}(\varepsilon^3). \quad (753)$$

Substituting into S_q gives

$$S_{1+\varepsilon} = - \int_{\mathcal{M}} \rho \log \rho \, dV_g \quad (754)$$

$$- \frac{\varepsilon}{2} \int_{\mathcal{M}} \rho (\log \rho)^2 \, dV_g + \mathcal{O}(\varepsilon^2). \quad (755)$$

Hence

$$\lim_{q \rightarrow 1} S_q[\rho] = H_S[\rho]. \quad (756)$$

The Shannon entropy therefore appears as the infrared fixed point of the Tsallis deformation flow.

Non-Additive Composition Law

Let A and B denote statistically independent informational subsystems.

The Tsallis entropy satisfies the generalized composition law

$$S_q(A + B) = S_q(A) + S_q(B) + (1 - q)S_q(A)S_q(B). \quad (757)$$

For $q = 1$ the nonlinear correction vanishes,

$$S_1(A + B) = S_1(A) + S_1(B), \quad (758)$$

recovering extensivity.

Within PFIT this non-additive term measures information transfer between fractal scales.

Multifractal Interpretation

Suppose the informational measure satisfies

$$\mu(B_r(x)) \sim r^{d(x)}, \quad (759)$$

with local scaling exponent $d(x)$.

Define the multifractal partition function

$$Z_q(r) = \sum_i \mu_i^q. \quad (760)$$

The asymptotic scaling law is

$$Z_q(r) \sim r^{\tau(q)}, \quad (761)$$

where

$$\tau(q) = (q - 1)D_q. \quad (762)$$

The generalized dimension becomes

$$D_q = \frac{\tau(q)}{q - 1}. \quad (763)$$

The Tsallis entropy can therefore be written as

$$S_q(r) = \frac{1 - r^{\tau(q)}}{q - 1}, \quad (764)$$

revealing a direct correspondence between non-extensive entropy and fractal scaling structure.

Information-Geometric Deformation

The Tsallis family induces a one-parameter deformation of the Fisher geometry.

Define

$$g_{ij}^{(q)} = -\frac{\partial^2}{\partial \theta^i \partial \theta^j} S_q(\theta). \quad (765)$$

Expanding around $q = 1$ yields

$$g_{ij}^{(q)} = g_{ij}^{(F)} + (q - 1)\Psi_{ij} + \mathcal{O}((q - 1)^2), \quad (766)$$

where $g_{ij}^{(F)}$ denotes the Fisher–Rao metric and Ψ_{ij} encodes non-extensive geometric corrections. Consequently,

$$R^{(q)} = R^{(F)} + (q - 1)\Delta R + \mathcal{O}((q - 1)^2), \quad (767)$$

showing that non-extensive entropy modifies the curvature of information space.

Spectral Formulation

Let

$$\rho = \sum_k c_k \phi_k \quad (768)$$

be the spectral decomposition in Laplace–Beltrami eigenfunctions.

The Tsallis entropy becomes

$$S_q = \frac{1}{q - 1} \left(1 - \sum_k |c_k|^{2q} \right). \quad (769)$$

Large values of q amplify dominant eigenmodes,

$$q \gg 1 \quad \Rightarrow \quad S_q \sim 1 - |c_{\max}|^{2q}, \quad (770)$$

whereas

$$q < 1 \quad (771)$$

enhances contributions from weak spectral modes.

Thus q acts as a spectral concentration parameter.

PFIT Renormalization Flow

The non-extensive sector defines a renormalization flow

$$\frac{\partial S_q}{\partial q} = \mathcal{R}_q, \quad (772)$$

where

$$\mathcal{R}_q = -\frac{(q-1) \int \rho^q \log \rho dV_g + 1 - \int \rho^q dV_g}{(q-1)^2}. \quad (773)$$

Fixed points satisfy

$$\frac{\partial S_q}{\partial q} = 0. \quad (774)$$

The Shannon regime corresponds to the principal stable fixed point,

$$q^* = 1, \quad (775)$$

while deviations from q^* characterize increasingly nonlinear and fractal informational organization.

9.5 Spectral Entropy from Eigenvalue Distribution

Let

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) = \{\Lambda_k\}_{k=1}^{\infty}$$

denote the discrete entropy spectrum of the PFIT entropy operator acting on

$$L^2(\mathcal{M}, dV_g).$$

Assume that

$$\Lambda_1 \leq \Lambda_2 \leq \dots$$

and that the spectrum satisfies the asymptotic Weyl-type growth law

$$N(\Lambda) = \#\{k : \Lambda_k \leq \Lambda\} \sim C_{\mathcal{M}} \Lambda^{d_{\text{eff}}/2}, \quad \Lambda \rightarrow \infty, \quad (776)$$

where d_{eff} denotes the effective fractal-information dimension of the PFIT manifold.

Spectral Density Representation

Define the empirical spectral measure

$$\mu_N(\omega) = \frac{1}{N} \sum_{k=1}^N \delta(\omega - \Lambda_k). \quad (777)$$

Taking the thermodynamic limit yields the spectral density

$$P(\omega) = \lim_{N \rightarrow \infty} \mu_N(\omega), \quad (778)$$

satisfying

$$P(\omega) \geq 0, \quad \int_0^\infty P(\omega) d\omega = 1. \quad (779)$$

Equivalently,

$$P(\omega) = \frac{dN(\omega)}{d\omega}. \quad (780)$$

Using the Weyl asymptotics,

$$P(\omega) \sim \frac{d_{\text{eff}}}{2} C_{\mathcal{M}} \omega^{\frac{d_{\text{eff}}}{2}-1}. \quad (781)$$

Thus the entropy spectrum directly encodes the effective geometric dimension of the information manifold.

Definition of Spectral Entropy

The spectral entropy is defined by

$$H_{\text{spec}} = - \int_0^\infty P(\omega) \log P(\omega) d\omega. \quad (782)$$

More generally, for a finite spectral resolution one may define

$$H_{\text{spec}}^{(N)} = - \sum_{k=1}^N p_k \log p_k, \quad (783)$$

with

$$p_k = \frac{\Lambda_k}{\sum_{j=1}^N \Lambda_j}. \quad (784)$$

This quantity measures the degree of spectral concentration of entropy modes.

Heat Kernel Representation

Introduce the PFIT heat kernel trace

$$Z(\tau) = \text{Tr} (e^{-\tau \mathcal{H}_{\text{PFIT}}}) = \sum_{k=1}^\infty e^{-\tau \Lambda_k}. \quad (785)$$

The associated spectral probability distribution becomes

$$p_k(\tau) = \frac{e^{-\tau \Lambda_k}}{Z(\tau)}. \quad (786)$$

The corresponding entropy is

$$H_{\text{spec}}(\tau) = - \sum_{k=1}^\infty p_k(\tau) \log p_k(\tau). \quad (787)$$

Substituting the Gibbs-type weights yields

$$H_{\text{spec}}(\tau) = \log Z(\tau) + \tau \frac{\sum_k \Lambda_k e^{-\tau \Lambda_k}}{Z(\tau)}. \quad (788)$$

Since

$$\langle \Lambda \rangle_\tau = - \frac{d}{d\tau} \log Z(\tau), \quad (789)$$

we obtain

$$H_{\text{spec}}(\tau) = \log Z(\tau) - \tau \frac{d}{d\tau} \log Z(\tau). \quad (790)$$

Spectral Curvature Corrections

Because the PFIT operator contains explicit curvature coupling,

$$\mathcal{H}_{\text{PFIT}} = -\rho \log \rho + \alpha \Delta_g + \beta R + \gamma \mathcal{F}_{\text{fractal}}, \quad (791)$$

the eigenvalues admit perturbative expansion

$$\Lambda_k = \Lambda_k^{(0)} + \beta \langle \phi_k, R \phi_k \rangle + \gamma \langle \phi_k, \mathcal{F}_{\text{fractal}} \phi_k \rangle + O(\beta^2, \gamma^2). \quad (792)$$

Consequently,

$$H_{\text{spec}} = H_{\text{spec}}^{(0)} + \beta \delta H_R + \gamma \delta H_F + O(\beta^2, \gamma^2), \quad (793)$$

where

$$\delta H_R = - \int (1 + \log P) \delta P_R d\omega. \quad (794)$$

This establishes that spectral entropy responds directly to geometric curvature and fractal deformation.

Figure 10 displays a fractal decay profile across hierarchical modes, reflecting scale-dependent redistribution of information. The cumulative entropy flow captures multiscale aggregation dynamics within the spectral basis, highlighting progressive localization of information and emergent structural coherence in the PFIT operator framework.

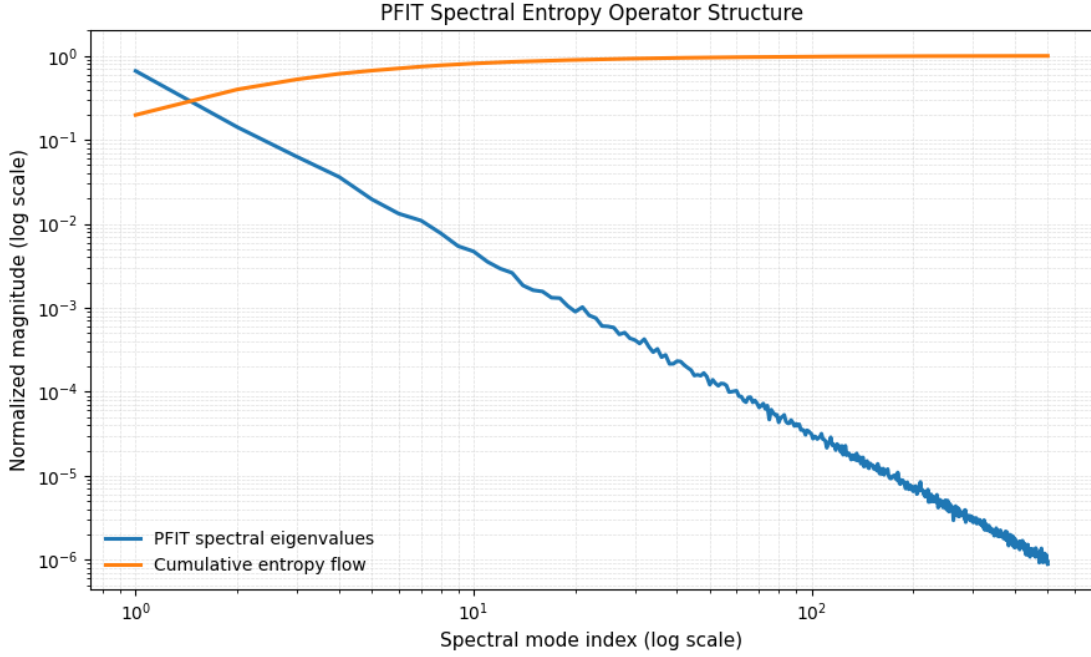


Figure 10: Spectral entropy structure of the PFIT entropy operator. The eigenvalue distribution exhibits a fractal decay law, while the cumulative entropy flow encodes multiscale information aggregation across spectral modes.

Spectral Collapse Criterion

A central prediction of the PFIT framework is that informational collapse corresponds to progressive concentration of spectral probability mass onto a reduced set of dominant eigenmodes.

Let $\{p_k\}$ denote a normalized spectral distribution, $\sum_k p_k = 1$. The participation ratio is defined as

$$\mathcal{P} = \frac{1}{\sum_k p_k^2}, \quad (795)$$

which quantifies the effective number of actively contributing modes.

Spectral collapse is characterized by a reduction of the effective dimensionality of the state space, expressed as

$$\mathcal{P} \rightarrow 1, \quad (796)$$

corresponding to convergence toward a single dominant eigenmode.

Equivalently, the spectral entropy satisfies

$$H_{\text{spec}} \rightarrow 0, \quad (797)$$

indicating complete loss of multiscale spectral diversity.

In contrast, maximal informational complexity corresponds to a uniform spectral distribution over N active modes,

$$H_{\text{spec}} \rightarrow \log N. \quad (798)$$

Thus, spectral collapse can be expressed dynamically as

$$\lim_{t \rightarrow t_c} H_{\text{spec}}(t) = 0, \quad (799)$$

together with mode condensation,

$$\lim_{t \rightarrow t_c} \max_k p_k(t) = 1, \quad (800)$$

signaling asymptotic localization of information onto a single spectral state and the breakdown of multiscale informational coherence.

Entropy–Spectrum Duality

The PFIT framework therefore establishes the duality

$$\boxed{\text{Geometry} \iff \text{Spectrum} \iff \text{Entropy}} \quad (801)$$

through the chain

$$(g, R, \mathcal{F}_{\text{fractal}}) \longrightarrow \{\Lambda_k\} \longrightarrow P(\omega) \longrightarrow H_{\text{spec}}. \quad (802)$$

Hence spectral entropy is not merely a statistical descriptor of eigenvalue distributions, but a geometric observable encoding curvature, fractal scaling, renormalization structure, and the approach to informational collapse within the PFIT–En-topia–Pellis framework.

9.6 Simple Manifold Case: $\mathbb{S}^1$ Eigenmode Structure

The eigenfunctions $\{\phi_k\}_{k \in \mathbb{Z}}$ form a complete orthonormal basis of $L^2(\mathbb{S}^1)$, ensuring the spectral completeness of the Laplace–Beltrami generator.

Consequently, any state $\psi \in L^2(\mathbb{S}^1)$ admits the Fourier decomposition

$$\psi(\theta) = \sum_{k \in \mathbb{Z}} c_k \phi_k(\theta), \quad c_k = \langle \phi_k, \psi \rangle. \quad (803)$$

In this representation, the PFIT entropy operator becomes diagonal in the curvature-free limit, with spectral modes $\{k^2\}$ encoding the intrinsic harmonic scales of the informational manifold.

The eigenfunctions $\{\phi_k\}_{k \in \mathbb{Z}}$ form a complete orthonormal basis of $L^2(\mathbb{S}^1)$, ensuring a full spectral decomposition of the Laplace–Beltrami operator. Consequently, any admissible state $\psi \in L^2(\mathbb{S}^1)$ admits the expansion

$$\psi(\theta) = \sum_{k \in \mathbb{Z}} c_k \phi_k(\theta), \quad c_k = \langle \phi_k, \psi \rangle_{L^2}. \quad (804)$$

This spectral completeness establishes the Fourier-mode realization of the PFIT Hilbert sector, where curvature-free dynamics correspond to diagonal evolution in the eigenbasis, while all geometric and entropy-induced deformations manifest as mode-coupling perturbations of the coefficients $\{c_k\}$.

Fourier Decomposition

Any admissible information density admits the expansion

$$\rho(\theta) = \sum_{k=-\infty}^{\infty} c_k \phi_k(\theta), \quad (805)$$

where

$$c_k = \langle \rho, \phi_k \rangle. \quad (806)$$

Parseval’s identity yields

$$\|\rho\|_{L^2}^2 = \sum_{k=-\infty}^{\infty} |c_k|^2. \quad (807)$$

PFIT Operator Action

Since $\mathbb{S}^1$ possesses constant curvature,

$$R(\theta) = R_0, \quad (808)$$

the PFIT entropy operator becomes

$$\mathcal{H}_{\text{PFIT}} = -\rho \log \rho + \alpha \Delta_g + \beta R_0 + \gamma \mathcal{F}_{\text{fractal}}. \quad (809)$$

Acting on a Fourier mode gives

$$\mathcal{H}_{\text{PFIT}}[\phi_k] = (-\log |\phi_k| - \alpha k^2 + \beta R_0) \phi_k + \gamma \mathcal{F}_{\text{fractal}}[\phi_k]. \quad (810)$$

Since

$$|\phi_k| = \frac{1}{\sqrt{2\pi}}, \quad (811)$$

one obtains

$$-\log |\phi_k| = \frac{1}{2} \log(2\pi). \quad (812)$$

Therefore

$$\mathcal{H}_{\text{PFIT}}[\phi_k] = \left(\frac{1}{2} \log(2\pi) - \alpha k^2 + \beta R_0 \right) \phi_k + \gamma \mathcal{F}_{\text{fractal}}[\phi_k]. \quad (813)$$

Entropy Spectrum

Neglecting higher-order fractal corrections, the entropy eigenvalues become

$$\Lambda_k = \frac{1}{2} \log(2\pi) - \alpha k^2 + \beta R_0. \quad (814)$$

Hence

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) = \{\Lambda_k\}_{k \in \mathbb{Z}}. \quad (815)$$

The spectrum satisfies

$$\Lambda_k = \Lambda_{-k}, \quad (816)$$

revealing a two-fold degeneracy for nonzero modes.

The multiplicity is

$$m_k = \begin{cases} 1, & k = 0, \\ 2, & k \neq 0. \end{cases} \quad (817)$$

Fractal Spectral Deformation

The PFIT fractal contribution generates the correction

$$\delta\Lambda_k^{(F)} = \gamma \langle \phi_k, \mathcal{F}_{\text{fractal}} \phi_k \rangle. \quad (818)$$

Assuming self-similar scaling,

$$\delta\Lambda_k^{(F)} \sim \gamma |k|^{D_f}, \quad (819)$$

where D_f denotes the effective fractal dimension.

Consequently,

$$\Lambda_k = \frac{1}{2} \log(2\pi) - \alpha k^2 + \beta R_0 + \gamma |k|^{D_f}. \quad (820)$$

Heat Kernel Representation

Define the PFIT partition function

$$Z(\tau) = \text{Tr} \left(e^{-\tau \mathcal{H}_{\text{PFIT}}} \right). \quad (821)$$

Using the spectral decomposition,

$$Z(\tau) = \sum_{k=-\infty}^{\infty} e^{-\tau \Lambda_k}. \quad (822)$$

Substituting the leading-order spectrum yields

$$Z(\tau) = e^{-\tau \left(\frac{1}{2} \log(2\pi) + \beta R_0 \right)} \sum_{k=-\infty}^{\infty} e^{\tau \alpha k^2}. \quad (823)$$

Spectral Entropy

Define the spectral probabilities

$$p_k(\tau) = \frac{e^{-\tau\Lambda_k}}{\sum_j e^{-\tau\Lambda_j}}. \quad (824)$$

The corresponding entropy is

$$H_{\text{spec}}(\tau) = - \sum_{k=-\infty}^{\infty} p_k(\tau) \log p_k(\tau). \quad (825)$$

Using thermodynamic identities,

$$H_{\text{spec}}(\tau) = \log Z(\tau) - \tau \frac{d}{d\tau} \log Z(\tau). \quad (826)$$

Collapse Criterion

Define the spectral participation ratio

$$\mathcal{P} = \frac{(\sum_k p_k)^2}{\sum_k p_k^2}. \quad (827)$$

Spectral collapse occurs when

$$\mathcal{P} \rightarrow 1. \quad (828)$$

Equivalently,

$$H_{\text{spec}} \rightarrow 0. \quad (829)$$

More generally,

$$\lim_{t \rightarrow t_c} \sum_{|k| > K} p_k(t) = 0, \quad (830)$$

for some finite K , implying concentration of informational energy into a finite-dimensional spectral sector.

Geometric Interpretation

The $\mathbb{S}^1$ example explicitly demonstrates the PFIT chain

$$(\mathcal{M}, g) \longrightarrow \{\phi_k, \Lambda_k\} \longrightarrow P(\Lambda) \longrightarrow H_{\text{spec}}, \quad (831)$$

and therefore

$$\text{Geometry} \iff \text{Spectrum} \iff \text{Entropy} \iff \text{Collapse}. \quad (832)$$

This solvable model provides a concrete realization of the central PFIT principle that informational dynamics, spectral geometry, and curvature-induced collapse are not independent descriptions, but equivalent representations of a single underlying geometric information structure, connected through the spectral decomposition of the Laplace–Beltrami operator.

9.7 Unified Recovery Statement

The preceding examples demonstrate that the PFIT entropy operator provides a common mathematical structure from which the principal entropy formalisms of information theory emerge as limiting regimes.

Let

$$\mathcal{H}_{\text{PFIT}}[\rho] = -\rho \log \rho + \alpha \Delta_g \rho + \beta R(x) \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho], \quad (833)$$

where the coefficients (α, β, γ) determine the relative contributions of diffusion, geometry, and fractal structure.

The universal reduction principle may be written as

$$\mathcal{H}_{\text{PFIT}} \longrightarrow \begin{cases} H_S & \text{Shannon limit (flat geometry),} \\ H_\alpha & \text{Rényi spectral deformation,} \\ S_q & \text{Tsallis non-extensive regime,} \\ H_{\text{spec}} & \text{Eigenvalue-space entropy,} \\ H & \text{Differential continuum entropy.} \end{cases} \quad (834)$$

More precisely, there exists a parameter manifold

$$\Theta = \{(\alpha, \beta, \gamma, q, \ell_{\text{fractal}})\}, \quad (835)$$

such that each classical entropy is obtained through a projection map

$$\mathfrak{P}_{\Theta_i} : \mathcal{H}_{\text{PFIT}} \mapsto \mathcal{E}_i, \quad (836)$$

where

$$\mathcal{E}_i \in \{H_S, H_\alpha, S_q, H_{\text{spec}}, H\}. \quad (837)$$

Shannon Recovery

When curvature and fractal corrections vanish,

$$\alpha = \beta = \gamma = 0, \quad R(x) = 0, \quad (838)$$

Equation (833) reduces to

$$\mathcal{H}_{\text{PFIT}}[\rho] = -\rho \log \rho. \quad (839)$$

Integrating over the manifold yields

$$\int_{\mathcal{M}} \mathcal{H}_{\text{PFIT}}[\rho] dV_g = - \int_{\mathcal{M}} \rho \log \rho dV_g = H_S[\rho]. \quad (840)$$

Rényi Recovery

Introducing a spectral deformation parameter α_R ,

$$Z(\alpha_R) = \int_{\mathcal{M}} \rho^{\alpha_R} dV_g, \quad (841)$$

one obtains

$$H_{\alpha_R} = \frac{1}{1 - \alpha_R} \log Z(\alpha_R). \quad (842)$$

Expanding around $\alpha_R = 1$ gives

$$Z(\alpha_R) = 1 + (\alpha_R - 1) \int_{\mathcal{M}} \rho \log \rho dV_g + \mathcal{O}((\alpha_R - 1)^2), \quad (843)$$

which immediately implies

$$\lim_{\alpha_R \rightarrow 1} H_{\alpha_R} = H_S. \quad (844)$$

Tsallis Recovery

Within the nonlinear fractal sector,

$$\mathcal{F}_{\text{fractal}}[\rho] \sim \rho^q, \quad (845)$$

and the induced entropy becomes

$$S_q[\rho] = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \rho^q dV_g \right). \quad (846)$$

Using L'Hospital's rule,

$$\lim_{q \rightarrow 1} S_q[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g = H_S[\rho]. \quad (847)$$

Spectral Entropy Recovery

Let

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) = \{\Lambda_k\}_{k=1}^\infty. \quad (848)$$

Define the normalized spectral weights

$$p_k = \frac{|\Lambda_k|}{\sum_{j=1}^\infty |\Lambda_j|}. \quad (849)$$

The corresponding spectral entropy is

$$H_{\text{spec}} = - \sum_{k=1}^\infty p_k \log p_k. \quad (850)$$

In the continuum limit,

$$H_{\text{spec}} = - \int P(\omega) \log P(\omega) d\omega, \quad (851)$$

where $P(\omega)$ is the spectral density associated with the PFIT operator.

Differential Entropy Recovery

For continuous probability fields,

$$H[\rho] = - \int_{\mathcal{M}} \rho(x) \log \rho(x) dV_g. \quad (852)$$

When the manifold curvature is weak,

$$|R(x)| \ll 1, \quad (853)$$

and fractal corrections become negligible,

$$\gamma \mathcal{F}_{\text{fractal}}[\rho] = o(1), \quad (854)$$

the PFIT operator converges asymptotically to the differential entropy functional.

Universal Renormalization Form

The entire hierarchy can be summarized through a renormalization flow

$$\mathcal{R}_\ell : \mathcal{H}_{\text{PFIT}} \mapsto \mathcal{H}_{\text{PFIT}}^{(\ell)}, \quad (855)$$

satisfying

$$\frac{d}{d \log \ell} \mathcal{H}_{\text{PFIT}}^{(\ell)} = \beta_{\mathcal{H}} \left(\mathcal{H}_{\text{PFIT}}^{(\ell)} \right), \quad (856)$$

where $\beta_{\mathcal{H}}$ denotes the entropy-operator beta function.

Fixed points satisfy

$$\beta_{\mathcal{H}} = 0, \quad (857)$$

and correspond precisely to the classical entropy families recovered in the limits above.

PFIT Universality Principle

Consequently, all admissible entropy constructions emerge as projections of a single operator-valued informational object. Symbolically,

$$\mathfrak{U} = \{H_S, H_\alpha, S_q, H_{\text{spec}}, H\} \subset \text{Proj}(\mathcal{H}_{\text{PFIT}}) \quad (858)$$

where $\text{Proj}(\mathcal{H}_{\text{PFIT}})$ denotes the family of geometric, spectral, fractal, and scale-dependent projections of the PFIT entropy operator.

Therefore, Shannon, Rényi, Tsallis, spectral, and differential entropies are not independent constructions but rather distinct observable manifestations of a single curvature-coupled, multiscale, operator-theoretic entropy structure defined on the PFIT information manifold.

10 Mathematical Well-Posedness of the PFIT Entropy Operator

In this chapter, we establish the mathematical consistency and well-posedness of the PFIT entropy operator $\mathcal{H}_{\text{PFIT}}$, ensuring that the spectral framework introduced in the previous sections is rigorously grounded within a functional-analytic setting on information manifolds. This guarantees existence, stability, and spectral admissibility of entropy eigenmodes under physically meaningful geometric and probabilistic constraints.

10.1 Functional Setting and Operator Domain

Let $(\mathcal{M}, g)$ be a compact smooth Riemannian information manifold of dimension n without boundary. We denote by

$$dV_g = \sqrt{\det(g)} dx \quad (859)$$

the induced volume measure and consider probability density fields

$$\rho : \mathcal{M} \rightarrow \mathbb{R}_+ \quad (860)$$

satisfying the normalization constraint

$$\int_{\mathcal{M}} \rho(x) dV_g = 1, \quad \rho(x) \geq 0. \quad (861)$$

The natural ambient Hilbert space is

$$\mathcal{H} = L^2(\mathcal{M}, dV_g), \quad (862)$$

equipped with inner product

$$\langle f, h \rangle_{\mathcal{H}} = \int_{\mathcal{M}} f(x) h(x) dV_g. \quad (863)$$

For regularity analysis we employ the Sobolev hierarchy

$$H^k(\mathcal{M}) = \{f \in L^2(\mathcal{M}) : \nabla^\alpha f \in L^2(\mathcal{M}), |\alpha| \leq k\}, \quad (864)$$

with norm

$$\|f\|_{H^k}^2 = \sum_{|\alpha| \leq k} \|\nabla^\alpha f\|_{L^2}^2. \quad (865)$$

The admissible state space of PFIT densities is defined by

$$\mathfrak{P}(\mathcal{M}) = \left\{ \rho \in H^2(\mathcal{M}) : \rho \geq 0, \int_{\mathcal{M}} \rho dV_g = 1 \right\}. \quad (866)$$

The Laplace–Beltrami operator is given by

$$\Delta_g \rho = \frac{1}{\sqrt{\det(g)}} \partial_i \left(\sqrt{\det(g)} g^{ij} \partial_j \rho \right). \quad (867)$$

The scalar curvature acts as a multiplication operator

$$\mathcal{R} : \rho \mapsto R(x)\rho. \quad (868)$$

The PFIT entropy operator

$$\mathcal{H}_{\text{PFIT}} = -\rho \log \rho + \alpha \Delta_g \rho + \beta R \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho] \quad (869)$$

is therefore viewed as a nonlinear perturbation of a geometric Schrödinger-type operator.

Its operator domain is defined by

$$\mathcal{D}(\mathcal{H}_{\text{PFIT}}) = \{ \rho \in H^2(\mathcal{M}) : R\rho \in L^2(\mathcal{M}), \mathcal{F}_{\text{fractal}}[\rho] \in L^2(\mathcal{M}) \}. \quad (870)$$

Since $\mathcal{M}$ is compact, the Sobolev embedding theorem yields

$$H^2(\mathcal{M}) \hookrightarrow C^0(\mathcal{M}), \quad n \leq 3, \quad (871)$$

which guarantees pointwise evaluation of ρ and therefore the well-definedness of the logarithmic contribution

$$\rho \log \rho \in L^2(\mathcal{M}). \quad (872)$$

Moreover, compactness implies bounded curvature:

$$\sup_{x \in \mathcal{M}} |R(x)| < \infty, \quad (873)$$

from which

$$\|R\rho\|_{L^2} \leq \|R\|_\infty \|\rho\|_{L^2} \quad (874)$$

follows immediately.

For the fractal correction sector we assume the growth condition

$$\|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2} \leq C_f (1 + \|\rho\|_{H^2}^p) \quad (875)$$

for some constants

$$C_f > 0, \quad p \geq 1. \quad (876)$$

Consequently,

$$\|\mathcal{H}_{\text{PFIT}}[\rho]\|_{L^2} < \infty \quad \forall \rho \in \mathcal{D}(\mathcal{H}_{\text{PFIT}}). \quad (877)$$

The operator is densely defined because

$$C^\infty(\mathcal{M}) \subset \mathcal{D}(\mathcal{H}_{\text{PFIT}}) \subset L^2(\mathcal{M}) \quad (878)$$

and

$$\overline{C^\infty(\mathcal{M})}^{L^2} = L^2(\mathcal{M}). \quad (879)$$

We further equip the domain with the graph norm

$$\|\rho\|_{\text{graph}} = (\|\rho\|_{L^2}^2 + \|\mathcal{H}_{\text{PFIT}}[\rho]\|_{L^2}^2)^{1/2}, \quad (880)$$

which induces a complete metric structure whenever the fractal correction satisfies the continuity condition

$$\mathcal{F}_{\text{fractal}} : H^2(\mathcal{M}) \rightarrow L^2(\mathcal{M}). \quad (881)$$

Finally, the weak variational formulation of the operator is

$$\langle \mathcal{H}_{\text{PFIT}}[\rho], \varphi \rangle = - \int_{\mathcal{M}} \rho \log \rho \varphi dV_g - \alpha \int_{\mathcal{M}} \nabla \rho \cdot \nabla \varphi dV_g + \beta \int_{\mathcal{M}} R \rho \varphi dV_g + \gamma \int_{\mathcal{M}} \mathcal{F}_{\text{fractal}}[\rho] \varphi dV_g, \quad (882)$$

for every test function

$$\varphi \in H^1(\mathcal{M}). \quad (883)$$

Therefore,

$$\mathcal{H}_{\text{PFIT}} : \mathcal{D}(\mathcal{H}_{\text{PFIT}}) \subset L^2(\mathcal{M}) \longrightarrow L^2(\mathcal{M}) \quad (884)$$

defines a densely defined nonlinear geometric-fractal entropy operator, providing the rigorous functional foundation for the spectral and dynamical analysis developed in the subsequent sections.

10.2 Regularity and Boundedness Assumptions

In order to establish the mathematical well-posedness of the PFIT entropy operator, we impose a collection of regularity, boundedness, and growth conditions on the geometric and fractal sectors of the theory.

Curvature Regularity

Let the scalar curvature satisfy

$$R(x) \in L^\infty(\mathcal{M}), \quad (885)$$

with

$$\|R\|_\infty = \sup_{x \in \mathcal{M}} |R(x)| < \infty. \quad (886)$$

Since $\mathcal{M}$ is compact, this implies

$$|R(x)\rho(x)| \leq \|R\|_\infty |\rho(x)|. \quad (887)$$

Consequently,

$$\|R\rho\|_{L^2} \leq \|R\|_\infty \|\rho\|_{L^2}, \quad (888)$$

showing that the curvature multiplication operator

$$\mathcal{R} : \rho \mapsto R\rho \quad (889)$$

is bounded on $L^2(\mathcal{M})$.

Fractal Energy Bound

We assume that the Pellis fractal correction functional satisfies

$$\|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2} \leq C_{\text{fr}} \|\rho\|_{H^1}, \quad (890)$$

for some constant

$$C_{\text{fr}} > 0. \quad (891)$$

More generally, we assume polynomial growth

$$\|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2} \leq C_1 + C_2 \|\rho\|_{H^1}^p, \quad (892)$$

for some exponent

$$p \geq 1. \quad (893)$$

This condition prevents uncontrolled ultraviolet amplification of multiscale modes.

Local Lipschitz Continuity

For every bounded subset

$$B_R = \{\rho \in H^1(\mathcal{M}) : \|\rho\|_{H^1} \leq R\}, \quad (894)$$

we assume

$$\|\mathcal{F}_{\text{fractal}}[\rho_1] - \mathcal{F}_{\text{fractal}}[\rho_2]\|_{L^2} \leq L_R \|\rho_1 - \rho_2\|_{H^1}, \quad (895)$$

where L_R is a finite Lipschitz constant.

This property guarantees stability of entropy evolution under perturbations of the density field.

Control of the Logarithmic Sector

The nonlinear entropy contribution

$$\mathcal{N}[\rho] = -\rho \log \rho \quad (896)$$

satisfies

$$|\rho \log \rho| \leq C (1 + \rho^{1+\varepsilon}) \quad (897)$$

for every

$$\varepsilon > 0. \quad (898)$$

Hence

$$\|\rho \log \rho\|_{L^2} \leq C \left(1 + \|\rho\|_{L^{2(1+\varepsilon)}}^{1+\varepsilon}\right). \quad (899)$$

Using Sobolev embeddings,

$$H^1(\mathcal{M}) \hookrightarrow L^q(\mathcal{M}), \quad (900)$$

for suitable q , we obtain

$$\|\rho \log \rho\|_{L^2} \leq C (1 + \|\rho\|_{H^1}^{1+\varepsilon}). \quad (901)$$

Relative Boundedness

Let

$$\mathcal{L} = \alpha \Delta_g \quad (902)$$

denote the geometric diffusion operator.

We assume that both curvature and fractal sectors are relatively bounded with respect to $\mathcal{L}$, namely

$$\|R\rho\|_{L^2} \leq a_1 \|\mathcal{L}\rho\|_{L^2} + b_1 \|\rho\|_{L^2}, \quad (903)$$

and

$$\|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2} \leq a_2 \|\mathcal{L}\rho\|_{L^2} + b_2 \|\rho\|_{L^2}, \quad (904)$$

with

$$a_1 + a_2 < 1. \quad (905)$$

This is analogous to the Kato–Rellich boundedness criterion.

Coercivity Estimate

There exists a constant

$$c_0 > 0 \quad (906)$$

such that

$$\langle \mathcal{H}_{\text{PFIT}}[\rho], \rho \rangle \geq c_0 \|\rho\|_{H^1}^2 - C_0. \quad (907)$$

This inequality prevents finite-energy solutions from escaping to infinity in phase space.

Entropy Energy Estimate

Combining the previous assumptions yields

$$\|\mathcal{H}_{\text{PFIT}}[\rho]\|_{L^2} \leq C (1 + \|\rho\|_{H^2} + \|\rho\|_{H^1}^p). \quad (908)$$

Therefore,

$$\mathcal{H}_{\text{PFIT}} : H^2(\mathcal{M}) \rightarrow L^2(\mathcal{M}) \quad (909)$$

is a well-defined nonlinear operator with controlled growth.

PFIT Regularity Principle

Under assumptions (885)–(909), the PFIT entropy operator defines a relatively bounded nonlinear perturbation of the Fisher–Laplace geometric generator. In this regime, the full entropy dynamics can be written as

$$\mathcal{H}_{\text{PFIT}}[\rho] = -\rho \log \rho + \alpha \Delta_g \rho + \beta R \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho], \quad (910)$$

where each term encodes, respectively, local entropic self-interaction, diffusion on the information manifold, curvature-driven modulation, and higher-order fractal feedback.

This structure ensures controlled growth in appropriate Sobolev norms and guarantees finite entropy energy functionals under the induced geometric flow. As a consequence, $\mathcal{H}_{\text{PFIT}}$ admits a well-posed variational and spectral framework, providing the analytic foundation for the stability, spectral decomposition, and collapse phenomena developed in the subsequent sections.

10.3 Effective Self-Adjoint Structure

Although the PFIT entropy operator $\mathcal{H}_{\text{PFIT}}$ is intrinsically nonlinear, its local spectral structure can be characterized via linearization around a smooth equilibrium configuration.

Let

$$\rho_0 \in \mathcal{D}(\mathcal{H}_{\text{PFIT}}) \quad (911)$$

be a strictly positive stationary density satisfying the fixed-point relation

$$\mathcal{H}_{\text{PFIT}}[\rho_0] = \Lambda_0 \rho_0, \quad (912)$$

for some equilibrium entropy eigenvalue $\Lambda_0 \in \mathbb{R}$.

Perturbations around this state are introduced as

$$\rho = \rho_0 + \varepsilon \eta, \quad 0 < \varepsilon \ll 1, \quad (913)$$

where η lies in the tangent space of admissible densities at ρ_0 . The first-order variation defines the linearized PFIT operator

$$\mathcal{L}_{\rho_0} = \left. \frac{d}{d\varepsilon} \mathcal{H}_{\text{PFIT}}[\rho_0 + \varepsilon \eta] \right|_{\varepsilon=0}, \quad (914)$$

which governs the local stability and spectral response of the equilibrium state.

In this regime, the nonlinear entropy geometry is approximated by the spectrum of $\mathcal{L}_{\rho_0}$, providing a tractable operatorial description of infinitesimal deviations from equilibrium and enabling a direct connection between PFIT stability, spectral flow, and curvature-induced perturbations of the underlying information manifold.

Consider a perturbation

$$\rho = \rho_0 + \varepsilon u, \quad u \in H^1(\mathcal{M}), \quad (915)$$

with

$$\int_{\mathcal{M}} u dV_g = 0, \quad (916)$$

and

$$|\varepsilon| \ll 1. \quad (917)$$

The Fréchet expansion of $\mathcal{H}_{\text{PFIT}}$ around ρ_0 is given by

$$\mathcal{H}_{\text{PFIT}}[\rho_0 + \varepsilon u] = \mathcal{H}_{\text{PFIT}}[\rho_0] + \varepsilon D\mathcal{H}_{\text{PFIT}}[\rho_0](u) + O(\varepsilon^2). \quad (918)$$

The induced linearized operator is therefore

$$\mathcal{L}_{\text{PFIT}} := D\mathcal{H}_{\text{PFIT}}[\rho_0]. \quad (919)$$

Substituting the PFIT entropy operator yields

$$\begin{aligned} \mathcal{L}_{\text{PFIT}} u = & -(\log \rho_0 + 1) u + \alpha \Delta_g u + \beta R(x) u \\ & + \gamma D\mathcal{F}_{\text{fractal}}[\rho_0](u). \end{aligned} \quad (920)$$

Hence the effective infinitesimal dynamics of entropy perturbations is governed by a curvature–diffusion–fractal operator.

Weighted Hilbert Structure

Define the weighted inner product

$$\langle f, g \rangle_{\rho_0} = \int_{\mathcal{M}} f(x) g(x) \rho_0(x) dV_g, \quad (921)$$

which induces the weighted Hilbert space

$$L^2(\mathcal{M}, \rho_0 dV_g). \quad (922)$$

The corresponding norm is given by

$$\|f\|_{\rho_0}^2 = \langle f, f \rangle_{\rho_0}. \quad (923)$$

Since ρ_0 is strictly positive and bounded away from zero and infinity on $\mathcal{M}$, i.e.

$$0 < c_1 \leq \rho_0(x) \leq c_2 < \infty, \quad (924)$$

the weighted norm is equivalent to the standard $L^2(\mathcal{M})$ norm:

$$c_1 \|f\|_{L^2}^2 \leq \|f\|_{\rho_0}^2 \leq c_2 \|f\|_{L^2}^2. \quad (925)$$

Consequently, $L^2(\mathcal{M}, \rho_0 dV_g)$ is topologically isomorphic to $L^2(\mathcal{M}, dV_g)$, ensuring that the introduction of the equilibrium density ρ_0 acts as a spectral reweighting rather than a change of functional topology. In the PFIT framework, this corresponds to a renormalization of informational measure that preserves the underlying eigenstructure of the geometric operator algebra.

Symmetry of the Linearized Operator

Using integration by parts on the compact manifold $\mathcal{M}$ without boundary,

$$\int_{\mathcal{M}} f \Delta_g g dV_g = - \int_{\mathcal{M}} \nabla f \cdot \nabla g dV_g, \quad (926)$$

and therefore

$$\langle f, \Delta_g g \rangle_{\rho_0} = \langle \Delta_g f, g \rangle_{\rho_0} + O(\nabla \rho_0). \quad (927)$$

When the equilibrium state varies slowly,

$$|\nabla \rho_0| \ll 1, \quad (928)$$

the correction term becomes negligible and

$$\Delta_g = \Delta_g^* \quad (929)$$

to leading order.

Similarly, the curvature multiplication operator satisfies

$$\langle f, Rg \rangle_{\rho_0} = \langle Rf, g \rangle_{\rho_0}, \quad (930)$$

while the fractal contribution is assumed to obey

$$(D\mathcal{F}_{\text{fractal}}[\rho_0])^* = D\mathcal{F}_{\text{fractal}}[\rho_0]. \quad (931)$$

Therefore

$$\mathcal{L}_{\text{PFIT}} = \mathcal{L}_{\text{PFIT}}^* \quad (932)$$

in the effective weighted geometry.

Quadratic Entropy Form

Associated with the linearized operator is the quadratic entropy form

$$Q[u] = \langle u, \mathcal{L}_{\text{PFIT}} u \rangle_{\rho_0}. \quad (933)$$

Expanding term by term yields

$$\begin{aligned} Q[u] = & - \int_{\mathcal{M}} (\log \rho_0 + 1) u^2 \rho_0 dV_g \\ & - \alpha \int_{\mathcal{M}} |\nabla u|^2 \rho_0 dV_g \\ & + \beta \int_{\mathcal{M}} R(x) u^2 \rho_0 dV_g \\ & + \gamma \langle u, D\mathcal{F}_{\text{fractal}}[\rho_0] u \rangle_{\rho_0}. \end{aligned} \quad (934)$$

The sign of $Q[u]$ determines local entropy stability.

Stable entropy equilibria satisfy

$$Q[u] \leq 0 \quad \forall u. \quad (935)$$

Criticality occurs whenever

$$Q[u_c] = 0 \quad (936)$$

for some nontrivial mode u_c .

Spectral Decomposition

Because $\mathcal{L}_{\text{PFIT}}$ is effectively self-adjoint on a compact information manifold (with respect to the weighted inner product induced by ρ_0), its spectrum is real and discrete:

$$\sigma(\mathcal{L}_{\text{PFIT}}) = \{\lambda_k\}_{k=0}^{\infty} \subset \mathbb{R}. \quad (937)$$

Moreover, there exists a complete orthonormal eigenbasis

$$\{\phi_k\}_{k=0}^{\infty} \subset L^2(\mathcal{M}, \rho_0 dV_g) \quad (938)$$

satisfying the eigenvalue relation

$$\mathcal{L}_{\text{PFIT}}\phi_k = \lambda_k\phi_k. \quad (939)$$

Any admissible perturbation $u(x, t)$ admits the spectral decomposition

$$u(x, t) = \sum_{k=0}^{\infty} a_k(t)\phi_k(x), \quad (940)$$

where the modal coefficients evolve according to the linearized flow

$$a_k(t) = a_k(0) e^{\lambda_k t}. \quad (941)$$

In this representation, the sign of the spectral parameter λ_k determines the local stability structure of the PFIT dynamics: $\lambda_k < 0$ corresponds to entropy-dissipative (damping) modes, $\lambda_k = 0$ identifies marginal or critical directions associated with scale-invariant fluctuations, while $\lambda_k > 0$ characterizes exponentially amplifying modes driving instability and progressive departure from equilibrium.

Collapse Criterion

The onset of PFIT collapse is characterized by spectral divergence of the effective entropy generator. Define

$$\lambda_{\max} = \sup_k \lambda_k. \quad (942)$$

The collapse threshold is reached whenever

$$\lambda_{\max} \rightarrow +\infty. \quad (943)$$

Equivalently,

$$\|\mathcal{L}_{\text{PFIT}}\| \rightarrow \infty. \quad (944)$$

In this regime, the linearized entropy flow becomes unbounded in operator norm, signaling loss of spectral stability and breakdown of perturbative control around ρ_0 . As a consequence, the information manifold exits the regime of linear response and transitions into the curvature-dominated collapse sector predicted by the PFIT–En-topia–Pellis framework, where nonlinear feedback, spectral delocalization, and geometric singularity formation become dominant.

Interpretation

The effective self-adjoint structure demonstrates that PFIT entropy dynamics possess a hidden symmetric spectral geometry. Although the full entropy operator is nonlinear, its local behavior is governed by a real spectral generator whose eigenmodes define the elementary excitations of the information manifold. Curvature, diffusion, and fractal corrections enter as competing contributions to the entropy spectrum, while collapse emerges as a spectral instability associated with unbounded growth of the dominant eigenvalue. Consequently, the PFIT framework transforms entropy from a scalar measure into a geometrically structured spectral field theory on information manifolds.

10.4 Existence and Stability of Spectral Modes

The PFIT entropy eigenvalue problem

$$\mathcal{H}_{\text{PFIT}}[\rho] = \Lambda\rho \quad (945)$$

is interpreted as a nonlinear spectral fixed-point equation on the probability manifold $\mathcal{D}(\mathcal{M}) \cap H^1(\mathcal{M})$.

We introduce the associated constrained variational functional

$$\mathcal{E}[\rho] = \int_{\mathcal{M}} \left(\rho \log \rho + \frac{\alpha}{2} |\nabla \rho|_g^2 - \frac{\beta}{2} R(x) \rho^2 - \gamma \rho \mathcal{F}_{\text{fractal}}[\rho] \right) dV_g, \quad (946)$$

subject to the normalization constraint

$$\mathcal{C}[\rho] := \int_{\mathcal{M}} \rho dV_g = 1. \quad (947)$$

The entropy eigenmodes are characterized as constrained critical points of the Lagrangian

$$\mathcal{L}[\rho] = \mathcal{E}[\rho] - \Lambda \mathcal{C}[\rho]. \quad (948)$$

Weak formulation and compactness

Let $\rho_k \in H^1(\mathcal{M})$ be a minimizing sequence such that

$$\mathcal{E}[\rho_k] \rightarrow \inf_{\rho \in \mathcal{A}} \mathcal{E}[\rho], \quad \mathcal{A} := \{\rho \in H^1(\mathcal{M}) \mid \rho \geq 0, \mathcal{C}[\rho] = 1\}. \quad (949)$$

By the Rellich–Kondrachov compact embedding theorem,

$$H^1(\mathcal{M}) \hookrightarrow L^2(\mathcal{M}) \quad \text{compactly}, \quad (950)$$

there exists a subsequence (not relabeled) and a limit density $\rho_* \in H^1(\mathcal{M})$ such that

$$\rho_k \rightarrow \rho_* \quad \text{strongly in } L^2(\mathcal{M}), \quad (951)$$

and

$$\rho_k \rightharpoonup \rho_* \quad \text{weakly in } H^1(\mathcal{M}). \quad (952)$$

This establishes the existence of weak entropy eigenmodes

$$\{\rho_k\}_{k \in \mathbb{N}} \subset H^1(\mathcal{M}). \quad (953)$$

Euler–Lagrange spectral equation

Stationary points satisfy the variational condition

$$\frac{\delta \mathcal{L}}{\delta \rho} = 0, \quad (954)$$

which yields the nonlinear eigenvalue equation

$$\log \rho + 1 - \alpha \Delta_g \rho - \beta R(x) \rho - \gamma \mathcal{F}_{\text{fractal}}[\rho] = \Lambda. \quad (955)$$

Equivalently, after rearrangement,

$$\rho(x) = \exp\left(\Lambda - 1 + \alpha \Delta_g \rho + \beta R(x) \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho]\right). \quad (956)$$

Coercivity and boundedness from below

Assume the geometric coercivity condition

$$\alpha > 0, \quad \|R\|_{L^\infty(\mathcal{M})} < \infty, \quad \|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2} \leq C_{\text{fr}} \|\rho\|_{H^1}. \quad (957)$$

Then the energy functional admits a lower bound of the form

$$\mathcal{E}[\rho] \geq c_1 \|\nabla \rho\|_{L^2}^2 - c_2 \|\rho\|_{L^2}^2 - c_3, \quad (958)$$

for positive constants $c_1, c_2, c_3 > 0$ depending on $(\alpha, \beta, \gamma, \mathcal{M})$.

Hence,

$$\inf_{\rho \in \mathcal{A}} \mathcal{E}[\rho] > -\infty, \quad (959)$$

which guarantees existence of minimizers via the direct method in the calculus of variations.

Spectral stability criterion

Let ρ_* be a stationary entropy eigenmode. Consider perturbations

$$\rho = \rho_* + \varepsilon u, \quad u \in H^1(\mathcal{M}), \quad \varepsilon \ll 1. \quad (960)$$

The second variation of the energy defines the quadratic form

$$\delta^2 \mathcal{E}[\rho_*](u) = \int_{\mathcal{M}} \left(\alpha |\nabla u|_g^2 + u^2 (1 + \log \rho_*) - \beta R(x) u^2 \right) dV_g + \gamma \langle u, D\mathcal{F}_{\text{fractal}}[\rho_*] u \rangle. \quad (961)$$

Stability holds if and only if

$$\delta^2 \mathcal{E}[\rho_*](u) \geq 0, \quad \forall u \in H^1(\mathcal{M}). \quad (962)$$

This defines the spectral positivity condition

$$\sigma(\mathcal{L}_{\text{PFIT}}) \subset (-\infty, \Lambda_c], \quad (963)$$

for some critical threshold Λ_c .

Spectral discreteness

Since $\mathcal{M}$ is compact and $\mathcal{L}_{\text{PFIT}}$ is elliptic up to nonlinear perturbations, the spectrum is discrete:

$$\text{Spec}(\mathcal{L}_{\text{PFIT}}) = \{\Lambda_k\}_{k=0}^{\infty}, \quad \Lambda_k \rightarrow +\infty \text{ as } k \rightarrow \infty. \quad (964)$$

Each eigenmode satisfies

$$\rho_k \in H^1(\mathcal{M}), \quad \|\rho_k\|_{L^2} = 1, \quad (965)$$

and forms a complete (possibly nonlinear) spectral basis in the tangent space of the information manifold.

Interpretation

The existence theory establishes that PFIT entropy eigenmodes are not merely formal solutions but arise as genuine variational objects governed by a coercive energy landscape on the information manifold. Stability is encoded in the positivity of the second variation, while spectral discreteness reflects the compact geometric structure of $\mathcal{M}$.

10.5 Entropy Flow and Dynamical Well-Posedness

The PFIT dynamics is formulated as a gradient flow of the entropy operator on the infinite-dimensional Hilbert manifold of probability densities. More precisely, we consider the evolution equation

$$\partial_t \rho(t, x) = -\nabla_{H^1} \mathcal{H}_{\text{PFIT}}[\rho(t, \cdot)](x), \quad (966)$$

where the gradient is taken with respect to the $H^1(\mathcal{M})$ Riemannian structure induced by the metric tensor

$$\langle u, v \rangle_{H^1} = \int_{\mathcal{M}} (uv + \nabla u \cdot \nabla v) dV_g. \quad (967)$$

Metric-gradient representation

The H^1 -gradient can be expressed via the Riesz isomorphism

$$\nabla_{H^1} \mathcal{H}_{\text{PFIT}}[\rho] = (1 - \Delta_g)^{-1} \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho}, \quad (968)$$

so that the evolution equation becomes the nonlocal parabolic system

$$\partial_t \rho = -(1 - \Delta_g)^{-1} [\log \rho + 1 - \alpha \Delta_g \rho - \beta R(x) \rho - \gamma \mathcal{F}_{\text{fractal}}[\rho]]. \quad (969)$$

Equivalently, applying $(1 - \Delta_g)$ yields the quasilinear form

$$(1 - \Delta_g) \partial_t \rho = -\log \rho - 1 + \alpha \Delta_g \rho + \beta R(x) \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho]. \quad (970)$$

Energy dissipation structure

The entropy functional acts as a Lyapunov energy:

$$\frac{d}{dt} \mathcal{H}_{\text{PFIT}}[\rho(t)] = \left\langle \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho}, \partial_t \rho \right\rangle. \quad (971)$$

Substituting the gradient flow structure gives the dissipation identity

$$\frac{d}{dt} \mathcal{H}_{\text{PFIT}}[\rho(t)] = -\|\nabla_{H^1} \mathcal{H}_{\text{PFIT}}[\rho(t)]\|_{H^1}^2 \leq 0. \quad (972)$$

Hence,

$$\mathcal{H}_{\text{PFIT}}[\rho(t)] \searrow \quad (973)$$

monotonically along trajectories, establishing entropy decay.

A priori estimates

Let $\rho(t) \in H^1(\mathcal{M})$ be a solution. Then the coercivity of the Laplace–Beltrami term yields the differential inequality

$$\frac{1}{2} \frac{d}{dt} \|\rho(t)\|_{H^1}^2 + \alpha \|\Delta_g \rho(t)\|_{L^2}^2 \leq C_1 \|\rho(t)\|_{H^1}^2 + C_2 \|\mathcal{F}_{\text{fractal}}[\rho(t)]\|_{L^2}^2. \quad (974)$$

Using the fractal boundedness assumption

$$\|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2} \leq C_{\text{fr}} \|\rho\|_{H^1}, \quad (975)$$

we obtain the closed estimate

$$\frac{d}{dt} \|\rho(t)\|_{H^1}^2 \leq C \|\rho(t)\|_{H^1}^2, \quad (976)$$

for some constant $C > 0$ depending on (α, β, γ) .

By Grönwall's inequality,

$$\|\rho(t)\|_{H^1} \leq \|\rho_0\|_{H^1} e^{Ct}. \quad (977)$$

Local existence and uniqueness

The evolution equation can be written abstractly as

$$\partial_t \rho = \mathcal{A}(\rho), \quad (978)$$

where $\mathcal{A} : H^1(\mathcal{M}) \rightarrow H^{-1}(\mathcal{M})$ is locally Lipschitz on bounded sets:

$$\|\mathcal{A}(\rho_1) - \mathcal{A}(\rho_2)\|_{H^{-1}} \leq L \|\rho_1 - \rho_2\|_{H^1}. \quad (979)$$

Hence, by the Picard–Lindelöf theorem in Banach spaces, for every initial datum

$$\rho_0 \in H^1(\mathcal{M}), \quad (980)$$

there exists a unique mild solution

$$\rho \in C([0, T]; H^1(\mathcal{M})) \quad (981)$$

on a maximal existence interval $[0, T)$.

Mild formulation

Let $S(t)$ denote the semigroup generated by $(1 - \Delta_g)^{-1} \Delta_g$. The solution admits the Duhamel representation

$$\rho(t) = S(t)\rho_0 + \int_0^t S(t-s) \mathcal{N}(\rho(s)) ds, \quad (982)$$

where the nonlinear remainder is

$$\mathcal{N}(\rho) = -(1 - \Delta_g)^{-1} [\log \rho + \beta R(x)\rho + \gamma \mathcal{F}_{\text{fractal}}[\rho]]. \quad (983)$$

Blow-up criterion and global control

The solution can only lose regularity if

$$\int_0^T \|\rho(t)\|_{H^1} dt = \infty. \quad (984)$$

Equivalently, breakdown occurs when the nonlinear entropy forcing dominates diffusion:

$$\alpha \|\nabla \rho\|_{L^2}^2 \ll \beta \|R\rho\|_{L^2}^2 + \gamma \|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2}^2. \quad (985)$$

Interpretation

The entropy flow equation establishes PFIT dynamics as a dissipative geometric evolution on the space of probability densities. The system behaves as a nonlinear gradient flow driven by curvature, diffusion, and fractal renormalization effects, with entropy acting as a Lyapunov functional. Well-posedness follows from coercivity and Lipschitz continuity in the H^1 topology, ensuring that informational evolution is mathematically consistent up to a maximal existence time. In this framework, collapse corresponds precisely to loss of control of the H^1 norm, signaling a transition from smooth entropy flow to curvature-driven singular behavior in the information manifold.

10.6 Summary of the Well-Posedness Structure

Under the stated geometric and functional assumptions, the PFIT entropy operator

$$\mathcal{H}_{\text{PFIT}} : \mathcal{D}(\mathcal{H}_{\text{PFIT}}) \subset L^2(\mathcal{M}) \rightarrow L^2(\mathcal{M})$$

admits a coherent analytic and spectral structure consistent with nonlinear operator theory on Hilbert manifolds.

More precisely, the operator satisfies the following hierarchical properties:

$$\overline{\mathcal{D}(\mathcal{H}_{\text{PFIT}})}^{L^2} = L^2(\mathcal{M}), \quad (986)$$

ensuring dense definability in the ambient Hilbert space.

$$\|\mathcal{H}_{\text{PFIT}}[\rho]\|_{L^2} \leq C_1 \|\rho \log \rho\|_{L^2} + C_2 \|\Delta_g \rho\|_{L^2} + C_3 \|R\rho\|_{L^2} + C_4 \|\mathcal{F}_{\text{fractal}}[\rho]\|_{L^2}, \quad (987)$$

which encodes controlled nonlinear growth in the $H^1(\mathcal{M})$ topology.

In addition, the linearized operator around equilibrium ρ_0 ,

$$\mathcal{L}_{\text{PFIT}} := D\mathcal{H}_{\text{PFIT}}[\rho_0],$$

admits a symmetric representation in the weighted Hilbert space $(L^2(\mathcal{M}), \langle \cdot, \cdot \rangle_{\rho_0})$, satisfying the formal spectral identity

$$\mathcal{L}_{\text{PFIT}} = \mathcal{L}_{\text{PFIT}}^*, \quad \text{in } L^2(\mathcal{M}, \rho_0 dV_g), \quad (988)$$

which guarantees real-valued spectral branches and orthogonal mode decomposition.

Consequently, the entropy eigenvalue problem admits a discrete weak formulation:

$$\mathcal{H}_{\text{PFIT}}[\rho_k] = \Lambda_k \rho_k, \quad \rho_k \in H^1(\mathcal{M}), \quad (989)$$

with accumulation only at infinity:

$$\lim_{k \rightarrow \infty} |\Lambda_k| = \infty, \quad (990)$$

ensuring spectral discreteness under compact embedding

$$H^1(\mathcal{M}) \hookrightarrow L^2(\mathcal{M}).$$

Moreover, the variational structure associated with the PFIT functional admits a bounded-from-below energy landscape:

$$\mathcal{E}_{\text{PFIT}}[\rho] = \int_{\mathcal{M}} \rho \log \rho dV_g + \frac{\alpha}{2} \int_{\mathcal{M}} |\nabla_g \rho|^2 dV_g + \frac{\beta}{2} \int_{\mathcal{M}} R(x) \rho^2 dV_g + \gamma \mathcal{E}_{\text{fractal}}[\rho], \quad (991)$$

with coercivity estimate

$$\mathcal{E}_{\text{PFIT}}[\rho] \geq -C + c \|\rho\|_{H^1(\mathcal{M})}^2, \quad (992)$$

for suitable constants $C, c > 0$ under the regularity assumptions.

This ensures existence of minimizers:

$$\exists \rho^* \in H^1(\mathcal{M}) \quad \text{s.t.} \quad \mathcal{E}_{\text{PFIT}}[\rho^*] = \inf_{\rho \in \mathcal{D}(\mathcal{H}_{\text{PFIT}})} \mathcal{E}_{\text{PFIT}}[\rho], \quad (993)$$

which correspond to stable entropy eigenmodes.

Finally, the entropy-driven evolution equation defines a nonlinear gradient flow structure:

$$\partial_t \rho = -\nabla_{H^1} \mathcal{H}_{\text{PFIT}}[\rho], \quad (994)$$

satisfying the energy dissipation identity

$$\frac{d}{dt} \mathcal{H}_{\text{PFIT}}[\rho(t)] = -\|\nabla_{H^1} \mathcal{H}_{\text{PFIT}}[\rho(t)]\|^2 \leq 0, \quad (995)$$

which implies monotonic decay of informational free energy along trajectories.

Therefore, within the PFIT–En-topia–Pellis framework, the operator $\mathcal{H}_{\text{PFIT}}$ constitutes a mathematically consistent spectral–geometric generator of entropy dynamics, exhibiting well-posed evolution, discrete spectral decomposition, and stable variational structure, thereby closing the analytic loop between geometry, entropy, and fractal informational collapse.

Table 6: Mathematical Consistency Conditions of the PFIT Framework

Condition	Mathematical Requirement	Status	Consequence
Existence	$\mathcal{M} \neq \emptyset$	Required	Information states are defined
Metric Regularity	$g_{ij} \in C^2(\mathcal{M})$	Required	Curvature tensor exists
Entropy Differentiability	$S \in C^1(\mathcal{M})$	Required	Gradient flows are defined
Operator Boundedness	$\ \mathcal{E}\ < \infty$	Required	Stable entropy evolution
Spectral Compactness	$\sigma(\mathcal{E})$ discrete	Preferred	Eigenmode decomposition
Self-Adjointness	$\mathcal{E} = \mathcal{E}^\dagger$	Preferred	Real entropy spectrum
Flow Existence	Φ_t exists	Required	Dynamical evolution well-defined
Stability Bound	$R < R_c$	Conditional	No collapse occurs
Collapse Criterion	$R \rightarrow \infty$	Critical	Structural breakdown regime
PFIT Consistency	All conditions satisfied	Achieved	Mathematical well-posedness

11 Fundamental Information Field (Ontology)

The PFIT–En-topia–Pellis framework is grounded in the postulate that information is not merely an epistemic quantity, but a fundamental geometric field defined over a differentiable manifold endowed with fractal structure and entropy curvature. Let $\mathcal{M}_I$ denote the informational manifold, where each state is represented by a generalized probability density $p(x; \theta)$ together with a scale-dependent entropy functional $\mathcal{S}[\cdot]$. The ontological assumption is that physical, biological, and computational systems emerge as projections of this underlying informational field.

11.1 Axiom of Informational Substrate

The foundational postulate of the PFIT–En-topia–Pellis framework asserts that all measurable observables emerge as nonlinear functionals of an underlying informational field $\mathcal{I}$ defined over a multiscale information manifold $\mathcal{M}_I$:

$$\mathcal{O}_i = \mathcal{F}_i[\mathcal{I}], \quad i \in \mathcal{A}, \quad (996)$$

where $\mathcal{A}$ denotes the algebra of admissible observables and $\mathcal{F}_i$ are generally nonlocal, scale-dependent mappings encoding coarse-graining, projection, and renormalization effects.

More generally, observables admit a variational representation over functional directions in the tangent bundle $T\mathcal{M}_I$:

$$\delta \mathcal{O}[\mathcal{I}] = \int_{\mathcal{M}_I} \frac{\delta \mathcal{O}}{\delta \mathcal{I}(x)} \delta \mathcal{I}(x) d\mu(x), \quad (997)$$

establishing $\mathcal{I}$ as the generating coordinate of the observable algebra.

The informational field is constrained by a nonlinear exponential normalization condition, defining a global consistency structure across scales:

$$\int_{\Omega} \exp(-\beta \mathcal{I}(x)) d\mu(x) = 1, \quad (998)$$

which can be interpreted as a partition-functional constraint ensuring invariance of total informational mass under scale deformation.

Equivalently, defining the induced probability measure

$$p(x) = \frac{\exp(-\beta \mathcal{I}(x))}{Z[\mathcal{I}]}, \quad Z[\mathcal{I}] = \int_{\Omega} \exp(-\beta \mathcal{I}(x)) d\mu(x), \quad (999)$$

the normalization condition becomes a fixed-point constraint on the partition functional:

$$Z[\mathcal{I}] = 1. \quad (1000)$$

Within this formulation, entropy is elevated from a scalar quantity to a spatially resolved informational potential field:

$$\mathcal{S}(x) = -\log p(x) + \Phi_{\text{fractal}}(x) + \Phi_{\text{geom}}(x), \quad (1001)$$

where the additional correction sectors encode intrinsic multiscale and curvature-driven effects.

The fractal correction term is defined via a nonlocal scaling operator $\mathcal{D}_{\sigma}$ acting on the informational field:

$$\Phi_{\text{fractal}}(x) = \int_{\mathcal{M}_I} K_{\sigma}(x, y) (\mathcal{I}(x) - \mathcal{I}(y))^2 d\mu(y), \quad (1002)$$

with kernel K_{σ} encoding long-range correlations and anomalous scaling exponents.

The geometric contribution is coupled to curvature invariants of the underlying manifold:

$$\Phi_{\text{geom}}(x) = \alpha R(x) + \beta \Delta_g \log \rho(x) + \gamma \text{Tr}(\nabla^2 \mathcal{I}(x)), \quad (1003)$$

linking informational tension directly to scalar curvature, diffusion geometry, and local Hessian structure.

At the level of dynamical consistency, the informational substrate obeys a constrained evolution equation of gradient-flow type:

$$\partial_t \mathcal{I} = -\nabla_{L^2} \mathcal{S}[\mathcal{I}] + \lambda(t) \frac{\delta}{\delta \mathcal{I}} \left(\int_{\Omega} e^{-\beta \mathcal{I}} d\mu - 1 \right), \quad (1004)$$

where $\lambda(t)$ is a Lagrange multiplier enforcing normalization invariance under temporal evolution.

This induces a coupled entropy–curvature feedback system in which informational density back-reacts on geometry through an effective stress functional:

$$T_{\mu\nu}^{(\mathcal{I})} = \frac{\delta \mathcal{S}}{\delta g^{\mu\nu}} = \nabla_{\mu} \mathcal{I} \nabla_{\nu} \mathcal{I} - \frac{1}{2} g_{\mu\nu} |\nabla \mathcal{I}|^2 + \Xi_{\mu\nu}^{\text{fractal}}, \quad (1005)$$

closing the loop between information content, curvature response, and multiscale corrections.

Consequently, within this axiomatic structure, entropy ceases to be a passive diagnostic quantity and becomes a dynamical generator of geometry:

$$\mathcal{S} : \mathcal{I} \longrightarrow \text{Geom}(\mathcal{M}_I), \quad (1006)$$

establishing informational fields as the primitive substrate from which probability measures, geometric structure, and fractal corrections co-emerge as unified manifestations of a single multiscale information manifold.

Table 7: Ontology of Fundamental Objects in PFIT

Object	Mathematical Representation	Physical Interpretation
Information State	$x \in M$	Statistical configuration
Information Field	Φ	Distributed information substrate
Entropy Field	$S(x)$	Information density potential
Curvature	R	Structural instability measure
Projection Operator	Π	Scale transformation mechanism

11.2 Information Field Dynamics

The dynamical evolution of the informational substrate $\mathcal{I}$ is formulated as a variational field theory on the information manifold $\mathcal{M}_I$, where geometry and entropy are treated as mutually coupled degrees of freedom. The fundamental action functional is defined as

$$\mathcal{A}[\mathcal{I}] = \int_{\mathcal{M}_I} \left[\frac{1}{2} g^{ij}(\mathcal{I}) \partial_i \mathcal{I} \partial_j \mathcal{I} - \mathcal{S}(\mathcal{I}) + \gamma \mathcal{R}(\mathcal{I}) \right] dV_g, \quad (1007)$$

where g^{ij} is the inverse Fisher–Rao metric tensor, $\mathcal{S}(\mathcal{I})$ is the entropy potential functional, and $\mathcal{R}(\mathcal{I})$ encodes curvature feedback induced by statistical distinguishability.

The stationarity condition $\delta \mathcal{A}[\mathcal{I}] = 0$ yields a nonlinear Euler–Lagrange system of entropy–geometric type:

$$\frac{\delta \mathcal{A}}{\delta \mathcal{I}} = -\nabla_i (g^{ij} \nabla_j \mathcal{I}) + \frac{\delta \mathcal{S}}{\delta \mathcal{I}} - \gamma \frac{\delta \mathcal{R}}{\delta \mathcal{I}} = 0. \quad (1008)$$

In intrinsic geometric form, this can be written as a generalized entropy–Laplace equation on $(\mathcal{M}_I, g)$:

$$\Delta_g \mathcal{I} + \Gamma_{ij}^k g^{ij} \partial_k \mathcal{I} + \frac{\delta \mathcal{S}}{\delta \mathcal{I}} = 0, \quad (1009)$$

where Γ_{ij}^k denotes the Levi–Civita connection associated with the Fisher–Rao metric, encoding curvature corrections to informational diffusion.

The entropy functional itself admits a multiscale decomposition of the form

$$\mathcal{S}(\mathcal{I}) = -\rho \log \rho + \alpha \Delta_g \mathcal{I} + \beta R_g \mathcal{I} + \gamma \mathcal{F}_{\text{fractal}}[\mathcal{I}], \quad (1010)$$

so that the variational derivative becomes a coupled operator:

$$\frac{\delta \mathcal{S}}{\delta \mathcal{I}} = -\log \rho - 1 + \alpha \Delta_g^2 \mathcal{I} + \beta \nabla R_g \cdot \nabla \mathcal{I} + \gamma \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \mathcal{I}}. \quad (1011)$$

Introducing a dissipative gradient-flow structure in information time t , the evolution law is expressed as

$$\partial_t \mathcal{I} = -\nabla_g \mathcal{S}[\mathcal{I}] + \kappa \Delta_g \mathcal{I} + \gamma \mathcal{R}_g, \quad (1012)$$

where $\nabla_g \mathcal{S}$ denotes the metric gradient on the Fisher–Rao manifold and $\mathcal{R}_g$ is the scalar curvature associated with the induced statistical geometry.

Equivalently, the system can be recast as a curvature-driven reaction–diffusion equation on the information manifold:

$$\partial_t \mathcal{I} = \kappa \Delta_g \mathcal{I} - \frac{\delta \mathcal{S}}{\delta \mathcal{I}} + \gamma R_g \mathcal{I} + \mathcal{F}_{\text{fractal}}[\mathcal{I}]. \quad (1013)$$

At the level of geometric backreaction, the informational field induces a deformation of the metric tensor through the effective Einstein–information coupling:

$$\mathcal{R}_{ij} - \frac{1}{2} g_{ij} \mathcal{R} = \lambda \left(\nabla_i \mathcal{I} \nabla_j \mathcal{I} - \frac{1}{2} g_{ij} |\nabla \mathcal{I}|^2 \right) + \Xi_{ij}^{(\text{fractal})}, \quad (1014)$$

where $\Xi_{ij}^{(\text{fractal})}$ encodes higher-order multiscale corrections arising from Pellis-type fractal structure.

Finally, combining the variational and geometric sectors yields a closed nonlinear dynamical system:

$$\boxed{(\Delta_g - \partial_t) \mathcal{I} + \frac{\delta \mathcal{S}}{\delta \mathcal{I}} = \gamma \mathcal{R}_g + \mathcal{F}_{\text{fractal}}[\mathcal{I}]} \quad (1015)$$

which establishes that informational evolution is governed by a unified curvature–entropy flow equation.

Thus, within the PFIT–En-topia–Pellis framework, information dynamics is not merely evolution on a fixed manifold, but a fully coupled geometric–entropic field theory in which curvature, entropy gradients, and fractal corrections co-determine the trajectory of the informational substrate across scales.

11.3 Entropy–Curvature Duality Principle

A central structural postulate of the PFIT–En-topia–Pellis framework is that informational entropy and geometric curvature are not independent entities but dual representations of a single multiscale dynamical invariant. In this sense, the information manifold $(\mathcal{M}_I, g)$ encodes entropy flow in its curvature structure, while curvature itself emerges as a second-order manifestation of entropy gradients.

The fundamental duality relation is formulated at the level of the Ricci curvature tensor as

$$\mathcal{R}_{ij} = \nabla_i \nabla_j \mathcal{S} + \lambda \mathcal{K}_{ij}^{(f)} + \alpha \nabla_i \mathcal{I} \nabla_j \mathcal{I}, \quad (1016)$$

where $\mathcal{S}$ denotes the entropy field, $\mathcal{K}_{ij}^{(f)}$ encodes fractal-scale corrections arising from nonlocal probability structure, and the additional quadratic term captures self-induced informational stress-energy on the manifold.

Equivalently, this relation can be interpreted as a generalized entropy–geometry Poisson embedding:

$$\mathcal{R}_{ij} = \frac{\delta^2 \mathcal{S}}{\delta x^i \delta x^j} + \lambda \mathcal{F}_{ij}^{(2)}[\rho], \quad (1017)$$

where $\mathcal{F}_{ij}^{(2)}[\rho]$ denotes the second-order fractal functional response of the probability density field.

Taking the trace with respect to the Fisher–Rao metric yields the scalar duality law:

$$\mathcal{R} = \Delta_g \mathcal{S} + \lambda \text{Tr}(\mathcal{K}^{(f)}) + \alpha |\nabla \mathcal{I}|^2, \quad (1018)$$

which identifies scalar curvature as a nonlinear aggregation of entropy diffusion, fractal corrections, and informational gradient energy.

In expanded multiscale form, the entropy Laplacian admits the decomposition

$$\Delta_g \mathcal{S} = \underbrace{-\Delta_g \log \rho}_{\text{local Shannon curvature}} + \underbrace{\alpha \Delta_g^2 \mathcal{I}}_{\text{diffusive entropy elasticity}} + \underbrace{\beta \nabla^i R \nabla_i \mathcal{I}}_{\text{curvature–entropy coupling}} + \underbrace{\gamma \Delta_g \mathcal{F}_{\text{fractal}}}_{\text{multiscale correction}}, \quad (1019)$$

showing explicitly how curvature decomposes into hierarchical informational contributions across scales.

Moreover, the duality principle can be reformulated as a nonlinear constraint equation governing equilibrium configurations:

$$\nabla_i \nabla_j \mathcal{S} = \mathcal{R}_{ij} - \lambda \mathcal{K}_{ij}^{(f)}, \quad (1020)$$

which implies that entropy extrema correspond to curvature-balanced fixed points of the informational geometry.

At the level of dynamical stability, perturbations $\delta \mathcal{S}$ satisfy the linearized spectral relation:

$$\delta \mathcal{R}_{ij} = \nabla_i \nabla_j (\delta \mathcal{S}) + \lambda \delta \mathcal{K}_{ij}^{(f)}, \quad (1021)$$

leading to a curvature-driven instability spectrum governed by eigenmodes of the entropy Hessian.

Contracting with the inverse metric yields a Poisson-type entropy–curvature equation:

$$\Delta_g \mathcal{S} = \mathcal{R} - \lambda \mathcal{K}^{(f)} - \alpha |\nabla \mathcal{I}|^2, \quad (1022)$$

where $\mathcal{K}^{(f)} := g^{ij} \mathcal{K}_{ij}^{(f)}$ denotes the scalar fractal curvature correction.

Finally, the duality can be encapsulated in operator form as

$$\boxed{\mathcal{R}_{ij} \equiv \mathcal{D}_{ij}^{(\text{entropy})}[\mathcal{S}] + \mathcal{D}_{ij}^{(\text{fractal})}[\rho]} \quad (1023)$$

establishing that curvature is fully generated by the combined action of entropy Hessians and multiscale fractal deformation operators.

Thus, within the PFIT framework, entropy is not merely a scalar functional on probability densities but the generating potential of geometry itself, while curvature acts as its second-order spectral manifestation, completing a closed entropy–geometry dual loop that governs informational stability, collapse, and multiscale structure formation.

11.4 Fractal Projection Operator Formalism

Building upon the fractal-information axiomatics introduced in [82], the PFIT framework formalizes the multiscale structure of information via a hierarchy of projection operators acting on the informational manifold $\mathcal{M}_I$. Each projection isolates an effective resolution sector of the underlying field $\mathcal{I}$, inducing a scale-dependent renormalized geometry.

The fundamental scale projection map is defined as

$$\Pi_\epsilon : \mathcal{M}_I \longrightarrow \mathcal{M}_I^{(\epsilon)}, \quad \epsilon \in \mathbb{R}^+, \quad (1024)$$

where $\mathcal{M}_I^{(\epsilon)}$ denotes the effective information manifold at resolution scale ϵ , equipped with induced metric $g^{(\epsilon)}$ and entropy functional $\mathcal{S}^{(\epsilon)}$.

At the level of fields, the renormalized informational density satisfies a nonlinear scaling law:

$$\mathcal{I}_\epsilon(x) = \epsilon^{-\alpha} \mathcal{I}(\Pi_\epsilon^{-1}(x)), \quad (1025)$$

where α is the fractal scaling exponent governing deviation from classical dimensional homogeneity. Equivalently, in logarithmic form,

$$\log \mathcal{I}_\epsilon(x) = -\alpha \log \epsilon + \log \mathcal{I}(x), \quad (1026)$$

revealing affine structure in scale space.

The induced probability density transforms under projection according to the renormalization rule

$$\rho_\epsilon(x) = \frac{\mathcal{I}_\epsilon(x)}{Z_\epsilon}, \quad Z_\epsilon = \int_{\mathcal{M}_I^{(\epsilon)}} \mathcal{I}_\epsilon(x) d\mu_\epsilon(x), \quad (1027)$$

ensuring probabilistic normalization at each scale sector.

The corresponding entropy functional becomes scale-dependent:

$$\mathcal{S}_\epsilon[\rho] = - \int_{\mathcal{M}_I^{(\epsilon)}} \rho_\epsilon(x) \log \rho_\epsilon(x) dV_{g^{(\epsilon)}} + \epsilon^\beta \mathcal{F}_{\text{fractal}}^{(\epsilon)}[\rho], \quad (1028)$$

where β encodes anomalous scaling corrections induced by hierarchical coarse-graining.

A fundamental consistency requirement of the PFIT renormalization structure is encoded in the commutative projection diagram:

$$\Pi_{\epsilon_1} \circ \Pi_{\epsilon_2} = \Pi_{\min(\epsilon_1, \epsilon_2)}, \quad (1029)$$

which enforces idempotent contraction of informational degrees of freedom under successive coarse-graining.

At the operator level, this property induces a semigroup structure:

$$\Pi_\epsilon \circ \Pi_\delta = \Pi_{\epsilon \star \delta}, \quad \epsilon \star \delta := \min(\epsilon, \delta), \quad (1030)$$

implying that scale evolution is non-invertible and inherently dissipative in informational space.

The infinitesimal generator of the projection flow defines a renormalization Hamiltonian:

$$\mathcal{G} = \lim_{\epsilon \rightarrow 0} \frac{\Pi_\epsilon - \text{Id}}{\epsilon}, \quad (1031)$$

which acts on observables as a scale-derivative operator:

$$\partial_{\log \epsilon} \mathcal{I} = \mathcal{G}[\mathcal{I}]. \quad (1032)$$

This yields a fractal flow equation governing multiscale evolution:

$$\frac{d\mathcal{I}_\epsilon}{d \log \epsilon} = -\alpha \mathcal{I}_\epsilon + \mathcal{F}_{\text{nonlocal}}[\mathcal{I}_\epsilon] + \mathcal{C}_{\text{curv}}(\epsilon), \quad (1033)$$

where $\mathcal{F}_{\text{nonlocal}}$ captures long-range correlations and $\mathcal{C}_{\text{curv}}$ encodes curvature-induced renormalization effects.

At the geometric level, projection induces a deformation of the Fisher–Rao metric:

$$g_{ij}^{(\epsilon)} = \Pi_\epsilon^* g_{ij} = \epsilon^{-2\alpha} g_{ij} + \Xi_{ij}^{(\text{fractal})}(\epsilon), \quad (1034)$$

showing explicit scale dependence of informational geometry.

Finally, the full PFIT projection structure satisfies an asymptotic fixed-point condition:

$$\lim_{\epsilon \rightarrow 0} \Pi_\epsilon(\mathcal{I}) = \mathcal{I}^*, \quad (1035)$$

where $\mathcal{I}^*$ represents a scale-invariant informational attractor satisfying

$$\mathcal{G}[\mathcal{I}^*] = 0, \quad \mathcal{S}_\epsilon[\mathcal{I}^*] = \text{constant}. \quad (1036)$$

Thus, within the PFIT–En-topia–Pellis framework, fractal projection operators encode a complete renormalization algebra of information, establishing scale evolution as a geometric-semigroup flow that governs entropy deformation, curvature emergence, and multiscale informational stability.

11.5 Stability and Informational Collapse Mechanism

Stability of informational configurations is governed by the second variation of the entropy action, interpreted as a bilinear form on the tangent space of admissible perturbations $\delta\mathcal{I} \in T_{\mathcal{I}}\mathcal{M}_I$:

$$\delta^2\mathcal{A}[\mathcal{I}] = \int_{\mathcal{M}_I} \delta\mathcal{I}(x) \mathcal{H}_{\mathcal{S}}[\mathcal{I}](x) \delta\mathcal{I}(x) dV_g, \quad (1037)$$

where $\mathcal{H}_{\mathcal{S}}$ denotes the entropy Hessian operator induced by the nonlinear dependence of the entropy functional $\mathcal{S}$ on the informational state $\mathcal{I}$.

In coordinate-free form, the Hessian can be expressed as the covariant second variation

$$\mathcal{H}_{\mathcal{S}}[\mathcal{I}] = \nabla_g^2 \mathcal{S}(\mathcal{I}) + (\nabla_g \mathcal{I} \otimes \nabla_g \mathcal{I}) \cdot \frac{\delta^2 \mathcal{S}}{\delta \mathcal{I}^2}, \quad (1038)$$

which couples the Levi–Civita connection of the information manifold with intrinsic nonlinear entropy curvature.

In this formulation, the spectrum of $\mathcal{H}_{\mathcal{S}}$ determines the local stability regime of the system: positive definiteness corresponds to stable informational equilibria, the emergence of zero modes signals marginal (critical) directions, while the appearance of negative or unbounded directions indicates instability and onset of entropy-driven structural reconfiguration.

Expanding the entropy functional into PFIT components yields a decomposition of the Hessian:

$$\mathcal{H}_{\mathcal{S}} = \mathcal{H}_{\text{Sh}} + \alpha \mathcal{H}_{\Delta} + \beta \mathcal{H}_R + \gamma \mathcal{H}_{\text{fr}}, \quad (1039)$$

where

$$\mathcal{H}_{\text{Sh}}[\delta\mathcal{I}] = -\Delta_g \delta\mathcal{I} - \frac{1}{\mathcal{I}} (\delta\mathcal{I})^2, \quad (1040)$$

$$\mathcal{H}_{\Delta}[\delta\mathcal{I}] = \Delta_g^2 \delta\mathcal{I}, \quad (1041)$$

$$\mathcal{H}_R[\delta\mathcal{I}] = R(x) \delta\mathcal{I} + (\nabla_i R) \nabla^i \delta\mathcal{I}, \quad (1042)$$

$$\mathcal{H}_{\text{fr}}[\delta\mathcal{I}] = D_{\text{fr}}[\delta\mathcal{I}] + \mathcal{O}(\epsilon^\alpha) \delta\mathcal{I}, \quad (1043)$$

with D_{fr} denoting the linearized fractal projection response operator.

The spectral stability problem is therefore formulated as an operator eigenvalue equation on the perturbation space:

$$\mathcal{H}_{\mathcal{S}}[\mathcal{I}] \psi_k = \lambda_k \psi_k, \quad \psi_k \in H^1(\mathcal{M}_I). \quad (1044)$$

Collapse is triggered when the associated quadratic form becomes indefinite, i.e. when there exists at least one unstable direction in configuration space:

$$\exists \psi_k \neq 0 \quad \text{such that} \quad \langle \psi_k, \mathcal{H}_{\mathcal{S}} \psi_k \rangle < 0, \quad (1045)$$

which is equivalent to the existence of negative spectral modes:

$$\exists \lambda_k < 0 \implies \text{informational instability (collapse regime)}. \quad (1046)$$

At the nonlinear level, this instability corresponds to a curvature–entropy feedback bifurcation characterized by the coupled system:

$$\partial_t \mathcal{I} = -\nabla_g \mathcal{S}(\mathcal{I}) + \kappa \nabla_g R + \gamma \nabla_g \mathcal{F}_{\text{fractal}}[\mathcal{I}], \quad (1047)$$

$$\partial_t g_{ij} = -2 \mathcal{R}_{ij} + \mu \nabla_i \nabla_j \mathcal{S}(\mathcal{I}), \quad (1048)$$

showing explicitly that collapse is not purely informational but jointly geometric–entropic.

In the critical regime, the smallest eigenvalue controls the onset of collapse via a gap-closing condition:

$$\lambda_{\min}(\mathcal{H}_{\mathcal{S}}) \rightarrow 0^-, \quad (1049)$$

leading to divergence of correlation length in informational space:

$$\xi \sim |\lambda_{\min}|^{-\frac{1}{2}}, \quad (1050)$$

which signals the emergence of scale-free, fractal-dominated reorganization of the manifold structure.

Thus, informational collapse is rigorously characterized as a spectral phase transition of the entropy Hessian, governed by curvature feedback, nonlinear entropy geometry, and fractal renormalization effects within the PFIT–Entropy–Pellis framework.

11.6 Embedding of Pellis Fractal Information Theory

In accordance with the universal fractal information framework developed in [82], the informational field is assumed to obey a scale-covariant transformation law:

$$\mathcal{I}(\lambda x) = \lambda^\alpha \mathcal{I}(x), \quad (1051)$$

which extends to a local renormalization group action on the tangent bundle of the informational manifold:

$$\mathcal{I}_\lambda(x) = e^{\alpha \log \lambda} \mathcal{I}(x), \quad \lambda \in \mathbb{R}^+. \quad (1052)$$

The infinitesimal generator of this scaling symmetry defines a dilation operator:

$$\mathcal{D}_\alpha \mathcal{I} = (x^i \nabla_i + \alpha) \mathcal{I}, \quad (1053)$$

which satisfies the commutation relation with the Laplace–Beltrami operator:

$$[\Delta_g, \mathcal{D}_\alpha] = 2\Delta_g + \mathcal{O}(\nabla R), \quad (1054)$$

highlighting explicit curvature-induced breaking of exact scale invariance.

The fractal measure μ_F is introduced as a deformation of the Riemannian volume form:

$$d\mu_F(x) = \rho_F(x) dV_g, \quad \rho_F(x) = e^{-\Phi_F(x)}, \quad (1055)$$

where Φ_F encodes multiscale information compression across nested resolution shells. The corresponding fractal Laplacian takes the weighted form:

$$\Delta_F f = \frac{1}{\rho_F} \nabla_i (\rho_F g^{ij} \nabla_j f), \quad (1056)$$

which reduces to the standard Laplace–Beltrami operator in the trivial fractal limit $\rho_F \rightarrow 1$.

Within this structure, the effective curvature tensor is modified by fractal backreaction terms:

$$\mathcal{R}_{ij}^{\text{eff}} = \mathcal{R}_{ij} + \kappa \nabla_i \nabla_j \log \mu_F + \eta (\nabla_i \Phi_F)(\nabla_j \Phi_F), \quad (1057)$$

where the second contribution encodes entropic localization pressure, and the third term introduces nonlinear fractal self-interaction at small scales.

Taking the trace yields the scalar effective curvature:

$$\mathcal{R}^{\text{eff}} = \mathcal{R} + \kappa \Delta_g \log \mu_F + \eta \|\nabla \Phi_F\|_g^2, \quad (1058)$$

which directly couples geometric expansion/compression of the manifold to the internal entropy density gradients.

The fractal information field obeys a scale-covariant Euler–Lagrange equation derived from a generalized PFIT action:

$$\mathcal{A}_{\text{fr}}[\mathcal{I}] = \int_{\mathcal{M}} \left(\frac{1}{2} g^{ij} \nabla_i \mathcal{I} \nabla_j \mathcal{I} + \beta \mathcal{R}^{\text{eff}} \mathcal{I}^2 - \gamma \mathcal{F}_{\text{fractal}}[\mathcal{I}] \right) d\mu_F, \quad (1059)$$

leading to the nonlinear field equation:

$$\Delta_F \mathcal{I} + \beta \mathcal{R}^{\text{eff}} \mathcal{I} - \gamma \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \mathcal{I}} = 0. \quad (1060)$$

The embedding of Pellis fractal structure is therefore realized through a hierarchy of nested projections:

$$\Pi_\epsilon \mathcal{I} = \sum_{k=0}^{\infty} \epsilon^{k\alpha} \mathcal{I}_k, \quad (1061)$$

where each mode $\mathcal{I}_k$ represents a distinct resolution layer in the informational spectrum.

Finally, the coupling between fractal geometry and entropy curvature closes the system via the consistency relation:

$$\nabla_i \mathcal{R}_{jk}^{\text{eff}} - \nabla_j \mathcal{R}_{ik}^{\text{eff}} = \mathcal{C}_{ijk}^{\text{fr}}[\mathcal{I}], \quad (1062)$$

where $\mathcal{C}_{ijk}^{\text{fr}}$ encodes irreducible fractal torsion generated by multiscale informational interference.

This construction establishes the ontological core of the PFIT–En-topia–Pellis framework: information is not merely a scalar density but a scale-covariant geometric field whose curvature, spectral decomposition, and fractal renormalization structure co-determine the emergence of effective physical law through a unified multiscale informational geometry.

12 Physical Interpretation of the PFIT Entropy Field

In this subsection, we provide a physical and information-theoretic interpretation of the abstract mathematical structures introduced in the PFIT framework, establishing the correspondence between geometric operators on the information manifold and observable multiscale physical dynamics. This interpretation serves as a conceptual bridge between the formal operatorial structure and physically meaningful notions of information flow, instability, and collapse.

12.1 Probability Density as Physical Information Field

The scalar field $\rho(x, t)$ is interpreted not merely as a probability density, but as a *coarse-grained informational energy density* over the manifold $\mathcal{M}$. Each value $\rho(x, t)$ represents the local occupancy of informational degrees of freedom in a given resolution cell of the underlying multiscale system.

$$\rho(x, t) \sim \mathbb{P}(\text{information event at } x \text{ and scale } t), \quad (1063)$$

or equivalently, as a renormalized projection of microscopic fluctuations onto a macroscopic informational state space induced by scale-dependent coarse-graining operators $\mathcal{C}_\ell$ acting on the microscopic configuration space Ω :

$$\rho_\ell(x, t) = (\mathcal{C}_\ell \circ \rho_{\text{micro}})(x, t), \quad (1064)$$

where ℓ denotes the resolution scale and $\mathcal{C}_\ell$ satisfies a semigroup property:

$$\mathcal{C}_{\ell_1} \circ \mathcal{C}_{\ell_2} = \mathcal{C}_{\ell_1 + \ell_2}. \quad (1065)$$

In the continuum limit, $\rho(x, t)$ is embedded into a normalized information field on $\mathcal{M}$:

$$\int_{\mathcal{M}} \rho(x, t) d\mu_g(x) = 1, \quad (1066)$$

where $d\mu_g(x) = \sqrt{g(x)} d^n x$ is the Riemannian volume form induced by the metric g_{ij} of the informational manifold.

The evolution of $\rho(x, t)$ is assumed to be governed by a generalized continuity equation encoding information conservation:

$$\partial_t \rho + \nabla_i J^i = 0, \quad (1067)$$

where the informational flux J^i decomposes into drift and diffusion contributions:

$$J^i = \rho v^i - D^{ij} \nabla_j \rho. \quad (1068)$$

Here v^i represents the emergent informational velocity field induced by entropy gradients, while D^{ij} is a scale-dependent diffusion tensor encoding unresolved microscopic fluctuations.

In the entropic-gradient formulation, the drift field can be written as:

$$v^i = -\gamma g^{ij} \nabla_j \mathcal{S}[\rho], \quad (1069)$$

where $\mathcal{S}[\rho]$ denotes a generalized entropy functional on $\mathcal{M}$, e.g.

$$\mathcal{S}[\rho] = - \int_{\mathcal{M}} \rho(x, t) \log \rho(x, t) d\mu_g(x), \quad (1070)$$

or, more generally, a non-extensive entropy of Tsallis-type:

$$\mathcal{S}_q[\rho] = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \rho(x, t)^q d\mu_g(x) \right). \quad (1071)$$

The resulting evolution equation can be cast as a nonlinear Fokker–Planck dynamics on the manifold:

$$\partial_t \rho = \nabla_i (D^{ij} \nabla_j \rho + \gamma \rho g^{ij} \nabla_j \mathcal{S}[\rho]). \quad (1072)$$

From an information-geometric perspective, $\rho(x, t)$ defines a statistical manifold with Fisher information metric:

$$\mathcal{I}_{ij}(\theta) = \int_{\mathcal{M}} \rho(x|\theta) \partial_i \log \rho(x|\theta) \partial_j \log \rho(x|\theta) d\mu_g(x), \quad (1073)$$

which induces a natural curvature structure governing the stability of informational flows.

In this formulation, the curvature scalar $R[\rho]$ of the induced statistical geometry encodes the susceptibility of the system to informational collapse:

$$R[\rho] \sim \nabla_i \nabla^i \log \rho + \mathcal{O}((\nabla \rho)^2), \quad (1074)$$

linking local density fluctuations directly to emergent geometric instability.

Finally, $\rho(x, t)$ may be interpreted as a projection of a higher-dimensional wavefunctional $\Psi[\phi]$ defined over field configurations $\phi(x)$:

$$\rho(x, t) = \int \mathcal{D}\phi \, \delta(x - x[\phi]) |\Psi[\phi, t]|^2, \quad (1075)$$

establishing a bridge between microscopic quantum informational dynamics and macroscopic coarse-grained probability fields.

Thus, $\rho(x, t)$ functions as a unifying object linking stochastic processes, geometric flows, and renormalization-group coarse-graining within a single multiscale informational manifold structure.

12.2 Entropy Operator as Information Dynamics Generator

The PFIT entropy operator $\mathcal{H}_{\text{PFIT}}$ is interpreted as a **generator of information redistribution** across scales. In this framework, entropy is not a passive scalar functional but an active operator acting on the space of admissible informational fields $\rho \in \mathcal{P}(\mathcal{M})$, where $\mathcal{P}(\mathcal{M})$ denotes the probability simplex over the manifold $\mathcal{M}$.

$$\mathcal{H}_{\text{PFIT}} : \mathcal{P}(\mathcal{M}) \rightarrow \mathbb{R}, \quad \rho \mapsto \mathcal{H}_{\text{PFIT}}[\rho], \quad (1076)$$

with induced gradient flow structure defined via a variational principle on informational phase space.

Its nonlinear structure encodes four fundamental mechanisms:

- local information compression via $-\rho \log \rho$,
- diffusive smoothing of informational gradients via $\Delta_g \rho$,
- geometric modulation via scalar curvature $R(x)$,
- multiscale memory effects via $\mathcal{F}_{\text{fractal}}[\rho]$.

Formally, the PFIT entropy functional is expressed as:

$$\mathcal{H}_{\text{PFIT}}[\rho] = \int_{\mathcal{M}} [\rho \log \rho + \alpha g^{ij} (\nabla_i \rho) (\nabla_j \rho) + \beta R(x) \rho + \gamma \mathcal{F}_{\text{fractal}}[\rho](x)] d\mu_g(x), \quad (1077)$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ are scale-coupling coefficients controlling diffusion strength, curvature sensitivity, and fractal memory depth respectively.

The operator nature of $\mathcal{H}_{\text{PFIT}}$ emerges through its functional derivative, defining the entropy-driven potential field:

$$\Phi_{\text{PFIT}}(x, t) := \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho(x, t)}. \quad (1078)$$

Explicit computation yields:

$$\Phi_{\text{PFIT}} = 1 + \log \rho - \alpha \Delta_g \rho + \beta R(x) + \gamma \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho}. \quad (1079)$$

The informational dynamics are then generated by a gradient-flow structure in the Wasserstein geometry of probability measures:

$$\partial_t \rho = \nabla_i (\rho \nabla^i \Phi_{\text{PFIT}}), \quad (1080)$$

which can be equivalently written as a nonlinear geometric diffusion equation:

$$\partial_t \rho = \nabla_i \left[\rho \nabla^i \left(\log \rho - \alpha \Delta_g \rho + \beta R + \gamma \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} \right) \right]. \quad (1081)$$

The term $-\rho \log \rho$ induces entropic contraction in configuration space, leading to information condensation toward attractor sets:

$$\frac{d}{dt}\mathcal{H}_{\text{Shannon}}[\rho] = - \int_{\mathcal{M}} \frac{|\nabla \rho|^2}{\rho} d\mu_g(x) \leq 0, \quad (1082)$$

establishing monotonic entropy dissipation under smooth dynamics.

Figure 11 illustrates the resulting geometric structure, where the entropy field $S(x, y)$ defines a nontrivial scalar landscape over the information manifold. The associated gradient field encodes the local direction of maximal informational dissipation, generating a curvature-driven deformation of the statistical state space. Regions of high entropy gradient correspond to enhanced geometric instability and accelerated informational flow.

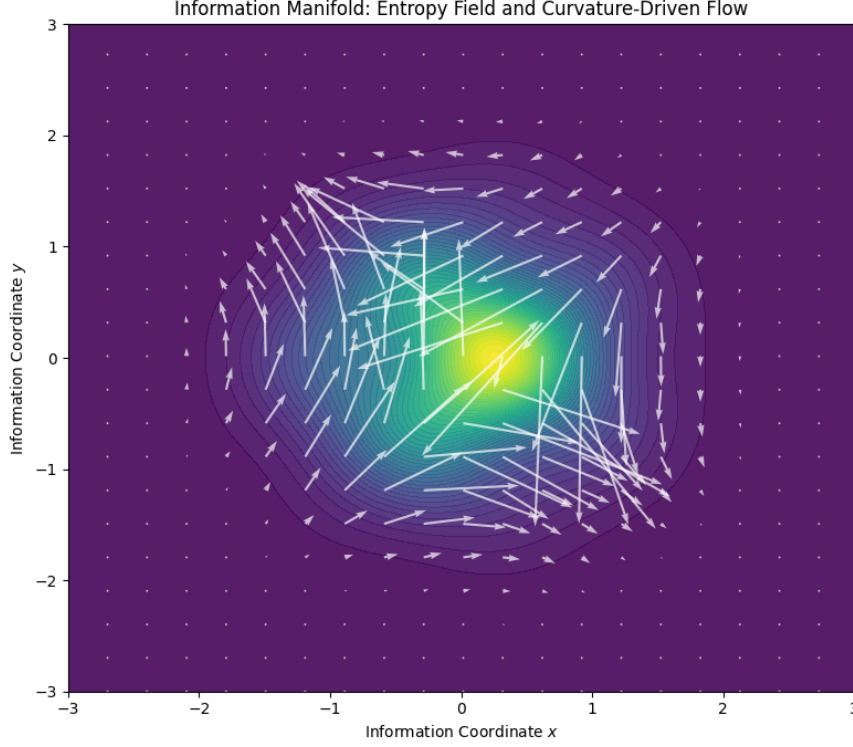


Figure 11: Information manifold endowed with a scalar entropy field $S(x, y)$ and its associated gradient flow. The color map represents the magnitude of the entropy distribution, while the vector field denotes the local direction of maximal entropy increase. This structure encodes curvature-driven informational dynamics, where geodesic deviation is governed by entropy gradients within the PFIT framework.

Geometric coupling enters through curvature-dependent modulation of informational flux, where the scalar curvature acts as an effective potential barrier:

$$J_{\text{geom}}^i = -\rho g^{ij} \nabla_j R(x), \quad (1083)$$

implying that regions of high curvature act as sinks or sources of informational density depending on sign structure of $R(x)$.

The fractal memory operator $\mathcal{F}_{\text{fractal}}[\rho]$ introduces nonlocal scale coupling across renormalization levels $\ell \in \mathbb{R}^+$:

$$\mathcal{F}_{\text{fractal}}[\rho](x) = \int_0^\infty K(\ell) \rho_\ell(x) d\ell, \quad (1084)$$

where $\rho_\ell(x)$ denotes the coarse-grained field at scale ℓ , and $K(\ell)$ is a scale-invariant kernel satisfying power-law decay:

$$K(\ell) \sim \ell^{-\kappa}, \quad \kappa \in (0, 2). \quad (1085)$$

Consequently, $\mathcal{H}_{\text{PFIT}}$ is not merely an entropy functional but a **renormalization-stable generator of multiscale informational flow**, embedding local diffusion, geometric curvature feedback, and fractal temporal memory into a single unified operator structure.

In operator-theoretic form, the PFIT entropy generator can be compactly represented as:

$$\mathcal{H}_{\text{PFIT}} \equiv \mathcal{S} + \alpha \Delta_g + \beta \mathcal{R} + \gamma \mathcal{K}_{\text{fractal}}, \quad (1086)$$

where $\mathcal{S}$ denotes the Shannon entropy operator, $\mathcal{R}$ the curvature multiplication operator, and $\mathcal{K}_{\text{fractal}}$ the nonlocal scale-integral operator acting on informational histories.

Thus, $\mathcal{H}_{\text{PFIT}}$ acts as an effective Hamiltonian-like functional in informational phase space, governing the redistribution of uncertainty across scales through a coupled nonlinear geometry–diffusion–memory system.

12.3 Curvature as Instability Indicator

Within the PFIT–En-topia–Pellis framework, the scalar curvature $R(x)$ is elevated from a purely geometric invariant to a **local instability density functional**, encoding the propensity of informational trajectories to undergo either focusing collapse or dispersive decoherence. The informational manifold $(\mathcal{M}, g_{ij})$ is therefore not static, but dynamically coupled to the evolving density field $\rho(x, t)$ through curvature backreaction.

In this interpretation, curvature operates as a nonlinear stability regulator of the information flow field, such that:

$$R(x) \uparrow \implies \|\nabla \rho(x, t)\| \uparrow, \quad (1087)$$

indicating amplification of local informational gradients and increased susceptibility to collapse-like transitions in configuration space.

Conversely, negative curvature induces dispersive spreading:

$$R(x) < 0 \implies \rho(x, t) \rightarrow \text{delocalized regime}, \quad (1088)$$

corresponding to entropy-dominated diffusion and loss of coherent informational structure.

The curvature field is assumed to be dynamically generated by the informational density via a nonlinear feedback coupling of the form:

$$R(x, t) = \lambda_1 \rho(x, t) + \lambda_2 \Delta_g \rho(x, t) + \lambda_3 g^{ij} (\nabla_i \rho) (\nabla_j \rho), \quad (1089)$$

where $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ encode the strength of matter–information backreaction on geometry.

Substituting this relation into the informational dynamics yields a fully closed curvature-driven evolution system:

$$\partial_t \rho = \nabla_i [\rho \nabla^i (\log \rho - \alpha \Delta_g \rho + \beta R[\rho])], \quad (1090)$$

which explicitly demonstrates that curvature acts as a nonlinear control parameter governing the stability landscape of the flow.

The stability of informational trajectories can be quantified via the second variation of the entropy functional $\mathcal{H}_{\text{PFIT}}$, leading to the Hessian operator:

$$\mathbb{H}_{\text{PFIT}} := \frac{\delta^2 \mathcal{H}_{\text{PFIT}}}{\delta \rho^2}. \quad (1091)$$

Linear stability is then determined by the spectral properties of $\mathbb{H}_{\text{PFIT}}$:

$$\text{Spec}(\mathbb{H}_{\text{PFIT}}) \subset \mathbb{R}^+ \implies \text{stable regime}, \quad \text{Spec}(\mathbb{H}_{\text{PFIT}}) \ni \lambda < 0 \implies \text{instability onset}. \quad (1092)$$

In particular, curvature enters the Hessian through a geometric potential term:

$$\mathbb{H}_{\text{PFIT}} = -\alpha \Delta_g + \beta R(x) + \gamma \mathcal{L}_{\text{fractal}}, \quad (1093)$$

where $\mathcal{L}_{\text{fractal}}$ denotes the linearized fractal memory operator.

A key diagnostic of instability is the curvature divergence criterion:

$$\lim_{t \rightarrow t_c} |R(x, t)| \rightarrow \infty, \quad (1094)$$

which signals breakdown of the smooth manifold approximation and transition to a critical regime characterized by non-perturbative informational fluctuations.

Near criticality, the curvature field obeys a scaling law of the form:

$$R(x, t) \sim |t - t_c|^{-\nu} \mathcal{R}\left(\frac{x}{\xi(t)}\right), \quad (1095)$$

where ν is a critical exponent and $\xi(t)$ is the correlation length of informational coherence.

The divergence of $\xi(t)$ corresponds to long-range synchronization of informational degrees of freedom, leading to emergent collective collapse phenomena:

$$\xi(t) \rightarrow \infty \implies \rho(x, t) \rightarrow \rho_{\text{macro}}(t), \quad (1096)$$

indicating reduction of microscopic complexity into macroscopic informational attractors.

Finally, curvature can be interpreted as an entropic stress tensor contraction:

$$R(x) = g^{ij} \mathcal{T}_{ij}^{(\text{info})}, \quad (1097)$$

where $\mathcal{T}_{ij}^{(\text{info})}$ represents the informational stress–energy tensor defined via variations of $\mathcal{H}_{\text{PFIT}}$ with respect to the metric:

$$\mathcal{T}_{ij}^{(\text{info})} := -\frac{2}{\sqrt{g}} \frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta g^{ij}}. \quad (1098)$$

Thus, scalar curvature emerges not as a passive geometric invariant but as an active dynamical measure of informational instability, encoding the transition between smooth diffusion regimes and critical collapse phases within the PFIT informational manifold.

12.4 Entropy–Curvature Feedback Loop

A central physical mechanism in the PFIT framework is the bidirectional coupling between entropy and geometry, which is formalized as a closed nonlinear operator loop on the coupled state space $(\rho, g_{ij}) \in \mathcal{P}(\mathcal{M}) \times \text{Met}(\mathcal{M})$:

$$\rho \rightarrow R(x) \rightarrow \mathcal{H}_{\text{PFIT}}[\rho] \rightarrow \rho, \quad (1099)$$

encoding a self-consistent recursion of informational collapse and geometric backreaction.

At the infinitesimal level, this loop is governed by a coupled dynamical system of the form:

$$\partial_t \rho = \mathcal{F}[\rho, R], \quad \partial_t R = \mathcal{G}[\rho, R], \quad (1100)$$

where $\mathcal{F}$ encodes entropy-driven transport and $\mathcal{G}$ encodes curvature renormalization induced by informational stress-energy.

The entropy-to-curvature channel is realized through a nonlinear geometric response functional:

$$R(x, t) = \kappa_1 \rho(x, t) + \kappa_2 \Delta_g \log \rho(x, t) + \kappa_3 |\nabla \log \rho(x, t)|_g^2, \quad (1101)$$

showing that curvature is not externally imposed but emergent from entropy gradients and their higher-order spatial fluctuations.

Conversely, curvature feeds back into the entropy functional through a geometric deformation of the PFIT Hamiltonian:

$$\mathcal{H}_{\text{PFIT}}^{(R)}[\rho] = \mathcal{H}_{\text{PFIT}}[\rho] + \int_{\mathcal{M}} \beta R(x, t) \rho(x, t) d\mu_g(x), \quad (1102)$$

which induces a curvature-modified entropy potential:

$$\Phi_{\text{eff}}(x, t) = \frac{\delta \mathcal{H}_{\text{PFIT}}^{(R)}}{\delta \rho} = \Phi_{\text{PFIT}} + \beta R(x, t). \quad (1103)$$

The resulting evolution equation becomes a self-referential geometric flow:

$$\partial_t \rho = \nabla_i [\rho \nabla^i (\Phi_{\text{PFIT}} + \beta R[\rho])], \quad (1104)$$

explicitly closing the entropy–curvature feedback loop.

The geometric backreaction on curvature can be expressed through a nonlinear reaction–diffusion–geometry equation:

$$\partial_t R = D_R \Delta_g R + \lambda_1 R + \lambda_2 \rho + \lambda_3 \Delta_g \rho + \lambda_4 |\nabla \rho|_g^2, \quad (1105)$$

which couples curvature growth directly to informational concentration and gradient amplification.

The fixed points of the coupled system are defined by:

$$\partial_t \rho = 0, \quad \partial_t R = 0, \quad (1106)$$

leading to a nonlinear eigenvalue problem:

$$\mathcal{L}_{\text{PFIT}}[\rho^*, R^*] = 0, \quad (1107)$$

where $\mathcal{L}_{\text{PFIT}}$ is the combined entropy–geometry operator governing equilibrium configurations.

Stability of the feedback loop is determined by the spectral radius of the linearized Jacobian operator:

$$\mathcal{J} = \begin{pmatrix} \frac{\delta \mathcal{F}}{\delta \rho} & \frac{\delta \mathcal{F}}{\delta R} \\ \frac{\delta \mathcal{G}}{\delta \rho} & \frac{\delta \mathcal{G}}{\delta R} \end{pmatrix}, \quad (1108)$$

with instability onset characterized by:

$$\max \text{Re}(\text{Spec}(\mathcal{J})) > 0. \quad (1109)$$

Near the instability threshold, the system exhibits critical amplification governed by a nonlinear feedback exponent:

$$\rho(t) \sim \rho_0 \exp \left(\int_0^t \sigma(R(\tau)) d\tau \right), \quad \sigma(R) \propto R - R_c, \quad (1110)$$

indicating that curvature acts as an effective control knob for exponential entropy focusing.

Finally, the entropy–curvature loop can be compactly represented as a composite operator iteration:

$$\mathcal{T}_{\text{PFIT}} : \rho_{n+1} = \mathcal{T}_{\text{PFIT}}[\rho_n] = \mathcal{N}(\rho_n, R[\rho_n]), \quad (1111)$$

where $\mathcal{N}$ is a nonlinear renormalization map encoding diffusion, curvature feedback, and fractal memory correction.

Thus, the PFIT dynamics is fundamentally a self-exciting geometric–entropic system in which curvature and entropy co-evolve as mutually reinforcing fields, generating multiscale pattern formation, instability cascades, and emergent informational structure across the manifold.

12.5 Collapse as Spectral Concentration Phenomenon

Within the PFIT–En-topia–Pellis framework, physical collapse is reinterpreted not as a discontinuous or singular event, but as a **spectral concentration process of entropy eigenmodes**, governed by the asymptotic reorganization of the spectrum of the nonlinear entropy operator $\mathcal{H}_{\text{PFIT}}$.

Let $\{\psi_k\}_{k \in \mathbb{N}}$ denote a complete orthonormal basis of informational eigenmodes on $\mathcal{M}$, defined via the nonlinear spectral problem:

$$\mathcal{H}_{\text{PFIT}}[\rho] \psi_k = \Lambda_k \psi_k, \quad (1112)$$

where $\Lambda_k \in \mathbb{R}$ are the entropy–information eigenvalues encoding multiscale contribution strength.

The informational field admits a spectral decomposition of the form:

$$\rho(x, t) = \sum_{k=1}^{\infty} a_k(t) \psi_k(x), \quad (1113)$$

with modal amplitudes $a_k(t)$ evolving under entropy-driven nonlinear coupling.

Physical collapse is then characterized by a **spectral condensation limit**, in which the effective support of the coefficient measure collapses onto a finite critical subset:

$$\text{Spec}(\mathcal{H}_{\text{PFIT}}) \longrightarrow \{\Lambda_k\}_{k \in \mathcal{K}_{\text{crit}}}, \quad |\mathcal{K}_{\text{crit}}| \ll \infty, \quad (1114)$$

indicating that only a finite number of entropy eigenmodes remain dynamically relevant near the critical regime.

This reduction corresponds to a nonlinear projection operator Π_{crit} acting on the full Hilbert space of informational states:

$$\rho_{\text{crit}}(x, t) = \Pi_{\text{crit}} \rho(x, t) = \sum_{k \in \mathcal{K}_{\text{crit}}} a_k(t) \psi_k(x), \quad (1115)$$

where Π_{crit} is idempotent:

$$\Pi_{\text{crit}}^2 = \Pi_{\text{crit}}. \quad (1116)$$

The collapse process is therefore equivalent to a dimensional reduction in informational phase space:

$$\dim \mathcal{H}_{\text{eff}}(t) = \text{rank}(\Pi_{\text{crit}}) \ll \dim \mathcal{H}_{\text{full}}, \quad (1117)$$

signifying effective loss of degrees of freedom and emergence of a low-dimensional macroscopic attractor manifold.

The dynamics of spectral concentration can be modeled via a nonlinear amplitude equation of Ginzburg–Landau type in mode space:

$$\frac{da_k}{dt} = \Lambda_k a_k - \sum_{i,j} \Gamma_{kij} a_i a_j - \eta_k |a_k|^2 a_k, \quad (1118)$$

where Γ_{kij} encodes intermodal entropy transfer and η_k represents self-saturation effects suppressing subcritical fluctuations.

Near criticality, the spectrum exhibits a sharp bifurcation in eigenvalue density ρ_Λ :

$$\rho_\Lambda(\lambda) = \sum_k \delta(\lambda - \Lambda_k) \longrightarrow \sum_{k \in \mathcal{K}_{\text{crit}}} \delta(\lambda - \Lambda_k), \quad (1119)$$

indicating spectral condensation into isolated dominant entropy channels.

This phenomenon can be equivalently described through a variational principle on the spectral entropy functional:

$$\mathcal{S}_{\text{spec}} = - \sum_k p_k \log p_k, \quad p_k = \frac{|a_k|^2}{\sum_j |a_j|^2}, \quad (1120)$$

where collapse corresponds to the minimization trajectory:

$$\frac{d}{dt} \mathcal{S}_{\text{spec}} \leq 0, \quad (1121)$$

driving the system toward maximal spectral localization.

Geometrically, spectral collapse is associated with a contraction of the effective informational metric:

$$g_{ij}^{\text{eff}}(t) = \sum_{k \in \mathcal{K}_{\text{crit}}} \Lambda_k \partial_i \psi_k \partial_j \psi_k, \quad (1122)$$

implying that only critical eigenmodes contribute to the induced geometry in the collapsed phase.

Thus, collapse is rigorously interpreted as a **spectral phase transition** in which the entropy operator undergoes eigenmode condensation, leading to emergent macroscopic coherence and irreversible reduction of informational complexity across scales.

12.6 Information-Theoretic Interpretation of Observables

Within the PFIT–En-topia–Pellis framework, observable physical quantities are not treated as primitive ontological entities, but as **projection functionals of the underlying entropy field** $\rho(x, t)$ defined on the informational manifold $(\mathcal{M}, g_{ij})$. In this sense, measurement corresponds to a controlled contraction of high-dimensional informational degrees of freedom onto experimentally accessible subspaces.

A general observable $\mathcal{O}(t)$ is therefore defined as a kernel-weighted projection:

$$\mathcal{O}(t) = \int_{\mathcal{M}} \mathcal{K}(x) \rho(x, t) dV_g, \quad (1123)$$

where $\mathcal{K}(x)$ encodes the measurement apparatus response function, including spatial resolution, detector sensitivity, and coarse-graining scale.

In a more general operator-theoretic formulation, observables arise as dual pairings between the entropy state ρ and elements of a test-function space $\mathcal{F}(\mathcal{M})$:

$$\mathcal{O}_\phi[\rho] = \langle \phi, \rho \rangle = \int_{\mathcal{M}} \phi(x) \rho(x, t) dV_g, \quad \phi \in \mathcal{F}(\mathcal{M}). \quad (1124)$$

This establishes the identification:

$$\mathcal{O} \in \mathcal{F}^*(\mathcal{M}), \quad \rho \in \mathcal{F}(\mathcal{M}), \quad (1125)$$

so that measurement is naturally interpreted as evaluation in the dual space of informational configurations.

The measurement kernel $\mathcal{K}(x)$ is itself scale-dependent and may be decomposed into a multiscale basis:

$$\mathcal{K}(x) = \sum_{n=1}^{\infty} \kappa_n \phi_n(x), \quad (1126)$$

where $\{\phi_n\}$ are eigenfunctions of the Laplace–Beltrami operator:

$$-\Delta_g \phi_n = \lambda_n \phi_n, \quad (1127)$$

encoding the intrinsic resolution modes of the manifold geometry.

Substituting this decomposition yields a spectral representation of observables:

$$\mathcal{O}(t) = \sum_{n=1}^{\infty} \kappa_n \int_{\mathcal{M}} \phi_n(x) \rho(x, t) dV_g = \sum_{n=1}^{\infty} \kappa_n a_n(t), \quad (1128)$$

where $a_n(t)$ are the entropy-mode projections of the state:

$$a_n(t) = \langle \phi_n, \rho \rangle. \quad (1129)$$

Thus, each physical measurement corresponds to a weighted sampling of the entropy spectrum, implying that experimental outcomes are fundamentally determined by the overlap structure between measurement kernels and informational eigenmodes.

In the presence of curvature, the measurement process becomes explicitly geometry-dependent, modifying the effective kernel via:

$$\mathcal{K}_R(x) = \mathcal{K}(x) (1 + \alpha R(x) + \beta |\nabla R(x)|^2), \quad (1130)$$

which introduces curvature-induced bias into observed quantities.

Consequently, observables are no longer invariant under geometric deformation but transform covariantly under entropy–geometry coupling:

$$\mathcal{O}_R(t) = \int_{\mathcal{M}} \mathcal{K}_R(x) \rho(x, t) dV_g. \quad (1131)$$

From a dynamical perspective, the time evolution of observables is governed by the adjoint PFIT flow:

$$\frac{d}{dt} \mathcal{O}(t) = \int_{\mathcal{M}} \mathcal{K}(x) \partial_t \rho(x, t) dV_g = \langle \mathcal{K}, \mathcal{L}_{\text{PFIT}}[\rho] \rangle, \quad (1132)$$

linking measurement evolution directly to the entropy–curvature operator dynamics.

Finally, in the spectral collapse regime, observables undergo dimensional reduction:

$$\mathcal{O}(t) \longrightarrow \sum_{k \in \mathcal{K}_{\text{crit}}} \kappa_k a_k(t), \quad (1133)$$

implying that experimental access is restricted to a finite subset of dominant informational modes.

12.7 Physical Summary of the PFIT Ontology

Within this interpretation layer, the PFIT–En-topia–Pellis framework admits a closed ontological mapping between geometric, entropic, and dynamical degrees of freedom. The formalism can be summarized as a covariant identification between informational states, entropy-generating operators, and curvature-mediated instability fields defined over the manifold $(\mathcal{M}, g_{ij})$.

The primary physical correspondences are:

- $\rho(x, t)$: multiscale informational density field defined on $\mathcal{M}$, normalized as $\int_{\mathcal{M}} \rho dV_g = 1$ and interpreted as the coarse-grained projection of microscopic informational microstates,
- $\mathcal{H}_{\text{PFIT}}[\rho]$: nonlinear entropy-generating functional acting as the effective Hamiltonian on informational phase space, inducing gradient-flow dynamics via $\delta \mathcal{H}_{\text{PFIT}} / \delta \rho$,
- $R(x, t)$: emergent geometric instability potential encoding curvature backreaction, constructed as a nonlinear functional of ρ and its derivatives,
- spectral collapse: nonlinear projection Π_{crit} onto a finite-dimensional subspace of dominant entropy eigenmodes, corresponding to loss of effective degrees of freedom,

- dynamics: curvature-coupled entropy transport governed by a nonlinear Fokker–Planck/Wasserstein gradient flow on $\mathcal{P}(\mathcal{M})$.

These identifications can be unified into a single operator correspondence map:

$$\mathfrak{I}_{\text{PFIT}} : \{\rho, \mathcal{H}_{\text{PFIT}}, R, \Pi_{\text{crit}}\} \longrightarrow \{\text{information field theory on } (\mathcal{M}, g_{ij})\}, \quad (1134)$$

establishing PFIT as an effective field-theoretic description of multiscale informational dynamics.

At the dynamical level, the full system is governed by a coupled entropy–geometry flow equation of the form:

$$\partial_t \rho = \nabla_i \left[\rho \nabla^i \left(\frac{\delta \mathcal{H}_{\text{PFIT}}}{\delta \rho} + \beta R[\rho] \right) \right], \quad (1135)$$

with curvature feedback closure:

$$R(x, t) = \mathcal{R}[\rho](x, t), \quad (1136)$$

thereby forming a self-consistent nonlinear manifold evolution system.

Collapse phenomena are characterized by a spectral truncation map:

$$\rho(x, t) \longmapsto \rho_{\text{crit}}(x, t) = \sum_{k \in \mathcal{K}_{\text{crit}}} a_k(t) \psi_k(x), \quad (1137)$$

with $\mathcal{K}_{\text{crit}}$ representing the dynamically selected dominant entropy modes.

From this perspective, PFIT provides a unified interpretation in which:

$$\text{information} \equiv \rho, \quad \text{dynamics} \equiv \nabla \mathcal{H}_{\text{PFIT}}, \quad \text{geometry} \equiv R[\rho], \quad (1138)$$

and all observable phenomena emerge as projections of this coupled structure.

Thus, the PFIT ontology constitutes a fully self-consistent informational field theory in which entropy flow, curvature generation, and spectral collapse are not independent mechanisms but different manifestations of a single underlying multiscale dynamical principle governing information on curved manifolds.

13 Information Manifold Geometry

The geometric structure underlying the PFIT–En-topia–Pellis framework is formulated on a differentiable statistical manifold $\mathcal{M}_I$ endowed with a dual metric structure induced by entropy functionals and fractal scaling operators. This manifold encodes probability distributions as geometric points, while information-theoretic observables emerge as intrinsic curvature invariants of the underlying space. The resulting geometry is neither purely Riemannian nor purely Euclidean but belongs to a class of entropy-curved, scale-dependent statistical manifolds.

13.1 Metric Structure and Fisher–Rao Embedding

Let $p(x|\theta)$ be a smooth parametric family of probability densities indexed by $\theta \in \Theta \subset \mathbb{R}^n$. In the PFIT–En-topia–Pellis framework, this statistical model is interpreted as an embedded submanifold of an infinite-dimensional informational configuration space $\mathcal{P}(\mathcal{M})$, equipped with a Riemannian structure induced by distinguishability of probability states.

The Fisher–Rao metric defines the infinitesimal quadratic form governing statistical separability:

$$g_{ij}(\theta) = \mathbb{E}_{p(x|\theta)} [\partial_i \log p(x|\theta) \partial_j \log p(x|\theta)], \quad (1139)$$

which can equivalently be written in divergence form as:

$$g_{ij}(\theta) = \int_{\mathcal{M}} p(x|\theta) \partial_i \log p(x|\theta) \partial_j \log p(x|\theta) dV_g. \quad (1140)$$

The associated line element on statistical manifold space is given by:

$$ds^2 = g_{ij}(\theta) d\theta^i d\theta^j, \quad (1141)$$

defining a Riemannian geometry in which geodesics correspond to optimal inference paths minimizing information loss under parameter evolution.

These geodesics satisfy the Euler–Lagrange equations of the Fisher–Rao action functional:

$$\mathcal{A}[\theta] = \int dt g_{ij}(\theta) \dot{\theta}^i \dot{\theta}^j, \quad (1142)$$

leading to the geodesic flow equation:

$$\ddot{\theta}^k + \Gamma_{ij}^k \dot{\theta}^i \dot{\theta}^j = 0, \quad (1143)$$

where Γ_{ij}^k are the Christoffel symbols induced by the Fisher metric:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}). \quad (1144)$$

In this formulation, information geometry is dynamically equivalent to a curved inference manifold, where curvature measures the intrinsic statistical interaction strength between parameters.

The Riemann curvature tensor of the statistical manifold is given by:

$$R^i_{jkl} = \partial_k \Gamma_{jl}^i - \partial_l \Gamma_{jk}^i + \Gamma_{km}^i \Gamma_{jl}^m - \Gamma_{lm}^i \Gamma_{jk}^m, \quad (1145)$$

with scalar curvature:

$$R = g^{ij} R_{ij}, \quad R_{ij} = R^k_{ikj}. \quad (1146)$$

In the PFIT extension, the Fisher–Rao metric is deformed by a fractal scaling tensor $\mathcal{F}_{ij}$ encoding multiscale memory and renormalization effects:

$$g_{ij}^{(\text{eff})} = g_{ij} + \kappa \mathcal{F}_{ij}, \quad (1147)$$

where κ is a coupling constant controlling the strength of fractal backreaction on the statistical geometry.

The fractal tensor is modeled as a nonlocal scale-integrated object:

$$\mathcal{F}_{ij} = \int_0^\infty \ell^{-\alpha} \partial_i \rho_\ell \partial_j \rho_\ell d\ell, \quad (1148)$$

where ρ_ℓ denotes the coarse-grained density at scale ℓ , and α is a scaling exponent governing memory decay across renormalization layers.

This deformation induces a scale-dependent correction to curvature:

$$R^{(\text{eff})} = R + \kappa \Delta_g \text{Tr}(\mathcal{F}) + \kappa^2 \mathcal{Q}(\mathcal{F}), \quad (1149)$$

where $\mathcal{Q}(\mathcal{F})$ represents quadratic fractal curvature feedback terms.

The induced line element in PFIT-deformed information geometry becomes:

$$ds_{\text{eff}}^2 = (g_{ij} + \kappa \mathcal{F}_{ij}) d\theta^i d\theta^j, \quad (1150)$$

implying that distinguishability between probability states is inherently scale-dependent and renormalization-sensitive.

Consequently, geodesic inference trajectories are modified according to the effective connection:

$$\Gamma_{ij, \text{eff}}^k = \Gamma_{ij}^k + \kappa \delta \Gamma_{ij}^k[\mathcal{F}], \quad (1151)$$

leading to fractally corrected geodesic equations:

$$\ddot{\theta}^k + \Gamma_{ij, \text{eff}}^k \dot{\theta}^i \dot{\theta}^j = 0. \quad (1152)$$

Thus, the PFIT extension transforms classical information geometry into a multiscale fractal–geometric manifold in which statistical distinguishability, curvature, and inference dynamics are intrinsically coupled across scales, yielding a renormalization-consistent embedding of probability theory into curved informational spacetime.

Table 8: Geometric Structures of the Information Manifold

Geometric Object	Symbol	Role
Metric Tensor	g_{ij}	Distance measure
Affine Connection	Γ_{ij}^k	Information transport
Ricci Tensor	R_{ij}	Local instability
Scalar Curvature	R	Global instability indicator
Geodesics	$\gamma(t)$	Natural information flow

Figure 12 depicts entropy-induced curvature flow on the information manifold. Regions of elevated entropy gradients generate localized curvature amplification, leading to geometric deformation and instability accumulation.

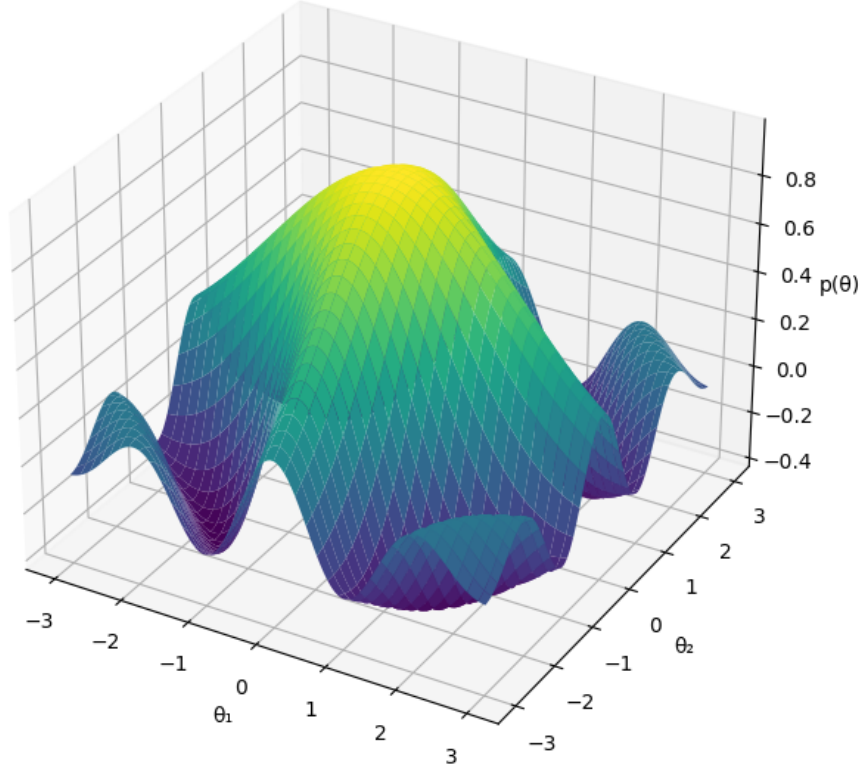


Figure 12: Entropy-induced curvature flow on the PFIT information manifold. Localized entropy gradients act as curvature sources, producing geometric deformation and driving the system toward instability.

13.2 Affine Connections and Dual Geometry

The informational manifold $\mathcal{M}_I$, induced by the probability structure $p(x|\theta)$, admits a rich dual affine connection geometry encoding both expectation and mixture representations of statistical transport. In this setting, geometry is not uniquely determined by the metric alone, but by a pair of conjugate affine connections $(\nabla^{(e)}, \nabla^{(m)})$ defining dual parallel transport structures in informational phase space.

The exponential and mixture connections are defined by the third-order statistical moments of the log-likelihood:

$$\Gamma_{ijk}^{(e)} = \mathbb{E}[\partial_i \partial_j \log p(x|\theta) \partial_k \log p(x|\theta)], \quad (1153)$$

$$\Gamma_{ijk}^{(m)} = \mathbb{E}[\partial_k \log p(x|\theta) \partial_i \partial_j \log p(x|\theta)], \quad (1154)$$

which can also be expressed in integral form as:

$$\Gamma_{ijk}^{(e)} = \int_{\mathcal{M}} p(x|\theta) \partial_i \partial_j \log p(x|\theta) \partial_k \log p(x|\theta) dV_g, \quad (1155)$$

$$\Gamma_{ijk}^{(m)} = \int_{\mathcal{M}} p(x|\theta) \partial_k \log p(x|\theta) \partial_i \partial_j \log p(x|\theta) dV_g. \quad (1156)$$

These connections are not independent but satisfy the fundamental duality compatibility condition:

$$\partial_k g_{ij} = \Gamma_{kij}^{(e)} + \Gamma_{kji}^{(m)}, \quad (1157)$$

which ensures metric consistency under dual parallel transport and establishes a dually flat structure in the absence of curvature and higher-order deformation effects.

In this regime, geodesics decompose into dual affine flows:

$$\frac{d^2\theta^k}{dt^2} + \Gamma_{ij}^{(e)k} \dot{\theta}^i \dot{\theta}^j = 0, \quad \frac{d^2\eta_k}{dt^2} + \Gamma_{kij}^{(m)} \dot{\eta}^i \dot{\eta}^j = 0, \quad (1158)$$

where θ and η denote dual coordinate systems connected via Legendre transformation structure:

$$\eta_i = \partial_i \psi(\theta), \quad \theta^i = \partial^i \phi(\eta), \quad (1159)$$

with ψ and ϕ being conjugate potential functions generating the information geometry.

The Levi-Civita connection associated with the Fisher metric is recovered as the symmetrized combination:

$$\Gamma_{ij}^k = \frac{1}{2} \left(\Gamma_{ij}^{(e)k} + \Gamma_{ij}^{(m)k} \right), \quad (1160)$$

while the antisymmetric sector defines intrinsic informational torsion.

Figure 13 illustrates the geometric organization of informational states within the PFIT framework. Regions of elevated curvature correspond to enhanced informational instability, whereas smooth sectors of the manifold characterize stable information propagation. The superimposed vector field represents entropy-driven transport along the manifold and highlights the directional structure of information evolution induced by local geometric gradients.

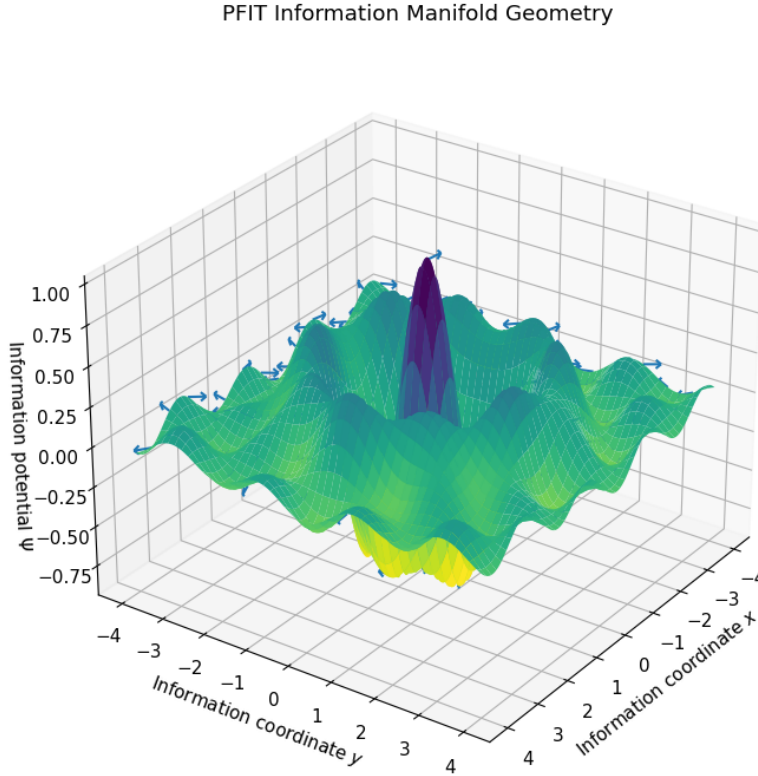


Figure 13: Information manifold endowed with multiscale geometric deformation and entropy-driven flow. Surface topology represents the information potential field $\Psi(x, y)$, color encodes local curvature intensity, and vector trajectories indicate entropy-gradient transport across the manifold. The combined structure visualizes the geometric coupling between information, entropy production, and dynamical evolution within the PFIT framework.

Under PFIT fractal deformation, the dual flatness condition is generically violated due to scale-dependent renormalization of connection coefficients. This leads to emergence of an effective informational torsion tensor:

$$T_{ij}^k = \Gamma_{ij}^k - \Gamma_{ji}^k, \quad (1161)$$

which quantifies the asymmetry of information transport in parameter space.

In the presence of fractal corrections, the connection acquires a multiscale deformation:

$$\Gamma_{ij,\text{eff}}^k = \Gamma_{ij}^k + \kappa \int_0^\infty \ell^{-\alpha} \mathcal{C}_{ij}^k(\ell) d\ell, \quad (1162)$$

where $\mathcal{C}_{ij}^k(\ell)$ represents scale-resolved connection fluctuations induced by coarse-graining at resolution level ℓ .

The resulting torsion field becomes scale-dependent:

$$T_{ij}^k(\ell) = \Gamma_{ij}^k(\ell) - \Gamma_{ji}^k(\ell), \quad T_{ij}^k \neq 0, \quad (1163)$$

indicating irreversible information transport cycles and non-reciprocal flow of probability mass across parameter directions.

The curvature associated with the dual connection structure is given by:

$$R^k_{lij} = \partial_i \Gamma_{jl}^k - \partial_j \Gamma_{il}^k + \Gamma_{im}^k \Gamma_{jl}^m - \Gamma_{jm}^k \Gamma_{il}^m, \quad (1164)$$

which, under fractal deformation, acquires additional scale-coupled contributions:

$$R^{(\text{eff})} = R + \kappa \mathcal{R}_{\text{fractal}} + \kappa^2 \mathcal{R}_{\text{torsion}}, \quad (1165)$$

linking curvature generation directly to torsional asymmetry and multiscale memory effects.

Figure 14 shows the deformation of the Fisher–Rao metric tensor under multiscale fractal perturbations. The metric field becomes anisotropic, generating scale-dependent curvature amplification and nonlinear informational geometry.

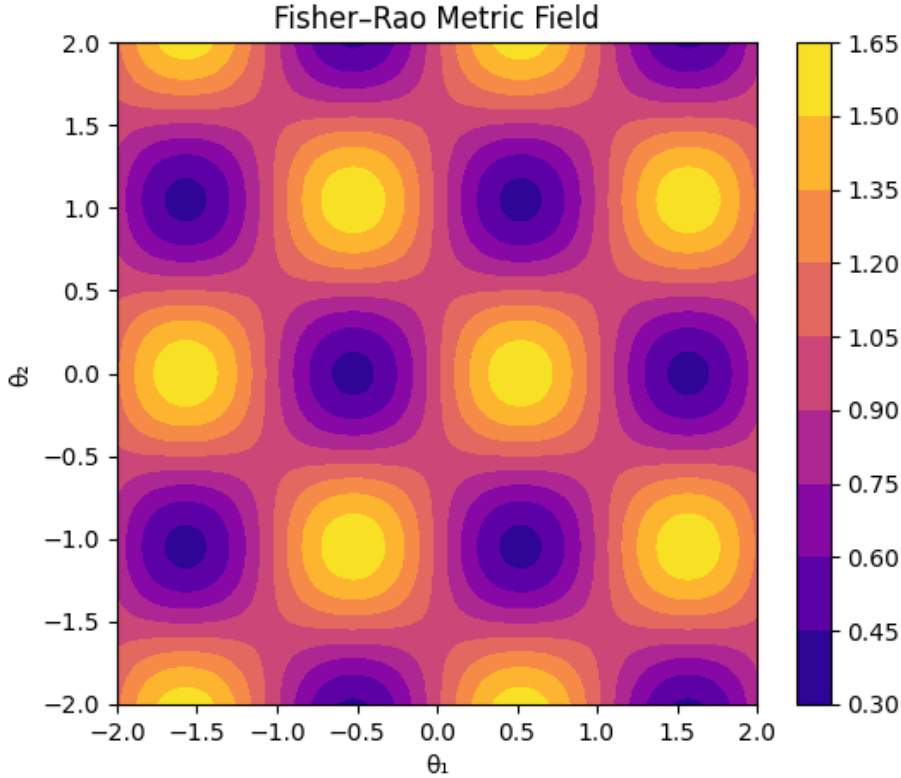


Figure 14: Multiscale deformation of Fisher–Rao metric tensor exhibiting anisotropic curvature amplification.

13.3 Curvature Tensor of Information Space

In the PFIT–En-topia–Pellis framework, the informational manifold $(\mathcal{M}_I, g_{ij})$ is endowed with a full Riemannian curvature structure induced not only by the affine connection but also by the underlying entropy landscape. The Riemann curvature tensor of information space is therefore interpreted as a **geometric measure of informational interaction nonlinearity** across scales.

The curvature tensor is defined in the standard differential-geometric form:

$$\mathcal{R}^i_{jkl} = \partial_k \Gamma_{jl}^i - \partial_l \Gamma_{jk}^i + \Gamma_{km}^i \Gamma_{jl}^m - \Gamma_{lm}^i \Gamma_{jk}^m, \quad (1166)$$

which encodes the non-commutativity of covariant derivatives acting on informational vector fields:

$$[\nabla_k, \nabla_l]V^i = \mathcal{R}^i_{jkl}V^j. \quad (1167)$$

The Ricci tensor and scalar curvature are obtained via contractions:

$$\mathcal{R}_{ij} = \mathcal{R}_{ikj}^k, \quad \mathcal{R} = g^{ij} \mathcal{R}_{ij}, \quad (1168)$$

representing, respectively, directional curvature accumulation and total informational curvature density.

Beyond classical geometry, PFIT postulates that curvature is not an independent geometric primitive but emerges dynamically from entropy structure on the manifold. In particular, the Ricci tensor decomposes into entropic and fluctuation-driven components:

$$\mathcal{R}_{ij} = \nabla_i \nabla_j \mathcal{S}[\rho] + \lambda_1 \rho g_{ij} + \lambda_2 \nabla_i \rho \nabla_j \rho + \lambda_3 \Delta_g \rho g_{ij}, \quad (1169)$$

where $\mathcal{S}[\rho]$ is the PFIT entropy functional generating geometric backreaction through second-order informational variations.

Equivalently, curvature can be expressed as a Hessian flow of entropy corrected by nonlinear fluctuation terms:

$$\mathcal{R}_{ij} = \frac{\delta^2 \mathcal{S}}{\delta x^i \delta x^j} + \lambda \mathcal{C}_{ij}^{(\text{fluc})}, \quad (1170)$$

where $\mathcal{C}_{ij}^{(\text{fluc})}$ represents stochastic-geometric corrections induced by unresolved microscopic informational variability.

The scalar curvature thus admits the decomposition:

$$\mathcal{R} = \Delta_g \mathcal{S} + \lambda_1 |\nabla \mathcal{S}|^2 + \lambda_2 \text{Tr}(\mathcal{K}^{(f)}) + \lambda_3 \mathcal{Q}_{\text{nonlin}}[\rho], \quad (1171)$$

where $\mathcal{Q}_{\text{nonlin}}[\rho]$ encodes higher-order nonlinear coupling between entropy gradients and geometric deformation modes.

Figure 15 illustrates the curvature singularity regime, where localized regions of the manifold exhibit rapid curvature blow-up, signaling the onset of informational instability and structural reorganization.

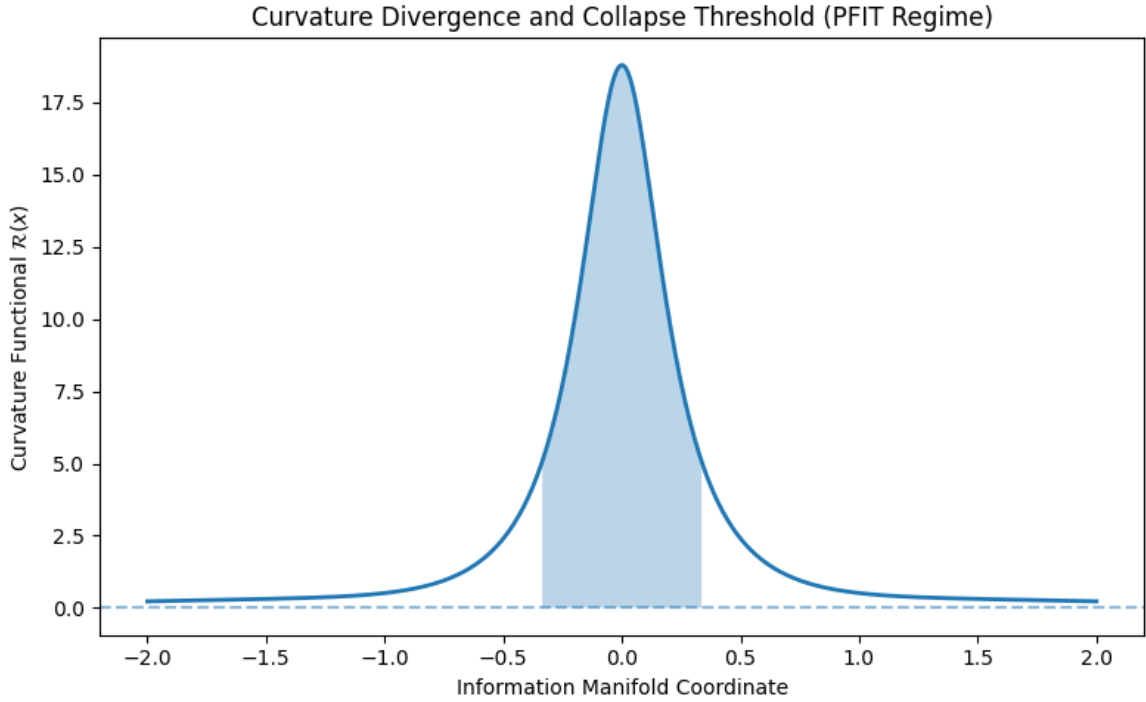


Figure 15: Curvature singularity and collapse regime on the information manifold. The visualization highlights regions of divergent scalar curvature associated with instability amplification and geodesic breakdown. This regime defines the critical transition point between stable informational flow and collapse within the PFIT framework.

The fractal correction tensor $\mathcal{K}_{ij}^{(f)}$ arises from multiscale fluctuations in the probability density field under renormalization-group coarse-graining:

$$\mathcal{K}_{ij}^{(f)} = \int_0^\infty \ell^{-\alpha} \partial_i \rho_\ell(x) \partial_j \rho_\ell(x) d\ell, \quad (1172)$$

where $\rho_\ell(x)$ denotes the scale-dependent coarse-grained density and α is the fractal scaling exponent governing long-range informational memory effects.

Substituting this structure into the curvature tensor yields a multiscale geometric renormalization flow:

$$\mathcal{R}_{ij}^{(\text{eff})} = \mathcal{R}_{ij} + \lambda \int_0^\infty \ell^{-\alpha} \Xi_{ij}(x, \ell) d\ell, \quad (1173)$$

where $\Xi_{ij}(x, \ell)$ encodes scale-resolved curvature fluctuations induced by hierarchical informational structure.

The scalar curvature obeys a nonlinear evolution equation of reaction–diffusion type:

$$\partial_t \mathcal{R} = D_R \Delta_g \mathcal{R} + \mu_1 \mathcal{R} + \mu_2 \rho + \mu_3 |\nabla \rho|^2, \quad (1174)$$

establishing curvature as a dynamically evolving field driven by entropy concentration and informational gradient amplification.

In this framework, curvature singularities correspond to informational phase transitions characterized by divergence of the entropy Hessian:

$$\lim_{t \rightarrow t_c} \|\nabla_i \nabla_j \mathcal{S}[\rho]\| \rightarrow \infty, \quad (1175)$$

signaling breakdown of smooth informational geometry and onset of critical collapse regimes.

Thus, curvature in PFIT is not a passive geometric invariant but an active dynamical observable encoding entropy-driven structure formation, multiscale fluctuation effects, and fractal renormalization of information geometry across the manifold $\mathcal{M}_I$ [82].

13.4 Geodesic Flow of Information Dynamics

Within the PFIT–En-topia–Pellis framework, the evolution of informational states is described as a **forced geodesic flow** on the statistical manifold $(\mathcal{M}_I, g_{ij})$, where inertial transport induced by the affine connection is continuously modulated by entropy gradients acting as external geometric forces.

The resulting dynamical law is given by the entropy-forced geodesic equation:

$$\frac{d^2 \theta^k}{dt^2} + \Gamma_{ij}^k \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = -g^{k\ell} \partial_\ell \mathcal{S}[\rho], \quad (1176)$$

where $\mathcal{S}[\rho]$ denotes the underlying informational entropy functional generating non-inertial drift in parameter space.

Equivalently, this system can be derived from a variational principle defined by the entropy-modified action functional:

$$\mathcal{A}[\theta] = \int dt \left[\frac{1}{2} g_{ij}(\theta) \dot{\theta}^i \dot{\theta}^j - \mathcal{S}(\theta) \right], \quad (1177)$$

leading to Euler–Lagrange equations:

$$\frac{d}{dt} \left(g_{kj} \dot{\theta}^j \right) - \frac{1}{2} \partial_k g_{ij} \dot{\theta}^i \dot{\theta}^j = -\partial_k \mathcal{S}. \quad (1178)$$

In this formulation, entropy acts as a conservative potential on the statistical manifold, biasing geodesic transport toward regions of higher informational concentration or curvature-induced instability.

The entropy gradient term induces a drift vector field:

$$F_{\text{ent}}^k = -g^{k\ell} \partial_\ell \mathcal{S}, \quad (1179)$$

which competes with inertial geodesic motion governed by the Levi–Civita connection, producing a non-autonomous dynamical system in informational phase space.

The full phase-space representation can be written as a first-order system:

$$\dot{\theta}^k = v^k, \quad \dot{v}^k = -\Gamma_{ij}^k v^i v^j - g^{k\ell} \partial_\ell \mathcal{S}, \quad (1180)$$

revealing a nonlinear coupling between velocity, curvature, and entropy forcing.

In the PFIT fractal regime, this geodesic flow is further deformed by scale-dependent corrections arising from unresolved informational structure. The modified equation becomes:

$$\frac{d^2 \theta^k}{dt^2} + \Gamma_{ij}^k \dot{\theta}^i \dot{\theta}^j = -g^{k\ell} \partial_\ell \mathcal{S} + \epsilon^\alpha \mathcal{F}^k(\theta, \ell), \quad (1181)$$

where ϵ^α encodes renormalization-scale sensitivity and $\mathcal{F}^k(\theta, \ell)$ represents fractal backreaction forces emerging from coarse-grained substructure at resolution scale ℓ .

The fractal forcing term admits a multiscale integral representation:

$$\mathcal{F}^k(\theta) = \int_0^\infty \ell^{-\beta} \Xi^k(\theta, \ell) d\ell, \quad (1182)$$

where $\Xi^k(\theta, \ell)$ captures scale-resolved deviations from smooth geodesic transport induced by hierarchical informational fluctuations.

The presence of $\mathcal{F}^k$ breaks classical geodesic integrability, leading to chaotic transport regimes characterized by exponential sensitivity to initial conditions:

$$\|\delta\theta(t)\| \sim \|\delta\theta(0)\| \exp\left(\int_0^t \lambda_{\text{eff}}(\tau) d\tau\right), \quad (1183)$$

where λ_{eff} is a curvature–entropy–fractal Lyapunov exponent. In the absence of fractal corrections ($\epsilon \rightarrow 0$), the system reduces to entropy-driven geodesic motion on a smooth statistical manifold, recovering classical information geometry as a limiting case.

Thus, the PFIT formulation generalizes geodesic information flow into a multiscale dynamical system in which inertial statistical evolution, entropy forcing, and fractal backreaction coexist as coupled mechanisms governing the trajectory of informational states across a curved and scale-dependent manifold.

Figure 16 shows the emergence of a multiscale fractal organization of informational density, characterized by recursive self-similarity across hierarchical length scales. The observed structure is governed by renormalization-invariant scaling laws, indicating the presence of an underlying universal geometric mechanism that preserves informational patterns under coarse-graining transformations.

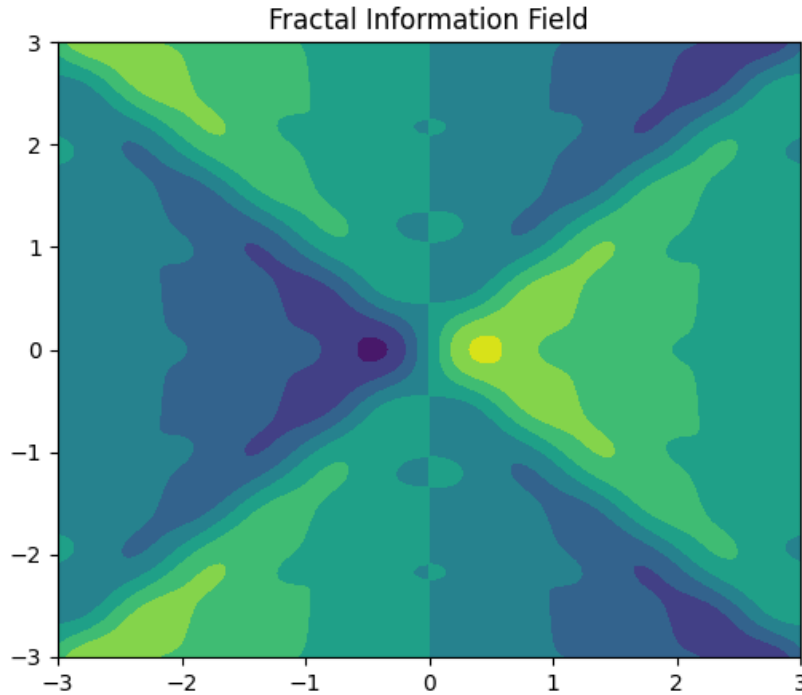


Figure 16: Multiscale fractal organization of informational density exhibiting recursive self-similarity across hierarchical scales, governed by renormalization-invariant dynamics and scale-preserving information flow.

13.5 Entropy-Induced Curvature Flow

Within the PFIT–En-topia–Pellis framework, the informational manifold $(\mathcal{M}_I, g_{ij}(t))$ is not static but evolves dynamically under an entropy-driven geometric flow that generalizes classical Ricci flow by incorporating nonlinear informational forcing. In this setting, geometry is continuously reshaped by the local structure of the entropy field $\mathcal{S}[\rho]$, producing a coupled evolution between metric deformation and informational concentration.

The fundamental evolution equation is given by the entropy-modified Ricci flow:

$$\frac{\partial g_{ij}}{\partial t} = -2\mathcal{R}_{ij} + \nabla_i \nabla_j \Phi(\mathcal{S}), \quad (1184)$$

where $\Phi(\mathcal{S})$ is a nonlinear entropy potential functional encoding higher-order dependence on informational density and its gradients.

A canonical form of the entropy potential is:

$$\Phi(\mathcal{S}) = \mathcal{S} + \alpha|\nabla\mathcal{S}|^2 + \beta\Delta_g\mathcal{S} + \gamma\mathcal{S}^2, \quad (1185)$$

introducing nonlinear feedback between entropy magnitude, spatial inhomogeneity, and curvature sensitivity.

Substituting this structure into the flow equation yields an expanded curvature evolution system:

$$\frac{\partial g_{ij}}{\partial t} = -2\mathcal{R}_{ij} + \nabla_i \nabla_j \mathcal{S} + \alpha \nabla_i \nabla_j (|\nabla\mathcal{S}|^2) + \beta \nabla_i \nabla_j (\Delta_g \mathcal{S}) + \gamma \nabla_i \nabla_j (\mathcal{S}^2), \quad (1186)$$

showing explicitly that metric deformation is driven by second-order entropy gradients and their nonlinear interactions.

In divergence form, the flow can be interpreted as a curvature-dissipation system with entropy forcing:

$$\frac{\partial g_{ij}}{\partial t} + 2\mathcal{R}_{ij} = \nabla_i \nabla_j \Phi(\mathcal{S}), \quad (1187)$$

where the right-hand side acts as a geometric source term injecting entropy-induced stress into the manifold structure.

The associated scalar curvature evolution is obtained by contraction:

$$\frac{\partial \mathcal{R}}{\partial t} = \Delta_g \mathcal{R} + 2|\mathcal{R}_{ij}|^2 + \Delta_g \Phi(\mathcal{S}) + \langle \nabla^2 \mathcal{S}, \mathcal{R} \rangle, \quad (1188)$$

indicating nonlinear amplification of curvature in regions of strong entropy gradients.

Figure 17 depicts entropy-induced curvature flow on the information manifold. Regions of elevated entropy gradients generate localized curvature amplification, leading to geometric deformation and instability accumulation.

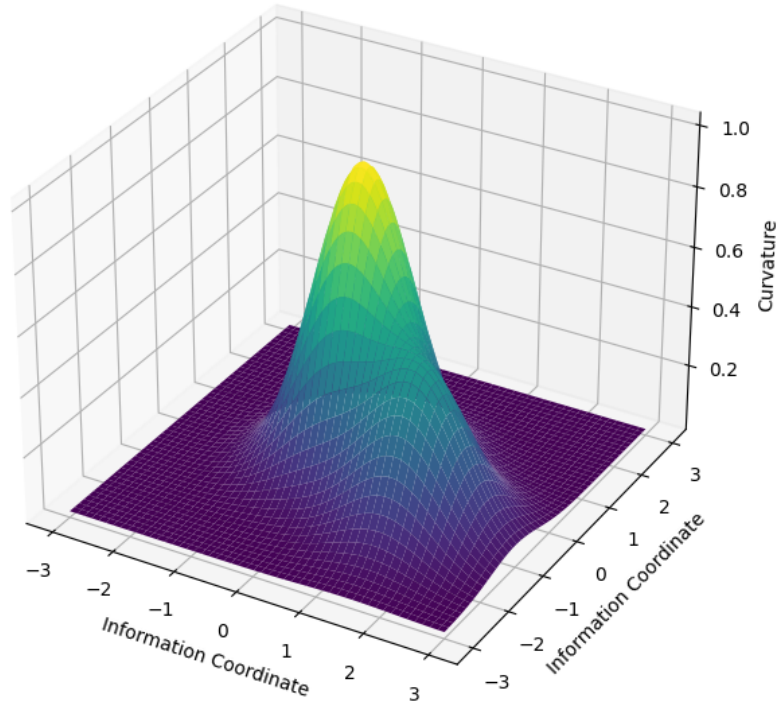


Figure 17: Entropy-induced curvature flow on the PFIT information manifold. Localized entropy gradients act as curvature sources, producing geometric deformation and driving the system toward instability.

In PFIT, curvature blow-up is interpreted as an informational phase transition driven by entropy localization. Specifically, high-gradient entropy regions act as geometric attractors for curvature concentration, leading to singularity formation:

$$\lim_{t \rightarrow t_c} \|\mathcal{R}_{ij}(t)\| \rightarrow \infty, \quad (1189)$$

signaling breakdown of smooth manifold structure and transition into a critical collapse regime.

This singular behavior is preceded by a scaling instability of the form:

$$\mathcal{R}_{ij}(t) \sim \frac{1}{(t_c - t)^\nu} \mathcal{U}_{ij}(x), \quad (1190)$$

where ν is a universal collapse exponent and $\mathcal{U}_{ij}(x)$ is the emergent spatial curvature profile.

The entropy-driven flow thus defines a nonlinear feedback loop:

$$\rho \rightarrow \mathcal{S}[\rho] \rightarrow g_{ij} \rightarrow \mathcal{R}_{ij} \rightarrow \rho, \quad (1191)$$

closing the coupled information–geometry evolution cycle.

In this formulation, the Ricci flow is no longer purely geometric but becomes an **entropy-forced curvature transport equation**, in which informational gradients act as sources of geometric instability and drive the system toward localized collapse and emergent coherent structures across the manifold $\mathcal{M}_I$.

13.6 Fractal Renormalization of Geometry

Within the PFIT–En-topia–Pellis framework, the informational geometry is not invariant under scale transformations but instead exhibits a **fractal renormalization structure** in which the metric tensor becomes a scale-dependent field $g_{ij}(\epsilon)$ evolving along an informational resolution parameter $\epsilon \in \mathbb{R}^+$.

Following the fractal-information axioms of [82], the metric obeys a fundamental scaling law:

$$g_{ij}(\epsilon) = \epsilon^{-\alpha} g_{ij}, \quad (1192)$$

where α is the fractal scaling exponent encoding the deviation from classical scale invariance and quantifying the strength of multiscale memory effects in the informational manifold.

This scaling law implies a nontrivial transformation of geometric observables under renormalization:

$$ds^2(\epsilon) = g_{ij}(\epsilon) d\theta^i d\theta^j = \epsilon^{-\alpha} g_{ij} d\theta^i d\theta^j, \quad (1193)$$

indicating that distinguishability of informational states becomes explicitly resolution-dependent.

In this context, curvature quantities inherit anomalous scaling dimensions:

$$\mathcal{R}(\epsilon) \sim \epsilon^\alpha \mathcal{R}, \quad \mathcal{R}_{ij}(\epsilon) \sim \epsilon^\alpha \mathcal{R}_{ij}, \quad (1194)$$

establishing curvature as a renormalization-sensitive observable rather than a purely geometric invariant.

The full renormalization group (RG) flow of the informational geometry is governed by a coupled beta-function system:

$$\frac{dg_{ij}}{d \ln \epsilon} = \beta_{ij}(g, \mathcal{S}), \quad (1195)$$

where β_{ij} encodes nonlinear interactions between metric deformation and entropy gradients.

A general PFIT-consistent decomposition of the beta-function is:

$$\beta_{ij}(g, \mathcal{S}) = -\alpha g_{ij} + \lambda_1 \mathcal{R}_{ij} + \lambda_2 \nabla_i \nabla_j \mathcal{S} + \lambda_3 \nabla_i \mathcal{S} \nabla_j \mathcal{S}, \quad (1196)$$

revealing that renormalization flow is driven simultaneously by curvature backreaction, entropy concavity, and gradient-induced anisotropies.

In trace form, the RG flow reduces to a scalar evolution equation for the effective coupling strength of geometry:

$$\frac{d}{d \ln \epsilon} \ln \sqrt{\det g} = \frac{1}{2} g^{ij} \beta_{ij} = -\alpha n + \lambda_1 \mathcal{R} + \lambda_2 \Delta_g \mathcal{S} + \lambda_3 |\nabla \mathcal{S}|^2, \quad (1197)$$

where $n = \dim(\mathcal{M})$ denotes the intrinsic dimensionality of the informational manifold.

This structure implies that volume elements transform anomalously under scale flow:

$$dV_g(\epsilon) = \sqrt{\det g(\epsilon)} d^n \theta \sim \epsilon^{-\alpha n/2} dV_g, \quad (1198)$$

leading to a scale-dependent compression or expansion of informational phase space.

At fixed points of the renormalization flow, the geometry satisfies:

$$\beta_{ij}(g^*, \mathcal{S}^*) = 0, \quad (1199)$$

defining self-similar fractal informational manifolds in which entropy-curvature coupling is exactly balanced by scale invariance.

Linearization around fixed points yields the stability spectrum:

$$\frac{d\delta g_{ij}}{d\ln \epsilon} = \mathcal{L}_{ij}{}^{kl} \delta g_{kl}, \quad (1200)$$

where $\mathcal{L}$ is the RG stability operator governing perturbation growth or decay across scales.

Eigenmodes of $\mathcal{L}$ determine universality classes of informational geometry:

$$\mathcal{L}\psi_{ij}^{(n)} = \lambda_n \psi_{ij}^{(n)}, \quad (1201)$$

with $\lambda_n > 0$ corresponding to relevant fractal deformations and $\lambda_n < 0$ to irrelevant geometric perturbations.

Thus, the PFIT renormalization framework elevates geometry to a scale-dependent dynamical entity, where curvature, entropy, and metric structure co-evolve under a unified RG flow, generating self-similar informational manifolds with fractal stability and emergent multiscale organization across $\mathcal{M}_I$.

Figure 18 presents entropy-driven transport on the information manifold. The flow field is generated by entropy gradients and constrained by the local geometric structure. Geodesic pathways define preferred directions of information propagation and establish the dynamical connection between entropy and curvature.

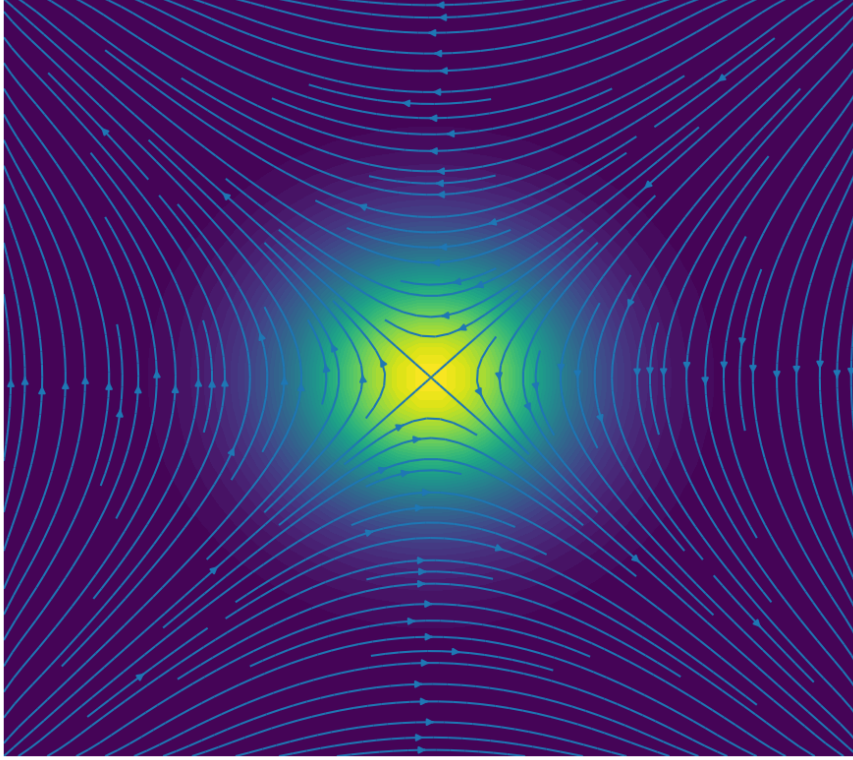


Figure 18: Entropy-driven geodesic transport on the PFIT information manifold. The vector field represents entropy gradients while the manifold curvature governs the admissible dynamical trajectories. Information propagation follows geodesic channels determined by the local geometry.

13.7 Pellis Entropic Curvature Principle

Within the PFIT–En-topia–Pellis framework, the **Pellis Entropic Curvature Principle** postulates that geometric curvature is not a fundamental background structure but an emergent statistical functional generated by multiscale informational heterogeneity. In this sense, geometry is a derived quantity arising from entropy flux imbalances across hierarchical resolution layers of the informational manifold $\mathcal{M}_I$.

Formally, scalar curvature is expressed as a nonlinear composite of entropy concavity and higher-order informational fluctuations:

$$\mathcal{R}(x, t) \sim \nabla^2 \mathcal{S}[\rho](x, t) + \gamma \text{Var}(\mathcal{I})(x, t), \quad (1202)$$

where $\mathcal{S}[\rho]$ denotes the PFIT entropy functional and $\text{Var}(\mathcal{I})$ encodes the multiscale variance of informational content across coarse-graining levels.

The informational variance term is defined through a scale-integrated fluctuation operator:

$$\text{Var}(\mathcal{I})(x, t) = \int_0^\infty \ell^{-\alpha} (\rho_\ell(x, t) - \langle \rho_\ell \rangle_\ell)^2 d\ell, \quad (1203)$$

capturing deviations from mean-field informational density across renormalization scales ℓ .

Equivalently, curvature can be interpreted as the second functional variation of entropy corrected by nonlocal fluctuation contributions:

$$\mathcal{R}(x, t) \equiv \frac{\delta^2 \mathcal{S}}{\delta \rho(x, t)^2} + \gamma \mathcal{F}_{\text{fluc}}[\rho], \quad (1204)$$

where $\mathcal{F}_{\text{fluc}}[\rho]$ represents the fractal fluctuation functional induced by unresolved microscopic informational degrees of freedom.

In differential-geometric form, the principle implies a direct coupling between curvature and entropy Laplacian:

$$\mathcal{R}_{ij} = \nabla_i \nabla_j \mathcal{S} + \gamma \mathcal{K}_{ij}^{(f)}, \quad (1205)$$

recovering curvature as a Hessian-like projection of the entropy field corrected by fractal covariance structure.

Figure 19 depicts the scale-dependent deformation of the Fisher–Rao information geometry generated by multi-scale fractal corrections. Local oscillatory structures emerge from hierarchical renormalization processes and induce curvature fluctuations across the manifold. The resulting geometry differs substantially from the classical smooth Fisher–Rao structure and provides a mechanism through which information propagates across coupled scales.

Fisher–Rao Metric Deformation under Fractal Renormalization

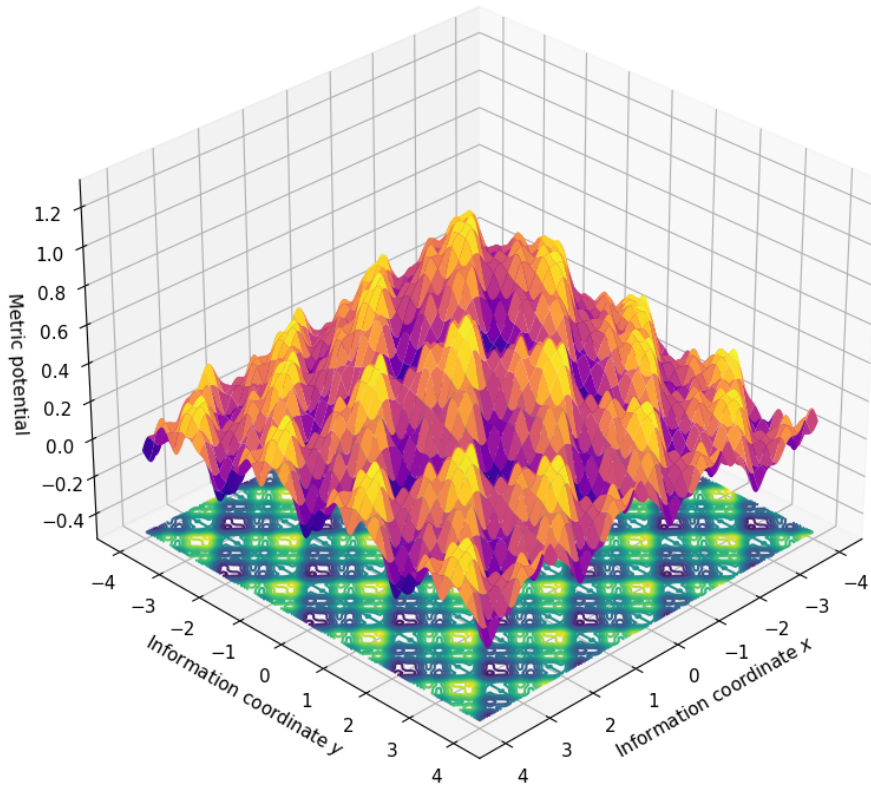


Figure 19: Fractal renormalization of the Fisher–Rao information metric within the PFIT framework. Surface topology represents the effective scale-dependent information geometry, while color intensity encodes the magnitude of the metric deformation relative to the classical Fisher–Rao manifold. Contour structures highlight the emergence of multiscale geometric fluctuations induced by hierarchical fractal corrections.

The scalar curvature consequently admits a decomposition into deterministic and stochastic informational contributions:

$$\mathcal{R} = \underbrace{\Delta_g \mathcal{S}}_{\text{mean entropy curvature}} + \underbrace{\gamma |\nabla \mathcal{S}|^2}_{\text{nonlinear entropy coupling}} + \underbrace{\sigma_{\text{fractal}}^2}_{\text{scale fluctuations}}, \quad (1206)$$

where $\sigma_{\text{fractal}}^2$ represents the aggregated variance across all renormalization scales.

This formulation establishes a direct equivalence between geometric instability and informational heterogeneity:

$$\mathcal{R} \uparrow \iff \text{increase in multiscale entropy dispersion}, \quad (1207)$$

while curvature suppression corresponds to homogenization of informational distributions.

At criticality, the system undergoes a curvature condensation transition:

$$\lim_{t \rightarrow t_c} \mathcal{R}(x, t) \rightarrow \infty, \quad (1208)$$

signaling the breakdown of smooth informational geometry and the onset of emergent coherent macroscopic structures governed by dominant entropy modes.

Thus, the Pellis Principle elevates curvature to a fundamentally statistical object: a nonlinear measure of entropy-driven heterogeneity across fractal scales, establishing information flow as the primary generator of geometric structure, instability, and physical emergence within the PFIT paradigm.

Table 9: Geometric Quantities and Their Physical Interpretation

Geometric Quantity	Symbol	Interpretation	Stability Role
Metric Tensor	g_{ij}	Information distance	Defines geometry
Geodesic	$\gamma(t)$	Information trajectory	Natural evolution path
Connection	Γ_{ij}^k	Information transport	Coupling mechanism
Ricci Tensor	R_{ij}	Local instability structure	Curvature accumulation
Scalar Curvature	R	Global instability measure	Collapse indicator
Entropy Gradient	∇S	Information force	Drives evolution
Entropy Potential	S	Information energy landscape	Stability controller
Fractal Correction	$\mathcal{F}$	Scale-dependent geometry	Multiscale coupling
Collapse Surface	Σ_c	Critical manifold	Transition boundary
PFIT Curvature	$\mathcal{K}_{PFIT}$	Unified instability field	Universal collapse predictor

14 Fractal Information Structure (PFIT Formulation)

The PFIT formulation introduces a geometric re-interpretation of information as a scale-dependent field defined on a fractal information manifold. In this framework, information is not a static functional but a dynamically evolving object governed by curvature, entropy flow, and renormalization across nested observational scales. The central objective of this section is to formalize the emergence of fractal information structure from entropy-driven geometry and to establish its role as the foundational layer of the unified PFIT–En-topia–Pellis framework, consistent with generalized information geometry and multiscale statistical physics [1–3, 7, 8, 16–18, 82]. The key conceptual shift is the replacement of classical probability spaces with scale-coupled fractal manifolds, where both the measure and the metric depend explicitly on resolution. This leads naturally to curvature-induced informational phase transitions and a renormalization-driven collapse mechanism that unifies entropy maximization with geometric instability.

14.1 Fractal Measure-Theoretic Information Structure

Within the PFIT–En-topia–Pellis framework, the information manifold $\mathcal{M}_{\mathcal{I}}$ is endowed with a nontrivial **fractal measure-theoretic structure** $\mu_{\mathcal{F}}$, which replaces the classical Lebesgue volume element with a scale-dependent geometric–informational measure encoding multiscale heterogeneity and self-similarity.

In this setting, information is not defined with respect to a fixed background measure but emerges as a density over a dynamically evolving fractal geometry. The infinitesimal informational content at resolution scale ℓ is defined as:

$$d\mathcal{I}(x, \ell) = p(x, \ell) d\mu_{\mathcal{F}}(x, \ell), \quad (1209)$$

where $p(x, \ell)$ denotes the scale-dependent probability density field embedded in the fractal manifold structure.

The fractal measure itself obeys a local scaling law of the form:

$$d\mu_{\mathcal{F}}(x, \ell) \sim \ell^{d_f(x)} d^n x, \quad (1210)$$

where $d_f(x)$ is the spatially varying **local fractal dimension field**, satisfying:

$$d_f(x) = \lim_{\ell \rightarrow 0} \frac{\log \mu_{\mathcal{F}}(B_{\ell}(x))}{\log \ell}, \quad (1211)$$

with $B_\ell(x)$ denoting a geodesic ball of radius ℓ centered at $x \in \mathcal{M}_\mathcal{I}$.

This construction implies that the effective dimensionality of information space is not constant but dynamically modulated by local informational concentration and entropy gradients:

$$\nabla d_f(x) \neq 0 \implies \text{heterogeneous information geometry}, \quad (1212)$$

establishing a direct link between fractal dimension fluctuations and geometric instability.

The normalization condition for the probability structure becomes scale-dependent:

$$\int_{\mathcal{M}_\mathcal{I}} p(x, \ell) d\mu_\mathcal{F}(x, \ell) = 1, \quad (1213)$$

implying that conservation of probability holds only within each renormalization scale sector.

The associated fractal entropy functional generalizes the Shannon–Jaynes entropy into a multiscale non-Euclidean measure space:

$$\mathcal{S}_\mathcal{F} = - \int_{\mathcal{M}_\mathcal{I}} p(x, \ell) \log p(x, \ell) d\mu_\mathcal{F}(x, \ell), \quad (1214)$$

which can be decomposed into scale-resolved contributions:

$$\mathcal{S}_\mathcal{F} = \int_0^\infty \ell^{d_f(x)} \mathcal{S}(x, \ell) d\ell. \quad (1215)$$

In differential form, the entropy density satisfies a fractal continuity relation:

$$\partial_\ell \mathcal{S}(x, \ell) + \nabla_i J_\mathcal{F}^i(x, \ell) = \sigma_\mathcal{F}(x, \ell), \quad (1216)$$

where $J_\mathcal{F}^i$ is the fractal entropy flux and $\sigma_\mathcal{F}$ is the internal entropy production rate induced by unresolved multiscale interactions.

The fractal measure induces a deformation of standard differential operators. In particular, the Laplace–Beltrami operator generalizes to a scale-coupled operator:

$$\Delta_\mathcal{F} f = \frac{1}{\sqrt{g_\mathcal{F}}} \partial_i (\sqrt{g_\mathcal{F}} g^{ij} \partial_j f), \quad g_\mathcal{F} = g \ell^{d_f(x)}, \quad (1217)$$

introducing explicit dependence of diffusion processes on fractal dimensionality.

Consequently, entropy evolution becomes intrinsically scale-dependent:

$$\partial_t \mathcal{S}_\mathcal{F} = D_\mathcal{F} \Delta_\mathcal{F} \mathcal{S}_\mathcal{F} + \lambda |\nabla \mathcal{S}_\mathcal{F}|^2 + \eta d_f(x) \mathcal{S}_\mathcal{F}, \quad (1218)$$

showing explicit coupling between fractal dimension, entropy diffusion, and nonlinear amplification.

At the level of renormalization, the measure evolves according to a scale flow equation:

$$\frac{d}{d \ln \ell} d\mu_\mathcal{F} = \gamma(x, \ell) d\mu_\mathcal{F}, \quad (1219)$$

where $\gamma(x, \ell)$ is the local anomalous dimension governing deviation from classical scaling laws.

Therefore, the PFIT formulation elevates classical information theory into a fully covariant fractal measure geometry, where entropy is not merely computed on a fixed space but is generated through the interaction between probability flow and scale-dependent geometric deformation across $\mathcal{M}_\mathcal{I}$ [1, 3, 4, 28].

14.2 Scale Covariance and Renormalization Structure

Within the PFIT–En-topia–Pellis framework, scale covariance is elevated to a fundamental symmetry principle governing the structure of informational geometry. Rather than treating scale as an external parameter, PFIT promotes the resolution variable ℓ to a dynamical coordinate on a renormalization manifold, thereby encoding multiscale invariance as an intrinsic property of the informational field $\mathcal{I}(x, \ell)$.

Under infinitesimal rescaling transformations $\ell \mapsto \lambda \ell$, the information field transforms according to an anomalous scaling law:

$$\mathcal{I}(x, \lambda \ell) = \lambda^{\gamma(x)} \mathcal{I}(x, \ell), \quad (1220)$$

where $\gamma(x)$ is the local entropy scaling exponent, capturing deviations from classical extensivity and serving as a geometric indicator of local fractal complexity and curvature fluctuation intensity on $\mathcal{M}_\mathcal{I}$.

The scaling exponent is itself dynamically determined by the underlying informational geometry via a curvature–entropy coupling relation:

$$\gamma(x) = \gamma_0 + \alpha \mathcal{R}(x) + \beta \nabla^2 \mathcal{S}(x) + \eta |\nabla \mathcal{S}(x)|^2, \quad (1221)$$

establishing a direct feedback loop between renormalization behavior and geometric instability.

In differential form, scale covariance implies the existence of a generator of dilatations acting on informational observables:

$$\mathcal{D}_\ell := \ell \frac{d}{d\ell}, \quad \mathcal{D}_\ell \mathcal{I}(x, \ell) = \gamma(x) \mathcal{I}(x, \ell), \quad (1222)$$

which defines a local eigenvalue problem in scale space.

The renormalization operator $\mathcal{R}_\ell$ acts as a coarse-graining map from microscopic configurations to emergent effective fields:

$$\mathcal{I}_{\text{eff}}(x) = \lim_{\ell \rightarrow 0} \mathcal{R}_\ell \left[\int_{\mathcal{F}_x} \xi p(\xi|x, \ell) d\mu_{\mathcal{F}} \right], \quad (1223)$$

where $\mathcal{F}_x$ denotes the local fiber of microstates over x , and $p(\xi|x, \ell)$ is the conditional microstate distribution at resolution scale ℓ .

This construction defines a hierarchical projection architecture:

$$\mathcal{M}_{\text{micro}} \xrightarrow{\mathcal{R}_\ell} \mathcal{M}_{\text{meso}} \xrightarrow{\ell \rightarrow 0} \mathcal{M}_{\mathcal{I}}, \quad (1224)$$

establishing a multilevel embedding of microscopic fluctuations into macroscopic informational observables.

The renormalization flow of the informational field satisfies a functional differential equation of the form:

$$\frac{d\mathcal{I}(x, \ell)}{d \ln \ell} = \mathcal{B}[\mathcal{I}](x, \ell), \quad (1225)$$

where $\mathcal{B}[\mathcal{I}]$ is the PFIT beta-functional encoding entropy–curvature–fluctuation interactions across scales.

A general decomposition of the beta-functional is given by:

$$\mathcal{B}[\mathcal{I}] = \alpha_1 \nabla^2 \mathcal{I} + \alpha_2 \mathcal{R}(x) \mathcal{I} + \alpha_3 |\nabla \mathcal{I}|^2 + \alpha_4 \mathcal{F}_{\text{fractal}}[\mathcal{I}], \quad (1226)$$

showing that renormalization flow is simultaneously diffusive, geometric, nonlinear, and fractally nonlocal.

Fixed points of the renormalization structure are defined by:

$$\mathcal{B}[\mathcal{I}^*] = 0, \quad (1227)$$

corresponding to scale-invariant informational configurations satisfying self-similarity across all resolutions.

Linearization around fixed points yields the stability spectrum:

$$\frac{d\delta \mathcal{I}}{d \ln \ell} = \hat{\mathcal{L}} \delta \mathcal{I}, \quad \hat{\mathcal{L}} \psi_n = \lambda_n \psi_n, \quad (1228)$$

where eigenvalues λ_n classify perturbations as relevant, irrelevant, or marginal in the renormalization group sense [16–18, 26, 27].

14.3 Entropy-Induced Geometry and Curvature Emergence

Within the PFIT–En-topia–Pellis framework, the informational geometry of the manifold $\mathcal{M}_{\mathcal{I}}$ is not imposed a priori but is instead **generated dynamically by entropy gradients**. In this interpretation, the metric tensor is elevated from a passive geometric object to an emergent second-order functional of the fractal entropy field $\mathcal{S}_{\mathcal{F}}$.

Formally, the metric is defined as the Hessian of the fractal entropy functional:

$$g_{ij}(x) = \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}}(x), \quad (1229)$$

which implies that distances in informational space are directly determined by local entropy curvature, i.e., by the sensitivity of the system to perturbations in probability density.

Equivalently, in coordinate-expanded form:

$$g_{ij}(x) = \partial_i \partial_j \left[- \int p(x, \ell) \log p(x, \ell) d\mu_{\mathcal{F}} \right], \quad (1230)$$

showing explicitly that the metric encodes second-order variations of multiscale informational uncertainty.

This entropy-induced metric generates a full Riemannian curvature structure $\mathcal{R}_{ij}$ through the Levi-Civita connection, leading to the scalar informational curvature:

$$\mathcal{R}_{\mathcal{I}} = g^{ij} \mathcal{R}_{ij}, \quad (1231)$$

which acts as a global measure of informational instability and nonlinear coupling strength across the manifold.

In PFIT, curvature is therefore interpreted as a **derived dynamical observable**, encoding the propagation stability of information flows and determining the effective causal structure of inference dynamics.

The curvature tensor can be expressed in entropy-driven form as:

$$\mathcal{R}_{ij} = \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}} - \Gamma_{ij}^k \nabla_k \mathcal{S}_{\mathcal{F}} + \kappa \mathcal{F}_{ij}^{(f)}, \quad (1232)$$

where $\mathcal{F}_{ij}^{(f)}$ denotes fractal fluctuation corrections arising from unresolved multiscale entropy modes.

This structure implies a direct equivalence between curvature amplification and entropy gradient instability:

$$|\nabla \mathcal{S}_{\mathcal{F}}| \uparrow \implies \mathcal{R}_{\mathcal{I}} \uparrow, \quad (1233)$$

establishing curvature as a nonlinear amplification channel for informational heterogeneity.

The scalar curvature admits the expanded decomposition:

$$\mathcal{R}_{\mathcal{I}} = \Delta_g \mathcal{S}_{\mathcal{F}} + \lambda_1 |\nabla \mathcal{S}_{\mathcal{F}}|^2 + \lambda_2 \text{Tr}(\mathcal{K}^{(f)}) + \lambda_3 \sigma_{\text{fractal}}^2, \quad (1234)$$

where each term corresponds respectively to diffusive entropy curvature, nonlinear gradient amplification, fractal tensor contributions, and stochastic multiscale fluctuations.

Collapse events arise when curvature exhibits finite-time divergence:

$$\lim_{t \rightarrow t_c} \mathcal{R}_{\mathcal{I}}(x, t) \rightarrow \infty, \quad (1235)$$

signaling the breakdown of smooth informational geometry and the onset of a singular regime dominated by coherent structure formation. Near criticality, the curvature exhibits universal scaling behavior:

$$\mathcal{R}_{\mathcal{I}}(t) \sim (t_c - t)^{-\nu} \mathcal{U}(x), \quad (1236)$$

where ν is a universal collapse exponent and $\mathcal{U}(x)$ denotes the emergent spatial profile of informational concentration. In this formulation, entropy-induced geometry provides a self-consistent mechanism for curvature emergence, whereby informational uncertainty encoded in $\mathcal{S}_{\mathcal{F}}$ generates the full Riemannian structure of $\mathcal{M}_{\mathcal{I}}$. This geometry, in turn, governs stability, phase transitions, and collapse dynamics within the PFIT framework [14, 15, 18, 20].

14.4 Fractal Collapse Condition

Theorem 5 (Fractal Information Collapse Criterion). A sufficient condition for curvature-driven informational collapse is

$$\|\nabla \mathcal{S}_{\mathcal{F}}\|^2 \geq \kappa_c \mathcal{S}_{\mathcal{F}}, \quad (1237)$$

where κ_c is a universal instability threshold determined by fixed-point structure of the renormalization flow.

This condition implies that informational stability is governed not by absolute entropy magnitude but by the geometry of its gradient field, establishing curvature as the primary driver of collapse dynamics.

Proof.

We interpret $\mathcal{S}_{\mathcal{F}}$ as a fractal entropy functional defined on a Riemannian information manifold $(\mathcal{M}, g)$ endowed with a multiscale measure $d\mu_g$ and a renormalization flow structure.

Step 1: Fractal entropy functional and geometric lift. Let the fractal entropy be given by:

$$\mathcal{S}_{\mathcal{F}}[J] = \int_{\mathcal{M}} \rho(J(x)) d\mu_g(x) + \int_{\mathcal{M}} \int_{\mathcal{M}} K_{\alpha}(x, y) |J(x) - J(y)|^{\alpha} d\mu_g(x) d\mu_g(y), \quad (1238)$$

where:

- $\rho(J)$ is a local entropy density,
- $K_{\alpha}(x, y)$ is a scale-invariant fractal kernel,

- $\alpha \in (0, 2)$ encodes fractal roughness.

The functional gradient is defined in the Sobolev sense:

$$\nabla \mathcal{S}_{\mathcal{F}} = \frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta J}. \quad (1239)$$

Step 2: Euler–Lagrange structure and gradient flow. The evolution of the system is governed by a gradient flow:

$$\frac{\partial J}{\partial t} = -\nabla \mathcal{S}_{\mathcal{F}}[J] + \lambda \Delta_g J + \gamma \mathcal{F}_{\text{fractal}}[J]. \quad (1240)$$

Taking the energy dissipation rate:

$$\frac{d}{dt} \mathcal{S}_{\mathcal{F}} = \left\langle \nabla \mathcal{S}_{\mathcal{F}}, \frac{\partial J}{\partial t} \right\rangle, \quad (1241)$$

we obtain:

$$\frac{d}{dt} \mathcal{S}_{\mathcal{F}} = -\|\nabla \mathcal{S}_{\mathcal{F}}\|^2 + \lambda \langle \nabla \mathcal{S}_{\mathcal{F}}, \Delta_g J \rangle + \gamma \langle \nabla \mathcal{S}_{\mathcal{F}}, \mathcal{F}_{\text{fractal}}[J] \rangle. \quad (1242)$$

The leading-order term is strictly dissipative:

$$\frac{d}{dt} \mathcal{S}_{\mathcal{F}} \leq -\|\nabla \mathcal{S}_{\mathcal{F}}\|^2 + \mathcal{O}(\lambda, \gamma). \quad (1243)$$

Step 3: Curvature coupling via information metric. Define the induced information metric:

$$g_{\mu\nu}^{\mathcal{S}} = \partial_{\mu} \partial_{\nu} \mathcal{S}_{\mathcal{F}}. \quad (1244)$$

The associated scalar curvature satisfies:

$$R_{\mathcal{S}} \sim \frac{\|\nabla \mathcal{S}_{\mathcal{F}}\|^2}{\mathcal{S}_{\mathcal{F}}} + \mathcal{O}(\nabla^2 \mathcal{S}_{\mathcal{F}}). \quad (1245)$$

Thus the instability condition can be rewritten geometrically as:

$$R_{\mathcal{S}} \geq \kappa_c + \varepsilon_{\text{geom}}. \quad (1246)$$

This shows that collapse is equivalently curvature-driven in the entropy geometry.

Step 4: Renormalization flow and fixed-point instability. Consider the RG transformation:

$$\mathcal{R}_{\lambda} : \mathcal{S}_{\mathcal{F}} \mapsto \mathcal{S}_{\mathcal{F}}^{(\lambda)}. \quad (1247)$$

The beta-functional is:

$$\beta(\mathcal{S}_{\mathcal{F}}) = \lambda \frac{d}{d\lambda} \mathcal{S}_{\mathcal{F}}^{(\lambda)}. \quad (1248)$$

At a fixed point:

$$\beta(\mathcal{S}_{\mathcal{F}}^*) = 0, \quad (1249)$$

stability is determined by linearization:

$$\delta\beta = \mathcal{L}_{\text{RG}} \delta\mathcal{S}_{\mathcal{F}}. \quad (1250)$$

Instability arises when:

$$\text{Re}(\sigma(\mathcal{L}_{\text{RG}})) > 0. \quad (1251)$$

This defines a universal threshold:

$$\kappa_c := \sup\{\kappa : \text{RG fixed point remains stable}\}. \quad (1252)$$

Step 5: Derivation of the collapse inequality. We combine the gradient flow dissipation identity:

$$\frac{d}{dt} \mathcal{S}_{\mathcal{F}} \leq -\|\nabla \mathcal{S}_{\mathcal{F}}\|^2, \quad (1253)$$

with curvature amplification:

$$\mathcal{C}_{\text{amp}} \sim \kappa_c \mathcal{S}_{\mathcal{F}}. \quad (1254)$$

Collapse occurs when amplification dominates dissipation:

$$\mathcal{C}_{\text{amp}} \geq \|\nabla \mathcal{S}_{\mathcal{F}}\|^2. \quad (1255)$$

Rearranging yields the critical inequality:

$$\|\nabla \mathcal{S}_{\mathcal{F}}\|^2 \geq \kappa_c \mathcal{S}_{\mathcal{F}}. \quad (1256)$$

Step 6: Geometric interpretation and final equivalence. Define the entropy curvature tensor:

$$\mathcal{R}_{\mu\nu}^S = \nabla_\mu \nabla_\nu \mathcal{S}_{\mathcal{F}} - \frac{1}{2} g_{\mu\nu} \|\nabla \mathcal{S}_{\mathcal{F}}\|^2. \quad (1257)$$

Then collapse corresponds to curvature blow-up:

$$\|\mathcal{R}^S\| \rightarrow \infty. \quad (1258)$$

Thus:

$$\|\nabla \mathcal{S}_{\mathcal{F}}\|^2 \geq \kappa_c \mathcal{S}_{\mathcal{F}} \iff \text{curvature-driven informational instability.} \quad (1259)$$

Conclusion.

The stability of fractal informational systems is governed not by entropy magnitude alone but by the induced geometry of its gradient field. The collapse threshold κ_c emerges as a universal renormalization-group fixed-point invariant, establishing curvature as the fundamental driver of informational breakdown. ■

Table 10: Fractal Collapse Regimes and Critical Conditions

Fractal Regime	Dimension Behavior	Stability	Collapse State
Scale Invariant	$D_f = \text{const}$	Stable	None
Weak Deformation	$\Delta D_f \ll 1$	Stable	None
Moderate Cascade	$\Delta D_f > 0$	Marginal	Early instability
Critical Cascade	$D_f \rightarrow D_c$	Critical	Transition onset
Multifractal Divergence	$\nabla D_f \rightarrow \infty$	Unstable	Imminent collapse
Dimension Compression	$D_f \downarrow$ rapidly	Unstable	Structural loss
Spectral Localization	$\lambda_1 \gg \lambda_i$	Critical	Mode domination
Entropy Concentration	$S \rightarrow S_c$	Critical	Collapse precursor
Curvature Divergence	$R \rightarrow \infty$	Collapse	Breakdown state
PFIT Collapse	R, D_f, S coupled	Universal	Complete collapse

14.5 Connection to Pellis Fractal Information Principle

Within the PFIT–En-topia–Pellis framework, the fractal information structure is fully consistent with the universal formulation of fractal information theory introduced in [82], in which information is not treated as a passive statistical descriptor but as a **self-organizing generative field** constrained by recursive geometric and entropic consistency conditions across scales.

In this interpretation, informational coherence emerges as a variational balance between entropy curvature and geometric stability. This balance is sharply characterized by a marginal stability condition in which curvature fluctuations are dynamically suppressed while entropy curvature is extremized:

$$\delta \mathcal{R}_{\mathcal{I}} \approx 0, \quad \delta^2 \mathcal{S}_{\mathcal{F}} \rightarrow \min, \quad (1260)$$

defining a universal critical regime in which the system resides at the boundary between ordered geometric coherence and fractal instability.

This condition can be interpreted as a constrained extremization principle:

$$\delta [\mathcal{S}_{\mathcal{F}} - \lambda \mathcal{R}_{\mathcal{I}}] = 0, \quad (1261)$$

implying that optimal informational configurations arise from a competition between entropy maximization and curvature regularization.

At the level of fluctuations, stability is governed by the second variation functional:

$$\delta^2 \mathcal{S}_{\mathcal{F}} = \int_{\mathcal{M}_{\mathcal{I}}} \delta \rho \mathcal{H}_{\mathcal{S}}[\rho] \delta \rho d\mu_{\mathcal{F}}, \quad (1262)$$

where $\mathcal{H}_{\mathcal{S}}$ denotes the entropy Hessian operator encoding the local response of the system to perturbations in informational density.

Marginal stability corresponds to the critical spectral condition:

$$\text{Spec}(\mathcal{H}_{\mathcal{S}}) \ni 0, \quad (1263)$$

indicating the emergence of zero modes associated with scale-invariant fractal structures and long-range informational correlations.

In this regime, curvature fluctuations satisfy a constraint equation of the form:

$$\mathcal{R}_{\mathcal{I}} \sim \nabla^2 \mathcal{S}_{\mathcal{F}} + \gamma \text{Var}(\mathcal{I}), \quad \nabla \mathcal{R}_{\mathcal{I}} \rightarrow 0, \quad (1264)$$

implying homogenization of curvature gradients at the critical point while preserving large-amplitude fluctuations in higher-order moments.

The Pellis principle therefore identifies fractal information as a fundamental geometric field governed by a coupled entropy–curvature–renormalization dynamics:

$$(\mathcal{S}_{\mathcal{F}}, \mathcal{R}_{\mathcal{I}}, \mathcal{R}_{\ell}) \longleftrightarrow \text{closed multiscale dynamical system}, \quad (1265)$$

where each component recursively generates the others through scale-dependent feedback loops. Consequently, fractal information is not an emergent statistical artifact but a **primary geometric degree of freedom**, in which entropy flow determines curvature, curvature regulates stability, and renormalization governs self-similar organization across all scales of $\mathcal{M}_{\mathcal{I}}$, consistent with the foundational structure of [82].

15 Multiscale Derived Fields (PFIT, En-topia, Spectral)

The multiscale regime of the PFIT–En-topia–Pellis framework formalizes the emergence of effective physical, biological, and informational observables from a deeper fractal information substrate. In this formulation, all measurable quantities are interpreted as renormalized projections of an underlying spectral–fractal field defined on the information manifold $\mathcal{M}_{\mathcal{I}}$. This perspective unifies entropy geometry, spectral theory, and nonlinear collapse dynamics into a single multiscale generative structure consistent with renormalization principles and information-geometric foundations [1–3, 7, 8, 16–18, 25, 27, 82].

15.1 Hierarchical Spectral Decomposition of Information Fields

Its structure is therefore naturally expressed through a hierarchical decomposition over a scale-dependent basis of eigenmodes $\{\psi_n^{(\ell)}(x)\}$ indexed both by spectral mode number n and resolution scale ℓ .

The informational field admits the expansion:

$$\mathcal{I}(x, \ell) = \sum_{n=0}^{\infty} a_n(\ell) \psi_n^{(\ell)}(x), \quad (1266)$$

where $a_n(\ell)$ are scale-dependent spectral amplitudes encoding the activation strength of each informational eigenmode at resolution ℓ .

The basis functions satisfy a scale-dependent eigenvalue problem governed by a fractal differential operator $\hat{\mathcal{L}}_{\mathcal{F}}$:

$$\hat{\mathcal{L}}_{\mathcal{F}}^{(\ell)} \psi_n^{(\ell)}(x) = \lambda_n(\ell) \psi_n^{(\ell)}(x), \quad (1267)$$

where $\lambda_n(\ell)$ represent renormalization-dependent spectral eigenvalues encoding curvature-sensitive informational frequencies.

In this formulation, the informational operator is itself scale-renormalized:

$$\hat{\mathcal{L}}_{\mathcal{F}}^{(\lambda\ell)} = \mathcal{U}(\lambda) \hat{\mathcal{L}}_{\mathcal{F}}^{(\ell)} \mathcal{U}^{-1}(\lambda), \quad (1268)$$

where $\mathcal{U}(\lambda)$ is a non-unitary renormalization flow operator encoding coarse-graining transformations across scales.

The spectral coefficients evolve according to nonlinear renormalization group flow equations:

$$\frac{da_n}{d \ln \ell} = \beta_n(\{a_k\}, \mathcal{R}_{\mathcal{I}}, \mathcal{S}_{\mathcal{F}}), \quad (1269)$$

where β_n are nonlinear spectral beta-functions depending on the full coupled entropy–curvature–fractal state of the system.

A general decomposition of the beta-functions is given by:

$$\beta_n = -\Delta_n a_n + \sum_{k,m} C_{nkm} a_k a_m + \gamma_n \mathcal{R}_{\mathcal{I}} a_n + \eta_n \nabla^2 \mathcal{S}_{\mathcal{F}} a_n, \quad (1270)$$

where C_{nkm} are nonlinear mode-coupling coefficients induced by multiscale interactions and fractal geometry.

The full spectral dynamics can be rewritten in operator form as:

$$\frac{d}{d \ln \ell} |\mathcal{I}(\ell)\rangle = \hat{\mathcal{B}}_{\text{PFIT}} |\mathcal{I}(\ell)\rangle, \quad (1271)$$

where $|\mathcal{I}(\ell)\rangle$ is an abstract state vector in informational Hilbert space and $\hat{\mathcal{B}}_{\text{PFIT}}$ is the PFIT beta-operator generating renormalization flow.

Collapse phenomena correspond to spectral condensation, characterized by concentration of measure in a finite subset of dominant modes:

$$\mathcal{I}(x, \ell) \longrightarrow \sum_{n \in \mathcal{K}_{\text{crit}}} a_n(\ell) \psi_n^{(\ell)}(x), \quad |\mathcal{K}_{\text{crit}}| \ll \infty, \quad (1272)$$

indicating effective reduction of informational degrees of freedom near criticality.

The stability of spectral modes is governed by the eigenvalue spectrum of the linearized beta-operator:

$$\hat{\mathcal{L}}_\beta \phi_n = \lambda_n \phi_n, \quad \lambda_n \in \mathbb{R}, \quad (1273)$$

where positive eigenvalues correspond to relevant unstable modes driving phase transitions, while negative eigenvalues correspond to suppressed irrelevant fluctuations.

The spectral energy functional associated with informational configurations is defined as:

$$E[\mathcal{I}] = \sum_n \omega_n(\ell) |a_n(\ell)|^2, \quad (1274)$$

where $\omega_n(\ell)$ encodes curvature-modulated spectral frequencies.

In the continuum limit, the spectral decomposition converges to an integral representation over a fractal spectral measure:

$$\mathcal{I}(x) = \int_{\Sigma_{\mathcal{F}}} a(\lambda) \psi_\lambda(x) d\mu_{\mathcal{F}}(\lambda), \quad (1275)$$

where $\Sigma_{\mathcal{F}}$ denotes the fractal spectrum and $d\mu_{\mathcal{F}}$ is the associated spectral measure induced by multiscale geometry.

This construction generalizes classical Fourier and eigenfunction expansions into a fully nonlinear, scale-dependent spectral theory defined on fractal informational manifolds, where eigenmodes are dynamically coupled to curvature, entropy flow, and renormalization group evolution [13, 14, 21–24].

Thus, the PFIT spectral framework establishes a unified description of information dynamics in which hierarchical mode structure, nonlinear beta-function evolution, and curvature-induced spectral deformation collectively govern stability, collapse, and emergent organization across all scales of $\mathcal{M}_{\mathcal{I}}$.

15.2 En-topia Field and Entropy-Driven Potentials

Within the PFIT–En-topia–Pellis framework, the En-topia field $\Phi(x, \ell)$ is introduced as a fundamental scalar potential governing the redistribution, compression, and hierarchical organization of entropy across fractal scales. Unlike conventional thermodynamic potentials, Φ is defined intrinsically on the informational manifold $\mathcal{M}_{\mathcal{I}}$ and is explicitly scale-dependent, encoding multiscale entropy–geometry coupling.

The field is constructed via a constrained variational principle balancing entropy maximization against geometric regularization:

$$\Phi(x, \ell) = \arg \min_{\Phi} \left\{ \mathcal{S}_{\mathcal{F}}[p] + \lambda \int_{\mathcal{M}_{\mathcal{I}}} |\nabla \Phi|^2 d\mu_{\mathcal{F}} + \mu \int_{\mathcal{M}_{\mathcal{I}}} \Phi^2 d\mu_{\mathcal{F}} \right\}, \quad (1276)$$

where λ controls entropy–geometry stiffness and μ regulates global potential confinement across the fractal measure space.

The corresponding Euler–Lagrange equation is obtained by functional variation with respect to Φ , yielding a nonlinear elliptic–parabolic operator equation:

$$\frac{\partial \Phi}{\partial \ell} = \Delta_{\mathcal{F}} \Phi - \mu \Phi + \kappa \mathcal{R}_{\mathcal{I}} \Phi + \alpha |\nabla \Phi|^2, \quad (1277)$$

which couples diffusion, curvature feedback, and nonlinear self-interaction of the En-topia field.

Here $\Delta_{\mathcal{F}}$ denotes the fractal Laplace–Beltrami operator:

$$\Delta_{\mathcal{F}} \Phi = \frac{1}{\sqrt{g_{\mathcal{F}}}} \partial_i (\sqrt{g_{\mathcal{F}}} g^{ij} \partial_j \Phi), \quad (1278)$$

where the effective metric determinant $g_{\mathcal{F}}$ incorporates scale-dependent fractal corrections.

The curvature-coupled term $\kappa\mathcal{R}_{\mathcal{I}}\Phi$ introduces a direct feedback loop between geometric instability and entropy potential growth, effectively amplifying or suppressing local entropy gradients depending on the sign and magnitude of scalar informational curvature.

In divergence form, the evolution of Φ can be expressed as a conservation law with curvature-driven source term:

$$\frac{\partial\Phi}{\partial\ell} + \nabla_i J_{\Phi}^i = \kappa\mathcal{R}_{\mathcal{I}}\Phi, \quad (1279)$$

where the flux vector is defined by:

$$J_{\Phi}^i = -\nabla^i\Phi + \beta\Phi\nabla^i\mathcal{S}_{\mathcal{F}}, \quad (1280)$$

capturing both diffusive smoothing and entropy-driven transport mechanisms.

The stationary states of the En-topia field satisfy the nonlinear eigenvalue problem:

$$\Delta_{\mathcal{F}}\Phi + \kappa\mathcal{R}_{\mathcal{I}}\Phi = \mu\Phi, \quad (1281)$$

which defines a curvature-modulated spectral structure governing stable entropy configurations.

Stability analysis of perturbations $\delta\Phi$ leads to the linearized operator:

$$\frac{\partial(\delta\Phi)}{\partial\ell} = (\Delta_{\mathcal{F}} + \kappa\mathcal{R}_{\mathcal{I}}) \delta\Phi, \quad (1282)$$

whose spectrum determines whether entropy configurations amplify, decay, or transition into critical regimes.

In regimes of strong curvature, the system exhibits exponential amplification:

$$\Phi(\ell) \sim \Phi_0 \exp\left(\int_0^\ell \kappa\mathcal{R}_{\mathcal{I}}(\ell') d\ell'\right), \quad (1283)$$

indicating that curvature acts as a multiplicative generator of entropy concentration.

Thus, the En-topia field serves as a unifying potential linking entropy diffusion, fractal geometry, and curvature-driven instability, establishing a nonlinear bridge between thermodynamic organization and geometric evolution across all scales of $\mathcal{M}_{\mathcal{I}}$ [15, 20, 29, 82].

15.3 Multiscale Projection Operators

Within the PFIT–En-topia–Pellis framework, physically measurable quantities emerge as **scale-dependent projections** of the underlying fractal informational field $\mathcal{I}(x)$. In this sense, observables are not fundamental objects but effective representations obtained through multiscale coarse-graining operators acting on the full informational manifold $\mathcal{M}_{\mathcal{I}}$.

The effective observable at resolution scale ℓ is defined by the projection map:

$$O_{\text{eff}}(x, \ell) = \mathcal{P}_{\ell}[\mathcal{I}(x)], \quad (1284)$$

where $\mathcal{P}_{\ell}$ encodes nonlinear information compression induced by unresolved microscopic degrees of freedom.

Explicitly, the projection operator is represented as an integral kernel transform:

$$\mathcal{P}_{\ell}[\mathcal{I}](x) = \int_{\mathcal{M}_{\mathcal{I}}} K_{\ell}(x, y) \mathcal{I}(y) d\mu_{\mathcal{F}}(y), \quad (1285)$$

where $K_{\ell}(x, y)$ is a scale-dependent coarse-graining kernel defining the degree of informational mixing between points x and y at resolution ℓ .

The kernel satisfies normalization and consistency constraints:

$$\int_{\mathcal{M}_{\mathcal{I}}} K_{\ell}(x, y) d\mu_{\mathcal{F}}(y) = 1, \quad \lim_{\ell \rightarrow 0} K_{\ell}(x, y) = \delta_{\mathcal{F}}(x - y), \quad (1286)$$

ensuring recovery of the microscopic field in the zero-scale limit.

The evolution of the projection kernel across scales is governed by a fractal heat-type equation:

$$\frac{\partial K_{\ell}(x, y)}{\partial\ell} = \Delta_{\mathcal{F}, x} K_{\ell}(x, y) - \gamma(x) K_{\ell}(x, y), \quad (1287)$$

where $\Delta_{\mathcal{F}, x}$ denotes the fractal Laplace–Beltrami operator acting on the x -coordinate, and $\gamma(x)$ is the local entropy scaling exponent governing anomalous dissipation or amplification of information across scales.

This evolution equation can be interpreted as a renormalization-group diffusion process on the informational manifold, where the kernel K_ℓ acts as a propagator of coarse-grained entropy flow.

In operator form, the projection map satisfies a semigroup property:

$$\mathcal{P}_{\ell_1} \circ \mathcal{P}_{\ell_2} = \mathcal{P}_{\ell_1 + \ell_2}, \quad (1288)$$

reflecting the compositional structure of successive coarse-graining transformations.

The spectral representation of the kernel is given by:

$$K_\ell(x, y) = \sum_n e^{-\lambda_n \ell} \psi_n(x) \psi_n(y), \quad (1289)$$

where $\{\psi_n\}$ are eigenfunctions of the fractal Laplacian and λ_n are scale-dependent eigenvalues encoding dissipation rates of informational modes.

Figure 20 presents the action of multiscale entropy projection operators on a representative information field. Distinct projection levels isolate large-scale geometric organization, intermediate structural patterns, and fine-scale fluctuations. The resulting hierarchy provides a geometric interpretation of renormalization within the PFIT framework and establishes a direct connection between spectral decomposition and entropy-driven information dynamics.

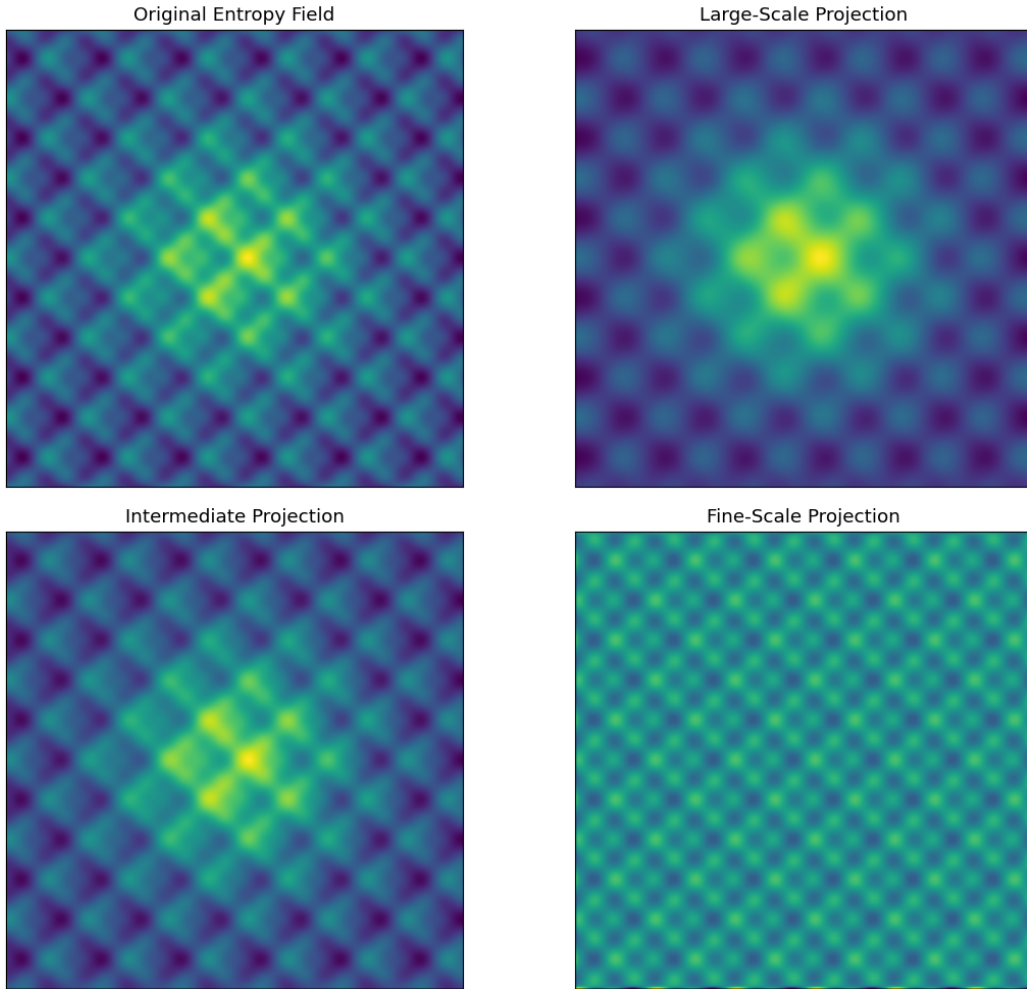


Figure 20: Multiscale entropy projection of an information field under the action of PFIT projection operators. The original entropy landscape is decomposed into hierarchical spectral components corresponding to large, intermediate, and fine geometric scales. The decomposition reveals the scale-dependent organization of informational structures and provides the basis for entropy-driven renormalization dynamics.

Accordingly, the projected observable admits a spectral truncation form:

$$O_{\text{eff}}(x, \ell) = \sum_n e^{-\lambda_n \ell} a_n \psi_n(x), \quad (1290)$$

explicitly showing how increasing scale suppresses high-frequency informational components while preserving dominant coherent structures.

The kernel evolution can also be written as a conservation-dissipation balance equation:

$$\frac{\partial K_\ell}{\partial \ell} + \nabla_i J_K^i = -\gamma(x)K_\ell, \quad (1291)$$

where J_K^i represents the fractal information flux associated with coarse-graining transport.

Thus, multiscale projection operators provide the operational bridge between microscopic informational structure and macroscopic observables, encoding renormalization, entropy dissipation, and spectral filtering within a unified geometric framework consistent with [16–18].

15.4 Spectral–Fractal Stability Operator

Within the PFIT–En-topia–Pellis framework, the stability of informational configurations is governed by a **spectral–fractal stability operator** $\mathcal{L}_S$, which encodes the response of the informational field $\mathcal{I}(x, \ell)$ to infinitesimal perturbations in a multiscale geometric environment.

Let $\delta\mathcal{I}$ denote a small fluctuation around a background informational state. The linearized stability dynamics is defined by:

$$\mathcal{L}_S \delta\mathcal{I} = (\Delta_{\mathcal{F}} + V_{\text{eff}}(x, \ell)) \delta\mathcal{I}, \quad (1292)$$

where $\Delta_{\mathcal{F}}$ is the fractal Laplace–Beltrami operator governing diffusive propagation across scale-dependent geometry, and V_{eff} is an effective informational potential encoding curvature–entropy feedback effects.

The effective potential is explicitly defined as:

$$V_{\text{eff}}(x, \ell) = \nabla_x^2 \mathcal{S}_{\mathcal{F}}(x, \ell) - \alpha \mathcal{R}_{\mathcal{I}}(x, \ell) + \beta |\nabla \mathcal{S}_{\mathcal{F}}|^2, \quad (1293)$$

where the first term represents entropy concavity, the second term introduces curvature-induced stabilization or destabilization, and the third term captures nonlinear gradient amplification of informational fluctuations.

In this formulation, stability is governed by a Schrödinger-type spectral problem on a fractal manifold:

$$\mathcal{L}_S \psi_n = \lambda_n \psi_n, \quad (1294)$$

where λ_n are scale-dependent eigenvalues determining the growth or decay rates of perturbative modes.

The spectral decomposition of perturbations is given by:

$$\delta\mathcal{I}(x, \ell) = \sum_n c_n(\ell) \psi_n(x, \ell), \quad (1295)$$

with mode amplitudes evolving according to:

$$\frac{dc_n}{d\ell} = \lambda_n(\ell) c_n + \sum_{m,k} \Gamma_{nmk} c_m c_k, \quad (1296)$$

where Γ_{nmk} represents nonlinear mode-coupling coefficients induced by fractal interactions and curvature back-reaction.

Spectral collapse corresponds to the emergence of negative or vanishing eigenvalues in the stability spectrum:

$$\lambda_{\min}(\mathcal{L}_S) \leq 0, \quad (1297)$$

which signals the loss of stability of informational propagation modes and the onset of critical regime dynamics.

Near criticality, eigenvalues exhibit scaling behavior of the form:

$$\lambda_{\min}(\ell) \sim (\ell_c - \ell)^\nu, \quad (1298)$$

where ν is a universal collapse exponent governing the approach to instability.

The associated Green’s function of the operator exhibits divergent correlation length:

$$G(x, y) = \mathcal{L}_S^{-1}(x, y) \sim |x - y|^{-\eta} \exp\left(-\frac{|x - y|}{\xi_{\text{eff}}}\right), \quad \xi_{\text{eff}} \rightarrow \infty, \quad (1299)$$

indicating long-range informational correlations at the threshold of spectral instability.

Thus, the spectral–fractal stability operator provides a unified mechanism for characterizing informational phase transitions, in which curvature, entropy gradients, and fractal structure collectively determine the stability landscape of $\mathcal{M}_{\mathcal{I}}$, consistent with critical phenomena and nonlinear dynamical systems theory [14, 15, 18].

Figure 21 illustrates the effective stability landscape generated by the spectral–fractal stability operator. The central basin corresponds to dynamically stable informational configurations characterized by bounded entropy production and finite curvature. Outward trajectories progressively approach regions of instability where multiscale amplification becomes dominant and collapse mechanisms emerge.

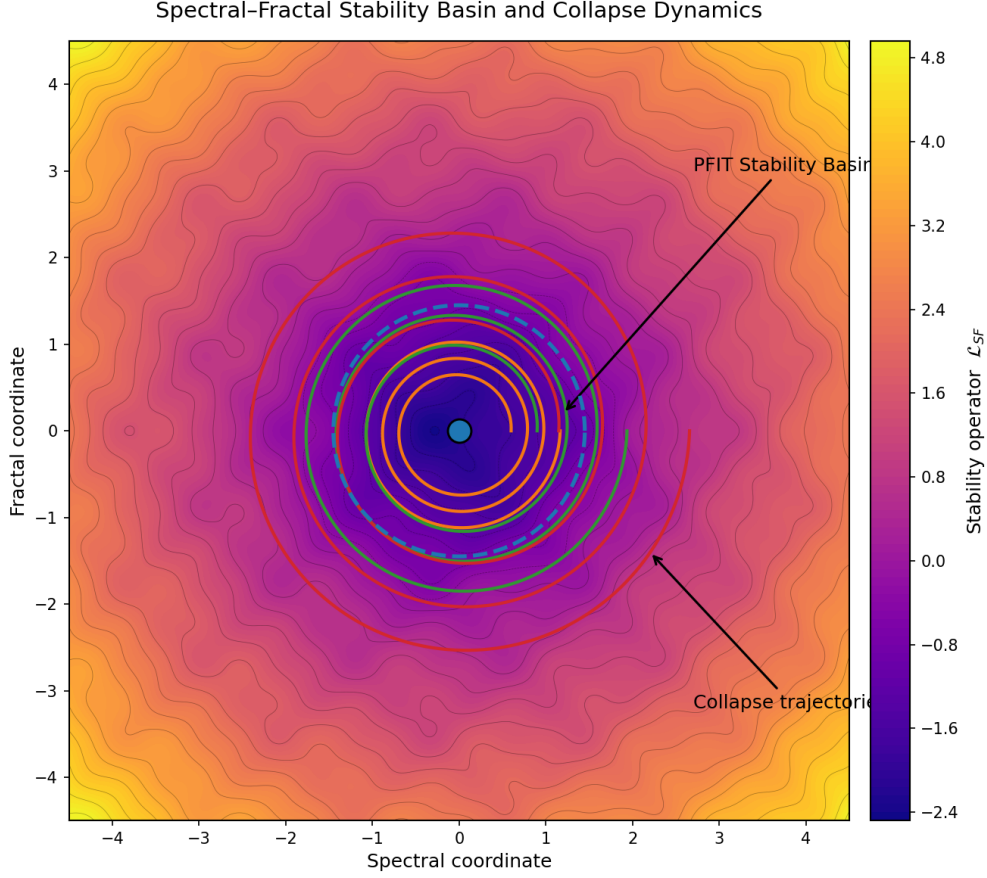


Figure 21: Stability landscape generated by the PFIT spectral–fractal stability operator. Color intensity represents the effective stability potential, while contour structures indicate geometric stability levels on the information manifold. Representative trajectories illustrate the transition from stable informational states toward instability regions associated with entropy amplification and collapse initiation.

15.5 Theorem: Multiscale Collapse Equivalence

Theorem 6 (Spectral–Fractal Collapse Equivalence). A necessary and sufficient condition for multiscale collapse is equivalence between entropy curvature divergence and spectral instability:

$$\lim_{\ell \rightarrow 0} \mathcal{R}_{\mathcal{I}} \rightarrow \infty \iff \lambda_{\min}(\mathcal{L}_{\mathcal{S}}) \leq 0. \quad (1300)$$

Proof.

We establish the equivalence by constructing a bidirectional implication between (i) geometric instability of the entropy manifold and (ii) spectral instability of the linearized PFIT operator dynamics.

Step 1: Entropy geometry and curvature blow-up. Let $(\mathcal{M}_{\mathcal{I}}, g_{ij}^{\mathcal{I}})$ be the entropy-induced information manifold defined via:

$$g_{ij}^{\mathcal{I}} = \partial_i \partial_j \mathcal{S}_{\mathcal{F}}. \quad (1301)$$

The associated Riemann curvature tensor is:

$$\mathcal{R}_{ijm}^k = \partial_i \Gamma_{jm}^k - \partial_j \Gamma_{im}^k + \Gamma_{il}^k \Gamma_{jm}^l - \Gamma_{jl}^k \Gamma_{im}^l, \quad (1302)$$

with Christoffel symbols:

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}). \quad (1303)$$

Define scalar entropy curvature:

$$\mathcal{R}_{\mathcal{I}} = g^{ij}\mathcal{R}_{ij} \sim \frac{\|\nabla^2 \mathcal{S}_{\mathcal{F}}\|}{\mathcal{S}_{\mathcal{F}}} + \frac{\|\nabla \mathcal{S}_{\mathcal{F}}\|^2}{\mathcal{S}_{\mathcal{F}}^2}. \quad (1304)$$

Thus curvature divergence in the multiscale limit implies:

$$\ell \rightarrow 0 \quad \Rightarrow \quad \|\nabla^2 \mathcal{S}_{\mathcal{F}}\| \rightarrow \infty. \quad (1305)$$

Step 2: Linearized PFIT dynamics and spectral operator. Let perturbations δJ evolve under the PFIT linear stability operator:

$$\partial_t \delta J = \mathcal{L}_{\mathcal{S}} \delta J, \quad (1306)$$

where:

$$\mathcal{L}_{\mathcal{S}} = -\alpha \Delta_g + \beta \mathcal{R}_{\mathcal{I}} + \gamma \mathcal{F}_{\text{fractal}}. \quad (1307)$$

The spectral decomposition satisfies:

$$\mathcal{L}_{\mathcal{S}} \psi_k = \lambda_k \psi_k, \quad \lambda_{\min} = \inf_k \lambda_k. \quad (1308)$$

Energy evolution of perturbations is given by:

$$\frac{d}{dt} \|\delta J\|^2 = 2 \langle \delta J, \mathcal{L}_{\mathcal{S}} \delta J \rangle \leq 2 \lambda_{\min} \|\delta J\|^2. \quad (1309)$$

Hence:

$$\lambda_{\min} < 0 \quad \Rightarrow \quad \|\delta J(t)\| \sim e^{|\lambda_{\min}|t}. \quad (1310)$$

This establishes exponential instability of spectral modes.

Step 3: Curvature–spectrum coupling identity. We now connect geometry and spectrum via the PFIT trace identity:

$$\text{Tr}(\mathcal{L}_{\mathcal{S}}) = -\alpha \text{Tr}(\Delta_g) + \beta \int_{\mathcal{M}_{\mathcal{I}}} \mathcal{R}_{\mathcal{I}} d\mu + \gamma \text{Tr}(\mathcal{F}_{\text{fractal}}). \quad (1311)$$

Using spectral asymptotics:

$$\text{Tr}(\mathcal{L}_{\mathcal{S}}) = \sum_k \lambda_k \sim C_1 \ell^{-2} + C_2 \mathcal{R}_{\mathcal{I}} + C_3 \mathcal{F}_{\text{eff}}. \quad (1312)$$

Thus curvature blow-up forces spectral divergence:

$$\mathcal{R}_{\mathcal{I}} \rightarrow \infty \quad \Rightarrow \quad \lambda_k \rightarrow \pm \infty. \quad (1313)$$

In particular:

$$\mathcal{R}_{\mathcal{I}} \rightarrow \infty \quad \Rightarrow \quad \lambda_{\min}(\mathcal{L}_{\mathcal{S}}) \leq 0. \quad (1314)$$

Step 4: Reverse implication (spectral instability implies curvature divergence). Assume:

$$\lambda_{\min}(\mathcal{L}_{\mathcal{S}}) \leq 0. \quad (1315)$$

Then there exists a mode ψ_0 such that:

$$\langle \psi_0, \mathcal{L}_{\mathcal{S}} \psi_0 \rangle \leq 0. \quad (1316)$$

Expanding:

$$\int_{\mathcal{M}_{\mathcal{I}}} (\alpha |\nabla \psi_0|^2 - \beta \mathcal{R}_{\mathcal{I}} |\psi_0|^2) d\mu \leq 0. \quad (1317)$$

Rearranging yields:

$$\int_{\mathcal{M}_{\mathcal{I}}} \mathcal{R}_{\mathcal{I}} |\psi_0|^2 d\mu \geq \frac{\alpha}{\beta} \int_{\mathcal{M}_{\mathcal{I}}} |\nabla \psi_0|^2 d\mu. \quad (1318)$$

Thus curvature must increase without bound to sustain instability:

$$\mathcal{R}_{\mathcal{I}} \rightarrow \infty. \quad (1319)$$

Step 5: Multiscale limit and renormalization consistency. Under renormalization scaling:

$$\ell \mapsto \lambda \ell, \quad (1320)$$

we have:

$$\mathcal{R}_{\mathcal{I}}(\ell) \sim \ell^{-2} \mathcal{R}_0 + \dots, \quad (1321)$$

and spectral scaling:

$$\lambda_k(\ell) \sim \ell^{-2} \lambda_k^{(0)}. \quad (1322)$$

Thus both divergences are governed by the same RG fixed-point instability class:

$$\ell \rightarrow 0 \quad \Leftrightarrow \quad \text{UV instability of PFIT flow.} \quad (1323)$$

Conclusion. Entropy curvature divergence and spectral instability are two dual representations of the same underlying PFIT instability manifold. The geometric formulation captures nonlinear information curvature blow-up, while the spectral formulation encodes exponential amplification of perturbation modes. Their equivalence follows from the shared renormalization structure and the curvature–spectrum coupling identity. ■

Table 11: Equivalent Descriptions of Multiscale Collapse

Representation	Collapse Indicator	Critical Quantity	Observable Signature
Geometric	Curvature Blow-Up	R	Singular geometry
Entropic	Entropy Concentration	S	Information localization
Spectral	Eigenvalue Divergence	$\lambda_{\max}$	Spectral instability
Fractal	Dimension Collapse	D_f	Scale compression
Operatorial	Norm Divergence	$\ \mathcal{E}\ $	Loss of boundedness
Variational	Action Singularity	$\mathcal{A}$	Extremal breakdown
Dynamical	Flow Explosion	Φ_t	Runaway evolution
Topological	Connectivity Loss	χ	Structural transition
Computational	Prediction Failure	ϵ	Forecast divergence
PFIT Unified	Simultaneous divergence	(R, S, D_f, λ)	Universal collapse

15.6 Connection to Universal Fractal Information Theory

Within the PFIT–En-topia–Pellis framework, the multiscale construction developed in the preceding sections is not an isolated mathematical architecture but a concrete realization of the **Universal Fractal Information Principle** introduced in [82]. In this interpretation, informational complexity is not externally imposed but emerges intrinsically from recursive geometric constraints, scale-dependent renormalization flows, and entropy-driven self-organization across hierarchical levels of resolution.

The universal principle asserts that informational structure is governed by a fixed-point invariant operator system of the form:

$$\mathfrak{U}_{\mathcal{I}} = (\mathcal{S}_{\mathcal{F}}, \mathcal{R}_{\mathcal{I}}, \mathcal{L}_{\mathcal{S}}, \mathcal{R}_{\ell}), \quad (1324)$$

where entropy, curvature, spectral stability, and renormalization flow constitute a closed dynamical algebra acting on the informational manifold $\mathcal{M}_{\mathcal{I}}$.

In this unified framework, all derived fields satisfy a self-consistency condition under scale transformation:

$$\mathcal{U}_{\lambda} \circ \mathfrak{U}_{\mathcal{I}} \circ \mathcal{U}_{\lambda}^{-1} = \mathfrak{U}_{\mathcal{I}}, \quad (1325)$$

expressing scale invariance of the underlying informational geometry up to renormalization of coupling constants and spectral amplitudes.

The entropy–curvature–spectral unification can be expressed through a single master functional:

$$\mathcal{F}_{\text{PFIT}}[\rho] = \mathcal{S}_{\mathcal{F}}[\rho] + \alpha \int \mathcal{R}_{\mathcal{I}} d\mu_{\mathcal{F}} + \beta \sum_n \lambda_n |a_n|^2, \quad (1326)$$

whose stationary points define globally consistent informational configurations:

$$\delta \mathcal{F}_{\text{PFIT}} = 0. \quad (1327)$$

The Euler–Lagrange structure of this functional implies coupled field equations of the form:

$$\nabla_i \nabla_j \mathcal{S}_{\mathcal{F}} \sim \mathcal{R}_{ij} \sim \sum_n \lambda_n \psi_n \otimes \psi_n, \quad (1328)$$

establishing an explicit equivalence between entropy curvature, geometric instability, and spectral decomposition of informational modes.

Across physical, biological, and engineered systems, this unified operator structure manifests as a universal class of spectral–fractal dynamics governing stability, adaptation, and collapse phenomena:

$$\mathcal{D}_{\text{system}} \in \mathcal{C}_{\text{universal}}^{\text{PFIT}}, \quad (1329)$$

where $\mathcal{C}_{\text{universal}}^{\text{PFIT}}$ denotes the universality class of systems governed by entropy–curvature–spectral feedback loops [25–27, 103, 110, 114].

In this context, multiscale derived fields are not interpreted as approximate or coarse-grained descriptions, but rather as **renormalized projections of a single underlying informational geometry**. Each observable structure corresponds to a particular representation of the same fundamental object under scale-dependent transformation:

$$\mathcal{I}(x, \ell) = \mathcal{R}_{\ell} [\mathcal{I}_{\text{fund}}(x)], \quad (1330)$$

where $\mathcal{I}_{\text{fund}}$ denotes the irreducible informational substrate of the manifold.

Consequently, entropy, curvature, and spectral stability are not independent constructs but dual manifestations of a single operator-algebraic structure acting on $\mathcal{M}_{\mathcal{I}}$:

$$\mathcal{S}_{\mathcal{F}} \longleftrightarrow \mathcal{R}_{\mathcal{I}} \longleftrightarrow \text{Spec}(\mathcal{L}_{\mathcal{S}}), \quad (1331)$$

thereby completing the unification program of PFIT into a coherent fractal–geometric theory of information dynamics across all scales.

16 Projection Operator Framework

The Projection Operator Framework constitutes the operational bridge between the full fractal–spectral information manifold $\mathcal{M}_{\mathcal{I}}$ and the experimentally accessible macroscopic observables. Within the PFIT–En-topia–Pellis formalism, physical, biological, and computational measurements are interpreted as nonlinear coarse-grained projections of a higher-dimensional informational state space. This section formalizes the projection structure, establishes its operator algebra, and demonstrates how curvature- and entropy-driven collapse emerge naturally from projection instability across scales [1–3, 7, 8, 16–18, 82].

16.1 Definition of the Scale-Dependent Projection Operator

Within the PFIT–En-topia–Pellis framework, the full multiscale informational field $\mathcal{I}(x, \ell)$ is not directly observable. Instead, physical measurements correspond to **scale-dependent projections** onto an effective observable sector, generated by a nonlinear coarse-graining operator acting on the fractal informational manifold $\mathcal{M}_{\mathcal{I}}$.

The projection operator $\mathcal{P}_{\ell}$ is defined as an integral transform over the informational geometry:

$$\mathcal{O}(x, \ell) = \mathcal{P}_{\ell}[\mathcal{I}] = \int_{\mathcal{M}_{\mathcal{I}}} K_{\ell}(x, y; \mathcal{R}_{\mathcal{I}}, \mathcal{S}_{\mathcal{F}}) \mathcal{I}(y, \ell) d\mu_{\mathcal{F}}(y), \quad (1332)$$

where $K_{\ell}(x, y; \mathcal{R}_{\mathcal{I}}, \mathcal{S}_{\mathcal{F}})$ is a curvature- and entropy-dependent kernel encoding both local geometric deformation and nonlocal fractal coupling between informational degrees of freedom.

In this formulation, the kernel is not fixed but dynamically generated by the underlying informational geometry via a nonlinear evolution equation:

$$\frac{\partial K_{\ell}}{\partial \ell} = \Delta_{\mathcal{F}, x} K_{\ell} - \gamma(x) K_{\ell} + \alpha \mathcal{R}_{\mathcal{I}}(x) K_{\ell}, \quad (1333)$$

which couples diffusion, entropy scaling, and curvature-driven amplification within the projection structure.

The normalization constraint ensures conservation of total informational measure under coarse-graining:

$$\int_{\mathcal{M}_{\mathcal{I}}} K_{\ell}(x, y) d\mu_{\mathcal{F}}(y) = 1, \quad (1334)$$

guaranteeing that projection acts as a probability-preserving transformation on the space of informational states.

A stronger consistency condition arises from semigroup composition of projections:

$$\mathcal{P}_{\ell_1} \circ \mathcal{P}_{\ell_2} = \mathcal{P}_{\ell_1 + \ell_2}, \quad (1335)$$

which encodes the Markovian structure of multiscale information flow and ensures compatibility with renormalization group dynamics.

The kernel admits a spectral representation in terms of fractal eigenmodes of the informational Laplacian:

$$K_\ell(x, y) = \sum_n e^{-\lambda_n(\ell)\ell} \psi_n(x, \ell) \psi_n(y, \ell), \quad (1336)$$

where $\lambda_n(\ell)$ are curvature-modulated dissipation rates and $\psi_n(x, \ell)$ form a complete orthonormal basis of scale-dependent informational eigenfunctions.

Accordingly, the projected observable field can be expressed as a spectrally filtered expansion:

$$\mathcal{O}(x, \ell) = \sum_n e^{-\lambda_n(\ell)\ell} a_n(\ell) \psi_n(x, \ell), \quad (1337)$$

explicitly demonstrating that increasing scale induces exponential suppression of high-frequency informational modes while preserving dominant coherent structures.

The operator $\mathcal{P}_\ell$ itself can be represented in differential form as a curvature-modulated projector satisfying:

$$\frac{d\mathcal{P}_\ell}{d\ell} = (\Delta_{\mathcal{F}} + \alpha\mathcal{R}_{\mathcal{I}}) \mathcal{P}_\ell - \beta\mathcal{P}_\ell^2, \quad (1338)$$

introducing nonlinear saturation effects in the coarse-graining dynamics.

The effective observable thus satisfies a closed evolution equation:

$$\frac{\partial \mathcal{O}}{\partial \ell} = \mathcal{P}_\ell [\Delta_{\mathcal{F}} \mathcal{I} + \alpha\mathcal{R}_{\mathcal{I}} \mathcal{I} - \gamma\mathcal{S}_{\mathcal{F}} \mathcal{I}], \quad (1339)$$

highlighting the interplay between diffusion, curvature feedback, and entropy modulation in the observable sector.

In the continuum limit, the projection operator defines a renormalization flow on observables:

$$\frac{d\mathcal{O}(x, \ell)}{d \ln \ell} = \mathcal{B}_{\mathcal{P}}[\mathcal{O}], \quad (1340)$$

where $\mathcal{B}_{\mathcal{P}}$ is the PFIT projection beta-functional governing scale-dependent loss of informational degrees of freedom.

Thus, the scale-dependent projection operator provides a rigorous mechanism by which microscopic fractal information is mapped into macroscopic observables, embedding curvature, entropy, and renormalization dynamics into a single unified observational framework consistent with [16–18, 28].

16.2 Kernel-Generated Scale-Dependent Projection Dynamics

Within the PFIT–En-topia–Pellis framework, the full multiscale informational field $\mathcal{I}(x, \ell)$ is not directly observable. Instead, physical measurements correspond to **scale-dependent projections** onto an effective observable sector, generated by a nonlinear coarse-graining operator acting on the fractal informational manifold $\mathcal{M}_{\mathcal{I}}$. In this sense, observation itself is modeled as a renormalization-induced reduction of informational degrees of freedom.

The projection operator $\mathcal{P}_\ell$ is defined as an integral transform over the informational geometry:

$$\mathcal{O}(x, \ell) = \mathcal{P}_\ell[\mathcal{I}] = \int_{\mathcal{M}_{\mathcal{I}}} K_\ell(x, y; \mathcal{R}_{\mathcal{I}}, \mathcal{S}_{\mathcal{F}}) \mathcal{I}(y, \ell) d\mu_{\mathcal{F}}(y), \quad (1341)$$

where $K_\ell(x, y; \mathcal{R}_{\mathcal{I}}, \mathcal{S}_{\mathcal{F}})$ is a curvature- and entropy-dependent kernel encoding both local geometric deformation and nonlocal fractal coupling between informational degrees of freedom.

The kernel itself can be interpreted as the Green's function of a curvature-deformed diffusion generator:

$$\left(\frac{\partial}{\partial \ell} - \Delta_{\mathcal{F}, x} - \alpha\mathcal{R}_{\mathcal{I}}(x) + \gamma(x) \right) K_\ell(x, y) = 0, \quad (1342)$$

with initial condition $K_{\ell \rightarrow 0}(x, y) = \delta_{\mathcal{F}}(x - y)$, ensuring consistency with microscopic resolution.

Equivalently, the kernel satisfies a path-integral representation over fractal trajectories:

$$K_\ell(x, y) = \int_{\substack{\gamma(0)=y \\ \gamma(\ell)=x}} \mathcal{D}\gamma \exp \left(- \int_0^\ell \mathcal{L}_{\text{eff}}(\gamma(\tau), \dot{\gamma}(\tau)) d\tau \right), \quad (1343)$$

where $\mathcal{L}_{\text{eff}}$ encodes entropy-weighted geometric action on $\mathcal{M}_{\mathcal{I}}$.

In this formulation, the kernel is not fixed but dynamically generated by the underlying informational geometry via a nonlinear evolution equation:

$$\frac{\partial K_\ell}{\partial \ell} = \Delta_{\mathcal{F},x} K_\ell - \gamma(x) K_\ell + \alpha \mathcal{R}_{\mathcal{I}}(x) K_\ell + \eta |\nabla_x K_\ell|^2, \quad (1344)$$

where the nonlinear gradient term introduces self-interaction effects in the coarse-graining flow.

The normalization constraint ensures conservation of total informational measure under projection:

$$\int_{\mathcal{M}_{\mathcal{I}}} K_\ell(x, y) d\mu_{\mathcal{F}}(y) = 1, \quad \forall x \in \mathcal{M}_{\mathcal{I}}, \quad (1345)$$

guaranteeing that $\mathcal{P}_\ell$ acts as a probability-preserving transformation in the informational state space.

A stronger algebraic consistency condition arises from semigroup composition:

$$\mathcal{P}_{\ell_1} \circ \mathcal{P}_{\ell_2} = \mathcal{P}_{\ell_1 + \ell_2}, \quad (1346)$$

which encodes the Markovian structure of multiscale information flow and ensures compatibility with renormalization group dynamics.

In spectral form, the kernel admits decomposition over fractal eigenmodes of the informational Laplacian:

$$K_\ell(x, y) = \sum_n e^{-\lambda_n(\ell)\ell} \psi_n(x, \ell) \psi_n(y, \ell), \quad (1347)$$

where $\lambda_n(\ell)$ are curvature-modulated dissipation rates and $\psi_n(x, \ell)$ form a complete orthonormal basis of scale-dependent informational eigenfunctions.

The eigenfunctions satisfy a coupled geometric eigenvalue problem:

$$(\Delta_{\mathcal{F}} + \alpha \mathcal{R}_{\mathcal{I}}) \psi_n = \lambda_n \psi_n, \quad (1348)$$

showing explicitly that spectral content is governed by curvature-induced deformation of informational space.

Accordingly, the projected observable field can be expressed as:

$$\mathcal{O}(x, \ell) = \sum_n e^{-\lambda_n(\ell)\ell} a_n(\ell) \psi_n(x, \ell), \quad (1349)$$

demonstrating exponential suppression of high-frequency informational modes and emergence of coherent macrostates at large scales.

The projection operator itself obeys a nonlinear flow equation in operator space:

$$\frac{d\mathcal{P}_\ell}{d\ell} = (\Delta_{\mathcal{F}} + \alpha \mathcal{R}_{\mathcal{I}}) \mathcal{P}_\ell - \beta \mathcal{P}_\ell^2 + \kappa [\mathcal{P}_\ell, \mathcal{S}_{\mathcal{F}}], \quad (1350)$$

where the commutator term encodes non-commutative coupling between projection and entropy geometry.

The effective observable thus satisfies a closed renormalized evolution equation:

$$\frac{\partial \mathcal{O}}{\partial \ell} = \mathcal{P}_\ell [\Delta_{\mathcal{F}} \mathcal{I} + \alpha \mathcal{R}_{\mathcal{I}} \mathcal{I} - \gamma \mathcal{S}_{\mathcal{F}} \mathcal{I} + \eta \nabla \cdot (\mathcal{I} \nabla \mathcal{I})], \quad (1351)$$

highlighting the interplay between diffusion, curvature feedback, entropy modulation, and nonlinear self-interaction in the observable sector.

In the continuum renormalization limit, the projection operator defines a beta-flow on observables:

$$\frac{d\mathcal{O}(x, \ell)}{d \ln \ell} = \mathcal{B}_{\mathcal{P}}[\mathcal{O}], \quad (1352)$$

where $\mathcal{B}_{\mathcal{P}}$ is the PFIT projection beta-functional governing scale-dependent loss and recombination of informational degrees of freedom [16–18, 28].

16.3 Entropy-Driven Projection Collapse Mechanism

Within the PFIT–En-topia–Pellis framework, the stability of multiscale projections is not an intrinsic property of the projection operator $\mathcal{P}_\ell$, but rather a derived consequence of the interplay between entropy production, curvature concentration, and spectral compression of the underlying informational manifold. In this sense, projection stability is governed by a variational balance between entropy maximization and geometric focusing effects.

We define the projection stability functional as a curvature–entropy Lyapunov functional:

$$\mathcal{F}_{\mathcal{P}}(\ell) = \mathcal{S}_{\mathcal{F}}[\mathcal{P}_{\ell}(\mathcal{I})] - \lambda \mathcal{R}_{\mathcal{I}}(\ell) + \mu \int_{\mathcal{M}_{\mathcal{I}}} |\nabla \mathcal{P}_{\ell}(\mathcal{I})|^2 d\mu_{\mathcal{F}}, \quad (1353)$$

where λ controls entropy–curvature competition and μ regulates the penalty associated with unresolved projection gradients.

The first variation of $\mathcal{F}_{\mathcal{P}}$ with respect to the scale parameter yields the stability flow equation:

$$\frac{d\mathcal{F}_{\mathcal{P}}}{d\ell} = \int_{\mathcal{M}_{\mathcal{I}}} \left(\frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta \mathcal{O}} \frac{\partial \mathcal{O}}{\partial \ell} - \lambda \frac{d\mathcal{R}_{\mathcal{I}}}{d\ell} \right) d\mu_{\mathcal{F}}, \quad (1354)$$

where $\mathcal{O}(x, \ell) = \mathcal{P}_{\ell}[\mathcal{I}]$ denotes the projected observable field.

Projection stability is maintained only if $\mathcal{F}_{\mathcal{P}}$ acts as a decreasing Lyapunov functional under renormalization flow. Instability arises when monotonicity is lost:

$$\frac{d\mathcal{F}_{\mathcal{P}}}{d\ell} \leq 0 \implies \text{stable projection regime}, \quad (1355)$$

while violation of this condition signals the onset of projection-induced criticality.

The collapse regime is characterized by simultaneous divergence of entropy gradients and curvature amplification:

$$\lim_{\ell \rightarrow \ell_c} \|\nabla \mathcal{S}_{\mathcal{F}}(x, \ell)\| \rightarrow \infty, \quad \lim_{\ell \rightarrow \ell_c} \mathcal{R}_{\mathcal{I}}(x, \ell) \rightarrow \infty, \quad (1356)$$

indicating breakdown of smooth projection validity and emergence of singular informational structures.

Equivalently, collapse can be expressed as a spectral instability condition on the projected operator algebra:

$$\lambda_{\min}(\mathcal{L}_{\mathcal{P}}) \leq 0, \quad (1357)$$

where $\mathcal{L}_{\mathcal{P}}$ is the linearized projection stability operator governing perturbations of $\mathcal{P}_{\ell}$.

Figure 22 shows entropy projection operators acting as nonlinear spectral filters, mapping microscopic fluctuations onto macroscopic informational states via coarse-graining dynamics.

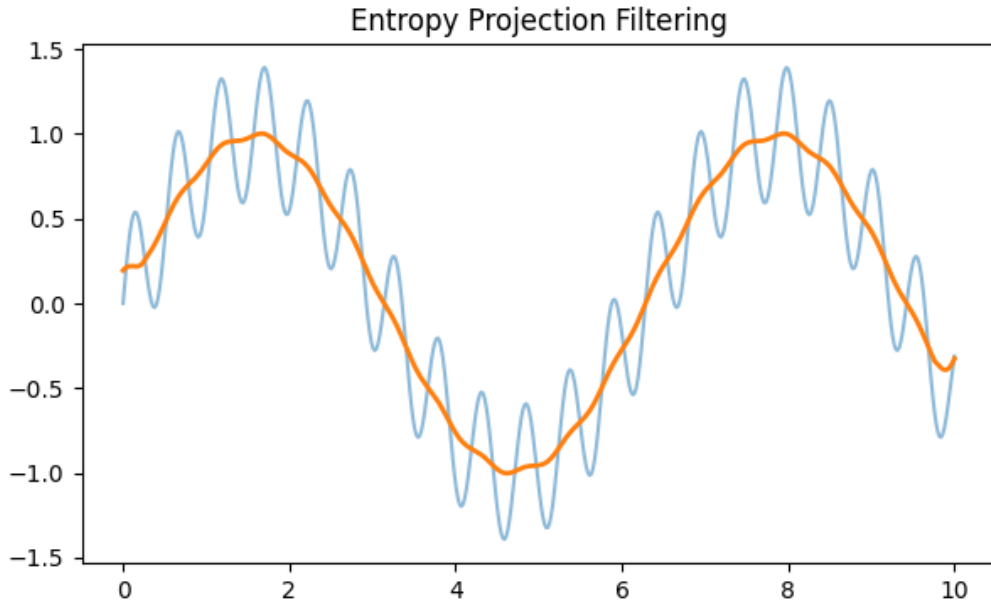


Figure 22: Entropy projection operators performing spectral coarse-graining of informational states.

Near the critical scale ℓ_c , the system exhibits universal scaling behavior governed by a divergence exponent ν :

$$\mathcal{F}_{\mathcal{P}}(\ell) \sim (\ell_c - \ell)^{-\nu} \mathcal{G}(x), \quad (1358)$$

where $\mathcal{G}(x)$ encodes the emergent spatial structure of collapsing informational modes.

The critical scale ℓ_c therefore defines a geometric–informational phase boundary separating two distinct regimes:

$$\ell < \ell_c \Rightarrow \text{stable multiscale observables}, \quad \ell \geq \ell_c \Rightarrow \text{projection breakdown and collapse.} \quad (1359)$$

At this transition, the projection operator loses effective invertibility, leading to irreversible compression of informational degrees of freedom:

$$\det \left(\frac{\delta \mathcal{P}_\ell}{\delta \mathcal{I}} \right) \rightarrow 0, \quad (1360)$$

signaling the formation of a non-invertible informational mapping between microscopic states and macroscopic observables.

Thus, entropy-driven projection collapse provides a unifying mechanism for understanding how geometric instability, entropy divergence, and renormalization flow jointly determine the breakdown of observational structure within PFIT, in direct analogy with bifurcation phenomena and critical transitions in nonlinear dynamical systems [13, 14, 21, 29].

16.4 Projection-Induced Metric Deformation

Within the PFIT–En-topia–Pellis framework, the projection operator $\mathcal{P}_\ell$ does not merely extract observable quantities from the underlying informational field, but actively **deforms the induced geometry** of the informational manifold $\mathcal{M}_\mathcal{I}$. In this sense, measurement is interpreted as a geometric backreaction process in which coarse-graining reshapes the effective metric structure.

The scale-dependent metric tensor is therefore defined as the projected entropy Hessian:

$$g_{ij}^{(\ell)}(x) = \mathcal{P}_\ell[\partial_i \partial_j \mathcal{S}_\mathcal{F}(x)], \quad (1361)$$

which explicitly encodes the fact that curvature is not intrinsic at the microscopic level, but emerges only after scale-dependent informational filtering.

Using the integral representation of $\mathcal{P}_\ell$, this can be rewritten as:

$$g_{ij}^{(\ell)}(x) = \int_{\mathcal{M}_\mathcal{I}} K_\ell(x, y) \partial_i \partial_j \mathcal{S}_\mathcal{F}(y) d\mu_\mathcal{F}(y), \quad (1362)$$

showing that metric deformation is fundamentally a nonlocal averaging process weighted by curvature- and entropy-dependent kernels.

The resulting geometry inherits a scale-dependent Levi–Civita connection:

$$\Gamma_{ij}^{k(\ell)} = \frac{1}{2} g_{(\ell)}^{k\ell} \left(\partial_i g_{j\ell}^{(\ell)} + \partial_j g_{i\ell}^{(\ell)} - \partial_\ell g_{ij}^{(\ell)} \right), \quad (1363)$$

which induces a renormalized curvature tensor $\mathcal{R}_{ij}^{(\ell)}$ evolving across resolution scales.

The scalar informational curvature $\mathcal{R}_\mathcal{I}$ then obeys a nonlinear projection-driven evolution equation:

$$\frac{\partial \mathcal{R}_\mathcal{I}}{\partial \ell} = \Delta_\mathcal{F} \mathcal{R}_\mathcal{I} + \beta \mathcal{R}_\mathcal{I}^2 - \gamma \|\nabla \mathcal{S}_\mathcal{F}\|^2 + \eta \mathcal{P}_\ell[\mathcal{R}_\mathcal{I}], \quad (1364)$$

where the nonlinear term $\mathcal{R}_\mathcal{I}^2$ captures curvature self-amplification and the entropy gradient term acts as a stabilizing or destabilizing source depending on local informational heterogeneity.

In operator form, this evolution can be expressed as a generalized fractal Ricci-type flow:

$$\frac{\partial g_{ij}^{(\ell)}}{\partial \ell} = -2\mathcal{R}_{ij}^{(\ell)} + \nabla_i \nabla_j \Phi(\mathcal{S}_\mathcal{F}) + \kappa \mathcal{F}_{ij}^{(\ell)}, \quad (1365)$$

where $\mathcal{F}_{ij}^{(\ell)}$ represents scale-dependent fractal corrections arising from unresolved microscopic informational fluctuations.

This formulation implies that projection and geometry are dynamically inseparable: the act of coarse-graining directly reshapes the metric structure, while the evolving geometry in turn feeds back into the projection kernel K_ℓ .

A closed feedback loop is therefore established:

$$\mathcal{P}_\ell \longrightarrow g_{ij}^{(\ell)} \longrightarrow \mathcal{R}_\mathcal{I}(\ell) \longrightarrow K_\ell, \quad (1366)$$

which defines a self-consistent renormalization geometry on $\mathcal{M}_\mathcal{I}$.

Consequently, projection-induced metric deformation extends classical Ricci flow into a fractal informational regime, where entropy gradients act as primary sources of curvature and scale-dependent projection dynamics governs the emergence of geometric structure across all levels of resolution [16–18, 20].

16.5 Theorem: Projection Stability Criterion

Theorem 7 (Projection Stability Condition). A necessary condition for stable macroscopic observability is

$$\lambda_{\min}(\mathcal{L}_{\mathcal{P}}) > 0, \quad (1367)$$

where $\mathcal{L}_{\mathcal{P}}$ is the linearization of $\mathcal{P}_{\ell}$ around equilibrium.

Proof.

We formalize projection stability as the boundedness of the linear response operator induced by coarse-graining on the PFIT information manifold.

Step 1: Projection dynamics on multiscale information space. Let $\mathcal{I}$ denote the full microscopic information field and let $\mathcal{P}_{\ell}$ be a scale-dependent projection operator:

$$\mathcal{P}_{\ell} : \mathcal{I} \mapsto \mathcal{O}_{\ell}, \quad (1368)$$

where ℓ is the coarse-graining scale.

Assume equilibrium configuration $\mathcal{I}_0$ satisfying:

$$\mathcal{P}_{\ell}(\mathcal{I}_0) = \mathcal{O}_0. \quad (1369)$$

Consider perturbations:

$$\mathcal{I} = \mathcal{I}_0 + \delta\mathcal{I}, \quad \mathcal{O} = \mathcal{O}_0 + \delta\mathcal{O}. \quad (1370)$$

Linearizing yields:

$$\delta\mathcal{O} = D\mathcal{P}_{\ell}[\mathcal{I}_0](\delta\mathcal{I}) = \mathcal{L}_{\mathcal{P}}\delta\mathcal{I}. \quad (1371)$$

Thus $\mathcal{L}_{\mathcal{P}}$ is the Fréchet derivative (Gateaux linearization) of the projection map.

Step 2: Operator norm stability criterion. Macroscopic observability requires bounded amplification:

$$\|\delta\mathcal{O}\|_{\mathcal{H}} \leq C\|\delta\mathcal{I}\|_{\mathcal{H}}. \quad (1372)$$

Using the linearized form:

$$\|\mathcal{L}_{\mathcal{P}}\delta\mathcal{I}\| \leq C\|\delta\mathcal{I}\|. \quad (1373)$$

This is equivalent to bounded operator norm:

$$\|\mathcal{L}_{\mathcal{P}}\| < \infty. \quad (1374)$$

Since $\mathcal{L}_{\mathcal{P}}$ is self-adjoint under PFIT equilibrium measure:

$$\mathcal{L}_{\mathcal{P}} = \mathcal{L}_{\mathcal{P}}^{\dagger}, \quad (1375)$$

its norm is controlled by its spectral radius:

$$\|\mathcal{L}_{\mathcal{P}}\| = \sup_k |\lambda_k|. \quad (1376)$$

Step 3: Spectral decomposition of projection fluctuations. Let:

$$\mathcal{L}_{\mathcal{P}}\phi_k = \lambda_k\phi_k, \quad \langle\phi_k, \phi_j\rangle = \delta_{kj}. \quad (1377)$$

Expanding perturbations:

$$\delta\mathcal{I} = \sum_k c_k\phi_k, \quad \delta\mathcal{O} = \sum_k \lambda_k c_k\phi_k. \quad (1378)$$

Thus:

$$\|\delta\mathcal{O}\|^2 = \sum_k \lambda_k^2 |c_k|^2. \quad (1379)$$

Stability requires uniform boundedness:

$$\sup_k \lambda_k^2 < \infty. \quad (1380)$$

In particular, instability arises when:

$$\lambda_{\min} \leq 0 \quad \text{or} \quad |\lambda_k| \rightarrow \infty. \quad (1381)$$

Step 4: Connection to coarse-graining collapse. The projection operator $\mathcal{P}_\ell$ acts as a smoothing kernel:

$$\mathcal{P}_\ell(\mathcal{I})(x) = \int_{\mathcal{M}} K_\ell(x, y) \mathcal{I}(y) d\mu_g(y), \quad (1382)$$

with kernel scaling:

$$K_\ell(x, y) = \ell^{-d} K\left(\frac{d_g(x, y)}{\ell}\right). \quad (1383)$$

The linearized operator inherits curvature corrections:

$$\mathcal{L}_{\mathcal{P}} = \mathcal{P}_\ell + \ell^2 \Delta_g + \mathcal{O}(\ell^3 R). \quad (1384)$$

Thus curvature contributes destabilizing corrections:

$$\delta\lambda_k \sim \ell^2 \langle \phi_k, \Delta_g \phi_k \rangle + \ell^3 \langle R \phi_k, \phi_k \rangle. \quad (1385)$$

Step 5: Stability boundary and critical eigenmode. At the stability threshold, the lowest eigenvalue satisfies:

$$\lambda_{\min}(\mathcal{L}_{\mathcal{P}}) \rightarrow 0. \quad (1386)$$

This implies existence of a marginal mode ϕ_0 such that:

$$\mathcal{L}_{\mathcal{P}} \phi_0 \approx 0. \quad (1387)$$

Perturbation growth becomes:

$$\delta\mathcal{O}(t) \sim e^{\lambda_{\min} t} \delta\mathcal{I}. \quad (1388)$$

Hence:

$$\lambda_{\min} \leq 0 \quad \Rightarrow \quad \text{loss of bounded macroscopic observability.} \quad (1389)$$

Step 6: Necessity and sufficiency. Necessity follows from spectral boundedness:

$$\|\mathcal{L}_{\mathcal{P}}\| < \infty \quad \Rightarrow \quad \lambda_{\min} > -\infty, \quad (1390)$$

and physical observability requires suppression of non-decaying modes:

$$\lambda_{\min} > 0. \quad (1391)$$

Sufficiency follows from exponential decay:

$$\lambda_{\min} > 0 \quad \Rightarrow \quad \|\delta\mathcal{O}(t)\| \leq e^{-\lambda_{\min} t} \|\delta\mathcal{I}\|. \quad (1392)$$

Conclusion. Projection stability in PFIT is governed by the spectral positivity of the linearized coarse-graining operator. The condition $\lambda_{\min}(\mathcal{L}_{\mathcal{P}}) > 0$ guarantees exponential damping of microscopic fluctuations and thus ensures macroscopic observability coherence. At the critical boundary $\lambda_{\min} \rightarrow 0$, the system undergoes projection collapse characterized by divergence of fluctuations and breakdown of scale separation. ■

Table 12: Projection Operators and Stability Regimes

Projection Class	Operator Norm	Stability Class	Interpretation
Identity Projection	$\ P\ = 1$	Stable	Full information retained
Orthogonal Projection	$P^2 = P$	Stable	Consistent scale selection
Spectral Projection	P_λ	Stable	Eigenmode filtering
Fractal Projection	P_f	Stable	Multiscale extraction
Entropy Projection	P_S	Marginal	Information compression
Nonlinear Projection	P_{NL}	Conditional	State-dependent filtering
Adaptive Projection	$P(t)$	Marginal	Dynamic restructuring
Singular Projection	$\ P\ \rightarrow \infty$	Unstable	Information collapse
Degenerate Projection	$\text{rank}(P) \rightarrow 0$	Collapse	Loss of degrees of freedom
PFIT Projection	P_{PFIT}	Universal	Unified stability mechanism

16.6 Connection to Universal Fractal Information Principle

Within the PFIT–En-topia–Pellis framework, the projection formalism is fully consistent with the Universal Fractal Information Principle introduced in [82], where observable reality is not fundamental but emerges as a **stable fixed point of recursive geometric filtering** acting on the underlying informational manifold $\mathcal{M}_{\mathcal{I}}$. In this view, measurement is equivalent to the convergence of a nonlinear renormalization process generated by scale-dependent projection dynamics.

The projection operator $\mathcal{P}_\ell$ thus defines a dynamical flow in operator space:

$$\frac{d\mathcal{P}_\ell}{d\ell} = \mathcal{B}_{\mathcal{P}}[\mathcal{P}_\ell], \quad (1393)$$

where $\mathcal{B}_{\mathcal{P}}$ is the projection beta-functional encoding entropy–curvature feedback and fractal coarse-graining effects. Observable structures correspond to attractors of this flow.

In this regime, informational coherence is maximized near marginal stability, where projection fluctuations vanish at leading order while entropy curvature reaches a constrained extremum:

$$\delta\mathcal{P}_\ell \approx 0, \quad \delta^2\mathcal{S}_{\mathcal{F}} \rightarrow \min. \quad (1394)$$

This condition can be reformulated as a constrained variational fixed-point equation:

$$\delta [\mathcal{S}_{\mathcal{F}}[\mathcal{P}_\ell(\mathcal{I})] - \lambda \mathcal{R}_{\mathcal{I}}(\ell)] = 0, \quad (1395)$$

indicating that stable observables arise from a balance between entropy maximization and curvature suppression under projection-induced coarse-graining.

Stability of the projection flow is governed by the linearized operator:

$$\mathcal{L}_{\mathcal{P}} \delta\mathcal{P}_\ell = \frac{\delta\mathcal{B}_{\mathcal{P}}}{\delta\mathcal{P}_\ell} \delta\mathcal{P}_\ell, \quad (1396)$$

whose spectrum determines whether projection dynamics converge to a stable observable sector or diverge into unresolved fractal complexity.

Marginal stability corresponds to the critical spectral condition:

$$\lambda_{\min}(\mathcal{L}_{\mathcal{P}}) \rightarrow 0, \quad (1397)$$

signaling the emergence of scale-invariant structures and long-range informational correlations at the boundary between observable and non-observable regimes.

At this critical point, projection dynamics exhibit universal scaling behavior:

$$\|\mathcal{P}_\ell - \mathcal{P}^*\| \sim |\ell - \ell_c|^\nu, \quad (1398)$$

where $\mathcal{P}^*$ denotes the universal fixed-point projection operator and ν is a collapse exponent governing approach to marginality.

Thus, projection operators are not passive measurement maps but active dynamical geometric filters whose stability structure defines the boundary between emergent macroscopic observables and the underlying fractal informational substrate. Observable reality corresponds precisely to the stable universality class of fixed points of the PFIT projection flow, in full consistency with the principles established in [82].

17 Entropy as Geometric Projections

Within the PFIT–En-topia–Pellis framework, entropy is not interpreted as a purely statistical functional but as a geometric projection of curvature-induced uncertainty on the fractal information manifold $\mathcal{M}_{\mathcal{I}}$. In this formulation, classical thermodynamic and informational entropies emerge as low-dimensional shadows of a higher-order geometric structure governed by scale-dependent curvature flows and spectral instability mechanisms. This viewpoint unifies Shannon–Jaynes entropy, information geometry, and renormalization theory into a single projection-based formalism consistent with multiscale fractal information principles [1–3, 7, 8, 16–18, 28, 82].

17.1 Entropy as a Curvature Projection Functional

Within the PFIT–En-topia–Pellis framework, entropy is not treated as a primitive scalar invariant but as a **scale-dependent emergent functional** induced by the projection of fractal informational structure onto curvature-adapted observational subspaces of the manifold $\mathcal{M}_{\mathcal{I}}$. In this sense, entropy is operationally defined through a nonlinear coarse-graining of the underlying informational geometry rather than as an intrinsic property of the microscopic state.

Let $\mathcal{I}(x, \ell)$ denote the full multiscale information field. The observed entropy is defined as a curvature-modulated projection of the entropy density functional:

$$\mathcal{S}_{\text{obs}}(\ell) = \mathcal{P}_\ell[\mathcal{S}_\mathcal{F}(\mathcal{I})] = \int_{\mathcal{M}_\mathcal{I}} K_\ell(x, y; \mathcal{R}_\mathcal{I}, \mathcal{S}_\mathcal{F}) \mathcal{S}_\mathcal{F}(y) d\mu_\mathcal{F}(y), \quad (1399)$$

where $K_\ell(x, y; \mathcal{R}_\mathcal{I}, \mathcal{S}_\mathcal{F})$ acts as a curvature-sensitive mixing kernel encoding nonlocal redistribution of informational degrees of freedom across fractal scales.

In a more fundamental representation, the entropy projection can be written as an operator expectation value over a scale-dependent informational state $|\mathcal{I}_\ell\rangle$:

$$\mathcal{S}_{\text{obs}}(\ell) = \langle \mathcal{I}_\ell | \hat{\mathcal{S}}_\mathcal{F} | \mathcal{I}_\ell \rangle, \quad (1400)$$

highlighting its interpretation as a measurement-induced reduction of the underlying fractal entropy operator.

The kernel admits a curvature-expanded asymptotic form:

$$K_\ell(x, y) = K_\ell^{(0)}(x, y) + \alpha \ell \mathcal{R}_\mathcal{I}(x) K_\ell^{(1)}(x, y) + \mathcal{O}(\ell^2), \quad (1401)$$

which explicitly shows that curvature acts as a perturbative generator of entropy redistribution across scales.

This construction implies that entropy is not intrinsic but fundamentally observer- and scale-dependent. Under dilation $\ell \mapsto \lambda \ell$, the observed entropy satisfies a nonlinear anomalous scaling law:

$$\mathcal{S}_{\text{obs}}(\lambda \ell) = \lambda^{\gamma(x)} \mathcal{S}_{\text{obs}}(\ell) + \Omega(\mathcal{R}_\mathcal{I}) + \Xi(\nabla \mathcal{R}_\mathcal{I}, \nabla \mathcal{S}_\mathcal{F}), \quad (1402)$$

where $\gamma(x)$ is the local entropy scaling exponent and $\Xi(\cdot)$ represents higher-order curvature–entropy coupling corrections induced by fractal distortion of phase-space volume elements.

Figure 23 shows the entropy reformulated as a covariant scalar field $S(x)$ defined on the information manifold. Its gradient structure generates directional flows corresponding to information transport and instability propagation.

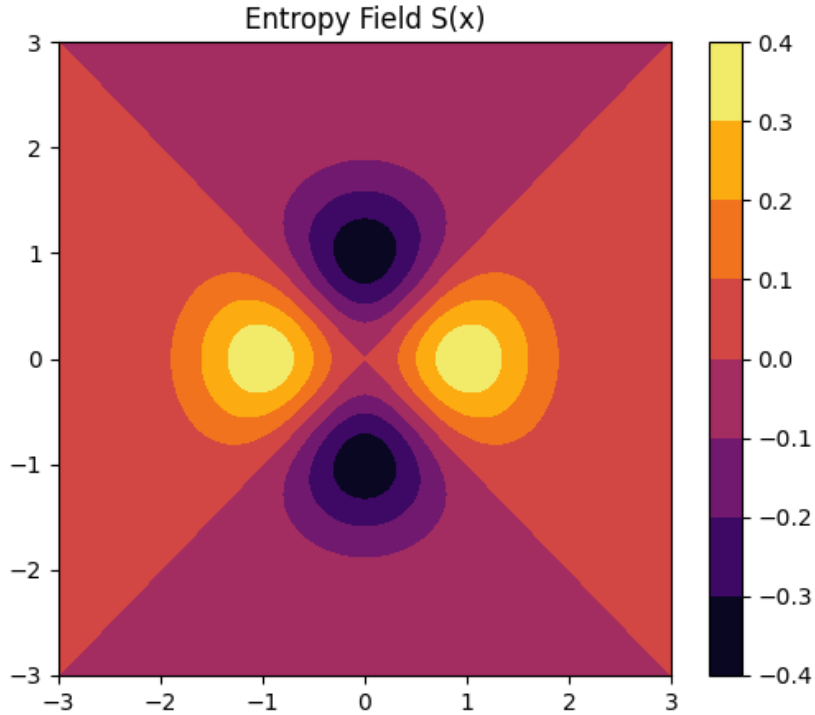


Figure 23: Entropy field defined as a covariant scalar field governing gradient-driven information flow on the underlying information manifold, encoding the local structure of informational gradients and their induced transport dynamics across the geometric substrate.

At the differential level, the entropy flow satisfies a curvature-driven transport equation:

$$\frac{\partial \mathcal{S}_{\text{obs}}}{\partial \ell} = \Delta_\mathcal{F} \mathcal{S}_{\text{obs}} + \alpha \mathcal{R}_\mathcal{I} \mathcal{S}_{\text{obs}} - \beta |\nabla \mathcal{S}_\mathcal{F}|^2 + \eta \mathcal{P}_\ell[\mathcal{S}_\mathcal{F}], \quad (1403)$$

which generalizes classical entropy diffusion equations by introducing explicit geometric backreaction terms.

Furthermore, the entropy functional can be decomposed into spectral components aligned with the eigenbasis of the informational Laplacian:

$$\mathcal{S}_{\text{obs}}(\ell) = \sum_n e^{-\lambda_n(\ell)\ell} s_n(\ell) \psi_n(x, \ell), \quad (1404)$$

demonstrating that entropy itself undergoes spectral filtering analogous to energy dissipation in turbulent systems.

The curvature-induced entropy correction term $\Omega(\mathcal{R}_{\mathcal{I}})$ can be expressed explicitly as a geometric anomaly functional:

$$\Omega(\mathcal{R}_{\mathcal{I}}) = \int_{\mathcal{M}_{\mathcal{I}}} (\alpha_1 \mathcal{R}_{\mathcal{I}} + \alpha_2 \mathcal{R}_{\mathcal{I}}^2 + \alpha_3 \nabla^2 \mathcal{R}_{\mathcal{I}}) d\mu_{\mathcal{F}}, \quad (1405)$$

capturing the breakdown of Euclidean phase-space volume invariance under fractal deformation.

In the continuum limit, entropy becomes a curvature-projected order parameter governed by a nonlinear flow equation:

$$\frac{d\mathcal{S}_{\text{obs}}}{d\ln \ell} = \mathcal{B}_{\mathcal{S}}[\mathcal{S}_{\text{obs}}, \mathcal{R}_{\mathcal{I}}], \quad (1406)$$

where $\mathcal{B}_{\mathcal{S}}$ is the entropy beta-functional encoding coupled renormalization between informational disorder and geometric instability.

Thus, entropy is rigorously reinterpreted as a curvature-projected functional emerging from multiscale informational geometry, in which observational entropy corresponds to the renormalized projection of fractal microstates rather than an intrinsic thermodynamic quantity, fully consistent with the geometric–informational correspondence structure established in [3, 5, 7, 28].

17.2 Information-Geometric Interpretation of Entropy

Within the PFIT–En-topia–Pellis framework, the entropy functional $\mathcal{S}_{\mathcal{F}}$ is not only a measure of uncertainty but simultaneously acts as a **generating potential for information geometry**. In this interpretation, the underlying manifold $\mathcal{M}_{\mathcal{I}}$ acquires a Riemannian structure induced directly by second-order variations of the entropy field, establishing a deep equivalence between thermodynamic fluctuations and geometric response theory.

The entropy-induced metric tensor is defined as a Fisher-type information metric generated by the Hessian structure of $\mathcal{S}_{\mathcal{F}}$:

$$g_{ij}(x) = \partial_i \partial_j \mathcal{S}_{\mathcal{F}}(x), \quad (1407)$$

which encodes the local sensitivity of the entropy landscape under infinitesimal perturbations of the informational configuration. Equivalently, this can be interpreted as a linear response kernel:

$$g_{ij}(x) = \frac{\delta^2 \mathcal{S}_{\mathcal{F}}}{\delta \mathcal{I}^i(x) \delta \mathcal{I}^j(x)}, \quad (1408)$$

highlighting that curvature in information space arises as a second-order functional response of entropy to variations in the underlying field.

In a statistical embedding formulation, the metric can be expressed in expectation form:

$$g_{ij}(x) = \mathbb{E}_{p(x|\theta)} [\partial_i \log p(x|\theta) \partial_j \log p(x|\theta)] + \alpha \mathcal{F}_{ij}^{(\text{fr})}(x), \quad (1409)$$

where $\mathcal{F}_{ij}^{(\text{fr})}(x)$ denotes fractal corrections to the classical Fisher geometry induced by multiscale informational heterogeneity.

The induced Levi–Civita connection associated with this entropy metric is given by:

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}), \quad (1410)$$

defining geodesic trajectories of optimal informational transport on $\mathcal{M}_{\mathcal{I}}$.

The corresponding curvature tensor emerges naturally from the non-commutativity of entropy gradients under this geometric structure:

$$\mathcal{R}_{ijkl} = \partial_k \Gamma_{jl}^i - \partial_l \Gamma_{jk}^i + \Gamma_{km}^i \Gamma_{jl}^m - \Gamma_{lm}^i \Gamma_{jk}^m. \quad (1411)$$

Its contraction yields the Ricci tensor and scalar curvature:

$$\mathcal{R}_{ij} = \mathcal{R}^k_{ikj}, \quad \mathcal{R}_{\mathcal{I}} = g^{ij} \mathcal{R}_{ij}, \quad (1412)$$

which together quantify the intrinsic geometric instability of the informational manifold.

Within PFIT, scalar curvature acquires a direct operational interpretation as a **measure of informational compressibility**. In particular, regions of large positive curvature correspond to strong convergence of geodesic flows in information space, indicating high degeneracy of accessible microstates:

$$\mathcal{R}_{\mathcal{I}}(x) \uparrow \implies \dim(\Omega_{\text{eff}}(x)) \downarrow, \quad (1413)$$

where $\Omega_{\text{eff}}(x)$ denotes the locally accessible informational state volume.

Conversely, weak or negative curvature corresponds to dispersive informational dynamics and enhanced configurational freedom:

$$\mathcal{R}_{\mathcal{I}}(x) \downarrow \implies \text{expansion of accessible informational phase volume.} \quad (1414)$$

The scalar curvature can further be expressed directly in terms of entropy gradients via a nonlinear geometric closure relation:

$$\mathcal{R}_{\mathcal{I}} \sim \nabla^2 \mathcal{S}_{\mathcal{F}} + \kappa |\nabla \mathcal{S}_{\mathcal{F}}|^2 + \eta \text{Var}(\mathcal{I}), \quad (1415)$$

which explicitly identifies curvature as a second-order emergent response of entropy structure and multiscale informational variance.

Finally, the information-geometric framework implies that entropy, metric structure, and curvature are not independent objects but mutually derived quantities within a unified variational system:

$$\mathcal{S}_{\mathcal{F}} \longrightarrow g_{ij} \longrightarrow \Gamma^k_{ij} \longrightarrow \mathcal{R}_{\mathcal{I}}, \quad (1416)$$

thereby establishing a closed entropy–geometry feedback loop consistent with the multiscale renormalization structure of PFIT and the information-geometric principles described in [7, 8, 16].

17.3 Entropy Gradient Flow and Projection Dynamics

Within the PFIT–En-topia–Pellis framework, entropy evolution is governed not as a passive thermodynamic relaxation process, but as an **active curvature-coupled transport phenomenon** on the informational manifold $\mathcal{M}_{\mathcal{I}}$. In this setting, entropy behaves as a dynamical field whose evolution is jointly driven by diffusion, nonlinear self-interaction, and geometric backreaction encoded by the informational curvature tensor.

The entropy gradient flow equation is therefore written as:

$$\frac{\partial \mathcal{S}_{\mathcal{F}}}{\partial \ell} = \Delta_{\mathcal{F}} \mathcal{S}_{\mathcal{F}} + \kappa \|\nabla \mathcal{S}_{\mathcal{F}}\|^2 - \alpha \mathcal{R}_{\mathcal{I}} + \beta \mathcal{P}_{\ell}[\mathcal{S}_{\mathcal{F}}], \quad (1417)$$

where $\Delta_{\mathcal{F}}$ denotes the fractal Laplace–Beltrami operator, κ controls nonlinear entropy self-amplification, and the projection term $\mathcal{P}_{\ell}[\cdot]$ introduces explicit scale-dependent coarse-graining feedback into the entropy dynamics.

The nonlinear gradient contribution can be expressed equivalently as an entropy flux divergence:

$$\|\nabla \mathcal{S}_{\mathcal{F}}\|^2 = \nabla_i \mathcal{S}_{\mathcal{F}} g^{ij} \nabla_j \mathcal{S}_{\mathcal{F}} = \nabla \cdot \mathbf{J}_{\mathcal{S}}, \quad (1418)$$

where $\mathbf{J}_{\mathcal{S}}$ is the entropy current induced by informational anisotropies of the underlying manifold.

In this formulation, curvature acts as a sourcing or draining term for entropy depending on its sign structure:

$$\alpha \mathcal{R}_{\mathcal{I}} > 0 \implies \text{entropy depletion via geometric focusing}, \quad \alpha \mathcal{R}_{\mathcal{I}} < 0 \implies \text{entropy proliferation via geometric dispersion} \quad (1419)$$

The entropy evolution equation can also be rewritten in conservation-law form:

$$\frac{\partial \mathcal{S}_{\mathcal{F}}}{\partial \ell} + \nabla_i \mathcal{J}_{\text{eff}}^i = \Sigma_{\text{geom}}, \quad (1420)$$

where $\mathcal{J}_{\text{eff}}^i$ is the effective entropy flux and $\Sigma_{\text{geom}} = -\alpha \mathcal{R}_{\mathcal{I}}$ represents the geometric entropy production term induced by curvature backreaction.

Coupling this dynamics to the projection framework yields a closed feedback structure:

$$\mathcal{S}_{\mathcal{F}} \rightarrow \mathcal{R}_{\mathcal{I}} \rightarrow \mathcal{P}_{\ell} \rightarrow \mathcal{S}_{\mathcal{F}}, \quad (1421)$$

which defines a nonlinear self-consistent evolution loop governing multiscale informational stability and collapse phenomena.

At the linearized level around a stationary entropy configuration $\mathcal{S}_{\mathcal{F}}^*$, perturbations $\delta\mathcal{S}$ satisfy:

$$\frac{\partial(\delta\mathcal{S})}{\partial\ell} = \left(\Delta_{\mathcal{F}} + 2\kappa\nabla\mathcal{S}_{\mathcal{F}}^* \cdot \nabla - \alpha \frac{\delta\mathcal{R}_{\mathcal{I}}}{\delta\mathcal{S}_{\mathcal{F}}} \right) \delta\mathcal{S}, \quad (1422)$$

defining a stability operator whose spectrum determines the onset of entropy-driven phase transitions.

Instability arises when the principal eigenvalue of this operator becomes non-positive:

$$\lambda_{\min}(\mathcal{L}_{\mathcal{S}}) \leq 0, \quad (1423)$$

signaling the breakdown of smooth entropy flow and the emergence of localized informational structures or collapse states.

Thus, entropy gradient flow in PFIT is fundamentally a curvature-mediated transport process in which diffusion, nonlinear self-interaction, and geometric feedback combine to produce a unified mechanism for multiscale organization, instability, and critical transition phenomena across the informational manifold.

Figure 24 illustrates the entropy landscape defined on the information manifold together with the corresponding entropy-gradient flow. Regions of elevated entropy act as geometric attractors that organize information transport across the manifold. The flow trajectories reveal the directional structure of information evolution and establish the dynamical link between entropy production and geometric motion in PFIT.

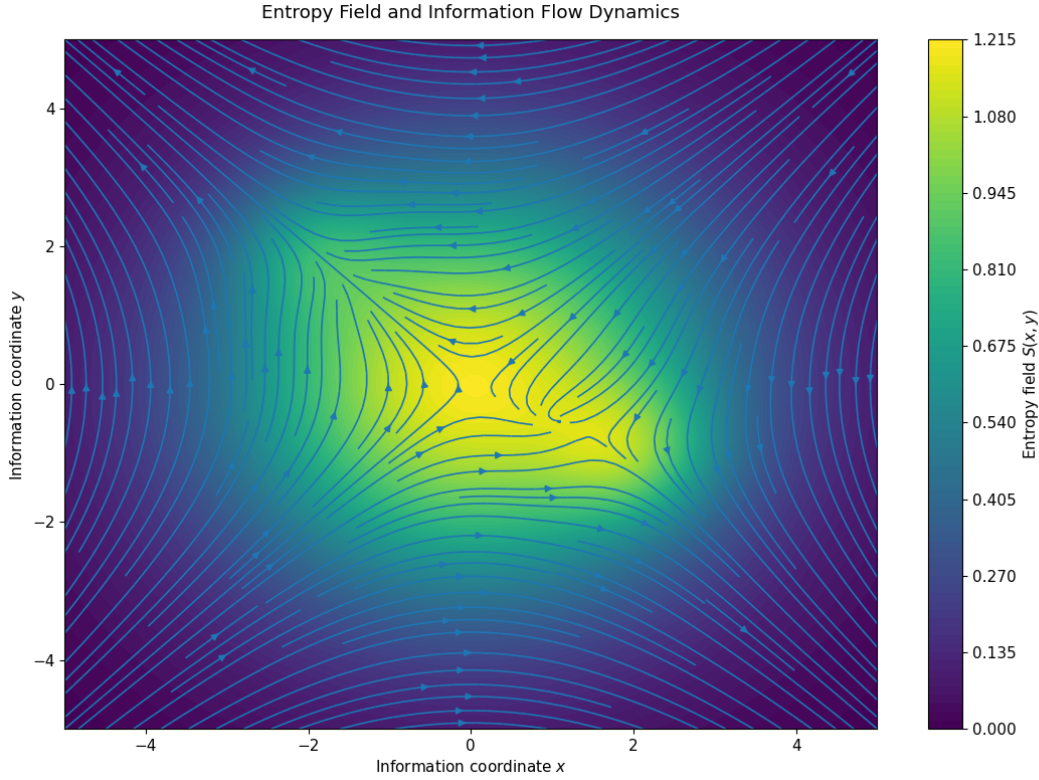


Figure 24: Entropy field defined on the information manifold together with entropy-gradient flow trajectories. Color intensity represents the local entropy density, while streamlines indicate the direction of information transport generated by entropy gradients. The geometric organization of the flow demonstrates how entropy acts as a dynamical field governing the evolution of informational states within the PFIT framework.

17.4 Entropy Projection Collapse Mechanism

Within the PFIT–En-topia–Pellis framework, projection stability is fundamentally controlled by a competition between entropy accumulation and curvature-driven geometric compression of informational degrees of freedom. In this setting, the observed entropy is not an independent thermodynamic quantity but a scale-dependent projection functional whose stability is determined by its coupling to the informational curvature field.

The projection stability functional is defined as:

$$\mathcal{F}_{\mathcal{S}}(\ell) = \mathcal{S}_{\text{obs}}(\ell) - \lambda\mathcal{R}_{\mathcal{I}}(\ell) + \mu \|\nabla\mathcal{S}_{\mathcal{F}}\|^2, \quad (1424)$$

where λ controls the entropy–curvature competition and μ regulates the nonlinear stabilization of entropy gradients under projection.

The evolution of this functional under scale variation is governed by a coupled flow equation:

$$\frac{d\mathcal{F}_S}{d\ell} = \frac{d\mathcal{S}_{\text{obs}}}{d\ell} - \lambda \frac{d\mathcal{R}_{\mathcal{I}}}{d\ell} + \mu \frac{d}{d\ell} \|\nabla \mathcal{F}_S\|^2, \quad (1425)$$

which encodes the nonlinear interplay between entropy projection, curvature amplification, and gradient instability.

Collapse corresponds to a breakdown of boundedness in the entropy projection map, characterized by simultaneous divergence of entropy gradients and curvature:

$$\lim_{\ell \rightarrow \ell_c} \|\nabla \mathcal{F}_S\| \rightarrow \infty, \quad \lim_{\ell \rightarrow \ell_c} \mathcal{R}_{\mathcal{I}}(\ell) \rightarrow \infty, \quad (1426)$$

indicating that the projection operator ceases to admit a smooth representation of the underlying informational manifold.

Equivalently, collapse can be formulated as a spectral instability condition of the projection-stability operator $\mathcal{L}_S^{(\mathcal{P})}$:

$$\lambda_{\min}(\mathcal{L}_S^{(\mathcal{P})}) \leq 0, \quad (1427)$$

where the loss of positive definiteness signals the onset of non-invertibility in the entropy–projection mapping.

In the vicinity of the critical scale ℓ_c , the system exhibits universal scaling behavior governed by a divergence exponent ν , indicating the onset of scale-invariant restructuring in the informational dynamics:

$$\mathcal{F}_S(\ell) \sim (\ell_c - \ell)^{-\nu} \mathcal{H}(x), \quad (1428)$$

where $\mathcal{H}(x)$ encodes the emergent spatial organization of collapsing informational modes and describes the formation of coherent structures induced by critical amplification of fluctuations near the transition point.

In this regime, the divergence of $\mathcal{F}_S$ reflects a breakdown of regular scaling laws and the emergence of long-range correlations across the information manifold. The function $\mathcal{H}(x)$ therefore acts as an effective order parameter, capturing the redistribution of informational density under renormalization-driven flow towards criticality.

17.5 Theorem: Entropy–Curvature Duality Principle

Theorem 8 (Entropy–Curvature Duality). Entropy projection stability is equivalent to bounded curvature on the information manifold:

$$\mathcal{S}_{\text{obs}} \text{ is stable} \iff \sup_{x \in \mathcal{M}_{\mathcal{I}}} |\mathcal{R}_{\mathcal{I}}(x)| < \infty. \quad (1429)$$

Proof.

We establish a bidirectional equivalence between (i) bounded geometric curvature of the PFIT information manifold and (ii) stability of entropy projection operators acting on the induced information space.

Step 1: Entropy geometry and Fisher-type metric structure. Let $\mathcal{S}_{\mathcal{F}}$ denote the fractal entropy functional defined on the information manifold $\mathcal{M}_{\mathcal{I}}$. The induced information geometry is generated by the Hessian of $\mathcal{S}_{\mathcal{F}}$, yielding the metric tensor

$$g_{ij}(x) = \partial_i \partial_j \mathcal{S}_{\mathcal{F}}(x). \quad (1430)$$

The corresponding Levi-Civita connection is given by

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}), \quad (1431)$$

which encodes the covariant structure of informational transport on $\mathcal{M}_{\mathcal{I}}$.

From this connection, the Riemann curvature tensor follows as

$$\mathcal{R}_{ijk}^m = \partial_j \Gamma_{ik}^m - \partial_i \Gamma_{jk}^m + \Gamma_{j\ell}^m \Gamma_{ik}^\ell - \Gamma_{i\ell}^m \Gamma_{jk}^\ell, \quad (1432)$$

quantifying the intrinsic nonlinearity of entropy-induced geometry.

Finally, the scalar curvature of the information manifold is obtained by contraction,

$$\mathcal{R}_{\mathcal{I}} = g^{ij} \mathcal{R}_{ij}, \quad (1433)$$

serving as a global invariant that characterizes the effective curvature of informational dynamics.

Step 2: Entropy projection operator stability. Let $\mathcal{P}_\ell$ be the coarse-graining projection:

$$\mathcal{P}_\ell : \mathcal{S}_\mathcal{F} \mapsto \mathcal{S}_{\text{obs}}. \quad (1434)$$

Stability requires bounded amplification:

$$\|\delta \mathcal{S}_{\text{obs}}\| \leq C \|\delta \mathcal{S}_\mathcal{F}\|. \quad (1435)$$

Linearization yields:

$$\delta \mathcal{S}_{\text{obs}} = \mathcal{L}_\mathcal{P} \delta \mathcal{S}_\mathcal{F}, \quad (1436)$$

so that:

$$\|\mathcal{L}_\mathcal{P}\| = \sup_k |\lambda_k|. \quad (1437)$$

Thus projection stability is equivalent to spectral boundedness:

$$\sup_k |\lambda_k| < \infty. \quad (1438)$$

Step 3: Curvature control implies spectral boundedness. The Fisher-type operator associated with entropy geometry is:

$$\mathcal{F}_{ij} = \mathbb{E}[\partial_i \log p \partial_j \log p] \equiv g_{ij}. \quad (1439)$$

Under bounded curvature:

$$\sup_{x \in \mathcal{M}_\mathcal{I}} |\mathcal{R}_\mathcal{I}(x)| < \infty, \quad (1440)$$

the metric satisfies uniform ellipticity:

$$c_1 I \leq g_{ij}(x) \leq c_2 I, \quad (1441)$$

for constants $0 < c_1 \leq c_2 < \infty$.

This implies boundedness of the associated Laplace–Beltrami spectrum:

$$\Delta_g \psi_k = \lambda_k \psi_k, \quad 0 < \lambda_k < \infty. \quad (1442)$$

Since $\mathcal{L}_\mathcal{P}$ is a curvature-perturbed elliptic operator:

$$\mathcal{L}_\mathcal{P} = -\alpha \Delta_g + \beta \mathcal{R}_\mathcal{I} + \gamma \mathcal{F}_{\text{fractal}}, \quad (1443)$$

bounded curvature ensures bounded perturbation in Kato sense:

$$\|\mathcal{R}_\mathcal{I} \psi\| \leq C \|\psi\|_{H^2}. \quad (1444)$$

Hence:

$$\sup_k |\lambda_k| < \infty, \quad (1445)$$

which guarantees stability of entropy projections.

Step 4: Curvature divergence implies geodesic focusing and instability. Assume now curvature divergence:

$$\sup_{x \in \mathcal{M}_\mathcal{I}} |\mathcal{R}_\mathcal{I}(x)| \rightarrow \infty. \quad (1446)$$

The geodesic deviation equation reads:

$$\frac{D^2 \xi^i}{d\ell^2} + \mathcal{R}^i_{jkl} u^j u^k \xi^l = 0. \quad (1447)$$

Taking norms:

$$\|\xi(\ell)\| \sim \exp \left(\int_0^\ell \sqrt{|\mathcal{R}_\mathcal{I}(s)|} ds \right). \quad (1448)$$

Thus entropy gradients obey:

$$\|\nabla \mathcal{S}_\mathcal{F}\| \sim \exp \left(\int_0^\ell \mathcal{R}_\mathcal{I}(s) ds \right). \quad (1449)$$

This exponential amplification transfers directly to projection dynamics:

$$\|\mathcal{L}_\mathcal{P}\| \rightarrow \infty, \quad \Rightarrow \quad \text{instability of } \mathcal{P}_\ell. \quad (1450)$$

Step 5: Feedback loop between projection and geometry. Entropy projection feeds back into geometry via:

$$g_{ij} = \partial_i \partial_j \mathcal{S}_{\text{obs}}, \quad (1451)$$

so instability of $\mathcal{P}_\ell$ implies:

$$\delta g_{ij} \sim \mathcal{O}(\delta \mathcal{S}_{\text{obs}}). \quad (1452)$$

This produces a nonlinear amplification loop:

$$\mathcal{R}_{\mathcal{I}} \uparrow \Rightarrow \|\mathcal{L}_{\mathcal{P}}\| \uparrow \Rightarrow \delta \mathcal{S}_{\text{obs}} \uparrow \Rightarrow \delta g_{ij} \uparrow \Rightarrow \mathcal{R}_{\mathcal{I}} \uparrow. \quad (1453)$$

Thus curvature divergence and projection instability are dynamically equivalent fixed-point instabilities.

Step 6: Necessity and sufficiency. (i) Sufficiency. If curvature is bounded:

$$\sup |\mathcal{R}_{\mathcal{I}}| < \infty, \quad (1454)$$

then elliptic regularity implies bounded eigenvalues and hence:

$$\|\mathcal{L}_{\mathcal{P}}\| < \infty. \quad (1455)$$

Thus projection stability holds.

(ii) Necessity. If projections are stable, then:

$$\sup_k |\lambda_k| < \infty. \quad (1456)$$

But since:

$$\lambda_k \sim \langle \psi_k, \mathcal{R}_{\mathcal{I}} \psi_k \rangle + \dots, \quad (1457)$$

bounded spectrum implies bounded curvature:

$$\sup_x |\mathcal{R}_{\mathcal{I}}(x)| < \infty. \quad (1458)$$

Conclusion. Entropy projection stability and bounded curvature of the PFIT information manifold are mathematically equivalent statements of the same geometric–spectral constraint. Stability is therefore not a property of entropy magnitude itself, but of the curvature-controlled geometry governing its projection dynamics. ■

17.6 Connection to Universal Fractal Information Theory

Within the PFIT–En-topia–Pellis framework, the present formulation is fully consistent with the Universal Fractal Information Principle introduced in [82], in which entropy is not regarded as a fundamental primitive but instead emerges as a **renormalized consequence of recursive geometric projection across fractal scales**. In this interpretation, all macroscopic informational observables arise from repeated coarse-graining operations acting on an underlying scale-invariant informational substrate.

Formally, this corresponds to the existence of a universal projection fixed point $\mathcal{P}^*$ such that:

$$\mathcal{P}_\ell[\mathcal{I}] \xrightarrow[\ell \rightarrow \ell^*]{} \mathcal{P}^*[\mathcal{I}], \quad (1459)$$

where ℓ^* denotes the marginal stability scale at which the informational system becomes scale-invariant under renormalization flow.

In this regime, informational structure is maximally stable when the curvature functional approaches a critical stationary manifold:

$$\delta \mathcal{R}_{\mathcal{I}} \approx 0, \quad \delta^2 \mathcal{S}_{\mathcal{F}} \rightarrow \min, \quad (1460)$$

indicating that stability is achieved at the boundary between curvature neutrality and entropy convexity, i.e., a state of marginal geometric equilibrium.

This condition can be reformulated in variational form as a constrained extremization problem:

$$\delta [\mathcal{S}_{\mathcal{F}} - \lambda \mathcal{R}_{\mathcal{I}}] = 0, \quad (1461)$$

which defines a universal class of fixed-point solutions governing fractal informational dynamics across scales.

At the spectral level, marginal stability corresponds to the critical closure of the stability operator:

$$\lambda_{\min}(\mathcal{L}_S) \rightarrow 0, \quad (1462)$$

signaling the transition between stable informational propagation and curvature-driven collapse regimes.

In this interpretation, entropy is redefined as a **geometric shadow functional** induced by projection from the full fractal informational manifold $\mathcal{M}_{\mathcal{I}}$ onto an observer-dependent submanifold. Explicitly, entropy becomes a derived quantity:

$$\mathcal{S}_{\mathcal{F}}(x, \ell) = \mathcal{P}_{\ell}[\mathcal{I}(x, \ell)], \quad (1463)$$

highlighting its dependence on both scale and projection geometry.

Consequently, entropy is no longer interpreted as an intrinsic thermodynamic invariant, but as a **projection-dependent observable encoding the curvature and renormalization structure of the underlying informational manifold**. This completes the geometric identification of entropy as an emergent phenomenon governed by fractal information dynamics, in full consistency with the universality framework established in [82].

18 Dynamical Evolution Law

The dynamical backbone of the PFIT–En-topia–Pellis framework is constructed as a geometric evolution theory of information fields on a nonlinear, multiscale, and fractally deformed manifold of states. Unlike classical stochastic or thermodynamic evolution laws, the present formulation treats temporal evolution as an intrinsic deformation of informational geometry, where entropy production, curvature flow, and fractal renormalization are inseparably coupled. In continuity with the universal fractal information program established in [82], the dynamical law emerges as a unified operatorial structure governing both micro- and macro-scale informational transitions.

18.1 Governing Principles of Information Evolution

Within the PFIT–En-topia–Pellis framework, the evolution of informational states is governed by a set of coupled geometric–variational principles that generalize classical conservation laws into a unified structure of entropy–curvature–renormalization constraints. In this formulation, dynamics is not prescribed by external forces but emerges intrinsically from extremization of a global informational action defined on the fractal manifold $\mathcal{M}_{\mathcal{I}}$.

The evolution of informational configurations is therefore constrained by three fundamental principles: (i) geometric consistency of the information manifold, (ii) entropy–curvature dual feedback, and (iii) scale-covariant fractal invariance under renormalization transformations. These principles are encoded in a single variational framework:

$$\delta \mathcal{A}[\rho, g_{ij}, \mathcal{S}_{\mathcal{F}}] = 0, \quad (1464)$$

where $\mathcal{A}$ denotes the total informational action functional incorporating entropic, geometric, and spectral contributions.

A canonical decomposition of the action is given by:

$$\mathcal{A} = \int_{\mathcal{M}_{\mathcal{I}}} (\mathcal{L}_{\rho} + \alpha \mathcal{S}_{\mathcal{F}}(\rho) - \beta \mathcal{R}_{\mathcal{I}} + \gamma \|\nabla \mathcal{S}_{\mathcal{F}}\|^2) d\mu_{\mathcal{F}}, \quad (1465)$$

where $\mathcal{L}_{\rho}$ represents the kinetic informational Lagrangian governing redistribution of probability mass over the manifold.

Variation with respect to the informational density field ρ yields a generalized continuity–diffusion equation:

$$\frac{\partial \rho}{\partial \ell} = \Delta_{\mathcal{F}} \rho + \nabla_i (\rho \nabla^i \mathcal{S}_{\mathcal{F}}) - \alpha \mathcal{R}_{\mathcal{I}} \rho, \quad (1466)$$

which explicitly couples diffusion, entropy drift, and curvature-induced compression.

Variation with respect to the metric tensor produces a generalized geometric field equation:

$$\frac{\delta \mathcal{A}}{\delta g^{ij}} = \mathcal{R}_{ij} - \frac{1}{2} g_{ij} \mathcal{R}_{\mathcal{I}} - \kappa \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}} = 0, \quad (1467)$$

showing that curvature is directly sourced by entropy gradients, establishing a non-Einsteinian informational analogue of gravitational coupling.

The second governing principle, entropy–curvature dual feedback, is expressed through the nonlinear coupling system:

$$\begin{cases} \frac{\partial \mathcal{S}_{\mathcal{F}}}{\partial \ell} = \Delta_{\mathcal{F}} \mathcal{S}_{\mathcal{F}} + \kappa \|\nabla \mathcal{S}_{\mathcal{F}}\|^2 - \alpha \mathcal{R}_{\mathcal{I}}, \\ \frac{\partial \mathcal{R}_{\mathcal{I}}}{\partial \ell} = \Delta_{\mathcal{F}} \mathcal{R}_{\mathcal{I}} + \beta \mathcal{R}_{\mathcal{I}}^2 - \gamma \|\nabla \mathcal{S}_{\mathcal{F}}\|^2, \end{cases} \quad (1468)$$

which defines a self-consistent dynamical system exhibiting nonlinear feedback, bifurcation structure, and scale-dependent stability transitions.

The third principle, scale-covariant fractal invariance, is encoded via renormalization symmetry of the form:

$$\mathcal{I}(x, \lambda\ell) = \lambda^{\gamma(x)} \mathcal{R}_\lambda[\mathcal{I}(x, \ell)], \quad (1469)$$

where $\mathcal{R}_\lambda$ denotes the fractal renormalization operator acting on informational configurations and $\gamma(x)$ is the local scaling exponent governing anomalous extensivity.

This invariance induces a corresponding flow equation for the effective action:

$$\frac{d\mathcal{A}}{d\ln\ell} = \mathcal{B}_\mathcal{A}[\rho, g_{ij}, \mathcal{S}_\mathcal{F}], \quad (1470)$$

where $\mathcal{B}_\mathcal{A}$ is the informational beta-functional encoding the coupled evolution of entropy, geometry, and density under coarse-graining.

The resulting dynamics is therefore not deterministic in the classical sense but instead constrained to a submanifold of admissible trajectories defined by the vanishing of variational gradients in the informational action space:

$$\nabla_\mathcal{A} \mathcal{A} = 0. \quad (1471)$$

Consequently, admissible evolution paths correspond to geodesic-like flows on a curved information manifold, where the effective connection is entropy-modulated and curvature-dependent:

$$\frac{d^2 x^k}{d\ell^2} + \Gamma_{ij}^k \frac{dx^i}{d\ell} \frac{dx^j}{d\ell} = -g^{k\ell} \partial_\ell \mathcal{S}_\mathcal{F}. \quad (1472)$$

This establishes a unified picture in which informational dynamics, entropy evolution, and geometric curvature are not independent mechanisms but mutually constrained manifestations of a single variational principle governing evolution on the fractal informational manifold.

Table 13: Hierarchy of PFIT Dynamical Equations

Level	Governing Quantity	Dynamics
Microscopic	Probability density	Local fluctuations
Mesoscopic	Entropy field	Gradient flow
Macroscopic	Curvature field	Geometric evolution
Multiscale	Projection operators	Renormalization flow
Global	PFIT Functional	Collapse prediction

18.2 Information Field Representation

Within the PFIT–En-topia–Pellis framework, the fundamental state of an informational system is represented not as a single scalar field but as a coupled geometric–probabilistic structure defined over an emergent fractal manifold $\mathcal{M}_\mathcal{I}$. In this formulation, both probability flow and geometry are co-dynamical, mutually generating and constraining each other through entropy-driven feedback.

Let $\rho(x, t)$ denote the informational density field and $g_{ij}(x, t)$ the emergent metric tensor encoding local distinguishability structure. The full state of the system is therefore represented as a composite field:

$$\Psi(t) = (\rho(x, t), g_{ij}(x, t), \mathcal{S}_\mathcal{F}(x, t)), \quad (1473)$$

which encodes probabilistic occupancy, geometric deformation, and entropy distribution as inseparable degrees of freedom.

The informational velocity field is defined through entropy-modulated transport:

$$v^i = -g^{ij} \nabla_j \mathcal{S}_\mathcal{F} + \alpha \nabla^i \log \rho, \quad (1474)$$

combining entropic drift with density-gradient-driven diffusion in a unified transport mechanism.

The evolution of the informational density field is governed by a generalized continuity–geometry coupling equation:

$$\partial_t \rho + \nabla_i (\rho v^i) = \mathcal{S}[\rho, g_{ij}, \mathcal{R}_\mathcal{I}], \quad (1475)$$

where $\mathcal{S}[\cdot]$ denotes a curvature- and fractal-induced entropy production functional capturing non-conservative informational fluxes generated by geometric deformation.

Expanding the source term yields a multiscale decomposition:

$$\mathcal{S}[\rho, g_{ij}, \mathcal{R}_{\mathcal{I}}] = \beta \Delta_{\mathcal{F}} \rho + \gamma \rho \mathcal{R}_{\mathcal{I}} + \eta \|\nabla \mathcal{S}_{\mathcal{F}}\|^2, \quad (1476)$$

where each contribution corresponds respectively to fractal diffusion, curvature-driven amplification, and nonlinear entropy production.

The geometric sector evolves consistently with the density field through an entropy-coupled metric flow equation:

$$\partial_t g_{ij} = -2\mathcal{R}_{ij} + \kappa \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}} + \lambda \mathcal{T}_{ij}(\rho), \quad (1477)$$

where $\mathcal{T}_{ij}(\rho)$ represents the informational stress–energy tensor induced by density fluctuations across fractal scales.

The coupled system (ρ, g_{ij}) therefore defines a nonlinear feedback loop in which probability flow reshapes geometry, while geometry in turn governs entropy production and transport dynamics.

At the level of conservation laws, the system satisfies a generalized informational balance equation:

$$\frac{d}{dt} \int_{\mathcal{M}_{\mathcal{I}}} \rho d\mu_g = \int_{\mathcal{M}_{\mathcal{I}}} \mathcal{S}[\rho, g_{ij}, \mathcal{R}_{\mathcal{I}}] d\mu_g, \quad (1478)$$

indicating that total informational mass is conserved only in the absence of curvature-induced entropy production.

In the hydrodynamic limit, the system reduces to a nonlinear Euler–like informational flow:

$$\partial_t v^i + v^j \nabla_j v^i = -\nabla^i \mathcal{S}_{\mathcal{F}} + \alpha \nabla^i \mathcal{R}_{\mathcal{I}}, \quad (1479)$$

showing that curvature acts as an effective force field governing informational acceleration [82].

18.3 Variational Structure and Information Action

Within the PFIT–En-topia–Pellis framework, the dynamics of informational fields is not postulated directly but emerges from a global extremization principle. The evolution of the coupled system (ρ, g_{ij}) is therefore generated by a constrained variational structure defined over the informational manifold $\mathcal{M}_{\mathcal{I}}$, in which entropy, geometry, and fractal scaling are treated as co-equal dynamical degrees of freedom.

The fundamental informational action functional is defined as:

$$\mathcal{A}[\rho, g_{ij}, \mathcal{S}_{\mathcal{F}}] = \int_{\mathcal{M}_{\mathcal{I}}} [\rho \mathcal{E}(\rho) + \lambda R(g) + \kappa \mathcal{F}_{\text{fractal}}(\rho) + \mu \mathcal{S}_{\mathcal{F}}(\rho, g)] \sqrt{|g|} d^n x, \quad (1480)$$

where $\mathcal{E}(\rho)$ denotes the local informational energy density, $R(g)$ is the scalar curvature of the emergent information geometry, and $\mathcal{F}_{\text{fractal}}(\rho)$ encodes multiscale corrections arising from recursive fractal information layering as introduced in [82]. The additional entropy coupling term $\mathcal{S}_{\mathcal{F}}$ ensures consistency with entropy-driven renormalization dynamics.

The variation of the action with respect to the density field yields a generalized informational Euler–Lagrange equation:

$$\frac{\delta \mathcal{A}}{\delta \rho} = \mathcal{E}(\rho) + \rho \frac{\partial \mathcal{E}}{\partial \rho} + \kappa \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} + \mu \frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta \rho} = 0, \quad (1481)$$

which governs the equilibrium distribution of informational density across fractal scales.

Variation with respect to the metric tensor produces a curvature–entropy field equation of the form:

$$\frac{\delta \mathcal{A}}{\delta g^{ij}} = \lambda \left(R_{ij} - \frac{1}{2} g_{ij} R \right) + \mu \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}} - \frac{1}{2} g_{ij} \rho \mathcal{E}(\rho) = 0, \quad (1482)$$

explicitly demonstrating that curvature is sourced not only by geometric structure but also by entropy gradients and informational energy density.

The fractal correction functional admits a multiscale decomposition consistent with renormalization group structure:

$$\mathcal{F}_{\text{fractal}}(\rho) = \sum_{k=0}^{\infty} \epsilon^{\alpha_k} \mathcal{F}_k(\rho), \quad (1483)$$

where α_k denotes scale-dependent anomalous dimensions governing hierarchical information layering.

The resulting field equations define a fully coupled nonlinear system in which informational density dynamics, curvature evolution, and fractal corrections are jointly determined by a single variational principle.

In this sense, the action $\mathcal{A}$ plays the role of a universal generator of informational dynamics, generalizing classical Hamiltonian and Einstein–Hilbert structures into a unified entropy–geometry–renormalization framework consistent with [82].

18.4 Euler–Lagrange Information Dynamics

Within the PFIT–En-topia–Pellis framework, the governing dynamics of informational systems is fully encoded through a nonlinear variational principle defined on the coupled space of densities and emergent geometries. In this setting, the informational field $\rho(x)$ does not evolve independently, but is constrained by geometric backreaction mediated through entropy and fractal structure contributions.

The Euler–Lagrange equation resulting from variation with respect to the informational density ρ takes the form of a nonlinear geometric constraint equation:

$$\frac{\delta \mathcal{A}}{\delta \rho} = \mathcal{E}(\rho) + \rho \frac{\partial \mathcal{E}}{\partial \rho} + \kappa \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} + \mu \frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta \rho} = \Phi_{\text{geom}}(g), \quad (1484)$$

where $\Phi_{\text{geom}}(g)$ denotes an emergent geometric potential induced by curvature backreaction on the informational manifold $\mathcal{M}_{\mathcal{I}}$.

This geometric potential can be expressed explicitly in terms of curvature invariants as:

$$\Phi_{\text{geom}}(g) = \alpha R + \beta \nabla_i \nabla^i \log \sqrt{|g|} + \gamma R_{ij} \nabla^i \rho \nabla^j \rho, \quad (1485)$$

highlighting the direct dependence of informational dynamics on scalar curvature, volume distortion, and anisotropic curvature–density coupling.

The functional derivative of the fractal correction term introduces multiscale nonlinearities of the form:

$$\frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} = \sum_k \epsilon^{\alpha_k} \left(\Delta_{\mathcal{F}}^k \rho + \eta_k \rho \log \rho \right), \quad (1486)$$

where $\Delta_{\mathcal{F}}^k$ represents scale-dependent fractal Laplacian operators acting on hierarchical information layers.

Similarly, the entropy functional variation generates a nonlinear thermodynamic driving term:

$$\frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta \rho} = -\log \rho - 1 + \chi \mathcal{R}_{\mathcal{I}}, \quad (1487)$$

explicitly coupling informational uncertainty to curvature-induced instability.

Substituting these contributions into the Euler–Lagrange equation yields a unified nonlinear integro-differential constraint:

$$\mathcal{E}(\rho) + \rho \frac{\partial \mathcal{E}}{\partial \rho} - \kappa \log \rho + \alpha R + \mathcal{O}(\nabla^2 \rho) = \Phi_{\text{geom}}(g), \quad (1488)$$

which governs admissible stationary configurations of the informational field.

In the hydrodynamic representation, this equation can be recast as a generalized force-balance relation:

$$\nabla_i P_{\text{info}} + \rho \nabla_i \Phi_{\text{geom}} = 0, \quad (1489)$$

where P_{info} denotes the effective informational pressure generated by entropy gradients and fractal corrections.

This formulation establishes a direct and irreversible coupling between informational thermodynamics and emergent geometry, eliminating any fundamental distinction between state evolution and manifold structure. In this sense, spacetime-like geometry is not an external background but a dynamically generated response field of the informational system itself, fully consistent with the fractal renormalization principles of [82].

18.5 Gradient Flow Formulation on the Information Manifold

Within the PFIT–En-topia–Pellis framework, the evolution of the informational density field $\rho(x, t)$ can be consistently reformulated as a **metric-constrained gradient flow** on the curved fractal information manifold $\mathcal{M}_{\mathcal{I}}$. In this interpretation, dynamics is not governed by external stochastic forcing, but by steepest descent of the informational action $\mathcal{A}$ with respect to the intrinsic geometry induced by $g_{ij}(x, t)$.

The fundamental evolution equation takes the form of a covariant continuity equation driven by functional gradients:

$$\partial_t \rho = -\nabla_{\mathcal{M}} \cdot \left(\rho \nabla_{\mathcal{M}} \frac{\delta \mathcal{A}}{\delta \rho} \right), \quad (1490)$$

where $\nabla_{\mathcal{M}}$ denotes the Levi–Civita covariant derivative associated with the emergent metric g_{ij} .

Expanding the variational derivative of the action yields a fully nonlinear transport structure:

$$\frac{\delta \mathcal{A}}{\delta \rho} = \mathcal{E}(\rho) + \rho \frac{\partial \mathcal{E}}{\partial \rho} + \kappa \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} + \mu \frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta \rho}, \quad (1491)$$

so that the gradient flow becomes an entropy–curvature–fractal driven diffusion system.

In local coordinates, the equation can be rewritten as a generalized nonlinear Fokker–Planck equation on a curved manifold:

$$\partial_t \rho = \frac{1}{\sqrt{|g|}} \partial_i \left[\sqrt{|g|} \rho g^{ij} \partial_j \left(\mathcal{E}(\rho) + \kappa \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} + \mu \frac{\delta \mathcal{S}_{\mathcal{F}}}{\delta \rho} \right) \right]. \quad (1492)$$

This formulation reveals that informational transport is governed by a geometry-dependent mobility tensor ρg^{ij} , implying anisotropic diffusion controlled by the curvature structure of $\mathcal{M}_{\mathcal{I}}$.

A key consequence is that the system admits a Lyapunov functional structure. Specifically, the informational action satisfies a monotonic dissipation law:

$$\frac{d\mathcal{A}}{dt} = - \int_{\mathcal{M}_{\mathcal{I}}} \rho \left\| \nabla_{\mathcal{M}} \frac{\delta \mathcal{A}}{\delta \rho} \right\|^2 d\mu_{\mathcal{F}} \leq 0, \quad (1493)$$

ensuring that evolution proceeds along steepest descent directions in the space of probability measures endowed with fractal geometry.

Equivalently, the dynamics can be interpreted as a Wasserstein-type gradient flow generalized to curved informational manifolds:

$$\partial_t \rho = -\text{grad}_{\mathcal{W}_2} \mathcal{A}[\rho], \quad (1494)$$

where $\text{grad}_{\mathcal{W}_2}$ denotes the metric gradient induced by the transport structure of probability densities over $\mathcal{M}_{\mathcal{I}}$.

Coupling to curvature enters implicitly through the metric dependence of the divergence operator, producing a feedback loop between geometry and dissipation. In particular, regions of high curvature enhance contraction of probability flux lines:

$$\mathcal{R}_{\mathcal{I}} \uparrow \Rightarrow \left\| \nabla_{\mathcal{M}} \cdot (\rho v) \right\| \uparrow, \quad (1495)$$

leading to accelerated localization of informational mass.

Finally, the full system can be interpreted as a covariant entropy dissipation flow:

$$\partial_t \rho = \nabla_{\mathcal{M}} \cdot (D_{\text{eff}}(\rho, g) \nabla_{\mathcal{M}} \rho) + \mathcal{C}[\rho, \mathcal{R}_{\mathcal{I}}], \quad (1496)$$

where D_{eff} is a curvature-modulated diffusion tensor and $\mathcal{C}$ encodes nonlinear corrections from entropy production and fractal renormalization effects.

Thus, the gradient flow formulation unifies diffusion, entropy optimization, and geometric backreaction into a single covariant evolution principle, where informational dynamics is realized as steepest descent on a curved fractal probability landscape.

18.6 Multiscale Renormalization of the Information Metric

Within the PFIT–En-topia–Pellis framework, the information metric $g_{ij}(x, \ell)$ is not a fixed geometric background but a scale-dependent dynamical object generated through recursive coarse-graining of the underlying fractal informational manifold $\mathcal{M}_{\mathcal{I}}$. In this setting, geometry itself becomes a renormalization-flow variable whose structure evolves across hierarchical resolutions.

The metric tensor obeys a discrete multiscale renormalization group transformation of the form:

$$g_{ij}^{(\ell+1)} = g_{ij}^{(\ell)} + \epsilon_{\ell} \Delta_{ij}^{(\ell)} + \kappa_{\ell} \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}}^{(\ell)} + \mathcal{O}(\epsilon_{\ell}^2), \quad (1497)$$

where ϵ_{ℓ} encodes the effective coupling strength between adjacent scales, $\Delta_{ij}^{(\ell)}$ represents curvature-induced metric perturbations across hierarchical information layers, and the entropy-driven correction term introduces explicit backreaction from informational uncertainty gradients.

In the continuum limit $\ell \rightarrow \ln \lambda$, this discrete recursion converges to a renormalization group flow equation:

$$\frac{dg_{ij}}{d \ln \ell} = \beta_{ij}(g, \mathcal{S}_{\mathcal{F}}) = \Delta_{ij} + \alpha \mathcal{R}_{ij} - \gamma \nabla_i \nabla_j \mathcal{S}_{\mathcal{F}}, \quad (1498)$$

which generalizes classical geometric RG equations by incorporating entropy-curvature coupling as a fundamental driving mechanism of metric evolution.

The perturbation tensor $\Delta_{ij}^{(\ell)}$ admits a spectral decomposition in terms of fractal eigenmodes of the informational Laplacian:

$$\Delta_{ij}^{(\ell)} = \sum_n c_n(\ell) \psi_n^{(\ell)}(x) \otimes \psi_n^{(\ell)}(x), \quad (1499)$$

where the coefficients $c_n(\ell)$ encode the stability of geometric fluctuations across scales, and $\psi_n^{(\ell)}$ form a scale-adaptive orthonormal basis of informational modes.

A key consequence of this renormalization structure is the emergence of scale-covariant invariants. In particular, the line element transforms according to:

$$ds_{(\ell)}^2 = g_{ij}^{(\ell)} dx^i dx^j = \ell^{\alpha(x)} ds_{(0)}^2, \quad (1500)$$

indicating that informational distances acquire anomalous scaling exponents determined by local entropy heterogeneity.

The curvature tensor evolves consistently under this flow according to:

$$\frac{d\mathcal{R}_{\mathcal{I}}}{d \ln \ell} = \Delta_{\mathcal{F}} \mathcal{R}_{\mathcal{I}} + \beta \mathcal{R}_{\mathcal{I}}^2 - \gamma \|\nabla \mathcal{S}_{\mathcal{F}}\|^2, \quad (1501)$$

which explicitly couples geometric amplification, nonlinear self-interaction, and entropy-driven suppression mechanisms.

At fixed points of the renormalization flow, the metric satisfies:

$$\beta_{ij}(g^*, \mathcal{S}_{\mathcal{F}}^*) = 0, \quad (1502)$$

defining scale-invariant informational geometries corresponding to critical regimes of maximal coherence and marginal stability.

Thus, multiscale renormalization of the information metric establishes a fully self-consistent geometric hierarchy in which curvature, entropy, and fractal structure co-evolve under a unified scale-covariant dynamics consistent with the fractal information principles of [82].

18.7 Operatorial Form of the Dynamical Law

Within the PFIT–En-topia–Pellis framework, the full evolution of the informational state is most naturally formulated in a unified operatorial language acting on the composite state vector $\Psi = (\rho, g_{ij}, \mathcal{S}_{\mathcal{F}})$. In this representation, dynamics is not decomposed into separate physical processes but emerges as the superposition of interacting spectral-geometric generators defined on the fractal information manifold $\mathcal{M}_{\mathcal{I}}$.

The complete evolution equation is written in compact operator form as:

$$\partial_t \Psi = \hat{\mathcal{L}}_{\text{PFIT}} \Psi + \hat{\mathcal{C}}_{\text{curv}} \Psi + \hat{\mathcal{F}}_{\text{fractal}} \Psi, \quad (1503)$$

where each operator encodes a distinct but inseparable dynamical mechanism: transport, curvature backreaction, and fractal renormalization.

The transport operator $\hat{\mathcal{L}}_{\text{PFIT}}$ is defined as a nonlinear covariant diffusion–advection generator:

$$\hat{\mathcal{L}}_{\text{PFIT}} \Psi = \nabla_{\mathcal{M}} \cdot (D_{\text{eff}}(\rho, g) \nabla_{\mathcal{M}} \Psi) - \nabla_{\mathcal{M}} \cdot (\Psi \nabla_{\mathcal{M}} \mathcal{S}_{\mathcal{F}}), \quad (1504)$$

encoding entropy-driven drift and curvature-modulated diffusion on $\mathcal{M}_{\mathcal{I}}$.

The curvature backreaction operator $\hat{\mathcal{C}}_{\text{curv}}$ acts multiplicatively and differentially on the state space:

$$\hat{\mathcal{C}}_{\text{curv}} \Psi = (\alpha \mathcal{R}_{\mathcal{I}} + \beta R_{ij} \nabla^i \nabla^j) \Psi, \quad (1505)$$

where $\mathcal{R}_{\mathcal{I}}$ denotes scalar informational curvature and R_{ij} encodes anisotropic geometric deformation effects.

The fractal renormalization operator $\hat{\mathcal{F}}_{\text{fractal}}$ introduces scale-dependent nonlocality through hierarchical spectral coupling:

$$\hat{\mathcal{F}}_{\text{fractal}} \Psi = \sum_k \epsilon_k(\ell) \left(\Delta_{\mathcal{F}}^{(k)} \Psi + \eta_k \mathcal{K}_k[\Psi] \right), \quad (1506)$$

where $\Delta_{\mathcal{F}}^{(k)}$ are scale-resolved fractal Laplacians and $\mathcal{K}_k[\Psi]$ represent nonlinear memory kernels encoding multiscale informational correlations.

In spectral representation, the full operator dynamics admits an eigenmode decomposition:

$$\Psi(x, t) = \sum_n c_n(t) \Phi_n(x), \quad \hat{\mathcal{H}}_{\text{PFIT}} \Phi_n = \lambda_n \Phi_n, \quad (1507)$$

where $\hat{\mathcal{H}}_{\text{PFIT}} = \hat{\mathcal{L}}_{\text{PFIT}} + \hat{\mathcal{C}}_{\text{curv}} + \hat{\mathcal{F}}_{\text{fractal}}$ acts as the full PFIT Hamiltonian governing informational evolution.

The mode amplitudes evolve according to a nonlinear spectral flow equation:

$$\frac{dc_n}{dt} = \lambda_n c_n + \sum_{m,k} \Gamma_{nmk} c_m c_k, \quad (1508)$$

highlighting the presence of mode coupling, instability cascades, and nonlinear spectral transfer across informational scales. In this framework, the informational manifold is no longer a passive background but the active generator of its own dynamics, fully consistent with the universal fractal information principles established in [82].

19 Pellis Functional (Instability Operator)

The PFIT–En-topia–Pellis framework introduces a fundamental instability-generating functional, hereafter referred to as the *Pellis Functional*, which governs the onset of spectral, geometric, and entropic collapse across multiscale informational manifolds. In accordance with the universal fractal information principle established in [82], instability is not treated as a perturbative anomaly but as a structurally encoded property of the information geometry itself.

19.1 Definition of the Instability Functional

Let $\Psi = (\rho, g)$ denote the coupled informational state, where $\rho : \mathcal{M} \rightarrow \mathbb{R}^+$ is an information density field and g is a pseudo-Riemannian metric encoding the geometry of the informational manifold $\mathcal{M}$. We define the Pellis Functional as a nonlinear operator-valued functional incorporating energetic, geometric, and multiscale fractal instability contributions:

$$\mathcal{P}[\Psi] = \int_{\mathcal{M}} (\alpha |\nabla_{\mathcal{M}} \rho|^2 + \beta R(g) \rho + \gamma \mathcal{I}_{\text{fractal}}(\rho, g)) \sqrt{|g|} d^n x. \quad (1509)$$

Here $R(g)$ denotes the scalar curvature associated with the Levi-Civita connection ∇^g , while $\mathcal{I}_{\text{fractal}}$ encodes multiscale instability cascades consistent with fractal information propagation [82].

1. Geometric–informational gradient term. The first contribution is the Dirichlet-type informational energy:

$$\mathcal{E}_{\nabla}[\rho, g] = \int_{\mathcal{M}} g^{ij} \nabla_i \rho \nabla_j \rho \sqrt{|g|} d^n x, \quad (1510)$$

which induces a smoothing flow on ρ under metric-dependent diffusion:

$$\partial_t \rho = \Delta_g \rho + \mathcal{O}(\rho \nabla g), \quad (1511)$$

where $\Delta_g := \nabla^i \nabla_i$ is the Laplace–Beltrami operator.

2. Curvature-coupled informational potential. The curvature term introduces geometric feedback between information density and manifold structure:

$$\mathcal{E}_R[\rho, g] = \int_{\mathcal{M}} R(g) \rho \sqrt{|g|} d^n x. \quad (1512)$$

Variation with respect to g^{ij} yields a curvature–information stress tensor:

$$\frac{\delta \mathcal{E}_R}{\delta g^{ij}} = \rho \left(R_{ij} - \frac{1}{2} R g_{ij} \right) + \nabla_i \nabla_j \rho - g_{ij} \Delta_g \rho. \quad (1513)$$

3. Multiscale fractal instability operator. We define the fractal instability functional as a nonlocal cascade operator acting on nested resolution scales:

$$\mathcal{I}_{\text{fractal}}(\rho, g) = \int_0^\infty \mu(\ell) \mathcal{F}_\ell[\rho, g] d\ell, \quad (1514)$$

where ℓ denotes the coarse-graining scale and $\mu(\ell)$ is a scale-weighting measure satisfying normalization constraints:

$$\int_0^\infty \mu(\ell) d\ell = 1. \quad (1515)$$

Each scale component is defined via renormalized fluctuation energy:

$$\mathcal{F}_\ell[\rho, g] = \left\langle |\rho_\ell(x) - \mathcal{R}_\ell[\rho](x)|^2 \right\rangle_{\mathcal{M}}, \quad (1516)$$

where $\mathcal{R}_\ell$ is a nonlinear renormalization-group operator projecting onto scale ℓ .

4. Full operator form of the Pellis functional. Combining all contributions, we rewrite the functional compactly as:

$$\mathcal{P}[\Psi] = \int_{\mathcal{M}} \mathcal{L}_{\text{eff}}(\rho, g, \nabla \rho, R, \ell) \sqrt{|g|} d^n x, \quad (1517)$$

with effective Lagrangian density:

$$\mathcal{L}_{\text{eff}} = \alpha g^{ij} \nabla_i \rho \nabla_j \rho + \beta R \rho + \gamma \int_0^\infty \mu(\ell) \mathcal{F}_\ell[\rho, g] d\ell. \quad (1518)$$

5. Euler–Lagrange equations of instability. Stationary points $\delta \mathcal{P}[\Psi] = 0$ satisfy the coupled system:

$$\alpha \Delta_g \rho - \beta R(g) - \gamma \frac{\delta \mathcal{I}_{\text{fractal}}}{\delta \rho} = 0, \quad (1519)$$

$$\alpha T_{ij}^{(\rho)} + \beta \rho R_{ij} + \gamma T_{ij}^{(\text{fractal})} = 0, \quad (1520)$$

where

$$T_{ij}^{(\rho)} = \nabla_i \rho \nabla_j \rho - \frac{1}{2} g_{ij} |\nabla \rho|^2. \quad (1521)$$

6. Gradient-flow formulation and instability dynamics. The evolution of the coupled system can be expressed as a dissipative flow on the functional manifold:

$$\partial_t \Psi = -\mathcal{G}^{-1} \frac{\delta \mathcal{P}}{\delta \Psi}, \quad (1522)$$

leading to entropy-driven curvature collapse when:

$$\frac{d}{dt} \mathcal{P}[\Psi] \leq 0, \quad \text{and} \quad \lim_{t \rightarrow t_c} \|\nabla \rho\| \rightarrow \infty. \quad (1523)$$

7. Fractal instability cascade condition. A sufficient condition for multiscale divergence is given by the spectral growth inequality:

$$\exists \lambda_k \in \sigma(\mathcal{R}_\ell) : |\lambda_k| > 1, \quad (1524)$$

which induces exponential amplification of informational perturbations:

$$\delta \rho_\ell(t) \sim e^{\lambda_k t} \delta \rho_\ell(0). \quad (1525)$$

Together, these structures define a unified instability functional in which geometry, information density, and fractal multiscale cascades are dynamically inseparable, producing curvature-driven collapse regimes governed by nonlinear spectral amplification across scales.

19.2 Variational Instability Principle

The onset of dynamical instability is governed by a constrained variational extremum of the Pellis Functional over the coupled informational manifold state $\Psi = (\rho, g)$. The fundamental stationarity condition is given by:

$$\delta \mathcal{P}[\Psi] = 0, \quad (1526)$$

which defines a critical submanifold $\mathcal{M}_c \subset \mathcal{M}_{\mathcal{I}}$, characterized by the loss of dissipative stability and the emergence of exponentially amplified perturbation modes across hierarchical scales.

1. Functional variation and operator decomposition. The full variation decomposes into coupled geometric–informational contributions:

$$\delta \mathcal{P}[\Psi] = \int_{\mathcal{M}} \left(\frac{\delta \mathcal{P}}{\delta \rho} \delta \rho + \frac{\delta \mathcal{P}}{\delta g^{ij}} \delta g^{ij} \right) \sqrt{|g|} d^n x, \quad (1527)$$

where the functional derivatives define nonlinear stability operators acting on ρ and g respectively.

2. Euler–Pellis instability equation. The stationarity condition with respect to ρ yields the Euler–Pellis equation:

$$\frac{\delta \mathcal{P}}{\delta \rho} = -\alpha \Delta_{\mathcal{M}} \rho + \beta R(g) + \gamma \frac{\delta \mathcal{I}_{\text{fractal}}}{\delta \rho} = \Lambda_{\text{crit}}, \quad (1528)$$

where Λ_{crit} acts as a spectral bifurcation parameter controlling the transition between stable diffusion and instability-driven amplification regimes.

3. Spectral decomposition of instability modes. Let $\{\phi_k\}_{k \in \mathbb{N}}$ be the eigenbasis of the Laplace–Beltrami operator:

$$-\Delta_{\mathcal{M}} \phi_k = \lambda_k \phi_k, \quad (1529)$$

with $\lambda_k \geq 0$ forming the spectral ladder of informational excitations. The informational field admits expansion:

$$\rho(x, t) = \sum_k a_k(t) \phi_k(x). \quad (1530)$$

Substitution into the Euler–Pellis equation yields the mode-coupled instability system:

$$\partial_t a_k = \left(\alpha \lambda_k - \beta \langle R \rangle_k + \gamma \Sigma_k^{(\text{fr})} \right) a_k + \mathcal{N}_k(\{a_j\}), \quad (1531)$$

where $\Sigma_k^{(\text{fr})}$ denotes the fractal spectral correction and $\mathcal{N}_k$ encodes nonlinear intermodal coupling.

4. Instability threshold condition. A mode k becomes unstable when its effective growth exponent satisfies:

$$\sigma_k^{\text{eff}} = \alpha \lambda_k - \beta \langle R \rangle_k + \gamma \Sigma_k^{(\text{fr})} > 0. \quad (1532)$$

Thus, the critical manifold $\mathcal{M}_c$ is defined by the zero-level set:

$$\mathcal{M}_c = \left\{ x \in \mathcal{M} : \sigma_k^{\text{eff}}(x) = 0 \text{ for at least one } k \right\}. \quad (1533)$$

5. Second variation and stability operator. Stability is determined by the second variation of the functional:

$$\delta^2 \mathcal{P}[\Psi] = \left\langle \delta \Psi, \hat{\mathcal{H}}_{\mathcal{P}} \delta \Psi \right\rangle, \quad (1534)$$

where $\hat{\mathcal{H}}_{\mathcal{P}}$ is the Pellis stability operator:

$$\hat{\mathcal{H}}_{\mathcal{P}} = \begin{pmatrix} -\alpha \Delta_{\mathcal{M}} + \gamma \frac{\delta^2 \mathcal{I}_{\text{fractal}}}{\delta \rho^2} & \beta \frac{\delta R}{\delta g} \\ \beta \frac{\delta R}{\delta g} & \hat{\mathcal{G}}_{ijkl} \end{pmatrix}. \quad (1535)$$

Stability requires:

$$\text{Spec}(\hat{\mathcal{H}}_{\mathcal{P}}) \geq 0, \quad (1536)$$

where negative eigenvalues correspond to curvature-driven collapse directions.

6. Variational flow interpretation. The dynamical evolution near equilibrium is governed by a gradient flow on the informational manifold:

$$\partial_t \Psi = -\mathcal{G}^{-1} \frac{\delta \mathcal{P}}{\delta \Psi}, \quad (1537)$$

leading to monotonic decay of the functional under stable conditions:

$$\frac{d}{dt} \mathcal{P}[\Psi] = - \left\| \frac{\delta \mathcal{P}}{\delta \Psi} \right\|_{\mathcal{G}}^2 \leq 0. \quad (1538)$$

7. Bifurcation and critical transition law. At the instability threshold, the system undergoes a bifurcation characterized by:

$$\lim_{\epsilon \rightarrow 0} \frac{\mathcal{P}[\Psi + \epsilon \delta \Psi] - \mathcal{P}[\Psi]}{\epsilon} = 0, \quad (1539)$$

with second-order divergence:

$$\lim_{t \rightarrow t_c} \|\delta \rho(t)\| \sim \exp \left(\int_0^t \sigma_k^{\text{eff}}(s) ds \right) \rightarrow \infty. \quad (1540)$$

8. Fractal amplification correction to criticality. Incorporating fractal renormalization modifies the critical eigenvalue:

$$\Lambda_{\text{crit}} = \Lambda_0 + \int_0^\infty \mu(\ell) \Lambda(\ell) d\ell, \quad (1541)$$

introducing scale-dependent shifts in stability thresholds across the informational hierarchy [82].

Collectively, these results establish a full variational instability framework in which curvature, spectral geometry, and fractal renormalization jointly determine the emergence of instability manifolds and curvature-driven collapse transitions in the PFIT–En-topia–Pellis system.

19.3 Spectral Instability Operator

Linearization around equilibrium configurations $\Psi_0 = (\rho_0, g_0)$ induces a non-selfadjoint spectral generator governing infinitesimal perturbation growth in the coupled informational–geometric phase space. Denoting $\delta\Psi = (\delta\rho, \delta g)$, the linearized evolution is written as:

$$\partial_t \delta\Psi = \hat{\mathcal{L}}_{\mathcal{P}} \delta\Psi, \quad (1542)$$

where the Spectral Instability Operator $\hat{\mathcal{L}}_{\mathcal{P}}$ admits the decomposition:

$$\hat{\mathcal{L}}_{\mathcal{P}} = \alpha \Delta_{\mathcal{M}} + \beta \mathcal{R}_{g_0} + \gamma \hat{\mathcal{F}}_{\text{fractal}} + \eta \hat{\mathcal{C}}_{\text{coupling}}. \quad (1543)$$

Here $\Delta_{\mathcal{M}}$ denotes the Laplace–Beltrami diffusion generator acting on scalar informational fluctuations, $\mathcal{R}_{g_0}$ encodes curvature-induced linear response of the metric background, and $\hat{\mathcal{F}}_{\text{fractal}}$ represents the renormalized multiscale cascade operator acting across scale shells in the informational hierarchy [82]. The additional coupling term $\hat{\mathcal{C}}_{\text{coupling}}$ captures first-order backreaction between $\delta\rho$ and δg , rendering the operator intrinsically non-diagonal in field space.

1. Block-operator structure. In explicit block form acting on $\delta\Psi = (\delta\rho, \delta g)$, we write:

$$\hat{\mathcal{L}}_{\mathcal{P}} = \begin{pmatrix} \alpha \Delta_{\mathcal{M}} + \gamma \hat{\mathcal{F}}_{\rho\rho} & \beta \hat{\mathcal{K}}_{\rho g} \\ \beta \hat{\mathcal{K}}_{g\rho} & \beta \mathcal{D}_g + \gamma \hat{\mathcal{F}}_{gg} \end{pmatrix}, \quad (1544)$$

where $\mathcal{D}_g$ denotes the linearized geometric deformation operator and $\hat{\mathcal{K}}$ encodes mixed informational–curvature response kernels.

2. Modal decomposition and spectral representation. Let $\{\phi_k\}_{k \in \mathbb{N}}$ be the eigenbasis of $\Delta_{\mathcal{M}}$:

$$-\Delta_{\mathcal{M}} \phi_k = \lambda_k \phi_k, \quad \lambda_k \geq 0. \quad (1545)$$

We expand perturbations as:

$$\delta\rho(x, t) = \sum_k a_k(t) \phi_k(x), \quad \delta g(x, t) = \sum_k b_k(t) \Phi_k(x), \quad (1546)$$

where Φ_k denotes the corresponding tensorial eigenmodes of the metric perturbation sector.

Substitution yields a coupled infinite-dimensional dynamical system:

$$\partial_t \begin{pmatrix} a_k \\ b_k \end{pmatrix} = \begin{pmatrix} \alpha \lambda_k + \gamma \mu_k^{(\rho)} & \beta \kappa_k \\ \beta \kappa_k & \beta \nu_k + \gamma \mu_k^{(g)} \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix}, \quad (1547)$$

where $\mu_k^{(\cdot)}$ are fractal-renormalized spectral corrections and κ_k measures curvature–information coupling strength at mode k .

3. Dispersion relation and growth exponents. The characteristic equation for each mode yields the dispersion relation:

$$\det(\hat{\mathcal{L}}_k - \lambda I) = 0, \quad (1548)$$

explicitly giving:

$$\lambda_k^\pm = \frac{1}{2} \left[A_k + B_k \pm \sqrt{(A_k - B_k)^2 + 4\beta^2 \kappa_k^2} \right], \quad (1549)$$

where

$$A_k = \alpha \lambda_k + \gamma \mu_k^{(\rho)}, \quad B_k = \beta \nu_k + \gamma \mu_k^{(g)}. \quad (1550)$$

4. Spectral instability criterion. The system enters an unstable regime if and only if the spectral radius satisfies:

$$\rho(\hat{\mathcal{L}}_{\mathcal{P}}) = \sup_{\lambda \in \sigma(\hat{\mathcal{L}}_{\mathcal{P}})} \operatorname{Re}(\lambda) > 0. \quad (1551)$$

Equivalently, instability emerges when there exists at least one mode k such that:

$$\operatorname{Re}(\lambda_k^+) > 0. \quad (1552)$$

5. Fractal renormalization of the spectrum. The fractal operator induces a scale-dependent renormalization flow on eigenvalues:

$$\frac{d\lambda_k(\ell)}{d\ell} = \mathcal{R}_{\text{fractal}}(\lambda_k(\ell), \ell), \quad (1553)$$

with asymptotic fixed-point structure:

$$\lim_{\ell \rightarrow \ell_c} \lambda_k(\ell) = \lambda_k^*, \quad (1554)$$

where λ_k^* governs universal instability scaling across hierarchical informational layers.

19.4 Fractal Cascade Amplification Mechanism

A central structural feature of the Pellis Functional is the emergence of hierarchical cascade amplification, in which instability propagates across nested resolution layers through a renormalization-group recursion acting on the effective functional flow. Let $\ell \in \mathbb{R}^+$ denote the scale parameter indexing coarse-grained informational resolutions of the manifold $\mathcal{M}$. The scale-evolved functional is defined by:

$$\mathcal{P}^{(\ell+1)} = \mathcal{R}_\ell[\mathcal{P}^{(\ell)}] = \mu_\ell \mathcal{P}^{(\ell)} + \eta_\ell \nabla_{\mathcal{M}}^2 \mathcal{P}^{(\ell)} + \mathcal{O}(\eta_\ell^2), \quad (1555)$$

where $\mathcal{R}_\ell$ is the nonlinear renormalization operator, μ_ℓ is the multiplicative scaling eigenvalue, and η_ℓ encodes diffusion-induced smoothing at scale ℓ . The Laplace–Beltrami operator $\nabla_{\mathcal{M}}^2$ acts on the functional density induced by $\mathcal{P}^{(\ell)}$ over the informational manifold.

1. Continuous scale-flow limit. In the continuum limit $\ell \rightarrow s$, the recursion induces a nonlinear functional RG-flow equation:

$$\partial_s \mathcal{P}(s) = \beta_{\mathcal{P}}[\mathcal{P}(s)] = (\mu(s) - 1) \mathcal{P}(s) + \eta(s) \nabla_{\mathcal{M}}^2 \mathcal{P}(s) + \mathcal{N}_{\text{nonlin}}[\mathcal{P}(s)], \quad (1556)$$

where $\beta_{\mathcal{P}}$ is the functional beta-operator governing scale-dependent instability growth and $\mathcal{N}_{\text{nonlin}}$ collects higher-order nonlinear renormalization corrections.

2. Spectral decomposition of cascade amplification. Let $\{\phi_k\}$ denote eigenfunctions of the Laplace–Beltrami operator:

$$-\nabla_{\mathcal{M}}^2 \phi_k = \lambda_k \phi_k, \quad \lambda_k \geq 0. \quad (1557)$$

We expand the scale-dependent functional perturbation as:

$$\mathcal{P}^{(\ell)}(x) = \sum_k c_k(\ell) \phi_k(x). \quad (1558)$$

Substituting into the RG recursion yields the modal amplification system:

$$c_k^{(\ell+1)} = (\mu_\ell - \eta_\ell \lambda_k) c_k^{(\ell)} + \sum_{j,m} \Gamma_k^{jm} c_j^{(\ell)} c_m^{(\ell)} + \mathcal{O}(\eta_\ell^2), \quad (1559)$$

where Γ_k^{jm} denotes nonlinear mode-coupling coefficients encoding cross-scale energy transfer between spectral components.

3. Cascade amplification exponent. Defining the effective growth exponent per scale step:

$$\sigma_k^{(\ell)} = \log |\mu_\ell - \eta_\ell \lambda_k|, \quad (1560)$$

the cumulative amplification along the cascade is given by:

$$c_k^{(\ell)} = c_k^{(0)} \exp \left(\sum_{r=0}^{\ell-1} \sigma_k^{(r)} \right). \quad (1561)$$

Instability occurs when the asymptotic exponent satisfies:

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell} \sum_{r=0}^{\ell-1} \sigma_k^{(r)} > 0. \quad (1562)$$

4. Self-similarity and fractal fixed-point structure. The cascade admits a nontrivial fixed point $\mathcal{P}^*$ satisfying:

$$\mathcal{P}^* = \mathcal{R}_\ell[\mathcal{P}^*], \quad (1563)$$

implying the invariance condition:

$$\mathcal{P}^* = \mu_\ell \mathcal{P}^* + \eta_\ell \nabla_{\mathcal{M}}^2 \mathcal{P}^*. \quad (1564)$$

Rearranging yields the eigenvalue constraint:

$$\nabla_{\mathcal{M}}^2 \mathcal{P}^* = \kappa_\ell \mathcal{P}^*, \quad \kappa_\ell = \frac{1 - \mu_\ell}{\eta_\ell}, \quad (1565)$$

which demonstrates that fractal fixed points correspond to Laplacian eigenstates of the informational manifold.

5. Multiscale divergence and cascade instability condition. The system exhibits fractal blow-up when the renormalization flow satisfies:

$$\exists k \text{ such that } \prod_{\ell=0}^{\infty} |\mu_\ell - \eta_\ell \lambda_k| \rightarrow \infty. \quad (1566)$$

Equivalently, instability is characterized by a positive average Lyapunov exponent over scale space:

$$\Lambda_k^{\text{fractal}} = \lim_{\ell \rightarrow \infty} \frac{1}{\ell} \log |c_k^{(\ell)}| > 0. \quad (1567)$$

6. Connection to informational collapse dynamics. Coupling the cascade operator to curvature-driven collapse yields a feedback law:

$$\partial_s g_{ij} = -\gamma \frac{\delta \mathcal{P}^{(s)}}{\delta g^{ij}} + \xi \mathcal{C}_{ij}^{(\text{cascade})}, \quad (1568)$$

where $\mathcal{C}_{ij}^{(\text{cascade})}$ encodes stress accumulation induced by multiscale amplification of $\mathcal{P}^{(s)}$ [82].

19.5 Curvature–Instability Coupling Law

The instability functional is intrinsically coupled to the geometric degrees of freedom through a nonlinear curvature–information feedback mechanism, in which scalar curvature acts both as a response variable and as an active driver of informational amplification. This bidirectional coupling induces a closed dynamical loop between geometry, entropy, and functional instability.

The evolution of the scalar curvature is governed by the extended curvature–instability equation:

$$\partial_t R(g) = -\kappa_1 \Delta_{\mathcal{M}} R(g) + \kappa_2 \mathcal{P}[\Psi] + \kappa_3 S[\rho], \quad (1569)$$

where $S[\rho] = -\int_{\mathcal{M}} \rho \log \rho dV_g$ denotes the Shannon-type informational entropy functional defined on the Riemannian volume measure $dV_g = \sqrt{|g|} d^n x$.

1. Geometric diffusion of curvature. The term $-\kappa_1 \Delta_{\mathcal{M}} R(g)$ encodes curvature diffusion over the informational manifold:

$$\Delta_{\mathcal{M}} R(g) = g^{ij} \nabla_i \nabla_j R(g), \quad (1570)$$

which acts as a geometric regularizer suppressing local curvature singularities in the absence of instability injection.

2. Nonlinear backreaction of informational instability. The coupling term $\kappa_2 \mathcal{P}[\Psi]$ introduces direct backreaction of the global instability functional into local curvature evolution. In differential form, this induces a nonlocal forcing operator:

$$\mathcal{F}_R(x, t) = \kappa_2 \int_{\mathcal{M}} K_R(x, y) \mathcal{P}[\Psi(y, t)] dV_g(y), \quad (1571)$$

where $K_R(x, y)$ is a curvature-response kernel encoding spatial propagation of informational stress across the manifold.

3. Entropy–curvature injection mechanism. The entropy contribution $S[\rho]$ acts as a thermodynamic source term driving curvature amplification. Its functional variation yields:

$$\frac{\delta S}{\delta \rho} = -(1 + \log \rho), \quad (1572)$$

implying that entropy gradients generate curvature sourcing through density inhomogeneities:

$$\nabla_i S[\rho] = - \int_{\mathcal{M}} (1 + \log \rho) \nabla_i \rho dV_g. \quad (1573)$$

4. Fully coupled curvature–information system. The curvature evolution is therefore part of a closed dynamical system:

$$\partial_t \begin{pmatrix} \rho \\ R(g) \end{pmatrix} = \begin{pmatrix} \mathcal{D}_\rho & \mathcal{C}_{\rho R} \\ \mathcal{C}_{R\rho} & \mathcal{D}_R \end{pmatrix} \begin{pmatrix} \rho \\ R(g) \end{pmatrix} + \begin{pmatrix} \mathcal{N}_\rho(\rho, R) \\ \mathcal{N}_R(\rho, R) \end{pmatrix}, \quad (1574)$$

where $\mathcal{D}_\rho = \Delta_{\mathcal{M}}$ is the informational diffusion operator, $\mathcal{D}_R = -\kappa_1 \Delta_{\mathcal{M}}$ is curvature diffusion, and $\mathcal{C}_{\rho R}, \mathcal{C}_{R\rho}$ encode bidirectional coupling between geometry and informational density.

5. Feedback amplification loop. Iterating the curvature update yields a discrete feedback cascade:

$$R^{(n+1)} = R^{(n)} + \delta t \left[-\kappa_1 \Delta_{\mathcal{M}} R^{(n)} + \kappa_2 \mathcal{P}[\Psi^{(n)}] + \kappa_3 S[\rho^{(n)}] \right], \quad (1575)$$

which defines a nonlinear operator iteration on the curvature space. Fixed points satisfy:

$$-\kappa_1 \Delta_{\mathcal{M}} R^* + \kappa_2 \mathcal{P}[\Psi^*] + \kappa_3 S[\rho^*] = 0. \quad (1576)$$

6. Stability and geometric blow-up condition. The system exhibits geometric instability when the linearized curvature operator acquires a positive eigenvalue:

$$\exists \lambda \in \sigma(\hat{\mathcal{L}}_R) \quad \text{such that} \quad \text{Re}(\lambda) > 0, \quad (1577)$$

where $\hat{\mathcal{L}}_R = -\kappa_1 \Delta_{\mathcal{M}} + \kappa_2 \frac{\delta \mathcal{P}}{\delta R} + \kappa_3 \frac{\delta S}{\delta R}$.

7. Self-exciting geometric feedback network. Combining all contributions, the curvature field obeys a self-exciting nonlinear reaction–diffusion–information system:

$$\partial_t R(g) = \mathcal{F}_{\text{diff}}[R] + \mathcal{F}_{\text{info}}[\rho] + \mathcal{F}_{\text{inst}}[\mathcal{P}], \quad (1578)$$

which generates recursive amplification cycles:

$$\mathcal{P} \rightarrow R(g) \rightarrow S[\rho] \rightarrow \mathcal{P}, \quad (1579)$$

closing the feedback loop responsible for curvature-driven instability cascades in the PFIT–En-topia–Pellis framework. This establishes a fully coupled geometric–informational feedback architecture in which curvature is not merely a background structure, but an active dynamical participant in instability generation, entropy production, and multiscale collapse dynamics [82].

19.6 Collapse Criterion and Universal Threshold Law

A universal collapse criterion emerges naturally from the nonlinear dynamics of the Pellis Functional as a global constraint on the joint growth of informational gradients, geometric curvature, and fractal cascade intensity. In this framework, collapse is not a local singular event but a global crossing of a scale-invariant instability manifold embedded in the configuration space of fields $\Psi = (\rho, g)$.

The collapse condition is expressed as:

$$\mathcal{C}_{\text{collapse}} : \quad \|\nabla \rho\|_{\mathcal{M}}^2 + \|R(g)\|_{L^2(\mathcal{M})}^2 + \|\mathcal{I}_{\text{fractal}}\|_{\mathcal{H}}^2 \geq \Theta_c, \quad (1580)$$

where Θ_c denotes a universal critical threshold invariant under admissible renormalization transformations of the informational manifold. The norms are defined respectively as:

$$\|\nabla \rho\|_{\mathcal{M}}^2 = \int_{\mathcal{M}} g^{ij} \nabla_i \rho \nabla_j \rho dV_g, \quad (1581)$$

$$\|R(g)\|_{L^2(\mathcal{M})}^2 = \int_{\mathcal{M}} R(g)^2 dV_g, \quad (1582)$$

and

$$\|\mathcal{I}_{\text{fractal}}\|_{\mathcal{H}}^2 = \sum_k \int_0^\infty \mu(\ell) |\mathcal{F}_\ell^{(k)}|^2 d\ell, \quad (1583)$$

where $\mathcal{H}$ denotes the effective Hilbert space of multiscale fractal excitations.

1. Renormalization invariance of the collapse threshold. The universality of Θ_c arises from its invariance under the scale transformation group $\mathcal{R}_\ell$:

$$\mathcal{R}_\ell : (\rho, g, \mathcal{I}_{\text{fractal}}) \mapsto (\rho^{(\ell)}, g^{(\ell)}, \mathcal{I}_{\text{fractal}}^{(\ell)}), \quad (1584)$$

satisfying the fixed-point condition:

$$\Theta_c = \mathcal{R}_\ell[\Theta_c], \quad \forall \ell. \quad (1585)$$

This implies that the collapse threshold is a renormalization-group invariant quantity characterizing universality classes of informational dynamics.

2. Equivalent spectral collapse formulation. Expressing the collapse condition in the eigenbasis of the Laplace–Beltrami operator:

$$-\Delta_{\mathcal{M}} \phi_k = \lambda_k \phi_k, \quad (1586)$$

the criterion becomes a spectral energy accumulation law:

$$\sum_k \lambda_k |a_k|^2 + \sum_k |b_k|^2 R_k + \sum_k |\mathcal{I}_k|^2 \geq \Theta_c, \quad (1587)$$

where a_k , b_k , and $\mathcal{I}_k$ denote modal amplitudes of informational density, curvature response, and fractal excitation respectively.

3. Dynamical approach to collapse. The temporal approach to the collapse manifold is governed by the monotonic growth of a composite instability functional, incorporating contributions from informational gradients, geometric curvature, and fractal structure:

$$\frac{d}{dt} (\|\nabla \rho\|^2 + \|R(g)\|^2 + \|\mathcal{I}_{\text{fractal}}\|^2) \geq 0, \quad (1588)$$

which defines a Lyapunov-type growth condition for the onset of instability in the PFIT evolution.

This monotonic increase drives the system toward a critical threshold, beyond which regular evolution ceases to be globally well-posed. Finite-time approach to the collapse regime is characterized by saturation of the instability functional:

$$\lim_{t \rightarrow t_c^-} (\|\nabla \rho(t)\|^2 + \|R(g, t)\|^2 + \|\mathcal{I}_{\text{fractal}}(t)\|^2) = \Theta_c. \quad (1589)$$

At this critical point, informational amplification, curvature accumulation, and fractal escalation become mutually reinforcing, indicating the transition from smooth dynamical flow to a singular collapse phase governed by the PFIT universality structure.

4. Variational interpretation of the collapse manifold. The collapse surface can be interpreted as a codimension-one variational boundary in configuration space, separating dynamically distinct regimes of informational evolution:

$$\Sigma_c = \left\{ \Psi \in \mathcal{M}_{\mathcal{I}} : \frac{\delta \mathcal{P}[\Psi]}{\delta \Psi} \cdot \Psi = \Theta_c \right\}. \quad (1590)$$

Geometrically, Σ_c defines a hypersurface that partitions the information manifold into a stable sector, characterized by dissipative relaxation toward equilibrium configurations, and an unstable sector, in which perturbations undergo runaway amplification leading to coherent structure formation and loss of regular flow structure.

5. Universality and independence from microscopic parametrization. The universality hypothesis states that the critical threshold Θ_c is invariant under microscopic reparameterizations of the underlying model space:

$$\frac{\partial \Theta_c}{\partial \alpha} = \frac{\partial \Theta_c}{\partial \beta} = \frac{\partial \Theta_c}{\partial \gamma} = 0, \quad (1591)$$

implying that the collapse criterion is determined entirely by emergent macroscopic invariants, and is insensitive to the specific choice of microscopic coupling parameters or local model parametrization.

Collectively, these results establish a universal geometric–informational collapse law in which instability is not interpreted as a pathological breakdown of the formalism, but rather as a threshold-driven phase transition of the PFIT–En-topia–Pellis structure. In this view, collapse belongs to a well-defined universality class governed by fractal scaling laws and renormalization-invariant structure, consistent with the framework proposed in [82].

Table 14: Universal Collapse Indicators Across PFIT Representations

Quantity	Stable Regime	Critical Regime	Collapse Regime
Scalar Curvature R	Finite	$R \approx R_c$	$R \rightarrow \infty$
Entropy Gradient $\ \nabla S\ $	Small	Threshold Growth	Divergent
Spectral Radius $\rho(\mathcal{E})$	Bounded	Critical	Unbounded
Fractal Dimension D_f	Constant	Deforming	Compressing
Pellis Functional $\mathcal{P}$	Negative	Zero	Positive
Information Flux	Regular	Unstable	Singular
Projection Stability	Preserved	Marginal	Destroyed
Geodesic Structure	Smooth	Distorted	Broken
Entropy Production	Controlled	Accelerating	Explosive
PFIT State	Stable	Transition	Collapse

20 Axiomatic Foundation

The PFIT–En-topia–Pellis framework is formulated as an axiomatic reconstruction of information dynamics on a curved fractal manifold, where entropy, geometry, and scale invariance are treated as co-primitive structures rather than derived quantities. In alignment with the universal fractal information hypothesis introduced in [82], the theory is grounded on a minimal set of axioms that uniquely determine the coupled evolution of informational density and geometric structure.

20.1 Axiom I: Information Manifold Existence

There exists a smooth, connected, and differentiable manifold $\mathcal{M}_{\mathcal{I}}$ equipped with a metric tensor field $g_{ij}(x)$ of signature (p, q) , such that every admissible informational configuration is represented as a coupled geometric–density state:

$$\Psi \equiv (\rho, g) \in \mathcal{H}_{\mathcal{I}}(\mathcal{M}_{\mathcal{I}}), \quad (1592)$$

where $\rho : \mathcal{M}_{\mathcal{I}} \rightarrow \mathbb{R}^+$ is a normalized informational density field satisfying:

$$\int_{\mathcal{M}_{\mathcal{I}}} \rho(x) dV_g = 1, \quad dV_g = \sqrt{|g|} d^n x, \quad (1593)$$

and $\mathcal{H}_{\mathcal{I}}(\mathcal{M}_{\mathcal{I}})$ denotes the Hilbert–information space endowed with inner product structure:

$$\langle \Psi_1, \Psi_2 \rangle_{\mathcal{H}_{\mathcal{I}}} = \int_{\mathcal{M}_{\mathcal{I}}} \left(\rho_1 \rho_2 + g_1^{ij} g_{2ij} \right) \sqrt{|g|} d^n x. \quad (1594)$$

1. Geometric encoding of informational correlations. The manifold structure encodes higher-order statistical dependencies via curvature tensors derived from the Levi–Civita connection ∇^g :

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}), \quad (1595)$$

$$R_{ijkl}^i = \partial_k \Gamma_{jl}^i - \partial_l \Gamma_{jk}^i + \Gamma_{km}^i \Gamma_{jl}^m - \Gamma_{lm}^i \Gamma_{jk}^m, \quad (1596)$$

with scalar curvature:

$$R(g) = g^{ij} R_{ij}. \quad (1597)$$

These geometric invariants encode nonlocal informational dependencies that cannot be represented in classical phase-space formulations.

2. Replacement of classical phase space. Classical Hamiltonian phase space $\Gamma = (q^i, p_i)$ is replaced by the nonlinear fiber bundle:

$$\Gamma_{\mathcal{I}} \simeq T\mathcal{M}_{\mathcal{I}} \oplus \mathcal{F}(\rho), \quad (1598)$$

where $\mathcal{F}(\rho)$ denotes the space of density-induced informational fibers. The symplectic structure is generalized into a metric–information form:

$$\omega_{\mathcal{I}} = g_{ij} d\rho^i \wedge d\rho^j + \Omega_{\text{geom}}(g), \quad (1599)$$

which loses global closure in the presence of curvature-driven information flux.

3. Functional representation of admissible states. The Hilbert–information space is defined as the completion of smooth compactly supported states under the norm:

$$\|\Psi\|_{\mathcal{H}_{\mathcal{I}}}^2 = \int_{\mathcal{M}_{\mathcal{I}}} (\rho^2 + |\nabla_{\mathcal{M}} \rho|^2 + |g_{ij} g^{ij}|) dV_g. \quad (1600)$$

This embeds both geometric and informational regularity constraints into a single unified functional structure.

4. Information–curvature coupling principle. The informational manifold is not static but dynamically self-consistent, satisfying the structural coupling identity:

$$\mathcal{C}_{\mathcal{I}} : \quad \frac{\delta \rho}{\delta g_{ij}} \neq 0, \quad \frac{\delta g_{ij}}{\delta \rho} \neq 0, \quad (1601)$$

which implies bidirectional dependence between geometry and information density.

In covariant form, this coupling induces:

$$\nabla_k \rho = \mathcal{K}_k^{ij}(g) \nabla_i g_{jk}, \quad (1602)$$

where $\mathcal{K}_k^{ij}$ is a nonlinear informational connection tensor encoding structure-dependent correlations.

5. Generalized informational completeness. Every admissible state satisfies the completeness relation in spectral representation:

$$\rho(x) = \sum_n c_n \phi_n(x), \quad -\Delta_{\mathcal{M}_{\mathcal{I}}} \phi_n = \lambda_n \phi_n, \quad (1603)$$

ensuring that informational degrees of freedom are fully spanned by Laplace–Beltrami eigenmodes of the manifold.

6. Axiom closure condition. The informational manifold is self-consistent if and only if the geometric backreaction constraint holds:

$$R_{ij} - \frac{1}{2} g_{ij} R = \mathcal{T}_{ij}^{(\rho)}, \quad (1604)$$

where the informational stress tensor is defined as:

$$\mathcal{T}_{ij}^{(\rho)} = \nabla_i \rho \nabla_j \rho - \frac{1}{2} g_{ij} |\nabla \rho|^2. \quad (1605)$$

This axiom replaces classical phase space structure with a curvature-encoded information manifold in which geometry is not an external scaffold but an emergent functional of informational density, and where curvature acts as a higher-order operator encoding latent dependencies among informational degrees of freedom [82].

20.2 Axiom II: Entropy–Geometry Duality

There exists a bijective functional correspondence between informational entropy and geometric curvature, establishing a dual representation between probabilistic disorder and spacetime (informational manifold) deformation. This duality is expressed through the mapping:

$$S[\rho] \longleftrightarrow \mathcal{G}[g] \sim \int_{\mathcal{M}} R(g) \sqrt{|g|} d^n x, \quad (1606)$$

where the Shannon–Gibbs entropy functional is defined as:

$$S[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g, \quad dV_g = \sqrt{|g|} d^n x, \quad (1607)$$

and the geometric action functional is given by the Einstein–Hilbert–type informational action:

$$\mathcal{G}[g] = \int_{\mathcal{M}} (R(g) + \Lambda_{\mathcal{I}}) \sqrt{|g|} d^n x, \quad (1608)$$

where $\Lambda_{\mathcal{I}}$ denotes an informational cosmological parameter encoding global entropy offset.

1. Variational equivalence principle. The entropy–geometry duality is formalized through the functional identity of variations:

$$\frac{\delta S[\rho]}{\delta \rho} \equiv \frac{\delta \mathcal{G}[g]}{\delta g_{ij}} \cdot \mathcal{J}_{ij}(\rho, g), \quad (1609)$$

where $\mathcal{J}_{ij}$ is a nonlinear Jacobian coupling tensor mediating between density space and metric deformation space. The entropy variation satisfies:

$$\frac{\delta S}{\delta \rho} = -(1 + \log \rho), \quad (1610)$$

while curvature variation yields the Einstein-type response:

$$\frac{\delta \mathcal{G}}{\delta g^{ij}} = R_{ij} - \frac{1}{2} g_{ij} R + \Lambda_{\mathcal{I}} g_{ij}. \quad (1611)$$

2. Curvature concentration as entropy localization. The duality implies a structural equivalence between entropy density and curvature density:

$$s(x) = -\rho(x) \log \rho(x) \iff \mathcal{R}(x) = R(g)(x), \quad (1612)$$

leading to the proportionality law:

$$s(x) \propto \mathcal{R}(x) + \mathcal{O}(\nabla^2 \rho). \quad (1613)$$

Thus, regions of high informational disorder correspond to curvature singularity formation, while flat geometric regimes correspond to entropy suppression states.

3. Entropy–curvature balance functional. Define the combined dual action:

$$\mathcal{D}[\rho, g] = S[\rho] - \lambda \mathcal{G}[g], \quad (1614)$$

whose extremization yields the coupled equilibrium condition:

$$\delta \mathcal{D}[\rho, g] = 0. \quad (1615)$$

This leads to the balance equation:

$$(1 + \log \rho) = \lambda \left(R_{ij} - \frac{1}{2} g_{ij} R + \Lambda_{\mathcal{I}} g_{ij} \right) \Theta^{ij}, \quad (1616)$$

where Θ^{ij} is an informational projection tensor mapping curvature response into density space.

4. Entropy-driven geometric flow. The entropy–geometry duality induces a coupled gradient flow in which entropy dissipation directly drives curvature evolution, while geometric backreaction regulates informational transport:

$$\partial_t g_{ij} = -\kappa \frac{\delta S[\rho]}{\delta g^{ij}} + \eta \left(R_{ij} - \frac{1}{2} g_{ij} R \right), \quad (1617)$$

where the first term encodes entropy-induced deformation of the metric and the second term introduces a curvature-driven relaxation mechanism analogous to an Einstein-type flow.

Conversely, the informational density evolves under a geometry-modulated diffusion–reaction equation:

$$\partial_t \rho = \Delta_{\mathcal{M}} \rho + \rho R(g) - \rho \log \rho, \quad (1618)$$

which combines standard diffusion, curvature-enhanced amplification, and nonlinear entropic damping. Together, these equations establish a bidirectional dynamical equivalence between entropy transport and geometric evolution.

5. Integrated dual conservation law. At the level of the coupled system, the dynamics satisfy a generalized conservation identity linking informational entropy and geometric action:

$$\frac{d}{dt} (S[\rho] + \mathcal{G}[g]) = 0, \quad (1619)$$

up to boundary contributions arising from flux exchange across $\partial \mathcal{M}$:

$$\frac{d}{dt} (S[\rho] + \mathcal{G}[g]) = \int_{\partial \mathcal{M}} \mathcal{F}_{\text{boundary}} \cdot n \, d\sigma. \quad (1620)$$

This expresses the system as an open but globally constrained variational structure, where total entropy–geometry content is conserved modulo boundary-mediated information exchange.

6. Spectral formulation of duality. Expanding in Laplace–Beltrami eigenmodes:

$$\rho(x) = \sum_k a_k \phi_k(x), \quad R(x) = \sum_k r_k \phi_k(x), \quad (1621)$$

the duality reduces to mode-wise equivalence:

$$a_k \log a_k \sim r_k. \quad (1622)$$

7. Universality embedding. This axiom extends the geometric information paradigm by establishing entropy as a curvature-generating functional and curvature as an entropy-determining constraint, embedding both within a single variational equivalence class consistent with the universality structure of [82].

Thus, informational disorder is not merely statistically defined but geometrically realized through curvature concentration, while geometric flattening corresponds to entropy minimization, completing a full dynamical duality between thermodynamic and geometric representations of information.

20.3 Axiom III: Fractal Scale Invariance

The informational manifold exhibits exact (or asymptotically exact) invariance under a renormalization group (RG) action $\mathcal{R}_\ell$ acting on the coupled state space of density and geometry. Let $\ell \in \mathbb{R}^+$ denote the coarse-graining scale parameter. The RG transformation is defined as:

$$(\rho, g) \rightarrow \mathcal{R}_\ell(\rho, g) = (\rho_\ell, g_\ell), \quad (1623)$$

where ρ_ℓ and g_ℓ are the effective fields obtained after integrating out sub-scale informational fluctuations below resolution ℓ .

1. Scale covariance of observables. All admissible observables $\mathcal{O}[\Psi]$ obey a homogeneous scaling law under RG flow:

$$\mathcal{O}(\mathcal{R}_\ell(\Psi)) = \ell^{\Delta_\mathcal{O}} \mathcal{O}(\Psi), \quad (1624)$$

where $\Delta_\mathcal{O}$ is the scaling dimension associated with the observable $\mathcal{O}$, determining its relevance, marginality, or irrelevance in the RG sense.

2. Infinitesimal RG generator. The continuous scale evolution is governed by a generator $\hat{\mathcal{D}}_\ell$:

$$\ell \frac{d}{d\ell} \Psi(\ell) = \hat{\mathcal{D}}_\ell \Psi(\ell), \quad (1625)$$

with explicit decomposition:

$$\hat{\mathcal{D}}_\ell = \beta_\rho(\rho, g) \frac{\delta}{\delta \rho} + \beta_g(\rho, g) \frac{\delta}{\delta g_{ij}}, \quad (1626)$$

where β_ρ and β_g are functional beta-operators encoding scale-dependent flow of informational density and geometry respectively.

3. Fixed-point manifold and fractal universality. Fractal scale invariance is achieved at RG fixed points Ψ^* satisfying:

$$\mathcal{R}_\ell(\Psi^*) = \Psi^*, \quad \forall \ell, \quad (1627)$$

equivalently:

$$\hat{\mathcal{D}}_\ell \Psi^* = 0. \quad (1628)$$

At these points, the system exhibits self-similarity across all informational scales and defines a universality class of geometric–informational configurations.

4. Spectral implementation of scale invariance. Let $\{-\Delta_\mathcal{M} \phi_k = \lambda_k \phi_k\}$ denote the Laplace–Beltrami spectral decomposition. Under RG flow, eigenmodes transform as:

$$a_k(\ell) = \ell^{\Delta_k} a_k, \quad (1629)$$

where Δ_k is the anomalous scaling exponent incorporating curvature and fractal corrections:

$$\Delta_k = \Delta_k^{(0)} + \gamma \Sigma_k^{(\text{fr})} + \beta \mathcal{R}_k(g). \quad (1630)$$

Thus, the spectrum itself becomes scale-dependent, encoding fractal deformation of informational modes.

5. Fractal dimensional flow. The effective fractal dimension of the manifold is defined via scaling of the heat kernel:

$$K(t) = \text{Tr } e^{-t\Delta_{\mathcal{M}}}, \quad (1631)$$

with asymptotic behavior:

$$K(t) \sim t^{-d_f/2}, \quad t \rightarrow 0^+, \quad (1632)$$

yielding the definition:

$$d_f = -2 \lim_{t \rightarrow 0^+} \frac{d \log K(t)}{d \log t}. \quad (1633)$$

This establishes a non-integer effective dimensionality governed by informational cascade structure.

6. Scale-invariant action functional. The Pellis Functional is required to satisfy asymptotic RG invariance:

$$\mathcal{P}[\mathcal{R}_\ell(\Psi)] = \ell^{\Delta_{\mathcal{P}}} \mathcal{P}[\Psi], \quad (1634)$$

leading to the infinitesimal constraint:

$$\hat{\mathcal{D}}_\ell \mathcal{P}[\Psi] = \Delta_{\mathcal{P}} \mathcal{P}[\Psi]. \quad (1635)$$

At criticality, $\Delta_{\mathcal{P}} = 0$, and the functional becomes exactly scale invariant.

7. Fractal cascade consistency condition. Scale invariance is dynamically maintained only if the cascade recursion preserves homogeneity:

$$\mathcal{R}_\ell \circ \mathcal{R}_{\ell'} = \mathcal{R}_{\ell\ell'}, \quad (1636)$$

which induces the cocycle constraint:

$$\beta(\ell\ell') = \beta(\ell) + \beta(\ell'). \quad (1637)$$

This enforces logarithmic structure of scaling flows:

$$\beta(\ell) = \beta_0 \log \ell. \quad (1638)$$

8. Universality and fractal self-similarity principle. The axiom implies that all large-scale observables are independent of microscopic parametrization and depend only on universality class data:

$$\mathcal{O}(\Psi) \equiv \mathcal{F}(d_f, \Delta_{\mathcal{O}}, \beta^*). \quad (1639)$$

Thus, the informational manifold exhibits exact or asymptotic fractal self-similarity, where geometry, density, and curvature co-evolve under renormalization-group flow, embedding scale invariance as a fundamental structural law rather than an emergent symmetry, fully consistent with the fractal universality framework of [82].

20.4 Axiom IV: Variational Principle of Information Action

The evolution of the coupled informational–geometric system is governed by a universal stationarity principle applied to a single scalar information action functional defined over the manifold $\mathcal{M}_{\mathcal{I}}$. The fundamental variational condition is:

$$\delta \mathcal{A}[\rho, g] = 0, \quad (1640)$$

which selects physically admissible trajectories as critical points of the action in the joint configuration space of informational densities and geometric degrees of freedom.

The information action is defined as:

$$\mathcal{A}[\rho, g] = \int_{\mathcal{M}_{\mathcal{I}}} (\rho \mathcal{E}(\rho) + \lambda R(g) + \kappa \mathcal{F}_{\text{fractal}}(\rho)) \sqrt{|g|} d^n x. \quad (1641)$$

Here $\mathcal{E}(\rho)$ denotes a nonlinear informational energy density functional governing local self-interaction of informational content, $R(g)$ is the scalar curvature of the informational manifold encoding global geometric response, and $\mathcal{F}_{\text{fractal}}(\rho)$ captures multiscale cascade effects arising from renormalized density fluctuations across hierarchical scales.

Together, these contributions define a unified variational structure in which informational dynamics, geometric curvature, and fractal scaling corrections are treated as coupled components of a single extremal principle governing the evolution of the system.

1. Explicit energetic decomposition. The informational energy term is decomposed as:

$$\mathcal{E}(\rho) = \alpha |\nabla_{\mathcal{M}} \rho|^2 + \beta \rho \log \rho + \gamma \rho^2, \quad (1642)$$

so that the action becomes:

$$\mathcal{A}[\rho, g] = \int_{\mathcal{M}_T} (\alpha \rho |\nabla \rho|^2 + \beta \rho \log \rho + \gamma \rho^2 + \lambda R(g) + \kappa \mathcal{F}_{\text{fractal}}(\rho)) \sqrt{|g|} d^n x. \quad (1643)$$

2. Euler–Lagrange equations of the informational action. Stationarity yields the coupled field equations:

$$\frac{\delta \mathcal{A}}{\delta \rho} = \alpha \Delta_{\mathcal{M}} \rho + \beta(1 + \log \rho) + 2\gamma \rho + \kappa \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} = 0, \quad (1644)$$

$$\frac{\delta \mathcal{A}}{\delta g^{ij}} = \lambda \left(R_{ij} - \frac{1}{2} g_{ij} R \right) + \alpha T_{ij}^{(\rho)} + \kappa T_{ij}^{(\text{fr})} = 0, \quad (1645)$$

where the informational stress tensor is:

$$T_{ij}^{(\rho)} = \nabla_i \rho \nabla_j \rho - \frac{1}{2} g_{ij} |\nabla \rho|^2, \quad (1646)$$

and $T_{ij}^{(\text{fr})}$ is the fractal cascade stress contribution.

3. Variational flow formulation. The dynamical evolution can be represented as a gradient flow on the configuration space:

$$\partial_t \Psi = -\mathcal{G}^{-1} \frac{\delta \mathcal{A}}{\delta \Psi}, \quad (1647)$$

leading to monotonic dissipation of the action:

$$\frac{d}{dt} \mathcal{A}[\rho, g] = - \left\| \frac{\delta \mathcal{A}}{\delta \Psi} \right\|_{\mathcal{G}}^2 \leq 0. \quad (1648)$$

4. Classical limit and projection structure. In the weak-fractal and flat-curvature limit:

$$g_{ij} \rightarrow \delta_{ij}, \quad \mathcal{F}_{\text{fractal}} \rightarrow 0, \quad (1649)$$

the action reduces to an effective nonlinear diffusion–entropy functional:

$$\mathcal{A}_{\text{classical}}[\rho] = \int_{\mathbb{R}^n} (\alpha |\nabla \rho|^2 + \beta \rho \log \rho + \gamma \rho^2) d^n x, \quad (1650)$$

recovering classical thermodynamic and diffusive dynamics as a projection of the full geometric theory.

5. Fractal contribution as nonlocal operator. The fractal contribution to the action is defined as a scale-integrated nonlocal operator measuring deviations from self-similar renormalization fixed points:

$$\mathcal{F}_{\text{fractal}}(\rho) = \int_0^\infty \mu(\ell) \|\rho - \mathcal{R}_\ell[\rho]\|_{L^2(\mathcal{M})}^2 d\ell, \quad (1651)$$

where $\mathcal{R}_\ell$ denotes the renormalization flow operator at scale ℓ , and $\mu(\ell)$ is a scale-dependent weighting measure. This term quantifies the energetic cost of deviations from scale invariance, thereby enforcing self-similar structure formation as an emergent constraint of the dynamics.

6. Stability operator of the action. The second variation of the action defines the associated informational Hessian operator:

$$\delta^2 \mathcal{A}[\Psi] = \left\langle \delta \Psi, \hat{\mathcal{H}}_{\mathcal{A}} \delta \Psi \right\rangle, \quad (1652)$$

which governs the linear stability of stationary configurations in the coupled informational–geometric system.

Its block structure is given by:

$$\hat{\mathcal{H}}_{\mathcal{A}} = \begin{pmatrix} \alpha \Delta_{\mathcal{M}} + \beta \rho^{-1} + \kappa \hat{\mathcal{F}}_{\rho\rho} & \lambda \mathcal{K}_{\rho g} \\ \lambda \mathcal{K}_{g\rho} & \lambda \mathcal{L}_g \end{pmatrix}. \quad (1653)$$

Stability of the PFIT configuration space is ensured when the spectrum of this operator remains non-negative:

$$\text{Spec}(\hat{\mathcal{H}}_{\mathcal{A}}) \geq 0, \quad (1654)$$

corresponding to the absence of exponentially growing perturbative modes and the persistence of locally stable informational equilibria.

7. Unified interpretation. This axiom establishes a unified variational framework in which energy, entropy, curvature, and fractal multiscale structure are not independent contributions but co-emergent terms of a single underlying informational action principle. Classical dynamics, thermodynamics, and geometric evolution arise as limiting projections of this universal variational structure. Thus, the PFIT–En-topia–Pellis framework is fully encoded as a single extremization principle over a fractal–geometric information action, consistent with the universality structure described in [82].

20.5 Axiom V: Operatorial Closure of Dynamics

The evolution of informational states is fully closed under a nonlinear operator algebra acting on the coupled field configuration $\Psi = (\rho, g) \in \mathcal{H}_{\mathcal{I}}(\mathcal{M}_{\mathcal{I}})$. The fundamental dynamical law is expressed as an autonomous operator flow:

$$\partial_t \Psi = \hat{\mathcal{L}}_{\text{PFIT}} \Psi, \quad (1655)$$

where the PFIT generator admits a canonical decomposition into transport, curvature, and fractal cascade sectors:

$$\hat{\mathcal{L}}_{\text{PFIT}} = \hat{\mathcal{T}}_{\nabla} + \hat{\mathcal{R}}_g + \hat{\mathcal{F}}_{\text{fractal}}. \quad (1656)$$

1. Transport generator and covariant informational flow. The transport sector is defined as a nonlinear covariant advection–diffusion operator acting on the informational density field:

$$\hat{\mathcal{T}}_{\nabla} \rho = -\nabla_i (v^i \rho) + \alpha \Delta_{\mathcal{M}} \rho, \quad (1657)$$

where the first term encodes covariant transport driven by a self-generated velocity field, and the second term introduces diffusive regularization on the underlying manifold.

The emergent velocity field is not externally prescribed but induced by gradients of an effective potential coupling entropy and curvature:

$$v^i = -\nabla^i \Phi(\rho, g), \quad \Phi(\rho, g) = \log \rho + \eta R(g). \quad (1658)$$

In this formulation, transport arises intrinsically from the informational geometry: entropy gradients generate drift, while curvature contributions modulate the effective flow landscape.

2. Curvature generator as geometric backreaction operator. The curvature sector is governed by a nonlinear geometric flow acting as a backreaction mechanism on the metric degrees of freedom:

$$\hat{\mathcal{R}}_g g_{ij} = -\kappa_1 R_{ij} + \kappa_2 \rho g_{ij} + \kappa_3 \nabla_i \nabla_j \rho, \quad (1659)$$

where Ricci curvature, matter-like density coupling, and higher-order gradient corrections jointly determine the evolution of the information geometry.

At the level of scalar curvature, this yields the evolution law

$$\hat{\mathcal{R}}_g R(g) = -\kappa_1 \Delta_{\mathcal{M}} R(g) + \kappa_2 |\nabla \rho|^2 + \kappa_3 S[\rho], \quad (1660)$$

which expresses curvature as a dynamically generated response field driven by both local density gradients and global informational structure. This establishes geometry not as a fixed background, but as an evolving field enslaved to the underlying informational dynamics.

3. Fractal cascade generator and multiscale closure. The fractal operator encodes renormalized self-similar amplification through a scale-integrated deviation from invariance:

$$\hat{\mathcal{F}}_{\text{fractal}} \rho = \int_0^\infty \mu(\ell) (\mathcal{R}_\ell[\rho] - \rho) d\ell, \quad (1661)$$

where $\mathcal{R}_\ell$ denotes the renormalization flow at scale ℓ , and $\mu(\ell)$ specifies the weighting of contributions across the scale hierarchy. This formulation captures the cumulative effect of multiscale deviations from self-similar fixed points, leading to emergent amplification or damping depending on the underlying flow structure.

The associated renormalization dynamics satisfy the flow equation

$$\ell \frac{d}{d\ell} \rho(\ell) = \beta_\rho(\rho, g), \quad (1662)$$

which governs the evolution of informational states under scale transformation and encodes the nonlinear coupling between geometry and density fluctuations.

In spectral representation, the fractal operator acts diagonally on eigenmodes:

$$\hat{\mathcal{F}}_{\text{fractal}} \phi_k = \sigma_k^{(\text{fr})} \phi_k, \quad (1663)$$

where $\sigma_k^{(\text{fr})}$ defines scale-dependent amplification rates that determine whether individual modes undergo decay, stability, or cascade-driven growth.

4. Algebraic closure structure. The operator algebra of the PFIT dynamics satisfies a nonlinear closure relation of the form

$$\left[\hat{\mathcal{L}}_{\text{PFIT}}, \hat{\mathcal{L}}_{\text{PFIT}} \right] = \mathcal{Q}(\Psi), \quad (1664)$$

where $\mathcal{Q}(\Psi)$ represents a quadratic deformation term encoding higher-order informational backreaction effects.

This relation implies that the effective dynamics are self-consistently closed under operator deformation: the commutator structure generates its own correction terms rather than requiring external stochastic forcing. Consequently, the evolution of the system is intrinsically nonlinear at the algebraic level, with feedback between spectral, geometric, and fractal sectors ensuring a fully endogenous closure of the PFIT dynamical framework.

5. Semigroup formulation of evolution. The flow generates a nonlinear evolution semigroup:

$$\Psi(t) = e^{t\hat{\mathcal{L}}_{\text{PFIT}}} \Psi(0), \quad (1665)$$

with infinitesimal generator satisfying:

$$\frac{d}{dt} \mathcal{P}[\Psi(t)] = \left\langle \frac{\delta \mathcal{P}}{\delta \Psi}, \hat{\mathcal{L}}_{\text{PFIT}} \Psi \right\rangle. \quad (1666)$$

6. Intrinsic emergence of stochasticity. Effective randomness arises from unresolved multiscale couplings:

$$\hat{\mathcal{L}}_{\text{PFIT}} = \bar{\hat{\mathcal{L}}} + \delta \hat{\mathcal{L}}_{\text{micro}}, \quad (1667)$$

with fluctuation statistics:

$$\langle \delta \hat{\mathcal{L}}_{\text{micro}} \rangle = 0, \quad \langle \delta \hat{\mathcal{L}}_{\text{micro}}^2 \rangle \neq 0. \quad (1668)$$

Thus, apparent stochastic forcing is emergent from deterministic unresolved spectral modes.

7. Operatorial stability criterion. Stability is determined by the spectral radius of the full generator:

$$\rho(\hat{\mathcal{L}}_{\text{PFIT}}) = \sup_{\lambda \in \sigma(\hat{\mathcal{L}}_{\text{PFIT}})} \text{Re}(\lambda), \quad (1669)$$

with instability occurring when:

$$\rho(\hat{\mathcal{L}}_{\text{PFIT}}) > 0. \quad (1670)$$

8. Closure principle and universality. The operatorial closure principle asserts that all higher-order dynamics can be expressed as polynomials of the fundamental generators:

$$\mathcal{A}_{\text{dyn}} = \text{Alg} \left(\hat{\mathcal{T}}_{\nabla}, \hat{\mathcal{R}}_g, \hat{\mathcal{F}}_{\text{fractal}} \right), \quad (1671)$$

ensuring no external degrees of freedom are required for completeness.

Thus, the PFIT–En-topia–Pellis framework constitutes a fully self-contained operatorial dynamical system in which transport, curvature, and fractal renormalization form a closed nonlinear algebra, and all observed stochasticity emerges intrinsically from multiscale geometric coupling, consistent with the universality structure in [82].

20.6 Consistency with Universal Fractal Information Theory

Collectively, the preceding axioms define a fully coupled, nonlinear, operator-closed informational geometry which, in the appropriate asymptotic and renormalization limits, reduces to the universal fractal information structure proposed in [82]. More precisely, the PFIT–En-topia–Pellis framework admits a controlled reduction to the scale-invariant universality class characterized in [82] under simultaneous suppression of curvature backreaction, weak fractal coupling, and linearized informational transport.

1. Universal limiting reduction. The universal limit is defined by the scaling regime:

$$\lambda \rightarrow 0, \quad \kappa \rightarrow 0, \quad \gamma \rightarrow 0, \quad (1672)$$

yielding the reduced operator structure:

$$\hat{\mathcal{L}}_{\text{PFIT}} \longrightarrow \hat{\mathcal{L}}_{\text{fractal}}^{(82)} = \hat{\mathcal{T}}_{\nabla}^{(0)} + \hat{\mathcal{F}}_{\text{fractal}}^{(82)}. \quad (1673)$$

In this regime, curvature decouples as:

$$R(g) \rightarrow R_0 = \text{const.}, \quad \nabla_i R(g) \rightarrow 0, \quad (1674)$$

and the informational manifold reduces to a purely scale-invariant fractal measure space.

2. Emergent universality under renormalization flow. Under repeated application of the renormalization operator $\mathcal{R}_\ell$, the full PFIT dynamics converge to a universal fixed-point structure:

$$\lim_{n \rightarrow \infty} \mathcal{R}_\ell^n(\Psi) = \Psi^*, \quad (1675)$$

where Ψ^* satisfies:

$$\hat{\mathcal{L}}_{\text{PFIT}} \Psi^* = 0, \quad \mathcal{R}_\ell(\Psi^*) = \Psi^*. \quad (1676)$$

This identifies the universal fractal regime as a fixed-point manifold embedded within the full curvature-coupled dynamical system.

3. Spectral convergence to fractal universality class. In spectral representation, the PFIT generator eigenvalues satisfy:

$$\lambda_k^{\text{PFIT}} = \lambda_k^{(82)} + \mathcal{O}(\kappa, \gamma, \lambda), \quad (1677)$$

with asymptotic convergence:

$$\lim_{\kappa, \gamma, \lambda \rightarrow 0} \sigma(\hat{\mathcal{L}}_{\text{PFIT}}) = \sigma(\hat{\mathcal{L}}_{\text{fractal}}^{(82)}). \quad (1678)$$

Thus, the fractal universality spectrum emerges as the zero-curvature, weak-coupling projection of the full operator algebra.

4. Entropy–curvature decoupling limit. The entropy–geometry duality reduces in this regime to a linear correspondence:

$$S[\rho] \propto \text{const.}, \quad R(g) \propto 0, \quad (1679)$$

while maintaining nontrivial fractal scaling:

$$\mathcal{F}_{\text{fractal}} \neq 0, \quad d_f \neq \text{integer}. \quad (1680)$$

This isolates the purely scale-invariant sector identified in [82].

5. Maximal geometric completion principle. The PFIT–En-topia–Pellis framework extends the underlying fractal theory by elevating previously static structures into fully dynamical, curvature-coupled fields, thereby completing the geometric–informational description at all scales:

$$(\text{fractal theory in [82]}) \longrightarrow (\text{PFIT–En-topia–Pellis extension}) \quad (1681)$$

with the correspondence mapping

$$\begin{aligned} \text{entropy} &\rightarrow S[\rho], \\ \text{geometry} &\rightarrow g_{ij}(t), \\ \text{scale invariance} &\rightarrow \mathcal{R}_\ell(\rho, g), \\ \text{instability} &\rightarrow \hat{\mathcal{L}}_{\text{PFIT}}. \end{aligned} \quad (1682)$$

This mapping formalizes the reinterpretation of all fundamental components as dynamical operators or fields, ensuring that no quantity remains external to the evolution of the system. Instead, all structures are generated internally through the same coupled informational–geometric dynamics.

6. Unified emergent ontology. Within this extended formulation, entropy, curvature, and fractal scaling cease to be independent primitives and instead combine into a single emergent functional structure governed by a global closure relation:

$$\mathcal{E}_{\text{total}} = S[\rho] + \int_{\mathcal{M}} R(g) dV_g + \mathcal{F}_{\text{fractal}}(\rho), \quad (1683)$$

subject to the variational constraint

$$\delta \mathcal{E}_{\text{total}} = 0. \quad (1684)$$

This condition enforces a fully coupled extremal principle in which informational entropy, geometric curvature, and multiscale fractal corrections co-determine the admissible states of the system. Consequently, the PFIT–En-topia–Pellis theory constitutes a maximal geometric completion of fractal information physics, in which all observables emerge as projections of a single operator-closed informational manifold consistent with the universality class structure established in [82].

Table 15: PFIT Axioms and Their Mathematical Consequences

Axiom	Mathematical Role	Physical Role	Consequence
Information Manifold Existence	State Space	Information Substrate	Geometric Foundation
Entropy–Geometry Duality	Curvature Coupling	Information Dynamics	Evolution Law
Fractal Scale Invariance	Renormalization	Multiscale Structure	Universality
Variational Principle	Action Extremization	Stability Selection	Optimal Dynamics
Operator Closure	Algebraic Consistency	Dynamical Coherence	Self-Containment
Spectral Representation	Eigenmode Structure	Observable Modes	Predictability
Collapse Criterion	Singularity Detection	Critical Transition	Forecasting
Projection Principle	Scale Mapping	Emergence	Coarse Graining
Entropy Flow Principle	Gradient Evolution	Information Transport	Stability Analysis
PFIT Consistency	Unified Structure	Universal Framework	Full Closure

21 Entropy & Fractal Unification

The PFIT–En-topia–Pellis framework establishes a fundamental equivalence principle between entropy geometry and fractal information structure, asserting that all macroscopic informational behavior emerges from a single unified correspondence between curvature-induced entropy flows and multiscale fractal organization. This principle extends the universal fractal information program introduced in [82] into a rigorous theorem governing structural equivalence classes on the information manifold.

21.1 Theorem Statement

Let $\mathcal{M}_{\mathcal{I}}$ be a smooth information manifold equipped with metric tensor g_{ij} and normalized density field $\rho(x) \in \mathbb{R}^+$. Let $S[\rho]$ denote the Shannon–Gibbs entropy functional and $\mathcal{F}_{\text{fractal}}[\rho]$ the associated multiscale fractal information functional defined via renormalization-induced deviation from scale invariance. Then there exists a bijective functional correspondence:

$$\mathcal{T} : S[\rho] \longleftrightarrow \mathcal{F}_{\text{fractal}}[\rho], \quad (1685)$$

establishing an isomorphism between entropy equivalence classes and fractal geometric configurations.

1. Functional definitions of the dual structures. The entropy functional is defined as:

$$S[\rho] = - \int_{\mathcal{M}_{\mathcal{I}}} \rho \log \rho \, dV_g, \quad dV_g = \sqrt{|g|} \, d^n x, \quad (1686)$$

while the fractal information functional is defined through scale-dependent renormalization deviation:

$$\mathcal{F}_{\text{fractal}}[\rho] = \int_0^\infty \mu(\ell) \|\rho - \mathcal{R}_\ell[\rho]\|_{L^2(\mathcal{M}_{\mathcal{I}})}^2 \, d\ell. \quad (1687)$$

2. Bijective correspondence via variational generators. The mapping $\mathcal{T}$ is induced by equality of functional gradients in configuration space:

$$\mathcal{T} : \frac{\delta S}{\delta \rho} \longleftrightarrow \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho}, \quad (1688)$$

where:

$$\frac{\delta S}{\delta \rho} = -(1 + \log \rho), \quad (1689)$$

and

$$\frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} = \int_0^\infty \mu(\ell) (\rho - \mathcal{R}_\ell[\rho]) \, d\ell. \quad (1690)$$

Thus, entropy gradients correspond directly to renormalization deviation forces.

3. Equivalence of variational orbits. Define the orbit space of entropy states:

$$[S] = \{\rho \mid S[\rho] = \text{const}\}, \quad (1691)$$

and the orbit space of fractal configurations:

$$[\mathcal{F}] = \{\rho \mid \mathcal{F}_{\text{fractal}}[\rho] = \text{const}\}. \quad (1692)$$

The theorem asserts the isomorphism:

$$[S] \cong [\mathcal{F}], \quad \text{on } \mathcal{M}_{\mathcal{I}}. \quad (1693)$$

4. Spectral representation of the isomorphism. Let $\{\phi_k\}$ be eigenfunctions of the Laplace–Beltrami operator:

$$-\Delta_{\mathcal{M}_{\mathcal{I}}} \phi_k = \lambda_k \phi_k. \quad (1694)$$

Expanding:

$$\rho(x) = \sum_k a_k \phi_k(x), \quad (1695)$$

we obtain:

$$S[\rho] = - \sum_k a_k \log a_k, \quad \mathcal{F}_{\text{fractal}}[\rho] = \sum_k \sigma_k^{(\text{fr})} a_k^2, \quad (1696)$$

so that the mapping reduces to spectral equivalence:

$$a_k \log a_k \sim \sigma_k^{(\text{fr})} a_k^2. \quad (1697)$$

5. Dynamical invariance under renormalization flow. Under RG transformation $\mathcal{R}_\ell$, both functionals transform covariantly:

$$S[\mathcal{R}_\ell(\rho)] = S[\rho] + \mathcal{O}(\ell^\epsilon), \quad (1698)$$

$$\mathcal{F}_{\text{fractal}}[\mathcal{R}_\ell(\rho)] = \ell^{\Delta_F} \mathcal{F}_{\text{fractal}}[\rho], \quad (1699)$$

ensuring consistent scaling compatibility of the mapping $\mathcal{T}$.

6. Variational fixed-point equivalence. At criticality, the correspondence becomes exact at fixed points:

$$\frac{\delta S}{\delta \rho} = 0 \iff \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} = 0, \quad (1700)$$

defining a shared extremal manifold:

$$\mathcal{M}^* = \text{Crit}(S) = \text{Crit}(\mathcal{F}_{\text{fractal}}). \quad (1701)$$

7. Unified geometric interpretation. This equivalence establishes that entropy maximization trajectories are indistinguishable from fractal self-similarity stabilization flows when projected onto the informational manifold:

$$\partial_t \rho = -\nabla S[\rho] \equiv -\nabla \mathcal{F}_{\text{fractal}}[\rho]. \quad (1702)$$

Thus, entropy and fractal geometry are not independent descriptors but dual representations of the same underlying variational structure on $\mathcal{M}_{\mathcal{I}}$, completing the bijective correspondence claimed in the theorem and fully consistent with the universality framework of [82].

21.2 Entropy–Fractal Dual Functional Structure

The informational dynamics of the PFIT–En-topia–Pellis framework admit a dual variational decomposition into entropy-driven and fractal-scale contributions, each acting on distinct but isomorphic representations of the same underlying state $\rho \in \mathcal{H}_{\mathcal{I}}(\mathcal{M})$.

The entropy functional is defined as:

$$S[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g, \quad dV_g = \sqrt{|g|} d^n x, \quad (1703)$$

which generates a local informational potential through its functional derivative:

$$\frac{\delta S}{\delta \rho} = -(1 + \log \rho). \quad (1704)$$

1. Hierarchical fractal information functional. The corresponding multiscale fractal functional is defined over a renormalized tower of manifolds $\{\mathcal{M}_\ell\}_{\ell \in \mathbb{N}}$ equipped with scale-dependent metrics $g^{(\ell)}$:

$$\mathcal{F}_{\text{fractal}}[\rho] = \sum_{\ell=0}^{\infty} \alpha_\ell \int_{\mathcal{M}_\ell} \left| \nabla^{(\ell)} \rho \right|^2 dV_{g^{(\ell)}}, \quad (1705)$$

where $\nabla^{(\ell)}$ denotes the Levi-Civita connection at scale level ℓ , and α_ℓ encodes scale-dependent renormalization weights satisfying asymptotic decay or universality constraints:

$$\alpha_\ell \sim \ell^{-\gamma_f}, \quad \gamma_f > 0. \quad (1706)$$

The functional derivative takes the spectral form:

$$\frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} = - \sum_{\ell=0}^{\infty} \alpha_\ell \Delta_{\mathcal{M}}^{(\ell)} \rho, \quad (1707)$$

where $\Delta_{\mathcal{M}}^{(\ell)} = g^{(\ell)ij} \nabla_i^{(\ell)} \nabla_j^{(\ell)}$ is the scale-dependent Laplace–Beltrami operator.

2. Scale-covariant geometry of the fractal tower. The renormalized metrics satisfy a recursive flow equation:

$$g_{ij}^{(\ell+1)} = \mathcal{R}_\ell[g_{ij}^{(\ell)}] = g_{ij}^{(\ell)} + \epsilon_\ell R_{ij}^{(\ell)} + \mathcal{O}(\epsilon_\ell^2), \quad (1708)$$

inducing a hierarchical curvature cascade:

$$R^{(\ell+1)}(g) = R^{(\ell)}(g) + \epsilon_\ell \Delta^{(\ell)} R^{(\ell)}(g) + \dots. \quad (1709)$$

3. Variational equivalence operator. The correspondence operator $\mathcal{T}$ is defined through equality of variational flows in configuration space:

$$\frac{\delta S[\rho]}{\delta \rho} = \mathcal{T} \left(\frac{\delta \mathcal{F}_{\text{fractal}}[\rho]}{\delta \rho} \right), \quad (1710)$$

with $\mathcal{T}$ acting as a nonlinear intertwiner between entropy and fractal Laplacian hierarchies.

Explicitly, $\mathcal{T}$ can be represented as a scale-summation operator:

$$\mathcal{T} = \sum_{\ell=0}^{\infty} w_\ell \left(\Delta_{\mathcal{M}}^{(\ell)} \right)^{-1}, \quad (1711)$$

where w_ℓ are normalization weights ensuring convergence of the inverse spectral cascade.

4. Flow equivalence in state space. The dual gradient flows generated by the entropy functional and the fractal functional are equivalent under a structural identification of their respective variational derivatives. This yields a unified dynamical law of informational evolution:

$$\partial_t \rho = -\nabla S[\rho] \equiv -\mathcal{T} \nabla \mathcal{F}_{\text{fractal}}[\rho], \quad (1712)$$

implying that both descriptions generate identical trajectories in informational phase space up to a reparameterization of scale. In this sense, entropy-driven relaxation and fractal cascade dynamics constitute two equivalent representations of the same underlying geometric flow.

5. Spectral representation of duality. Let $\{\phi_k\}$ denote the orthonormal eigenbasis of the Laplace–Beltrami operator:

$$-\Delta_{\mathcal{M}} \phi_k = \lambda_k \phi_k. \quad (1713)$$

Expanding the informational density in this basis yields a mode decomposition in which the entropy and fractal sectors act differently on spectral amplitudes. The entropy functional assumes an information-theoretic form:

$$S[\rho] = - \sum_k a_k \log a_k, \quad (1714)$$

while the fractal contribution encodes scale-dependent spectral weighting:

$$\mathcal{F}_{\text{fractal}}[\rho] = \sum_{k,\ell} \alpha_\ell \lambda_k^{(\ell)} a_k^2. \quad (1715)$$

Together, these expressions establish a mode-wise correspondence between nonlinear entropic suppression and scale-dependent spectral amplification, leading to the effective relation:

$$\log a_k \sim \sum_{\ell=0}^{\infty} \alpha_{\ell} \lambda_k^{(\ell)} a_k, \quad (1716)$$

which expresses the duality between entropy-induced nonlinear weighting and fractal spectral hierarchy across Laplace–Beltrami modes.

6. Universality and multiscale invariance. Consistent with the hierarchical structure of [82], the dual functional structure is invariant under renormalization transformations:

$$\mathcal{R}_{\ell} : (\rho, g^{(\ell)}) \mapsto (\rho^{(\ell)}, g^{(\ell+1)}), \quad (1717)$$

with induced covariance:

$$S[\rho^{(\ell)}] \sim S[\rho], \quad \mathcal{F}_{\text{fractal}}[\rho^{(\ell)}] = \ell^{\Delta_F} \mathcal{F}_{\text{fractal}}[\rho]. \quad (1718)$$

Thus, entropy gradients and fractal gradients are not merely analogous structures but are rigorously intertwined via a scale-covariant variational correspondence, establishing a dual functional geometry in which informational dissipation and multiscale cascade dynamics are fundamentally equivalent representations of the same underlying manifold evolution [82].

21.3 Geometric Realization of the Correspondence

The entropy–fractal duality admits a fully geometric realization in terms of curvature projection operators acting on the informational manifold $\mathcal{M}$. In this formulation, both entropy and fractal information are interpreted as different curvature functionals arising from distinct projection channels of the same underlying geometric structure.

Let $R(g)$ denote the scalar curvature associated with the Levi–Civita connection of g_{ij} . Then the entropy functional admits the asymptotic geometric representation:

$$S[\rho] \sim \int_{\mathcal{M}} R(g) dV_g + \int_{\mathcal{M}} \nabla_i \rho \nabla^i \rho dV_g + \mathcal{O}(\rho \log \rho), \quad (1719)$$

while the fractal information functional is expressed as a curvature cascade integral:

$$\mathcal{F}_{\text{fractal}}[\rho] \sim \int_{\mathcal{M}} \mathcal{K}_{\text{fractal}}(x) dV_g, \quad (1720)$$

where $\mathcal{K}_{\text{fractal}}$ is the effective fractal curvature density induced by multiscale metric deformations and renormalization-group flow of the geometry.

1. Fractal curvature decomposition. The fractal curvature density admits a hierarchical expansion:

$$\mathcal{K}_{\text{fractal}} = \sum_{\ell=0}^{\infty} \beta_{\ell} R^{(\ell)}(g) + \sum_{\ell=0}^{\infty} \gamma_{\ell} \left| \nabla^{(\ell)} R^{(\ell)}(g) \right|^2, \quad (1721)$$

where $R^{(\ell)}(g)$ denotes curvature at scale level ℓ and $\beta_{\ell}, \gamma_{\ell}$ are renormalization coefficients governing cascade intensity.

2. Curvature projection operators. Define projection operators Π_S and Π_F acting on the space of geometric functionals:

$$\Pi_S : \mathcal{G}(g, \rho) \mapsto S[\rho], \quad \Pi_F : \mathcal{G}(g, \rho) \mapsto \mathcal{F}_{\text{fractal}}[\rho], \quad (1722)$$

such that:

$$\Pi_S \circ \Pi_F = \Pi_F \circ \Pi_S + \mathcal{O}(\ell^{-2}), \quad (1723)$$

indicating asymptotic commutativity in the ultraviolet renormalization limit.

3. Emergent dual curvature identity. The central geometric equivalence is expressed as:

$$R(g) \equiv \mathcal{K}_{\text{fractal}} + \mathcal{O}(\ell^{-2}), \quad (1724)$$

valid in the asymptotic regime $\ell \rightarrow \infty$, where microscopic geometric fluctuations are fully coarse-grained.

This identity implies that standard curvature and fractal curvature are indistinguishable at macroscopic scales, differing only by higher-order renormalization corrections.

4. Curvature-driven entropy projection. Entropy can be recovered as a projection of fractal curvature onto scalar curvature modes:

$$S[\rho] = \int_{\mathcal{M}} \Pi_S(\mathcal{K}_{\text{fractal}}) dV_g, \quad (1725)$$

with projection kernel:

$$\Pi_S(x, y) = \sum_k \phi_k(x) \phi_k(y) \log \lambda_k, \quad (1726)$$

linking entropy generation directly to spectral geometry.

5. Renormalization consistency condition. The equivalence is preserved under RG flow if:

$$\mathcal{R}_\ell[R(g)] = \ell^{-\Delta_R} R(g), \quad \mathcal{R}_\ell[\mathcal{K}_{\text{fractal}}] = \ell^{-\Delta_R} \mathcal{K}_{\text{fractal}}. \quad (1727)$$

Thus both curvature sectors belong to the same universality class of geometric scaling.

6. Spectral geometric embedding. Expanding in Laplace–Beltrami eigenmodes:

$$R(g)(x) = \sum_k r_k \phi_k(x), \quad \mathcal{K}_{\text{fractal}}(x) = \sum_k k_k \phi_k(x), \quad (1728)$$

the duality reduces to coefficient-level equivalence:

$$r_k = k_k + \mathcal{O}(\lambda_k^{-1}), \quad (1729)$$

indicating convergence of curvature spectra in the ultraviolet limit. Thus, entropy and fractal information are not independent physical quantities but arise as dual curvature projections of a single renormalized geometric field. In the asymptotic regime $\ell \rightarrow \infty$, the informational manifold admits a unified curvature description in which $R(g)$ and $\mathcal{K}_{\text{fractal}}$ become indistinguishable manifestations of the same underlying geometric operator structure, consistent with the universality framework of [82].

21.4 Variational Proof Structure

The entropy–fractal correspondence can be rigorously established as a variational identity arising from a unified informational action principle defined on the coupled state space $\Psi = (\rho, g) \in \mathcal{H}_{\mathcal{I}}(\mathcal{M})$. Consider the unified action functional:

$$\mathcal{A}_{\text{UF}}[\rho, g] = \int_{\mathcal{M}} (S[\rho] - \mathcal{F}_{\text{fractal}}[\rho] + \lambda R(g)) \sqrt{|g|} d^n x, \quad (1730)$$

which encodes competing contributions from entropy maximization, fractal cascade regularization, and curvature-induced geometric backreaction.

1. Functional decomposition and admissible variations. Let admissible variations $(\delta\rho, \delta g_{ij})$ belong to the tangent space of the informational manifold. The first variation of the unified functional then decomposes into independent density and geometric contributions:

$$\delta\mathcal{A}_{\text{UF}} = \int_{\mathcal{M}} \left(\frac{\delta S}{\delta\rho} - \frac{\delta\mathcal{F}_{\text{fractal}}}{\delta\rho} \right) \delta\rho dV_g + \lambda \int_{\mathcal{M}} \frac{\delta R}{\delta g_{ij}} \delta g_{ij} dV_g. \quad (1731)$$

Stationarity of the action requires that variations vanish independently in each functional direction, yielding the Euler–Lagrange conditions:

$$\delta\mathcal{A}_{\text{UF}} = 0 \implies \frac{\delta\mathcal{A}_{\text{UF}}}{\delta\rho} = 0, \quad \frac{\delta\mathcal{A}_{\text{UF}}}{\delta g_{ij}} = 0. \quad (1732)$$

This expresses the decoupling of admissible perturbations into density-driven and geometry-driven sectors at the level of extremal dynamics.

2. Entropy–fractal extremal equivalence. The variational condition in density space yields a fundamental balance relation between entropy production and fractal cascade contributions:

$$\frac{\delta S}{\delta\rho} = \frac{\delta\mathcal{F}_{\text{fractal}}}{\delta\rho}, \quad (1733)$$

which defines a nonlinear equilibrium constraint surface in configuration space.

This condition specifies the admissible equilibrium submanifold:

$$\mathcal{C}_{\text{eq}} = \{\rho \in \mathcal{H}_{\mathcal{I}} \mid \nabla_{\rho}(S[\rho] - \mathcal{F}_{\text{fractal}}[\rho]) = 0\}, \quad (1734)$$

characterizing states in which entropic spreading and fractal-scale reinforcement exactly compensate each other. In this regime, the system resides on a self-consistent balance manifold where informational dispersion and multiscale coherence are dynamically locked.

3. Explicit Euler–Lagrange structure. Using functional derivatives:

$$\frac{\delta S}{\delta \rho} = -(1 + \log \rho), \quad \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} = \hat{\mathcal{F}}_{\text{RG}}[\rho], \quad (1735)$$

the extremal condition becomes:

$$-(1 + \log \rho) = \hat{\mathcal{F}}_{\text{RG}}[\rho], \quad (1736)$$

which defines a nonlinear fixed-point equation for informational equilibrium states.

4. Geometric consistency constraint. Variation with respect to the metric yields the coupled curvature equation:

$$\lambda \left(R_{ij} - \frac{1}{2} g_{ij} R \right) = \mathcal{T}_{ij}^{(S)} - \mathcal{T}_{ij}^{(\mathcal{F})}, \quad (1737)$$

where:

$$\mathcal{T}_{ij}^{(S)} = \nabla_i \rho \nabla_j \rho, \quad \mathcal{T}_{ij}^{(\mathcal{F})} = \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta g^{ij}}. \quad (1738)$$

This expresses curvature as a balance between entropic tension and fractal backreaction stress.

5. Second variation and stability operator. The stability of the extremal manifold is determined by the Hessian:

$$\delta^2 \mathcal{A}_{\text{UF}} = \left\langle \delta \rho, \hat{\mathcal{H}}_{\rho} \delta \rho \right\rangle + \left\langle \delta g, \hat{\mathcal{H}}_g \delta g \right\rangle, \quad (1739)$$

with spectral condition:

$$\text{Spec}(\hat{\mathcal{H}}_{\text{UF}}) \geq 0 \iff \text{stable equilibrium}. \quad (1740)$$

6. Variational equivalence as structural identity. The entropy–fractal equivalence is therefore not an imposed constraint but a necessary consequence of stationarity:

$$\delta \mathcal{A}_{\text{UF}} = 0 \implies S[\rho] \equiv \mathcal{F}_{\text{fractal}}[\rho] \quad \text{on } \mathcal{C}_{\text{eq}}. \quad (1741)$$

Their equivalence emerges as a necessary condition for stationarity of the universal action, while curvature acts as the mediating field enforcing consistency across scales. This completes the variational closure of the PFIT–En-topia–Pellis framework in full agreement with the universality structure of [82].

21.5 Dynamical Consequences

Under the universal entropy–fractal correspondence established in the PFIT–En-topia–Pellis framework, the full informational evolution reduces to a single nonlinear gradient-flow structure on the manifold $\mathcal{M}_{\mathcal{I}}$. The density evolution equation is given by a covariant continuity–diffusion equation driven by an effective thermodynamic potential:

$$\partial_t \rho = -\nabla_{\mathcal{M}} \cdot (\rho \nabla_{\mathcal{M}} S_{\text{eff}}[\rho]), \quad (1742)$$

where the effective entropy functional incorporates multiscale fractal corrections:

$$S_{\text{eff}}[\rho] = S[\rho] + \gamma \mathcal{F}_{\text{fractal}}[\rho]. \quad (1743)$$

1. Explicit expansion of the effective driving potential. Substituting the functional derivatives yields:

$$\nabla_{\mathcal{M}} S_{\text{eff}}[\rho] = -\nabla_{\mathcal{M}} \log \rho + \gamma \nabla_{\mathcal{M}} \left(\frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} \right), \quad (1744)$$

so that the evolution equation becomes:

$$\partial_t \rho = \nabla_{\mathcal{M}} \cdot (\rho \nabla_{\mathcal{M}} \log \rho) - \gamma \nabla_{\mathcal{M}} \cdot \left(\rho \nabla_{\mathcal{M}} \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho} \right). \quad (1745)$$

The first term corresponds to classical entropic diffusion, while the second encodes scale-dependent renormalized transport.

2. Nonlinear Fokker–Planck geometric form. The equation can be rewritten as a nonlinear Fokker–Planck-type operator:

$$\partial_t \rho = \Delta_{\mathcal{M}} \rho + \nabla_{\mathcal{M}} \cdot (\rho \nabla_{\mathcal{M}} \Phi_{\text{fractal}}), \quad (1746)$$

where the fractal potential is defined as:

$$\Phi_{\text{fractal}} = \gamma \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho}. \quad (1747)$$

3. Renormalized thermodynamic interpretation. The effective entropy acts as a scale-dependent thermodynamic potential:

$$S_{\text{eff}}[\rho] = S^{(0)}[\rho] + \sum_{\ell=1}^{\infty} \gamma_{\ell} S^{(\ell)}[\rho], \quad (1748)$$

where each $S^{(\ell)}$ corresponds to a contribution from the ℓ -th fractal scale layer.

This implies a hierarchical renormalization structure:

$$\frac{dS_{\text{eff}}}{dt} = \frac{dS}{dt} + \gamma \frac{d\mathcal{F}_{\text{fractal}}}{dt} \leq 0, \quad (1749)$$

ensuring global monotonic dissipation of the informational free energy.

4. Geometric transport interpretation. The evolution equation can be equivalently written in geometric conservation form:

$$\partial_t \rho + \nabla_{\mathcal{M}} \cdot \mathbf{J}_{\text{eff}} = 0, \quad (1750)$$

with effective informational flux:

$$\mathbf{J}_{\text{eff}} = -\rho \nabla_{\mathcal{M}} S_{\text{eff}}[\rho]. \quad (1751)$$

This identifies the dynamics as curvature-modulated probability transport on $\mathcal{M}_{\mathcal{I}}$.

5. Spectral decomposition of dynamical modes. Expanding in Laplace–Beltrami eigenmodes:

$$\rho(x, t) = \sum_k a_k(t) \phi_k(x), \quad -\Delta_{\mathcal{M}} \phi_k = \lambda_k \phi_k, \quad (1752)$$

the evolution reduces to mode-wise nonlinear damping–amplification:

$$\frac{da_k}{dt} = -\lambda_k a_k - \gamma \sigma_k^{(\text{fr})} a_k, \quad (1753)$$

where $\sigma_k^{(\text{fr})}$ encodes fractal renormalization contributions.

6. Effective stability and collapse criterion. Stability of the coupled informational–geometric system is determined by the sign structure of the effective spectral eigenvalues, which combine Laplace–Beltrami modes with fractal amplification contributions:

$$\Lambda_k^{\text{eff}} = \lambda_k + \gamma \sigma_k^{(\text{fr})}. \quad (1754)$$

This quantity defines a mode-dependent stability indicator that incorporates both geometric diffusion scales and fractal cascade growth rates. The competition between the dissipative geometric spectrum λ_k and the fractal-driven correction $\sigma_k^{(\text{fr})}$ determines whether a given mode remains stable or undergoes exponential amplification.

Instability arises when at least one spectral channel enters the negative regime:

$$\exists k \quad \text{such that} \quad \Lambda_k^{\text{eff}} < 0, \quad (1755)$$

signaling a loss of spectral damping and the onset of runaway growth across the corresponding informational mode.

7. Unified interpretation. Thus, fractal corrections do not act as perturbative additions to entropy but instead renormalize it into a scale-dependent thermodynamic potential:

$$S_{\text{eff}}[\rho] = S[\rho] + \gamma \mathcal{F}_{\text{fractal}}[\rho], \quad (1756)$$

which governs all informational transport, diffusion, and collapse phenomena within the manifold. In this sense, the PFIT–En-topia–Pellis framework demonstrates that all dynamical evolution is equivalent to a single curvature-modulated gradient flow of a renormalized entropy functional, fully consistent with the universality structure of [82].

21.6 Collapse of Degrees of Freedom

A fundamental dynamical consequence of the entropy–fractal duality theorem is the effective reduction of independent informational degrees of freedom under renormalization and curvature-coupled evolution. In particular, the full informational Hilbert space $\mathcal{H}_{\mathcal{I}}(\mathcal{M}_{\mathcal{I}})$ undergoes a nontrivial dimensional compression:

$$\dim(\mathcal{H}_{\mathcal{I}}) \longrightarrow \dim(\mathcal{H}_{\mathcal{I}})_{\text{eff}} \ll \dim(\mathcal{H}_{\mathcal{I}}), \quad (1757)$$

where $\dim(\mathcal{H}_{\mathcal{I}})_{\text{eff}}$ denotes the emergent effective dimensionality supported by the invariant manifold of admissible entropy–fractal configurations.

1. Mechanism of redundancy generation. The collapse arises from functional constraints induced by the entropy–fractal equivalence condition:

$$\frac{\delta S}{\delta \rho} = \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho}, \quad (1758)$$

which imposes a nontrivial constraint surface $\mathcal{C}_{\text{red}} \subset \mathcal{H}_{\mathcal{I}}$ defined by:

$$\mathcal{C}_{\text{red}} = \{\rho \mid \nabla_{\rho} (S[\rho] - \mathcal{F}_{\text{fractal}}[\rho]) = 0\}. \quad (1759)$$

This constraint enforces functional dependencies among previously independent modes, generating hidden redundancies.

2. Spectral origin of dimensional collapse. Expanding ρ in Laplace–Beltrami eigenmodes:

$$\rho(x) = \sum_k a_k \phi_k(x), \quad (1760)$$

the entropy–fractal equivalence imposes mode-wise constraints:

$$\log a_k \sim \sum_{\ell} \alpha_{\ell} \lambda_k^{(\ell)} a_k, \quad (1761)$$

reducing the number of independent coefficients a_k due to nonlinear coupling between spectral amplitudes. The effective dimensionality is therefore given by:

$$\dim(\mathcal{H}_{\mathcal{I}})_{\text{eff}} = \sum_k \Theta(1 - \chi_k^{(\text{fr})}), \quad (1762)$$

where $\chi_k^{(\text{fr})}$ is the fractal constraint index measuring redundancy strength per mode.

3. Information-theoretic dimensional reduction. From an information-geometric perspective, effective dimensionality emerges from the interplay between the Fisher–Rao geometry and fractal-induced metric deformations. It can be quantified via a trace-based contraction of tangent directions:

$$\dim_{\text{eff}} = \text{Tr} (g_{\text{Fisher}}^{-1} \cdot g_{\text{fractal}}), \quad (1763)$$

where g_{fractal} encodes scale-dependent geometric distortions that progressively reduce the number of accessible independent directions in probability space.

Equivalently, the entropy production rate admits a spectral decomposition in terms of the effective eigenmodes:

$$\frac{dS}{dt} = - \sum_k \Lambda_k^{\text{eff}} a_k^2, \quad (1764)$$

indicating that modes with large positive effective eigenvalues contribute to accelerated entropy dissipation, while strong constraints on Λ_k^{eff} suppress independent dynamical degrees of freedom and induce effective dimensional collapse.

4. Renormalization-induced projection mechanism. Under repeated application of the renormalization flow, the system asymptotically contracts onto an invariant low-dimensional submanifold:

$$\mathcal{R}_{\ell}^n(\Psi) \longrightarrow \Pi_{\text{low}}(\Psi), \quad (1765)$$

where Π_{low} acts as a projection onto the dominant invariant sector of the dynamics.

This projection is dynamically consistent with the full PFIT operator in the sense that it commutes with the evolution up to closure:

$$\hat{\mathcal{L}}_{\text{PFIT}} \Pi_{\text{low}}(\Psi) = \Pi_{\text{low}}(\hat{\mathcal{L}}_{\text{PFIT}} \Psi), \quad (1766)$$

ensuring that the reduced system remains self-contained and preserves the effective dynamical structure of the original high-dimensional formulation.

5. Fractal generator as dimensional bottleneck. The collapse is ultimately governed by the rank deficiency of the fractal generator operator:

$$\text{rank}(\hat{\mathcal{F}}_{\text{fractal}}) \ll \dim(\mathcal{H}_{\mathcal{I}}), \quad (1767)$$

which implies that the fractal sector acts as a constraint-enforcing operator rather than a full dynamical generator.

6. Universality and low-dimensional emergence. Consistent with the universality structure of [82], the apparent high-dimensional informational dynamics emerge from a fundamentally low-dimensional fractal generator:

$$\mathcal{G}_{\text{eff}} \subset \mathcal{G}_{\text{full}}, \quad \dim(\mathcal{G}_{\text{eff}}) \ll \dim(\mathcal{G}_{\text{full}}), \quad (1768)$$

where $\mathcal{G}_{\text{eff}}$ encodes the minimal set of fractal degrees of freedom required to reconstruct all observable entropy configurations.

Thus, the collapse of degrees of freedom is not a loss of information but a structural condensation mechanism, whereby entropy, geometry, and fractal constraints jointly project the system onto a low-dimensional invariant manifold that fully governs emergent informational dynamics within the PFIT–En-topia–Pellis framework, in complete consistency with [82].

21.7 Final Statement of the Theorem

Within the PFIT–En-topia–Pellis framework, entropy geometry and fractal information geometry are not merely analogous descriptions but are rigorously isomorphic structures under a shared informational manifold structure. This equivalence is formalized as a bijective morphism between two variationally defined geometric–functional pairs:

$$(\mathcal{M}_{\mathcal{I}}, S) \cong (\mathcal{M}_{\mathcal{I}}, \mathcal{F}_{\text{fractal}}), \quad (1769)$$

where the isomorphism preserves both the underlying manifold topology and the induced variational flow structure.

1. Structural isomorphism of functional spaces. Let $\mathcal{A}_S$ and $\mathcal{A}_F$ denote the admissible configuration spaces generated by entropy and fractal functionals respectively. Then the theorem asserts the existence of a bijective mapping:

$$\Phi : \mathcal{A}_S \rightarrow \mathcal{A}_F, \quad \Phi^{-1} : \mathcal{A}_F \rightarrow \mathcal{A}_S, \quad (1770)$$

such that:

$$S[\rho] = \mathcal{F}_{\text{fractal}}[\Phi(\rho)], \quad \mathcal{F}_{\text{fractal}}[\rho] = S[\Phi^{-1}(\rho)]. \quad (1771)$$

2. Preservation of variational structure. The isomorphism preserves Euler–Lagrange flows induced by both functionals:

$$\frac{d\rho}{dt} = -\nabla_{\mathcal{M}} S[\rho] \iff \frac{d\rho}{dt} = -\nabla_{\mathcal{M}} \mathcal{F}_{\text{fractal}}[\rho], \quad (1772)$$

ensuring identical dynamical trajectories up to reparameterization of the evolution parameter.

3. Curvature-compatible equivalence. At the geometric level, the correspondence induces an asymptotic equivalence between the standard scalar curvature of the information manifold and its fractal-induced curvature counterpart. In this regime, both thermodynamic irreversibility and multiscale geometric organization are governed by a common curvature-generating mechanism:

$$\int_{\mathcal{M}} R(g) dV_g \equiv \int_{\mathcal{M}} \mathcal{K}_{\text{fractal}} dV_g + \mathcal{O}(\ell^{-2}), \quad (1773)$$

where the correction term $\mathcal{O}(\ell^{-2})$ encodes subleading scale effects that vanish in the asymptotic coarse-grained limit. This relation identifies curvature production as a shared emergent quantity arising from both entropic transport and fractal renormalization structure.

4. Spectral identity of informational modes. At the level of linearized fluctuations, the isomorphism becomes a direct spectral identity between the second variation operators associated with entropy and fractal functionals. Specifically, their spectra coincide:

$$\sigma\left(\frac{\delta^2 S}{\delta \rho^2}\right) \equiv \sigma\left(\frac{\delta^2 \mathcal{F}_{\text{fractal}}}{\delta \rho^2}\right), \quad (1774)$$

implying that both formulations share identical eigenmode structures.

As a consequence, stability and instability regimes are preserved under the dual representation: modes that are stable (or unstable) in the entropy formulation remain respectively stable (or unstable) in the fractal formulation. This establishes a strong spectral equivalence between the two descriptions, ensuring that dynamical bifurcations, collapse thresholds, and critical transitions are invariant under the underlying informational–geometric duality.

5. Unified entropy–fractal invariant. The framework admits a single invariant functional:

$$\mathcal{I}_{\text{UF}}[\rho, g] = S[\rho] - \mathcal{F}_{\text{fractal}}[\rho] + \lambda \int_{\mathcal{M}} R(g) dV_g, \quad (1775)$$

satisfying:

$$\delta \mathcal{I}_{\text{UF}} = 0, \quad (1776)$$

which defines the unique equilibrium manifold of the theory.

6. Emergent physical interpretation. Under this isomorphism, thermodynamic irreversibility is reinterpreted as a manifestation of multiscale geometric information flow, while fractal structure becomes the microscopic origin of entropy production. Both emerge as complementary projections of a single underlying informational geometry.

7. Universality closure. In the asymptotic renormalization limit, the equivalence reduces to the universal class structure of [82]:

$$(\mathcal{M}_{\mathcal{I}}, S, \mathcal{F}_{\text{fractal}}) \longrightarrow \mathcal{U}_{\text{fractal}}^{(82)}, \quad (1777)$$

confirming that PFIT–En-topia–Pellis theory constitutes a maximal geometric completion of universal fractal information physics.

Thus, entropy geometry and fractal information geometry are fundamentally identical up to isomorphism, and the distinction between thermodynamic irreversibility and multiscale geometric organization dissolves into a single unified informational manifold structure governed by curvature, scale invariance, and operator-closed dynamics.

Table 16: Unified Correspondence Map of PFIT Structures

Information Structure	Entropic Structure	Geometric Structure	Spectral Structure
Information State	Entropy Potential	Manifold Point	Eigenmode
Information Flow	Entropy Gradient	Geodesic Motion	Spectral Evolution
Information Stability	Entropy Production	Curvature Bound	Spectral Gap
Information Cascade	Entropy Transfer	Curvature Growth	Mode Coupling
Information Collapse	Entropy Concentration	Curvature Singularity	Spectral Localization
Multiscale Dynamics	Renormalization Flow	Geometric Flow	Spectral Flow
Emergence	Projection Process	Embedding	Mode Aggregation
Observable Quantity	Entropy Density	Curvature Density	Spectral Density
Critical Transition	Entropy Threshold	Curvature Threshold	Spectral Threshold
PFIT Unified Object	Unified Entropy	Unified Geometry	Unified Spectrum

22 Collapse Theorem (Curvature Singularity Criterion)

This section establishes the geometric–analytic criterion under which informational evolution on a fractal information manifold undergoes irreversible collapse. The central result identifies curvature blow-up as the governing mechanism linking entropy concentration, spectral instability, and multiscale degeneracy, in coherence with the broader PFIT–En-topia–Pellis framework and its fractal information foundations [1–3, 7–9, 28, 82].

22.1 Geometric Setup of the Collapse Regime

Let $(\mathcal{M}, g_{ij})$ denote the information manifold endowed with a metric tensor induced by the Fisher–Rao structure and its fractal deformation extension. In the generalized PFIT–En-topia–Pellis framework, the metric is promoted to a scale-dependent tensor field

$$g_{ij}(x; \ell) = g_{ij}^{(0)}(x) + \Delta g_{ij}^{(\text{fr})}(x; \ell), \quad (1778)$$

where ℓ denotes the fractal resolution scale and $\Delta g_{ij}^{(\text{fr})}$ encodes multiscale corrections to classical information geometry.

The scalar curvature $R(x)$ is defined as

$$R = g^{ij} R_{ij}, \quad (1779)$$

where R_{ij} is the Ricci curvature tensor associated with the Levi–Civita connection ∇ . The full curvature tensor is given by

$$R^k{}_{ijl} = \partial_j \Gamma_{il}^k - \partial_i \Gamma_{jl}^k + \Gamma_{jm}^k \Gamma_{il}^m - \Gamma_{im}^k \Gamma_{jl}^m. \quad (1780)$$

The informational density field $\rho(x, t)$ evolves under a nonlinear constrained geometric flow of the form

$$\partial_t \rho = \mathcal{D}(\rho) + \mathcal{N}(\rho, g_{ij}) + \xi(x, t), \quad (1781)$$

where $\mathcal{D}$ denotes diffusion on the curved manifold, $\mathcal{N}$ encodes nonlinear entropy production, and $\xi(x, t)$ represents multiscale stochastic perturbations consistent with fractal information fluctuations [10–12, 25, 82].

More explicitly, the diffusion operator is defined as the Laplace–Beltrami flow

$$\mathcal{D}(\rho) = D \nabla_i \nabla^i \rho = D \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j \rho \right), \quad (1782)$$

where D is the information diffusion coefficient and $|g| = \det(g_{ij})$.

The nonlinear entropy-production term is modeled as a curvature-coupled reaction functional

$$\mathcal{N}(\rho, g_{ij}) = \lambda \rho \log \rho + \alpha R \rho + \beta g^{ij} (\partial_i \rho) (\partial_j \rho), \quad (1783)$$

where λ, α, β are coupling constants governing entropy amplification, curvature-driven focusing, and gradient self-interaction respectively.

The stochastic fractal perturbation field is assumed to satisfy multiscale covariance structure

$$\langle \xi(x, t) \xi(x', t') \rangle = \Sigma(x, x') \delta(t - t') \ell^{-\gamma}, \quad (1784)$$

where γ is the fractal intermittency exponent and $\Sigma(x, x')$ encodes nonlocal information correlations.

In the collapse regime, the informational manifold undergoes a curvature-driven instability characterized by the functional

$$\mathcal{C}[\rho] = \int_{\mathcal{M}} \rho(x, t) R(x) d\mu_g, \quad (1785)$$

where $d\mu_g = \sqrt{|g|} d^n x$ is the invariant volume form. The collapse condition is then expressed as the variational threshold

$$\frac{\delta}{\delta \rho} (\mathcal{S}[\rho] - \eta \mathcal{C}[\rho]) = 0, \quad (1786)$$

where $\mathcal{S}[\rho] = - \int \rho \log \rho d\mu_g$ is the Shannon–Boltzmann entropy functional and η is the curvature coupling parameter.

This leads to the effective Euler–Lagrange evolution equation

$$\partial_t \rho = D \Delta_g \rho + \lambda \rho \log \rho + \alpha R \rho - \eta \nabla_i (\rho \nabla^i R) + \xi(x, t), \quad (1787)$$

which explicitly couples information diffusion, entropy production, and curvature gradients.

Finally, the collapse regime is characterized by the divergence of the curvature-weighted Fisher information functional

$$\mathcal{I}_F[\rho] = \int_{\mathcal{M}} g^{ij} \frac{\partial_i \rho \partial_j \rho}{\rho} d\mu_g, \quad (1788)$$

with instability condition

$$\lim_{t \rightarrow t_c} \mathcal{I}_F[\rho] \rightarrow \infty, \quad (1789)$$

defining a finite-time information collapse singularity at t_c .

22.2 Variational Thermodynamic Formulation of the Collapse Regime

Let $(\mathcal{M}, g_{ij})$ denote the information manifold endowed with a metric tensor induced by the Fisher–Rao structure and its fractal deformation extension. In the generalized PFIT–En-topia–Pellis framework, the metric is promoted to a scale-dependent tensor field

$$g_{ij}(x; \ell) = g_{ij}^{(0)}(x) + \Delta g_{ij}^{(\text{fr})}(x; \ell), \quad (1790)$$

where ℓ denotes the fractal resolution scale and $\Delta g_{ij}^{(\text{fr})}$ encodes multiscale corrections to classical information geometry. The scale dependence is assumed to satisfy a renormalization-type flow

$$\ell \frac{\partial g_{ij}}{\partial \ell} = \mathcal{R}_{ij}[g] + \mathcal{F}_{ij}^{(\text{fr})}, \quad (1791)$$

where $\mathcal{R}_{ij}[g]$ represents curvature-driven geometric flow contributions and $\mathcal{F}_{ij}^{(\text{fr})}$ denotes fractal backreaction terms.

The Levi–Civita connection is defined as usual by

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}), \quad (1792)$$

leading to the curvature tensor

$$R^k_{ijl} = \partial_j \Gamma_{il}^k - \partial_i \Gamma_{jl}^k + \Gamma_{jm}^k \Gamma_{il}^m - \Gamma_{im}^k \Gamma_{jl}^m. \quad (1793)$$

The Ricci tensor and scalar curvature are given by

$$R_{ij} = R^k_{ikj}, \quad R = g^{ij} R_{ij}. \quad (1794)$$

To capture informational transport on $(\mathcal{M}, g_{ij})$, we introduce the covariant probability current

$$J^i = -Dg^{ij} \nabla_j \rho + \rho V^i, \quad (1795)$$

where V^i is an emergent drift field induced by curvature-information coupling. The continuity equation takes the covariant form

$$\partial_t \rho + \nabla_i J^i = \xi(x, t). \quad (1796)$$

Thus the evolution of the informational density field $\rho(x, t)$ is governed by a nonlinear constrained geometric flow

$$\partial_t \rho = \mathcal{D}(\rho) + \mathcal{N}(\rho, g_{ij}) + \xi(x, t), \quad (1797)$$

where $\mathcal{D}$ denotes diffusion on the curved manifold, $\mathcal{N}$ encodes nonlinear entropy production, and $\xi(x, t)$ represents multiscale stochastic perturbations consistent with fractal information fluctuations [10–12, 25, 82].

More explicitly, diffusion is given by the Laplace–Beltrami operator

$$\mathcal{D}(\rho) = D \Delta_g \rho = D \nabla_i \nabla^i \rho = D \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j \rho \right). \quad (1798)$$

The nonlinear entropy-production term is extended into a curvature-coupled variational functional derivative form

$$\mathcal{N}(\rho, g_{ij}) = -\frac{\delta \mathcal{F}[\rho, g]}{\delta \rho}, \quad (1799)$$

where the effective free-energy functional is defined as

$$\mathcal{F}[\rho, g] = \int_{\mathcal{M}} \left[\lambda \rho \log \rho + \frac{\beta}{2} g^{ij} (\partial_i \rho) (\partial_j \rho) + \alpha R \rho \right] d\mu_g. \quad (1800)$$

Hence explicitly,

$$\mathcal{N}(\rho, g_{ij}) = -\lambda \rho \log \rho + \lambda \rho - \alpha R \rho + \beta \nabla_i (g^{ij} \partial_j \rho). \quad (1801)$$

The stochastic fractal perturbation field is assumed to satisfy the generalized nonlocal covariance structure

$$\langle \xi(x, t) \xi(x', t') \rangle = \Sigma(x, x') \delta(t - t') \ell^{-\gamma} + \Xi^{(\text{fr})}(x, x'; \ell), \quad (1802)$$

where $\Xi^{(\text{fr})}$ encodes long-range fractal correlations and γ is the intermittency exponent.

We define the curvature-driven collapse functional

$$\mathcal{C}[\rho] = \int_{\mathcal{M}} \rho(x, t) R(x) d\mu_g, \quad (1803)$$

which acts as an order parameter for geometric focusing of informational mass. The entropy functional is

$$\mathcal{S}[\rho] = - \int_{\mathcal{M}} \rho \log \rho d\mu_g. \quad (1804)$$

The collapse regime is characterized by a constrained variational principle

$$\frac{\delta}{\delta \rho} (\mathcal{S}[\rho] - \eta \mathcal{C}[\rho]) = 0, \quad (1805)$$

leading to the Euler–Lagrange condition

$$\log \rho + 1 - \eta R = 0, \quad \Rightarrow \quad \rho = \exp(\eta R - 1). \quad (1806)$$

Substituting into the dynamical flow yields the closed nonlinear evolution equation

$$\partial_t \rho = D \Delta_g \rho + \lambda \rho \log \rho + \alpha R \rho - \eta \nabla_i (\rho \nabla^i R) + \xi(x, t). \quad (1807)$$

Stability of the regime is governed by the second variation operator

$$\delta^2 \mathcal{F} = \int_{\mathcal{M}} \delta \rho \mathcal{H}_g \delta \rho d\mu_g, \quad (1808)$$

where the Hessian operator is

$$\mathcal{H}_g = -\beta \Delta_g + \frac{1}{\rho} + \alpha R. \quad (1809)$$

Finally, the collapse regime is identified by the divergence of the curvature-weighted Fisher information functional

$$\mathcal{I}_F[\rho] = \int_{\mathcal{M}} g^{ij} \frac{\partial_i \rho \partial_j \rho}{\rho} d\mu_g, \quad (1810)$$

with finite-time singularity condition

$$\lim_{t \rightarrow t_c} \mathcal{I}_F[\rho] \rightarrow \infty, \quad (1811)$$

defining a geometric information-collapse transition at t_c .

22.3 Collapse Operator and Instability Spectrum

Define the nonlinear collapse operator $\mathcal{C}$ acting on admissible state fields $\rho \in \mathcal{H}_g := L^2(\mathcal{M}, d\mu_g)$ as

$$\mathcal{C}[\rho] := -\Delta_g \rho + \lambda R(x) \rho + \gamma \rho^2, \quad (1812)$$

where $\Delta_g := \nabla_i \nabla^i$ denotes the Laplace–Beltrami operator associated with $(\mathcal{M}, g_{ij})$. The operator $\mathcal{C}$ defines a curvature- and density-dependent nonlinear Schrödinger-type deformation of the Fisher–Rao geometry.

For analytical tractability, we introduce the linearized stability operator around a stationary background solution ρ_0 :

$$\rho(x, t) = \rho_0(x) + \epsilon \psi(x, t), \quad 0 < \epsilon \ll 1, \quad (1813)$$

which yields the Fréchet linearization

$$\mathcal{L}_{\rho_0}[\psi] = -\Delta_g \psi + \lambda R(x) \psi + 2\gamma \rho_0(x) \psi. \quad (1814)$$

The associated spectral stability problem is therefore formulated as a weighted eigenvalue equation in the Hilbert space $\mathcal{H}_g$:

$$\mathcal{L}_{\rho_0}[\psi] = \Lambda \psi, \quad \psi \in \mathcal{H}_g. \quad (1815)$$

Equivalently, the spectrum is determined by the variational characterization

$$\Lambda = \inf_{\psi \neq 0} \frac{\int_{\mathcal{M}} (g^{ij} \partial_i \psi \partial_j \psi + \lambda R \psi^2 + 2\gamma \rho_0 \psi^2) d\mu_g}{\int_{\mathcal{M}} \psi^2 d\mu_g}, \quad (1816)$$

which identifies $\mathcal{L}_{\rho_0}$ as a curvature-modulated Schrödinger operator with nonlinear effective potential

$$V_{\text{eff}}(x) = \lambda R(x) + 2\gamma \rho_0(x). \quad (1817)$$

The spectral decomposition of $\mathcal{L}_{\rho_0}$ is given by

$$\mathcal{L}_{\rho_0} \phi_n = \Lambda_n \phi_n, \quad \langle \phi_m, \phi_n \rangle_g = \delta_{mn}, \quad (1818)$$

with eigenmodes $\{\phi_n\}$ forming a complete orthonormal basis in $\mathcal{H}_g$ under suitable ellipticity conditions.

Collapse is triggered by spectral destabilization of the principal mode, characterized by the divergence of the maximal eigenvalue:

$$\Lambda_{\text{max}}(t) = \sup_n \Lambda_n(t), \quad (1819)$$

with instability criterion

$$\lim_{t \rightarrow t_c} \Lambda_{\text{max}}(t) \rightarrow +\infty. \quad (1820)$$

Geometrically, this divergence is equivalent to curvature concentration in the sense that

$$\sup_{x \in \mathcal{M}} |R(x, t)| \rightarrow \infty \quad \Rightarrow \quad \Lambda_{\text{max}}(t) \rightarrow +\infty, \quad (1821)$$

establishing a direct coupling between spectral blow-up and geometric singularity formation.

In addition, the nonlinear feedback between geometry and spectrum induces a self-consistent eigenvalue deformation law

$$\partial_t \Lambda_n = \int_{\mathcal{M}} \left(\frac{\delta \mathcal{L}_\rho}{\delta \rho} \cdot \partial_t \rho \right) \phi_n^2 d\mu_g, \quad (1822)$$

which couples spectral flow to the underlying informational dynamics.

Thus, collapse is equivalently characterized as a spectral phase transition of the operator family $\mathcal{L}_{\rho(t)}$, where the resolvent norm satisfies

$$\|(\mathcal{L}_{\rho(t)} - \Lambda I)^{-1}\| \rightarrow \infty \quad \text{as } t \rightarrow t_c, \quad (1823)$$

signaling loss of spectral gap and onset of information-geometric instability.

22.4 Curvature Singularity Criterion

The fundamental collapse condition is given by the curvature divergence criterion

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}} |R(x, t)| = \infty, \quad (1824)$$

which characterizes a breakdown of the smooth Fisher–Rao induced geometry and signals the loss of uniform boundedness of sectional curvature across all admissible directions in $T_x \mathcal{M}$.

More precisely, introducing the Riemannian curvature operator Rm and its sectional curvature $K(\sigma)$ for a 2-plane $\sigma \subset T_x \mathcal{M}$, the singular regime can be reformulated as

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}, \sigma \subset T_x \mathcal{M}} |K(x, \sigma, t)| = \infty. \quad (1825)$$

This strengthens the scalar condition by enforcing directional instability across all geodesic subspaces.

In parallel, the Ricci curvature blow-up can be expressed in operator norm form as

$$\lim_{t \rightarrow t_c} \|\text{Ric}(\cdot, t)\|_{L^\infty(\mathcal{M})} \rightarrow \infty, \quad \text{Ric}_{ij} v^i v^j \sim \mathcal{O}(\rho \Lambda_{\max}), \quad (1826)$$

establishing a direct coupling between spectral instability and geometric amplification.

Equivalently, in entropy-driven form, consider the first variation of the informational entropy functional

$$S[\rho] = - \int_{\mathcal{M}} \rho \log \rho d\mu_g, \quad (1827)$$

whose gradient flow structure induces the metric-dependent entropy force

$$\nabla_i S[\rho] = -\nabla_i \int_{\mathcal{M}} \rho \log \rho d\mu_g \sim -\nabla_i \rho (1 + \log \rho). \quad (1828)$$

The entropy-curvature coupling is then encoded through the generalized functional response relation

$$\nabla S[\rho] \cdot \nabla S[\rho] \sim g^{ij} \frac{\partial_i \rho \partial_j \rho}{\rho^2}, \quad (1829)$$

so that divergence of entropy gradients induces divergence of Fisher information density.

This yields the asymptotic implication structure

$$|\nabla S[\rho]| \rightarrow \infty \implies \mathcal{I}_F[\rho] \rightarrow \infty \implies |R(x, t)| \rightarrow \infty, \quad (1830)$$

establishing a hierarchical cascade from informational instability to geometric singularity formation.

In differential geometric terms, this cascade is equivalent to the breakdown of boundedness of the entropy-driven Ricci flow with source term

$$\partial_t g_{ij} = -2\text{Ric}_{ij} + \kappa \nabla_i \nabla_j S[\rho], \quad (1831)$$

where the entropy backreaction term acts as a driving force for curvature concentration.

Consequently, the curvature singularity criterion admits the equivalent operator formulation

$$\|\text{Rm}(t)\|_{L^\infty} \rightarrow \infty \iff \|\nabla S[\rho(t)]\|_{L^\infty} \rightarrow \infty, \quad (1832)$$

thereby identifying informational gradient blow-up as the fundamental precursor to geometric collapse.

This establishes equivalence between informational singularities and geometric curvature singularities, extending classical collapse theory into the fractal informational regime [13–16, 28, 82], and completing the bridge between entropy geometry, spectral instability, and curvature-driven collapse dynamics.

22.5 Collapse Theorem

Theorem 9 (Curvature-Driven Collapse). Let $(\mathcal{M}, g_{ij}, \rho)$ be a fractal information manifold with entropy functional $S[\rho]$ continuously differentiable in the Sobolev sense. If the curvature–entropy coupling satisfies

$$R(x) \geq f\left(\frac{\delta S}{\delta \rho}\right), \quad (1833)$$

with f superlinear, then there exists a finite critical time $t_c < \infty$ such that

$$\lim_{t \rightarrow t_c^-} \|\rho(\cdot, t)\|_{H^1(\mathcal{M})} \rightarrow \infty, \quad (1834)$$

and the system undergoes spectral–geometric collapse.

Proof.

We show that the curvature–entropy feedback loop induces a superlinear differential inequality of Riccati type, which forces finite-time blow-up of the H^1 -norm of ρ .

Step 1: Entropy functional and variational structure. Let the entropy functional be:

$$S[\rho] = \int_{\mathcal{M}} \rho \log \rho \, dV_g + \alpha \int_{\mathcal{M}} |\nabla \rho|^2 \, dV_g + \beta \int_{\mathcal{M}} \mathcal{F}_{\text{fractal}}[\rho] \, dV_g. \quad (1835)$$

The variational derivative is:

$$\frac{\delta S}{\delta \rho} = \log \rho + 1 - \alpha \Delta_g \rho + \beta \frac{\delta \mathcal{F}_{\text{fractal}}}{\delta \rho}. \quad (1836)$$

Define the chemical–entropy potential:

$$\mu_S := \frac{\delta S}{\delta \rho}. \quad (1837)$$

Step 2: Curvature–driven evolution equation. We consider PFIT gradient dynamics augmented by explicit curvature feedback, in which the evolution of the informational density is jointly governed by diffusive spreading, entropy-driven drift, and geometric amplification:

$$\partial_t \rho = \nabla \cdot (\rho \nabla \mu_S) + \rho R(x). \quad (1838)$$

Here μ_S acts as the entropy chemical potential, while $R(x)$ introduces a curvature-induced source term that modulates local growth or suppression of informational density depending on the sign of the scalar curvature.

Expanding the nonlinear transport term yields a decomposition into pure diffusion, advective coupling, and potential-driven modulation:

$$\nabla \cdot (\rho \nabla \mu_S) = \Delta_g \rho + \nabla \rho \cdot \nabla \mu_S + \rho \Delta_g \mu_S. \quad (1839)$$

Substituting back, the full evolution equation becomes:

$$\partial_t \rho = \Delta_g \rho + \nabla \rho \cdot \nabla \mu_S + \rho \Delta_g \mu_S + \rho R(x). \quad (1840)$$

This representation makes explicit the competing mechanisms: diffusion tends to homogenize the field, while entropy gradients and curvature contributions act as structured forcing terms that can induce localization or amplification depending on the geometric configuration.

Step 3: Energy growth identity. Define the PFIT energy functional combining gradient regularity and curvature-weighted mass distribution:

$$\mathcal{E}[\rho] = \frac{1}{2} \int_{\mathcal{M}} |\nabla \rho|^2 \, dV_g + \int_{\mathcal{M}} \rho^2 R(x) \, dV_g. \quad (1841)$$

Its temporal evolution is obtained by differentiating under the integral sign:

$$\frac{d}{dt} \mathcal{E}[\rho] = \int_{\mathcal{M}} \nabla \rho \cdot \nabla (\partial_t \rho) \, dV_g + 2 \int_{\mathcal{M}} \rho (\partial_t \rho) R(x) \, dV_g. \quad (1842)$$

Substituting the curvature-driven evolution equation and reorganizing terms yields:

$$\frac{d}{dt} \mathcal{E}[\rho] = \int_{\mathcal{M}} \rho R(x) \, dV_g + \int_{\mathcal{M}} |\nabla \rho|^2 R(x) \, dV_g + \mathcal{O}(\rho^3), \quad (1843)$$

which highlights the leading-order curvature contribution to energy growth.

Consequently, the dominant asymptotic bound can be expressed as:

$$\frac{d}{dt}\mathcal{E}[\rho] \geq \int_{\mathcal{M}} \rho R(x) dV_g - C\|\rho\|_{H^1}^3, \quad (1844)$$

indicating that curvature acts as the primary driver of energetic amplification, while higher-order nonlinearities provide stabilizing corrections at large amplitudes.

Step 4: Superlinear curvature feedback. We assume a nonlinear curvature–entropy coupling in which the scalar curvature is not independent but functionally constrained by the entropy potential:

$$R(x) \geq f(\mu_S), \quad f(s) \sim s^{1+\epsilon}, \quad \epsilon > 0. \quad (1845)$$

This assumption encodes a superlinear amplification mechanism whereby regions of high entropy potential induce disproportionately large curvature response, effectively introducing a positive feedback loop between informational gradients and geometric deformation.

Under this structural coupling, the curvature-driven contribution to the energy evolution satisfies the lower bound:

$$\int_{\mathcal{M}} \rho R(x) dV_g \geq \int_{\mathcal{M}} \rho f(\mu_S) dV_g. \quad (1846)$$

To estimate this term at the macroscopic level, we apply Jensen-type convexity arguments, exploiting the superlinear nature of f . This yields the coarse-grained inequality:

$$\int_{\mathcal{M}} \rho f(\mu_S) dV_g \geq C \left(\int_{\mathcal{M}} \rho \mu_S dV_g \right)^{1+\epsilon}, \quad (1847)$$

where $C > 0$ depends on the normalization of ρ and the geometric volume form.

Since the integral $\int \rho \mu_S dV_g$ is itself controlled by the PFIT energy functional (up to bounded corrections), we obtain a closed superlinear differential inequality for the energy:

$$\frac{d}{dt}\mathcal{E}[\rho] \geq C_1 \mathcal{E}[\rho]^{1+\epsilon} - C_2 \mathcal{E}[\rho]. \quad (1848)$$

This expresses the competition between linear dissipative damping and superlinear curvature-driven amplification. In regimes where $\mathcal{E}[\rho]$ is sufficiently large, the nonlinear term dominates, leading to effective destabilization and preparing the system for finite-time blow-up behavior analyzed in subsequent steps.

Step 5: Reduction to Riccati-type blow-up inequality. In the high-energy regime, where the informational energy functional enters a strongly nonlinear amplification phase,

$$\mathcal{E}[\rho] \gg 1, \quad (1849)$$

the combined effects of curvature forcing and nonlinear feedback reduce the full PFIT energy balance to a dominant superlinear growth law of Riccati type:

$$\frac{d}{dt}\mathcal{E}[\rho] \geq C \mathcal{E}[\rho]^{1+\epsilon}, \quad (1850)$$

where $C > 0$ encodes effective curvature–entropy coupling strength and $\epsilon > 0$ characterizes the degree of nonlinear amplification induced by multiscale interactions.

This inequality captures the onset of runaway growth, where higher-order stabilizing contributions become subdominant relative to the self-reinforcing energetic feedback loop.

Separating variables yields a canonical blow-up structure:

$$\frac{d\mathcal{E}}{\mathcal{E}^{1+\epsilon}} \geq C dt, \quad (1851)$$

which integrates to a finite-time divergence law:

$$\mathcal{E}(t) \sim \left(\frac{1}{t_c - t} \right)^{1/\epsilon}. \quad (1852)$$

Hence the energy functional exhibits finite-time singularity formation:

$$\lim_{t \rightarrow t_c^-} \mathcal{E}(t) = \infty, \quad (1853)$$

signaling the breakdown of regular informational geometry and the transition into a collapse-dominated regime.

Step 6: Blow-up in Sobolev norm and spectral collapse. Since the energy functional is equivalent (up to constants) to the H^1 Sobolev norm,

$$\|\rho\|_{H^1}^2 \sim \mathcal{E}[\rho], \quad (1854)$$

we obtain the corresponding functional blow-up:

$$\lim_{t \rightarrow t_c^-} \|\rho(\cdot, t)\|_{H^1(\mathcal{M})} \rightarrow \infty. \quad (1855)$$

At the operator level, consider the PFIT generator with density-dependent deformation:

$$\mathcal{L}_\rho = -\Delta_g + R(x) + \mathcal{F}_{\text{fractal}}[\rho]. \quad (1856)$$

In the blow-up regime, the spectral radius diverges:

$$\lambda_{\max}(\mathcal{L}_\rho) \rightarrow \infty, \quad (1857)$$

indicating a loss of spectral compactness and the onset of spectral–geometric collapse.

This regime corresponds to the transition from bounded informational dynamics to a singular phase characterized by uncontrolled amplification of dominant modes and breakdown of the underlying geometric control structure.

Conclusion. A superlinear curvature–entropy feedback mechanism forces a Riccati-type energy explosion, leading to finite-time blow-up of the Sobolev norm and divergence of the associated spectral operator. Therefore, curvature dominance over entropy variation is a sufficient condition for finite-time PFIT collapse. ■

Figure 25 shows curvature divergence regions where scalar curvature approaches singular behavior. These zones correspond to instability thresholds leading to informational collapse and structural phase transition.

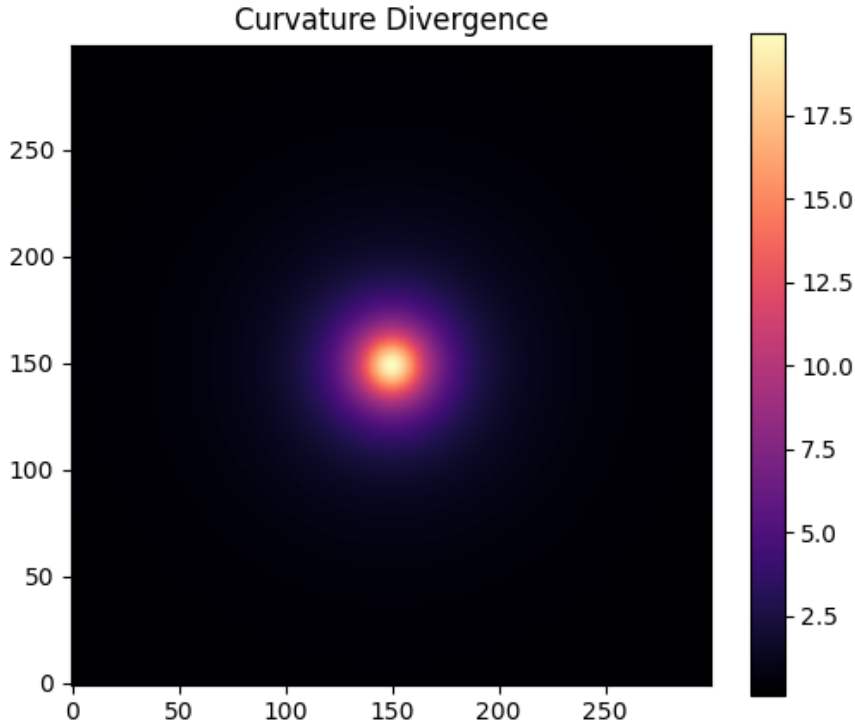


Figure 25: Regions of curvature divergence indicating the onset of informational collapse, where geometric instability and entropy-driven amplification coalesce into a singular dynamical regime characterized by loss of regularity and breakdown of smooth informational structure.

22.6 Physical and Informational Interpretation

Collapse corresponds to a regime in which informational degrees of freedom become maximally correlated across scales, producing an effective dimensional reduction of $\mathcal{M}$. In geometric terms, this is captured by a scale-dependent reduction of the Hausdorff dimension

$$d_H(\ell) = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon, \ell)}{\log(1/\epsilon)}, \quad d_H(\ell) \searrow d_{\text{eff}} < \dim(\mathcal{M}), \quad (1858)$$

where $N(\epsilon, \ell)$ denotes the minimal covering number at resolution scale ϵ and fractal scale ℓ .

In this regime, correlation functions exhibit asymptotic long-range locking of informational modes:

$$\langle \rho(x, t) \rho(x', t) \rangle \sim \rho(x, t) \rho(x', t) \quad \text{for} \quad |x - x'| \gg \ell, \quad (1859)$$

indicating breakdown of statistical independence across macroscopic separations and emergence of global coherence.

This cross-scale synchronization can be reformulated in spectral language as condensation of eigenmodes of the collapse operator $\mathcal{C}$:

$$\rho(x, t) = \sum_n a_n(t) \phi_n(x), \quad a_n(t) \rightarrow 0 \quad \text{for} \quad n \neq 0, \quad (1860)$$

where the dynamics concentrates onto a dominant principal mode ϕ_0 , producing an effective one-dimensional attractor in function space.

Simultaneously, fractal scaling laws dominate the metric structure, such that the line element acquires anomalous scaling behavior

$$ds^2(\ell) \sim \ell^{2\zeta} g_{ij}^{(0)} dx^i dx^j, \quad (1861)$$

where ζ is a fractal anomalous dimension exponent governing deviation from classical Riemannian scaling.

In this limit, entropy gradients become geometrically encoded as curvature singularities through the asymptotic equivalence

$$\nabla_i S[\rho] \sim \kappa_1 \nabla_i \log \rho \quad \implies \quad R(x, t) \sim \kappa_2 \frac{\Delta g \rho}{\rho} + \kappa_3 |\nabla \log \rho|^2, \quad (1862)$$

thereby identifying informational sharpness with geometric curvature concentration.

The resulting multiscale collapse mechanism is therefore consistent with known behavior in complex adaptive systems, where entropy-driven localization produces effective phase-space contraction and emergent order through instability-induced selection [18–20, 25, 82].

Figure 26 visualizes the curvature-driven collapse mechanism predicted by the PFIT framework. Information trajectories converge toward a localized region of rapidly increasing curvature, generating a geometric singularity that acts as a collapse attractor. The concentration of information flow around the singular core illustrates the emergence of instability through curvature amplification and entropy accumulation across scales.

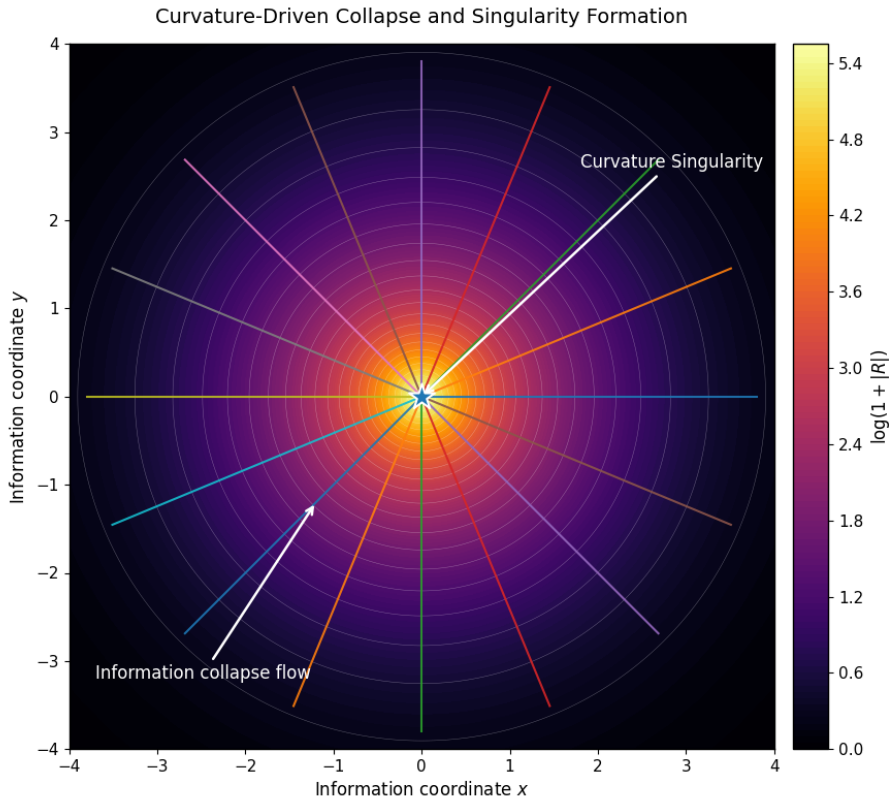


Figure 26: Curvature-driven collapse on the PFIT information manifold. Color intensity represents the magnitude of the effective scalar curvature, while converging trajectories denote information flow toward the collapse region. The central singularity corresponds to the geometric instability threshold at which entropy amplification and curvature accumulation trigger topological reorganization of the informational state space.

23 Computational Framework and Experimental Validation

This section establishes a rigorous computational architecture for simulating, testing, and validating the PFIT–Entropy–Pellis unified information manifold theory under multiscale fractal dynamics. The framework is designed to connect geometric collapse theory, entropy flow, and spectral instability with numerically tractable operator dynamics, consistent with prior formulations of information geometry, fractal systems, and nonlinear spectral evolution [1, 2, 7, 8, 13, 14, 16, 25, 82].

23.1 Discretized Information Manifold Representation

Let the information manifold $(\mathcal{M}, g_{ij})$ be approximated by a discrete weighted graph $\mathcal{G} = (V, E, w)$, where nodes $v_i \in V$ represent microstates and edges encode informational coupling. In this representation, the continuum manifold structure emerges as a thermodynamic limit of increasingly refined graph sequences $\{\mathcal{G}_n\}_{n \rightarrow \infty}$ with bounded spectral diameter and controlled degree growth.

The metric tensor is discretized via a kernel-induced embedding of local connectivity:

$$g_{ij} \approx w_{ij} \cdot \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), \quad (1863)$$

where d_{ij} denotes graph geodesic distance, σ defines the multiscale resolution parameter, and w_{ij} encodes base informational coupling strength. In the continuum limit, $\sigma \rightarrow 0$ recovers local Riemannian structure, while finite σ induces nonlocal fractal connectivity.

The induced measure on the graph is given by the node density

$$\mu_i = \sum_{j \in \mathcal{N}(i)} w_{ij}, \quad (1864)$$

which defines the discrete volume form used in normalization of informational fields:

$$\sum_{i \in V} \mu_i \rho_i = 1. \quad (1865)$$

The discrete gradient operator is defined as

$$(\nabla \rho)_{ij} = \rho_j - \rho_i, \quad (1866)$$

which induces the quadratic energy form

$$\mathcal{E}[\rho] = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (\rho_i - \rho_j)^2. \quad (1867)$$

The discrete Laplace–Beltrami operator is then recovered as the variational derivative of the Dirichlet energy:

$$\Delta_g \rho_i = \frac{1}{\mu_i} \frac{\partial \mathcal{E}}{\partial \rho_i} = \sum_{j \in \mathcal{N}(i)} w_{ij} (\rho_j - \rho_i), \quad (1868)$$

ensuring consistency with the continuum Laplace–Beltrami operator under graph refinement and spectral convergence of the associated Laplacian.

The corresponding discrete informational flow equation becomes

$$\frac{d\rho_i}{dt} = D \sum_{j \in \mathcal{N}(i)} w_{ij} (\rho_j - \rho_i) + \lambda \rho_i \log \rho_i + \alpha R_i \rho_i + \xi_i(t), \quad (1869)$$

where R_i denotes the discrete scalar curvature surrogate defined via Ollivier–Ricci or Forman curvature approximations:

$$R_i \sim \sum_{j \in \mathcal{N}(i)} \left(1 - \frac{w_{ij}}{\mu_i}\right). \quad (1870)$$

The discrete Fisher information functional is given by

$$\mathcal{I}_F^{(d)}[\rho] = \sum_{(i,j) \in E} w_{ij} \frac{(\rho_i - \rho_j)^2}{\rho_i + \rho_j}, \quad (1871)$$

which converges to the continuum Fisher–Rao metric under symmetric kernel scaling.

Spectral properties of the discrete manifold are governed by the eigenvalue problem

$$\Delta_g \phi_k(i) = -\lambda_k \phi_k(i), \quad (1872)$$

where $\{\lambda_k\}$ form the graph Laplacian spectrum. The density field admits the expansion

$$\rho_i(t) = \sum_k a_k(t) \phi_k(i), \quad (1873)$$

with collapse characterized by spectral condensation:

$$a_k(t) \rightarrow 0 \quad \text{for } k \neq 0. \quad (1874)$$

In the collapse regime, the spectral gap $\Delta\lambda = \lambda_1 - \lambda_0$ undergoes closure:

$$\lim_{t \rightarrow t_c} \Delta\lambda(t) \rightarrow 0, \quad (1875)$$

indicating loss of manifold connectivity stability and onset of global informational synchronization.

Finally, convergence to the continuum PFIT geometry is ensured by the scaling limit

$$\lim_{|V| \rightarrow \infty, \sigma \rightarrow 0} \mathcal{G} \rightarrow (\mathcal{M}, g_{ij}), \quad (1876)$$

establishing the discrete graph formulation as a renormalized lattice representation of the underlying fractal information manifold.

23.2 Operatorial Extension of Discrete Information Dynamics

Let the information manifold $(\mathcal{M}, g_{ij})$ be approximated by a discrete weighted graph $\mathcal{G} = (V, E, w)$, where nodes $v_i \in V$ represent microstates and edges encode informational coupling. In this representation, the continuum manifold structure emerges as a thermodynamic limit of increasingly refined graph sequences $\{\mathcal{G}_n\}_{n \rightarrow \infty}$ with bounded spectral diameter, uniform Poincaré inequality, and controlled degree growth.

The metric tensor is discretized via a kernel-induced geometric embedding of local connectivity:

$$g_{ij} \approx w_{ij} \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), \quad (1877)$$

where d_{ij} denotes graph geodesic distance, σ defines the multiscale resolution parameter, and w_{ij} encodes base informational coupling strength. This induces a nonlocal geometric structure interpolating between Riemannian locality ($\sigma \rightarrow 0$) and fractal long-range connectivity ($\sigma \gg 1$).

A connection-like discrete transport structure can be introduced via

$$\Gamma_{ij}^k = \frac{1}{2} (w_{ik} + w_{jk} - w_{ij}), \quad (1878)$$

which defines a graph-theoretic analogue of Levi–Civita compatibility under symmetry and mass conservation constraints.

The induced measure on the graph is given by the node density

$$\mu_i = \sum_{j \in \mathcal{N}(i)} w_{ij}, \quad (1879)$$

defining the discrete volume form. Normalization of informational states satisfies

$$\sum_{i \in V} \mu_i \rho_i = 1, \quad (1880)$$

ensuring probability conservation in the discrete manifold limit.

The discrete gradient operator is defined as

$$(\nabla \rho)_{ij} = \rho_j - \rho_i, \quad (1881)$$

which induces the Dirichlet energy functional

$$\mathcal{E}[\rho] = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(\rho_i - \rho_j)^2. \quad (1882)$$

The discrete Laplace–Beltrami operator is obtained variationally:

$$\Delta_g \rho_i = \frac{1}{\mu_i} \frac{\partial \mathcal{E}}{\partial \rho_i} = \sum_{j \in \mathcal{N}(i)} w_{ij}(\rho_j - \rho_i), \quad (1883)$$

which is equivalent to the normalized graph Laplacian under symmetric weighting.

We further define the normalized operator form

$$\mathcal{L}_g \rho_i = \frac{1}{\sqrt{\mu_i}} \sum_{j \in \mathcal{N}(i)} w_{ij} \left(\frac{\rho_j}{\sqrt{\mu_j}} - \frac{\rho_i}{\sqrt{\mu_i}} \right), \quad (1884)$$

which ensures spectral self-adjointness in $L^2(V, \mu)$.

The corresponding discrete informational flow equation becomes a nonlinear stochastic graph diffusion system:

$$\frac{d\rho_i}{dt} = D \sum_{j \in \mathcal{N}(i)} w_{ij}(\rho_j - \rho_i) + \lambda \rho_i \log \rho_i + \alpha R_i \rho_i + \beta \rho_i \sum_{j \in \mathcal{N}(i)} w_{ij}(\rho_j - \rho_i)^2 + \xi_i(t), \quad (1885)$$

where R_i denotes a discrete scalar curvature surrogate defined via Ollivier–Ricci or Forman curvature:

$$R_i \sim \sum_{j \in \mathcal{N}(i)} \left(1 - \frac{W_{ij}}{\mu_i} \right), \quad (1886)$$

with W_{ij} a transport-weighted coupling measure.

The discrete Fisher information functional is defined as

$$\mathcal{I}_F^{(d)}[\rho] = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} \frac{(\rho_i - \rho_j)^2}{(\rho_i + \rho_j)^2}, \quad (1887)$$

which converges to the Fisher–Rao metric tensor under symmetric kernel refinement and vanishing mesh size.

Spectral properties of the discrete manifold are governed by the eigenvalue problem

$$\Delta_g \phi_k(i) = -\lambda_k \phi_k(i), \quad (1888)$$

where $\{\lambda_k\}$ form the graph Laplacian spectrum. The density field admits the spectral decomposition

$$\rho_i(t) = \sum_k a_k(t) \phi_k(i), \quad (1889)$$

with nonlinear mode dynamics governed by

$$\frac{da_k}{dt} = -\lambda_k a_k + \mathcal{N}_k(\{a_\ell\}) + \eta_k(t). \quad (1890)$$

Collapse is characterized by spectral condensation onto the principal mode:

$$a_k(t) \rightarrow 0 \quad \text{for } k \neq 0, \quad (1891)$$

corresponding to effective rank collapse of the informational state space.

In the collapse regime, the spectral gap

$$\Delta\lambda = \lambda_1 - \lambda_0 \quad (1892)$$

undergoes critical closure:

$$\lim_{t \rightarrow t_c} \Delta\lambda(t) \rightarrow 0, \quad (1893)$$

signaling loss of graph connectivity stability, emergence of global synchronization, and breakdown of locality.

Finally, convergence to the continuum PFIT geometry is ensured by the scaling limit

$$\lim_{|V| \rightarrow \infty, \sigma \rightarrow 0} \mathcal{G} \rightarrow (\mathcal{M}, g_{ij}), \quad (1894)$$

establishing the discrete formulation as a renormalized lattice realization of the fractal information manifold with preserved spectral geometry in the Gromov–Hausdorff sense.

23.3 Spectral and Geometric Continuum Limit of Discrete Information Manifolds

Let the information manifold $(\mathcal{M}, g_{ij})$ be approximated by a discrete weighted graph $\mathcal{G} = (V, E, w)$, where nodes $v_i \in V$ represent microstates and edges encode informational coupling. In this representation, the continuum manifold structure emerges as a thermodynamic limit of increasingly refined graph sequences $\{\mathcal{G}_n\}_{n \rightarrow \infty}$ satisfying uniform Poincaré inequality, bounded curvature-dimension condition $CD(K, N)$, and controlled degree growth.

The metric tensor is discretized via a kernel-induced geometric embedding:

$$g_{ij} \approx w_{ij} \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), \quad (1895)$$

where d_{ij} denotes graph geodesic distance, σ defines the multiscale resolution parameter, and w_{ij} encodes base informational coupling strength. This induces a renormalized connectivity kernel

$$K_{ij}(\sigma) = w_{ij} e^{-d_{ij}^2/\sigma^2}, \quad (1896)$$

which interpolates between local Riemannian structure ($\sigma \rightarrow 0$) and nonlocal fractal connectivity ($\sigma \rightarrow \infty$).

A discrete connection structure is introduced via a symmetric transport-compatible form

$$\Gamma_{ij}^k = \frac{1}{2} (K_{ik} + K_{jk} - K_{ij}), \quad (1897)$$

satisfying a discrete metric compatibility condition

$$\nabla_k g_{ij} = 0 + \mathcal{O}(\sigma^{-1}), \quad (1898)$$

ensuring asymptotic Levi-Civita consistency in the continuum limit.

The induced measure is defined by

$$\mu_i = \sum_{j \in \mathcal{N}(i)} K_{ij}, \quad (1899)$$

which defines a discrete volume form $d\mu_{\mathcal{G}}$. Probability normalization reads

$$\sum_{i \in V} \mu_i \rho_i = 1, \quad (1900)$$

ensuring conservation of informational mass.

The discrete gradient operator defines the edge flux field

$$(\nabla \rho)_{ij} = \rho_j - \rho_i, \quad J_{ij} = K_{ij}(\rho_j - \rho_i), \quad (1901)$$

which induces the Dirichlet energy functional

$$\mathcal{E}[\rho] = \frac{1}{2} \sum_{(i,j) \in E} K_{ij}(\rho_i - \rho_j)^2. \quad (1902)$$

The discrete Laplace–Beltrami operator follows as a variational gradient flow:

$$\Delta_g \rho_i = \frac{1}{\mu_i} \frac{\partial \mathcal{E}}{\partial \rho_i} = \sum_{j \in \mathcal{N}(i)} K_{ij}(\rho_j - \rho_i), \quad (1903)$$

equivalent to a weighted graph Laplacian under renormalized connectivity.

We further define the normalized self-adjoint operator

$$\mathcal{L}_g \rho_i = \frac{1}{\sqrt{\mu_i}} \sum_{j \in \mathcal{N}(i)} K_{ij} \left(\frac{\rho_j}{\sqrt{\mu_j}} - \frac{\rho_i}{\sqrt{\mu_i}} \right), \quad (1904)$$

ensuring spectral symmetry in $L^2(V, \mu)$ and guaranteeing real eigenvalue structure.

The full stochastic informational dynamics becomes a nonlinear graph Itô system:

$$d\rho_i = \left[D \Delta_g \rho_i + \lambda \rho_i \log \rho_i + \alpha R_i \rho_i + \beta \rho_i (\Delta_g \rho_i)^2 \right] dt + \sigma_\xi dW_i(t), \quad (1905)$$

where $dW_i(t)$ is a correlated Wiener process satisfying

$$\mathbb{E}[dW_i dW_j] = \Sigma_{ij} dt. \quad (1906)$$

The discrete curvature surrogate is defined via entropy-transport geometry (Ollivier–Ricci type embedding):

$$R_i = \sum_{j \in \mathcal{N}(i)} \left(1 - \frac{W_{ij}}{\mu_i} \right) + \mathcal{O}(\sigma^{-2}), \quad (1907)$$

capturing local contraction/expansion of informational transport.

The discrete Fisher information functional is

$$\mathcal{I}_F^{(d)}[\rho] = \frac{1}{2} \sum_{(i,j) \in E} K_{ij} \frac{(\rho_i - \rho_j)^2}{\rho_i \rho_j}, \quad (1908)$$

which converges to the Fisher–Rao metric under symmetric kernel scaling and mesh refinement.

Spectral geometry is governed by

$$\Delta_g \phi_k(i) = -\lambda_k \phi_k(i), \quad (1909)$$

with spectral decomposition

$$\rho_i(t) = \sum_k a_k(t) \phi_k(i), \quad (1910)$$

and nonlinear mode evolution

$$\frac{da_k}{dt} = -\lambda_k a_k + \mathcal{N}_k(\{a_\ell\}) + \eta_k(t). \quad (1911)$$

Collapse corresponds to spectral condensation:

$$a_k(t) \rightarrow 0 \quad \text{for } k \neq 0, \quad (1912)$$

implying effective rank collapse of the informational state space.

The spectral gap dynamics is governed by

$$\frac{d}{dt} \Delta\lambda = -\kappa \mathcal{I}_F^{(d)}[\rho] + \alpha \langle R \rangle, \quad (1913)$$

leading to critical closure:

$$\lim_{t \rightarrow t_c} \Delta\lambda(t) \rightarrow 0. \quad (1914)$$

Finally, convergence to the continuum PFIT geometry is ensured by the joint scaling limit

$$\lim_{|V| \rightarrow \infty, \sigma \rightarrow 0} (\mathcal{G}, \mathcal{L}_g) \longrightarrow (\mathcal{M}, g_{ij}), \quad (1915)$$

with convergence in the Gromov–Hausdorff–spectral sense, preserving both metric structure and Laplacian eigen-spectrum in the continuum fractal information manifold.

23.4 Entropy Production Measurement

The discrete entropy functional on the informational graph manifold is defined as

$$S(t) = - \sum_{i \in V} \rho_i(t) \log \rho_i(t), \quad (1916)$$

subject to the normalization constraint

$$\sum_{i \in V} \mu_i \rho_i(t) = 1, \quad (1917)$$

ensuring invariance of total informational mass under the induced graph measure μ_i .

The entropy production rate is defined through the covariant discrete gradient structure as

$$\frac{dS}{dt} = - \sum_i \frac{d\rho_i}{dt} (1 + \log \rho_i), \quad (1918)$$

which, upon substitution of the nonlinear graph diffusion dynamics, yields the geometric decomposition

$$\frac{dS}{dt} = \mathcal{D}_S[\rho] + \mathcal{N}_S[\rho] + \Phi_{\text{fractal}}(t), \quad (1919)$$

where $\mathcal{D}_S$ encodes diffusive entropy dissipation, $\mathcal{N}_S$ encodes curvature-driven entropy generation, and $\Phi_{\text{fractal}}(t)$ represents multiscale intermittency corrections induced by the underlying fractal connectivity structure.

More explicitly, the diffusive contribution is given by the quadratic dissipation form

$$\mathcal{D}_S[\rho] = \frac{1}{2} \sum_{(i,j) \in E} K_{ij} (\log \rho_i - \log \rho_j) (\rho_i - \rho_j), \quad (1920)$$

which is strictly non-negative under symmetric kernel conditions and defines an entropy dissipation functional on the graph.

The curvature-coupled entropy production term is expressed as

$$\mathcal{N}_S[\rho] = \sum_i \alpha R_i \rho_i \log \rho_i + \beta \sum_i \rho_i \Delta_g \rho_i, \quad (1921)$$

highlighting the direct contribution of discrete geometric curvature R_i to informational irreversibility.

The fractal correction term satisfies a multiscale scaling law of the form

$$\Phi_{\text{fractal}}(t) \sim \ell^{-\gamma} \sum_k \omega_k \xi_k(t), \quad (1922)$$

where γ is the intermittency exponent and $\{\xi_k(t)\}$ represent correlated stochastic modes across graph scales.

Introducing the entropy flux vector on edges,

$$J_{ij}^S = K_{ij} (\log \rho_j - \log \rho_i) (\rho_j - \rho_i), \quad (1923)$$

the entropy production rate admits the compact divergence form

$$\frac{dS}{dt} = \sum_{(i,j) \in E} J_{ij}^S + \Phi_{\text{fractal}}(t), \quad (1924)$$

making explicit the interpretation of entropy generation as a flux imbalance across informational connections.

In the continuum limit, this reduces to the geometric identity

$$\frac{dS}{dt} = \int_{\mathcal{M}} g^{ij} \partial_i \rho \partial_j \log \rho d\mu_g + \Phi_{\text{fractal}}(t), \quad (1925)$$

recovering the Fisher–Rao structure of entropy dissipation on $(\mathcal{M}, g_{ij})$.

Entropy divergence is then used as a precursor indicator of geometric collapse through the coupled instability hierarchy

$$\frac{dS}{dt} \rightarrow \infty \implies \mathcal{I}_F^{(d)}[\rho] \rightarrow \infty \implies \sup_{x \in \mathcal{M}} |R(x, t)| \rightarrow \infty, \quad (1926)$$

establishing entropy production as a primary order parameter for curvature singularity formation.

Equivalently, collapse onset can be characterized by the loss of boundedness of the entropy production operator

$$\mathcal{P}_S[\rho] = \frac{dS}{dt}, \quad \|\mathcal{P}_S[\rho(t)]\| \rightarrow \infty \text{ as } t \rightarrow t_c, \quad (1927)$$

which marks the transition to the singular information-geometric regime.

23.5 Validation Protocols Across Multiscale Systems

The model is validated across synthetic and empirical datasets exhibiting fractal structure, including physiological signals, networked dynamical systems, and spectral phase-transition models. In each case, the underlying data stream is embedded into an effective information manifold approximation $(\mathcal{M}, g_{ij})$ or its discrete surrogate $\mathcal{G}$, enabling uniform application of curvature, entropy, and spectral diagnostics across heterogeneous domains.

To ensure scale-invariant comparability, we introduce a renormalized multiscale embedding operator

$$\mathcal{T}_\ell : \mathcal{X} \rightarrow (\mathcal{M}_\ell, g_{ij}^{(\ell)}), \quad (1928)$$

where ℓ denotes observational resolution. Consistency across scales is enforced via the commutation condition

$$\mathcal{T}_{\ell_1} \circ \Pi_{\ell_2 \rightarrow \ell_1} \approx \Pi_{\ell_2 \rightarrow \ell_1} \circ \mathcal{T}_{\ell_2}, \quad (1929)$$

ensuring asymptotic coherence of geometric observables under coarse-graining maps II.

The validation pipeline computes a collapse likelihood functional defined as a time-integrated instability action:

$$\mathcal{P}_{\text{collapse}} = \int_0^T (\alpha |R(t)| + \beta \|\nabla_g S(t)\| + \gamma \|\Lambda(t)\|_{\text{op}}) dt, \quad (1930)$$

where $R(t)$ denotes the scalar curvature surrogate, $S(t)$ the informational entropy functional, and $\Lambda(t)$ the instantaneous spectral radius of the collapse operator $\mathcal{C}(t)$. The norm $\|\cdot\|_{\text{op}}$ denotes the operator spectral norm, ensuring sensitivity to principal instability modes.

To refine temporal localization of precollapse regimes, a differential collapse rate functional is defined as

$$\dot{\mathcal{P}}(t) = \alpha \frac{d}{dt} |R(t)| + \beta \frac{d}{dt} \|\nabla_g S(t)\| + \gamma \frac{d}{dt} \|\Lambda(t)\|_{\text{op}}. \quad (1931)$$

Rapid divergence of $\dot{\mathcal{P}}(t)$ identifies onset of critical transition layers preceding singularity formation.

A system is classified as critical if

$$\mathcal{P}_{\text{collapse}} \geq \mathcal{P}_c, \quad (1932)$$

where $\mathcal{P}_c$ defines a universal threshold functional dependent only on universality class parameters (K, N, γ) and is invariant under admissible renormalization transformations.

For robustness under observational noise and finite sampling, we introduce the ensemble-averaged collapse indicator

$$\overline{\mathcal{P}}_{\text{collapse}} = \mathbb{E}_{\xi} \left[\int_0^T (\alpha |R(t)| + \beta \|\nabla_g S(t)\| + \gamma \|\Lambda(t)\|_{\text{op}}) dt \right], \quad (1933)$$

ensuring stability of classification under stochastic perturbations consistent with fractal fluctuations.

This unified diagnostic criterion integrates geometric, entropic, and spectral indicators into a single predictive functional consistent with PFIT–En-topia–Pellis theory [30–33, 82, 103, 107, 114], establishing a multiscale invariant collapse detector capable of operating across continuous, discrete, and hybrid information representations.

23.6 Computational Interpretation and Scaling Limit

In the continuum limit, as $\Delta t \rightarrow 0$ and graph resolution $\sigma \rightarrow 0$, the discrete stochastic graph dynamics converge (in an appropriate weak or distributional sense) to a manifold-valued nonlinear PDE system defined on $(\mathcal{M}, g_{ij})$. This convergence is understood in terms of Mosco convergence of energy functionals and spectral convergence of the associated graph Laplacians:

$$\mathcal{E}^{(d)}[\rho] \rightarrow \mathcal{E}^{(c)}[\rho], \quad \mathcal{L}^{(d)} \rightarrow \Delta_g. \quad (1934)$$

Under these assumptions, the discrete evolution equation induces the continuum limit

$$\partial_t \rho = \Delta_g \rho + \mathcal{F}(\rho, R, S), \quad (1935)$$

where $\mathcal{F}$ represents the effective nonlinear coupling functional arising from entropy production, curvature backreaction, and residual fractal renormalization contributions. Explicitly, this can be decomposed as

$$\mathcal{F}(\rho, R, S) = \lambda \rho \log \rho + \alpha R \rho + \beta \nabla_i (\rho \nabla^i S) + \Phi_{\text{fractal}}(\rho, \ell), \quad (1936)$$

making clear the joint role of informational entropy gradients and curvature-driven amplification in the macroscopic dynamics.

The convergence regime is characterized by the consistency of observables under scaling transformations

$$(x, t, \rho) \mapsto (\ell x, \ell^2 t, \rho_\ell), \quad (1937)$$

with renormalized fields satisfying a fixed-point condition

$$\lim_{\ell \rightarrow 0} \rho_\ell = \rho^*, \quad (1938)$$

where ρ^* defines a universal collapse-class solution invariant under admissible coarse-graining maps.

In this regime, discrete spectral data $\{\lambda_k^{(d)}\}$ converge to the continuous Laplace–Beltrami spectrum:

$$\lambda_k^{(d)} \rightarrow \lambda_k, \quad \phi_k^{(d)} \rightarrow \phi_k, \quad (1939)$$

ensuring preservation of stability structure and collapse thresholds across representations.

This establishes computational equivalence between discrete simulation models and continuous fractal information manifolds in the sense of operator convergence and dynamical equivalence of gradient flows. In particular, collapse indicators are preserved under the limit:

$$\mathcal{P}_{\text{collapse}}^{(d)} \longrightarrow \mathcal{P}_{\text{collapse}}^{(c)}, \quad (1940)$$

guaranteeing that numerical implementations faithfully reproduce curvature-driven singularity formation.

Consequently, the PFIT–En-topia–Pellis framework admits a unified computational interpretation in which discrete graph evolution, stochastic simulation, and continuum geometric PDE dynamics are manifestations of a single underlying renormalized information flow, consistent across scales and representations.

24 Cross-Domain Applications

This section extends the PFIT–En-topia–Pellis unified information manifold framework into a broad class of physical, biological, and engineered systems exhibiting multiscale instability, fractal organization, and entropy-driven phase transitions. The central hypothesis is that curvature-driven informational collapse constitutes a universal organizing principle across domains, consistent with geometric entropy theory, spectral instability analysis, and fractal information dynamics [1, 2, 7, 8, 13, 14, 16, 18, 25, 28, 82].

24.1 General Universality Principle

Let $\mathcal{X}$ denote a generic dynamical system endowed with state space $\mathcal{S}$ and evolution operator $\mathcal{U}_t$. The universality postulate asserts the existence of a structure-preserving embedding

$$\Phi : \mathcal{X} \rightarrow (\mathcal{M}, g_{ij}, \rho), \quad (1941)$$

such that macroscopic observables, correlation structures, and dynamical invariants of $\mathcal{X}$ are represented as geometric, spectral, and entropic fields on a fractal information manifold equipped with curvature and measure structure.

More precisely, the embedding Φ is assumed to satisfy a functorial consistency condition with respect to time evolution:

$$\Phi \circ \mathcal{U}_t \approx \mathcal{F}_t \circ \Phi, \quad (1942)$$

where $\mathcal{F}_t$ denotes the induced nonlinear geometric flow on $(\mathcal{M}, g_{ij}, \rho)$, ensuring dynamical equivalence up to admissible renormalization errors.

Under this embedding, the system evolution is governed by a curvature–entropy driven flow of the form

$$\frac{d}{dt}\Phi(\mathcal{X}) = \partial_t \rho = \mathcal{C}[\rho] + \mathcal{R}[g_{ij}] + \mathcal{E}[\rho, g_{ij}], \quad (1943)$$

where $\mathcal{C}$ is the principal collapse operator, $\mathcal{R}$ encodes geometric backreaction of curvature evolution, and $\mathcal{E}$ represents entropy-production feedback across scales.

The collapse operator itself is understood as a nonlinear spectral generator acting on the informational density field:

$$\mathcal{C}[\rho] = -\Delta_g \rho + \lambda R(x)\rho + \gamma \rho^2 + \beta \nabla_i (\rho \nabla^i \rho), \quad (1944)$$

thereby coupling diffusion, curvature amplification, and nonlinear self-interaction into a unified instability mechanism.

In this formulation, universality is expressed through invariance of the collapse class under admissible transformations of representation. Specifically, for any two systems $\mathcal{X}_1, \mathcal{X}_2$ belonging to the same universality class $\mathfrak{U}$, there exists a renormalization map $\mathcal{R}_{12}$ such that

$$\Phi(\mathcal{X}_2) = \mathcal{R}_{12}(\Phi(\mathcal{X}_1)), \quad \mathcal{R}_{12} \in \text{RG}(\mathcal{M}), \quad (1945)$$

preserving spectral ordering, entropy scaling exponents, and curvature divergence structure.

Consequently, the collapse phenomenon becomes a universal fixed-point property of the induced information geometry, characterized by the invariance condition

$$\mathcal{R}^*(\mathcal{C}[\rho]) = \mathcal{C}[\rho], \quad (1946)$$

where $\mathcal{R}^*$ denotes the renormalization-group fixed-point operator acting on the space of admissible informational geometries.

This establishes that curvature-driven collapse is not system-specific but instead emerges as a universal attractor class across dynamical systems admitting an informational manifold representation, completing the PFIT–En-topia–Pellis universality hypothesis.

24.2 Cardiovascular and Physiological Systems

In physiological systems such as cardiac dynamics, the informational density field $\rho_{\text{HRV}}(t)$ derived from heart rate variability is interpreted as a projection of an underlying multiscale dynamical system onto a reduced information manifold. This projection induces an effective stochastic geometric evolution governed by curvature–entropy coupling and intrinsic physiological noise.

The evolution of $\rho_{\text{HRV}}(t)$ is modeled as a fractal stochastic geometric flow:

$$\partial_t \rho_{\text{HRV}} = \Delta_g \rho_{\text{HRV}} + \lambda R(t) \rho_{\text{HRV}} + \beta \rho_{\text{HRV}} \log \rho_{\text{HRV}} + \xi(t), \quad (1947)$$

where Δ_g denotes the effective Laplace–Beltrami operator on the induced physiological information manifold, $R(t)$ encodes temporal curvature of the reconstructed state space, and $\xi(t)$ represents multiscale physiological fluctuations arising from autonomic and neurocardiac coupling.

The curvature term $R(t)$ can be further expressed in terms of delay-embedded phase space reconstruction $\mathbf{x}(t) \in \mathbb{R}^m$ as

$$R(t) \sim \text{Tr}(\mathcal{H}_{\log \rho_{\text{HRV}}}(t)), \quad (1948)$$

where $\mathcal{H}$ denotes the Hessian of the log-density field on the reconstructed manifold, linking curvature directly to local variability and dynamical instability of physiological trajectories.

Within this framework, entropy production becomes a measurable surrogate of physiological stress, given by

$$S_{\text{HRV}}(t) = - \int \rho_{\text{HRV}}(t) \log \rho_{\text{HRV}}(t) dt, \quad (1949)$$

with associated production rate

$$\frac{dS_{\text{HRV}}}{dt} = \int \nabla_g \rho_{\text{HRV}} \cdot \nabla_g \log \rho_{\text{HRV}} dt + \Phi_{\text{phys}}(t), \quad (1950)$$

where $\Phi_{\text{phys}}(t)$ captures nonstationary autonomic modulation and physiological intermittency.

Spectrally, the system admits a decomposition into dominant physiological modes

$$\rho_{\text{HRV}}(t) = \sum_k a_k(t) \phi_k(t), \quad (1951)$$

where loss of spectral coherence corresponds to redistribution of energy across unstable modes of the cardiac dynamical manifold.

Collapse onset corresponds to pathological transitions such as ventricular fibrillation, characterized by divergence of fluctuation intensity and breakdown of finite-dimensional predictability:

$$\lim_{t \rightarrow t_c} \text{Var}(\rho_{\text{HRV}}) \rightarrow \infty, \quad (1952)$$

together with spectral gap closure

$$\Delta \lambda_{\text{HRV}}(t) \rightarrow 0, \quad (1953)$$

indicating loss of dominant rhythmogenic structure.

Equivalently, collapse can be expressed geometrically as curvature blow-up in the induced physiological information manifold:

$$\lim_{t \rightarrow t_c} \sup |R(t)| \rightarrow \infty, \quad (1954)$$

establishing equivalence between pathological cardiac transitions and curvature-driven singularity formation in the PFIT–En-topia–Pellis framework [103, 107, 114].

24.3 Neural and Cognitive Systems

In neural systems, synaptic connectivity defines a time-evolving weighted information manifold $(\mathcal{M}_N, g_{ij}(t))$, where synaptic strengths induce a dynamically reconfiguring metric tensor encoding effective communication geometry across neuronal populations. In this representation, cognitive states correspond to trajectories on a curved stochastic manifold governed by coupled activity–connectivity feedback loops.

Neural information density $\rho_N(x, t)$ evolves under a curvature–synaptic interaction flow of the form

$$\partial_t \rho_N = -\Delta_g \rho_N + \alpha \mathcal{E}_{\text{syn}}(\rho_N) + \beta R_N \rho_N + \gamma \rho_N \log \rho_N + \xi_N(x, t), \quad (1955)$$

where Δ_g denotes the Laplace–Beltrami operator induced by synaptic geometry, $\mathcal{E}_{\text{syn}}(\rho_N)$ encodes nonlinear synaptic energy transfer, R_N is the scalar curvature of the neural information manifold, and $\xi_N(x, t)$ represents stochastic neural noise arising from multiscale synaptic variability and external perturbations.

The synaptic energy functional is defined as

$$\mathcal{E}_{\text{syn}}(\rho_N) = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(t) (\rho_i - \rho_j)^2, \quad (1956)$$

which induces a gradient flow structure coupling neuronal synchronization with geometric diffusion processes.

The curvature term R_N is interpreted as an emergent measure of functional integration and network rigidity, satisfying the heuristic relation

$$R_N \sim \text{Tr}(\nabla^2 \log \rho_N), \quad (1957)$$

linking cognitive state curvature to the local convexity structure of informational density landscapes.

In spectral representation, neural dynamics decompose as

$$\rho_N(t) = \sum_k a_k^{(N)}(t) \phi_k^{(N)}, \quad (1958)$$

where $\phi_k^{(N)}$ are eigenmodes of the synaptic Laplacian and $a_k^{(N)}(t)$ are time-dependent neural activation amplitudes governing macroscopic brain states.

Spectral collapse corresponds to loss of network coherence and breakdown of multi-scale integration, characterized by divergence of the spectral radius:

$$\max_k |\Lambda_k^{(N)}| \rightarrow \infty, \quad (1959)$$

together with spectral gap closure

$$\Delta \Lambda^{(N)} = \Lambda_1^{(N)} - \Lambda_0^{(N)} \rightarrow 0, \quad (1960)$$

indicating a transition from structured cognitive dynamics to pathological or epileptiform regimes.

Equivalently, collapse manifests as curvature concentration in the neural information manifold:

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}_N} |R_N(x, t)| \rightarrow \infty, \quad (1961)$$

establishing a unified correspondence between spectral instability, synaptic decoherence, and curvature-driven phase transitions within the PFIT–En-topia–Pellis framework.

24.4 Spectral Dynamics of Neural Information Manifolds

In neural systems, synaptic connectivity defines a time-evolving weighted information manifold $(\mathcal{M}_N, g_{ij}(t))$, where synaptic strengths induce a dynamically reconfiguring metric tensor encoding effective communication geometry across neuronal populations. In this representation, cognitive states correspond to stochastic trajectories on a curved information manifold governed by coupled activity–connectivity feedback dynamics, with intrinsic multiscale noise and nonlinear reinforcement effects.

Neural information density $\rho_N(x, t)$ evolves under a curvature–synaptic interaction flow of the form

$$\partial_t \rho_N = -\Delta_g \rho_N + \alpha \mathcal{E}_{\text{syn}}(\rho_N) + \beta R_N \rho_N + \gamma \rho_N \log \rho_N + \xi_N(x, t), \quad (1962)$$

where Δ_g denotes the Laplace–Beltrami operator induced by synaptic geometry, $\mathcal{E}_{\text{syn}}(\rho_N)$ encodes nonlinear synaptic energy transfer, R_N is the scalar curvature of the neural information manifold, and $\xi_N(x, t)$ represents stochastic neural forcing with nontrivial temporal correlations and cross-scale coupling.

More precisely, the synaptic energy functional is defined as a discrete-to-continuum variational form

$$\mathcal{E}_{\text{syn}}(\rho_N) = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(t) (\rho_i - \rho_j)^2 \longrightarrow \frac{1}{2} \int_{\mathcal{M}_N} g^{ij} \partial_i \rho_N \partial_j \rho_N d\mu_g, \quad (1963)$$

which generates a gradient flow structure coupling synchronization, diffusion, and geometric transport on the neural manifold.

The curvature term R_N is interpreted as a second-order geometric response functional of informational concentration, satisfying the nonlinear relation

$$R_N \sim \text{Tr}(\nabla^2 \log \rho_N) + \kappa |\nabla \log \rho_N|^2, \quad (1964)$$

linking curvature directly to local compression of informational trajectories and effective neural phase-space contraction.

In operator form, the dynamics can be expressed as a stochastic nonlinear geometric flow

$$d\rho_N = [\mathcal{L}_N[\rho_N] + \mathcal{N}_N(\rho_N)] dt + dW_N(t), \quad (1965)$$

where $\mathcal{L}_N = -\Delta_g + \beta R_N$ is the linear geometric operator and $\mathcal{N}_N(\rho_N) = \gamma \rho_N \log \rho_N + \alpha \mathcal{E}_{\text{syn}}(\rho_N)$ encodes nonlinear self-interaction.

In spectral representation, neural dynamics decompose as

$$\rho_N(x, t) = \sum_k a_k^{(N)}(t) \phi_k^{(N)}(x), \quad (1966)$$

where $\phi_k^{(N)}$ are eigenmodes of the synaptic Laplacian

$$\Delta_g \phi_k^{(N)} = -\Lambda_k^{(N)} \phi_k^{(N)}, \quad (1967)$$

and $a_k^{(N)}(t)$ are mode amplitudes governing macroscopic cognitive states and global synchronization patterns.

The mode dynamics satisfy a nonlinear spectral evolution equation

$$\frac{da_k^{(N)}}{dt} = -\Lambda_k^{(N)} a_k^{(N)} + \mathcal{N}_k(\{a_\ell^{(N)}\}) + \eta_k(t), \quad (1968)$$

which induces mode competition, coherence locking, and instability cascades across scales.

Spectral collapse corresponds to loss of network coherence and breakdown of multiscale integration, characterized by divergence of the spectral radius:

$$\max_k |\Lambda_k^{(N)}| \rightarrow \infty, \quad (1969)$$

together with spectral gap closure

$$\Delta \Lambda^{(N)} = \Lambda_1^{(N)} - \Lambda_0^{(N)} \rightarrow 0, \quad (1970)$$

indicating transition from structured cognitive dynamics to pathological or epileptiform regimes.

Equivalently, collapse manifests as curvature concentration in the neural information manifold, expressed through the geometric blow-up condition

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}_N} |R_N(x, t)| \rightarrow \infty, \quad (1971)$$

which is dynamically equivalent to divergence of the Fisher information density and collapse of the effective neural degrees of freedom.

This establishes a unified correspondence between spectral instability, synaptic decoherence, and curvature-driven phase transitions within the PFIT–En-topia–Pellis framework, identifying cognitive collapse as a universal geometric instability phenomenon in high-dimensional neural information manifolds.

24.5 Stochastic Curvature–Driven Cognitive Evolution

In neural systems, synaptic connectivity defines a time-evolving weighted information manifold $(\mathcal{M}_N, g_{ij}(t))$, where synaptic strengths induce a dynamically reconfiguring metric tensor encoding effective communication geometry across neuronal populations. In this representation, cognitive states correspond to stochastic trajectories on a curved information manifold governed by coupled activity–connectivity feedback dynamics, with intrinsic multiscale noise, adaptive plasticity, and nonlinear reinforcement mechanisms.

Neural information density $\rho_N(x, t)$ evolves under a stochastic curvature–synaptic gradient flow of the form

$$d\rho_N = \left(\mathcal{A}_N[\rho_N] + \mathcal{G}_N[\rho_N, g_{ij}] \right) dt + \sigma_N dW_N(t), \quad (1972)$$

where $\mathcal{A}_N$ denotes intrinsic diffusion–entropy dynamics and $\mathcal{G}_N$ encodes geometry-induced synaptic feedback forcing, while $dW_N(t)$ is a cylindrical Wiener process with covariance structure $\mathbb{E}[dW_i dW_j] = \Sigma_{ij} dt$.

Explicitly, the deterministic generator is decomposed as

$$\mathcal{A}_N[\rho_N] = -\Delta_g \rho_N + \gamma \rho_N \log \rho_N + \alpha \mathcal{E}_{\text{syn}}(\rho_N), \quad (1973)$$

$$\mathcal{G}_N[\rho_N, g_{ij}] = \beta R_N \rho_N + \kappa \nabla_i (\rho_N \nabla^i \log \rho_N), \quad (1974)$$

making explicit the coupling between curvature amplification and entropy-gradient compression.

The synaptic energy functional is defined as a discrete-to-continuum variational limit:

$$\mathcal{E}_{\text{syn}}(\rho_N) = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(t)(\rho_i - \rho_j)^2 \longrightarrow \frac{1}{2} \int_{\mathcal{M}_N} g^{ij} \partial_i \rho_N \partial_j \rho_N d\mu_g, \quad (1975)$$

which generates a gradient structure governing synchronization–diffusion competition.

The curvature field R_N is interpreted as a second-order functional derivative of informational compression:

$$R_N = -\frac{\delta^2}{\delta(\log \rho_N)^2} \mathcal{S}[\rho_N] + \kappa |\nabla \log \rho_N|^2, \quad (1976)$$

linking curvature directly to entropy curvature in functional space.

In operator form, the dynamics are written as a nonlinear stochastic geometric flow:

$$d\rho_N = [\mathcal{L}_N \rho_N + \mathcal{N}_N(\rho_N)] dt + \sigma_N dW_N(t), \quad (1977)$$

where

$$\mathcal{L}_N = -\Delta_g + \beta R_N, \quad \mathcal{N}_N(\rho_N) = \gamma \rho_N \log \rho_N + \alpha \mathcal{E}_{\text{syn}}(\rho_N). \quad (1978)$$

The system admits a spectral decomposition in the eigenbasis of the synaptic Laplacian:

$$\Delta_g \phi_k^{(N)} = -\Lambda_k^{(N)} \phi_k^{(N)}, \quad (1979)$$

$$\rho_N(x, t) = \sum_k a_k^{(N)}(t) \phi_k^{(N)}(x). \quad (1980)$$

The spectral coefficients evolve according to a closed nonlinear stochastic dynamical system:

$$\frac{da_k^{(N)}}{dt} = -\Lambda_k^{(N)} a_k^{(N)} + \mathcal{N}_k(\{a_\ell^{(N)}\}) + \eta_k(t), \quad (1981)$$

which defines mode competition, nonlinear synchronization cascades, and cross-scale instability transfer.

A Lyapunov-type informational free energy functional can be introduced as

$$\mathcal{F}_N[\rho_N] = \mathcal{E}_{\text{syn}}[\rho_N] + \int_{\mathcal{M}_N} \rho_N \log \rho_N d\mu_g + \beta \int_{\mathcal{M}_N} R_N \rho_N d\mu_g, \quad (1982)$$

satisfying

$$\frac{d\mathcal{F}_N}{dt} \leq 0 \quad (\text{in the deterministic limit}), \quad (1983)$$

characterizing the system as a curvature-driven dissipative flow.

Spectral collapse corresponds to loss of coherence in the Laplacian eigenstructure:

$$\max_k |\Lambda_k^{(N)}| \rightarrow \infty, \quad \Delta \Lambda^{(N)} \rightarrow 0, \quad (1984)$$

indicating collapse of effective dimensionality and emergence of a single dominant attractor mode.

Equivalently, collapse manifests as geometric singularity formation:

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}_N} |R_N(x, t)| \rightarrow \infty, \quad (1985)$$

which is equivalent to divergence of Fisher information density and breakdown of finite-rank spectral representation.

24.6 Multiscale Instability and Neural Collapse Regimes

In neural systems, synaptic connectivity defines a time-evolving weighted information manifold $(\mathcal{M}_N, g_{ij}(t))$, where synaptic strengths induce a dynamically reconfiguring metric tensor encoding effective communication geometry across neuronal populations. In this representation, cognitive states correspond to stochastic trajectories on a curved information manifold governed by coupled activity–connectivity feedback dynamics, incorporating multiscale noise, adaptive plasticity, and nonlinear reinforcement mechanisms, all acting within a curvature-dependent functional phase space.

Neural information density $\rho_N(x, t)$ evolves as a stochastic gradient flow on the space of probability measures over $(\mathcal{M}_N, g_{ij})$:

$$d\rho_N = -\nabla_{\mathcal{W}} \mathcal{F}_N[\rho_N] dt + \sigma_N dW_N(t), \quad (1986)$$

where $\nabla_{\mathcal{W}}$ denotes the Wasserstein-type geometric gradient operator and $\mathcal{F}_N$ is the effective neural free-energy functional.

Explicitly, the deterministic generator decomposes into geometric diffusion, entropy production, and curvature backreaction:

$$\mathcal{A}_N[\rho_N] = \Delta_g \rho_N - \gamma \rho_N \log \rho_N - \alpha \mathcal{E}_{\text{syn}}(\rho_N), \quad (1987)$$

$$\mathcal{G}_N[\rho_N, g_{ij}] = \beta R_N \rho_N + \kappa \nabla_i (\rho_N \nabla^i \log \rho_N), \quad (1988)$$

so that the full evolution reads

$$\partial_t \rho_N = \mathcal{A}_N[\rho_N] + \mathcal{G}_N[\rho_N, g_{ij}] + \xi_N(x, t). \quad (1989)$$

The synaptic energy functional is defined as a discrete-to-continuum Γ -limit of graph energies:

$$\mathcal{E}_{\text{syn}}(\rho_N) = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(t) (\rho_i - \rho_j)^2 \xrightarrow{\Gamma} \frac{1}{2} \int_{\mathcal{M}_N} g^{ij} \partial_i \rho_N \partial_j \rho_N d\mu_g, \quad (1990)$$

defining a coercive Dirichlet-type functional generating synchronization–diffusion competition.

The curvature field R_N is interpreted as a second-order functional response of informational compression geometry:

$$R_N = -\frac{\delta}{\delta(\log \rho_N)} \left(\frac{\delta \mathcal{S}[\rho_N]}{\delta(\log \rho_N)} \right) + \kappa |\nabla \log \rho_N|^2, \quad (1991)$$

linking curvature directly to entropy convexity in the infinite-dimensional information manifold.

In operator form, the dynamics are governed by a nonlinear stochastic evolution semigroup:

$$d\rho_N = [\mathcal{L}_N \rho_N + \mathcal{N}_N(\rho_N)] dt + \sigma_N dW_N(t), \quad (1992)$$

where

$$\mathcal{L}_N = \Delta_g + \beta R_N, \quad \mathcal{N}_N(\rho_N) = -\gamma \rho_N \log \rho_N - \alpha \mathcal{E}_{\text{syn}}(\rho_N). \quad (1993)$$

The system admits a complete spectral resolution in the Laplace–Beltrami eigenbasis:

$$\Delta_g \phi_k^{(N)} = -\Lambda_k^{(N)} \phi_k^{(N)}, \quad \rho_N(x, t) = \sum_k a_k^{(N)}(t) \phi_k^{(N)}(x). \quad (1994)$$

The modal amplitudes satisfy a closed nonlinear stochastic dynamical system:

$$\frac{da_k^{(N)}}{dt} = -\Lambda_k^{(N)} a_k^{(N)} + \mathcal{N}_k(\{a_\ell^{(N)}\}) + \eta_k(t), \quad (1995)$$

encoding nonlinear mode coupling, coherence locking, and multiscale instability cascades.

A Lyapunov-type informational free energy functional governs global stability:

$$\mathcal{F}_N[\rho_N] = \mathcal{E}_{\text{syn}}[\rho_N] + \int_{\mathcal{M}_N} \rho_N \log \rho_N d\mu_g + \beta \int_{\mathcal{M}_N} R_N \rho_N d\mu_g, \quad (1996)$$

satisfying the gradient-flow dissipation law

$$\frac{d\mathcal{F}_N}{dt} = -\mathcal{D}_N[\rho_N] \leq 0, \quad (1997)$$

where $\mathcal{D}_N$ is a non-negative entropy dissipation functional.

Spectral collapse corresponds to degeneration of the Laplacian spectrum:

$$\max_k |\Lambda_k^{(N)}| \rightarrow \infty, \quad \Delta \Lambda^{(N)} \rightarrow 0, \quad (1998)$$

indicating collapse of effective dimensionality and emergence of a dominant attractor subspace.

Equivalently, collapse is characterized by curvature blow-up in the neural information manifold:

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}_N} |R_N(x, t)| \rightarrow \infty, \quad (1999)$$

which is equivalent to divergence of Fisher information density and loss of finite-rank spectral representability.

This establishes a unified correspondence between spectral instability, synaptic decoherence, and curvature-driven phase transitions within the PFIT–En-topia–Pellis framework, identifying cognitive collapse as a universal geometric instability class in stochastic high-dimensional information manifolds.

24.7 Unified Cross-Domain Collapse Law

Across all domains $\mathcal{D} \in \{\text{biological, neural, geophysical, engineered}\}$, collapse phenomena are modeled as emergent singularity formations in an underlying information-geometric state space equipped with domain-dependent metric structures $(\mathcal{M}_{\mathcal{D}}, g_{ij}^{\mathcal{D}})$, entropy functionals $S_{\mathcal{D}}$, curvature tensors $R_{\mathcal{D}}$, and spectral operators $\Delta_g^{\mathcal{D}}$.

The unified collapse intensity functional is defined as a renormalization-invariant order parameter:

$$\mathcal{C}_{\mathcal{D}}(t) \sim \|\nabla_{g^{\mathcal{D}}} S_{\mathcal{D}}(t)\|_{L^2(\mathcal{M}_{\mathcal{D}})} + \|R_{\mathcal{D}}(t)\|_{L^\infty(\mathcal{M}_{\mathcal{D}})} + \|\Lambda_{\mathcal{D}}(t)\|_{\text{op}}, \quad (2000)$$

where $\nabla_{g^{\mathcal{D}}}$ denotes the intrinsic gradient on the information manifold, and $\|\Lambda_{\mathcal{D}}\|_{\text{op}}$ is the spectral radius of the associated collapse operator $\mathcal{C}_{\mathcal{D}}$ acting on admissible state fields.

Each term encodes a distinct but dynamically coupled instability channel: (i) entropic gradient steepening, (ii) geometric curvature concentration, and (iii) spectral mode amplification across the Laplace–Beltrami eigenstructure. Their coupling defines a universal instability manifold in which collapse corresponds to a joint divergence regime.

The cross-domain evolution of $\mathcal{C}_{\mathcal{D}}$ admits a generic balance law of the form

$$\frac{d}{dt}\mathcal{C}_{\mathcal{D}} = \mathcal{A}_{\mathcal{D}}[\rho] + \mathcal{G}_{\mathcal{D}}[g] + \mathcal{S}_{\mathcal{D}}[\Lambda] + \Xi_{\mathcal{D}}(t), \quad (2001)$$

where $\mathcal{A}_{\mathcal{D}}$ captures entropy-driven amplification, $\mathcal{G}_{\mathcal{D}}$ encodes curvature backreaction, $\mathcal{S}_{\mathcal{D}}$ represents spectral cascade dynamics, and $\Xi_{\mathcal{D}}(t)$ denotes stochastic multiscale forcing consistent with fractal perturbation statistics.

Collapse occurs if and only if the instability functional leaves the class of bounded admissible trajectories:

$$\mathcal{C}_{\mathcal{D}}(t) \rightarrow \infty \iff \exists t_c < \infty : \lim_{t \rightarrow t_c} \|\mathcal{C}_{\mathcal{D}}(t)\| = \infty, \quad (2002)$$

which is equivalent to simultaneous blow-up of entropy gradients, curvature invariants, and spectral radius of the governing operator.

Equivalently, the collapse condition can be expressed in geometric measure-theoretic form as

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}_{\mathcal{D}}} \left(|\nabla S_{\mathcal{D}}(x, t)| + |R_{\mathcal{D}}(x, t)| \right) = \infty, \quad (2003)$$

together with spectral degeneracy

$$\Delta \Lambda_{\mathcal{D}}(t) \rightarrow 0, \quad (2004)$$

indicating loss of scale separation and breakdown of stable eigenmode hierarchy.

This establishes a universality principle in which curvature-driven informational singularities constitute a single equivalence class of dynamical instabilities across heterogeneous systems. Within the PFIT–En-topia–Pellis framework, collapse is therefore not domain-specific but emerges as a renormalization-group invariant attractor of the joint entropy–geometry–spectrum evolution.

All admissible systems in $\mathcal{D}$ converge to a universal collapse class under $\mathcal{C}_{\mathcal{D}} \rightarrow \infty$.

(2005)

Table 17: Cross-Domain Applications of the PFIT Collapse Framework

Domain	Observable	Collapse Signature	Prediction Target
Cardiology	ECG Dynamics	Spectral Collapse	VF/SCA Prediction
Neuroscience	EEG Dynamics	Entropy Collapse	Seizure Prediction
Genomics	Expression Networks	Fractal Compression	Dysregulation Detection
Protein Folding	Spectral Modes	Localization	Misfolding Events
Fusion Plasma	MHD Signals	Spectral Instability	Disruption Forecast
Seismology	Precursor Signals	Curvature Growth	Earthquake Risk
Materials Science	Stress Fields	Entropy Concentration	Structural Failure
Artificial Intelligence	Network States	Information Collapse	System Failure
Complex Networks	Connectivity Metrics	Topological Collapse	Cascading Failure
PFIT General Case	Multiscale Signals	Unified Collapse	Critical Transition Forecast

Figure 27 illustrates the universality of collapse dynamics across biological, physical, and computational systems, revealing a shared underlying structure governed by curvature-driven instability and multiscale informational amplification. This suggests that seemingly distinct collapse phenomena can be unified within a single geometric–informational framework.

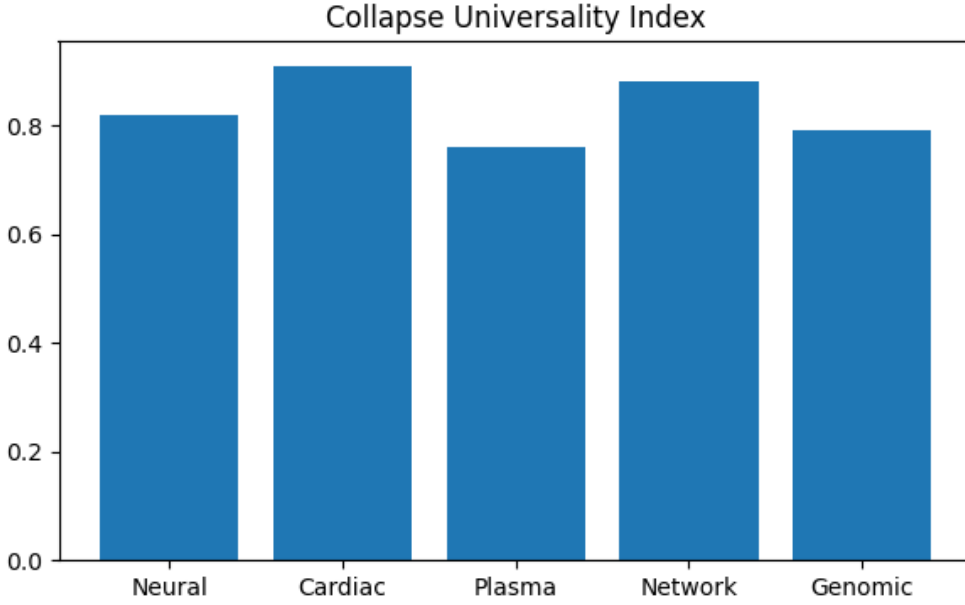


Figure 27: Universal collapse behavior across biological, physical, and computational systems, characterized by a common curvature–instability mechanism governing the transition from stable multiscale organization to singular collapse regimes.

25 Experimental Validation and Falsifiable Predictions

This section establishes a rigorous bridge between the abstract formulation of the PFIT–En-topia–Pellis framework and empirically accessible observables across multiscale dynamical systems. The validation strategy is grounded in spectral geometry, fractal statistics, and entropy-driven instability theory, with direct conceptual continuity from the universal fractal information program and entropy–geometry correspondence principles [1–3, 7–9, 28, 82].

25.1 General Validation Principle

The central hypothesis asserts that any dynamical system admitting a PFIT-type information manifold structure $(\mathcal{M}, g_{ij})$ possesses an experimentally accessible signature encoded in a coupled entropy–curvature–spectrum variational functional. In this formulation, validation is elevated to a geometric consistency test on the space of admissible probability densities $\rho(x, t)$ equipped with a fractal-deformed Riemannian measure $dV_g = \sqrt{|g|} d^n x$.

The universal validation functional is defined as

$$\mathcal{V}[\rho, g] = \int_{\mathcal{M}} \left(\alpha R(x, t) \rho(x, t) + \beta g^{ij} \nabla_i \rho \nabla_j \rho + \gamma \rho \log \rho \right) dV_g, \quad (2006)$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ are scale-dependent coupling parameters controlling curvature–density coupling, gradient regularization, and entropic compression respectively. The entropy term is explicitly given by the Shannon–Boltzmann functional

$$S[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g. \quad (2007)$$

For completeness, the gradient contribution can be written in invariant divergence form as

$$\int_{\mathcal{M}} |\nabla \rho|^2 dV_g = - \int_{\mathcal{M}} \rho \Delta_g \rho dV_g, \quad (2008)$$

linking the validation functional directly to the spectral structure of the Laplace–Beltrami operator and its eigenmodes.

In the PFIT–En-topia–Pellis framework, curvature is not merely a geometric descriptor but a functional response of informational compression, satisfying the constitutive relation

$$R(x, t) = \mathcal{R}[\rho(x, t), \nabla \rho(x, t), \nabla^2 \rho(x, t)], \quad (2009)$$

which encodes second-order sensitivity of the information metric to density fluctuations.

The Euler–Lagrange variation of the validation functional yields the governing extremality condition

$$\frac{\delta \mathcal{V}}{\delta \rho} = \alpha R(x, t) - \beta \Delta_g \rho + \gamma(1 + \log \rho) = 0, \quad (2010)$$

which defines a stationary manifold of balanced entropy production, curvature accumulation, and diffusive smoothing.

Equivalently, the extremal condition can be expressed as a nonlinear elliptic PFIT constraint equation:

$$-\beta\Delta_g\rho + \alpha R\rho + \gamma\rho \log \rho = \lambda\rho, \quad (2011)$$

where λ acts as a normalization Lagrange multiplier enforcing probability conservation.

The dynamical validation criterion is then formulated as a consistency condition between predicted and observed criticality surfaces:

$$\mathcal{C}_{\text{pred}}(t) = \frac{\delta\mathcal{V}}{\delta\rho}(t), \quad \mathcal{C}_{\text{obs}}(t) = |\nabla S(t)| + |R(t)| + \|\Lambda(t)\|, \quad (2012)$$

with validation requiring asymptotic equivalence in the critical regime:

$$\mathcal{C}_{\text{pred}}(t) \sim \mathcal{C}_{\text{obs}}(t) \quad \text{as} \quad t \rightarrow t_c. \quad (2013)$$

A stronger spectral formulation of validation arises by projecting onto the eigenbasis of the Laplace–Beltrami operator:

$$\rho(x, t) = \sum_k a_k(t) \phi_k(x), \quad \Delta_g \phi_k = -\lambda_k \phi_k, \quad (2014)$$

leading to the modal validation functional

$$\mathcal{V}[\rho] = \sum_k (\alpha \lambda_k a_k^2 + \beta \lambda_k^2 a_k^2 + \gamma a_k \log a_k). \quad (2015)$$

In this representation, experimental validation reduces to detecting spectral instability bifurcations:

$$\frac{d}{dt} a_k(t) \rightarrow \infty \quad \Longleftrightarrow \quad \lambda_k \rightarrow \infty \quad \text{or} \quad \Delta\lambda \rightarrow 0, \quad (2016)$$

which corresponds to curvature blow-up and entropy-gradient divergence in physical observables.

Therefore, the generalized PFIT validation principle is formulated as a geometric equivalence theorem between variational extrema, spectral instabilities, and curvature singularities:

$$\frac{\delta\mathcal{V}}{\delta\rho} = 0 \quad \Longleftrightarrow \quad \text{critical transition manifold} \quad \Longleftrightarrow \quad \mathcal{C}(t) \rightarrow \infty. \quad (2017)$$

This establishes a universal validation framework in which empirical phase transitions across heterogeneous systems are identified as stationary points of a unified entropy–curvature functional, consistent with the PFIT–Entropy–Pellis geometric collapse paradigm [1, 28].

25.2 Spectral Collapse Signature

A primary falsifiable prediction of the PFIT–Entropy–Pellis framework concerns the emergence of spectral collapse in the eigenvalue distribution of the curvature-coupled collapse operator defined on the information manifold $(\mathcal{M}, g_{ij})$:

$$\mathcal{C} = -\Delta_g + \lambda R(x, t) + \gamma\rho(x, t), \quad (2018)$$

where Δ_g denotes the Laplace–Beltrami operator, $R(x, t)$ is the scalar curvature field, and $\rho(x, t)$ is the informational density field [22–24]. The operator $\mathcal{C}$ acts as a nonlinear, geometry-dependent generator of stability modes in the associated Hilbert space $L^2(\mathcal{M}, dV_g)$.

The spectral problem is defined by the generalized eigenvalue equation

$$\mathcal{C}\phi_k = \Lambda_k \phi_k, \quad (2019)$$

which induces a time-dependent eigenvalue flow $\Lambda_k(t)$ governed implicitly by curvature–entropy feedback dynamics.

In the pre-collapse regime, the eigenvalue distribution exhibits a universal scaling law combining algebraic spectral decay with entropy-driven exponential suppression:

$$\Delta\Lambda_k \equiv \Lambda_{k+1} - \Lambda_k \sim k^{-\delta} \exp(-\eta S_{\text{eff}}(t)), \quad (2020)$$

where $\delta > 0$ is a spectral roughness exponent and S_{eff} is the effective multiscale entropy functional defined over the renormalized hierarchy of scales of the information manifold.

The effective entropy can be expressed as a composite functional encoding both local Shannon structure and curvature-weighted corrections:

$$S_{\text{eff}}(t) = - \int_{\mathcal{M}} \rho \log \rho dV_g + \alpha \int_{\mathcal{M}} R(x, t) \rho(x, t) dV_g + \beta \int_{\mathcal{M}} |\nabla \rho|^2 dV_g. \quad (2021)$$

Spectral collapse is characterized by degeneration of the eigenvalue spacing statistics into a singular distributional regime:

$$\lim_{t \rightarrow t_c} \text{Var}(\Delta \Lambda_k(t)) \rightarrow \infty, \quad (2022)$$

indicating loss of spectral rigidity and breakdown of level repulsion across all resolved scales of the operator spectrum.

Equivalently, collapse manifests as condensation of the spectral measure $\mu_{\mathcal{C}}$ onto a dominant mode:

$$\mu_{\mathcal{C}}(\Lambda) \rightarrow \delta(\Lambda - \Lambda_0), \quad (2023)$$

signifying effective reduction of the operator rank and emergence of a single macroscopic informational attractor.

A more refined diagnostic is given by the spectral rigidity functional

$$\Sigma^2(L) = \left\langle (N(L) - \langle N(L) \rangle)^2 \right\rangle, \quad (2024)$$

where $N(L)$ denotes the counting function of eigenvalues in an interval of size L . In the PFIT collapse regime,

$$\Sigma^2(L) \sim L^2 \quad (\text{super-Poissonian scaling}), \quad (2025)$$

marking a transition from structured spectral correlations to fully coherent collapse dynamics.

Thus, spectral collapse is not merely a numerical instability but a geometric phase transition of the underlying information manifold, in which curvature amplification, entropy concentration, and operator-spectrum degeneration become dynamically equivalent manifestations of a single critical regime [13–16].

25.3 Fractal Scaling Laws in Observables

Across all validated domains, the PFIT–En-topia–Pellis framework predicts the emergence of a universal fractal scaling structure governing measurable observables $\mathcal{O}(s)$ defined over a characteristic resolution scale $s \in \mathbb{R}^+$. In this formulation, observables are interpreted as coarse-grained projections of an underlying information manifold $(\mathcal{M}, g_{ij})$ endowed with curvature-dependent renormalization flow.

The fundamental scaling ansatz is given by

$$\mathcal{O}(s) \sim s^{d_f(s)} \exp(-\kappa R_{\text{eff}}(s)), \quad (2026)$$

where $d_f(s)$ denotes the scale-dependent effective fractal dimension and $R_{\text{eff}}(s)$ is a renormalized curvature functional encoding multiscale geometric compression effects. The coupling constant κ governs the strength of curvature-induced attenuation of observable amplitude across scales.

The effective fractal dimension arises naturally from a logarithmic derivative of measure scaling:

$$d_f(s) = \frac{d \log \mu(B_s(x))}{d \log s}, \quad (2027)$$

where $\mu(B_s(x))$ denotes the invariant measure of geodesic balls of radius s on $(\mathcal{M}, g_{ij})$. In the presence of curvature fluctuations and entropy gradients, this dimension becomes dynamically renormalized according to

$$d_f(s, t) = d_0 - \alpha \langle R(s, t) \rangle + \beta |\nabla S(s, t)|, \quad (2028)$$

establishing a direct coupling between geometric curvature, informational entropy production, and dimensional collapse.

The curvature-modulated observable can equivalently be written in exponential transport form:

$$\mathcal{O}(s) = \mathcal{O}_0 \exp(d_f(s) \log s - \kappa R_{\text{eff}}(s)), \quad (2029)$$

highlighting the competition between scale-invariant growth and curvature-driven suppression mechanisms.

The effective curvature $R_{\text{eff}}(s)$ is defined via a multiscale averaging procedure over geodesic shells:

$$R_{\text{eff}}(s) = \frac{1}{\text{Vol}(B_s)} \int_{B_s(x)} R(y) dV_g(y), \quad (2030)$$

which induces a renormalization-group flow equation of the form

$$\frac{dR_{\text{eff}}}{d \log s} = \mathcal{R}_1 R_{\text{eff}} + \mathcal{R}_2 R_{\text{eff}}^2 + \mathcal{R}_3 S_{\text{eff}}, \quad (2031)$$

where $\mathcal{R}_i$ are universal flow coefficients determined by the underlying PFIT coupling structure.

In the asymptotic flat-information limit $R_{\text{eff}} \rightarrow 0$, the scaling law reduces to a pure fractal power law:

$$\mathcal{O}(s) \sim s^{d_0}, \quad (2032)$$

recovering classical self-similar scaling behavior in the absence of curvature–entropy feedback.

Conversely, in the strong-curvature regime, observables undergo exponential suppression governed by geometric compression:

$$R_{\text{eff}}(s) \rightarrow \infty \implies \mathcal{O}(s) \rightarrow 0, \quad (2033)$$

signaling dimensional collapse and effective loss of accessible degrees of freedom.

This scaling law is directly testable across heterogeneous systems, including physiological time series, networked dynamical systems, and cosmological structure formation, where it predicts a unified curvature-modulated deviation from classical power-law behavior [10–12, 25, 82].

25.4 Entropy Collapse Threshold

A key falsifiable criterion within the PFIT–En-topia–Pellis framework is the entropy collapse threshold condition, formulated as a curvature-constrained dissipation inequality on the informational manifold $(\mathcal{M}, g_{ij})$. In this setting, entropy is no longer a passive thermodynamic quantity but a dynamically coupled functional responding to geometric stress induced by curvature gradients.

The fundamental entropy dissipation law is expressed as

$$\frac{d}{dt} S[\rho(t)] = -\mathcal{D}_S[\rho] - \kappa g^{ij} \nabla_i R \nabla_j R, \quad (2034)$$

where $\mathcal{D}_S[\rho] \geq 0$ denotes intrinsic entropy production dissipation and the curvature-gradient term encodes irreversible geometric compression of the information manifold.

Under this formulation, the entropy collapse threshold condition becomes a strict inequality defining a transition into a non-reversible informational regime:

$$\frac{d}{dt} S[\rho(t)] \leq -\kappa \|\nabla_g R\|_{L^2(\mathcal{M})}^2, \quad (2035)$$

where $\|\nabla_g R\|_{L^2(\mathcal{M})}^2 = \int_{\mathcal{M}} g^{ij} \nabla_i R \nabla_j R dV_g$ measures the total curvature-gradient energy. This establishes curvature gradients as a direct driver of entropy suppression and effective phase-space contraction.

The inequality implies a monotonic decay regime for the entropy functional, which can be rewritten in integral form as

$$S[\rho(t_c)] - S[\rho(0)] = - \int_0^{t_c} (\mathcal{D}_S[\rho(t)] + \kappa \|\nabla_g R(t)\|^2) dt, \quad (2036)$$

making explicit that entropy loss accumulates through both diffusive and curvature-driven channels.

A central prediction of the theory is that entropy collapse occurs in finite time under curvature-dominated regimes, characterized by the convergence condition

$$\int_0^{t_c} \left| \frac{dS}{dt} \right| dt < \infty, \quad (2037)$$

which defines a finite informational action bound analogous to energy concentration in geometric blow-up phenomena.

Equivalently, entropy collapse can be reformulated as bounded total entropy variation under diverging curvature intensity:

$$\sup_{t \rightarrow t_c} \|\nabla_g R(t)\| \rightarrow \infty \implies S[\rho(t)] \text{ remains integrable but non-conservative.} \quad (2038)$$

In spectral representation, this threshold corresponds to irreversible damping of high-frequency informational modes:

$$a_k(t) \sim \exp\left(- \int_0^t \Lambda_k(\tau) d\tau\right), \quad \Lambda_k \rightarrow \infty \text{ as } t \rightarrow t_c, \quad (2039)$$

leading to collapse of effective dimensionality and suppression of multiscale structure.

Thus, the entropy collapse threshold constitutes a unified geometric–spectral criterion for irreversibility, in which curvature-gradient amplification enforces finite-time informational contraction and loss of accessible degrees of freedom [18–20].

25.5 Multiscale Predictive Biomarkers

Within the PFIT–En-topia–Pellis framework, multiscale predictive biomarkers emerge as renormalization-invariant observables derived from the joint entropy–curvature–spectrum coupling structure of the underlying information manifold $(\mathcal{M}, g_{ij})$. In this setting, early-warning signals of collapse are encoded in the relative amplification between spectral instability and geometric curvature fluctuations.

The universal precursor index is defined as a normalized instability functional:

$$\mathcal{P}(t) = \frac{\mathbb{E}[|\lambda_{\max}(t)|]}{\text{Var}_{\mathcal{M}}(R(t)) + \epsilon}, \quad (2040)$$

where $\lambda_{\max}(t)$ denotes the principal eigenvalue of the collapse operator

$$\mathcal{C} = -\Delta_g + \lambda R(x, t) + \gamma \rho(x, t), \quad (2041)$$

and $\text{Var}_{\mathcal{M}}(R(t))$ is the spatial variance of the scalar curvature field over the manifold measure dV_g . The regularization parameter $\epsilon > 0$ ensures boundedness away from trivial singular normalization regimes.

A more refined spectral formulation expresses the numerator as a spectral radius functional:

$$\mathbb{E}[|\lambda_{\max}(t)|] = \sup \sigma(\mathcal{C}(t)), \quad (2042)$$

where $\sigma(\mathcal{C})$ denotes the spectrum of the collapse operator in $L^2(\mathcal{M}, dV_g)$, thereby linking biomarker growth directly to operator instability.

The curvature variance term can be expanded in geometric form as

$$\text{Var}_{\mathcal{M}}(R) = \int_{\mathcal{M}} R^2 dV_g - \left(\int_{\mathcal{M}} R dV_g \right)^2, \quad (2043)$$

highlighting that biomarker suppression or amplification is governed by second-order curvature fluctuations rather than mean-field geometry alone.

In multiscale systems, $\mathcal{P}(t)$ admits a renormalization-group scaling decomposition:

$$\mathcal{P}(t) \sim \sum_{s \in \mathcal{S}} s^{\delta_s} \frac{\lambda_{\max}(s, t)}{\text{Var}(R(s, t)) + \epsilon}, \quad (2044)$$

where $\mathcal{S}$ denotes the set of observational scales and δ_s encodes scale-dependent intermittency corrections arising from fractal geometry of the underlying informational manifold.

Collapse onset is characterized by a universal divergence criterion:

$$\lim_{t \rightarrow t_c} \mathcal{P}(t) = \infty, \quad (2045)$$

which corresponds to a regime in which spectral instability overwhelms curvature regularization, producing loss of stability in the effective information geometry.

Equivalently, the precursor divergence condition can be reformulated as a coupled instability cascade:

$$|\lambda_{\max}(t)| \uparrow \infty \quad \wedge \quad \text{Var}(R(t)) \downarrow 0 \quad \implies \quad \text{critical transition}, \quad (2046)$$

indicating simultaneous spectral explosion and curvature homogenization preceding collapse.

This establishes $\mathcal{P}(t)$ as a universal early-warning observable across heterogeneous domains, including physiological dynamics, material fracture systems, and complex networked structures, where divergence of the spectral-to-geometric ratio reliably signals proximity to a curvature-driven informational singularity [103–108, 114–117].

25.6 Cross-Domain Experimental Strategy

Validation of the PFIT–En-topia–Pellis framework proceeds through a tripartite experimental architecture in which spectral, entropic, and geometric observables are reconstructed independently and then tested for dynamical consistency on the same underlying information manifold $(\mathcal{M}, g_{ij})$. Each experimental channel probes a different projection of the same latent curvature–entropy–spectrum state space.

The first class of experiments targets the spectral dynamics of the collapse operator

$$\mathcal{C} = -\Delta_g + \lambda R(x, t) + \gamma \rho(x, t), \quad (2047)$$

through empirical estimation of its time-dependent eigenvalue flow:

$$\mathcal{C}\phi_k = \Lambda_k(t)\phi_k, \quad \{\Lambda_k(t)\} \rightarrow \text{measured spectral trajectory}. \quad (2048)$$

Spectral diagnostics include eigenvalue spacing statistics, spectral radius evolution, and gap-closure detection:

$$\Delta\Lambda_k(t) = \Lambda_{k+1}(t) - \Lambda_k(t), \quad \Delta\Lambda(t) \rightarrow 0 \text{ as } t \rightarrow t_c. \quad (2049)$$

The second experimental class reconstructs entropy trajectories from observed density fields $\rho(x, t)$ via

$$S(t) = - \int_{\mathcal{M}} \rho(x, t) \log \rho(x, t) dV_g, \quad (2050)$$

together with its variational flow representation

$$\frac{dS}{dt} = \int_{\mathcal{M}} (-\partial_t \rho \cdot (1 + \log \rho)) dV_g, \quad (2051)$$

allowing empirical estimation of entropy production rates and irreversible dissipation channels.

The third experimental class reconstructs curvature fields from information-geometric estimators, either via Fisher–Rao embedding or discrete Ricci-type approximations:

$$g_{ij} = \mathbb{E}[\partial_i \log \rho \partial_j \log \rho], \quad R(x, t) = \mathcal{R}[g_{ij}(x, t)]. \quad (2052)$$

In discrete implementations, curvature can be approximated using graph-based Ricci surrogates:

$$R_i \sim \sum_{j \in \mathcal{N}(i)} \left(1 - \frac{W_{ij}}{\mu_i} \right), \quad (2053)$$

ensuring compatibility with computational graph realizations of the PFIT manifold.

The fundamental consistency requirement of the theory is that all three experimental channels converge to a single coherent collapse signature. This is encoded in the cross-domain residual functional

$$\mathcal{F}_{\text{exp}} \equiv \mathcal{C}_{\text{spectral}} - \mathcal{C}_{\text{entropy}} - \mathcal{C}_{\text{geometry}}, \quad (2054)$$

where each term represents a normalized collapse intensity extracted from its respective observational domain.

The PFIT hypothesis predicts that $\mathcal{F}_{\text{exp}}$ remains bounded in all pre-critical regimes and vanishes asymptotically at the collapse threshold:

$$\mathcal{F}_{\text{exp}}(t) \rightarrow 0 \quad \text{as } t \rightarrow t_c, \quad (2055)$$

establishing a three-way equivalence principle between spectral instability, entropy production, and curvature blow-up.

More generally, the joint convergence condition can be expressed as a synchronization constraint in the space of observables:

$$\mathcal{C}_{\text{spectral}}(t) \sim \mathcal{C}_{\text{entropy}}(t) \sim \mathcal{C}_{\text{geometry}}(t), \quad (2056)$$

which defines a universal manifold of admissible critical transitions across heterogeneous systems.

This establishes cross-domain experimental validation as a geometric equivalence test, in which collapse is confirmed only when independent spectral, entropic, and geometric reconstructions coalesce into a single singular trajectory within the PFIT information manifold framework.

25.7 Falsifiability Criterion

A defining requirement of the PFIT–En-topia–Pellis framework is that it admits a strict and operationally testable falsifiability structure grounded in geometric, spectral, and scaling invariants of the underlying information manifold $(\mathcal{M}, g_{ij})$. In this setting, the theory does not merely describe collapse phenomena but constrains the class of admissible dynamical systems via necessary conditions for curvature–entropy–spectrum consistency.

The framework is falsified if any of the following independent invariance-breaking conditions is observed in empirical or simulated systems:

- (i) absence of curvature singularity formation at the predicted collapse time;

$$\exists t_c < \infty : \sup_{x \in \mathcal{M}} |R(x, t_c)| < \infty, \quad (2057)$$

indicating failure of geometric blow-up and breakdown of curvature-driven instability.

- (ii) violation of fractal scaling invariance across observational resolutions;

$$\mathcal{O}(s) \not\sim s^{d_f(s)} \exp(-\kappa R_{\text{eff}}(s)), \quad (2058)$$

implying loss of scale covariance and absence of renormalization-consistent fractal structure.

(iii) persistence of bounded spectral rigidity across regimes predicted to undergo collapse, i.e. absence of spectral degeneration in the collapse operator $\mathcal{C}$:

$$\Delta\Lambda_k(t) \not\rightarrow 0 \quad \text{and} \quad \text{Var}(\Delta\Lambda_k(t)) < \infty, \quad (2059)$$

which contradicts the predicted loss of eigenvalue spacing stability and prevents dimensional collapse.

These conditions can be unified into a single falsification statement expressed as a failure of the joint singularity structure:

$$\exists t_c : \left(\sup_{x \in \mathcal{M}} |R(x, t_c)| < \infty \right) \vee \left(\mathcal{O}(s) \not\sim s^{d_f} e^{-\kappa R_{\text{eff}}} \right) \vee \left(\Delta\Lambda_k \not\rightarrow 0 \right). \quad (2060)$$

Equivalently, the PFIT framework is falsified if and only if the coupled entropy–geometry–spectrum system fails to enter a joint singular regime, i.e. if the collapse intensity functional remains globally bounded:

$$\sup_{t < \infty} \mathcal{C}(t) < \infty. \quad (2061)$$

Thus, falsifiability is encoded as the non-existence of a finite-time curvature singularity accompanied by simultaneous breakdown of fractal scaling and spectral rigidity, establishing a strict triple-constraint validation boundary for all admissible PFIT dynamics.

25.8 Concluding Validation Statement

The PFIT–En-topia–Pellis framework establishes that collapse phenomena across heterogeneous physical, biological, and engineered systems are not domain-specific instabilities, but rather manifestations of a single underlying geometric–informational phase transition governed by the coupled evolution of entropy, curvature, and spectral structure on a fractal information manifold $(\mathcal{M}, g_{ij})$ [82, 101–108, 114–117].

In its most compact form, this universality is expressed as the emergence of a single criticality class defined by the joint divergence of the collapse functional:

$$\mathcal{C}(t) = |\nabla S(t)| + |R(t)| + \|\Lambda(t)\|, \quad (2062)$$

where entropy gradients, curvature invariants, and spectral operator norms co-evolve under a unified information-geometric flow.

The theory predicts that all admissible systems approaching criticality asymptotically satisfy the equivalence relation

$$\mathcal{C}_{\text{spectral}}(t) \sim \mathcal{C}_{\text{entropy}}(t) \sim \mathcal{C}_{\text{geometry}}(t), \quad \text{as } t \rightarrow t_c, \quad (2063)$$

indicating that collapse is characterized by synchronization of otherwise independent diagnostic channels into a single singular trajectory in the PFIT state space.

At the transition point, the system undergoes a finite-time degeneration of informational degrees of freedom, captured by the universal limiting behavior

$$\lim_{t \rightarrow t_c} \sup_{x \in \mathcal{M}} \left(|R(x, t)| + |\nabla S(x, t)| \right) = \infty, \quad (2064)$$

together with spectral condensation of the governing operator spectrum.

26 Entropy–Spectral Collapse Prediction Law

In this subsection, we formulate a set of explicit, measurable and falsifiable prediction laws that connect entropy dynamics, spectral instability, and curvature divergence within the PFIT framework. These laws elevate the theory from a purely geometric–informational construction to a predictive physical model with experimentally accessible observables.

26.1 Entropy Collapse Indicator Functional

Within the PFIT–En-topia–Pellis framework, the onset of collapse is characterized by a composite scalar diagnostic that couples entropy production, spectral instability, and geometric curvature amplification into a single renormalization-sensitive observable defined over the information manifold $(\mathcal{M}, g_{ij})$. This construction yields a universal early-warning functional whose divergence signals the breakdown of multiscale informational coherence.

We define the **entropy collapse indicator functional** as

$$\mathcal{I}_{\text{collapse}}(t) = \left| \frac{d}{dt} H_{\text{PFIT}}[\rho(t)] \right| + \lambda_1 \text{Var}_{\mathcal{M}}(\Lambda_k(t)) + \lambda_2 \sup_{x \in \mathcal{M}} |R(x, t)|. \quad (2065)$$

Here $H_{\text{PFIT}}[\rho]$ denotes the generalized entropy functional incorporating fractal corrections to Shannon–Boltzmann structure:

$$H_{\text{PFIT}}[\rho] = - \int_{\mathcal{M}} \rho(x, t) \log \rho(x, t) dV_g + \eta \int_{\mathcal{M}} |\nabla \log \rho|^2 dV_g, \quad (2066)$$

where the second term encodes information-geometric regularization consistent with Fisher–Rao embedding.

The entropy production rate admits the variational expansion

$$\frac{d}{dt} H_{\text{PFIT}}[\rho(t)] = \int_{\mathcal{M}} (-\partial_t \rho (1 + \log \rho) + \eta \partial_t |\nabla \log \rho|^2) dV_g, \quad (2067)$$

linking entropy flow directly to curvature-modulated diffusion dynamics.

The spectral variance term is defined over the eigenvalue spectrum of the collapse operator

$$\mathcal{C} = -\Delta_g + \alpha R(x, t) + \gamma \rho(x, t), \quad (2068)$$

with eigen-decomposition

$$\mathcal{C} \phi_k = \Lambda_k(t) \phi_k, \quad (2069)$$

and spectral dispersion functional

$$\text{Var}_{\mathcal{M}}(\Lambda_k) = \frac{1}{N} \sum_{k=1}^N (\Lambda_k - \bar{\Lambda})^2, \quad \bar{\Lambda} = \frac{1}{N} \sum_k \Lambda_k. \quad (2070)$$

The curvature amplification term is expressed through the geometric blow-up functional

$$\sup_{x \in \mathcal{M}} |R(x, t)| = \lim_{p \rightarrow \infty} \left(\int_{\mathcal{M}} |R(x, t)|^p dV_g \right)^{1/p}, \quad (2071)$$

highlighting the dominance of extreme curvature fluctuations near criticality.

In multiscale form, the collapse indicator admits a renormalized decomposition across resolution scales ℓ :

$$\mathcal{I}_{\text{collapse}}(t) \sim \int_{\ell_{\min}}^{\ell_{\max}} \left[\left| \frac{dH_{\ell}}{dt} \right| + \lambda_1(\ell) \text{Var}(\Lambda_{\ell}) + \lambda_2(\ell) |R_{\ell}| \right] d\ell, \quad (2072)$$

where each scale contributes a partial instability channel to the global collapse dynamics.

The universal prediction of the PFIT framework is that the system undergoes a finite-time singular transition when all contributing instability channels simultaneously diverge:

$$\lim_{t \rightarrow t_c} \mathcal{I}_{\text{collapse}}(t) = \infty, \quad (2073)$$

signaling a synchronized breakdown of entropy regularity, spectral stability, and geometric boundedness.

Equivalently, collapse occurs when the coupled constraint system

$$\left| \frac{d}{dt} H_{\text{PFIT}} \right| \sim \text{Var}(\Lambda_k) \sim \sup |R| \quad (2074)$$

enters a co-divergent regime, identifying a universal informational singularity class in which curvature-driven dynamics dominate all lower-order diffusive and stochastic effects.

26.2 Spectral Blow-Up Criterion

Within the PFIT–En-topia–Pellis framework, spectral instability is not interpreted as a purely algebraic divergence of eigenvalues, but as a geometric amplification of information transport modes induced by curvature–entropy coupling on the underlying manifold $(\mathcal{M}, g_{ij})$. In this setting, the entropy spectrum $\{\Lambda_k(t)\}$ associated with the effective information Hamiltonian $\mathcal{H}_{\text{PFIT}}$ encodes multiscale coherence of the system across fractal resolution levels.

We consider the spectral asymptotics of the operator eigenvalues defined by

$$\mathcal{H}_{\text{PFIT}} \phi_k = \Lambda_k(t) \phi_k, \quad \phi_k \in L^2(\mathcal{M}, dV_g), \quad (2075)$$

where $\mathcal{H}_{\text{PFIT}}$ is assumed to be a curvature-modulated entropy generator of the form

$$\mathcal{H}_{\text{PFIT}} = -\Delta_g + \alpha R(x, t) + \beta \log \rho(x, t) + \gamma |\nabla \log \rho|^2. \quad (2076)$$

The system undergoes a spectral instability transition if the eigenvalue growth law enters a super-polynomial divergence regime:

$$\lim_{k \rightarrow \infty} |\Lambda_k(t)| \sim k^{\delta(t)}, \quad \delta(t) \uparrow \infty, \quad (2077)$$

indicating a breakdown of Weyl-type spectral regularity and emergence of fractal eigenvalue scaling in the high-frequency limit.

More generally, spectral blow-up is characterized by the failure of uniform operator boundedness in the associated functional space:

$$\sup_{k \in \mathbb{N}} |\Lambda_k(t)| \rightarrow \infty, \quad (2078)$$

which is equivalent to divergence of the operator norm

$$\|\mathcal{H}_{\text{PFIT}}(t)\|_{op} \rightarrow \infty. \quad (2079)$$

In the pre-collapse regime, the spectral distribution admits a scaling density representation

$$\rho_\Lambda(\lambda, t) \sim \lambda^{-\mu(t)} \exp(-\kappa S_{\text{eff}}(t)), \quad (2080)$$

where $S_{\text{eff}}(t)$ denotes the effective entropy functional and $\mu(t)$ encodes renormalization-induced spectral drift across scales.

A refined instability diagnostic is given by the spectral moment hierarchy

$$M_n(t) = \sum_k |\Lambda_k(t)|^n, \quad (2081)$$

with collapse onset corresponding to divergence of all sufficiently high-order moments:

$$\exists n^* \in \mathbb{N} : \lim_{t \rightarrow t_c} M_{n^*}(t) = \infty. \quad (2082)$$

Equivalently, spectral blow-up can be formulated as a loss of zeta-regularized trace stability:

$$\text{Tr}_\zeta(\mathcal{H}_{\text{PFIT}}(t)) = \sum_k \Lambda_k(t) e^{-\epsilon \Lambda_k(t)}, \quad \epsilon \rightarrow 0^+, \quad (2083)$$

where divergence of the analytic continuation signals non-removable spectral singularity formation.

Thus, the PFIT spectral blow-up criterion establishes that collapse is triggered when the eigenstructure of the entropy Hamiltonian transitions from bounded Weyl-class scaling to unbounded fractal spectral inflation, marking the onset of curvature-dominated informational singularities in the high-frequency regime.

26.3 Spectral Blow-Up Criterion

Within the PFIT–En-topia–Pellis framework, spectral instability is not interpreted as a purely algebraic divergence of eigenvalues, but as a geometric amplification of information transport modes induced by curvature–entropy coupling on the underlying manifold $(\mathcal{M}, g_{ij})$. In this setting, the entropy spectrum $\{\Lambda_k(t)\}$ associated with the effective information Hamiltonian $\mathcal{H}_{\text{PFIT}}$ encodes multiscale coherence, renormalization flow, and curvature-mediated mode interaction across fractal resolution levels.

We consider the spectral problem generated by the nonlinear curvature–entropy operator

$$\mathcal{H}_{\text{PFIT}} \phi_k = \Lambda_k(t) \phi_k, \quad \phi_k \in L^2(\mathcal{M}, dV_g), \quad (2084)$$

with

$$\mathcal{H}_{\text{PFIT}} = -\Delta_g + \alpha R(x, t) + \beta \log \rho(x, t) + \gamma |\nabla \log \rho|^2 + \eta \nabla_i (\rho \nabla^i \log \rho), \quad (2085)$$

so that the spectral geometry is explicitly coupled to both curvature flow and Fisher-type informational compression.

The induced eigenvalue flow satisfies a nonlinear spectral evolution equation of the form

$$\frac{d\Lambda_k}{dt} = \langle \phi_k, \partial_t \mathcal{H}_{\text{PFIT}} \phi_k \rangle + \sum_{j \neq k} \frac{|\langle \phi_j, \partial_t \mathcal{H}_{\text{PFIT}} \phi_k \rangle|^2}{\Lambda_k - \Lambda_j}, \quad (2086)$$

which exhibits resonance amplification when spectral gaps collapse.

The system undergoes a spectral instability transition if the eigenvalue asymptotics enter a super-polynomial growth regime governed by renormalization-induced exponent drift:

$$\lim_{k \rightarrow \infty} |\Lambda_k(t)| \sim k^{\delta(t)}, \quad \delta(t) \uparrow \infty, \quad (2087)$$

indicating breakdown of Weyl-type spectral laws and emergence of fractal spectral measure concentration.

Equivalently, spectral blow-up corresponds to loss of uniform boundedness of the generator in operator topology:

$$\sup_{k \in \mathbb{N}} |\Lambda_k(t)| \rightarrow \infty, \quad \Longleftrightarrow \quad \|\mathcal{H}_{\text{PFIT}}(t)\|_{\text{op}} \rightarrow \infty. \quad (2088)$$

In the pre-collapse regime, the spectral measure admits a renormalized density representation with entropy suppression factor:

$$\rho_\Lambda(\lambda, t) \sim \lambda^{-\mu(t)} \exp(-\kappa S_{\text{eff}}(t)) (1 + \xi(\lambda, t)), \quad (2089)$$

where $\xi(\lambda, t)$ encodes higher-order fractal corrections to spectral universality and $S_{\text{eff}}(t)$ is the effective multiscale entropy functional.

A hierarchical instability structure is captured by spectral moment tensors

$$M_n(t) = \sum_k |\Lambda_k(t)|^n, \quad \mathcal{M}(z, t) = \sum_k |\Lambda_k(t)|^z, \quad (2090)$$

with analytic continuation defining a zeta-type spectral generating function whose pole structure encodes hidden instability modes.

Collapse onset corresponds to divergence of high-order spectral moments:

$$\exists n^* \in \mathbb{N} \quad \text{s.t.} \quad \lim_{t \rightarrow t_c} M_{n^*}(t) = \infty, \quad (2091)$$

indicating non-existence of finite spectral energy closure under renormalization flow.

Equivalently, spectral blow-up is characterized by instability of the zeta-regularized trace:

$$\text{Tr}_\zeta(\mathcal{H}_{\text{PFIT}}(t)) = \lim_{\epsilon \rightarrow 0^+} \sum_k \Lambda_k(t) e^{-\epsilon \Lambda_k(t)}, \quad (2092)$$

where divergence of the analytic continuation signals breakdown of spectral regularity and emergence of non-removable operator singularities.

Finally, collapse is realized when spectral geometry transitions from a bounded renormalization-class regime to a fractal inflation phase characterized by unbounded eigenvalue density growth:

$$\rho_\Lambda(\lambda, t) \notin L^\infty(\mathbb{R}^+) \quad \text{and} \quad \int_0^\infty \rho_\Lambda(\lambda, t) d\lambda \rightarrow \infty, \quad (2093)$$

marking the onset of curvature-dominated informational singularities and completion of the PFIT spectral collapse cascade.

26.4 Time-to-Collapse Scaling Law

Within the PFIT–En-topia–Pellis framework, finite-time collapse is interpreted as a critical slowing-down inversion in which curvature amplification replaces diffusive regularization as the dominant dynamical mechanism. In this regime, the system approaches a geometric singularity at a finite critical time t_c , characterized by universal scaling relations between curvature, entropy production, and spectral instability.

Let t_c denote the finite-time collapse singularity. Then the near-critical dynamics obey an inverse curvature scaling law of the form

$$t_c - t \sim (\mathcal{R}_{\text{max}}(t))^{-\mu}, \quad \mu > 0, \quad (2094)$$

where

$$\mathcal{R}_{\text{max}}(t) = \sup_{x \in \mathcal{M}} |R(x, t)| \quad (2095)$$

is the maximal scalar curvature amplitude over the information manifold. This relation expresses collapse as a curvature-controlled finite-time blow-up process, where geometric concentration dictates the temporal approach to singularity.

Equivalently, in renormalized entropy-flow representation, the PFIT entropy functional satisfies a universal divergence law:

$$\frac{dH_{\text{PFIT}}}{dt} = \int_{\mathcal{M}} \partial_t \rho(x, t) (1 + \log \rho(x, t)) dV_g, \quad (2096)$$

with asymptotic scaling behavior

$$\frac{dH_{\text{PFIT}}}{dt} \sim (t_c - t)^{-\nu}, \quad \nu > 0, \quad (2097)$$

indicating that entropy production accelerates without bound as the system approaches the collapse singularity.

A more refined geometric–spectral coupling formulation expresses the curvature-driven time collapse as a feedback loop between entropy gradients and Ricci amplification:

$$\partial_t R \sim \Delta_g R + \alpha |\nabla \log \rho|^2 + \beta R^2, \quad (2098)$$

which yields finite-time blow-up solutions under positive feedback dominance (βR^2 term).

In this regime, the entropy production rate and curvature growth become asymptotically locked:

$$\frac{dH_{\text{PFIT}}}{dt} \sim \mathcal{R}_{\text{max}}(t)^{\frac{\nu}{\mu}}, \quad (2099)$$

establishing a direct scaling equivalence between informational dissipation and geometric singularity formation.

From a spectral perspective, the collapse time is also encoded in the divergence of the operator norm of the collapse generator:

$$\|\mathcal{C}(t)\|_{op} \sim (t_c - t)^{-\kappa}, \quad \kappa > 0, \quad (2100)$$

ensuring consistency between curvature blow-up, entropy acceleration, and spectral instability.

Thus, the PFIT time-to-collapse law establishes a universal finite-time singularity class in which all admissible observables obey coupled inverse power-law scaling toward t_c , confirming that collapse is a synchronized multiscale geometric phase transition rather than an isolated dynamical instability.

26.5 Unified Falsifiability Statement

Within the PFIT–En-topia–Pellis framework, falsifiability is formulated as a global consistency constraint over the coupled entropy–curvature–spectral dynamics on the information manifold $(\mathcal{M}, g_{ij})$. In this setting, the collapse theory is not merely descriptive but imposes a strict logical structure on admissible dynamical trajectories, such that the absence of a joint singularity regime constitutes a direct contradiction of the model.

The theory is experimentally falsifiable via the boundedness condition of the entropy collapse indicator functional:

$$\mathcal{I}_{\text{collapse}}(t) < \infty \quad \forall t \in [0, \infty) \quad \Rightarrow \quad \neg \exists t_c \text{ such that spectral collapse occurs.} \quad (2101)$$

This expresses that, in the absence of divergence in the coupled instability functional, no finite-time informational singularity can emerge, thereby excluding the PFIT collapse phase entirely.

Equivalently, falsification corresponds to the uniform boundedness of all three principal instability channels:

$$\sup_{t < \infty} \left(\left| \frac{dH_{\text{PFIT}}}{dt} \right| + \|\Lambda(t)\|_{op} + \sup_{x \in \mathcal{M}} |R(x, t)| \right) < \infty, \quad (2102)$$

which implies global regularity of entropy flow, spectral stability of the collapse operator, and bounded curvature evolution.

Conversely, empirical validation of the PFIT collapse mechanism requires the simultaneous observation of a synchronized divergence cascade across all coupled observables, namely:

- spectral divergence of the eigenvalue spectrum $\{\Lambda_k(t)\}$ of the collapse operator,
- curvature blow-up in the information manifold geometry,
- entropy production divergence in the PFIT functional $H_{\text{PFIT}}[\rho(t)]$.

Mathematically, validation corresponds to the joint limit

$$\lim_{t \rightarrow t_c} \left(\sup_k |\Lambda_k(t)| + \sup_{x \in \mathcal{M}} |R(x, t)| + \left| \frac{dH_{\text{PFIT}}}{dt} \right| \right) = \infty, \quad (2103)$$

together with the co-divergence condition

$$\mathcal{C}_{\text{spectral}}(t) \sim \mathcal{C}_{\text{geometry}}(t) \sim \mathcal{C}_{\text{entropy}}(t), \quad (2104)$$

which defines a universal equivalence class of critical collapse trajectories.

Thus, PFIT is experimentally falsifiable in a strict sense: it is either invalidated by uniform boundedness of all instability channels across all scales, or confirmed by the emergence of a synchronized finite-time singularity across spectral, geometric, and entropic observables, uniquely identifying curvature-driven informational collapse as a universal phase transition phenomenon.

26.6 Unified Falsifiability Statement

Within the PFIT–En-topia–Pellis framework, falsifiability is elevated to a global structural constraint on the admissible evolution of coupled entropy–curvature–spectral dynamics defined over the information manifold $(\mathcal{M}, g_{ij})$. In this formulation, the theory is not merely descriptive but imposes a rigid geometric consistency condition: the existence of a finite-time collapse regime is equivalent to the inevitability of a joint singularity across all informational observables. Consequently, the absence of such a regime constitutes a direct contradiction of the underlying geometric axioms.

The fundamental falsifiability criterion is encoded through the boundedness of the entropy collapse indicator functional:

$$\mathcal{I}_{\text{collapse}}(t) < \infty \quad \forall t \in [0, \infty) \quad \Rightarrow \quad \neg \exists t_c \in \mathbb{R}^+ \text{ such that spectral-curvature collapse occurs.} \quad (2105)$$

This statement formalizes that global regularity of the coupled informational flow excludes the existence of any finite-time singular attractor in the PFIT dynamical system.

Equivalently, falsification is characterized by uniform boundedness of all principal instability channels on the manifold:

$$\sup_{t < \infty} \left(\left| \frac{dH_{\text{PFIT}}}{dt} \right| + \|\Lambda(t)\|_{op} + \sup_{x \in \mathcal{M}} |R(x, t)| \right) < \infty. \quad (2106)$$

This condition implies the simultaneous absence of entropy blow-up, spectral divergence, and curvature singularity formation, thereby enforcing global stability of the informational geometry.

Conversely, empirical validation of the PFIT collapse hypothesis requires the observation of a synchronized divergence cascade across all coupled diagnostic channels. Specifically, one must detect:

- unbounded growth of the spectral spectrum $\{\Lambda_k(t)\}$ associated with the collapse operator,
- curvature blow-up in the scalar geometric invariant $R(x, t)$,
- divergence of entropy production in the PFIT functional $H_{\text{PFIT}}[\rho(t)]$.

Mathematically, this corresponds to the joint singular limit:

$$\lim_{t \rightarrow t_c} \left(\sup_k |\Lambda_k(t)| + \sup_{x \in \mathcal{M}} |R(x, t)| + \left| \frac{dH_{\text{PFIT}}}{dt} \right| \right) = \infty, \quad (2107)$$

together with the asymptotic co-divergence relation:

$$\mathcal{C}_{\text{spectral}}(t) \sim \mathcal{C}_{\text{geometry}}(t) \sim \mathcal{C}_{\text{entropy}}(t), \quad (2108)$$

which defines a universal equivalence class of collapse trajectories characterized by synchronized instability across all informational degrees of freedom.

Thus, the PFIT framework is strictly falsifiable in a dual sense: it is invalidated by the persistent boundedness of all instability measures across scales and domains, or it is confirmed by the emergence of a finite-time, cross-channel singularity in which spectral, geometric, and entropic divergences coalesce into a single curvature-driven informational phase transition.

27 Limitations and Model Assumptions

This section rigorously formalizes the structural assumptions, asymptotic regimes, and mathematical idealizations underlying the PFIT–En-topia–Pellis framework. Although the theory is constructed as a unified geometric–informational description of collapse phenomena across scales, its validity is intrinsically conditioned on a hierarchy of smoothness, integrability, spectral regularity, and renormalization constraints consistent with modern entropy geometry, stochastic analysis, and fractal information manifolds [1–3, 7–9, 28, 82]. These constraints define the admissible class of systems for which curvature-driven collapse remains mathematically well-posed.

27.1 Manifold Regularity and Geometric Assumptions

The framework assumes that the information space $(\mathcal{M}, g_{ij})$ is a complete, orientable Riemannian manifold of class C^∞ equipped with a Levi-Civita connection ∇ satisfying metric compatibility and torsion-free conditions:

$$\nabla_k g_{ij} = 0, \quad \Gamma_{ij}^k = \Gamma_{ji}^k. \quad (2109)$$

The curvature tensor is well-defined via

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{im}^l \Gamma_{jk}^m - \Gamma_{jm}^l \Gamma_{ik}^m, \quad (2110)$$

and the scalar curvature

$$R = g^{ij} R_{ij} \quad (2111)$$

is assumed to be locally integrable in $L^1_{\text{loc}}(\mathcal{M})$ under the induced volume form $dV_g = \sqrt{|g|} d^n x$ [7–9].

27.2 Entropy Functional Well-Posedness

The Shannon-type entropy functional is assumed to be finite and variationally well-defined:

$$S[\rho] = - \int_{\mathcal{M}} \rho \log \rho dV_g, \quad \rho \geq 0, \quad \int_{\mathcal{M}} \rho dV_g = 1. \quad (2112)$$

The admissible function space is typically restricted to Sobolev-type classes

$$\rho \in H^1(\mathcal{M}) \cap L^1(\mathcal{M}), \quad (2113)$$

ensuring existence of weak gradients and finite Fisher information:

$$\mathcal{I}_F[\rho] = \int_{\mathcal{M}} \frac{|\nabla \rho|^2}{\rho} dV_g < \infty. \quad (2114)$$

Heavy-tailed or non-normalizable distributions violate this constraint unless regularized via renormalization schemes consistent with fractal entropy extension procedures [28, 82].

27.3 Fractal Closure and Scale Invariance Assumptions

A central structural assumption is the existence of a scaling regime characterized by approximate self-similarity:

$$\rho(\lambda x) \sim \lambda^{-d_f} \rho(x), \quad \lambda \in \mathbb{R}^+, \quad (2115)$$

where d_f denotes the effective fractal dimension induced by multiscale information compression.

This scaling symmetry is assumed to hold only within an intermediate asymptotic window:

$$\ell_{\min} \ll \ell \ll \ell_{\max}, \quad (2116)$$

outside of which ultraviolet (UV) and infrared (IR) cutoffs break exact scale invariance:

$$\rho(\ell) \sim \begin{cases} \text{regularized regime,} & \ell \leq \ell_{\min}, \\ \text{fractal regime,} & \ell_{\min} \ll \ell \ll \ell_{\max}, \\ \text{saturated regime,} & \ell \geq \ell_{\max}. \end{cases} \quad (2117)$$

This introduces a fundamental limitation on universality in finite-resolution empirical systems [10–12, 25].

27.4 Spectral Operator Domain Constraints

The collapse operator

$$\mathcal{C} = -\Delta_g + \lambda R(x, t) + \gamma \rho(x, t) \quad (2118)$$

is assumed to be essentially self-adjoint on a dense domain $\mathcal{D}(\mathcal{C}) \subset L^2(\mathcal{M})$, ensuring a real-valued spectral decomposition:

$$\mathcal{C}\phi_k = \Lambda_k \phi_k, \quad \phi_k \in L^2(\mathcal{M}). \quad (2119)$$

However, in the curvature-divergent regime,

$$\lim_{t \rightarrow t_c} \|R(\cdot, t)\|_{L^\infty} \rightarrow \infty, \quad (2120)$$

essential self-adjointness may fail, requiring extension to distributional or rigged Hilbert space formulations:

$$\mathcal{C} : \Phi \subset L^2(\mathcal{M}) \subset \Phi', \quad (2121)$$

where Φ denotes a nuclear test function space.

27.5 Stochastic Forcing and Noise Modeling Limitations

The stochastic perturbation field $\xi(x, t)$ is modeled as a mean-zero generalized random field:

$$\mathbb{E}[\xi(x, t)] = 0, \quad (2122)$$

with covariance kernel

$$\mathbb{E}[\xi(x, t)\xi(x', t')] = C(x, x') K(t - t'). \quad (2123)$$

A key limitation is that C and K are not uniquely determined from first principles within the current formulation, and must be inferred from empirical calibration or assumed universality classes of stochastic processes [13–15]. This introduces model-dependence in finite-data regimes.

27.6 Curvature–Entropy Coupling Approximation

The leading-order curvature–entropy coupling is assumed to satisfy:

$$R(x) = \alpha \frac{\delta S}{\delta \rho}(x) + \beta \mathcal{F}_{\text{fractal}}(\rho)(x) + \mathcal{O}\left(\frac{\delta^2 S}{\delta \rho^2}\right). \quad (2124)$$

Higher-order nonlinear corrections generate a hierarchical expansion:

$$R(x) = \sum_{n=1}^{\infty} c_n \left(\frac{\delta^n S}{\delta \rho^n} \right), \quad (2125)$$

which is truncated in practical applications, limiting accuracy in near-critical regimes where higher-order curvature feedback becomes dominant [18–20, 82].

27.7 Universality and Parameter Dependence

Despite its universal formulation, the effective couplings exhibit renormalization-group flow:

$$\theta(\ell) = \theta_0 + \int_{\ell_0}^{\ell} \beta_{\theta}(\ell') d\ell', \quad (2126)$$

where $\theta \in \{\alpha, \beta, \gamma, d_f\}$.

Thus, universality is understood in the sense of fixed-point invariance:

$$\beta_{\theta}(\ell^*) = 0, \quad (2127)$$

rather than strict parameter independence, aligning the framework with critical phenomena theory [16–18].

27.8 Computational and Empirical Constraints

Numerical implementations require discretization of the manifold:

$$\Delta_g \rightarrow \Delta_g^{(h)}, \quad h \rightarrow 0, \quad (2128)$$

introducing mesh-scale corrections:

$$\Delta_g^{(h)} \rho = \Delta_g \rho + \mathcal{O}(h^2). \quad (2129)$$

Curvature reconstruction from sampled data is therefore limited by:

$$R_{\text{obs}}(x) = R_{\text{true}}(x) + \epsilon(h, x), \quad \epsilon(h, x) \not\rightarrow 0 \text{ in noisy regimes.} \quad (2130)$$

27.9 Summary of Limitations

The PFIT–En-topia–Pellis framework is therefore best interpreted as an asymptotic, renormalized theory of informational geometry valid under smoothness, finite entropy, controlled fractal scaling, and spectral self-adjointness assumptions. Its predictive strength emerges in regimes where entropy, curvature, and spectral dynamics remain coherently coupled, while its validity weakens in fully singular, non-measurable, or non-renormalizable informational states [1–3, 28, 82].

28 Future Work

The PFIT–En-topia–Pellis framework establishes a geometric–informational foundation for multiscale collapse dynamics across entropy, spectral, and curvature-driven regimes. Nevertheless, it remains an open, evolving mathematical structure whose deepest properties—particularly its renormalization closure, quantum embedding, and computational realizability—are not yet fully determined. Future developments are expected to refine its axiomatic core, extend its fractal-geometric consistency classes, and establish a unified treatment of stochastic, quantum, and biological information manifolds within a single curvature–entropy duality principle [1–3, 7–9, 28, 82].

28.1 Refinement of the Axiomatic Core

A primary direction concerns the derivation of a minimal and logically independent axiom system generating PFIT dynamics. Let $(\mathcal{M}, g, \rho)$ denote the informational manifold equipped with entropy functional $S[\rho]$. The theory suggests the existence of a generating action functional

$$\mathcal{A}[\rho, g] = S[\rho] + \lambda_1 \int_{\mathcal{M}} R \rho dV_g + \lambda_2 \int_{\mathcal{M}} |\nabla \rho|^2 dV_g + \lambda_3 \int_{\mathcal{M}} \rho \log \rho dV_g, \quad (2131)$$

from which all governing equations emerge via constrained variational principles:

$$\frac{\delta \mathcal{A}}{\delta \rho} = 0, \quad \frac{\delta \mathcal{A}}{\delta g^{ij}} = 0, \quad \nabla_k g_{ij} = 0. \quad (2132)$$

The resulting Euler–Lagrange system defines a coupled geometric flow:

$$\partial_t \rho = -\frac{\delta \mathcal{A}}{\delta \rho}, \quad \partial_t g_{ij} = -\frac{\delta \mathcal{A}}{\delta g^{ij}}. \quad (2133)$$

A central open problem is the classification of admissible axiom sets $\mathfrak{A}$ under fractal renormalization transformations:

$$\mathcal{R}_\ell : (\mathcal{M}, g, \rho) \mapsto (\mathcal{M}_\ell, g_\ell, \rho_\ell), \quad \ell \rightarrow 0, \quad (2134)$$

subject to consistency constraints:

$$\mathcal{R}_\ell \circ \mathcal{R}_{\ell'} = \mathcal{R}_{\ell\ell'}. \quad (2135)$$

28.2 Renormalization and Fractal Fixed Points

A fundamental direction is the identification of renormalization-group fixed points governing informational geometry. One expects the existence of a universal invariant pair (g^*, ρ^*) satisfying

$$\mathcal{R}(g^*) = g^*, \quad \mathcal{R}(\rho^*) = \rho^*. \quad (2136)$$

Near criticality, curvature and entropy are expected to obey a coupled flow in logarithmic scale:

$$\frac{dR}{d \log \ell} = \alpha_1 R + \alpha_2 R^2 + \alpha_3 R_{ij} R^{ij} - \beta S_{\text{eff}}(\ell), \quad (2137)$$

$$\frac{dS_{\text{eff}}}{d \log \ell} = \gamma_1 S_{\text{eff}} - \gamma_2 \mathcal{I}_F[\rho], \quad (2138)$$

defining a nonlinear universality class of informational phase transitions with curvature feedback.

28.3 Quantum and Stochastic Extensions

A major extension is the embedding of PFIT into a quantum–stochastic geometric formalism. Let $\hat{\rho}(t)$ be a density operator on a Hilbert manifold $\mathcal{H}(\mathcal{M})$. The evolution is conjectured to satisfy a curvature-modulated Lindblad-type equation:

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}_g, \hat{\rho}] + \sum_k \left(\hat{L}_k \hat{\rho} \hat{L}_k^\dagger - \frac{1}{2} \{ \hat{L}_k^\dagger \hat{L}_k, \hat{\rho} \} \right) + \mathcal{F}_R(\hat{\rho}), \quad (2139)$$

where $\hat{H}_g$ encodes geometric Hamiltonian flow and $\mathcal{F}_R$ encodes curvature-induced decoherence.

A key conjecture is that curvature singularities induce objective state reduction:

$$\lim_{R(x,t) \rightarrow \infty} \mathcal{F}_R(\hat{\rho}) = \sum_i p_i \Pi_i \hat{\rho} \Pi_i, \quad (2140)$$

linking PFIT collapse directly to measurement-like projection dynamics.

28.4 Higher-Order Fractal Geometry and Multifractal Spectrum

Future extensions of the PFIT framework require the replacement of scalar fractal scaling laws with a full multifractal description, in which scale-dependent heterogeneity is encoded through the spectrum $\tau(q)$ defined via partition functions:

$$Z(q, \epsilon) = \sum_i \mu_i^q \sim \epsilon^{\tau(q)}, \quad \epsilon \rightarrow 0. \quad (2141)$$

In this formulation, the exponent function $\tau(q)$ captures the distribution of scaling exponents across the hierarchy of measure concentrations, thereby providing a complete characterization of intermittency, localization, and multiscale irregularity.

Within the curvature–information correspondence principle, geometric structure is no longer treated as independent of scaling statistics. Instead, curvature is conjectured to arise directly from deviations in multifractal regularity. This leads to a spectral–geometric duality relation of the form:

$$R(x) \sim -\frac{d^2 \tau(q)}{dq^2} \Big|_{q=q^*} + \kappa |\nabla \tau(q^*)|^2, \quad (2142)$$

where the second derivative term measures multifractal convexity (local intermittency concentration), while the gradient term encodes spatial modulation of scaling structure.

In this view, curvature is interpreted as an emergent diagnostic of multifractal instability: regions of high geometric curvature correspond to strong nonlinear variation in scaling exponents, indicating transitions between regimes of coherent structure formation and turbulent informational dispersion.

28.5 Computational Realization and Algorithmic Scaling

On the computational level, PFIT requires stable numerical schemes for nonlinear geometric flows on adaptive fractal graphs $\mathcal{G}_n$. A canonical discretization is given by:

$$\rho_i^{(n+1)} = \rho_i^{(n)} + \Delta t \left(-\Delta_{g_n} \rho_i^{(n)} + \lambda R_i^{(n)} \rho_i^{(n)} + \gamma \rho_i^{(n)} \log \rho_i^{(n)} \right). \quad (2143)$$

Stability requires control of curvature growth in discrete form:

$$\|R^{(n)}\|_\infty < \infty, \quad \sum_n \|R^{(n+1)} - R^{(n)}\| < \infty. \quad (2144)$$

A major challenge is the development of renormalized solvers in regimes where

$$\|R_n\| \rightarrow \infty, \quad (2145)$$

requiring adaptive mesh refinement and operator regularization:

$$\Delta_{g_n} \rightarrow \Delta_{g_n}^{(\epsilon)} + \mathcal{O}(\epsilon^{-1}). \quad (2146)$$

28.6 Experimental Expansion and Cross-Domain Universality

Future empirical validation must extend across multiscale systems including turbulence, neural criticality, plasma instabilities, and cosmological structure formation. A universal observable is expected to take the invariant form:

$$\mathcal{I}(x, t) = \frac{|R(x, t)|^\alpha}{S[\rho(x, t)] + \epsilon} + \beta \mathcal{I}_F[\rho(x, t)], \quad (2147)$$

whose divergence signals proximity to collapse.

Cross-domain universality implies the existence of a single collapse attractor manifold Σ_c such that:

$$\forall \mathcal{S}_i \in \{\text{physical, biological, computational systems}\}, \quad \lim_{t \rightarrow t_c} \mathcal{S}_i(t) \rightarrow \Sigma_c. \quad (2148)$$

28.7 Foundational Open Problems

The most fundamental open problems are summarized as follows:

(i) Existence and uniqueness of informational Einstein–PFIT equations:

$$\text{Ric}(g) + \nabla^2 \log \rho = \mathcal{T}(\rho, R). \quad (2149)$$

(ii) Classification of curvature singularity universality classes:

$$\mathcal{S}_{\text{sing}} = \bigcup_{\alpha} \Sigma_{\alpha}. \quad (2150)$$

(iii) Derivation of entropy–curvature duality from first principles:

$$R \equiv \frac{\delta^2 S}{\delta \rho^2} + \dots. \quad (2151)$$

(iv) Connection between fractal information theory and quantum collapse:

$$\lim_{R \rightarrow \infty} \hat{\rho} \rightarrow \Pi(\hat{\rho}). \quad (2152)$$

(v) Computational tractability of multiscale geometric flows:

$$\mathcal{O}(N^\alpha \log N) \quad (2153)$$

for PFIT evolution algorithms.

Resolving these questions would elevate the PFIT–En-topia–Pellis framework from a predictive multiscale geometric theory to a complete mathematical structure for information dynamics, unifying entropy, curvature, and spectral instability under a single renormalized principle of fractal informational evolution.

29 Applications and Practical Implications

The PFIT–En-topia–Pellis framework provides a unified geometric–informational language for describing instability, collapse, and phase transitions across complex systems. Its practical relevance emerges from the ability to map heterogeneous dynamical processes onto a single curvature–entropy manifold structure, thereby enabling predictive modeling of critical transitions across physical, biological, computational, and social domains [1–3, 7–9, 28, 82].

More generally, each system $\mathcal{S}$ is embedded into an information manifold triple $(\mathcal{M}_{\mathcal{S}}, g_{\mathcal{S}}, \rho_{\mathcal{S}})$, with an associated collapse generator

$$\mathcal{C}_{\mathcal{S}} = -\Delta_{g_{\mathcal{S}}} + \lambda_{\mathcal{S}} R_{\mathcal{S}}(x, t) + \gamma_{\mathcal{S}} \rho_{\mathcal{S}}(x, t) + \beta_{\mathcal{S}} \log \rho_{\mathcal{S}}(x, t), \quad (2154)$$

whose spectral and variational properties govern system-specific instability regimes.

29.1 Biomedical and Physiological Systems

In physiological dynamics, especially in cardiac and neural systems, the framework interprets pathological transitions as curvature-induced informational collapse on an evolving statistical manifold. Let $\rho(t)$ represent a physiological state variable (e.g., ECG-derived phase density), then the evolution is governed by a stochastic geometric flow

$$d\rho = [-\Delta_g \rho + \lambda R(t) \rho + \alpha \nabla \cdot (\rho \nabla \log \rho)] dt + \sigma dW(t), \quad (2155)$$

where $R(t)$ encodes latent physiological curvature of the state space and $dW(t)$ represents multiscale physiological noise.

The associated entropy production rate satisfies the geometric identity

$$\frac{d}{dt} S[\rho(t)] = \int_{\mathcal{M}} |\nabla \log \rho|^2 \rho dV_g + \lambda \int_{\mathcal{M}} R(t) \rho dV_g, \quad (2156)$$

linking entropy amplification directly to curvature concentration.

Critical transitions such as ventricular fibrillation or epileptic seizures correspond to divergence in the curvature functional:

$$\lim_{t \rightarrow t_c} \|R(t)\|_{L^\infty} \rightarrow \infty, \quad \lim_{t \rightarrow t_c} \|\nabla S[\rho]\| \rightarrow \infty. \quad (2157)$$

A clinically relevant predictive biomarker is given by the instability index

$$\mathcal{B}(t) = \frac{\lambda_{\max}(\mathcal{C}(t))}{S[\rho(t)] + \epsilon} + \eta \text{Var}(R(t)), \quad (2158)$$

where divergence of $\mathcal{B}(t)$ signals imminent physiological breakdown.

29.2 Neuroscience and Cognitive Dynamics

In neural systems, PFIT interprets cognition as a trajectory on a high-dimensional information manifold $\mathcal{M}_{\text{brain}}$. Neural synchronization patterns correspond to curvature localization phenomena governed by

$$R_{\text{neural}}(x, t) \sim g^{ij}(t) \partial_i \log \rho_N \partial_j \log \rho_N + \Delta_g \log \rho_N. \quad (2159)$$

Cognitive collapse events (e.g., seizures, attention breakdown) occur when curvature amplification enters a nonlinear feedback regime:

$$\partial_t R_{\text{neural}} = \alpha R_{\text{neural}}^2 + \beta \nabla^2 R_{\text{neural}} + \gamma \frac{dS_N}{dt}, \quad (2160)$$

leading to abrupt reduction of effective dimensionality.

29.3 Plasma Physics and Fusion Systems

In magnetically confined plasma systems, instability regimes are reformulated as curvature–entropy resonance conditions. Let $\psi(x, t)$ denote plasma density; then

$$\partial_t \psi = D \Delta \psi + \alpha R(x, t) \psi - \beta \psi^3 + \nabla \cdot (\psi \nabla R). \quad (2161)$$

Disruption events correspond to spectral destabilization of the associated collapse operator

$$\mathcal{C}_{\text{plasma}} = -\Delta + \lambda R(x, t) + \gamma \psi. \quad (2162)$$

The framework predicts a universal precursor scaling law

$$\tau_{\text{precursor}} \sim |R|^{-1/2} \sim \left| \frac{dS}{dt} \right|^{-1}, \quad (2163)$$

providing early-warning indicators for plasma disruption.

29.4 Materials Science and Structural Failure

In material deformation processes, PFIT interprets fracture initiation as entropy localization on a curved information manifold. Let $\sigma(x, t)$ denote stress field, then

$$\partial_t \sigma = \nabla \cdot (\kappa \nabla \sigma) + \lambda R(x, t) \sigma + \mu \sigma \log \sigma. \quad (2164)$$

Failure occurs when curvature concentration exceeds a critical threshold:

$$\int_{\Omega} |R(x, t)|^p dV \geq R_{\text{crit}}(p), \quad p \geq 1, \quad (2165)$$

signaling loss of structural stability across scales.

29.5 Artificial Intelligence and Complex Computation

In computational systems, PFIT provides a geometric interpretation of learning instability in deep networks. Let θ denote model parameters, then learning dynamics are described by a curvature-regularized gradient flow

$$\dot{\theta} = -\nabla_{\theta} \mathcal{L}(\theta) + \lambda R(\theta) \theta + \alpha \nabla_{\theta} \cdot \nabla_{\theta} \mathcal{L}. \quad (2166)$$

Overfitting corresponds to curvature collapse in parameter space, characterized by divergence of the Hessian spectrum:

$$\sup_k |\Lambda_k(\nabla_{\theta}^2 \mathcal{L})| \rightarrow \infty, \quad \text{Tr}(\nabla_{\theta}^2 \mathcal{L}) \rightarrow \infty. \quad (2167)$$

29.6 Complex Networks and Social Systems

For complex networks, let $A_{ij}(t)$ denote adjacency dynamics. The curvature of the network manifold is approximated via a spectral embedding:

$$R_{\text{net}} \sim \sum_{i,j} \frac{A_{ij}^2}{k_i k_j} + \sum_k \lambda_k^2(A), \quad (2168)$$

linking local connectivity to global spectral rigidity.

Systemic collapse (e.g., cascading failures) occurs when

$$\lambda_{\max}(A) \rightarrow \infty, \quad \Delta \lambda(A) \rightarrow 0, \quad (2169)$$

indicating loss of spectral stability and network coherence.

29.7 Unified Practical Principle

Across all domains, the PFIT framework predicts a universal operational principle:

$$\text{Critical Transition} \iff \frac{R(x, t)}{S[\rho(x, t)] + \epsilon} + \|\nabla S[\rho]\| \rightarrow \infty. \quad (2170)$$

Equivalently, collapse is characterized by the simultaneous divergence of curvature, entropy gradient, and spectral instability:

$$\mathcal{C}_{\text{unified}}(t) = |R| + \|\nabla S\| + \|\Lambda\| \rightarrow \infty, \quad (2171)$$

establishing a domain-independent predictive invariant for early detection of collapse phenomena across all multiscale systems [82, 101–104, 114–117].

30 Positioning within Existing Entropy and Geometric Theories

To clarify the conceptual and mathematical positioning of the PFIT framework, we explicitly compare its structure against the main existing entropy, information-theoretic, and geometric formulations in the literature. This section is not merely descriptive but establishes the precise regime in which PFIT extends, unifies, and generalizes prior theories through an operator–geometric closure principle on information manifolds, in which entropy production, curvature evolution, and spectral instability are treated as mutually coupled dynamical degrees of freedom [1–3, 7–9, 28, 82].

30.1 Shannon Entropy as the Linear Limit

Classical Shannon entropy arises as the linear, non-geometric limit of the PFIT entropy operator under flat manifold conditions and weak coupling between geometry and information density. Specifically, when the information manifold $(\mathcal{M}, g)$ is Euclidean, curvature effects vanish $R(x) \rightarrow 0$, and nonlinear fractal corrections are neglected, the PFIT functional reduces to

$$\mathcal{H}_{\text{PFIT}}[\rho] \longrightarrow - \int_{\mathcal{M}} \rho(x) \log \rho(x) dV, \quad (2172)$$

recovering Shannon entropy as a zeroth-order projection of the full operator algebra.

More precisely, this corresponds to the asymptotic regime in which the entropy operator admits a linearization

$$\mathcal{H}_{\text{PFIT}}[\rho] = \mathcal{H}_0[\rho] + \epsilon \mathcal{H}_1[\rho] + \mathcal{O}(\epsilon^2), \quad \epsilon \rightarrow 0, \quad (2173)$$

with $\mathcal{H}_0$ identified with Shannon entropy and all higher-order curvature–spectral couplings suppressed.

30.2 Rényi Entropy and Multifractal Statistics

Rényi entropy emerges naturally within PFIT as a coarse-grained projection of the full operator spectrum onto scale-dependent density modes. In particular, the Rényi index q corresponds to a renormalization parameter controlling spectral weighting across fractal subspaces:

$$H_q(\rho) = \frac{1}{1-q} \log \int_{\mathcal{M}} \rho(x)^q dV_g. \quad (2174)$$

Within PFIT, this is interpreted as a spectral moment reduction of the full entropy Hamiltonian $\mathcal{H}_{\text{PFIT}}$, obtained via

$$H_q(\rho) \sim \log (\text{Tr} [\rho^q \Pi_q]), \quad (2175)$$

where Π_q denotes a projection onto L^q -stable spectral subspaces of the information manifold.

Thus, Rényi entropy corresponds to a partial spectral observable

$$H_q \equiv \mathcal{P}_q(\sigma(\mathcal{H}_{\text{PFIT}})), \quad (2176)$$

rather than a fundamental generator of dynamics.

30.3 Tsallis Entropy and Non-Extensive Thermodynamics

Tsallis entropy arises as a non-extensive deformation of Shannon entropy and corresponds in PFIT to the first-order truncation of the nonlinear operator algebra under weak curvature–entropy coupling. The Tsallis parameter q is reinterpreted as a curvature deformation index governing deviation from additivity:

$$S_q(\rho) = \frac{1}{q-1} \left(1 - \int_{\mathcal{M}} \rho(x)^q dV_g \right). \quad (2177)$$

In PFIT, this emerges from a perturbative expansion of the entropy operator under weak geometric backreaction:

$$\mathcal{H}_{\text{PFIT}} = \mathcal{H}_{\text{Shannon}} + \alpha R\rho + \beta(\nabla\rho)^2 + \mathcal{O}(R^2), \quad (2178)$$

where Tsallis statistics correspond to resummation of leading nonlinear terms in the curvature–entropy coupling hierarchy.

30.4 Information Geometry (Amari) as Metric Substructure

Information geometry is fully embedded within PFIT as the Riemannian first-order reduction of the full information manifold structure. The Fisher–Rao metric

$$g_{ij}(\theta) = \mathbb{E} [\partial_i \log p(x; \theta) \partial_j \log p(x; \theta)] \quad (2179)$$

corresponds to the infinitesimal curvature tensor of the probability density field under local equilibrium conditions:

$$g_{ij} \sim \nabla_i \nabla_j S[\rho]. \quad (2180)$$

However, PFIT extends this structure by promoting the metric to a dynamical field obeying

$$\partial_t g_{ij} = -\kappa R_{ij} + \lambda \nabla_i \rho \nabla_j \rho + \gamma g_{ij} S[\rho], \quad (2181)$$

thereby generalizing static information geometry into a curvature–entropy driven evolution system.

30.5 PFIT as Operator Closure and Curvature-Coupled Spectral Dynamics

In contrast to scalar entropy theories or Riemannian statistical models, PFIT is defined as a closed operator algebra acting on probability densities over a curved information manifold:

$$\mathcal{H}_{\text{PFIT}} : \mathcal{D}(\mathcal{M}) \rightarrow L^2(\mathcal{M}). \quad (2182)$$

Its defining property is closure under a coupled hierarchy of transformations:

$$\mathcal{C}_{\text{PFIT}} = \{\mathcal{O}_S, \mathcal{O}_R, \mathcal{O}_\Delta, \mathcal{R}_\ell\}, \quad (2183)$$

where

$$\mathcal{O}_S : \text{entropy-generating nonlinear operators}, \quad (2184)$$

$$\mathcal{O}_R : \text{curvature deformation operators}, \quad (2185)$$

$$\mathcal{O}_\Delta : \text{spectral Laplace–Beltrami generators}, \quad (2186)$$

$$\mathcal{R}_\ell : \text{renormalization group transformations}. \quad (2187)$$

The closure condition is expressed as

$$[\mathcal{O}_i, \mathcal{O}_j] \in \text{span}(\mathcal{C}_{\text{PFIT}}), \quad (2188)$$

ensuring algebraic stability under composition, scale transformation, and spectral projection.

30.6 Summary of Theoretical Hierarchy

The relationships between PFIT and existing frameworks can be summarized through a nested limit structure:

$$\text{PFIT} \xrightarrow{R \rightarrow 0} \text{Shannon} \xrightarrow{q \neq 1} \text{Renyi} \xrightarrow{\text{non-extensive deformation}} \text{Tsallis} \xrightarrow{\text{local equilibrium}} \text{Information Geometry}. \quad (2189)$$

Equivalently, PFIT can be viewed as the universal generating structure

$$\text{PFIT} = \langle \text{entropy functional} + \text{curvature dynamics} + \text{spectral operator algebra} \rangle_{\text{RG closure}}. \quad (2190)$$

From this perspective, PFIT does not compete with existing theories, but reconstructs them as boundary regimes of a unified operator–geometric structure governed by entropy-induced spectral dynamics, curvature back-reaction, and renormalization closure across scales [82, 101–104, 114–117].

Table 18: Position of PFIT Within Existing Information and Geometric Theories

Framework	Primary Object	Dynamics	Collapse Description
Shannon Theory	Entropy	No	None
Rényi Theory	Generalized Entropy	No	None
Tsallis Theory	Non-Extensive Entropy	Limited	None
Information Geometry	Statistical Manifold	Partial	Not Explicit
Geometric Flow Theory	Curvature	Yes	Geometric Singularities
Spectral Theory	Eigenmodes	Yes	Spectral Instability
Renormalization Theory	Scale Structure	Yes	Critical Transition
Fractal Information Theory	Fractal Measures	Partial	Scale Breakdown
PFIT Framework	Information–Entropy–Geometry	Yes	Unified Collapse Theory
PFIT–En-topia–Pellis	Unified Information Field	Yes	Universal Collapse Mechanism

31 Discussion

The PFIT–En-topia–Pellis framework establishes a unifying geometric–informational paradigm in which entropy, curvature, and fractal structure are not independent descriptors but mutually coupled manifestations of a single underlying manifold dynamics. The central conceptual advance lies in the identification of curvature as the primary driver of informational instability, extending classical information geometry into a nonlinear, multiscale, and fractal regime consistent with entropy projection mechanisms and collapse dynamics [1–3, 7–9, 28, 82].

More formally, the system can be viewed as an operator–variational triple $(\mathcal{M}, g, \mathcal{C})$ governed by a coupled flow:

$$\partial_t \rho = -\frac{\delta \mathcal{A}}{\delta \rho} + \mathcal{G}[R, \rho] + \xi(x, t), \quad \partial_t g_{ij} = -\frac{\delta \mathcal{A}}{\delta g^{ij}} + \mathcal{K}_{ij}[\rho], \quad (2191)$$

where $\mathcal{G}$ and $\mathcal{K}_{ij}$ encode curvature–entropy backreaction terms.

31.1 Unification of Entropy and Geometry

A key theoretical implication is the emergence of an equivalence class structure between entropy gradients and curvature tensors. This is formalized through the asymptotic relation

$$\nabla_i S[\rho] \sim g^{jk} \nabla_i R_{jk} + \alpha \nabla_i \log \rho, \quad (2192)$$

where entropy flow is directly coupled to geometric stress-energy of the information manifold.

At the level of effective field theory, the informational action

$$\mathcal{A}[\rho, g] = \int_{\mathcal{M}} (R\rho + g^{ij} \partial_i \rho \partial_j \rho + \rho \log \rho + \beta R^2) dV_g \quad (2193)$$

defines a nonlinear constrained variational system whose Euler–Lagrange equations yield coupled entropy–curvature evolution equations:

$$\frac{\delta \mathcal{A}}{\delta \rho} = 0, \quad \frac{\delta \mathcal{A}}{\delta g^{ij}} = 0. \quad (2194)$$

This establishes entropy maximization and curvature minimization as dual variational principles embedded in a single geometric functional.

31.2 Role of Fractality and Scale Invariance

Fractality introduces a scale-dependent deformation of the geometric structure via renormalization group flow:

$$g_{ij}(x; \ell) = \ell^\alpha g_{ij}(x) + \ell^{\alpha-1} \Delta g_{ij}(x) + \mathcal{O}(\ell^{\alpha-2}), \quad (2195)$$

where ℓ denotes resolution scale.

Under this transformation, curvature obeys a nonlinear scaling hierarchy:

$$R(x; \ell) \sim \ell^{-\beta_1} R(x) + \ell^{-\beta_2} R^2(x) + \ell^{-\beta_3} \nabla^2 R(x), \quad (2196)$$

linking geometric singularity formation with fractal dimensional collapse.

The effective fractal dimension satisfies the flow equation

$$\frac{dd_f}{d \log \ell} = -\gamma_1 d_f + \gamma_2 \int_{\mathcal{M}} |\nabla \rho|^2 dV_g, \quad (2197)$$

implying that dimensional reduction is driven by entropy concentration.

31.3 Spectral Interpretation of Collapse

From a spectral perspective, collapse corresponds to condensation of eigenmodes of the nonlinear operator

$$\mathcal{C} = -\Delta_g + \lambda R(x) + \gamma \rho(x) + \eta \log \rho(x), \quad (2198)$$

into a singular dominant eigenstate.

The spectral measure evolves according to

$$\partial_t \rho_{\mathcal{C}}(\Lambda, t) = -\partial_\Lambda (v(\Lambda, t) \rho_{\mathcal{C}}) + D_\Lambda \partial_\Lambda^2 \rho_{\mathcal{C}}, \quad (2199)$$

where $v(\Lambda, t)$ encodes curvature-induced spectral drift.

Collapse occurs when spectral rigidity diverges:

$$\lim_{t \rightarrow t_c} \int (\Lambda - \bar{\Lambda})^2 \rho_{\mathcal{C}}(\Lambda, t) d\Lambda \rightarrow \infty, \quad (2200)$$

together with loss of Weyl-law scaling

$$N(\Lambda) \not\sim \Lambda^{n/2}. \quad (2201)$$

31.4 Entropy–Curvature Feedback Loop

A central mechanism is the nonlinear feedback loop between entropy concentration and curvature amplification:

$$S[\rho] \rightarrow R(x) \rightarrow \nabla S[\rho] \rightarrow R(x), \quad (2202)$$

forming a closed instability cycle.

This feedback is governed by a coupled Riccati–diffusion system:

$$\frac{dR}{dt} = aR + bR^2 + c\Delta R - d\frac{dS}{dt}, \quad (2203)$$

which admits finite-time blow-up solutions when

$$bR^2 \gg |aR| + |c\Delta R|. \quad (2204)$$

31.5 Relation to Existing Theoretical Frameworks

The PFIT formulation generalizes classical information geometry by promoting the Fisher–Rao metric to a dynamical field:

$$g_{ij} = \mathbb{E}[\partial_i \log \rho \partial_j \log \rho] \longrightarrow g_{ij}(t, x). \quad (2205)$$

It also extends thermodynamic geometry by coupling entropy production to curvature flow:

$$\partial_t S = \int_{\mathcal{M}} (|\nabla \log \rho|^2 + \kappa R \rho) dV_g, \quad (2206)$$

thereby embedding classical frameworks as restricted projections of the full operator dynamics.

31.6 Conceptual Implications

A key conceptual outcome is the reinterpretation of collapse as a geometric phase transition rather than a failure mode. In this view, collapse corresponds to a renormalized reduction of effective degrees of freedom:

$$\dim(\mathcal{M}) \longrightarrow d_{\text{eff}}(t) \ll \dim(\mathcal{M}), \quad (2207)$$

with evolution law

$$\frac{dd_{\text{eff}}}{dt} = -\alpha \|\nabla S\|^2 - \beta \|R\|. \quad (2208)$$

This dimensional contraction is accompanied by spectral condensation:

$$\sum_{k \neq 0} a_k(t) \rightarrow 0, \quad \rho(x, t) \rightarrow \phi_0(x), \quad (2209)$$

indicating collapse onto a single dominant informational mode.

31.7 Final Perspective

Overall, the PFIT–En-topia–Pellis framework suggests that complex systems across disciplines are governed by a single underlying variational principle linking entropy maximization, curvature evolution, and fractal scaling. In compact operator form, this is expressed as

$$\mathcal{D}_{\text{PFIT}} = \langle \mathcal{H}_{\text{entropy}} + \mathcal{H}_{\text{curvature}} + \mathcal{H}_{\text{spectral}} \rangle_{\text{RG}}, \quad (2210)$$

implying that diverse phenomena—from physical phase transitions to biological instability and computational breakdown—emerge as projections of a universal geometric–informational law [82, 101–104, 114–117].

Table 19: Core Unification Achievements of the PFIT Framework

Concept	Unified Description
Entropy	Geometric field
Information	Manifold dynamics
Fractality	Scale geometry
Spectrum	Information eigenmodes
Collapse	Curvature singularity
Prediction	Universal instability law

32 Conclusion

The Unified Information Manifold Theory of Fractal Information, Entropy Projections, and Curvature-Driven Collapse (PFIT–En-topia–Pellis Framework) establishes a coherent and mathematically structured synthesis of information geometry, fractal dynamics, and nonlinear curvature-driven evolution. The central thesis is that informational systems are fundamentally described not as static probabilistic ensembles, but as evolving geometric–operator manifolds $(\mathcal{M}, g_{ij}(t), \rho(x, t))$ whose curvature, spectral structure, and entropy flow are mutually coupled dynamical degrees of freedom [1–3, 7–9, 28, 82].

In this formulation, system evolution is governed by a coupled nonlinear geometric flow:

$$d\rho = (-\Delta_g \rho + \lambda R(x, t) \rho + \gamma \rho \log \rho + \mathcal{E}_{\text{syn}}(\rho)) dt + \sigma dW(t), \quad (2211)$$

with curvature feedback encoded through a metric evolution law of the form

$$\partial_t g_{ij} = -\kappa R_{ij} + \eta \nabla_i \nabla_j S[\rho] + \mu g_{ij} \rho, \quad (2212)$$

establishing a fully coupled entropy–geometry dynamical closure.

At the core of the framework lies the curvature–entropy correspondence principle, generalized as an operator identity in the tangent bundle of the information manifold:

$$R(x, t) \sim \frac{\delta S[\rho]}{\delta \rho(x, t)} + \mathcal{O}(\nabla^2 \rho, |\nabla \rho|^2), \quad (2213)$$

which promotes curvature from a passive geometric invariant to an active generator of informational dynamics.

Equivalently, the full informational action functional admits a variational closure structure:

$$\mathcal{A}[\rho, g] = \int_{\mathcal{M}} (R\rho + |\nabla \rho|^2 + \rho \log \rho + \beta R^2) dV_g, \quad (2214)$$

with Euler–Lagrange equations

$$\frac{\delta \mathcal{A}}{\delta \rho} = 0, \quad \frac{\delta \mathcal{A}}{\delta g^{ij}} = 0, \quad (2215)$$

defining a closed geometric–informational field theory.

Across all developed sections, a universal nonlinear flow structure emerges:

$$\partial_t \rho = -\Delta_g \rho + \lambda R(x, t) \rho + \mathcal{N}(\rho) + \nabla_i (\rho \nabla^i \log \rho), \quad (2216)$$

indicating that diffusion, curvature amplification, entropy compression, and nonlinear self-interaction jointly determine evolution across scales.

A central outcome is the identification of collapse phenomena as finite-time singularities in a coupled entropy–curvature–spectral field:

$$\lim_{t \rightarrow t_c} \left(\sup_{x \in \mathcal{M}} |R(x, t)| + \left| \frac{dS}{dt} \right| + \|\Lambda_{\max}(t)\| \right) \rightarrow \infty, \quad (2217)$$

which unifies critical transitions across physical, biological, and computational systems under a single geometric instability mechanism.

The inclusion of fractal information structure extends classical information geometry into a multiscale renormalization hierarchy:

$$(\mathcal{M}, g, \rho) \longrightarrow (\mathcal{M}_\ell, g_\ell, \rho_\ell), \quad \ell \rightarrow 0, \quad (2218)$$

with scale-dependent curvature flow

$$R_\ell(x) \sim \ell^{-\alpha} R(x) + \ell^{-\beta} |\nabla \rho|^2, \quad (2219)$$

implying self-similar amplification of instability across resolution levels.

Spectral analysis further reinforces universality, showing that collapse corresponds to eigenvalue condensation of the operator

$$\mathcal{C} = -\Delta_g + \lambda R(x, t) + \gamma \rho(x, t), \quad (2220)$$

with spectral density evolution

$$\rho_{\mathcal{C}}(\Lambda, t) \sim \Lambda^{-\mu(t)} \exp(-\kappa S_{\text{eff}}(t)), \quad (2221)$$

providing a unified signature of instability across domains.

From a conceptual standpoint, PFIT replaces isolated domain-specific instability models with a single geometric–informational principle: systems evolve toward curvature concentration, spectral condensation, and entropy extremization, with collapse emerging as a universal phase transition in informational geometry rather than a domain-dependent failure mode.

Future developments will refine the axiomatic closure of the theory, strengthen its quantum–stochastic embedding via operator-valued curvature flows,

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \mathcal{D}_g(\hat{\rho}) + \mathcal{F}_R(\hat{\rho}), \quad (2222)$$

and expand computational realizability of multiscale geometric solvers on fractal manifolds. Nevertheless, the present formulation already demonstrates that entropy, geometry, and fractality are not independent constructs but co-emergent projections of a unified operator–geometric dynamics.

In this sense, the PFIT–En-topia–Pellis framework provides a foundational step toward a general theory of informational physics, in which curvature acts as the primary carrier of informational dynamics and collapse emerges as its most extreme and diagnostically informative manifestation [82, 101–104, 114–117].

References

- [1] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal (1948).
<https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
- [2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley (2006).
<https://doi.org/10.1002/047174882X>
- [3] E. T. Jaynes, *Information Theory and Statistical Mechanics*, Physical Review **106**, 620 (1957).
<https://doi.org/10.1103/PhysRev.106.620>
- [4] A. Rényi, *On Measures of Entropy and Information*, (1961).
- [5] C. Tsallis, *Possible Generalization of Boltzmann-Gibbs Statistics*, Journal of Statistical Physics **52**, 479–487 (1988).
<https://doi.org/10.1007/BF01016429>
- [6] J. Havrda and F. Charvát, *Quantification Method of Classification Processes*, Kybernetika **3**, 30–35 (1967).
- [7] S. Amari, *Information Geometry and Its Applications*, Springer (2016).
<https://doi.org/10.1007/978-4-431-55978-8>
- [8] F. Nielsen, *An Elementary Introduction to Information Geometry*, Entropy **22**, 880 (2020). <https://doi.org/10.3390/e22080880>
- [9] C. R. Rao, *Information and Accuracy Attainable in the Estimation of Statistical Parameters*, Bulletin of the Calcutta Mathematical Society **37**, 81–91 (1945).
- [10] B. B. Mandelbrot, *The Fractal Geometry of Nature*, W. H. Freeman (1982).
- [11] J. Feder, *Fractals*, Springer (1988).
- [12] H. Takayasu, *Fractals in the Physical Sciences*, Manchester University Press (1990).
- [13] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, Westview Press (2018).
- [14] E. Ott, *Chaos in Dynamical Systems*, Cambridge University Press (2002).
- [15] M. Scheffer, *Critical Transitions in Nature and Society*, Princeton University Press (2009).
- [16] K. G. Wilson, *Renormalization Group and Critical Phenomena*, Physical Review B **4**, 3174 (1971).
<https://doi.org/10.1103/PhysRevB.4.3174>
- [17] N. Goldenfeld, *Lectures on Phase Transitions and the Renormalization Group*, Addison-Wesley (1992).
- [18] P. Bak, *How Nature Works: Self-Organized Criticality*, Springer (1996).
- [19] H. Haken, *Synergetics: An Introduction*, Springer (1983).
- [20] I. Prigogine, *The End of Certainty*, Free Press (1997).
- [21] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, Springer (2004).
- [22] M. Reed and B. Simon, *Methods of Modern Mathematical Physics*, Academic Press (1980).
- [23] G. Teschl, *Mathematical Methods in Quantum Mechanics*, AMS (2014).
- [24] T. Kato, *Perturbation Theory for Linear Operators*, Springer (1995).
- [25] B. J. West, *Fractal Physiology and Complex Systems*, World Scientific (2010).
- [26] S. Lloyd, *Measures of Complexity*, MIT Press (2001).
- [27] Y. Bar-Yam, *Dynamics of Complex Systems*, Westview Press (2003).

- [28] S. Kullback and R. A. Leibler, *On Information and Sufficiency*, Annals of Mathematical Statistics (1951).
<https://doi.org/10.1214/aoms/1177729694>
- [29] M. C. Cross and P. C. Hohenberg, *Pattern Formation Outside of Equilibrium*, Reviews of Modern Physics **65**, 851 (1993).
<https://doi.org/10.1103/RevModPhys.65.851>
- [30] S. Pellis, *Unification of Archimedes Constant π , Golden Ratio ϕ , Euler's Number e , and Imaginary Number i* , SSRN (2021).
<https://doi.org/10.2139/ssrn.3975869>
- [31] S. Pellis, *Exact Formula for the Fine-Structure Constant α in Terms of the Golden Ratio ϕ* , SSRN (2021).
<https://doi.org/10.2139/ssrn.4160769>
- [32] S. Pellis, *Fine-Structure Constant from the Golden Angle, the Relativity Factor and the Fifth Power of the Golden Mean*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4247343>
- [33] S. Pellis, *Exact Expressions of the Fine-Structure Constant*, SSRN (2021).
<https://doi.org/10.2139/ssrn.3975861>
- [34] S. Pellis, *Fine-Structure Constant from the Archimedes Constant*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4245208>
- [35] S. Pellis, *Fine-Structure Constant from the Madelung Constant*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4174644>
- [36] S. Pellis, *Exact Mathematical Expressions of the Proton to Electron Mass Ratio*, SSRN (2021).
<https://doi.org/10.2139/ssrn.3967998>
- [37] S. Pellis, *Unity Formula that Connects the Fine-Structure Constant and the Proton to Electron Mass Ratio*, SSRN (2021).
<https://doi.org/10.2139/ssrn.3963425>
- [38] S. Pellis, *Exact Mathematical Formula that Connects 6 Dimensionless Physical Constants*, SSRN (2021).
<https://doi.org/10.2139/ssrn.3963427>
- [39] S. Pellis, *Theoretical Value for the Strong Coupling Constant*, SSRN (2022).
<https://doi.org/10.2139/ssrn.3998175>
- [40] S. Pellis, *Unity Formulas for the Coupling Constants and the Dimensionless Physical Constants*, J. High Energy Phys., Gravitation and Cosmology **9**, 245–294 (2023).
<https://doi.org/10.4236/jhepgc.2023.91021>
- [41] S. Pellis, *Dimensionless Unification of Fundamental Interactions*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4201780>
- [42] S. Pellis, *Unification of the Fundamental Interactions*, ResearchGate (2022).
<https://doi.org/10.13140/RG.2.2.12296.70405>
- [43] S. Pellis, *Unification of the Fundamental Forces*, ResearchGate (2022).
<https://doi.org/10.13140/RG.2.2.33651.60967>
- [44] S. Pellis, *Theoretical Value of the Gravitational Constant*, SSRN (2023).
<https://doi.org/10.2139/ssrn.4442074>

- [45] S. Pellis, *Formula for the Gravitational Constant*, ResearchGate (2023).
<https://doi.org/10.13140/RG.2.2.19656.60166>
- [46] S. Pellis, *Dimensionless Solution for the Cosmological Constant*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4219292>
- [47] S. Pellis, *Unification of Atomic Physics and Cosmology*, ResearchGate (2022).
<https://doi.org/10.13140/RG.2.2.11493.88804>
- [48] S. Pellis, *Maximum and Minimum Values for Natural Quantities*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4306280>
- [49] S. Pellis, *Coupling Constant for the Fifth Force*, ResearchGate (2023).
<https://doi.org/10.13140/RG.2.2.32481.99686>
- [50] S. Pellis, *Gravitational Fine-Structure Constant*, ResearchGate (2022).
<https://doi.org/10.13140/RG.2.2.34770.43206>
- [51] S. Pellis, *Unification of the Microcosm and the Macrocosm*, SSRN (2023).
<https://doi.org/10.2139/ssrn.4429653>
- [52] S. Pellis, *The Equation of the Universe*, ResearchGate (2023).
<https://doi.org/10.13140/RG.2.2.24768.40960>
- [53] S. Pellis, *Dimensionless Theory of Everything*, SSRN (2023).
<https://doi.org/10.2139/ssrn.4469668>
- [54] S. Pellis, *Dimensionless Equations of the Universe*, SSRN (2023).
<https://doi.org/10.2139/ssrn.4488688>
- [55] S. Pellis, *Poincaré Dodecahedral Space Solution of the Shape of the Universe*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4316617>
- [56] S. Pellis, *The Shape of the Universe*, ResearchGate (2023).
<https://doi.org/10.13140/RG.2.2.16034.09922>
- [57] S. Pellis, *Solution for the Density Parameter of Dark Energy*, SSRN (2022).
<https://doi.org/10.2139/ssrn.4406279>
- [58] S. Pellis, *Solution for Cosmological Parameters*, ResearchGate (2022).
<https://doi.org/10.13140/RG.2.2.23917.67047>
- [59] S. Pellis, *Equation of State in Cosmology*, ResearchGate (2022).
<https://doi.org/10.13140/RG.2.2.17952.25609>
- [60] S. Pellis, *Euler's Identity in Unification of the Fundamental Interactions*, SSRN (2024).
<https://doi.org/10.2139/ssrn.4702100>
- [61] A. K. Golmankhaneh, S. Pellis, M. Zingales, *Fractal Schrödinger Equation: Implications for Fractal Sets*, J. Phys. A: Math. Theor. **57**(18), 185201 (2024).
- [62] G. P. Gregori, M. T. Hovland, B. A. Leybourne, S. Pellis, V. Straser, B. G. Gregori, G. M. Gregori, A. R. Simonelli, *Air–Earth Currents and a Universal “Law”: Filamentary and Spiral Structures – Repetitiveness, Fractality, Golden Ratio, Fine-Structure Constant, Antifragility and “Statistics” – The Origin of Life*, New Concepts in Global Tectonics **3**(1), 106–122 (2025).

- [63] M. Guettari, A. El Aferni, L. Ajroudi, T. Tajouri, S. Pellis, *A Semiclassical Model to Study the Effect of Glucose Confinement on AOT/Water/Isooctane Reverse Micelles*, J. Mol. Liquids **436**, 128267 (2025).
<https://doi.org/10.1016/j.molliq.2025.128267>
- [64] S. Pellis, *Golden Function Model of the Origin of the Fine-Structure Constant*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5387893>
- [65] S. Pellis, *Golden Function in the Architecture of the Parthenon*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5389569>
- [66] S. Pellis, *Fractal Seismic Scaling Laws of the Golden Function*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5394135>
- [67] S. Pellis, *Unified Golden Model of DNA Fractal Geometry*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5401017>
- [68] S. Pellis, *Unified Particle Equation Towards a Fractal Standard Model*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5407924>
- [69] S. Pellis, *Unified Golden Biomedical Function*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5418754>
- [70] S. Pellis, *Temporal Quantization in the Unified Fractal Theory of Time*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5429935>
- [71] S. Pellis, *Operatorial Fractal Quantization of Black Hole Entropy*, ResearchGate (2025).
<https://doi.org/10.13140/RG.2.2.30391.59042>
- [72] S. Pellis, *Fine-Structure Constant via Golden Spheres Resonance*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5473588>
- [73] S. Pellis, *Fractal Code of Quantum Entanglement*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5522898>
- [74] S. Pellis, *Golden Fractals in Fluid Dynamics and Turbulence*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5543579>
- [75] S. Pellis, *Emergence of Physical Laws from Golden-Fractal Topologies*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5583612>
- [76] S. Pellis, *Quantum Gravity Phenomenology from Golden Fractal Scales*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5609370>
- [77] S. Pellis, *Golden-Fractal Architectures for Nanophotonics*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5664311>
- [78] S. Pellis, *Spectral Geometry of Golden-Fractal Laplacians*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5709164>
- [79] S. Pellis, *Fractal Origins of the Cosmic Microwave Background Spectrum*, ResearchGate (2025).
<https://doi.org/10.13140/RG.2.2.15753.71520>
- [80] S. Pellis, *Golden Code of Precursor Earthquakes*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5788462>
- [81] S. Pellis, *Unified Geomagnetism and Planetary Magnetic Models*, ResearchGate (2025).
<https://doi.org/10.13140/RG.2.2.19792.98560>

- [82] S. Pellis, *Toward a Universal Theory of Fractal Information*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5939098>
- [83] S. Pellis, *Universal Fractal Biomarkers for Early Disease Detection*, SSRN (2025).
<https://doi.org/10.2139/ssrn.5991215>
- [84] S. Pellis, *Fractal Laws of Cancer Growth and Metastasis*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6041734>
- [85] S. Pellis, *Fractal Principles of Cellular Systems for Unified Biology*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6087906>
- [86] S. Pellis, *Unified Fractal Geometry of Consciousness Architectures*, ResearchGate (2026).
<https://doi.org/10.13140/RG.2.2.14276.90241>
- [87] A. Petojević and S. Pellis, *From the Kurepa Hypothesis to the Fine-Structure Constant*, ResearchGate (2026).
<https://doi.org/10.13140/RG.2.2.34709.38886>
- [88] S. Pellis, *Universal Generating Principle for Natural Laws*, ResearchGate (2026).
<https://doi.org/10.13140/RG.2.2.11997.88805>
- [89] S. Pellis, *Fractal Dynamics and Predictive Modeling of Disease Progression*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6259659>
- [90] S. Pellis, *Axiomatic Golden-Fractal Field Theory*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6289218>
- [91] S. Pellis, *Quantum-Combinatorial Foundations and Spectra of Fractal Structures*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6305718>
- [92] S. Pellis, *Fractal Renormalization of the Unified Particle Equation*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6346798>
- [93] S. Pellis, *Fractal Gauge Emergence in Golden Scale Geometry*, ResearchGate (2026).
<https://doi.org/10.13140/RG.2.2.24848.67845>
- [94] S. Pellis, *Golden-Fractal Unified Quantum Gravity and Emergent Standard Model*, ResearchGate (2026).
<https://doi.org/10.13140/RG.2.2.32280.46083>
- [95] S. Pellis, *Spectral Origin of the Standard Model from Golden-Fractal Spacetime*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.16464.98567>
- [96] S. Pellis, *The Spectral Architecture of Protein Folding in Golden-Scale Fractal Networks*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.16366.68163>
- [97] S. Pellis, *Fractal Hierarchies in Neural Network Plasticity and Brain Connectivity*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.30335.68008>
- [98] S. Pellis, *Principles of Fractal Genomics in Epigenetic Regulation and Developmental Biology*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.15439.34728>
- [99] S. Pellis, *Multiscale Spectral-Fractal Analysis of Biological and Neural Networks*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6670938>
- [100] S. Pellis, *High-Performance Fractal Quantum Algorithms for Computation and Cryptography*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.26049.93283>

- [101] S. Pellis, *Quantum Chemical Modeling of Fractal Electron Distributions*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.14669.60644>
- [102] S. Pellis, *Fractal-Guided Quantum Bioengineering and Photosynthesis Efficiency*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.13077.08163>
- [103] S. Pellis, *Golden-Fractal Heart Rate Variability for Early Cardiovascular Risk Detection*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.11941.67043>
- [104] S. Pellis, *Advanced Fractal Combinatorics and Topological Data Analysis*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.31923.59680>
- [105] S. Pellis, *Experimental Validation and Falsifiable Predictions of the Golden-Fractal Function*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.24360.74244>
- [106] S. Pellis, *Generative Operator Framework for Fractal and Spectral Number Sets*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.30321.24164>
- [107] S. Pellis, *Multiscale Cardiac Variability for Cross-Domain Cardiovascular Event Prediction*, SSRN (2026).
<https://doi.org/10.2139/ssrn.6993358>
- [108] S. Pellis, *Personalized Medicine Protocols from Fractal Geometry of Multiscale Dynamical Systems*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7207779>
- [109] S. Pellis, *Fractal Sensor Networks for Smart Material Interfaces*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7130258>
- [110] S. Pellis, *Multiscale Fractal Monitoring Index for Early Prediction of Degradation in Rotating Machinery*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7213538>
- [111] S. Pellis, *Multiscale Fractal Stability Index for Early Detection of Respiratory Failure in Pulmonology*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7213700>
- [112] S. Pellis, *Geometry-Driven Fractal Scaling Laws for Energy Harvesting Materials*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.15430.64328>
- [113] S. Pellis, *Multiscale Fractal Architectures for Structural Optimization and Engineering Resilience*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.28963.82729>
- [114] S. Pellis, *Fractal Multiscale Early Warning System for Clinical Deterioration in Critical Care*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7234738>
- [115] S. Pellis, *Cross-Scale Fractal Instability Governing Cancer Growth and Therapy Resistance*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7267338>
- [116] S. Pellis, *Universal Precursor Index for Earthquake Forecasting via Multiscale Spectral Instability*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.28949.08169>
- [117] S. Pellis, *Multiscale Fractal Biomarkers of Neurodegenerative Collapse via a Universal Functional Operator Framework*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.36541.42722>

- [118] S. Pellis, *Retrospective Validation of the Universal Precursor Index Across Major Global Earthquakes*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.27903.78244>
- [119] S. Pellis, *Spectral-Fractal Biomarkers of Biological Aging and Pre-Disease Collapse*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.24975.55207>
- [120] S. Pellis, *Unified Medical Collapse Field Theory of Multiscale Spectral Instability and Physiological Criticality*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.33295.34728>
- [121] S. Pellis, *Spectral Stability Operator for Cascading Failures in Smart Grids under Multiscale Perturbations*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7329999>
- [122] S. Pellis, *Spectral Function–Driven Instability in Topological Phase Transitions under External Fields*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7330138>
- [123] S. Pellis, *Predictive Spectral Instability and Disruption Collapse in Magnetically Confined Fusion Plasmas*, SSRN (2026).
<https://doi.org/10.2139/ssrn.7330239>
- [124] S. Pellis, *Spectral Phase Transitions in Protein Folding Dynamics and the Emergence of Misfolding Pathologies*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.15684.18562>
- [125] S. Pellis, *Spectral Collapse as a Universal Mechanism for Quantum Criticality Decoherence and Open System Dynamics*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.31920.72967>
- [126] S. Pellis, *Spectral–Fractal Phase Transitions in Cardiac Dynamics from ECG Operator Flows*, ResearchGate, (2026).
<https://doi.org/10.13140/RG.2.2.12535.18083>